

Compressing Global Dynamics of Deep Nonlinear Feature Learning

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Abstract

Can a deep network’s entire learning trajectory be represented by a much smaller state? We study Gaussian-initialized nonlinear networks with learned hidden features, fixed depth, at least two training examples, a positive initial feature-Gram gap, and small fixed nonzero labels. The common principle is to compress the forward and backward responses that generate learning, rather than approximate a fitted weight matrix. Legendre compression summarizes their temporal histories, using $n^{5/4+o(1)}$ moving coordinates while retaining fixed dense mixers. Two realizations of a shared reduced nonlinear dynamics also compress the neuron coordinate: Harmonic compression uses joint time–input regularity and retains $O((\log n)^{3d+2})$ real coordinates over the whole input sphere; Logarithmic compression uses residual-adapted temporal approximation and retains $O((\log n)^3)$ real coordinates on any fixed panel of inputs declared at initialization. Passive labels are not used. Each tracks a coupled dense trajectory, uniformly over all physical times including the fitted limit, with error negligible in probability relative to independent dense-run variability. Harmonic and Logarithmic attain absolute error Y/n , where Y is the fixed RMS label scale. The proof retains Gram dissipation and finite residual activity; a readout correction cancels the feedback that would otherwise give a poor stability estimate. We expose the trade between query scope, moving memory and retained state, and separate initialization-only information from the work needed to process it.

1 Introduction

Compressing a trained function does not necessarily compress its learning. Two parameter states can agree on every input while their backward responses differ, so that the same next update separates their predictions. A description of learning must retain enough state to continue the dynamics, not merely reproduce one fitted snapshot.

Our target keeps depth, nonlinear activations and trained hidden representations. Infinite-width descriptions organize these dynamics through population variables and responses to reused random matrices [Yang and Hu, 2021, Nguyen and Pham, 2023, Bordelon and Pehlevan, 2022]. They motivate a further question: how much finite state is needed to reproduce the actual learning trajectory of a finite network? Here both the reference and the compressed predictor remain finite. Accuracy includes inputs absent from the updates and all physical time. The query domain is an explicit part of the guarantee: the whole sphere, or a declared finite panel whose passive labels are never supplied.

An intrinsic accuracy benchmark. Independent initializations already produce different dense trajectories. Our common compression theorem makes the ratio of compression error to this actual discrepancy tend to zero in probability. All three methods therefore track their dense reference more closely than independent dense runs track one another, in their respective query domains. This compares to realized variability, not an analytic upper envelope or a fixed-factor target. No

measurement of dense variability is needed to initialize the compressed model. The dense lower bound is witnessed during early training; it is not an endpoint lower bound.

One object to compress, two reduction mechanisms. A learned matrix increment integrates paired forward and backward responses. The first reduction, Legendre, compresses the history while retaining every neuron. The second compresses the response space itself and realizes its geometry on a small set of neuron coordinates. Harmonic and Logarithmic are two source builders for this same reduced dynamics. Harmonic represents responses jointly in time and input; Logarithmic represents only the finitely many requested input curves and spends fewer temporal pieces as learning slows. The latter removes the dimension-dependent logarithmic exponent by giving up a uniform guarantee for inputs first revealed after setup.

Why feedback is the central obstacle. These are closed learning systems, not playback of an approximated dense trajectory. A response error changes the next update, which changes the responses used thereafter. A black-box Lipschitz estimate treats this loop as exponential growth over an infinite horizon. The proof instead keeps the dissipative sign of the training Gram, measures forcing against the finite amount of remaining learning, and, for the selected models, cancels the leading readout–residual feedback. The main text explains these mechanisms before the appendix supplies their quantitative estimates.

Scope. The results allow arbitrary fixed depth, compatible finite sphere data, and analytic activations with unbounded values. Labels are small but fixed as width grows. The feature-learning mobilities are retained rather than replaced by a frozen-kernel scaling. This permits hidden learning; it does not assert nonzero feature motion for every activation and dataset. The guarantees concern prediction fidelity, not risk against unknown test labels. Sufficient stochastic widths remain qualitative. The logarithmic storage counts real coordinates, not fixed-precision words; its finite initialization-jet construction establishes information provenance, not a practical setup-time bound.

2 Setting and reading conventions

There are m training pairs (x_a, y_a) , with $\|x_a\|_2 = \sqrt{d}$, width n , and $L \geq 2$ hidden layers. We use the paper’s layer notation:

$$\begin{aligned} z_a^{(1)} &= W^{(1)} x_a / \sqrt{d}, & z_a^{(\ell)} &= W^{(\ell)} h_a^{(\ell-1)}, & h_a^{(\ell)} &= \phi(z_a^{(\ell)}), \\ f_{n,a} &= \frac{w^\top h_a^{(L)}}{n}, & r_a &= f_{n,a} - y_a, & \mathcal{L} &= \frac{1}{m} \sum_a r_a^2. \end{aligned} \quad (1)$$

Here $w = W^{(L+1)}$ is the output weight; the activation may vary by layer. Backward responses exclude the residual:

$$\delta_a^{(L)} = w \odot \phi'(z_a^{(L)}), \quad \delta_a^{(\ell)} = \phi'(z_a^{(\ell)}) \odot (W^{(\ell+1)})^\top \delta_a^{(\ell+1)}.$$

With block mobilities $(n, 1, \dots, 1, n)$ the exact gradient flow is

$$\begin{aligned}\dot{W}^{(1)} &= -\frac{2}{m} \sum_a r_a \delta_a^{(1)} x_a^\top / \sqrt{d}, \\ \dot{W}^{(\ell)} &= -\frac{2}{nm} \sum_a r_a \delta_a^{(\ell)} (h_a^{(\ell-1)})^\top, \quad 2 \leq \ell \leq L, \\ \dot{w} &= -\frac{2}{m} \sum_a r_a h_a^{(L)}.\end{aligned}\tag{2}$$

First-layer entries are independent $\mathcal{N}(0, 1)$, hidden entries are independent $\mathcal{N}(0, 1/n)$, blocks are independent, and $w(0) = 0$. Activations are real on the real axis, analytic in a strip about it, and have bounded first derivative there. Their values need not be bounded.

Data and labels. The population initial feature Gram (an uncentered second moment) is defined by

$$Q_{ab}^{(0)} = x_a^\top x_b / d, \quad Q_{ab}^{(\ell)} = \mathbb{E}[\phi(Z_a) \phi(Z_b)], \quad Z \sim \mathcal{N}(0, Q^{(\ell-1)}).$$

Its gap and the label scale are

$$\gamma = \lambda_{\min}(Q^{(L)}) > 0, \quad Y = \|y\|_2 / \sqrt{m}.$$

The gap is not divided by m . The clean main-text bounds use $0 < Y \leq c\gamma/m$, for a sufficiently small activation/depth constant c . The appendix retains the larger original recurrence-defined allowance and its full certificates. Y is fixed, not width-dependent; zero labels give the exact zero predictor separately. Dense, Legendre and Harmonic require no orthogonality, centering or input-rank assumption. Neither does the new Logarithmic construction. For it, fix $p \geq 0$ further sphere inputs before initialization and write the full panel as $x_1, \dots, x_{m+p}$. Only the first m inputs have labels or enter the training sums.

One observable and one accuracy benchmark. For the promised query set $\mathcal{X}$, write

$$\|f - g\|_{\mathcal{X}} = \sup_{t \in [0, \infty]} \sup_{x \in \mathcal{X}} |f(t, x) - g(t, x)|.\tag{3}$$

Take $\mathcal{X}$ to be the whole sphere for Legendre and Harmonic, and the declared panel for Logarithmic. This includes fitted limits. Let f_n be the compressed model's coupled ordinary dense reference and $\tilde{f}_n$ an independently initialized ordinary dense run. Their discrepancy $\|f_n - \tilde{f}_n\|_{\mathcal{X}}$ is the random benchmark in the theorem. Compression error divided by this discrepancy tends to zero in probability. Thus every fixed positive relative-error tolerance is met with probability tending to one. For this ratio, declare failure (value $+\infty$) if a required fitted limit does not exist or the denominator vanishes; these exceptional events have probability tending to zero under the theorem's assumptions. All errors are absolute, with no implicit division by Y . We do not spend the label cap to reduce powers of m/γ : the learned-storage envelopes remain cubic for Legendre and quartic for Harmonic and in the leading Logarithmic term. The panel and sphere discrepancies need not agree: both sides of a comparison always use the same query set. In the detailed sphere proofs, $\|\cdot\|_*$ denotes the sphere instance of this norm.

Constants, confidence and limits. Main-text $c, C > 0$ depend only on activations and fixed depth and can change between occurrences. They hide no dependence on m, n, d, γ . The relative-error conclusion is convergence in probability. Finite-width absolute-error bounds hold at any fixed confidence $1 - \delta$ once the width is sufficiently large; the appendix retains the dependence on δ in the lower coefficient and width gates. At each sufficiently large individual width the stated comparisons hold, not on one event over infinitely many independent widths. The stochastic width thresholds are not quantified. Expressions with $o(1)$ fix the admissible data, depth, activation and confidence before $n \rightarrow \infty$; they are not uniform growing-data claims.

State and reference. Internal memory and approximation orders are optimized inside the constructions. They appear in the mechanism descriptions and appendix, not as inputs to the headline theorem. Moving state and complete retained state are distinguished. All three track a coupled dense reference. The implicit Harmonic implementation generates that reference jointly instead of materializing its hidden matrices. For Logarithmic, initialization jets determine a finite response-space construction; the resulting small model then evolves autonomously. Its temporal pieces are preparation devices, not a growing runtime history or a prescribed prediction trajectory.

3 Quantitative results

3.1 One theorem, with the query domain made explicit

The theorem prescribes the compressed sizes; internal orders have already been chosen. Its benchmark is the realized discrepancy of independent dense runs in the norm (3), not an upper bound or a fixed-confidence quantile of that discrepancy.

Theorem 3.1 (Compression with negligible relative error). *Fix the admissible data, activations and depth of Section 2, with $m \geq 2$, $Y > 0$ and $m/\gamma \geq 1$. Each construction below can be chosen with the stated storage so that, as $n \rightarrow \infty$,*

$$\frac{\|f_{\text{model}} - f_n\|_{\mathcal{X}}}{\|f_n - \tilde{f}_n\|_{\mathcal{X}}} \xrightarrow{\mathbb{P}} 0. \quad (4)$$

Here f_n is its coupled ordinary dense reference, $\tilde{f}_n$ is an independent dense run, and $\mathcal{X}$ is the query domain specified below. Both norms cover that entire domain, all physical training times and the fitted endpoint. In addition, Harmonic and Logarithmic attain

$$\|f_{\text{model}} - f_n\|_{\mathcal{X}} \leq \frac{Y}{n} \quad \text{with probability at least } 1 - \delta \quad (5)$$

for every fixed $0 < \delta < 1$ and sufficiently large individual width, with the same storage orders below.

The following are sufficient real-coordinate bounds. Legendre and Harmonic cover the whole sphere; their learned and total states, respectively, satisfy the bounds below. Here C depends only on activations and fixed depth.

$$\text{Legendre: } Cm \left(\frac{m}{\gamma}\right)^3 n^{5/4+o(1)}, \quad (6)$$

$$\text{Harmonic: } C(m+d)^2 + \left(\frac{C}{d}\right)^{d+1} \left(\frac{Ym}{\gamma}\right)^4 [\log(en)]^{3d+2}, \quad (7)$$

Legendre additionally retains $(L - 1)n^2$ fixed mixer coordinates. For Logarithmic, $\mathcal{X} = \{x_1, \dots, x_{m+p}\}$, and the total is

$$C(m + p)^2 \left[\left(\frac{Ym}{\gamma} \right)^4 [\log(en)]^3 + \left(\frac{m}{\gamma} \right)^2 [\log(en)]^2 [\log(e + \log(en))]^2 \right] + C(m + p)d. \quad (8)$$

Harmonic and Logarithmic include the data, selected mixers, metrics, initial copies and working arrays. The fixed activation evaluator's description and scratch, when not treated as primitives, are charged separately for each method. These are real-coordinate, not bit-complexity, bounds. The $n^{o(1)}$ in (6) can be taken as $[\log(en)]^2 \exp(\sqrt{\log(en)}/2)$, up to C .

No dense training trajectory or later trained weight is an input to their theoretical initialization procedure.

Dense itself retains $(L - 1)n^2 + n(d + 1)$ weight coordinates. Legendre reduces the changing part of this state, not its fixed dense environment. The two selected constructions remove that environment from the deployed model as well. Ordinary training-data storage is additional for Dense and Legendre. The bounds are constructive sufficient sizes, not claims of storage optimality. In particular, Legendre's proved width exponent is $5/4$, not 1.

The qualification is shared, not hidden. Sufficiently large means a finite threshold for each fixed admissible problem, confidence and, for a finite-width relative comparison, chosen positive relative-error tolerance. It covers the source event, stability and tail bounds, internal-order overheads, and dense lower-bound onset; no polynomial bound on that threshold is proved. The theorem does not assert one event over infinitely many independently initialized widths or a uniform limit with growing data. Its small-label hypothesis is fixed as n grows. In particular, the onset depends on the positive label size Y ; the displayed Y -dependence is not uniform as $Y \downarrow 0$. The explicit source gates can require astronomically large widths, and the dimension-count gates are also eventual. These certificates are asymptotic, not practical width predictions. The more general recurrence-defined label allowance and the exact finite-width statements remain in the appendix.

For the last construction, availability of the passive inputs is substantive: its state is tailored to their response curves. It can be evaluated elsewhere, but the theorem makes no accuracy claim there. The input dimension has not disappeared: it remains in the input map and data charge, and can affect the qualifying width. The two different sample/gap coefficients in (8) are retained rather than merging them into a misleading single prefactor.

3.2 Why comparison with actual variability is possible

Independent dense trajectories fluctuate on the $n^{-1/2}$ width scale, up to logarithmic and subpolynomial factors. The lower-bound witness already occurs at a training input, so it applies to both query domains. Its mechanism is elementary but important. Conditional on earlier layers, new Gaussian rows give a fresh empirical feature-Gram fluctuation. A positive feature-Gram gap prevents all label-weighted feature products from being deterministic: that would force their feature vector onto a single line. The zero initial readout turns this fluctuation into a fluctuating initial prediction velocity. Analytic-time estimates transfer it to an actual, positive-time prediction difference. This is not an endpoint lower bound; fitted trajectories agree on the training labels.

The compressed errors are made asymptotically smaller than this lower scale without measuring dense variability as an algorithmic input. Thus the theorem is not certified by comparing two upper

bounds. The panel lower-bound transfer and the assembly of all three optimized-size claims are given in Appendix F.7; the full dense upper and lower proofs are in the dense appendix.

3.3 What each reduction buys

At fixed admissible parameters, the three scales are $n^{5/4+o(1)}$ moving coordinates, $\log^{3d+2} n$ total real coordinates, and $\log^3 n$ total real coordinates. They form a sequence of explicit trade-offs, not increasingly accurate approximations to the same small network: compress history; compress neuron coordinates using uniform time–input regularity; then restrict the query family and adapt the temporal resolution to remaining learning.

The last step saves two logarithmic powers over a uniform-resolution finite-panel construction. It does not improve Harmonic’s whole-sphere exponent: that would require joint complex-input estimates on the enlarged time domain. Nor does initialization-only information imply inexpensive initialization. These distinctions are part of the result, not exceptions to its accuracy contract.

4 Compressing the responses that carry learning

The common object behind the constructions is the history of an interaction. A forward feature determines which incoming directions a training example activates; a backward response determines how a change at that layer reaches the output. The residual controls the strength and sign of their interaction. The matrix records the accumulated interactions, but its individual entries need not be the most economical state in which to describe them.

There are two reductions. Legendre compresses the temporal histories while keeping their original neuron coordinates. The selected-coordinate construction compresses the neuron spaces needed to realize the interactions. Harmonic and Logarithmic use that same selected architecture and optimizer; they build its source spaces for different query sets. Harmonic covers the whole sphere through joint time and input regularity. Logarithmic covers a finite panel declared before initialization, and uses the increasing temporal regularity of a fitting trajectory. All three models then evolve from time zero without consulting their coupled dense reference.

4.1 The exact memory written into a weight matrix

For $2 \leq \ell \leq L$, integration of Equation (2) gives

$$W^{(\ell)}(t) - W_0^{(\ell)} = -\frac{2}{mn} \sum_{a=1}^m \int_0^t r_a(s) \delta_a^{(\ell)}(s) h_a^{(\ell-1)}(s)^\top ds. \quad (9)$$

Each instantaneous update has rank at most m . Its integral can have much higher rank because both vectors turn during training. Compressibility must therefore come from control of their entire motion, rather than the rank of one update.

Write $\langle u, v \rangle_n = u^\top v / n$ for the empirical neuron pairing. Suppressing the layer superscripts in one hidden link, the two actions of (9) are

$$\begin{aligned} W(t)u &= W_0 u - \frac{2}{m} \sum_a \int_0^t r_a(s) \delta_a(s) \langle h_a(s), u \rangle_n ds, \\ W(t)^\top v &= W_0^\top v - \frac{2}{m} \sum_a \int_0^t r_a(s) h_a(s) \langle \delta_a(s), v \rangle_n ds. \end{aligned} \quad (10)$$

Forward propagation and backward propagation exchange the roles of the two histories. This identifies what a compression must preserve: their pairings, the initialized actions in both directions, and the nonlinear operations that create the next features and responses. Matching the final output alone does not supply these objects.

The initialized mixer can remain full rank. A smaller model needs its action on the relevant source vectors, not an accurate approximation to its action on every vector in $\mathbb{R}^n$. Conversely, replacing a historical pairing by a current-current pairing changes the evolution. The constructions below retain the historical dependence through explicit state or through source spaces whose selected realization supports autonomous learning.

4.2 Legendre: evolving moments of the two histories

Hats denote the Legendre model's own reconstructed network. Its activity clock satisfies $\dot{\tau} = \hat{\rho}$, $\tau(0) = 1$, where $\hat{\rho} = \|\hat{r}\|_2/\sqrt{m}$. Under the fitting estimates, $\int_0^\infty \hat{\rho}(t) dt$ is finite. Thus the entire physical trajectory occupies a finite interval of clock time. The clock slows when little error remains to drive the weights.

At one hidden link, write $h_a(\xi), b_a(\xi) \in \mathbb{R}^n$ for the histories defined, while the clock advances, by $h_a(\tau(t)) = \hat{h}_a(t)$ and $b_a(\tau(t)) = \hat{r}_a(t)\hat{\delta}_a(t)/\hat{\rho}(t)$. On the initial prefix $0 \leq \xi \leq 1$, set $h_a(\xi) = h_{0,a}$ and $b_a(\xi) = 0$, where $h_{0,a}$ is the initialized forward feature. This prefix gives a nonsingular starting interval and preserves the realized initialization. The quotient defines a history for the proof; the autonomous equations below contain no division by a residual.

Let $p_j^\tau(\xi) = P_j(2\xi/\tau - 1)$, where P_j is the Legendre polynomial with $P_j(1) = 1$. These polynomials have squared norm $\tau/(2j+1)$ in Lebesgue $L^2([0, \tau])$. Let Π_q^τ denote orthogonal projection onto polynomials of degree below q , applied to each neuron coordinate. The stored moments at each link are

$$\bar{h}_{a,j}^{(\ell-1)} = \int_0^\tau p_j^\tau(\xi) h_a^{(\ell-1)}(\xi) d\xi, \quad \bar{\delta}_{a,j}^{(\ell)} = \int_0^\tau p_j^\tau(\xi) b_a^{(\ell)}(\xi) d\xi, \quad 0 \leq j < q.$$

They are unnormalized history integrals. Each is still a vector in $\mathbb{R}^n$: the projection removes temporal modes, not neurons. Orthogonality gives the reconstructed mixers

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{mn\tau} \sum_{a=1}^m \sum_{j=0}^{q-1} (2j+1) \bar{\delta}_{a,j}^{(\ell)} \bar{h}_{a,j}^{(\ell-1)\top}. \quad (11)$$

Its correction is exactly the integral of $(\Pi_q^\tau b_a)(\Pi_q^\tau h_a)^\top$ with the factor $-2/(mn)$.

The current forward pass and backward responses are evaluated using these matrices. The first matrix and readout follow Equation (2) at that reconstructed state. Differentiating the moving-interval moment integrals gives the remaining autonomous equations:

$$\begin{aligned} \dot{\bar{h}}_{a,j}^{(\ell-1)} &= \hat{\rho} \hat{h}_a^{(\ell-1)} - \frac{\hat{\rho}}{\tau} \left(j \bar{h}_{a,j}^{(\ell-1)} + \sum_{i < j} (2i+1) \bar{h}_{a,i}^{(\ell-1)} \right), \\ \dot{\bar{\delta}}_{a,j}^{(\ell)} &= \hat{r}_a \hat{\delta}_a^{(\ell)} - \frac{\hat{\rho}}{\tau} \left(j \bar{\delta}_{a,j}^{(\ell)} + \sum_{i < j} (2i+1) \bar{\delta}_{a,i}^{(\ell)} \right). \end{aligned} \quad (12)$$

Initially $\bar{h}_{a,0}^{(\ell-1)} = h_{0,a}^{(\ell-1)}$ and every other moment is zero; the stored first matrix and readout share the dense initial values. The triangular sums account for rescaling the basis as the clock interval grows. At zero residual every velocity vanishes, including the clock velocity. No past trajectory is stored or replayed.

The strength of this reconstruction comes from projecting both histories with the same orthogonal projector. At a fixed clock endpoint, suppressing its superscript on Π_q , one has

$$\int_0^\tau b h^\top d\xi - \int_0^\tau (\Pi_q b)(\Pi_q h)^\top d\xi = \int_0^\tau [(I - \Pi_q)b][(I - \Pi_q)h]^\top d\xi. \quad (13)$$

The retained–discarded cross terms vanish coordinate by coordinate. For example, the average of $(v_0 + \xi v_1)(u_0 + \xi u_1)^\top$ on $[-1, 1]$ is $v_0 u_0^\top + v_1 u_1^\top / 3$. Dropping the linear modes loses their mutual interaction, with no cross term involving a retained constant mode. The same cancellation holds for arbitrary histories.

There is also an exact dynamical form. If $e_g = g(\tau) - (\Pi_q^\tau g)(\tau)$ denotes an endpoint projection error, differentiation of the reconstructed mixer gives

$$\dot{\hat{W}}^{(\ell)} = -\frac{2}{mn} \sum_a \hat{r}_a \hat{\delta}_a^{(\ell)} \hat{h}_a^{(\ell-1)\top} + \mathcal{E}_\ell, \quad \mathcal{E}_\ell = \frac{2\hat{\rho}}{mn} \sum_a e_{b,a}^{(\ell)} e_{h,a}^{(\ell-1)\top}. \quad (14)$$

This product of two tails is responsible for the essentially quadratic decay in order. It does not follow from a pointwise low-rank approximation of the weights. The moment histories are endogenous: changing the moments changes the network that writes their next values. The stability argument below controls that feedback.

The moving state has exactly $n(d+1) + 1 + 2(L-1)mnq$ real coordinates. The initialized hidden mixers add $(L-1)n^2$ fixed coordinates and still act during forward and backward propagation. Legendre therefore compresses the learned history while preserving an original-width nonlinear network. The leading error is q^{-2} times a coefficient that is subpolynomial in n at fixed parameters. Comparing with the intrinsic variability scale $n^{-1/2}$, with its subpolynomial and logarithmic factors, leads to $q = n^{1/4+o(1)}$. The theorem chooses the overhead to obtain negligible relative error; substitution into the mnq moment count gives $n^{5/4+o(1)}$ moving coordinates at fixed m, d, L .

4.3 A selected realization of the response spaces

The second reduction replaces each neuron space by a small set of actual neuron coordinates and a metric. Its input is a fixed finite-dimensional source space $E_\ell \subset \mathbb{R}^n$ for each layer. The spaces approximate the dense forward features and the initialized images needed for forward propagation, together with backward responses and initialized transpose images needed for training. Their construction depends on the promised query set. Exact initial training features and their forward images, the constant vector, and the first-weight columns are added.

An initialized image must be paired with its preimage. If a computed coefficient is u , its image coefficient is formed as $W_0 u$, or as $W_0^\top u$ in the reverse direction. Identical scalar approximation operations commute with these fixed matrices. Consequently the computed source spaces preserve the image relation exactly, while each member has its own error bound relative to the corresponding dense source. This is what lets one small mixer serve both directions of (10).

At layer ℓ , selection chooses an index set I_ℓ of size $q_\ell \leq 9 \dim E_\ell$ and a positive metric M_ℓ such that

$$\langle u, v \rangle_n = u_{I_\ell}^\top M_\ell v_{I_\ell}, \quad u, v \in E_\ell. \quad (15)$$

The metric is generally a full matrix. A positive diagonal comparison metric D_ℓ satisfies $D_\ell/4 \preceq M_\ell \preceq D_\ell$ and $\mathbf{1}^\top D_\ell \mathbf{1} \leq 4$. Exact isometry on E_ℓ preserves pairings, while the diagonal comparison bounds coordinatewise gates and converts coordinate approximation errors into metric errors. Neither property requires a lower bound on the smallest selection weight.

Coordinate selection also preserves the original nonlinear operation: $\phi(z)_{I_\ell} = \phi(z_{I_\ell})$. An arbitrary change of basis has no such identity. For example, averaging two coordinates before

applying tanh generally differs from averaging their activations. The selected model can therefore use the same coordinatewise activations as the dense network. Its activations need not remain in E_ℓ ; the metric controls their operations on the entire selected coordinate space, and trajectory comparison controls their deviation from the approximated dense sources.

Here is the common autonomous model. The subscript C denotes either selected construction. Its moving state is $(A_C, B_C^{(2)}, \dots, B_C^{(L)}, w_C, c_C)$, where $A_C \in \mathbb{R}^{q_1 \times d}$, $B_C^{(\ell)} \in \mathbb{R}^{q_\ell \times q_{\ell-1}}$, $w_C \in \mathbb{R}^{q_L}$, and $c_C \in \mathbb{R}^m$. For $v = x/\sqrt{d}$, compute $z_C^{(1)} = A_C v$, $z_C^{(\ell)} = B_C^{(\ell)} h_C^{(\ell-1)}$, and $h_C^{(\ell)} = \phi_\ell(z_C^{(\ell)})$. Write $\langle u, v \rangle_{M_\ell} = u^\top M_\ell v$. Let V_C have columns $h_{C,a}^{(L)}/\sqrt{m}$, and define $V_C^* = V_C^\top M_L$ and $Q_C = V_C^* V_C$. The readout used for predictions and backward propagation is

$$\begin{aligned} \hat{w}_C &= w_C + V_C Q_C^{-1} \left(\frac{y - c_C}{\sqrt{m}} - V_C^* w_C \right), \\ f_C(x) &= \langle \hat{w}_C, h_C^{(L)}(x) \rangle_{M_L}. \end{aligned} \quad (16)$$

The correction is the vector in the training-feature span that enforces $f_C(x_a) = y_a - c_{C,a}$ exactly. Thus c_C is the actual label deficit, with the opposite sign to the residual $r = f - y$. The raw readout retains its component orthogonal to the current training-feature span. This matters for passive predictions: fitting the training values does not determine the full readout. The feature gap keeps Q_C^{-1} well defined.

For a map $B_C^{(\ell+1)}$ between consecutive selected spaces, its metric adjoint is $(B_C^{(\ell+1)})^* = M_\ell^{-1} B_C^{(\ell+1)\top} M_{\ell+1}$. Set $k_{C,a}^{(L)} = \hat{w}_C$, $k_{C,a}^{(\ell)} = (B_C^{(\ell+1)})^* \delta_{C,a}^{(\ell+1)}$, and $\delta_{C,a}^{(\ell)} = \phi'_\ell(z_{C,a}^{(\ell)}) \odot k_{C,a}^{(\ell)}$. Define the unnormalized sample Gram

$$\begin{aligned} K_{C,ab} &= \langle h_{C,a}^{(L)}, h_{C,b}^{(L)} \rangle_{M_L} + \langle \delta_{C,a}^{(1)}, \delta_{C,b}^{(1)} \rangle_{M_1} v_a^\top v_b \\ &\quad + \sum_{\ell=2}^L \langle \delta_{C,a}^{(\ell)}, \delta_{C,b}^{(\ell)} \rangle_{M_\ell} \langle h_{C,a}^{(\ell-1)}, h_{C,b}^{(\ell-1)} \rangle_{M_{\ell-1}}. \end{aligned} \quad (17)$$

The updates are

$$\begin{aligned} \dot{A}_C &= \frac{2}{m} \sum_a c_{C,a} \delta_{C,a}^{(1)} v_a^\top, \quad \dot{B}_C^{(\ell)} = \frac{2}{m} \sum_a c_{C,a} \delta_{C,a}^{(\ell)} h_{C,a}^{(\ell-1)\top} M_{\ell-1}, \\ \dot{w}_C &= \frac{2}{m} \sum_a c_{C,a} h_{C,a}^{(L)}, \quad \dot{c}_C = -\frac{2}{m} K_C c_C. \end{aligned} \quad (18)$$

Only training samples enter these sums. These are specified update directions. A coordinate gate need not be self-adjoint in M_ℓ , so they are not asserted to be the gradients of the corrected predictor.

Initialization is also explicit. Choose a basis matrix U_ℓ of E_ℓ with $U_\ell^\top U_\ell/n = I$, and put $P_\ell = (U_\ell)_{I_\ell}$ and $A_0 = W^{(1)}(0)$. Set $A_C(0) = (A_0)_{I_1}$, $w_C(0) = 0$, $c_C(0) = y$, and

$$B_C^{(\ell)}(0) = P_\ell \frac{U_\ell^\top W_0^{(\ell)} U_{\ell-1}}{n} P_{\ell-1}^\top M_{\ell-1}. \quad (19)$$

Since $P_\ell^\top M_\ell P_\ell = I$, this mixer reproduces the initialized action on every paired source vector. Its metric adjoint reproduces the corresponding transpose action. The exact additions to the spaces make the initial training features equal the corresponding dense coordinates, and their full metric Gram equals the dense empirical Gram. All width- n bases, source vectors and dense arrays are then discarded.

This is not equivalent to drawing a fresh small Gaussian network. The selected coordinates and metric preserve interactions from the realized width- n reference, including its training feature Gram. A fresh small network resamples those interactions instead. Nor is the reduction a low-rank approximation to every initialized mixer: it preserves the paired actions needed by the response space and allows the selected features to keep changing nonlinearly. As a full-retention check, taking $E_\ell = \mathbb{R}^n$, every coordinate, and $M_\ell = I/n$ recovers (2): the deficit then equals $y - f_n$, the readout correction vanishes, and the factor $1/n$ in each hidden update comes from its metric.

For $q = \max_\ell q_\ell$, the moving coordinates number $dq_1 + \sum_{\ell=2}^L q_\ell q_{\ell-1} + q_L + m$. Here q is a neuron budget, distinct from the Legendre polynomial order. The fixed metrics and their inverses add $O(Lq^2)$ coordinates. The effective readout is computed from the state, not a second independent learned vector. This quadratic dependence on the selected width explains why a gain in source dimension is squared in the retained storage.

4.4 Harmonic: one source for the whole input sphere

Harmonic constructs E_ℓ using the four vector families $h_n^{(\ell)}(t, v)$, $W_0^{(\ell)} h_n^{(\ell-1)}(t, v)$, $\delta_n^{(\ell)}(t, v)$, and $W_0^{(\ell+1)\top} \delta_n^{(\ell+1)}(t, v)$, omitting nonexistent boundary-layer images. Here $(t, v) \in [0, T] \times S^{d-1}$, including passive inputs. For a passive input the backward field is evaluated through the trained network but supplies no update force. The source spaces are the real spans of finitely computed coefficient vectors for these families, together with the exact initialized additions described above.

The expansion separates two kinds of regularity. With $t = T(1 + \cos u)/2$, temporal cosines are Chebyshev polynomials in physical time. Real spherical harmonics describe input direction. On a circle, the spatial expansion is an ordinary sine and cosine series; in higher dimension it is organized by harmonic degree. Holomorphy in a complex time neighborhood and a complex neighborhood of the sphere makes both coefficient tails decay exponentially. A weighted cutoff in temporal and spatial degree retains only the coefficient vectors needed at the chosen source accuracy.

Bounded activation values are unnecessary here. If s is a common bound on $|\phi'_\ell|$ in the safe strip, integration from zero gives $|\phi_\ell(z)| \leq |\phi_\ell(0)| + s|z|$ on the safe strip, so operator and feature RMS bounds propagate through the depth even when individual activations are large. The normalized response energy controls the bulk motion. Coordinate maxima still enter when the source argument keeps every complex preactivation inside its analytic strip; the relevant maxima have order $\sqrt{\log(en)}$ at fixed parameters. This distinction explains why the available complex radii shrink with width although the RMS state remains bounded. Bounds on derivatives control the gates; they do not impose a bounded-activation hypothesis.

This explains the dimension-dependent logarithmic exponent. At fixed problem parameters and inverse-polynomial source accuracy, the sufficient horizon is of order $\log(en)$. The certified time and input radii are of order $1/\sqrt{\log(en)}$. The temporal degree is therefore $O(\log(en)^{5/2})$, while spherical degrees up to $O(\log(en)^{3/2})$ contribute $O(\log(en)^{3(d-1)/2})$ basis functions. The weighted cutoff refines this product count, with the same resulting source-dimension exponent:

$$\dim E_\ell = O(\log(en)^{3d/2+1}), \quad \text{retained state} = O(\log(en)^{3d+2}).$$

These counts fix data, labels, depth, activation and confidence. The full parameter dependence, including the learned-storage factor $(Ym/\gamma)^4$, is retained in the theorem and appendix. In dimension one the sphere consists of two points and only temporal expansions are needed.

The source spaces are computed during initialization, before the compact model begins its own training. The efficient explicit initializer advances certified local expansions using stored dense matrices. The implicit initializer generates their Gaussian forward and transpose actions

conditionally on earlier requests, preserving the joint reference law. It constructs a fresh coupled reference without materializing every hidden matrix. Both assemble the same kind of selected model at physical time zero. Their preprocessing costs are separate from the retained state and are stated in Section 5. The time and spherical coefficients are setup objects; inference applies the selected nonlinear network directly.

4.5 Logarithmic: a declared input panel and expanding time disks

Suppose the requested queries are the m training inputs and p passive inputs fixed before initialization. A finite panel permits a different source builder. Retain the forward feature curves and their initialized images at all $m + p$ inputs, and the backward response curves and their transpose images only at the m training inputs. There are at most $2(2m + p)$ vector curves per layer. No spatial harmonic expansion is needed, because the guarantee is a maximum over these declared inputs. The selected network remains evaluable elsewhere, but the panel certificate does not bound its error there.

Removing spatial approximation alone would leave the temporal cost of a uniform narrow analytic strip. The additional gain comes from fitting. As the real residual decays, a complex time excursion starts with less activity driving its parameter motion. The finite-query source argument uses this fact to certify larger analytic disks at later real times. The enlarged domain is proved together with the required source bounds; small real-time derivatives alone would not imply it.

The useful model is a damped mode $e^{-\kappa t}$, with $\kappa > 0$: an imaginary time increment rotates its phase without changing its magnitude. In the neural proof, freezing the real Gram matrix gives the same unitary evolution along imaginary time. Growth comes from variation of that Gram, not from exponentiating its entire norm as though it were unstable. That variation is paid for by the remaining residual activity. Later real anchors start with exponentially less residual, so they permit larger complex excursions before consuming the same activity budget. The certified radius first grows approximately exponentially and then linearly. Controlling the corresponding random responses on this enlarged domain is the difficult analytic step, not simply changing a plotting clock. In particular, flattening in real time does not by itself justify analyticity at the endpoint of an arbitrary compactified clock.

Use the source horizon $T_0 = 32(m/\gamma) \log(en)$ of the panel certificate. Let $\varrho(t) > 0$ be the certified disk radius at a real time $t \in [0, T_0]$. Cover this interval by predetermined temporal pieces, each using an anchor t_b with $|t - t_b| \leq \varrho(t_b)/2$. For each required vector curve g , use the degree- K Taylor polynomial

$$g(t) \approx \sum_{j=0}^K \frac{\varrho(t_b)^j g^{(j)}(t_b)}{j!} \left(\frac{t - t_b}{\varrho(t_b)} \right)^j \quad \text{on piece } b.$$

The vector coefficients from every piece enter one fixed space E_ℓ . No switch between temporal pieces occurs in the selected runtime. Each local polynomial can equally be written in Chebyshev form: the degree-preserving basis change is invertible, so it leaves the vector coefficient span and its paired initialized images unchanged. The saving comes from the larger analytic domain, not from renaming the polynomial basis. If J is their count and $B = 2m + d + 1$ bounds the exact initialized additions, their source-dimension bound is

$$\begin{aligned} \dim E_\ell &\leq B + 2(2m + p)J(K + 1), \\ J &= O\left(\sqrt{\log(en)} + \log \log(e^n)\right), \quad K + 1 = O(\log(en)). \end{aligned} \tag{20}$$

The orders here fix the problem parameters and use the source accuracy chosen for the theorem's trajectory target. The covering count follows from an integral of $1/\varrho(t)$: early narrow disks require

many short pieces, whereas the later widening disks cover the fitting tail with progressively longer pieces. The geometric Taylor tail on each half-radius piece explains why its degree is only logarithmic.

Squaring this source-dimension bound gives the cubic logarithm in retained storage. The coefficient of the square-root-logarithm term in J carries $(Ym/\gamma)^2$; its square retains the fourth label/gap power. The logarithmic-in-log n contribution has a separate gap-dependent coefficient, kept in the full bound. Thus the improvement removes the spatial degree count and reduces the temporal covering cost. It trades whole-sphere coverage for a panel specified before initialization; the finite-query flaring theorem does not itself enlarge Harmonic’s joint complex time/input domain.

There is an optional exact simplification of the input dimension. Let the columns of $U \in \mathbb{R}^{d \times k}$ be an orthonormal basis of the span of all panel inputs, with $k \leq \min(d, m + p)$. Use inputs $U^\top v_i$ and first weights $A_C \in \mathbb{R}^{q_1 \times k}$. All dense first-layer updates lie in this span, so the coupled dense network with first matrix $A_0 U$ has exactly the same panel trajectory. The Gram and mobilities are unchanged. Retaining U costs dk scalars; it replaces d by k in the source overhead and leaves ambient input storage linear in d .

Finally, the finite coefficients can be produced from initial derivatives and certified continuation, using the same scalar operations for each base/image pair. Their finite availability is not a cheap-setup or finite-precision guarantee. All coefficients, anchors, original-width arrays and continuation state are discarded after selection. The retained model is precisely (16)–(18), initialized by (19); it counts real coordinates. The complete flaring domain, coefficient construction, parameter dependence and panel norm are given in the panel-compression appendix.

4.6 Why the approximation survives continued learning

The global argument has two tasks: show that each approximate model keeps fitting, and show that it stays close to its dense reference. Accuracy does not follow merely from small spectral tails, because future source vectors depend on the approximate state. The small-label assumptions keep the feature Gram away from singularity while allowing hidden learning. The surviving gap damps the residual, making total residual activity finite. The stability proofs retain this sign information before estimating the nonlinear remainder.

For the selected models, every term of K_C is a positive semidefinite Gram matrix, including tensor-product Grams for hidden-layer updates. With $\rho_C = \|c_C\|_2/\sqrt{m}$, the optimizer gives the exact identity

$$-\frac{d}{dt}\rho_C^2 = \frac{4}{m^2}c_C^\top K_C c_C, \quad K_C/m \succeq Q_C.$$

The same quadratic expression is the squared speed of the raw parameters in their mobility-compatible metric. Consequently the preserved feature gap controls both residual decay and total parameter path length. This energy identity is algebraic; it does not require the optimizer to be gradient flow of the corrected predictor. The resulting small feature motion closes the gap estimate and keeps the readout solve nonsingular.

Source isometry then transfers approximation error to the interactions. If source vectors u, v have coordinate errors at most η and dense RMS bounds A_u, A_v , inserting their approximants into both pairings gives

$$|\langle u_I, v_I \rangle_M - \langle u, v \rangle_n| \leq 3\eta(A_u + A_v) + 9\eta^2.$$

The initialized paired actions have comparably small defects. Equation (10) integrates the learned action defects against training residuals, so their total forcing is governed by accumulated activity. No time derivative of a source approximation error is needed. This is why the same comparison accepts a global harmonic expansion or the temporal pieces of the panel construction.

One further cancellation is essential for the selected model. Let $c_n = y - f_n$ on the training inputs and $w_R = (w_n)_{I_L}$ be the restricted dense readout. Define $e = (c_C - c_n)/\sqrt{m}$ and $T_C = V_C Q_C^{-1}$. The raw-readout difference has a term $+2V_C e$ in its velocity. The deficit difference has the feature-Gram damping term $-2Q_C e$. Lifting the latter into feature space and adding the two errors gives

$$\zeta = w_C - w_R + T_C e, \quad 2V_C e - 2T_C Q_C e = 0. \quad (21)$$

This uses $T_C Q_C = V_C$, not an inverse on the entire feature space. The remaining terms involve changing current features, small hidden response contributions, and the source defects. They are controlled by the residual activity and readout path length. Only the actual selected feature map is differentiated. A signed estimate for the lifted deficit retains the damping needed to absorb the hidden contributions under the stated label allowance. The correction thus serves both exact training consistency and stable comparison with the dense readout. Preserving this cancellation permits the stated explicit gap dependence; estimating both leading coupling terms separately would discard that structure and repeatedly pay for the inverse feature gap.

Legendre has a different error mechanism. Its reconstruction is the dense gradient field plus $\mathcal{E}$ from (14). First, growing-interval projection energy gives an all-order fitting estimate. For a history g ,

$$\frac{d}{dA} \int_0^A \|g - \Pi_q^A g\|^2 d\xi = \|g(A) - (\Pi_q^A g)(A)\|^2.$$

Together with the defect's product structure, this controls its effect on the residual and preserves a damped fitting flow. It also turns the integral of the absolute matrix defect into a product of two history-tail norms. The forward tail is of order q^{-1} . For the backward tail the proof compares with the dense response expressed in the closure clock, uses residual decay near the clock endpoint, and absorbs the response difference created by the closure itself. This yields the $q^{-2}\sqrt{\log(eq)}$ comparison rate, rather than assuming an independently regular prescribed backward signal.

For the final comparison, put the parameters in physical coordinates:

$$\theta = (W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n}).$$

Subtracting the dense and reconstructed gradient fields produces a negative squared residual-difference term in $\frac{1}{2}d\|\hat{\theta} - \theta_n\|^2/dt$. Once its nonpositive sign is identified, that term is dropped for the upper bound. The Taylor remainder and Jacobian variation carry residual factors, leaving growth proportional to $\rho_n + 3\hat{\rho}$, plus the forcing $\mathcal{E}$. The residual-weighted growth coefficient has finite integral even as physical time tends to infinity. This signed parameter-energy argument is distinct from the selected model's readout-deficit cancellation. Both use the training flow's damping to control feedback, but they close different error variables.

The resulting estimates give finite amplification, not uniform contraction. At fixed problem parameters the comparison coefficient can still grow as $\exp(C\sqrt{\log(en)}) = n^{o(1)}$. The source tolerance is chosen tightly enough to absorb this factor. Its contribution to $\log(1/\eta)$ is only of order $\sqrt{\log(en)}$, so it does not change either selected construction's leading logarithmic storage exponent.

Finally, finite source construction is compatible with all-time fidelity. For either selected model, comparison through a horizon T bounds the error by a source tolerance times a stability coefficient. Beyond T , each model is compared to its own state at T using its convergent fitting tail. Both continue to evolve. The horizon is chosen so that this tail fits the requested error budget, and the same argument includes their limiting predictions. Legendre's projection comparison is uniform in the physical horizon and also passes to its fitted limit. These mechanisms control amplification by accumulated learning activity, while their width dependence and source preparation costs remain explicit in the guarantees.

5 What is saved, and what preparation still costs

A small deployed state, a cheap initialization, and a cheap query are different claims. Legendre starts from the supplied dense initialization and creates its moments directly. It saves changing memory but still multiplies by the fixed dense matrices, so it does not remove their quadratic cost. The selected models discard those matrices after setup. Their nonlinear forward and backward passes then use only the retained small matrices, metrics and training data. No width- n oracle is called during their subsequent evolution.

The common selected runtime. Both source builders lead to the same execution pattern: a small forward pass, metric-adjoint backward pass, and a training-feature Gram solve for the readout correction. The solve is part of training, not an uncounted oracle. Refreshing the corrected readout once per state permits later queries to use a plain small forward pass. Passive labels are absent from all these operations. For Logarithmic, the certified passive queries were declared during setup; the existence of an evaluable network on other inputs is not a guarantee of accuracy there.

Information available at initialization. Analyticity makes every finite coefficient list recoverable, to a prescribed tolerance, from finitely many derivatives at time zero. Those derivatives are obtained by differentiating the dense training vector field at its initial weights, using the training labels. Passive inputs only specify which response curves are represented. The continuation procedure in Appendix F reconstructs the later polynomial coefficients without access to later trained weights. This is a data-dependent compression of a specified learning problem, not a data-oblivious map.

That statement is about available information, not computational economy. Analytic continuation from one point can require a very large derivative order and high precision. A practical dense rollout can supply source coefficients more conveniently, but its work and full-horizon information must then be reported as setup. Such an implementation is not evidence for a cheap initialization-only algorithm.

Which efficient setup bounds are already available. Harmonic has two separately certified implementations. For the exact absolute target Y/n , the inverse certificate prescribes a longer source horizon and a tighter source tolerance; Appendix B gives their substitution into both setup bounds. At fixed positive label size and other problem parameters, the arithmetic work is n^2 times a polylogarithm in n for explicit dense continuation, and n times a polylogarithm in n for implicit Gaussian continuation. The latter reveals only the random matrix actions actually needed. These rates inherit the source and dimension width gates; they are not uniform as labels shrink or dimension grows. The latter samples a fresh coupled dense reference: it cannot read a supplied dense matrix in subquadratic time. Their temporary arrays are discarded before deployment.

The new Logarithmic theorem does not inherit either setup-time bound merely because its runtime is the same. Its enlarged source domain and finite-jet compiler prove the cubic-logarithmic retained state, not a corresponding fast preprocessing algorithm. Appendix B preserves the detailed existing execution ledgers, and the setup appendix proves the two Harmonic bounds. Real-coordinate arithmetic, evaluator cost, numerical precision and the number of integration steps remain separate resources throughout.

6 Related work and discussion

The comparison is between representations of a specified learning process, not a claim that other descriptions lack quantitative guarantees. Tensor Programs and multilayer mean-field theory provide feature-learning population frameworks [Yang and Hu, 2021, Nguyen and Pham, 2023]; dynamical mean-field descriptions organize temporal correlations and responses [Bordelon and Pehlevan, 2022, 2023]. Model assumptions, observables and solver representations must be matched before comparing their resource costs. Our certificates concern finite Gaussian networks, all physical times, with the promised query domain kept explicit.

A frozen neural tangent kernel describes a different limiting regime [Jacot et al., 2018, Lee et al., 2019]. Higher-order responses appear in the neural tangent hierarchy [Huang and Yau, 2020], whose truncation theorem has its own scaling and horizon. Online polynomial memory summarizes an externally given signal [Gu et al., 2020]; here the signals are endogenous, generated by the network reconstructed from those memories. Stabilizing that feedback is an additional obligation. Our second reduction preserves coordinatewise nonlinear gates together with response inner products; it is not ordinary projection of every layer onto an arbitrary linear subspace. No fresh-noise approximation replaces a reused trained matrix.

The trade-off is not one-dimensional. Legendre makes the evolving memory small but retains a quadratic fixed environment. Harmonic removes that environment from the retained state and has polylogarithmic query work at fixed dimension, with near-linear implicit setup. The new Logarithmic model uses the same selected runtime but specializes its response space to declared input curves. This removes the dimension-dependent logarithmic exponent, not the cost of preparing that space or the restriction on which inputs are certified. These are two different notions of generalization: accuracy on inputs absent from the updates, and uniform accuracy on inputs absent even from setup.

Open boundaries. The dense lower bound is transient and does not establish sharp sample/gap dependence. Small labels are a sufficient stability regime, not a sharp necessity or a width-dependent scaling. The analytic class includes unbounded values but not exact ReLU. Exact-real setup costs do not establish fixed-machine-precision performance or superiority at practical widths. A fixed program for an arbitrary analytic activation also needs a separately specified evaluation cost. The finite-panel third power does not yet imply a second-power neural construction or an efficient initialization-only compiler. Finally, trajectory fidelity alone does not establish out-of-sample label risk: the usefulness of the selected predictor remains a separate question.

Earlier empirical illustrations. Appendix A preserves existing Legendre and Harmonic illustrations and experiments for an older seeded finite-word decoder. That decoder is a separate supplementary construction, not the present finite-panel Logarithmic model. Those measurements are not empirical validation of the new cubic-logarithmic guarantee.

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A Earlier empirical illustrations

These experiments concern the earlier Harmonic implementation and seeded finite-word decoder. The latter is not the finite-panel Logarithmic model of Theorem 3.1; none of the decoder measurements below validates the new cubic-logarithmic theorem. We preserve the original measurements, controls and failures without reinterpreting their scope. In this appendix, $b_n(10^{-4})$ denotes the historical whole-sphere quantile

$$b_n(10^{-4}) = \inf\{b \geq 0 : \Pr(\|f_n - \tilde{f}_n\|_* \leq b) \geq 0.9999\}. \quad (22)$$

Here $\|\cdot\|_*$ is the whole-sphere instance of (3), with discrepancy $+\infty$ on a run lacking a required fitted limit. This quantile belongs to the supplementary seeded decoder comparison, not the main theorem’s negligible-relative-error claim. The label “Logarithmic” inside the preserved figures refers to the seeded decoder.

A.1 Trajectory fidelity and retained state

Earlier experiments test empirical Harmonic and seeded decoder implementations against independent dense-run variability. Error is the largest unseen-input RMS discrepancy at recorded common times; the practical threshold is three times the corresponding dense-pair discrepancy, not the historical quantile $b_n(10^{-4})$. Harmonic meets this threshold in all nine held-out circle cases ($n = 1024, 2048, 4096$, three seeds), with ratios 0.159–1.617. At $n = 4096$ it uses 11.7–11.8 times fewer model-tensor scalars. Single-seed checks in dimensions 3, 5, 10 also meet the threshold. On digits 3 versus 8, three width-4096 runs preserve the dense teacher’s 97.32% accuracy with about 4.6-fold smaller tensor state; their fidelity ratios are 1.85–2.44. The matched smaller networks have better fidelity on all three real-data runs. Thus compression success does not establish superiority over ordinary small networks.

The seeded decoder implementation retains 688,220 numerical words across the tested dimension-five widths 1024, 2048, 4096, a 24.4-fold persistent payload reduction at width 4096. All five dimension-five confirmation cases meet the empirical threshold, with ratios 0.909–1.781 against independent dense references. Its circle check fails, with ratio 6.95; a separate digits decoder check also fails (4.94). These results are not a universal success claim. Figure 2 shows both state and fidelity comparisons. Full-horizon source construction is charged as offline setup, so these experiments demonstrate no setup speedup. The implementations, query workspace, numerical checks and negative cases are detailed in Appendix A.2. Fresh labels have RMS one and need not satisfy the small-label theorem. The earlier Legendre trajectory in Figure 1 remains a separate illustration.

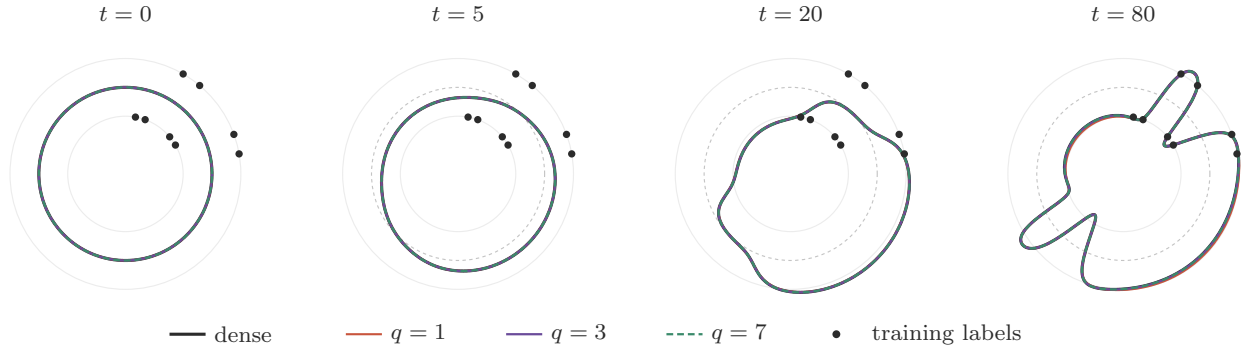


Figure 1: A trajectory, not only an endpoint. Existing paired-label circle experiment: two hidden tanh layers, width 2048, and Legendre orders $q = 1, 3, 7$. Predictions are saved at the same physical times $t = 0, 5, 20, 80$. Radius is $3 + f(\theta)$, the dashed ring marks zero, and black points are training labels. The source replay records 39 common times and uses float64 adaptive Heun integration at two tolerances. These are discrete observations, not a certificate between the saved times.

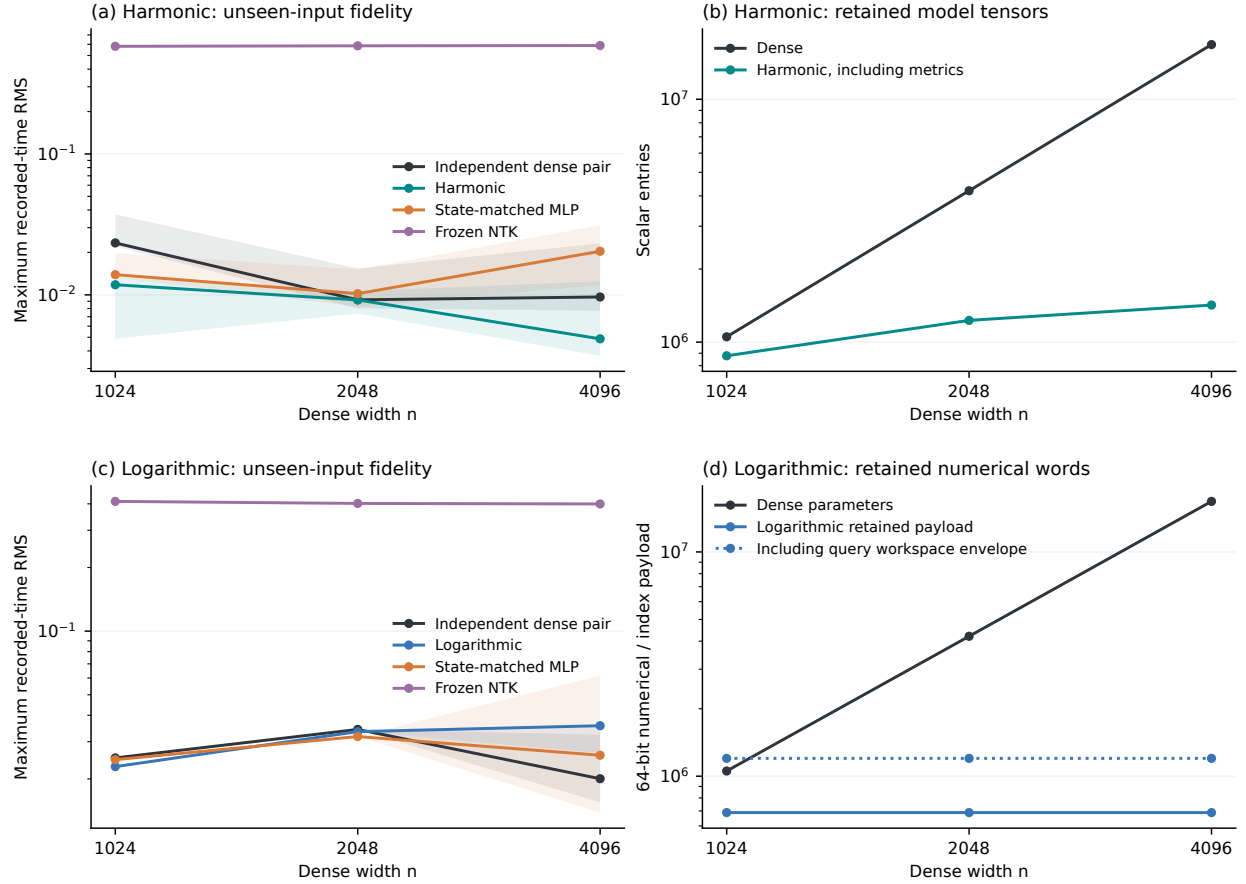


Figure 2: Retained state and finite-time prediction fidelity. Top: Harmonic circle confirmation, with three held-out seeds at each width and a coupled dense reference. Bottom: the dimension-five seeded decoder finite program, with seed 201 across widths and three seeds at width 4096, compared with independent dense references. Ordinary small-network controls match retained payload; the frozen initial kernel uses the dense initialization and physical clock. Ratios use a measured dense pair on the same times and queries. Storage means model-tensor scalars for Harmonic and persistent numerical payload for seeded decoder; setup and query workspace are additional (the dotted curve includes the latter’s numerical envelope). Lines show medians and shading shows the full observed seed range, not confidence intervals. The fixed finite program’s flat payload across these widths is not an asymptotic scaling estimate.

A.2 Protocol and additional results

Observable and split. For recorded physical times $\mathcal{T}$ and unseen inputs $\mathcal{Q}$, use

$$E(g, h) = \max_{t \in \mathcal{T}} \left(\frac{1}{|\mathcal{Q}|} \sum_{x \in \mathcal{Q}} |g(t, x) - h(t, x)|^2 \right)^{1/2}, \quad R_{\text{emp}} = \frac{E(f_{\text{model}}, f_{\text{reference}})}{E(f_n, \tilde{f}_n)}.$$

The denominator uses independent dense initializations. A practical pass requires $R_{\text{emp}} \leq 3$ and resolved reference numerics; Harmonic’s combined coarse–fine sensitivity must also be at most 10% of that denominator. This is neither a confidence estimate for $b_n(10^{-4})$ nor a continuous-time, whole-sphere or asymptotic guarantee. Pilots use seeds 101–102; settings were frozen before seeds 201–203. Setup nodes are separate from test queries. Digits uses 32 training images, training-only PCA to dimension eight, normalization to unit directions, 64 tuning images and 261 held-out images.

Dynamics and empirical backends. All runs in this experiment use two tanh hidden layers, zero readout, Gaussian initialization and mobilities $(n, 1, n)$ from Equation (2); stored inputs are $v = x/\sqrt{d}$. Labels have RMS one. Harmonic and dense use float64 RK4, with steps 0.125 and 0.0625, through time 20. Harmonic’s full dense warmup is followed by Chebyshev/sphere-polynomial least squares and per-family SVD truncation before paired mixer images are formed. Mandatory initialization fields, BSS selection, full metrics and the corrected optimizer are retained. The circle uses family rank $\lceil 2 \log(en) \rceil$, time degree five and circle degree nine; dimension checks use rank $\lceil \log(en) \rceil$ and degrees three. Digits uses rank 19. These source approximations have no certified uniform error bound. All use nine time nodes; the circle, sphere checks and digits use respectively 64, 128 and 256 passive spatial nodes. The toy circle has seven training points and target $\sin(3\theta) + \frac{1}{2} \cos(5\theta)$; the sphere target on unit directions v is $\sqrt{d} v_1 + d^{3/2} v_1 v_2 v_3$. Training-label RMS normalizes the targets. Harmonic uses 256 test directions, twelve sphere training points, and times 0, 0.5, 1, 2, 5, 10, 20. seeded decoder uses ten Euler panels of length 0.5, noise 0.01, one member and 32 queries through time five. It retains both Gaussian orientations and covariance corrections, immutable row fields, causal metric replay and empirical query sums. Float64, a conventional counter PRNG and numerical-rank selection replace the certified finite-word backend; no bit, PRG or confidence certificate is asserted. Its fixed Euler program is not an integration-convergence study. Toy decoder checks use $m = \max(d, 3)$ and the common recorded times 0, 0.5, 1, 2, 5. Harmonic is coupled to its source; the decoder’s target is independent.

Controls, storage and costs. Small MLPs train all layers at the same canonical mobilities, with width chosen to fit the retained payload. At zero readout, the exact frozen initial NTK has only its readout block; its constant-kernel flow is solved spectrally without ridge or time rescaling. Harmonic counts $q_1 d + q_1 q_2 + q_2 + m$ moving and $2q_1^2 + q_2^2$ fixed tensor scalars. Common training data, measurement copies and query scratch are separate. seeded decoder counts data, seeds, selected fields, metric caches, acquired pairs, field instructions and immutable coefficients as 64-bit payload; Python overhead is excluded. In dimension five its retained-plus-batched-query workspace envelope is 1,201,027 words: about 14.0-fold smaller than the width-4096 dense parameters, but larger than the width-1024 dense parameters. Source setup, training, readout refresh and batched queries are timed separately. Runs use two RTX 3090 GPUs and CPU scalar programs, with 120-second toy and at most 300-second real-data model caps. These measurements establish no end-to-end speedup.

One circle confirmation also compares explicit LoRA increments with trained outer layers at width 2048. Its 313,344 moving scalars match Harmonic’s 315,481 from below, but its 4,194,304 fixed mixer scalars are additional. Orthonormal initial right factors and zero left factors preserve the dense initialization. The factor mobility multiplier 0.25 was selected from 0.25, 1, 4 on pilot seed 101; steps 0.01 and 0.005 resolve the held-out seed-201 comparison. LoRA’s discrepancy is 7.51 dense-pair units versus 0.479 for Harmonic. This is evidence against this matched-moving-state factorization and optimizer, not against every low-rank alternative.

For timing context, the width-4096 circle seed-201 Harmonic source rollout and spectral fit take 4.75 seconds, followed by 4.96 seconds of assembly. Its RK4 updates take 12.5 ms per step versus 27.7 ms for dense, at step 0.125; these figures exclude query evaluation and the separate refinement. For dimension-five seeded decoder seed 201 at the same width, CPU source and metric setup take $0.66 + 0.84$ seconds, ten causal training panels take 0.46 seconds, and all five batches of 32 passive queries take 3.31 seconds. The tasks,

clocks, solvers and hardware differ, so these are separate measured costs, not a cross-method speedup ratio. Device peaks in the records include other resident references; payload counts are not isolated process-RAM peaks.

Additional and negative results. Harmonic’s dimension-3, 5, 10 ratios are 0.751, 0.937, 0.997 at width 2048 (one seed each). The real-data smaller-MLP ratios are 0.646–1.123, below Harmonic’s 1.85–2.44; one also attains 97.70% accuracy. The frozen-kernel circle ratios are 15.6–76.6, so the measured trajectories depart substantially from that control. seeded decoder’s width-4096 dimension-three and dimension-ten checks give 1.84 and 1.26, while the circle gives 6.95 and fails. A dimension-ten pilot with twenty Euler panels exceeded the 120-second cap; the ten-panel pilot passed before its rule was frozen. A separate digits decoder experiment uses eight training images, training-only PCA dimension five and all 285 held-out images. The twenty-panel setting passed its tuning split but fails the single held-out confirmation: $R_{\text{emp}} = 4.94$, versus 3.02 for its payload-matched MLP. The decoder retains 6,608,623 words (2.54 times fewer than dense), and its retained-plus-batched-query envelope is 11,052,846 words. Setup, replay and queries take 67.9 seconds in total. This is not the 32-training-image Harmonic task and does not establish real-data decoder comparability. Figure 3 retains these scope distinctions. Gradient, metric, paired-action, posterior, replay and passive-query checks precede evaluation. The full-retention Harmonic trajectory error decreases approximately sixteenfold under RK4 step halving. Commands, hashes, inventories and failures are retained in `paper/figures/capture_trajectory.py` and the study records under `data/generated/compression_empirical_validation_20261007/`.

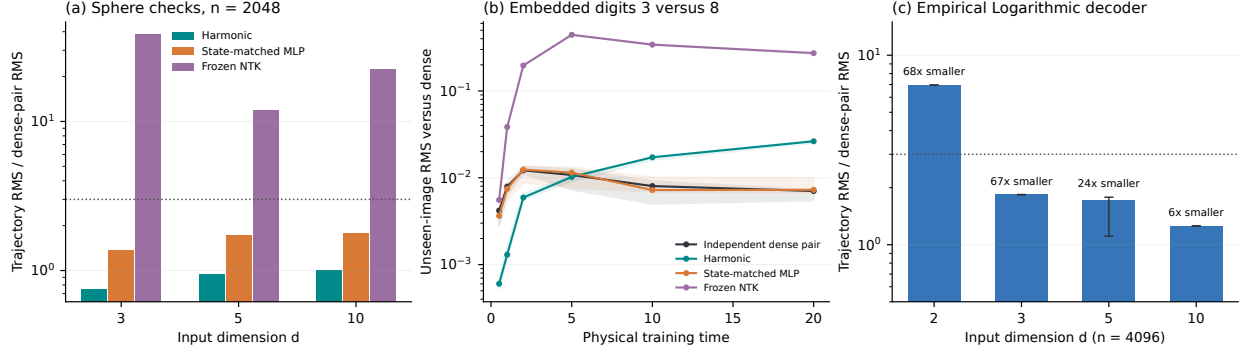


Figure 3: Dimension, real-data and decoder checks. Additional finite-time comparisons supplement the width experiments. Dimension checks use one held-out seed; digits uses three. The dimension-five decoder bar summarizes three seeds, the others one; its annotations count persistent payload only, excluding query workspace. Lines/bars show medians and ranges show all observed seeds. The empirical factor-three threshold is not a confidence statement. The circle decoder failure and matched small-network controls prevent interpreting the favorable cases as uniform superiority.

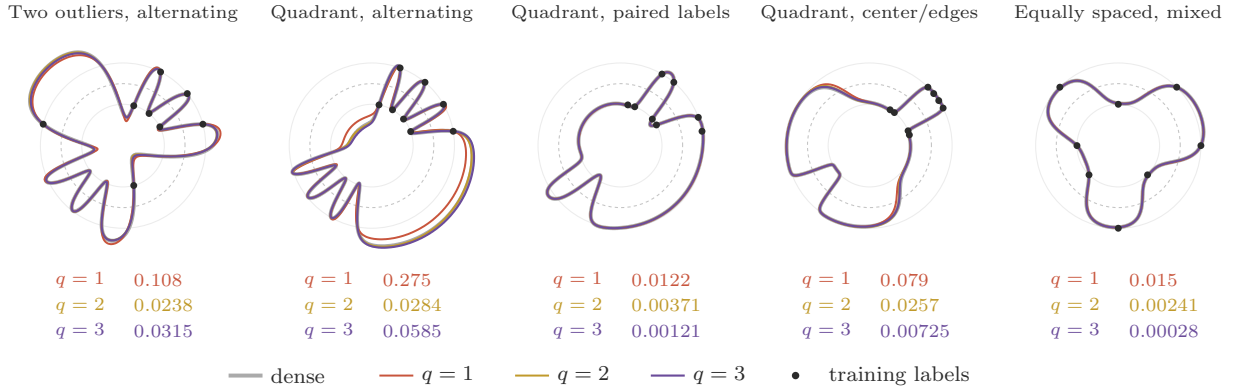


Figure 4: Earlier Legendre fitted-function illustration. Existing five-task comparison: three hidden tanh layers, width 4096, and Legendre orders $q = 1, 2, 3$, at each model's own training-MSE 10^{-3} endpoint, not a shared time. Grey curves are dense predictions, coloured curves are memory predictions, and black points are labels. Printed discrepancies are RMS differences on 8192 circle queries, not uniform errors or risk against unseen labels. Fixed initialized matrices remain additional to Legendre's moving state. This archived illustration is separate from the earlier confirmation runs above.

B Detailed execution costs and supplementary decoder costs

We cost the prescribed sphere models at dense width n : Legendre and Harmonic use the negligible-relative-error choices in Theorem 3.1, whereas the supplementary Seeded decoder retains its separate fixed-quantile certificate. Initialization includes every one-off operation needed to build a model. For the neural constructions, a training stage means one full-batch vector-field evaluation; a numerical update pays for all its stages. Initialization and training memory are total peaks, including data. Query workspace is additional to the loaded, inference-ready model. The supplementary seeded decoder, which is not the new finite-panel Logarithmic model, instead has a finite-word contract and a bound for its complete training schedule.

B.1 The neural constructions

For this section define the sufficient state envelopes

$$S_{\text{Leg}} = Cm \left(\frac{m}{\gamma} \right)^3 n^{5/4} [\log(en)]^2 e^{\sqrt{\log(en)}/2},$$

$$S_{\text{H}} = C(m+d)^2 + \left(\frac{C}{d} \right)^{d+1} \left(\frac{Ym}{\gamma} \right)^4 [\log(en)]^{3d+2}.$$

S_{Leg} bounds Legendre’s moving coordinates; its fixed dense matrices are additional. S_{H} bounds Harmonic’s complete retained state. These are the sizes selected by the sphere compression certificates, rather than separately supplied orders. Under the theorem’s eventual-width qualification and with $n \geq \max(m, d)$, their classical arithmetic costs are as follows; C retains the theorem’s activation/depth convention.

Operation	Work	Peak memory
Dense initialization	n^2	$n^2 + md$
Dense training stage	mn^2	$n^2 + mn + md$
Dense query	n^2	n additional
Legendre initialization	$mn^2 + S_{\text{Leg}}$	$n^2 + S_{\text{Leg}} + md$
Legendre training stage	$mn^2 + mS_{\text{Leg}}$	$n^2 + S_{\text{Leg}} + md$
Legendre query	$n^2 + S_{\text{Leg}}$	n additional
Harmonic training stage	mS_{H}	S_{H}
Harmonic query	S_{H}	$\sqrt{S_{\text{H}}}$ additional

Table 1: Execution of the models prescribed by the sphere compression certificates. Entries are upper bounds up to C ; scalar evaluator costs are charged below. Harmonic initialization is costed separately.

Legendre streams its correction factors and uses cumulative prefixes for the moment updates. Its $n^{5/4+o(1)}$ moving state therefore retains the fixed dense-matrix cost. Harmonic’s training row includes the moving feature-Gram solve and readout-cache refresh; the cubic solve is absorbed because $S_{\text{H}} \geq Cm^2$. Its query row uses that refreshed cache.

The table counts non-activation arithmetic. Dense and Legendre add Cmn activation-value and first-derivative calls per stage, and Cn value calls per query. Legendre initialization adds Cmn value calls. Harmonic adds $Cm\sqrt{S_{\text{H}}}$ value and first-derivative calls per stage and $C\sqrt{S_{\text{H}}}$ value calls per query. Add each routine’s work and scratch. Dense-reference generation also requires $nd + (L-1)n^2$

Gaussian draws; omit these draws when the reference is supplied. The complete execution inventory retains the unsimplified counts. These arithmetic bounds do not specify the number, precision, or accuracy of subsequent numerical training steps.

B.2 Two ways to initialize Harmonic compression

For $Y > 0$, both certified initializers support the Harmonic model chosen to have error at most Y/n . This needs the exact inverse substitution: the original choice $T = T_0$, $\eta = 1/n$ alone gives only $n^{-1+o(1)}$ error. Write $\ell = \log(en)$, $\lambda = \gamma/m$, $T_0 = 32\ell/\lambda$, and $\eta_0 = \min(1, Y, 16Y/\lambda)$. The explicit comparison coefficients $\mathcal{A}_n, \mathcal{D}$ in (93) and (82) give the all-time bound $\mathcal{A}_n\eta + \mathcal{D}e^{-\lambda T/4}$. Therefore set

$$\begin{aligned} \mathcal{E}_1 &= \mathcal{A}_n\eta_0 + \mathcal{D}e^{-8\ell}, & u &= \max\{0, \log(n\mathcal{E}_1/Y)\}, \\ T &= \frac{32\ell + 4u}{\lambda}, & \eta &= \eta_0 e^{-u}. \end{aligned} \quad (23)$$

Then $\mathcal{A}_n\eta + \mathcal{D}e^{-\lambda T/4} = \mathcal{E}_1 e^{-u} \leq Y/n$. Use the exact inverse's integer budget and its full-width fallback when required. For activation strip $|\operatorname{Im} z| < a$, recall the activation envelope

$$\beta = \max \left\{ 10, 1 + \max_j |\phi_j(0)|, 16/a, \max_{j, k \in \{1, 2\}} \sup_{|\operatorname{Im} z| \leq a/2} |\phi_j^{(k)}(z)| \right\}.$$

The numerical envelope in the inverse proof also gives the explicit estimate

$$u \leq \log n + 65\sqrt{\ell} + \log \max \left\{ 1, 4096e^{44}\beta^{42L} \frac{m}{\gamma} \left(1 + \sqrt{\frac{m}{\gamma}} \right) \right\}. \quad (24)$$

Thus $u = O(\ell)$, $T = O(\ell)$, and $\log(1/\eta) = \log(1/\eta_0) + u = O(\ell)$ when the positive problem parameters are fixed. No factor involving m/γ is omitted from the displayed substitutions.

For completeness, the source radii are $r_t = c_t/\sqrt{\ell}$, $r_q = c_q/\sqrt{\ell}$, with $c_t = a\gamma/(1024UmY^2)$ and $c_q = \min\{1/8, a/(8V)\}$; U, V are the source coefficients in (S.25). For $d \geq 2$, the inverse's exact mode cutoff obeys

$$\begin{aligned} \alpha_T^{-1} &= \frac{4T}{r_t} = 4096 \frac{U}{a} \left(\frac{Ym}{\gamma} \right)^2 (32\ell + 4u)\sqrt{\ell}, \\ H(T, \eta) &= H(T_0, \eta_0) + 2u + 2 \log \left(1 + \frac{u}{8\ell} \right). \end{aligned}$$

Consequently $\alpha_T^{-1} = O(\ell^{3/2})$ and $H(T, \eta) = O(\ell)$. Use the setup-local temporal and spherical degrees p, ℓ_* , node counts N_t, N_x , spatial basis size H_{sph} , joint mode count N , source-generator bound R , and actual rank r from the finite setup inventory. Here p is unrelated to the passive-panel size elsewhere in the paper. The exact mode counts and positive quadratures give

$$\begin{aligned} p + 1, N_t &= O(\ell^{5/2}), & \ell_* &= O(\ell^{3/2}), \\ N_x, H_{\text{sph}} &= O(\ell^{3(d-1)/2}), & N, R, r, q &= O(\ell^{3d/2+1}). \end{aligned}$$

At $d = 1$, use the certified two-point convention $N_x = H_{\text{sph}} = 2$; there is no spherical degree ℓ_* . The other displayed orders still hold. The nodal tolerance has $\log(1/\delta_{\text{node}}) = O(\ell)$. In the local-continuation recipe the distinct signed-stability coefficient $A_n = 4J^r Y/\lambda$ is $O(\sqrt{\ell})$; its factor e^{-A_n} in the defect allowance is retained. The recipe therefore gives panel count and activation-polynomial

Initializer	Arithmetic work	Peak real coordinates
Explicit dense continuation	$O(n^2[\log(en)]^{3d/2+1})$	$O(n^2)$
Implicit Gaussian continuation	$O(n[\log(en)]^{9d/2+3})$	$O(n[\log(en)]^{3d/2+1})$

Table 2: Certified initialization of the selected Harmonic model, with problem parameters fixed. Evaluator and sampler work remain charged.

degree $J, D = O(\ell^{3/2})$ and Taylor degree $K = O(\ell)$. Substitution in the supplied-order work and memory envelopes proves the following rates at the exact target Y/n . All problem parameters are fixed in this table; its constants may depend on them. The complete setup recipe and costs retain every structural parameter and internal order. Both initializers discard their temporary trajectory, source arrays and matrix-action history. Their activation-value counts are $O(nm + [\log(en)]^{3/2})$ in this fixed-problem specialization; add the evaluator’s actual work and scratch. Explicit initialization uses $nd + (L - 1)n^2$ Gaussian draws; implicit initialization uses at most $nd + Cn[\log(en)]^{3d/2+1}$ at fixed problem parameters. The displayed work rates include these draws only under unit-cost sampling. The explicit execution accepts a supplied dense realization. The implicit execution samples a fresh coupled reference and has no subquadratic conversion guarantee for a supplied dense array. Elementary scalar primitives (square roots, trigonometric functions, logarithms and exponentials) use the detailed inventory’s separate primitive-cost convention; unit-cost evaluation includes them in the displayed rates. For $Y = 0$, use the exact zero-predictor branch instead.

B.3 The finite-word decoder

The decoder counts word operations. Its explicit width gates bound the resource logarithm by $C \log(en)$ and its member moment order by $C \log(em)$, so the retained-word bound in the sphere compression certificates already displays its dependence on m, d, γ . Each word has $C \log(en)$ sufficient bits under the same activation/depth convention. With all problem parameters fixed, the complete operation bounds become

$$\begin{aligned}
&\text{initialization: } O(n[\log(en)]^{29/2}), \\
&\text{complete training: } O([\log(en)]^{12} \log(e + \log(en))), \\
&\text{one unseen query: } O(n[\log(en)]^{12} + [\log(en)]^{14}).
\end{aligned} \tag{25}$$

Training and live query memory fit the complete retained inventory, $O([\log(en)]^6 \log(e + \log(en)))$ words at fixed parameters. Initialization has peak $O(n[\log(en)]^{5/2} + [\log(en)]^{15/2})$ words. The full word-operation ledger keeps the structural prefactors and the separate temporary arrays. These query-memory counts are total memory, unlike the additional workspace in Table 1.

The finite tanh evaluator is included. For another analytic activation, add its supplied evaluator’s work at every recorded value call and its scratch, together with external data/query access and output. The finite training table is counted; other retained external descriptions require additional words. Analyticity supplies no evaluation-cost bound for an arbitrary function. The training row covers the complete prescribed schedule. A query streams the regenerated source rows, retaining the displayed width factor in its work.

Guide to the detailed statements and proofs

The appendices contain the complete parameter-explicit scientific statement and proof chain. The main text suppresses activation/depth factors, uses absolute prediction error, and internalizes the choice of approximation orders. The appendix restores activation/depth coefficients explicitly; its numerical C is distinguished from the activation/depth envelope denoted by C in the main text. Here $A = W^{(1)}$ when the first layer is written separately, and w is the same output weight as above. Local proof constants, source degrees and numerical orders are not new runtime model parameters.

The full label, width, confidence, storage and cost interfaces are followed by numerical certificates and proofs. The shared source construction, both efficient Harmonic initializers, the new residual-adapted panel proof, and the supplementary seeded decoder are included in the manuscript itself, not left as dependencies on research notes. The last is retained as a distinct result, not an alternative definition of the new Logarithmic model. Historical review records and superseded paper theorems are not premises.

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C Complete certificate interfaces and notation

Scope of this detailed library. This appendix contains the full dense, Legendre and Harmonic certificates, their common source and stability machinery, and a separate seeded finite-word decoder. The latter is supplementary: it is not the finite-panel Logarithmic construction of Theorem 3.1. Its notation f_{Seed} and its word counts refer only to that supplementary decoder. The finite-panel construction and the assembly of the current theorem appear in Appendices F and F.7, which invoke the shared lemmas below and prove the additional flaring-domain estimates in full. Unless explicitly qualified otherwise, the norm $\|\cdot\|_*$ in this library is the whole-sphere norm. In the panel appendix it is replaced by the declared-panel norm. Decoder-local p is a moment order, not the number of passive inputs in the main theorem.

The first-layer matrix denoted by A in the detailed formulas below is exactly $W^{(1)}$ in the main text. The hidden matrices $W^{(2)}, \dots, W^{(L)}$, readout w , physical time, and all normalizations are unchanged. In particular, $h^{(0)} = x/\sqrt{d}$, and the prediction is $w^\top h^{(L)}/n$. This correspondence avoids changing proof-local notation or introducing a second model.

Symbols introduced inside a proof or execution ledger are local to that section. In particular, $\ell = \log(en)$ is a logarithmic shorthand only where explicitly defined; it is not a layer index there. The horizon T_n , Legendre absorption threshold q_{abs} , extension radius b_{ext} , numerical restart radius r_{tube} , and variability quantile $b_n(\alpha)$ have distinct roles and retain these distinct names throughout.

C.1 Shared setup, notation and qualifications

The structural parameters remain separate: dense width n , sample count m , input dimension d , and hidden depth $L \geq 2$. Inputs satisfy $\|x_a\|_2 = \sqrt{d}$, and labels are arbitrary fixed real numbers. The common freely supplied compression order is q : Legendre memory order or Harmonic per-layer neuron budget. Seeded decoder compression has no free q ; its moment and field orders are prescribed by the width, problem and confidence certificate. The remaining global numerical symbols are label RMS Y , feature gap γ , activation envelope β , failure probability δ , target accuracy ε , and a generic numerical constant C . Different occurrences of C may denote different universal constants; they never conceal structural dependence. The detailed statements and proofs use locally defined coefficients, not extra model orders.

The dense forward pass is

$$z^{(1)} = Ax/\sqrt{d}, \quad z^{(j)} = W^{(j)}h^{(j-1)}, \quad h^{(j)} = \phi_j(z^{(j)}), \quad f_n = w^\top h^{(L)}/n.$$

The first weights have independent $N(0, 1)$ entries; hidden mixers have independent $N(0, 1/n)$ entries; blocks are independent; $w(0) = 0$. Training minimizes mean squared loss with mobilities $(n, 1, \dots, 1, n)$. The dense copy $\tilde{f}_n$ is independently initialized. Legendre and Harmonic use their realized reference initialization and the same physical time; the Seeded decoder instead compares a newly generated compressed model with an independent dense reference. For the implicit Harmonic initializer, that realization is latent: exact adaptive Gaussian actions couple the output to one ordinary dense reference without generating its full hidden matrices. This samples a fresh joint reference/output law; it is not conversion of a previously supplied matrix or a prescribed entrywise random seed.

Each activation is real on the real axis, holomorphic on a common strip $|\text{Im } z| < a$, with bounded first derivative there. Activation values may be unbounded. Set

$$\beta = \max \left\{ 10, 1 + \max_j |\phi_j(0)|, \frac{16}{a}, \max_{j, k \in \{1, 2\}} \sup_{|\text{Im } z| \leq a/2} |\phi_j^{(k)}(z)| \right\}, \quad Y = \frac{\|y\|_2}{\sqrt{m}}.$$

The covariance recursion and unweighted gap are

$$Q_{ab}^{(0)} = x_a^\top x_b/d, \quad Q_{ab}^{(j)} = \mathbb{E}[\phi_j(Z_a)\phi_j(Z_b)], \quad Z \sim N(0, Q^{(j-1)}), \quad \gamma = \lambda_{\min}(Q^{(L)}) > 0.$$

The gap is not divided by m . No orthogonality, centering, sign pattern or clipping is imposed. Dense, Legendre and Harmonic need no input-rank assumption. Only the Seeded decoder uses the already stated spanning condition and $m \geq d$.

Labels. The clean dense/Legendre envelopes below use the unchanged sufficient cap

$$0 < Y \leq \frac{\gamma}{m} \beta^{-30L}. \quad (26)$$

The larger original recurrence allowance is retained explicitly at the start of the detailed statements. The Harmonic forward theorem and its new beta-only storage envelope hold on that entire larger range. Zero labels give the stationary zero predictor and exact constant-zero compression separately.

Error. All comparisons use

$$\|f - g\|_* = \sup_{t \in [0, \infty]} \sup_{\|x\|_2 = \sqrt{d}} |f(t, x) - g(t, x)|. \quad (27)$$

This includes fitted limits and the whole sphere, a circle only when $d = 2$. It measures function fidelity, not unknown-label test risk.

Probability and width. Fix $0 < \delta < 1$. For the dense, Legendre and Harmonic results, each fixed admissible problem has a finite width threshold, depending on all its parameters and δ , above which the stated comparisons hold jointly with probability at least $1 - \delta$. It includes inherited source/initialization gates, the lower theorem’s eventual threshold when used, and the explicit analytic-tail and construction gates. Its stochastic part remains unquantified. Seeded decoder compression has instead the explicit, extremely conservative sufficient width stated below; it introduces no additional existential stochastic threshold. Every statement concerns each individual width, not one event over infinitely many independent initializations. Split failure budgets for joint statements. Fixed-confidence logarithms in the first three upper bounds are absorbed by their eventual-width convention, not by a claimed numerical failure-rate formula. Multiplying storage by a fixed constant is not claimed to produce exponential confidence amplification.

Model domains. Legendre permits every integer $q \geq 1$, with no separate accuracy-order condition. The Harmonic family permits every integer budget $q \geq 18(2m + d + 9)$, abbreviated $q \geq C(m + d)$ in the headline counts only. This is exact-initialization overhead: positive training Gram forces top width at least m , and exact first-weight columns force first width at least d when $n \geq d$. For $q \geq n$, the full selected space gives exact agreement with dense. Seeded decoder compression instead requires $m \geq d$, spanning training inputs, $0 < \delta < 1/4$, and the complete common fitting/source label allowance. It is a certificate-prescribed family indexed by n , not an arbitrary-budget family indexed by q .

C.2 Forward interface: supplied size/order gives storage and error

Dense, Legendre and Harmonic storage counts real coordinates, not bits. Seeded decoder storage counts finite numerical words and states its certified bits per word separately. The separate computational-cost section reports initialization, per-stage training and single-query costs; storage alone is not a runtime bound. For $n \geq d$, the clean counts are:

Detailed comparison

Model: Dense

Learned state	$O(Ln^2)$
All retained model storage	$O(Ln^2)$

Model: Legendre

Learned state	$O(n[d + Lmq])$
All retained model storage	$O(Ln^2 + Lmnq)$

Model: Harmonic

Learned state	$O(Lq^2)$
All retained model storage	$O(Lq^2)$

Model: Seeded decoder

Learned state	bounded by the complete-model envelope at right
All retained model storage	$O(\log^6(en) \log \log(e^e + n))$ numerical words at fixed problem parameters

Before simplification, the exact dense moving count is $(L-1)n^2 + n(d+1)$, the Legendre count is $n(d+1) + 1 + 2(L-1)mnq$, and Harmonic has at most $(L-1)q^2 + q(d+1) + m$. Legendre additionally retains $(L-1)n^2$ fixed mixers; reconstructed matrices are evaluation objects. Harmonic includes metrics, fixed copies, residual coordinates, data and solve caches. Original-width source arrays and jets are discarded. Ordinary data storage is additional for the dense/Legendre model-state counts if retained. The Seeded decoder count already includes its normalized training table, all ensemble members, seeds, selected packets, caches and current scalar state.

Legendre uses its original residual-RMS clock and prefixes. Harmonic uses the corrected-readout autonomous optimizer, not ordinary gradient flow on an arbitrary smaller network. All hidden arrays train. Neither has a second independent runtime approximation order.

C.2.1 Dense versus an independent dense copy

A simplified polynomial upper envelope is

$$\|f_n - \tilde{f}_n\|_* \leq C\beta^{CL}Y \left(1 + \frac{m}{\gamma}\right)^5 \sqrt{\frac{d}{n}} \log(en) e^{\sqrt{\log(en)}}. \quad (28)$$

The exact, sharper actual-label coefficient and confidence logarithm remain in the exact dense certificate. The sample/gap powers in this upper envelope are not claimed sharp.

For $m \geq 2$,

$$\|f_n - \tilde{f}_n\|_* \geq c_{\phi,L,\delta} \frac{Y\sqrt{\gamma}}{\sqrt{n} \log(en)^{5/2}}. \quad (29)$$

The positive coefficient depends only on activations, depth and confidence; its formula is in the exact lower theorem. It cannot be replaced by a confidence-independent universal constant. The actual-prediction witness occurs at a positive early time, possibly shrinking with width; this is not an endpoint lower bound. Deterministic exceptions prevent the general assertion at $m = 1$.

C.2.2 Legendre versus its realized dense reference

For every integer $q \geq 1$,

$$\|f_{\text{Leg},n,q} - f_n\|_* \leq C\beta^{CL}Y \left(1 + \frac{m}{\gamma}\right)^6 \frac{\sqrt{\log(en)\log(eq)}}{q^2} e^{\sqrt{\log(en)}}. \quad (30)$$

The stronger exact actual-label coefficient is unchanged: the exact all-order statement. Above the actual absorption threshold use the signed comparison; below it use the all-order fitting/predictor bound. Retaining the small label powers shows that the same coefficient covers both regions. No second error term or accuracy-order gate is needed on the explicit cap.

C.2.3 Harmonic versus its realized dense reference

For every Harmonic budget in the shared model domain,

$$\begin{aligned} \|f_{\text{Harm},n,q} - f_n\|_* &\leq C\beta^{CL}Y \frac{m}{\gamma} \left(1 + \sqrt{\frac{m}{\gamma}}\right) n e^{C\sqrt{\log(en)}} \\ &\quad \times \exp \left[-\frac{\sqrt{d}}{C\beta^{CL}} \left(\frac{q}{(Ym/\gamma)^2 \log(en)^{d/2}} \right)^{1/(d+1)} \right]. \end{aligned} \quad (31)$$

This is a genuine decreasing certificate in supplied q . The construction takes n, q as inputs; its internal source tolerance and horizon are determined by the budget and discarded after setup. There is no fixed error

floor. For $q \geq n$, exact retention improves the bound to zero. The growing comparison exponential has only numerical coefficient C ; structural factors in the negative exponential specify approximation efficiency, not unstable amplification.

The power $1/(d+1)$ comes from approximating the sphere and a horizon that grows with required accuracy. It is a construction rate, not an optimality lower bound. An initialization-only selected model covers smaller budgets within the shared domain using independent fitting; there is no hidden extra accuracy-dependent budget test.

The original construction at source tolerance $1/n$ remains valid with its sharper error $C\beta^{CL}Y(m/\gamma)(1 + \sqrt{m/\gamma})n^{-1}e^{C\sqrt{\log(en)}}$ and original $O(\log(en)^{3d+2})$ all-retained storage. The supplied-budget family preserves that specialization as a separate available construction.

C.2.4 Seeded decoder versus an independent dense reference

This method has no supplied order q . Define its implemented member moment order and common resource logarithm by

$$p = \max \left\{ 1, \left\lceil \frac{\log(2^{22}emL)}{\log(64e^2)} \right\rceil \right\}, \quad Z = \log(en) + \log \left(e + \frac{(m+d+2)\beta^{100L}(1+m/\gamma)}{\delta} \right). \quad (32)$$

The confidence-dependent order used only to certify the independent reference is

$$p_{\text{ref}} = \max \left\{ 1, \left\lceil \frac{\log(2^{22}emL/\delta)}{\log(64e^2)} \right\rceil \right\}. \quad (33)$$

A sufficient enclosing width is

$$n \geq \max \left\{ C_* \left[\frac{2^{20}\beta^{2000L}(1+m/\gamma)^4(m+d+p_{\text{ref}}+1)^4}{\delta} \right]^{1100}, Y^{-1}, C_{\text{num}} \right\}. \quad (34)$$

Here C_* is universal and $C_{\text{num}} = \max\{1, (A_{\text{num}}/32)^{1/9}\}$, where A_{num} is the absolute coefficient in the chosen numerical-error allocation. The sharper factorized gate and the few separately displayed implementation gates are retained in the detailed-statement appendix. This width is fully quantitative but extremely conservative.

Let

$$B_n = \frac{c_0 Y}{\sqrt{n}} e^{c_1 Y^2 \sqrt{\log(en)}} \sqrt{8 \log \frac{2048(n+1)(1+2n)^d}{\delta}} + c_{2,\text{mesh}} \frac{Y}{n}, \quad (35)$$

$$32 \leq c_{2,\text{mesh}} \leq C\beta^{4L}(1+m/\gamma).$$

The positive coefficients c_0, c_1 are the inherited fixed-problem coefficients of the dense comparison; this construction does not quantify them further. At every width satisfying the stated gates,

$$\|f_{\text{Seed},n} - f_n^{\text{ind}}\|_* \leq 3B_n \quad (36)$$

with probability at least $1 - \delta$. The event is simultaneous over all sphere inputs, all physical times, adaptively chosen queries, and the fitted endpoint. The compact model is built jointly with a fresh independent dense reference law; it does not encode a previously supplied dense matrix or seed.

Its learned and total retained storage are both at most a universal multiple of

$$p^2 \beta^{402L} (m+d+2)^2 (1+m/\gamma)^2 Z^6 [d+1 + \log(e + (d+1)Z)] \quad (37)$$

numerical words, each of sufficient length $C\beta^{110L}Z$ bits. This includes all retained data, seeds and members; the same word envelope also covers the live query workspace. The clean display uses $nY \geq 1$, already enforced by the width gate. For $0 < nY < 1$, replace Z everywhere in the numerical counts by $Z + \log_+(1/(nY))$ and retain the enlarged Gaussian-RMS gate.

The independent intrinsic certificate, with exactly the same construction, is $2b_n(\delta/32) + A_{\text{num}}Yn^{-10}$; see its complete proof. For $m \geq 2, Y > 0$ it is at most $3b_n(\delta/32)$ eventually. Neither this nor the additional analytic

bound $3B_n$ asserts Harmonic’s stronger matched-reference $n^{-1+o(1)}$ error or a vanishing random-discrepancy ratio.

C.2.5 Two efficient initializers for the same Harmonic family

The source coefficients can be constructed either by **explicit dense local continuation** or by **implicit Gaussian local continuation**. Both preserve the full original label interval, activation class, confidence event, prediction certificates, retained-state inventory and autonomous optimizer. Neither changes the final model’s physical time zero. The original finite initialization-jet construction remains a valid, separately costed alternative.

At source tolerance $1/n$ and source horizon $32(m/\gamma) \log(en)$, with each admissible structural parameter and confidence fixed separately, their sufficient bounds are:

Detailed comparison

Harmonic initializer: Explicit dense local continuation

Setup work $O(n^2 \log(en)^{3d/2+1})$

Peak setup memory $O(n^2)$

Harmonic initializer: Implicit Gaussian local continuation

Setup work $O(n \log(en)^{9d/2+3})$

Peak setup memory $O(n \log(en)^{3d/2+1})$

Both retain $O(\log(en)^{3d+2})$ coordinates and the same stronger $n^{-1+o(1)}$ whole-sphere/all-time error. These headline setup rows assume bounded-cost scalar activation, elementary-function and Gaussian primitives. The finite supplied-order formulas and complete parameter recipe expose all $m, d, L, \gamma, Y, \beta$, order and primitive-cost dependence; it is not suppressed in a universal constant.

Here a computationally cheap rollout reconstructs the required physical interval using restarted high-order local expansions. It is not a claim that an arbitrary short initial-time prefix suffices, or that a fixed number of ordinary Euler steps supplies the coefficients. The implicit method samples exactly the joint law of the explicit finite local initializer and its dense reference; it need not return the identical finite coefficient approximations of the older origin-jet algorithm. No dense array, query oracle, or setup history remains in either final compressed model.

C.3 Inverse interface: supplied accuracy determines size/order

The common target is

$$\mathbb{P}\{\|f_{\text{model}} - f_n\|_* \leq \varepsilon\} \geq 1 - \delta. \quad (38)$$

For each certificate choose the smallest permitted integer size/order whose right side is at most ε , with the common width gates enforced. This minimizes the increasing storage-budget bound within that certificate, not compression among all possible models. Exact scalar inversions, including loose targets and the full-width fallback, are in the detailed-statement appendix.

The following simpler prescriptions describe polynomial accuracy $\varepsilon = n^{-\Theta(1)}$, with structural parameters and confidence fixed and n sufficiently large: the accuracy may lie between two different fixed negative powers of n , not necessarily $1/n$. This does not identify or order $m, d, L, 1/\gamma$. The common eventual convention absorbs fixed coefficients inside logarithms. For the large-parameter simplification use $m/\gamma \geq 1$; otherwise replace that ratio by $1 + m/\gamma$ in the dense/Legendre prescriptions.

C.3.1 Dense: choose width before initialization

A sufficient eventual choice is

$$n = \left\lceil \frac{C\beta^{CL}Y^2d}{\varepsilon^2} \left(1 + \frac{m}{\gamma}\right)^{10} \log(e/\varepsilon)^2 e^{C\sqrt{\log(e/\varepsilon)}} \right\rceil, \quad \text{storage} = O(Ln^2). \quad (39)$$

Its all-retained model storage has the explicit sufficient envelope

$$\text{storage}(\tilde{f}_n) \leq CL\beta^{CL}Y^4d^2 \left(1 + \frac{m}{\gamma}\right)^{20} \varepsilon^{-4} \log(e/\varepsilon)^4 e^{C\sqrt{\log(e/\varepsilon)}}. \quad (40)$$

This is eventual as $\varepsilon \downarrow 0$: the selected width must exceed the common threshold. It is not a fully effective confidence-certified width, because the stochastic source threshold is unquantified. It selects a reference before initialization, not a replacement for an already prescribed reference.

C.3.2 Legendre: choose order for the given reference

The inverse of the forward bound has sufficient eventual form

$$q = \left\lceil C\beta^{CL} \left(\frac{m}{\gamma}\right)^3 \sqrt{\frac{Y}{\varepsilon}} [\log(en) \log(en/\varepsilon)]^{1/4} e^{\frac{1}{2}\sqrt{\log(en)}} \right\rceil. \quad (41)$$

Consequently

$$\text{learned storage} \leq Cnd + C\beta^{CL}Lmn \left(\frac{m}{\gamma}\right)^3 \sqrt{\frac{Y}{\varepsilon}} [\log(en) \log(en/\varepsilon)]^{1/4} e^{\frac{1}{2}\sqrt{\log(en)}}. \quad (42)$$

Fixed mixers remain additional. There is no separate order-admissibility test even before taking this eventual simplification. Keeping the two logarithms also keeps the coefficient uniform in the fixed polynomial accuracy exponent; replacing them by $\log(en)^2$ would hide a constant depending on that exponent. At the absolute root-width target $\varepsilon = n^{-1/2}$, this gives

$$\text{learned storage} \leq Cnd + C\beta^{CL}Lm\sqrt{Y} \left(\frac{m}{\gamma}\right)^3 n^{5/4} \sqrt{\log(en)} e^{\frac{1}{2}\sqrt{\log(en)}}. \quad (43)$$

The outer factor m is the response-memory multiplicity in the exact count $n(d+1) + 1 + 2(L-1)mnq$; it has not been removed. When $m \geq d$ and the positive problem parameters are fixed while n grows, the second term eventually dominates nd , and the clean form is

$$\text{learned storage} \leq C\beta^{CL}Lm\sqrt{Y} \left(\frac{m}{\gamma}\right)^3 n^{5/4+o(1)}. \quad (44)$$

At the label-scaled root target $\varepsilon = Y/\sqrt{n}$, used before the extra logarithmic enlargement that makes compression negligible relative to dense variability, the factor $\sqrt{Y}$ cancels. Thus, under the same $m \geq d$ convention,

$$\text{learned storage} \leq C\beta^{CL}Lm \left(\frac{m}{\gamma}\right)^3 n^{5/4+o(1)}. \quad (45)$$

C.3.3 Harmonic: choose width budget for the given reference

Inverting its stretched-exponential certificate gives

$$q = \left\lceil C(m+d) + \frac{\beta^{C(d+1)L}}{d^{(d+1)/2}} \left(\frac{Ym}{\gamma}\right)^2 \log(en)^{d/2} \log(en/\varepsilon)^{d+1} \right\rceil. \quad (46)$$

The corresponding all-retained bound, hence also the learned-state bound, is

$$\text{storage}(f_{\text{Harm},n,q}) \leq CL(m+d)^2 + \frac{L\beta^{C(d+1)L}}{d^{d+1}} \left(\frac{Ym}{\gamma}\right)^4 \log(en)^d \log(en/\varepsilon)^{2d+2}. \quad (47)$$

There is no factorial ratio: Stirling bounds $(d+3)^d/(d!)^2$ by C^{d+1}/d^{d+1} , and numerical powers are absorbed in the displayed activation-depth envelope. At polynomial accuracy this is $O(\log(en)^{3d+2})$ for fixed structural parameters, preserving the previous qualitative storage order.

For an arbitrary target outside that regime, keep the unshortened logarithm from the forward certificate: replace $\log(en/\varepsilon)$ in the sufficient budget by

$$\log(en) + \log \left(e + \frac{C\beta^{CL}Y(m/\gamma)(1 + \sqrt{m/\gamma})}{\varepsilon} \right). \quad (48)$$

Enlarging numerical constants absorbs $\sqrt{\log(en)}$. If the sufficient budget exceeds n , the exact full-width branch suffices instead. The exact inverse in the proof also handles targets already met by the initialization-only model. These are internal choices of one width budget, not extra runtime orders.

C.3.4 Seeded decoder: choose reference width for the target

There is no order to optimize after choosing n . For a requested ε , choose the smallest integer width satisfying

$$3B_n \leq \varepsilon \quad (49)$$

together with (34), and then run the prescribed finite-source construction. At fixed admissible problem parameters and confidence,

$$n(\varepsilon) = \varepsilon^{-2+o(1)} \quad (50)$$

suffices. Substitution into (37) gives

$$\text{storage}_{\text{Seed}} = O([\log(1/\varepsilon)]^6 \log \log(e^e + 1/\varepsilon)) \quad (51)$$

numerical words and

$$O([\log(1/\varepsilon)]^7 \log \log(e^e + 1/\varepsilon)) \quad (52)$$

bits. The inherited positive coefficients c_0, c_1 in B_n are not fully parameter-quantified, so this fixed-problem inverse rate is eventual; the separately displayed finite-source width itself is explicit.

C.3.5 Costs at the prescribed inverse widths/orders

In the next table, n and q mean the corresponding prescribed inverse choices above (or their exact the detailed-statement appendix versions), not new optimized quantities. Write $P = (L-1)n^2 + n(d+1)$ only as a cost-count abbreviation. The two local-continuation warmup envelopes $\mathcal{T}_{\text{HD}}, \mathcal{M}_{\text{HD}}$ and $\mathcal{T}_{\text{HI}}, \mathcal{M}_{\text{HI}}$ are defined in the detailed statements. Evaluate them at the inverse construction's source horizon and tolerance, using its prescribed mode set and the certified internal orders. The first pair uses explicit reference matrices; the second uses a fresh implicit Gaussian reference. A choice of q alone is not substituted for the distinct setup orders. The origin-jet envelopes remain available for that older alternative.

The table counts classical arithmetic, with scalar activation work and Gaussian sampling charged as specified in the cost contract. Training is one full-batch vector-field stage, or one fixed-stage explicit step up to its fixed stage multiplier. Warmup/training memory is total peak resident memory; query memory is additional peak workspace for an already loaded, inference-ready model.

Detailed comparison

Model and operation: Dense warmup

Arithmetic work after the indicated inverse substitution

Peak memory

$O(P)$, plus Gaussian draws

$O(P + m(d+1))$

Model and operation: Dense training stage

Arithmetic work after the indicated inverse substitution $O(mP)$
Peak memory $O(P + Lmn + m(d + 1))$

Model and operation: Dense query

Arithmetic work after the indicated inverse substitution $O(P)$
Peak memory $O(n)$ additional

Model and operation: Legendre warmup

Arithmetic work after the indicated inverse substitution $O(mP + Lmnq)$, plus Gaussian draws
Peak memory $O(P + Lmnq + m(d + 1))$

Model and operation: Legendre training stage

Arithmetic work after the indicated inverse substitution $O(mP + Lnm^2q + Lnmq)$
Peak memory $O(P + Lnmq + m(d + 1))$

Model and operation: Legendre query

Arithmetic work after the indicated inverse substitution $O(P + Lnmq)$
Peak memory $O(n)$ additional

Model and operation: Harmonic warmup, explicit dense local continuation

Arithmetic work after the indicated inverse substitution $O(\mathcal{T}_{\text{HD}})$, plus the specified primitive calls
Peak memory $O(\mathcal{M}_{\text{HD}})$

Model and operation: Harmonic warmup, implicit Gaussian local continuation

Arithmetic work after the indicated inverse substitution $O(\mathcal{T}_{\text{HI}})$, plus the specified primitive calls
Peak memory $O(\mathcal{M}_{\text{HI}})$

Model and operation: Harmonic training stage

Arithmetic work after the indicated inverse substitution $O(Lmq^2 + m^3)$
Peak memory $O(Lq^2)$

Model and operation: Harmonic query

Arithmetic work after the indicated inverse substitution $O(Lq^2)$

Detailed comparison

Peak memory $O(q)$ additional

The full-width exact fallback uses the dense row. Legendre rows use streamed correction factors and cumulative moment prefixes. Harmonic rows use fixed metric inverse caches, the necessary moving feature-Gram solve and a refreshed effective-readout cache. Cache refresh is charged to training/model preparation, not hidden in the single-query bound. These are sufficient implementation costs, not time-optimal algorithms or numerical-step accuracy guarantees.

The Seeded decoder uses finite-word rather than unit-cost real arithmetic, so its costs are stated separately. With p, Z as in (32), put

$$M_{\text{Seed}} = p^2 \beta^{402L} (m + d + 2)^2 (1 + m/\gamma)^2 Z^6 [d + 1 + \log(e + (d + 1)Z)].$$

Up to universal multiplicative constants:

Detailed comparison

Seeded decoder phase: Initialization

Word operations $n(d + 1)p^5 \beta^{1115L} (m + d + 2)^5 (1 + m/\gamma)^5 Z^{29/2}$

Peak numerical words $M_{\text{Seed}} + np \beta^{201L} (m + d + 2)(1 + m/\gamma) Z^{5/2} + p^3 \beta^{603L} (m + d + 2)^3 (1 + m/\gamma)^3 Z^{15/2}$

Seeded decoder phase: Complete compact training

Word operations $(d + 1)p^4 \beta^{914L} (m + d + 2)^4 (1 + m/\gamma)^4 Z^{12} \log(e + p \beta^{201L} (m + d + 2)(1 + m/\gamma) Z^{5/2})$

Peak numerical words M_{Seed}

Seeded decoder phase: One unseen-input query

Word operations $(d + 1)p^4 \beta^{914L} (m + d + 2)^4 (1 + m/\gamma)^4 [(L + 1)nZ^{12} + Z^{14}]$

Peak numerical words M_{Seed}

The training row counts all prescribed updates, not one vector-field stage. Each numerical word has $C\beta^{110L}Z$ sufficient bits. General analytic activation evaluation, external descriptions and their access work remain additional; the finite tanh evaluator is included. Initialization may inspect completed virtual training and has width-linear temporary storage. Querying regenerates virtual source rows, does not replay scalar training, and retains the displayed factor n .

C.4 Common accuracy-to-storage and computational-cost corollary

Fix data, activations, m, d, L, Y, γ , and confidence. Choose the dense reference before initialization to meet its own target: $n(\varepsilon) = \varepsilon^{-2+o(1)}$ suffices. Apply the two inverse orders at this same reference width for storage. The setup-cost specialization below uses a sufficient stronger-accuracy Harmonic budget with this same storage order, as specified there:

Detailed comparison

Model: Independent dense

Learned storage at error ε $\varepsilon^{-4+o(1)}$

All retained model storage $\varepsilon^{-4+o(1)}$

Model: Legendre

Learned storage at error ε $\varepsilon^{-5/2+o(1)}$

All retained model storage $\varepsilon^{-4+o(1)}$, including fixed mixers

Detailed comparison

Model: Harmonic

Learned storage at error ε $O(\log(1/\varepsilon)^{3d+2})$

All retained model storage $O(\log(1/\varepsilon)^{3d+2})$

Model: Seeded decoder

Learned storage at error ε bounded by the complete-model envelope at right

All retained model storage $O([\log(1/\varepsilon)]^6 \log \log(e^e + 1/\varepsilon))$ numerical words

These powers do not suppress growing structural parameters in the explicit bounds above. They are sufficient counts, not minimax storage lower bounds or a growing-data theorem. For comparison against an independent dense run, split accuracy and failure budgets between dense variability and compression; the triangle inequality gives the same exponents without event independence.

At the same reference width, use the original source-tolerance specialization $\eta = 1/n$, $T = 32(m/\gamma) \log(en)$ for the two efficient Harmonic warmup rows below. Its sufficient budget still has $q = O(\log(1/\varepsilon)^{3d/2+1})$ and gives error $\varepsilon^{2+o(1)} = o(\varepsilon)$. This is a sufficient possibly larger budget of the same headline storage order, not an assertion that the least integer inverse budget must support that stronger tolerance. The cost comparison is:

Detailed comparison

Model and operation: Dense warmup

Work $\varepsilon^{-4+o(1)}$

Peak memory $\varepsilon^{-4+o(1)}$

Model and operation: Dense training stage

Work $\varepsilon^{-4+o(1)}$

Peak memory $\varepsilon^{-4+o(1)}$

Model and operation: Dense query

Work $\varepsilon^{-4+o(1)}$

Peak memory $\varepsilon^{-2+o(1)}$ additional

Model and operation: Legendre warmup

Work $\varepsilon^{-4+o(1)}$

Peak memory $\varepsilon^{-4+o(1)}$

Model and operation: Legendre training stage

Work $\varepsilon^{-4+o(1)}$

Peak memory $\varepsilon^{-4+o(1)}$

Model and operation: Legendre query

Work $\varepsilon^{-4+o(1)}$

Peak memory $\varepsilon^{-2+o(1)}$ additional

Model and operation: Harmonic warmup, explicit dense local continuation

Work $\varepsilon^{-4+o(1)}$

Peak memory $\varepsilon^{-4+o(1)}$

Model and operation: Harmonic warmup, implicit Gaussian local continuation

Work $\varepsilon^{-2+o(1)}$

Detailed comparison

Peak memory $\varepsilon^{-2+o(1)}$

Model and operation: Harmonic training stage

Work $O(\log(1/\varepsilon)^{3d+2})$

Peak memory $O(\log(1/\varepsilon)^{3d+2})$

Model and operation: Harmonic query

Work $O(\log(1/\varepsilon)^{3d+2})$

Peak memory $O(\log(1/\varepsilon)^{3d/2+1})$ additional

Model and operation: Seeded decoder initialization

Work $\varepsilon^{-2+o(1)}$ word operations

Peak memory $\varepsilon^{-2+o(1)}$ numerical words

Model and operation: Seeded decoder complete training

Work $O([\log(1/\varepsilon)]^{12} \log \log(e^e + 1/\varepsilon))$ word operations

Peak memory retained model

Model and operation: Seeded decoder query

Work $\varepsilon^{-2+o(1)}$ word operations

Peak memory retained model

The displayed Harmonic rates follow by substituting $n = \varepsilon^{-2+o(1)}$ into the two proved local-continuation bounds, with the sufficient source orders just stated. Arbitrary inverse budgets retain the full finite cost formulas in the preceding table. The explicit implementation holds quadratic dense arrays during warmup; the implicit implementation does not generate them. Neither warmup is polylogarithmic merely because the retained model is. All asymptotic rows fix the dataset, depth, dimension, activations and confidence. Dense, Legendre and Harmonic use unit-cost scalar evaluation/sampling; their more general arithmetic-plus-primitive-call qualification is in the detailed-statement appendix. Those rows do not bound numerical precision or the number of subsequent compressed training steps. The local initializer’s own integration orders and numerical defect are controlled separately. Seeded rows instead use the finite-word contract, include their stated word length, and count the complete prescribed training schedule. Legendre’s smaller learned state does not eliminate its fixed dense-mixer work; Harmonic’s small runtime state does not bound the cost of constructing it. The Seeded decoder has a dimension-independent logarithmic word-storage exponent. Its exponent six is smaller than Harmonic’s coordinate exponent $3d + 2$ when $d \geq 2$, but not when $d = 1$; these also count different storage units. Unlike Harmonic, its certified query work is not polylogarithmic. Its error meets the actual dense-pair quantile scale; it is not proved negligible relative to a realized dense-pair discrepancy.

For $m \geq 2$ and nonzero labels, canonical independent-dense accuracy with fixed failure probability below one half requires, eventually,

$$n \log(en)^5 \gtrsim_{\phi, L, \delta} \frac{Y^2 \gamma}{\varepsilon^2}.$$

Thus, among widths meeting the lower theorem’s onset, its parameter count is $\Omega(\varepsilon^{-4}/\log(1/\varepsilon)^{10})$ at fixed problem. This argument does not exclude exceptional widths below that onset and is not a lower bound against arbitrary compressed representations.

C.5 Width scaling and actual dense variability

At root-width target $\varepsilon = Y/\sqrt{n}$, Legendre has order $n^{1/4+o(1)}$, moving state $n^{5/4+o(1)}$, and quadratic fixed mixers. Harmonic has width $O(\log(en)^{3d/2+1})$ and all-retained storage $O(\log(en)^{3d+2})$. At fixed positive error, Legendre moving storage is instead $n^{1+o(1)}$.

Harmonic can target n^{-1} , or use its original $n^{-1+o(1)}$ specialization, with the same polylogarithmic storage order. For Legendre, enlarge the root-target order by $\log(en)^{3/2}$; its error is then at most $CY/[\sqrt{n}\log(en)^3]$ eventually, without changing the moving-state exponent. For $m \geq 2$, the general dense lower bound gives for these choices

$$\frac{\|f_{\text{model}} - f_n\|_*}{\|f_n - \widetilde{f}_n\|_*} \xrightarrow{\mathbb{P}} 0.$$

This compares complete trajectory norms, not endpoint errors or pointwise-in-time ratios. Confidence dependence of the lower coefficient is retained; the arbitrary-failure-budget argument is given in Appendix F.7.

D Detailed statements and computational costs

The global quantities are exactly those defined in the certificate-interface appendix. The error and storage formulas in this part, rather than any choice of a headline constant, are the numerical certificates. Finite recurrence coefficients are defined locally where they are used; none is an independently adjustable parameter. The separate cost subsection exposes internal setup resolutions and qualifies its arithmetic big-O bounds explicitly. Dense, Legendre and Harmonic costs are not finite-bit certificates; Seeded costs use the separately defined finite-word model. None is a measured floating-point operation or wall-clock benchmark. The same physical-time model and the same whole-sphere, all-time norm apply throughout. Where a proof writes $v = x/\sqrt{d}$, its unit-sphere supremum is exactly the original input-sphere supremum.

D.1 One common label allowance

The complete original allowance is

$$0 < Y \leq \frac{\gamma}{m} \min \left\{ \frac{1}{8H_D\sqrt{F_D}}, \quad \frac{S_*^{\text{Leg}}}{8}, \quad \frac{1}{16H_c\sqrt{F_c}}, \quad \frac{S_*^{\text{src}}}{16} \right\}. \quad (53)$$

The four entries are fully defined in this document:

Entry	Exact definitions and proof
Dense fitting, H_D, F_D	Dense fitting coefficients
Every-order Legendre fitting, S_*^{Leg}	Legendre coefficients
Corrected Harmonic fitting, H_c, F_c	Harmonic fitting coefficients
Analytic source control, S_*^{src}	Source recurrences

The common sufficient specialization is exactly $0 < Y \leq (\gamma/m)\beta^{-30L}$. Its implication of all four entries is proved in the numerical coefficient ledgers, not assumed. The simpler dense/Legendre activation-power certificates use this smaller cap. Their full recurrence certificates retain the entire displayed interval. Harmonic's explicit full-range certificate and supplied-budget theorem use the entire interval. No target-dependent label assumption is introduced. If $Y = 0$, all four models use the identically zero predictor; the positive-label formulas are replaced by that exact stationary case.

D.2 One width and confidence convention

Each detailed statement is written at its individual confidence input $0 < \delta < 1$. For a joint assertion of dense upper, dense lower, Legendre and Harmonic, use confidence input $\delta/4$ in each statement, and take a width satisfying the union of their stated gates. The union bound gives joint success at least $1 - \delta$, without independence between compressed constructions. The lower assertion is included only for $m \geq 2$ and $Y > 0$.

A conservative common explicit initialization requirement is $n \geq N_{\text{fit}}(\delta/32)$, whose numerical formula appears below. Also take $n \geq d$ for the simplified storage counts.

To include the Seeded decoder in the same assertion, use input $\delta/5$ for each of the five assertions and take the union of their width requirements, including its explicit gate. The decoder’s additional spanning assumption applies only to that assertion. This is a joint-event allocation, not a new hypothesis for any standalone model theorem.

The remaining common gates have exactly these origins:

1. The source’s high-probability finite-deletion/moment event, proved in source foundations. Its sufficient width is existential and depends on the fixed problem and confidence.
2. The explicit source complex-radius and counting inequalities and the analytic-tail gates, including the dimension-one temporal count. These are finite formulas independent of the supplied model budget and requested tolerance.
3. When the lower bound is included, the eventual initialized-CLT and derivative-to-prediction thresholds in the lower theorem.

These finitely many eventual conditions have a common finite threshold for every fixed admissible problem and confidence, including the fixed positive label size. This assertion does not give a numerical value for its stochastic portion, or prove that the threshold is polynomial in sample count, gap or depth. The exact deterministic gates are listed and derived below; none is silently put inside an error constant. For a common canonical dense initialization at a width satisfying these gates, the same event covers every permitted Legendre order and Harmonic budget. The arbitrary-accuracy Harmonic construction introduces no extra stochastic event depending on its internal horizon or source tolerance. The implicit initializer transfers this statement to each prescribed coupled execution; independently restarted initializers do not share one success event.

When ε, δ are inputs to an inverse statement, its numerical order or storage prescription is effective **conditional on the reference width satisfying this convention**. The dense inverse must likewise meet the implicit source threshold. The eventual $\varepsilon \downarrow 0$ powers in the certificate-interface appendix do not quantify it.

D.3 Construction domains and what storage means

Legendre order is every integer $q \geq 1$. The Harmonic construction below covers every integer $q \geq 18(2m+d+9)$; its baseline is exact-initialization overhead, not an error floor. Full-coordinate selection at $q \geq n$ agrees exactly with dense. In eventual statements one may include $n \geq 18(2m+d+9)$ in the common width threshold so these domains overlap directly. At smaller widths the full-coordinate branch still exists but is not a compression claim.

Dense, Legendre and Harmonic storage counts are numbers of real coordinates. Seeded storage counts finite numerical words of the specified bit length. For dense and Legendre, state counts are distinguished from data and evaluation working memory. For Harmonic, the stated all-retained bound also includes fixed metrics/copies, data and prescribed solve caches. Setup jets and original-width arrays are discarded after construction. The separate cost analysis supplies arithmetic and peak-memory implementation bounds under each declared arithmetic model, not storage optimality over all possible representations. “Least certified order” means the least integer satisfying the particular displayed sufficient error bound; it is not an observed or minimax optimum.

D.4 Dense model, scope, and numerical certificates

This section uses width n , sample count m , input dimension d , and hidden depth $L \geq 2$. Set $v_a = x_a/\sqrt{d}$, so that $\|v_a\|_2 = 1$, and keep the labels y_a unchanged. The network is

$$z^{(1)}(v) = Av, \quad z^{(j)}(v) = W^{(j)}h^{(j-1)}(v), \quad h^{(j)}(v) = \phi_j(z^{(j)}(v)), \quad f_n(v) = \frac{w^T h^{(L)}(v)}{n}.$$

Here A is $n \times d$, the $L - 1$ hidden mixers are $n \times n$, and $w \in \mathbb{R}^n$. Initial entries of A are independent $N(0, 1)$; mixer entries are independent $N(0, 1/n)$; all blocks are independent; and $w(0) = 0$. The loss is $m^{-1} \sum_a r_a^2$, where $r_a = f_n(v_a) - y_a$, with mobilities $(n, 1, \dots, 1, n)$. Define

$$k_a^{(L)} = w, \quad \delta_a^{(j)} = \phi_j'(z_a^{(j)}) \odot k_a^{(j)}, \quad k_a^{(j)} = (W^{(j+1)})^T \delta_a^{(j+1)}.$$

The resulting physical-time flow is exactly

$$\dot{A} = -\frac{2}{m} \sum_a r_a \delta_a^{(1)} v_a^T, \quad \dot{W}^{(j)} = -\frac{2}{mn} \sum_a r_a \delta_a^{(j)} (h_a^{(j-1)})^T, \quad \dot{w} = -\frac{2}{m} \sum_a r_a h_a^{(L)}.$$

Every hidden array trains. The two independent dense runs use these same labels and the same physical time.

Activations are real on the real line and holomorphic on the common strip $|\operatorname{Im} z| < a$, with bounded first derivative on that strip. Values need not be bounded. Write

$$b = \max_j |\phi_j(0)|, \quad s = \max \left\{ 1, \max_j \sup_{|\operatorname{Im} z| \leq a/2} |\phi_j'(z)| \right\}, \quad t_2 = \max_j \sup_{|\operatorname{Im} z| \leq a/2} |\phi_j''(z)|,$$

$$\beta = \max\{10, 1 + b, 16/a, s, t_2\}, \quad Y = \|y\|_2 / \sqrt{m}.$$

Cauchy's integral formula applied to the full-strip first derivative bounds the higher derivatives on every smaller strip. Thus these quantities are finite. In particular $|\phi_j(z)| \leq b + s|z|$ on the half-strip, by integrating the first derivative along the segment from zero to z .

The initial population covariance and unweighted gap are

$$Q_{ab}^{(0)} = v_a^T v_b, \quad Q_{ab}^{(j)} = \mathbb{E}[\phi_j(Z_a) \phi_j(Z_b)], \quad Z \sim N(0, Q^{(j-1)}), \quad \gamma = \lambda_{\min}(Q^{(L)}) > 0.$$

There is no orthogonality, centering, sign, input-rank, or intermediate covariance nonsingularity assumption. The proof abbreviation $\lambda = \gamma/m$ is the normalized gap; it is not a redefinition of the global γ . For a standard real Gaussian G , define

$$\mu_{2,0} = 1, \quad \mu_{2,j} = \mathbb{E} \phi_j(\sqrt{\mu_{2,j-1}} G)^2, \quad H_D = \max(1, \sqrt{\mu_{2,1}}, \dots, \sqrt{\mu_{2,L}}).$$

These are the scalar moments called q_j in the component notes; the notation here keeps q available for a compressed model's order. Equal input norms imply $Q_{aa}^{(j)} = \mu_{2,j}$. Consequently $\lambda \leq H_D^2/m \leq H_D^2$.

The dense comparisons concern the single norm

$$\|f - g\|_* = \sup_{t \in [0, \infty]} \sup_{\|x\|_2 = \sqrt{d}} |f(t, x) - g(t, x)|.$$

The extra point $t = \infty$ denotes the fitted limit constructed below. An upper bound in this norm also bounds endpoints; the lower bound below has a positive-time witness and makes no endpoint claim.

The exact number of dense parameter coordinates per run is

$$S_D(n) = (L - 1)n^2 + n(d + 1).$$

All these coordinates are learned and retained. This counts real coordinates, not bits or preprocessing work. If data are retained, their ordinary $m(d + 1)$ coordinates are additional. Two independent runs require twice the per-run model storage.

D.5 Quantitative initialization and all-time dense fitting

Define the finite activation/depth coefficients

$$D_j^D = s(9s)^{L-j}, \quad U_1^D = D_1^D, \quad U_j^D = 2H_D D_j^D \quad (j \geq 2),$$

$$F_1^D = sU_1^D, \quad F_j^D = s(2H_D U_j^D + 9F_{j-1}^D), \quad F_D = F_L^D.$$

Unrolling the last recurrence gives the exact expression

$$F_D = s^2 \left[(9s)^{2L-2} + 4H_D^2 \sum_{j=0}^{L-2} (9s)^{2j} \right].$$

All coefficients are positive, and $F_D \geq \max(1, U_1^D, \dots, U_L^D, F_1^D, \dots, F_L^D)$.

For a confidence argument $0 < \alpha < 1$, set

$$A_D = s^2 + t_2(b + \sqrt{2}sH_D), \quad T_D = \sum_{j=0}^{L-1} A_D^j, \quad e_D = \min\{1/4, \gamma/(2m)\}, \quad M_D = 8b^4 + 96s^4 H_D^4,$$

$$h_D = \min\{1/2, H_D/[4(8s)^L]\}, \quad P_D = (1 + 2/h_D)^d,$$

$$N_{\text{fit}}(\alpha) = \left\lceil \max \left\{ 1, \frac{\log(8L/\alpha)}{8 - 2\log 9}, \frac{d \log 9 + \log(8/\alpha)}{8 - \log 9}, \frac{2L(m^2 + P_D)M_D T_D^2}{\alpha e_D^2} \right\} \right\rceil.$$

The denominators are positive. At every $n \geq N_{\text{fit}}(\alpha)$, with probability at least $1 - \alpha$,

$$\|A_0\|_{\text{op}}/\sqrt{n} \leq 8, \quad \max_{j \geq 2} \|W_0^{(j)}\|_{\text{op}} \leq 8, \quad \sup_{\|v\|_2=1, j} \|h_0^{(j)}(v)\|_2/\sqrt{n} \leq 3H_D/2,$$

$$\lambda_{\min} \left(\frac{H_0^T H_0}{mn} \right) \geq \lambda/2, \quad H_0 = [h_0^{(L)}(v_1), \dots, h_0^{(L)}(v_m)].$$

On this initialization event, the sole real-fitting label restriction is

$$0 < Y \leq \frac{\lambda}{8H_D \sqrt{F_D}}.$$

Under it the flow exists for all time, every parameter converges, and every training label is fitted. Simultaneously for all $t \geq 0$,

$$\|A(t)\|_{\text{op}}/\sqrt{n} < 9, \quad \max_{j \geq 2} \|W^{(j)}(t)\|_{\text{op}} < 9, \quad \sup_{\|v\|_2=1, j} \|h^{(j)}(t, v)\|_2/\sqrt{n} < 2H_D,$$

$$\lambda_{\min} \left(\frac{H(t)^T H(t)}{mn} \right) \geq \lambda/4, \quad \rho(t) := \|r(t)\|_2/\sqrt{m} \leq Y e^{-\lambda t/2}, \quad \int_0^\infty \rho(t) dt \leq 2Y/\lambda.$$

In the parameter norm

$$\|\theta\|_{\text{par}}^2 = \|A\|_F^2/n + \sum_{j=2}^L \|W^{(j)}\|_F^2 + \|w\|_2^2/n,$$

the path length and displacements satisfy

$$\int_0^\infty \|\dot{\theta}\|_{\text{par}} dt \leq 2Y/\sqrt{\lambda}, \quad \|w(t)\|_2/\sqrt{n} \leq 2Y/\sqrt{\lambda},$$

$$\frac{\|A(t) - A_0\|_F}{\sqrt{n}} \leq \frac{8U_1^D Y^2}{\lambda^{3/2}}, \quad \|W^{(j)}(t) - W_0^{(j)}\|_F \leq \frac{8U_j^D Y^2}{\lambda^{3/2}},$$

$$\sup_{\|v\|_2=1} \frac{\|h^{(j)}(t, v) - h_0^{(j)}(v)\|_2}{\sqrt{n}} \leq \frac{8F_j^D Y^2}{\lambda^{3/2}}.$$

The uniform endpoint tail is

$$\sup_{\|v\|_2=1} |f_n(\infty, v) - f_n(t, v)| \leq 16 \left(H_D^2 + \frac{F_D Y^2}{\lambda} \right) \frac{Y}{\lambda} e^{-\lambda t/2} \leq \frac{65}{4} H_D^2 \frac{Y}{\lambda} e^{-\lambda t/2}.$$

For $Y = 0$, all velocities vanish at initialization and the predictor remains identically zero. No division by Y is used in that case.

D.6 Explicit independent-dense upper certificates

The source foundation supplies activation/depth constants $S_*^{\text{src}}, K_{\text{src}} > 0$ and an eventual probability event for the actual trained carriers. The full dense allowance is

$$0 < Y \leq \lambda \min \left\{ \frac{1}{8H_D \sqrt{F_D}}, \frac{S_*^{\text{src}}}{16} \right\}.$$

When dense is compared jointly with compressed models, intersect this allowance with their stated fitting allowances. This intersection is the full common recurrence scope; none of the arguments here shrinks it to the beta-only cap below.

Put $S = 16Y/\lambda$, $\kappa = \lambda/2$, $R = 2Y/\sqrt{\lambda}$, and $u_n = \sqrt{\log(en)}$. The source event, including its deterministic post-horizon tail, gives

$$\max_{a,j,i,t \geq 0} |k_{a,i}^{(j)}(t)| \leq M_n, \quad M_n = 2K_{\text{src}} S u_n.$$

The factor two supplies the all-time tail margin. This bound concerns training carriers; no carrier maximum on an interpolating parameter state or on every passive query is assumed.

Define the following local comparison coefficients:

$$T = 9s, \quad F_z = 2H_D T^{L-1}, \quad B_\delta = sT^{L-1}R, \quad P = 1 + 2H_D(L-1), \quad G_* = PB_\delta + 2H_D,$$

$$D_\delta(M) = sT^{L-1}(1 + B_\delta) + Lt_2 F_z T^{L-1}M,$$

$$J(M) = PD_\delta(M) + [1 + (L-1)B_\delta]sF_z,$$

$$T_{\text{rem}}(M) = sF_z \sqrt{L} + LB_\delta sF_z + \frac{1}{2}L^2 t_2 F_z^2 M,$$

$$E_{\text{en}}(M) = 2[2T_{\text{rem}}(M) + \sqrt{L+1}J(M)]Y/\kappa,$$

$$C_Y(M) = \frac{2H_D}{\kappa} \left(\frac{16H_D G_* J(M)}{\kappa} + 2sF_z \right) + \frac{2sF_z}{\sqrt{\lambda}}, \quad \mathcal{L}_n = \frac{\sqrt{L+1}Y C_Y(M_n)}{\sqrt{n}} e^{E_{\text{en}}(M_n)},$$

$$K_t = \frac{8(H_D^2 + F_D Y^2/\lambda)Y}{\kappa}, \quad K_x = RT^L, \quad N_n = (n+1)(1+2n)^d.$$

For each fixed $0 < \delta < 1$, the full-range certificate is

$$\|f_n - \tilde{f}_n\|_* \leq E_D(n, \delta) := 2\mathcal{L}_n \sqrt{\log(4N_n/\delta)} + \frac{2(K_t + K_x)}{n}.$$

It holds with probability at least $1 - \delta$ once $n \geq N_{\text{fit}}(\delta/8)$ and the source failure, including its all-time tail gate, is at most $\delta/8$ for each path. The latter sufficient width is not quantified. All coefficients in the displayed deterministic bound are finite recurrences; the unknown width must not be interpreted as an explicit confidence-certified initialization rule.

For completeness, the original full-range unsigned comparison is also retained. It gives the same certificate with $\mathcal{L}_n$ replaced by

$$\mathcal{L}_n^{\text{old}} = \frac{(2H_D + RsF_z)\sqrt{L}}{\sqrt{n}} \exp \{2J(M_n)(1 + 4G_*^2/\kappa)Y/\kappa\}.$$

The signed-energy proof below also applies on the full recurrence range and retains an explicit factor Y in the newer prefactor.

On the unchanged simpler sufficient cap

$$0 < Y \leq \frac{\gamma}{m} \beta^{-30L},$$

let $B_D = \beta^{100L}$. A shorter fully numerical upper bound is

$$\begin{aligned} \|f_n - \tilde{f}_n\|_* &\leq B_D Y \left(1 + \frac{m}{\gamma}\right)^2 \left[1 + B_D \frac{Ym}{\gamma} \sqrt{\log(en)}\right] \\ &\times \left[1 + B_D \left(\frac{Ym}{\gamma}\right)^2 \left(e^{\sqrt{\log(en)}} - 1\right)\right] \sqrt{\frac{\log[8(n+1)(1+2n)^d/\delta]}{n}}. \end{aligned}$$

This keeps the actual label amplitude in every bracket. It is an $n^{-1/2+o(1)}$ upper bound at fixed problem parameters, not a constant-times- $n^{-1/2}$ theorem. The source probability qualification is exactly the same as for the full-range certificate.

D.7 Accuracy inversion and dense storage

For the full recurrence certificate, the exact scalar prescription is the smallest integer n meeting its fitting/source width requirements and $E_D(n, \delta) \leq \varepsilon$. Its storage is exactly $(L-1)n^2 + n(d+1)$. The source threshold remains a specified unquantified requirement, so this prescription is not a fully effective confidence-certified width algorithm. The admissible set is nonempty: at fixed problem parameters M_n is proportional to $\sqrt{\log(en)}$, the coefficients J, C_Y are affine in M_n , and E_{en} is affine in M_n . Thus $E_D(n, \delta) = n^{-1/2+o(1)} \rightarrow 0$.

Here is an explicit inversion of the simpler cap's numerical expression, separate from that stochastic qualification. Put

$$D_{d,\delta} = d + 1 + \log(16 \cdot 3^d / \delta), \quad A_{D,\delta} = B_D Y (1 + m/\gamma)^2 (1 + B_D Y m/\gamma) (1 + B_D (Y m/\gamma)^2) \sqrt{D_{d,\delta}}.$$

For every $n \geq 1$, the beta certificate is at most

$$A_{D,\delta} \frac{\log(en) e^{\sqrt{\log(en)}}}{\sqrt{n}}.$$

For $Y > 0$, define

$$a_\varepsilon = \max\{1, A_{D,\delta}/\varepsilon\}, \quad v_\varepsilon = \log(e + a_\varepsilon), \quad \bar{n}_\varepsilon = \left\lceil a_\varepsilon^2 v_\varepsilon^2 e^{8\sqrt{v_\varepsilon} + 16} \right\rceil.$$

Every $n \geq \bar{n}_\varepsilon$ makes the preceding envelope at most ε . Thus one may take the maximum of $\bar{n}_\varepsilon$, $N_{\text{fit}}(\delta/8)$, and the same eventual source threshold. At fixed nonzero labels and fixed problem parameters this gives $n = \varepsilon^{-2+o(1)}$ and dense storage $\varepsilon^{-4+o(1)}$ as $\varepsilon \downarrow 0$. These are sufficient parameter counts for choosing the reference before initialization; they do not replace an already prescribed dense reference.

D.8 Initialized Gram fluctuations, including singular covariances

This initialization statement needs only real C^2 activations with bounded first two derivatives and linear growth. Fix any finite query list, allowing duplicates, and index covariance coordinates by pairs $(a, b), (c, e)$. Let

$$K_{n,ab}^{(j)} = n^{-1}(h_0^{(j)}(v_a))^T h_0^{(j)}(v_b), \quad Q^{(j)} = \Psi_j(Q^{(j-1)}).$$

Here $\Psi_j(C)_{ab} = \mathbb{E}[\phi_j(Z_a)\phi_j(Z_b)]$ for $Z \sim N(0, C)$, and $Q^{(0)}$ is the input Gram on this query list. Identify symmetric matrices with their upper triangular coordinates. For $Z \sim N(0, Q^{(j-1)})$, define the innovation covariance

$$(B_j)_{ab,ce} = \mathbb{E}[\phi_j(Z_a)\phi_j(Z_b)\phi_j(Z_c)\phi_j(Z_e)] - Q_{ab}^{(j)}Q_{ce}^{(j)},$$

and define the linear map on symmetric perturbations by

$$(T_j E)_{ab} = \frac{1}{2}E_{aa}\mathbb{E}[\phi_j''(Z_a)\phi_j(Z_b)] + E_{ab}\mathbb{E}[\phi_j'(Z_a)\phi_j'(Z_b)] + \frac{1}{2}E_{bb}\mathbb{E}[\phi_j(Z_a)\phi_j''(Z_b)].$$

All these Gaussian integrals are finite, including at singular covariances. The joint limit at fixed depth and fixed query list is

$$\sqrt{n}(K_n^{(j)} - Q^{(j)}) \implies \Xi_j, \quad \Xi_0 = 0, \quad \Xi_j = T_j \Xi_{j-1} + G_j,$$

where the G_j are independent centered Gaussian symmetric matrices with covariances B_j . In particular, with C_j the covariance of Ξ_j ,

$$C_0 = 0, \quad C_j = T_j C_{j-1} T_j^T + B_j.$$

The transpose here is that of the finite-dimensional coordinate map on symmetric matrices. This is a theorem about initialized features, not an independence assertion for trained neurons.

D.9 Positive feature rank forces a last-layer innovation

Assume now $m \geq 2$ and $Y > 0$. Let

$$Z \sim N(0, Q^{(L-1)}), \quad H_a = \phi_L(Z_a), \quad Q = \mathbb{E}HH^T = Q^{(L)}, \quad S_y = y^T H,$$

$$\mu_4 = \mathbb{E}\phi_L(\sqrt{\mu_{2,L-1}}G)^4.$$

Both $\mu_{2,L}$ and μ_4 are positive and finite. Linear growth gives explicitly $\mu_4 \leq 8|\phi_L(0)|^4 + 24\|\phi_L'\|_\infty^4 \mu_{2,L-1}^2$. If $\mu_{2,L-1} = 0$, every coordinate of H would be the same deterministic constant, contradicting $Q \succ 0$ for $m \geq 2$. Thus the final preactivation marginal variance is positive as well.

The distribution-free moment inequality needed here is

$$\text{tr Cov}(HS_y) \geq \frac{(m-1)\gamma^3(m\mu_{2,L} - \gamma)}{4m^2\mu_4} \|y\|_2^2 \geq \frac{(m-1)^2\gamma^3\mu_{2,L}}{4m^2\mu_4} \|y\|_2^2 \geq \frac{\gamma^3\mu_{2,L}}{16\mu_4} \|y\|_2^2.$$

No independence between the m coordinates is used.

D.10 Finite-query complex time and its exact recurrence scope

This subsection invokes the local insertion interface in the source foundation, with the same claim level and eventual-width qualification as that interface. It records the finite-query modification required by the lower theorem; a whole-sphere complex-time radius cannot simply be replaced by a larger radius without this argument.

Write $H_j^{\text{src}}, f_j^{\text{src}}, q_j^{\text{src}}, T_Q^{\text{src}}$ for exactly the source coefficients called H_j, f_j, q_j, T_Q in that foundation. The superscript distinguishes its RMS and response coefficients from H_D , the Gaussian marginal moments, and the global compression order q . For activity $S = 16Y/\lambda$, define

$$U_1^{\text{fin}}(S) = 4sK_{\text{src}},$$

$$U_j^{\text{fin}}(S) = 2 \left\{ sK_{\text{src}} \left[(H_{j-1}^{\text{src}})^2 + (f_{j-1}^{\text{src}})^2 + ST_Q^{\text{src}} + S^2 H_{j-1}^{\text{src}} q_{j-1}^{\text{src}} \right] + 64q_{j-1}^{\text{src}} + 1 \right\} \quad (j \geq 2),$$

$$U_{\text{fin}}(S) = \max_j U_j^{\text{fin}}(S), \quad \chi(S) = \min \left\{ 1, \frac{a}{4S^2 U_{\text{fin}}(S)} \right\},$$

$$\chi_{\text{act}} = \min \left\{ 1, \frac{a}{4(S_*^{\text{src}})^2 U_{\text{fin}}(S_*^{\text{src}})} \right\} > 0.$$

Every coefficient in these polynomials is nonnegative, so $\chi(S) \geq \chi_{\text{act}}$ throughout the original source allowance $S \leq S_*^{\text{src}}$. The latter coefficient depends only on the fixed activations and depth.

Let $T_n = 32\lambda^{-1} \log(en)$, and choose

$$0 < c \leq \frac{a}{64Y S U_{\text{fin}}(S)} = \frac{a}{4\lambda S^2 U_{\text{fin}}(S)}, \quad r_n = c/\sqrt{\log(en)}.$$

For explicit domain gates, use the source's backward RMS coefficients τ_j^{src} , and define

$$\mathcal{K} = (H_L^{\text{src}})^2 + S^2 \left[(\tau_1^{\text{src}})^2 + \sum_{j=2}^L (\tau_j^{\text{src}})^2 (H_{j-1}^{\text{src}})^2 \right],$$

$$D_W = \max\{\tau_1^{\text{src}}, \max_{j \geq 2} \tau_j^{\text{src}} H_{j-1}^{\text{src}}\}.$$

The sufficient numerical gates are

$$n^{-1} \leq Y, \quad \sqrt{\log(en)} \geq c \max\{8, \lambda, 4\mathcal{K}/\log 2, 32Y S D_W\}.$$

There are localized source events with probability tending to one on which the predictions at every query in the fixed finite collection are holomorphic on a neighborhood of

$$[-r_n, T_n + r_n] + i[-r_n, r_n].$$

The forward RMS is at most H_j^{src} and the readout RMS at most $S H_L^{\text{src}}$. At any one of these queries v , put $g_n(t) = f_n(t, v) - \tilde{f}_n(t, v)$. On both copies' events,

$$|g_n(t)| \leq 2S(H_L^{\text{src}})^2 \leq 32\beta^{6L} Y/\lambda$$

throughout this rectangle. The last inequality uses $H_L^{\text{src}} \leq \beta^{3L}$, independently of the smaller beta-only label cap. All these localized probability conclusions retain the source interface's unquantified sufficient width.

D.11 Actual-trajectory lower bound and its confidence coefficient

For every fixed admissible problem with $m \geq 2$, nonzero labels, and $0 < \delta < 1$, at every sufficiently large individual width,

$$\mathbb{P} \left\{ \|f_n - \tilde{f}_n\|_* \geq c_{\phi, L, \delta} \frac{Y \sqrt{\gamma}}{\sqrt{n} [\log(en)]^{5/2}} \right\} \geq 1 - \delta,$$

where on the full recurrence range one can take

$$c_{\phi, L, \delta} = \frac{\chi_{\text{act}}}{128} \Phi^{-1}(1/2 + \delta/4) \sqrt{\frac{\mu_{2,L}}{\mu_4}}.$$

Here Φ is the standard normal distribution function. On the unchanged cap $Y \leq (\gamma/m)\beta^{-30L}$, one may replace χ_{act} by one. A sharper fixed-label version replaces it by $\chi(S)$. These are positive coefficients; neither a confidence-independent numerical constant nor a quantified success width is asserted.

The witness is the deterministic training query v_{a_*} selected by the innovation inequality and some strictly positive time

$$0 < t \leq \frac{\chi(S)m}{\gamma\sqrt{\log(en)}} \leq \frac{m}{\gamma\sqrt{\log(en)}}.$$

The witnessing time may depend on width and initialization. At that training query the fitted endpoints agree exactly with the prescribed label, so this theorem is not an endpoint lower bound.

D.11.1 Necessary storage within the independent dense family

For a nondegenerate fixed task with $m \geq 2$, fix $0 < \delta < 1/2$. Once the lower theorem's sufficient width is met, the accuracy condition $\Pr\{\|f_n - \tilde{f}_n\|_* \leq \varepsilon\} \geq 1 - \delta$ necessarily implies

$$n[\log(en)]^5 \geq \frac{c_{\phi,L,\delta}^2 Y^2 \gamma}{\varepsilon^2}, \quad S_D(n) \geq \frac{(L-1)c_{\phi,L,\delta}^4 Y^4 \gamma^2}{\varepsilon^4 [\log(en)]^{10}}.$$

The proof below establishes the resulting fixed-task storage order $\varepsilon^{-4}/\log(1/\varepsilon)^{10}$ up to a positive fixed-task coefficient, among widths beyond that onset. This is a statement about independent canonical dense runs, not arbitrary alternative representations or exceptional smaller widths.

D.12 Dependency and width boundary of the dense section

The quantitative fitting theorem, parameter comparison, Gaussian concentration, initialized Gram CLT, innovation inequality, confidence conversion, and scalar accuracy inversion have complete proofs below. The initialized CLT is qualitative in width; no Berry–Esseen constant or explicit CLT success threshold is available in these inputs.

The trained carrier event and finite-query holomorphic event depend on the shared source foundation's local Gaussian insertion theorem and its complex extension. Their finite activation/depth coefficients are the displayed source recurrences, not unspecified comparison constants. The maps, control-uniform stopping argument, nonlinear remainders, passive-query derivatives and scalar trace contractions are proved in (IC.1)–(IC.57), including the independent stopped-path extension (IC.stop). The complete complex query-tube source and initialized CLT still have eventual fixed-problem width quantifiers; no numerical full success threshold is asserted for those results. The concrete numerical initialization threshold and deterministic domain/remainder gates remain distinct from that qualitative width convention. The narrower finite training-source event has the separate explicit proof (IC.58)–(IC.90); its all-time training-carrier bridge suffices for the Seeded decoder's dense upper certificate, not for a quantitative lower-bound CLT.

The same statements concern each sufficiently large individual width. They do not provide one event over infinitely many independently initialized widths, a joint growing-data/depth limit, strict root-width upper concentration, or a general fitted-endpoint variability lower bound. A finite union of model comparisons uses split failure budgets; no independence of their good events is required.

D.13 Legendre compression: every order, both label ranges, and accuracy inversion

This section uses the dense model, Gaussian initialization, activation assumptions and covariance gap already specified in this document. In particular, $n, m, d \geq 1$, $L \geq 2$, $\|x_a\|_2 = \sqrt{d}$, $Y = \|y\|_2/\sqrt{m}$, and $\gamma = \lambda_{\min}(Q^{(L)}) > 0$. The loss is $m^{-1} \sum_a (f_n(x_a) - y_a)^2$, the mobilities are $(n, 1, \dots, 1, n)$, and the initial readout is zero. The Legendre model shares the realized initialization and the physical time t of its dense reference. Its order q is any positive integer.

The deterministic proofs below apply on the explicitly stated initialization and dense-carrier event. Its probability is supplied by the shared source argument, including its local Gaussian insertion proof. The source's sufficient stochastic width remains implicit. The initialization event itself has the effective threshold proved in dense fitting.

D.13.1 The stored state and its autonomous equations

Write $v_a = x_a/\sqrt{d}$. Superscripts indicate layers, including the stored readout $W^{(L+1)} \in \mathbb{R}^n$. For the dense model, and separately for the reconstructed model with hats, define

$$z^{(1)}(x) = W^{(1)}x/\sqrt{d}, \quad z^{(\ell)}(x) = W^{(\ell)}h^{(\ell-1)}(x), \quad h^{(\ell)}(x) = \phi_\ell(z^{(\ell)}(x)), \quad f_n(x) = \frac{(W^{(L+1)})^T h^{(L)}(x)}{n}.$$

The dense backward carriers and responses on training sample a are

$$k_a^{(L)} = W^{(L+1)}, \quad k_a^{(\ell)} = (W^{(\ell+1)})^T \delta_a^{(\ell+1)}, \quad \delta_a^{(\ell)} = \phi'_\ell(z_a^{(\ell)}) \odot k_a^{(\ell)}.$$

The symbol $\delta_a^{(\ell)}$ is a backward response; the unsubscripted $\delta \in (0, 1)$ remains the failure probability. The residual is $r_a = f_n(x_a) - y_a$, and $\rho = \|r\|_2/\sqrt{m}$. The dense physical equations are

$$\dot{W}^{(1)} = -\frac{2}{m} \sum_a r_a \delta_a^{(1)} v_a^T, \quad \dot{W}^{(\ell)} = -\frac{2}{mn} \sum_a r_a \delta_a^{(\ell)} (h_a^{(\ell-1)})^T, \quad \dot{W}^{(L+1)} = -\frac{2}{m} \sum_a r_a h_a^{(L)}. \quad (54)$$

The middle equation applies to $2 \leq \ell \leq L$.

For each such interface, training sample a , and $j = 0, \dots, q-1$, store two vectors in $\mathbb{R}^n$, $\bar{h}_{a,j}^{(\ell-1)}$ and $\bar{\delta}_{a,j}^{(\ell)}$. Together with $\widehat{W}^{(1)}, \widehat{W}^{(L+1)}$ and one scalar τ , these are the complete moving state. Reconstruct the remaining matrices by

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{mn\tau} \sum_{a=1}^m \sum_{j=0}^{q-1} (2j+1) \bar{\delta}_{a,j}^{(\ell)} (\bar{h}_{a,j}^{(\ell-1)})^T. \quad (55)$$

Compute its forward pass, backward responses, residuals and $\hat{\rho} = \|\hat{r}\|_2/\sqrt{m}$ from these matrices. Its first matrix and readout obey (54) evaluated at the reconstructed state. Its remaining equations are

$$\begin{aligned} \dot{\tau} &= \hat{\rho}, \quad \tau(0) = 1, \\ \dot{\bar{h}}_{a,j}^{(\ell-1)} &= \hat{\rho} \bar{h}_a^{(\ell-1)} - \frac{\hat{\rho}}{\tau} \left(j \bar{h}_{a,j}^{(\ell-1)} + \sum_{i < j} (2i+1) \bar{h}_{a,i}^{(\ell-1)} \right), \\ \dot{\bar{\delta}}_{a,j}^{(\ell)} &= \hat{\rho} \bar{\delta}_a^{(\ell)} - \frac{\hat{\rho}}{\tau} \left(j \bar{\delta}_{a,j}^{(\ell)} + \sum_{i < j} (2i+1) \bar{\delta}_{a,i}^{(\ell)} \right). \end{aligned} \quad (56)$$

Initially $\bar{h}_{a,0}^{(\ell-1)} = h_{0,a}^{(\ell-1)}$; all other moments are zero. Thus the forward histories have their constant initialized values on $[0, 1]$, while the backward histories are zero there. The runtime equations involve no division by a residual and are locally Lipschitz even at $\hat{\rho} = 0$. They then have zero velocity. The histories used in the proof are not additional stored arrays.

The exact counts in real coordinates are

$$\begin{aligned} S_{\text{moving}}(n, q) &= n(d+1) + 1 + 2(L-1)mnq, \quad S_{\text{fixed}}(n) = (L-1)n^2, \\ S_{\text{model}}(n, q) &= (L-1)n^2 + n(d+1) + 1 + 2(L-1)mnq. \end{aligned} \quad (57)$$

These are retained model-state counts; storing the input data and labels adds $m(d+1)$. Reconstructed dense matrices can be evaluated from the displayed state and are not counted as independently retained parameters. These inventory counts alone concern exact real coordinates, not bit precision or execution cost. The separate cost interface includes preprocessing, runtime and the specified working-memory caches.

D.13.2 Label allowances and the event used by the error certificates

The following coefficients are local to this section. Define

$$b = \max_{\ell} |\phi_{\ell}(0)|, \quad s = \max\{1, \max_{\ell} \sup_{|\operatorname{Im} z| \leq a/2} |\phi'_{\ell}(z)|\}, \quad t_2 = \max_{\ell} \sup_{|\operatorname{Im} z| \leq a/2} |\phi''_{\ell}(z)|.$$

The shared envelope is $\beta = \max\{10, 1 + b, 16/a, s, \max(1, t_2)\}$. Set $\sigma_0 = 1$ and

$$\sigma_{\ell}^2 = \mathbb{E}\phi_{\ell}(\sigma_{\ell-1}G)^2, \quad G \sim N(0, 1), \quad H = \max\{1, \sigma_1, \dots, \sigma_L\}.$$

These scalar moments exist by $|\phi_{\ell}(u)| \leq b + s|u|$. Every diagonal of $Q^{(\ell)}$ is σ_{ℓ}^2 , so $\gamma/m \leq H^2$. Define

$$\begin{aligned} D_{\ell} &= s(9s)^{L-\ell}, \quad F = s^2 \left[(9s)^{2L-2} + 4H^2 \sum_{j=0}^{L-2} (9s)^{2j} \right], \\ C_1 &= 4s^2 D_1^2, \quad C_{\ell} = 3s^2 (64H^4 D_{\ell}^2 + 82C_{\ell-1}), \quad E = 130 \sum_{\ell=2}^L D_{\ell} \sqrt{C_{\ell-1}}, \\ S_*^{\text{Leg}} &= \min \left\{ 1, \frac{1}{64H^2 D_1}, \frac{1}{\sqrt{8\sqrt{C_L}}}, \left(\frac{1}{8H^2 D_1 E} \right)^{2/7} \right\}. \end{aligned} \quad (58)$$

There is no unspecified constant in these recurrences. The source coefficients $S_*^{\text{src}}, K_{\text{src}}$ are exactly the finite recurrences defined and estimated in source foundations.

The larger label allowance needed by the Legendre comparison is

$$0 < Y \leq \frac{\gamma}{m} \min \left\{ \frac{1}{8H\sqrt{F}}, \frac{S_*^{\text{Leg}}}{8}, \frac{S_*^{\text{src}}}{16} \right\}. \quad (59)$$

In particular this holds throughout the original common allowance, whose additional Harmonic-fitting term only restricts the common range. The smaller, sufficient cap is exactly

$$0 < Y \leq \frac{\gamma}{m} \beta^{-30L}. \quad (60)$$

Neither label range is replaced by an error-dependent restriction.

The deterministic initialization event is

$$\begin{aligned} \frac{\|W_0^{(1)}\|_{\text{op}}}{\sqrt{n}} &\leq 8, \quad \max_{2 \leq \ell \leq L} \|W_0^{(\ell)}\|_{\text{op}} \leq 8, \quad \sup_{\|x\|_2 = \sqrt{d}, \ell} \frac{\|h_0^{(\ell)}(x)\|_2}{\sqrt{n}} \leq \frac{3H}{2}, \\ \frac{\mathbf{H}_0^T \mathbf{H}_0}{mn} &\succeq \frac{\gamma}{2m} I_m, \quad \mathbf{H}_0 = [h_{0,a}^{(L)}]_{a=1}^m. \end{aligned} \quad (61)$$

On this event dense fitting and the proof below give, for all orders, all sphere queries and all times, normalized first-matrix operator norm and hidden-matrix operator norms below nine, feature RMS below $2H$, and

$$\begin{aligned} \rho_D(t) &\leq Y e^{-\gamma t/(2m)}, \quad \hat{\rho}(t) \leq Y e^{-\gamma t/(4m)}, \\ \frac{\|W_D^{(L+1)}(t)\|_2}{\sqrt{n}} &\leq 2Y \sqrt{m/\gamma}, \quad \frac{\|\widehat{W}^{(L+1)}(t)\|_2}{\sqrt{n}} \leq 4Y \sqrt{m/\gamma}. \end{aligned} \quad (62)$$

Both systems converge in physical parameters and fit the training labels. The additional source event used for approximation is precisely

$$\max_{a, \ell, i} \sup_{t \geq 0} |k_{D,a,i}^{(\ell)}(t)| \leq M_n := 32K_{\text{src}} \frac{Ym}{\gamma} \sqrt{\log(en)}. \quad (63)$$

This involves training inputs only. The source argument supplies the all-time factor two relative to its finite-horizon carrier estimate. For every fixed $\delta \in (0, 1)$, split the failure budget between initialization and source events. Their intersection has probability at least $1 - \delta$ for each sufficiently large individual width. The source part of the width threshold remains implicit. One such event works for every order at that width and for every requested accuracy; no order or tolerance union bound is required.

D.13.3 Error from a supplied order

For any two predictors use the whole-trajectory, whole-sphere distance

$$\|f - g\|_* = \sup_{t \in [0, \infty]} \sup_{\|x\|_2 = \sqrt{d}} |f(t, x) - g(t, x)|.$$

The fitted endpoints are included. This is a function-fidelity statement, not a claim about unknown test labels.

On (60), define the exact actual-label coefficient

$$\begin{aligned} P_n &= 3\beta^{100L} \left(\frac{Ym}{\gamma} \right)^2 \left(1 + \frac{m}{\gamma} \right) \left[1 + \beta^{100L} \frac{Ym}{\gamma} \sqrt{\log(en)} \right] \\ &\quad \times \left[1 + \beta^{100L} \left(\frac{Ym}{\gamma} \right)^2 \left(e^{\sqrt{\log(en)}} - 1 \right) \right]. \end{aligned} \quad (64)$$

Simultaneously for every integer $q \geq 1$,

$$\|f_{\text{Leg}, n, q} - f_n\|_* \leq Y P_n \frac{\sqrt{\log(eq)}}{q^2}. \quad (65)$$

An explicit, weaker structural envelope, if a shorter expression is preferred, is

$$\|f_{\text{Leg}, n, q} - f_n\|_* \leq 12\beta^{300L} Y \left(1 + \frac{m}{\gamma} \right)^6 e^{\sqrt{\log(en)}} \frac{\sqrt{\log(en) \log(eq)}}{q^2}. \quad (66)$$

For the full range (59), use the following exact finite assembly instead. In these coefficient definitions only, put

$$\kappa = \frac{\gamma}{4m}, \quad R = 8Y\sqrt{m/\gamma}, \quad S = \frac{8Ym}{\gamma}, \quad a_0 = \frac{4Ym}{\gamma}, \quad T_0 = 9s, \quad B_\delta = sT_0^{L-1}R, \quad F_z = 2HT_0^{L-1},$$

$$P_0 = 1 + 2H(L-1), \quad B_d = sT_0^{L-1}(1 + B_\delta) + Lt_2F_zT_0^{L-1}M_n,$$

$$V_z = 2F_zB_\delta P_0, \quad V_\delta = T_0^{L-1}[4sH + 4sH(L-1)B_\delta^2 + Lt_2M_nV_z],$$

$$T_1 = 2D_1R, \quad T_\ell = 4HD_\ell R + 130D_\ell \sqrt{SC_{\ell-1}}R^2,$$

$$V_1 = sT_1, \quad V_\ell = s(2HT_\ell + 9V_{\ell-1}), \quad V_h = \max_{1 \leq \ell \leq L} V_\ell,$$

$$C_c = 4 \left(4H^2 + D_1^2 R^2 + 4H^2 R^2 \sum_{\ell=2}^L D_\ell^2 \right) + \frac{\gamma}{2m},$$

$$F_h = V_h a_0 \sqrt{(1 + a_0)/2}, \quad b_1 = \frac{2\sqrt{1 + a_0}}{\kappa} C_c B_\delta,$$

$$b_0 = \frac{\sqrt{1 + a_0}Y}{\kappa} V_\delta + 2B_\delta \sqrt{a_0}, \quad C_f = 2H + RsF_z. \quad (67)$$

Define segment feature and gradient coefficients by

$$\begin{aligned}
Q_1 &= b + 9s, & Q_\ell &= b + 9sQ_{\ell-1}, & Q &= \max\{1, Q_1, \dots, Q_L\}, & F_{\text{seg}} &= QT_0^{L-1}, \\
d_0 &= sT_0^{L-1}(1 + B_\delta), & d_1 &= Lt_2F_{\text{seg}}T_0^{L-1}, & p &= 1 + Q(L-1), \\
K_0 &= \sqrt{L+1}\{pd_0 + [1 + (L-1)B_\delta]sF_{\text{seg}}\}, & K_1 &= \sqrt{L+1}pd_1, \\
\mathcal{A}_n &= \sqrt{L+1} \exp \left\{ 14(K_0 + K_1M_n) \frac{Ym}{\gamma} \right\}.
\end{aligned} \tag{68}$$

The local scalar d_1 here is a gradient coefficient, not the input dimension d . Finally set

$$q_{\text{abs}} = 4(L-1)F_hB_d\sqrt{a_0}\mathcal{A}_n, \quad C_n = 4(L-1)C_f\mathcal{A}_nF_h(b_0 + b_1), \quad U = 12HY\sqrt{m/\gamma}. \tag{69}$$

Then, simultaneously for every integer $q \geq 1$,

$$\|f_{\text{Leg},n,q} - f_n\|_* \leq C_n \frac{\sqrt{\log(eq)}}{q^2} + U \left(\frac{q_{\text{abs}}}{q} \right)^4. \tag{70}$$

The order-independent bound $\|f_{\text{Leg},n,q} - f_n\|_* \leq U$ also holds. The simple-cap coefficient is not extended to larger labels; the displayed recurrence certificate preserves that larger range.

D.13.4 Accuracy and failure probability as inputs

Fix a permitted problem, $\varepsilon > 0$, $0 < \delta < 1$, and a reference width in the event's sufficient-width range. The width condition is the same one already stated above; no new accuracy-dependent stochastic threshold is introduced. The inverse below is therefore a deterministic order certificate with success probability at least $1 - \delta$, not an effective confidence-certified choice of reference width.

Under the simple cap and $Y > 0$, the least positive integer certified by (65) is exactly

$$q_{\min}(n, \varepsilon) = \min \left\{ q \in \mathbb{N} : q \geq 1, \quad \frac{YP_n}{\varepsilon} \sqrt{\log(eq)} \leq q^2 \right\}. \tag{71}$$

This minimizes the increasing storage formula within this certificate; it is not a minimum over all representations or the actual dynamics. The minimum exists and every larger integer works. For a closed sufficient choice put $r = YP_n/\varepsilon$ only in this inversion formula and take

$$q_{\text{suff}} = \begin{cases} 1, & r \leq 1, \\ \lceil 2\sqrt{r}[\log(e+r)]^{1/4} \rceil, & r > 1. \end{cases} \tag{72}$$

The second branch gives error at most $\varepsilon/2$.

A sufficient inverse using only the structural envelope is the same two-branch formula with the explicit ratio

$$r = \frac{12\beta^{300L}Y}{\varepsilon} \left(1 + \frac{m}{\gamma} \right)^6 \sqrt{\log(en)} e^{\sqrt{\log(en)}}. \tag{73}$$

Equivalently, when this ratio exceeds one, its order is

$$\left\lceil 4\sqrt{3}\beta^{150L} \left(1 + \frac{m}{\gamma} \right)^3 \sqrt{\frac{Y}{\varepsilon}} [\log(en) \log(e+r)]^{1/4} e^{\frac{1}{2}\sqrt{\log(en)}} \right\rceil. \tag{74}$$

The scalar r is only the fully displayed ratio above. Order one suffices when it is at most one; no asymptotic logarithm replacement is needed for these formulas.

For the full label allowance, define the exact least integer inverse

$$q_{\min}^{\text{full}}(n, \varepsilon) = \min \left\{ q \geq 1 : C_n \frac{\sqrt{\log(eq)}}{q^2} + U \left(\frac{q_{\text{abs}}}{q} \right)^4 \leq \varepsilon \right\}. \quad (75)$$

The braces here range over integers; this minimum exists. It equals one whenever $C_n + Uq_{\text{abs}}^4 \leq \varepsilon$. For a direct sufficient formula apply (72) with $r = 2C_n/\varepsilon$, call the resulting integer q_1 , and choose

$$q = \max \left\{ q_1, 1, \left\lceil q_{\text{abs}}(2U/\varepsilon)^{1/4} \right\rceil \right\}. \quad (76)$$

Both summands are then at most $\varepsilon/2$. The second entry is an accuracy allocation, not an absorption gate; before rounding it is below q_{abs} when $\varepsilon > 2U$. If using the additional uniform cap is desired, $\varepsilon \geq U$ already permits $q = 1$ in either range. This optional improvement is separate from the least integer inverse of the displayed decreasing envelope.

In all cases substitute the chosen order into (57). Conversely, a supplied moving-coordinate budget S_{budget} admits the largest order

$$q_{\text{budget}} = \left\lfloor \frac{S_{\text{budget}} - n(d+1) - 1}{2(L-1)mn} \right\rfloor. \quad (77)$$

If this is below one, the budget does not hold an order-one instance of the stipulated representation. Otherwise evaluate the applicable forward certificate at that order. For a total model-state budget first subtract $(L-1)n^2$, and also subtract $m(d+1)$ if data storage is included. Counts grow strictly with q , so this is the best certified decreasing error under that coordinate budget.

If $Y = 0$, all moment velocities, parameter velocities and predictions are zero. Every order is exact; the constant-zero predictor alone is an exact compression if only predictor fidelity is requested. No expression dividing by Y is used. If $\varepsilon = 0$ and $Y > 0$, the positive finite-order upper envelopes certify no finite order; this does not assert that a particular network cannot agree exactly. Negative accuracy or $\delta \notin (0, 1)$ is outside the stated interface. There is no exactness claim at $q = n$ for this temporal Legendre representation, and no need to constrain q by n .

The retained simple-cap prescription for a strict root-width target is

$$Q_n = \max\{3, n^{1/4}\sqrt{P_n}\}, \quad q_n = \left\lceil 4Q_n[\log(e + Q_n)]^{1/4} \right\rceil, \quad \|f_{\text{Leg}, n, q_n} - f_n\|_* \leq \frac{Y}{8\sqrt{n}}. \quad (78)$$

Here Q_n is only an auxiliary expression for selecting the one runtime order q_n , not a second approximation order.

At fixed admissible positive labels and other structural parameters, M_n is proportional to $\sqrt{\log(en)}$, and $\log C_n, \log q_{\text{abs}} = o(\log n)$. For a polynomial target $\varepsilon = Yn^{-\alpha}$ with fixed $\alpha > 0$, the sufficient orders are $n^{\alpha/2+o(1)}$ in both label ranges; the second full-range entry is only $n^{\alpha/4+o(1)}$. In particular $\varepsilon = Y/\sqrt{n}$ gives $q = n^{1/4+o(1)}$ and moving state $n^{5/4+o(1)}$, while the retained initialized mixers still have $(L-1)n^2$ entries. These are sufficient fixed-problem storage rates, with the probability and exact-real qualifications already stated.

D.14 Harmonic neural dynamics with a supplied neuron budget

The proof-local subscript C continues to denote the compressed Harmonic model; its coefficient labels are unchanged.

The Harmonic model has its own evolving first weights, hidden mixers, raw readout, and label deficit. Its effective readout is reconstructed from its current features. The construction below approximates the realized dense trajectory at the same physical times, on the whole input sphere, including the fitted limit. Its only size parameter is the integer neuron budget q . Source tolerances, approximation degrees, quadratures, initial or local jets, continuation states and matrix-action transcripts are setup objects and are discarded.

The dense architecture, Gaussian initialization with zero readout, loss, mobilities, activation strip and unweighted gap γ are the shared ones. In particular the residual convention remains $r = f - y$. The Harmonic state variable $c = y - f = -r$ is called the label deficit below. This section uses the source event and finite coefficients in the source foundations. It does not convert an eventual stochastic width into an effective

confidence bound. The deterministic harmonic approximation, coordinate selection, metric, initialization, runtime, and comparison arguments are supplied here.

For coefficient formulas and proofs only, write $\lambda = \gamma/m$, $z = Y/\lambda$, $\alpha = Y/\sqrt{\lambda}$, and $S = 16Y/\lambda$. These are explicit arithmetic expressions, not new parameters or a capped gap. Write $b = \max_j |\phi_j(0)|$, and let $s \geq 1$ and $t_2 \geq 1$ be the first and second derivative bounds on the safe half-strip. Thus $\beta = \max(10, 1 + b, 16/a, s, t_2)$. Superscript *src* distinguishes the source coefficients from the Harmonic coefficients defined next.

D.14.1 Exact label interval and runtime coefficients

Let H_d, F_d denote the dense fitting coefficients: if $v_0 = 1$ and $v_j = \mathbb{E}\phi_j(\sqrt{v_{j-1}}Z)^2$, $Z \sim N(0, 1)$, then

$$H_d = \max(1, \sqrt{v_1}, \dots, \sqrt{v_L}), \quad F_d = s^2 \left[(9s)^{2L-2} + 4H_d^2 \sum_{k=0}^{L-2} (9s)^{2k} \right]. \quad (79)$$

For the Harmonic runtime define

$$\begin{aligned} H_1^c &= 2b + 16s, & H_j^c &= 2b + 18sH_{j-1}^c, & H_c &= H_L^c, \\ d_j^c &= 2s(18s)^{L-j}, & U_1^c &= d_1^c, & U_j^c &= 2H_c d_j^c \ (j \geq 2), & F_1^c &= 2sU_1^c, & F_j^c &= 2s(2H_c U_j^c + 9F_{j-1}^c), \\ F_c &= F_L^c = (d_1^c)^2 + 4H_c^2 \sum_{j=2}^L (d_j^c)^2. \end{aligned} \quad (80)$$

The omitted maxima with one in these recurrences are inactive because $s \geq 1$. The full Harmonic label interval is

$$0 < Y \leq \frac{\gamma}{m} \min \left\{ \frac{1}{8H_d\sqrt{F_d}}, \frac{1}{16H_c\sqrt{F_c}}, \frac{S_*^{\text{src}}}{16} \right\}. \quad (81)$$

Every conclusion below holds on this entire interval. An additional Legendre allowance in a joint theorem can be intersected with this interval; no smaller Harmonic label cap is required. The scalar Gaussian moment recursion gives $\sqrt{\lambda} \leq H_c$. In particular $S \leq 1$, $z \leq 1/16$, and $\alpha \leq 1/(16\sqrt{F_c}) \leq 1/16$.

For later tail estimates define the following explicit functions of the same coefficients:

$$\begin{aligned} R_c &= 5\alpha, & G_c &= 4H_c^2 + R_c^2 F_c, \\ B_w^c &= 4H_c + 48R_c F_c z + 4G_c/\sqrt{\lambda}, & B_f^c &= 2H_c B_w^c + 2R_c^2 F_c, \\ \mathcal{K} &= (H_L^{\text{src}})^2 + S^2 \left[(\tau_1^{\text{src}})^2 + \sum_{j=2}^L (\tau_j^{\text{src}})^2 (H_{j-1}^{\text{src}})^2 \right], & \mathcal{D} &= z(16\mathcal{K} + 4B_f^c). \end{aligned} \quad (82)$$

D.14.2 Selected architecture, exact initialization and optimizer

Fix a source horizon T and coordinate tolerance η , chosen from the supplied budget in the theorem below. At layer j , define the source space $E_j \subset \mathbb{R}^n$ using the empirical pairing $u^T v/n$. The four possible vector-valued source families, indexed by $(t, v) \in [0, T] \times S^{d-1}$, are

$$h_n^{(j)}(t, v), \quad W_0^{(j)} h_n^{(j-1)}(t, v) \ (j \geq 2), \quad \delta_n^{(j)}(t, v), \quad W_0^{(j+1)T} \delta_n^{(j+1)}(t, v) \ (j < L). \quad (83)$$

Here $W_0^{(j)}$ is the initialized dense mixer. The forward-image family is absent at layer one, and the reverse-image family is absent at layer L . For a passive query the dense backward fields are

$$k_n^{(L)}(t, v) = w_n(t), \quad \delta_n^{(j)}(t, v) = \phi'_j(z_n^{(j)}(t, v)) \odot k_n^{(j)}(t, v), \quad k_n^{(j)}(t, v) = W_n^{(j+1)}(t)^T \delta_n^{(j+1)}(t, v) \quad (j < L).$$

A passive query is evaluated using the trained parameters but contributes no update force.

The space E_j is the real span of the coefficient vectors actually computed by the finite cosine-spherical-harmonic construction below for these families, together with the exact initialized vectors

$$\mathbf{1}_n, \quad \{h_n^{(j)}(0, v_a) : 1 \leq a \leq m\}, \quad \{W_0^{(j)} h_n^{(j-1)}(0, v_a) : 1 \leq a \leq m\} \quad (j \geq 2),$$

and the d columns of A_0 when $j = 1$. It uses certified finite coefficients, not unavailable exact integrals of a trained reference. They may be obtained from origin jets, or from the two certified local-continuation initializers below. The exact additions number at most $m+d+1$ at the first layer and $2m+1$ at each later layer. Thus $B = 2m + d + 1$ bounds their dimension uniformly. The construction's counts give $\dim E_j \leq B + 4N(T, \eta)$ for $d \geq 2$. For $d = 1$ the separate temporal expansions at $v = -1, 1$ give $\dim E_j \leq B + 8N_1(T, \eta)$. The functions N, N_1 are defined explicitly in the supplied-budget statement.

Use the same cutoff and scalar coefficient operations in every feature/image pair and response/reverse-image pair. In the local-continuation implementations, first accumulate the base coefficients and then form their exact initialized-mixer images; the stronger source-jet error estimate controls the image error separately. These scalar projection operations commute exactly with each initialized mixer. Consequently the computed approximants obey

$$p_h^{(j-1)} \in E_{j-1}, \quad W_0^{(j)} p_h^{(j-1)} \in E_j, \quad p_\delta^{(j+1)} \in E_{j+1}, \quad W_0^{(j+1)T} p_\delta^{(j+1)} \in E_j, \quad (84)$$

with a separate coordinate-error bound η for each member of each pair. These are the paired actions used in the comparison proof; no further source family or stored runtime order is being introduced.

At layer j , selection supplies an index set I_j of q_j coordinates and a positive metric M_j , with a positive diagonal comparison metric D_j . In addition to the bounds below, selection preserves the source inner product exactly:

$$u_{I_j}^T M_j v_{I_j} = \frac{u^T v}{n} \quad (u, v \in E_j).$$

$$D_j/4 \preceq M_j \preceq D_j, \quad \mathbf{1}^T M_j \mathbf{1} = 1, \quad \mathbf{1}^T D_j \mathbf{1} \leq 4. \quad (85)$$

Write $\langle u, v \rangle_{M_j} = u^T M_j v$. The moving state is

$$(A_C, B_C^{(2)}, \dots, B_C^{(L)}, w_C, c_C), \quad A_C \in \mathbb{R}^{q_1 \times d}, \quad B_C^{(j)} \in \mathbb{R}^{q_j \times q_{j-1}}, \quad w_C \in \mathbb{R}^{q_L}, \quad c_C \in \mathbb{R}^m. \quad (86)$$

For $v = x/\sqrt{d}$, set

$$z_C^{(1)}(v) = A_C v, \quad z_C^{(j)}(v) = B_C^{(j)} h_C^{(j-1)}(v), \quad h_C^{(j)}(v) = \phi_j(z_C^{(j)}(v)).$$

Let V_C have columns $h_C^{(L)}(v_a)/\sqrt{m}$, let $V_C^* = V_C^T M_L$, and put $Q_C = V_C^* V_C$. The effective readout and predictor are

$$\widehat{w}_C = w_C + V_C Q_C^{-1} [(y - c_C)/\sqrt{m} - V_C^* w_C], \quad f_C(v) = \langle \widehat{w}_C, h_C^{(L)}(v) \rangle_{M_L}. \quad (87)$$

Multiplication by V_C^* gives $f_C(v_a) = y_a - c_{C,a}$ exactly. For the specified training signals use

$$k_{C,a}^{(L)} = \widehat{w}_C, \quad \delta_{C,a}^{(j)} = \phi'_j(z_{C,a}^{(j)}) \odot k_{C,a}^{(j)}, \quad k_{C,a}^{(j)} = (B_C^{(j+1)})^* \delta_{C,a}^{(j+1)},$$

where $(B_C^{(j+1)})^* = M_j^{-1} (B_C^{(j+1)})^T M_{j+1}$. Define the unnormalized sample Gram by

$$K_{C,ab} = \langle h_{C,a}^{(L)}, h_{C,b}^{(L)} \rangle_{M_L} + \langle \delta_{C,a}^{(1)}, \delta_{C,b}^{(1)} \rangle_{M_1} v_a^T v_b + \sum_{j=2}^L \langle \delta_{C,a}^{(j)}, \delta_{C,b}^{(j)} \rangle_{M_j} \langle h_{C,a}^{(j-1)}, h_{C,b}^{(j-1)} \rangle_{M_{j-1}}. \quad (88)$$

The autonomous equations are

$$\begin{aligned} \dot{A}_C &= \frac{2}{m} \sum_a c_{C,a} \delta_{C,a}^{(1)} v_a^T, & \dot{B}_C^{(j)} &= \frac{2}{m} \sum_a c_{C,a} \delta_{C,a}^{(j)} h_{C,a}^{(j-1)T} M_{j-1}, \\ \dot{w}_C &= \frac{2}{m} \sum_a c_{C,a} h_{C,a}^{(L)}, & \dot{c}_C &= -2K_C c_C / m. \end{aligned} \quad (89)$$

These directions are specified by the optimizer. A coordinate gate need not be self-adjoint in M_j , so they are not asserted to be gradients of the corrected predictor.

To specify the initialized matrices, choose a basis matrix U_j of the source space with $U_j^T U_j / n = I$, and write $P_j = (U_j)_{I_j}$. The exact source isometry gives $P_j^T M_j P_j = I$. Initialize

$$\begin{aligned} A_C(0) &= (A_0)_{I_1}, & w_C(0) &= 0, & c_C(0) &= y, \\ B_C^{(j)}(0) &= P_j \frac{U_j^T W_0^{(j)} U_{j-1}}{n} P_{j-1}^T M_{j-1}. \end{aligned} \quad (90)$$

The source spaces include every initialized training feature and its initialized forward image, all first-weight columns, and the constant. Consequently this initialized forward pass and its training Gram equal the corresponding selected dense ones exactly. The initialized operator norms are at most eight. The basis matrices and dense arrays in this formula are discarded after setup.

D.14.3 Explicit full-range comparison coefficient

Here are finite recurrences for the comparison certificate. They use only the already defined source and fitting coefficients, and contain no unspecified numerical constant. Put

$$\begin{aligned} u_0 &= H_L^{\text{src}}, \quad \tau = \max_j \tau_j^{\text{src}}, \quad H_r = u_0 + 3, \quad H_C = 2H_c, \quad P_h = 6u_0 + 9, \quad P_\delta = 6\tau + 9, \\ A_f &= 18 + S^2(\tau + 3)P_h, \quad A_b = 18 + S^2 H_r P_\delta, \\ F_1^z &= 1, \quad F_1^h = 2s, \quad F_j^z = 9F_{j-1}^h + H_r + A_f, \quad F_j^h = 2sF_j^z, \quad F = \max_j F_j^h. \end{aligned} \quad (91)$$

Define

$$\begin{aligned} d_j^E &= 2s(9s)^{L-j} + 3H_c, \quad J_C = 5\sqrt{F_c}, \quad J_R = \left[(d_1^E)^2 + H_r^2 \sum_{j=2}^L (d_j^E)^2 \right]^{1/2}, \\ D_s &= P_h + 16zP_\delta[1 + (L-1)H_r^2] + (L-1)(16z)^2\tau^2P_h, \\ B_L &= 2s(1 + 6zF + 32zP_h) + 2t_2F_L^z, \quad B_j = 18sB_{j+1} + 2s(d_{j+1}^E zH_c + A_b) + 2t_2F_j^z, \\ B_h &= \sqrt{L} \left[H_C \max_j B_j + (\max_j d_j^E) zH_c F \right], \quad K_1 = 12F + 2B_h + 20z(J_C + J_R)B_h + 20D_s. \end{aligned} \quad (92)$$

The symbols J_C, J_R in this display are scalar norm coefficients; the velocity maps in the proof are written $\mathcal{J}_C, \mathcal{J}_R$. Set

$$\mathcal{B}_n = 400z^2F_c + 2zK_1 + 40zF + 64z^2K_1K_{\text{src}}\sqrt{\log(en)},$$

$$\begin{aligned}\mathcal{A}_n &= [H_C(1 + 6zF) + 3\alpha F](1 + \lambda^{-1/2})(e^{\mathcal{B}_n} - 1) \\ &\quad + 6H_C zF + 32H_C zP_h/\sqrt{\lambda} + 3\alpha F + 16zP_h.\end{aligned}\tag{93}$$

For any finite source horizon $T \geq 32(m/\gamma)\log(en)$ and any source coordinate tolerance $0 < \eta \leq \min(1, Y, 16Ym/\gamma)$, the constructed runtime satisfies

$$\sup_{0 \leq t \leq T, \|x\|=\sqrt{d}} |f_C - f_n| \leq \mathcal{A}_n \eta, \quad \|f_C - f_n\|_* \leq \mathcal{A}_n \eta + \mathcal{D}e^{-\gamma T/(4m)}.\tag{94}$$

The same construction also satisfies the fitting bound

$$\|f_C - f_n\|_* \leq (10H_c + 4H_d)Y\sqrt{m/\gamma}.\tag{95}$$

The two all-time bounds include the limiting predictors; the first bound is only on its displayed finite horizon. The factor 64 in $\mathcal{B}_n$ covers the all-time carrier extension. At the original horizon, before that extension is used, it can be replaced by 32.

D.14.4 Every supplied budget and its exact inverse

The following coefficient definitions expose all rounding and finite-width terms. Use the original source radii

$$\begin{aligned}c_t &= \frac{a}{64YSU}, \\ c_q &= \min\{1/8, a/(8V)\}, \\ r_t &= \frac{c_t}{\sqrt{\log(en)}}, \\ r_q &= \frac{c_q}{\sqrt{\log(en)}}, \\ T_0 &= 32\frac{m}{\gamma}\log(en), \\ \eta_0 &= \min(1, Y, S).\end{aligned}$$

For $T \geq T_0$, let $\alpha_T = r_t/(4T)$, and in every dimension put $M_n = 10\max(H_L^{\text{src}}, \tau)\sqrt{n}$. If $d \geq 2$, set

$$b_d = 2d - 2, \quad D_d = 2^{d+1}d^{d-2}, \quad P_T = \frac{18D_db_d!2^{b_d+1}}{\alpha_T r_q^{b_d+1}},$$

$$H(T, \eta) = 2\log \frac{16M_n P_T}{\eta}, \quad N(T, \eta) = \sum_{j=0}^{\lfloor H(T, \eta)/r_q \rfloor} \left[\binom{j+d-1}{d-1} - \binom{j+d-3}{d-1} \right] \left[1 + \left\lfloor \frac{H(T, \eta) - r_q j}{\alpha_T} \right\rfloor \right].\tag{96}$$

An impossible binomial coefficient is zero. Define

$$h_0 = \max\{8\log(en), H(T_0, \eta_0) + \alpha_{T_0} + (d-1)r_q\} \quad (d \geq 2).$$

For $d = 1$, use the two queries $-1, 1$ and instead define

$$H_1(T, \eta) = \log \frac{64M_n}{\alpha_T \eta}, \quad N_1(T, \eta) = 1 + \lfloor H_1(T, \eta)/\alpha_T \rfloor, \quad h_0 = \max\{8\log(en), H_1(T_0, \eta_0)\}.\tag{97}$$

In both cases set

$$\begin{aligned}B &= 2m + d + 1, \quad B_* = 2m + d + 9, \quad \mathcal{C}_d = \frac{2^{17}(9/4)^d}{d!} \frac{U}{a} c_q^{-(d-1)} \left(\frac{Ym}{\gamma} \right)^2 \log(en)^{d/2}, \\ \mathcal{E}_0 &= (10H_c + 4H_d)Y\sqrt{m/\gamma}, \quad \mathcal{E}_1 = \mathcal{A}_n \eta_0 + \mathcal{D}e^{-8\log(en)}, \quad \mathcal{E} = \max(\mathcal{E}_0, \mathcal{E}_1).\end{aligned}\tag{98}$$

All coefficients are evaluated at the actual label size Y .

For every integer $q < n$ with $q \geq 9B$, a Harmonic model with $q_j \leq q$ exists. Its certificate is the baseline $\mathcal{E}_0$ when $(\max(q/9 - B_*, 0)/\mathcal{C}_d)^{1/(d+1)} < h_0$. Otherwise choose

$$u = \left(\frac{q/9 - B_*}{\mathcal{C}_d} \right)^{1/(d+1)} - h_0, \quad T = T_0 + 4mu/\gamma, \quad \eta = \eta_0 e^{-u}. \quad (99)$$

The resulting error is at most $\mathcal{E}_1 e^{-u}$, as well as $\mathcal{E}_0$. In particular the uniform convenient certificate is

$$\|f_C - f_n\|_* \leq \mathcal{E} \min \left\{ 1, \exp \left[h_0 - \left(\frac{\max(q/9 - B_*, 0)}{\mathcal{C}_d} \right)^{1/(d+1)} \right] \right\}. \quad (100)$$

For $q \geq n$, retain every dense coordinate and take $M_j = I_n/n$; then the error is exactly zero. Thus the full-width branch remains available even if $n < 9B$. If $Y = 0$, the predictor is identically zero, and no formula dividing by Y is needed.

For the simpler domain $18B_* \leq q < n$, the certificate implies

$$\|f_C - f_n\|_* \leq \mathcal{E} \min \left\{ 1, e^{h_0 - (q/(18\mathcal{C}_d))^{1/(d+1)}} \right\}. \quad (101)$$

With the deterministic logarithmic gates below, a completely numerical activation-envelope version is

$$\begin{aligned} \|f_C - f_n\|_* &\leq 4096 e^{45} \beta^{42L} \frac{Ym}{\gamma} \left(1 + \sqrt{\frac{m}{\gamma}} \right) n e^{65\sqrt{\log(en)}} \\ &\times \exp \left[-\frac{\sqrt{d}}{294912\beta^{32L}} \left(\frac{q}{(Ym/\gamma)^2 \log(en)^{d/2}} \right)^{1/(d+1)} \right]. \end{aligned} \quad (102)$$

The numerical coefficient is conservative. The exact recurrence certificate is usually substantially smaller.

Here is an exact scalar inverse in the headline domain $q \geq 18B_*$, including loose targets and the full-width fallback. For a target $\varepsilon > 0$, if $\varepsilon \geq \mathcal{E}_0$, choose $q = \min(n, 18B_*)$. Otherwise put

$$u_\varepsilon = \max\{0, \log(\mathcal{E}_1/\varepsilon)\}, \quad q = \min \left\{ n, \max \left\{ 18B_*, \lceil 9[B_* + \mathcal{C}_d(h_0 + u_\varepsilon)^{d+1}] \rceil \right\} \right\}. \quad (103)$$

If the minimum chooses n , use exact retention. Otherwise use the analytic construction with $T = T_0 + 4mu_\varepsilon/\gamma$ and $\eta = \eta_0 e^{-u_\varepsilon}$. A sharper integer prescription replaces the ceiling in this formula by $9[B + 4N(T, \eta)]$ when $d \geq 2$, or $9[B + 8N_1(T, \eta)]$ when $d = 1$. These prescriptions invert explicit sufficient certificates. They do not assert optimal compression over other models or constructions. If one uses the sharper full construction domain $q \geq 9B$, the loose-target branch may instead use $\min(n, 9B)$, and the maximum with $18B_*$ may be omitted in the analytic branch. These are explicitly different domain choices. A selected full-width model uses width n even when it is below the headline baseline.

For the original source tolerance $1/n$ and horizon T_0 , replace 64 by 32 in $\mathcal{B}_n$ and consequently in $\mathcal{A}_n$; call the resulting explicit coefficient $\mathcal{A}_n^{(0)}$. Then

$$\|f_C - f_n\|_* \leq \frac{\mathcal{A}_n^{(0)}}{n} + \mathcal{D} e^{-8\log(en)}. \quad (104)$$

It needs the original source domain, without the analytic-extension gates. For $d \geq 2$ its source rank is at most B plus

$$\frac{2^{20} 9^d}{d!} \frac{U}{a} c_q^{-(d-1)} (Ym/\gamma)^2 \log(en)^{3d/2+1},$$

for $d = 1$, replace this expression by $8 \cdot 514 \cdot 1024(U/a)(Ym/\gamma)^2 \log(en)^{5/2}$. Substitution in the retained inventory preserves the original $\log(en)^{3d+2}$ storage order, including the exact initialization term. On the optional smaller cap $Y \leq (\gamma/m)\beta^{-30L}$, the same original construction also retains the numerical refinements

$$\|f_C - f_n\|_* \leq 10\beta^{40L} \frac{Ym}{\gamma} \left(1 + \sqrt{\frac{m}{\gamma}} \right) \frac{e^{2\sqrt{\log(en)}}}{n} \leq 250\beta^{40L} Y(1 + m/\gamma)^2 n^{-1/2}. \quad (105)$$

D.14.5 Learned coordinates, fixed storage and gates

For the actual selected dimensions, the exact moving-state count is

$$dq_1 + \sum_{j=2}^L q_j q_{j-1} + q_L + m \leq (L-1)q^2 + q(d+1) + m. \quad (106)$$

The effective readout is computed from this state and is not a second independent learned vector. The fixed symmetric metrics require $\sum_j q_j(q_j+1)/2$ real entries if packed; retaining diagonal comparison metrics uses another $\sum_j q_j$. Data cost $m(d+1)$. Optional metric inverses, initial copies, training feature/backward arrays and current Gram/solve caches are included in the following explicit upper inventory for an integer source-dimension bound R :

$$\text{storage} \leq 1020(L+1)R^2 + 10m(d+1). \quad (107)$$

This is an upper bound on the named inventory, not an exact equality for every possible cache layout. In the analytic branches one can take $R = B + 4N(T, \eta)$ or $R = B + 8N_1(T, \eta)$, respectively. Whenever the actual integer rank is at most $q/9$, the same inventory is at most $13(L+1)q^2 + 10m(d+1)$. Original-width coefficients, quadrature arrays, jets, bases and initialized dense arrays are absent from retained storage. Counts concern real coordinates, not bit precision, setup work, linear-solve work or a numerical ODE solver's chosen work space.

The width qualifications consist of the shared dense initialization and source probability event, the original deterministic source gates, and the following explicit additions. In particular retain $n^{-1} \leq \eta_0$, the source gate

$$\sqrt{\log(en)} \geq c_t \max\{8, \lambda, 4K/\log 2, 32YSD_W\}, \quad D_W = \max\{\tau_1^{\text{src}}, \max_{j \geq 2} \tau_j^{\text{src}} H_{j-1}^{\text{src}}\}. \quad (108)$$

Since $c_t = a\gamma/(1024UmY^2)$, even the first entry in this maximum requires

$$n \geq \exp\left\{\left(\frac{a\gamma}{128UmY^2}\right)^2 - 1\right\}.$$

This is a sufficient-width qualification of the certificate, not a necessary width for compression. In particular, its label-dependent storage coefficient is not a bound uniform as $Y \downarrow 0$. The following dimension gates likewise remain part of the certificate.

For $d \geq 2$, write

$$C_d^* = 16 \cdot 128 \cdot 18M_0 D_d b_d! 2^{b_d+1} \lambda^{-1} c_t^{-1} c_q^{-(b_d+1)}, \quad M_0 = M_n/\sqrt{n},$$

and retain

$$\log C_d^* \leq \log(en), \quad (d+1) \log \log(en) \leq \log(en), \quad (d-1)c_q/\sqrt{\log(en)} \leq 1, \quad \alpha_{T_0} \leq 1. \quad (109)$$

For $d = 1$, retain the temporal gate $\alpha_{T_0} \leq 1$ and use

$$\log \frac{8192M_0}{c_t \lambda} \leq \log(en). \quad (110)$$

Finally put

$$b_{\text{ext}} = \min \left\{ \frac{1}{4}, \frac{SH_L^{\text{src}}}{4}, \frac{a}{8\sqrt{n} \max_j P_j^{\text{src}}} \right\},$$

$$C_L^k = 1, \quad C_j^k = S\tau_{j+1}^{\text{src}} + 10sC_{j+1}^k + 10t_2 SP_{j+1}^{\text{src}} k_{j+1}^{\text{src}}, \quad C_{\text{carrier}} = \max_j C_j^k,$$

and impose

$$\left(2Y/\sqrt{\lambda} + 8Y\sqrt{K}r_t\right)(en)^{-16} < b_{\text{ext}}/2, \quad \frac{2C_{\text{carrier}}Y}{\sqrt{\lambda}} n(en)^{-16} \leq K_{\text{src}} S \sqrt{\log(en)}. \quad (111)$$

These inequalities are eventual at every fixed admissible problem. They are independent of q , T , and target accuracy. Thus one event for a common dense initialization at a fixed width supports all the Harmonic budgets and tolerances above; there is no union over new horizon-dependent Gaussian events. The implicit sampler gives the same probability guarantee for each prescribed execution, not a shared success event for independently restarted samplers.

D.15 Seeded decoder compression: explicit finite theorem

The Seeded decoder uses the same dense architecture, Gaussian initialization, zero readout, mean-square loss and mobilities as the preceding sections. Its additional data hypotheses are $m \geq d$ and $\text{span}\{x_1, \dots, x_m\} = \mathbb{R}^d$. It retains the full intersection of the fitting and analytic-source label allowances; it adds no smaller upper label cap. The branch $Y = 0$ is the exact zero predictor.

The method has three finite phases.

1. **Initialization.** Generate a virtual finite source independently of the dense reference, inspect its completed finite training computation, select row packets and a positive metric reproducing its required empirical contractions, and retain short seeds, selected packets, current scalar state, scalar-noise marks and finite instructions. Full row tables and future-answer tables are discarded. Setup is therefore offline but does not retain a trained dense oracle.
2. **Training.** Evolve the selected packets and scalar state using the retained metric. Each already created selected field and its exact metric product are cached. Creation-time scalar arguments are immutable.
3. **Querying.** Regenerate the source rows from the retained seeds, stream the empirical contractions for the new input, and apply the complete two-orientation conditional covariance correction. A median over complete source models gives one answer. Querying does not rerun scalar training.

No test input or test label is supplied during initialization or training. The construction produces a fresh joint law for the compressed model and an independent width- n dense reference. It is not a deterministic conversion of a supplied dense realization.

For $0 < \delta < 1/4$, define p, Z, p_{ref} by (32)–(33). The clean finite-word branch requires $nY \geq 1$. With a fixed absolute numerical allocation, it is sufficient that

$$n \geq \max \left\{ C_* \left[\frac{2^{20} \beta^{2000L} (1 + m/\gamma)^4 (m + d + p_{\text{ref}} + 1)^4}{\delta} \right]^{1100}, Y^{-1}, C_{\text{num}} \right\}. \quad (112)$$

Here

$$C_{\text{num}} = \max\{1, (A_{\text{num}}/32)^{1/9}\},$$

where A_{num} is the absolute coefficient assigned to the total Yn^{-10} numerical remainder. A sharper source-only alternative sets

$$\rho = 2^{-20} \delta, \quad \mathcal{A} = \beta^{2000L} (1 + m/\gamma)^4 (m + d + p_{\text{ref}} + 1)^4$$

and requires

$$n \geq \max \left\{ [2\mathcal{A}(2000 \log(2\mathcal{A}))^{16}]^{1000}, \frac{64mL}{\rho} \right\}. \quad (113)$$

When this sharper gate is used, retain separately

$$n \geq d, \quad n \geq \max\{1, e^{-1} \sqrt{1 + 66\beta^{100L} m/\gamma}\}, \quad n \geq \max\{1, (\beta^{23L}/8)^{1/15}\}, \quad (114)$$

$$n \geq C_0 \left[\frac{p_{\text{ref}} \beta^{201L} (m + d + 2)(1 + m/\gamma)}{\rho} \right]^2, \quad n \geq \max\{1, (A_{\text{num}}/32)^{1/9}\}. \quad (115)$$

All constants in these gates are universal or are the explicitly chosen numerical allocation. No stochastic eventual-width condition remains. The initialization input for this finite-width assertion is the exponential-concentration gate (IC.58)–(IC.61), proved in the finite source section. It replaces the older sufficient N_{fit} for this assertion; no exponential sphere-net cardinality is hidden in the power-1100 gate. The all-time independent-reference bridge is (IC.88)–(IC.90).

For the dense certificate, use the dense fitting quantities already defined above: the hidden-size envelope H_D , the derivative envelope F_D , the strip constant s , and $\lambda = \gamma/m$. One valid explicit mesh coefficient is

$$c_{2,\text{mesh}} = \frac{32H_D^2}{\lambda} + \frac{32F_D Y^2}{\lambda^2} + \frac{4(9s)^L}{\sqrt{\lambda}}, \quad 32 \leq c_{2,\text{mesh}} \leq C\beta^{4L}(1 + m/\gamma). \quad (116)$$

With the inherited positive fixed-problem leading coefficients c_0, c_1 , let B_n be (35). The additional absolute certificate is

$$\mathbb{P} \left\{ \sup_{t \in [0, \infty]} \sup_{\|x\|_2 = \sqrt{d}} |f_{\text{Seed},n}(t, x) - f_n^{\text{ind}}(t, x)| \leq 3B_n \right\} \geq 1 - \delta. \quad (117)$$

The event includes unseen and adaptively chosen inputs and the fitted endpoint. In addition, without using this analytic envelope, the same finite construction has error at most $2b_n(\delta/32) + A_{\text{num}}Yn^{-10}$ at confidence $1 - \delta$. The definition and full proof are in the intrinsic comparison. The original explicit gates suffice for the additive statement. The eventual pure factor-three conclusion for $m \geq 2, Y > 0$ inherits the lower theorem’s unquantified onset; it is not a new explicit-width claim.

The retained model uses at most a universal multiple of

$$M_{\text{Seed}} = p^2 \beta^{402L} (m + d + 2)^2 (1 + m/\gamma)^2 Z^6 [d + 1 + \log(e + (d + 1)Z)] \quad (118)$$

numerical words of sufficient length

$$w = \lceil C_{\text{word}} \beta^{110L} Z \rceil \quad \text{bits.} \quad (119)$$

This retained inventory includes every ensemble member, both seed levels, selected packets, metrics, caches, scalar state and the normalized training table. The same envelope covers the live query workspace. Some exact setup integers occupy several such words.

Up to universal multiplicative constants, the complete internal costs are

$$\begin{aligned} & \text{phase : initialization (word operations)} \\ & n(d+1)p^5 \beta^{1115L} (m+d+2)^5 (1+m/\gamma)^5 Z^{29/2} \\ & \text{phase : initialization (peak words)} \\ & M_{\text{Seed}} + np \beta^{201L} (m+d+2)(1+m/\gamma) Z^{5/2} \\ & \quad + p^3 \beta^{603L} (m+d+2)^3 (1+m/\gamma)^3 Z^{15/2} \\ & \text{phase : complete training (word operations)} \\ & (d+1)p^4 \beta^{914L} (m+d+2)^4 (1+m/\gamma)^4 Z^{12} \\ & \times \log(e + p \beta^{201L} (m+d+2)(1+m/\gamma) Z^{5/2}) \\ & \text{phase : complete training (peak words)} \\ & M_{\text{Seed}} \\ & \text{phase : one query (word operations)} \\ & (d+1)p^4 \beta^{914L} (m+d+2)^4 (1+m/\gamma)^4 [(L+1)nZ^{12} + Z^{14}] \\ & \text{phase : one query (peak words)} \\ & M_{\text{Seed}}. \end{aligned} \quad (120)$$

Sufficient activation-value call counts for those three rows are, respectively,

$$\begin{aligned}
N_{\phi,\text{init}} &\leq Cn(d+1)p\beta^{201L}(m+d+2)(1+m/\gamma)Z^{7/2}, \\
N_{\phi,\text{train}} &\leq C(d+1)p^2\beta^{402L}(m+d+2)^2(1+m/\gamma)^2Z^6, \\
N_{\phi,\text{query}} &\leq C(d+1)(L+1)[np\beta^{201L}(m+d+2)(1+m/\gamma)Z^{7/2} \\
&\quad + p^2\beta^{402L}(m+d+2)^2(1+m/\gamma)^2Z^6].
\end{aligned} \tag{121}$$

For tanh, the supplied finite evaluator is included. For a general analytic activation, multiply these counts by its actual finite-word work per call and add one evaluator’s workspace. Also add external data, certificate, query/time access and output work. If their retained descriptions require B_{ext} bits beyond the counted finite table, add $\lceil B_{\text{ext}}/w \rceil$ retained and peak words. Raw-label access requires at least

$$w + \lceil \log_2(16\sqrt{m}) \rceil + \lceil \log_2 \max(1, 1/Y) \rceil \tag{122}$$

fractional bits. Analytic regularity alone does not price these interfaces.

For $0 < nY < 1$, put

$$Z_Y = Z + \log_+ \frac{1}{nY}.$$

The explicit small-label recipe (FC.61)–(FC.62) in the finite construction enlarges the panel and Taylor orders as well as the precision, field, code and cost counts. Its constructed field/call count R_Y satisfies

$$R_Y \leq Cp\beta^{201L}(m+d+2)(1+m/\gamma)Z_Y^{5/2}, \quad n \geq 2\log(2^{20}R_Y).$$

The second inequality is the explicit enlarged finite Gaussian-RMS gate. Use Z_Y throughout the numerical storage/work envelopes in this branch. Changing word precision alone is not sufficient. There is no new lower label condition in the source probability theorem; the clean headline branch remains $nY \geq 1$, which is eventual for every fixed positive label scale.

D.16 Computational costs at supplied orders

This section distinguishes three operations. **Warmup** is all one-off work needed to create the model, including generation of its dense reference when not supplied. **Training** is one full-batch evaluation of the autonomous vector field, or a fixed-stage explicit numerical step, after the warmup-only arrays have been discarded. **Query** is one new passive input at a fixed, inference-ready model state. Warmup and training memory mean **total peak resident memory**; query memory means **additional peak workspace**, excluding the resident model and the supplied input. These definitions apply to every cost table. Warmup may compute across the whole finite source horizon. The efficient local schemes bound that one-off work; they do not assert that a physically short initial prefix alone determines the desired coefficients.

D.16.1 Arithmetic, activation and numerical qualifications

The bounds count classical dense scalar arithmetic, with scalar square root and the fixed elementary functions used in the continuation and geometric recurrences treated as unit primitives. They are upper bounds for the specified implementations, not optimal complexity or wall-clock measurements. In the full supplied-order cost envelopes, every big-O constant is numerical and implementation-dependent only: it hides no dependence on $n, m, d, L, q, K, p, J, N_x, N_t, \gamma, Y, \beta, \delta$, activation evaluation cost, solver stage count or working precision. The explicitly labelled fixed-parameter asymptotic specializations instead allow constants depending on those fixed structural parameters.

General activations are not unit-cost oracles by implication of analyticity. The tables count non-activation arithmetic; add the explicit activation calls and jet-composition costs stated below. Materializing a fresh dense reference requires exactly $nd + (L-1)n^2$ independent Gaussian draws. The corresponding arithmetic bounds include allocation/filling, but the sampling implementation’s time and scratch memory must be added. In the customary unit-cost sampling model this adds $O((L-1)n^2 + nd)$ work and $O(1)$ sampler scratch. If the reference is supplied, omit generation work but not its resident memory. The implicit Harmonic initializer

instead uses at most $nd + (L - 1)nk_*$ draws, with the full request count k_* defined in its own supplied-order interface. It neither accepts a prescribed entrywise realization for free nor stores a hidden dense array.

A numerical step with s vector-field stages has s times the per-stage work, plus state updates and any cache refresh. Only when s is fixed can it be absorbed in big-O; a fixed-stage method uses a fixed number of state-sized work arrays. Implicit solves, adaptive rejections, the number of subsequent training steps, roundoff and their numerical-integration accuracy are not bounded by the runtime table. The local setup algorithms separately have explicit finite orders and proved numerical defect bounds. In particular the trajectory-error theorems concern the continuous flows; this section does not promote them to finite-step error theorems. Feature-Gram solves are evaluated where the stated positive-gap conditions hold. Exact rank detection and exact arithmetic in coordinate selection are part of the operation model, not floating-point stability assertions.

For the clean common table take $n \geq \max(m, d)$, allowed by the shared eventual-width convention. Retain the existing Harmonic compressed domain $q \geq 18(2m + d + 9)$ when simplifying its runtime counts. No comparison between n and any internal setup order is assumed. Define, locally throughout this cost section,

$$P = (L - 1)n^2 + n(d + 1). \quad (123)$$

D.16.2 Common forward cost table: supplied n, q

The explicit-local and implicit-local Harmonic warmup envelopes $\mathcal{T}_{\text{HD}}, \mathcal{M}_{\text{HD}}$ and $\mathcal{T}_{\text{HI}}, \mathcal{M}_{\text{HI}}$ are defined below, retaining every supplied setup order and separate primitive costs. No inverse or optimized order is used in this table. The original initialization-jet envelopes $\mathcal{T}_{\text{H}}, \mathcal{M}_{\text{H}}$ are a separately costed alternative, not a lower bound on either new initializer.

Detailed comparison

Model and operation: Dense warmup

Arithmetic work $O(P)$, plus Gaussian draws

Peak memory in real words $O(P + m(d + 1))$

Model and operation: Dense training stage

Arithmetic work $O(mP)$

Peak memory in real words $O(P + Lmn + m(d + 1))$

Model and operation: Dense single query

Arithmetic work $O(P)$

Peak memory in real words $O(n)$ additional

Model and operation: Legendre warmup

Arithmetic work $O(P + mnd + (L - 2)mn^2 + (L - 1)mn + Lmnq)$, plus Gaussian draws

Peak memory in real words $O(P + Lmnq + m(d + 1))$

Model and operation: Legendre training stage

Arithmetic work $O(mP + Lnm^2q + Lnmq)$

Peak memory in real words $O(P + Lnmq + m(d + 1))$

Model and operation: Legendre single query, factor representation

Arithmetic work $O(P + Lnmq)$

Peak memory in real words $O(n)$ additional

Model and operation: Harmonic warmup, explicit dense local continuation

Arithmetic work $O(\mathcal{T}_{\text{HD}})$, plus specified primitive calls

Peak memory in real words $O(\mathcal{M}_{\text{HD}})$

Model and operation: Harmonic warmup, implicit Gaussian local continuation

Detailed comparison

Arithmetic work $O(\mathcal{T}_{\text{HI}})$, plus specified primitive calls
Peak memory in real words $O(\mathcal{M}_{\text{HI}})$

Model and operation: Harmonic training stage, including inference-cache refresh when required

Arithmetic work $O(Lmq^2 + m^3)$
Peak memory in real words $O(Lq^2)$

Model and operation: Harmonic single query, refreshed readout cache

Arithmetic work $O(Lq^2)$
Peak memory in real words $O(q)$ additional

Dense and Legendre training add at most a fixed numerical multiple of Lmn calls each to activation values and first derivatives. Their queries add Ln activation-value calls and no activation-derivative calls. Legendre warmup adds $(L - 1)mn$ activation-value calls; it need only compute initial features through layer $L - 1$, so at $L = 2$ the displayed warmup feature calculation contains no dense-mixer application. Harmonic training adds $m \sum_j q_j \leq Lmq$ calls each to values and first derivatives per vector-field evaluation, and a refresh adds a comparable number of value calls. Its query adds $\sum_j q_j \leq Lq$ value calls. Add the chosen routines' work and peak scratch for these calls; analyticity alone gives no uniform arithmetic bound on those routines.

Legendre applies its fixed mixers plus the mq correction factors without materializing reconstructed dense matrices. Moment-prefix sums are accumulated once, not recomputed for every prefix. Factor products can be streamed, avoiding an m^2q -word temporary. An optional fixed-snapshot cache of reconstructed dense mixers costs $O((L - 1)n^2mq)$ work to refresh and $O((L - 1)n^2)$ additional resident memory; afterward queries cost $O(P)$, with $O(n)$ additional workspace. Refresh whenever moments or the clock change. It is not one-off initialization if training continues.

The Harmonic inference-ready cache is the vector $M_L \hat{w}_C$, so a query is an ordinary hidden forward pass and a final dot product. Refresh it after any relevant model-state change. A refresh uses the current training features and feature-Gram solve and has at most the order of one training-stage cost. An implementation can charge a final refresh to training/model preparation; it cannot both omit refresh and claim an m -independent query cost. At initialization the cache is exactly zero.

D.16.3 Unsimplified Harmonic runtime and peak workspace

For the actual layer widths $q_j \leq q$, the training arithmetic is

$$O\left(m \left[q_1 d + \sum_{j=2}^L q_j q_{j-1} + \sum_{j=1}^L q_j^2 \right] + q_L m^2 + m^3\right). \quad (124)$$

Its total peak memory after discarding warmup-only arrays is

$$O\left(q_1 d + \sum_{j=2}^L q_j q_{j-1} + \sum_{j=1}^L q_j^2 + m \sum_{j=1}^L q_j + m^2 + m(d + 1)\right). \quad (125)$$

This includes model parameters, fixed metrics and inverse caches, data, feature/backward arrays, parameter directions and the moving feature-Gram factorization. It does not include any dense-reference array, jet, coefficient block or source basis. The m^3 term is the ordinary factorization of the current m -by- m feature Gram; it is not a hidden repeated inversion of the fixed layer metrics. Although $m^3 \leq mq^2$ in the compressed domain, it remains visible.

For a single inference-ready query, arithmetic and additional peak workspace are respectively

$$O\left(q_1 d + \sum_{j=2}^L q_j q_{j-1} + q_L\right), \quad O\left(\max_j q_j\right). \quad (126)$$

Use rolling feature buffers; normalize the first preactivation after multiplication by the supplied input, avoiding a separate normalized input copy. Fixed metrics are applied to batch vectors before forming weight directions. Their inverse caches are created during warmup, so there is no Lq^3 per-stage cost. The cost proof gives an exact matrix-free formula for the update-Gram action; it is not differentiation of the corrected predictor.

D.16.4 Efficient Harmonic setup: one construction, two executions

Keep the full common label interval, source event, width gates and original activations. Given an analytic-branch budget $q < n$, use exactly its previously prescribed source horizon $T \geq T_0$, tolerance $0 < \eta \leq \eta_0$, weighted mode set and exact initialized additions. The order recipe below supplies a finite numerical construction satisfying that source tolerance. Its resulting model obeys the unchanged certificate

$$\|f_H - f_n\|_* \leq \mathcal{A}_n \eta + \mathcal{D} e^{-\gamma T/(4m)}. \quad (127)$$

For the original $T = T_0, \eta = 1/n$ specialization, the available sharper $\mathcal{A}_n^{(0)}$ certificate remains valid. The coordinate-selection, metric, initial weights and corrected-readout optimizer are the ones already defined in the Harmonic runtime.

The **explicit dense local initializer** constructs the original reference arrays, or accepts their supplied realization, and advances computed anchors across $[0, T]$ by certified restarted Taylor panels. A temporary scalar polynomial approximates each activation during setup. The **implicit Gaussian local initializer** runs this same finite algorithm using exact adaptive forward/transpose Gaussian actions and factored anchor increments instead of materialized hidden matrices. Its output and latent dense reference have the same joint law as the explicit local execution, not merely the same marginal prediction law. No extra error or failure probability is introduced.

The implicit input contract is a freshly sampled reference. A supplied hidden weight array or prescribed entrywise generator seed does not inherit its near-linear conversion cost. Both executions use the actual training samples and labels for the gradient; designed spatial nodes are passive queries. They start the final model at original time zero with original activations. The temporary activation polynomials, numerical trajectory, source arrays, Gaussian transcript and increment factors are all discarded. No additional runtime approximation parameter or dense-reference access remains.

These statements concern finite exact-real setup with an explicitly controlled numerical defect. They do not require an oracle for an observed trained path, complex activation values or high activation derivatives. Population gaps, moments used to evaluate the original label allowance, and valid activation bounds are supplied analytic certificates; their determination for an arbitrary function description is not silently free computation.

Supplied orders and complete finite operation counts

All symbols in this subsection are local execution quantities, not additional retained model orders. They are inputs to the cost formulas; the subsequent recipe gives sufficient accuracy-certified choices when desired.

Symbol	Meaning
$J \geq 1$	Number of local time panels
$K \geq 1$	Taylor degree on each panel
$D \geq 1$	Temporary scalar activation-polynomial degree
p, ℓ_*	Largest temporal Chebyshev and spherical degrees
N_t, N_x	Temporal and spatial quadrature node counts
H_{sph}, N	Spatial basis size and retained joint time–sphere mode count

Symbol	Meaning
$R = 2m + d + 1 + 4N$	Per-layer source generator bound
$r \leq \min(n, R)$	Largest actual source-space rank

For $d \geq 2$, $H_{\text{sph}} = \sum_{\ell=0}^{\ell_*} h_\ell$, with the previously defined harmonic multiplicities; N uses the actual weighted simplex, not a rectangular enlargement. For $d = 1$, use $N_x = H_{\text{sph}} = 2$, $N = 2(p + 1) = 2N_1$, with no angular degree. The sufficient original budget has $9R \leq q$; actual rank can permit $9r \leq q$. Selection retains at most $9r$ coordinates per layer, without padding unused budget. In all costs one may use the actual largest selected support, or the larger supplied q . Arbitrary incompatible choices of orders and budget do not assert successful compression.

Recall $P = (L - 1)n^2 + n(d + 1)$. Define one request-count abbreviation:

$$k_* = m + 2J(m + N_x)(K + 1) + 2N + r. \quad (128)$$

This includes exact original initialization, all local forward/transpose actions, initialized images of completed coefficients, and final basis images. The Gaussian law theorem is invoked with the deterministic cap replacing r by $\min(n, R)$; its pathwise costs admit the smaller actual count above. Neither an initialized-matrix norm computation nor a good-event acceptance test is executed.

The following scalar geometry bounds specify one matched implementation: regenerate spatial values on each panel. With polar order N_θ and azimuthal order N_φ , use $N_x = N_\varphi N_\theta^{d-2}$ for $d \geq 3$, and $N_x = N_\varphi$ for $d = 2$. Put

$$G_{\text{time}} = \begin{cases} N_t + J, & d = 1, \\ N_t + N_x + JN_x(\ell_* + 1), & d = 2, \\ N_\theta^2 + dN_\theta + N_t + N_\varphi + d + d(\ell_* + 1)^2 + JdN_x[1 + (\ell_* + 1)^2 + H_{\text{sph}}], & d \geq 3, \end{cases}$$

$$G_{\text{memory}} = \begin{cases} N_t + 1, & d = 1, \\ N_t + N_x + \ell_* + 1, & d = 2, \\ dN_\theta + N_t + N_\varphi + d[1 + (\ell_* + 1)^2 + H_{\text{sph}}], & d \geq 3. \end{cases} \quad (129)$$

They include quadrature weights, node generation and separated harmonic recurrences. They are not uncharged geometric oracles.

For the explicit dense execution, define the sufficient arithmetic envelope

$$\begin{aligned} \mathcal{T}_{\text{HD}} = & P + L(D + 1)^2 + mP + nm^2 + m^3 + L + d + m + 1 \\ & + JP(m + N_x)(K + 1) + JLn(m + N_x)(K + 1)^2(n + K + 1) \\ & + JLn(m + N_x)(D + 1)(K + 1)^2 + (N_t + J)(p + 1)(K + 1) \\ & + LnJ(K + 1)(N_x H_{\text{sph}} + N) + Ln^2 N \\ & + LnRr + Ln r^3 + Ln^2 r + Lq^2 r + G_{\text{time}} + N_t \log(2 + N_t), \end{aligned} \quad (130)$$

and the matched peak real-word envelope

$$\begin{aligned} \mathcal{M}_{\text{HD}} = & P + Lmn(D + 1)(K + 1) + Lm^2(K + 1)^2 + LnR + Lq^2 + Ln(K + 1)H_{\text{sph}} \\ & + q(d + Lm + 1) + m^2 + L(D + 1) + (p + 1)(K + 1) + N_t + J + m(d + 1) \\ & + G_{\text{memory}} + L + d + m + 1. \end{aligned} \quad (131)$$

The optional exact initial training-Gram/solve-cache preparation is charged by $nm^2 + m^3$ and m^2 , rather than left implicit in final assembly. The source-image and final-basis terms $Ln^2 N, Ln^2 r$ charge the initialized matrices separately from the moving anchor matrices.

For the implicit execution, define

$$\begin{aligned}
\mathcal{T}_{\text{HI}} = & nd(1+m) + m^2d + nm^2 + m^3 + L + d + m + 1 + L(D+1)^2 \\
& + (L-1)nk_*^2 + (L-1)nm(m+N_x)J(J-1)K(K+1) \\
& + J[nd(m+N_x) + dmN_x] + JLn(m+N_x)(K+1)^2(n+K+1) \\
& + JLn(m+N_x)(D+1)(K+1)^2 + (N_t+J)(p+1)(K+1) \\
& + LnJ(K+1)(N_xH_{\text{sph}} + N) + LnRr + Lnr^3 + (L-1)nr^2 \\
& + L(qr^2 + r^3 + q^2r) + qd + G_{\text{time}} + N_t \log(2+N_t),
\end{aligned} \tag{132}$$

$$\begin{aligned}
\mathcal{M}_{\text{HI}} = & nd + (L-1)n[k_* + JmK] + Lmn(D+1)(K+1) \\
& + Lm^2(K+1)^2 + LnR + Ln(K+1)H_{\text{sph}} + Lq^2 \\
& + q(d+Lm+1) + m^2 + m(d+1) + L(D+1) + (p+1)(K+1) \\
& + N_t + J + G_{\text{memory}} + L + d + m + 1.
\end{aligned} \tag{133}$$

The $J(J-1)$ work term charges repeated application of all preceding update factors, not just their creation. Their peak storage is charged by $(L-1)nJmK$; the Gaussian direction bases and answers by $(L-1)nk_*$. No full hidden matrix or projector is formed.

Actual non-activation arithmetic and peak words are bounded by numerical, implementation-only constants times the respective envelopes. The bounds include one explicit n -by- d first matrix, data, source projection, rank-aware orthogonalization, conservative deterministic selection, final metrics/inverses and model caches. Gaussian draws and original activation value calls are separate:

	Explicit local	Implicit local	
standard Gaussian draws	$nd + (L-1)n^2$	$\leq nd + (L-1)nk_*$	(134)
original activation values	$Ln m + 4L(D+1)$	$Ln m + 4L(D+1)$	

At most L extra value calls compute $\phi_j(0)$ if needed. The original initialized features use exact values in the existing real-word model; the backend's permissible inexact value tolerance is specified below. There is no derivative-generation or trained-reference oracle.

For elementary functions, let the angular table size be zero at $d = 1$, N_x at $d = 2$, and $N_\theta + N_\varphi + d(\ell_* + 1)^2$ at $d \geq 3$. A sufficient call count is a numerical constant times

$$L + d + m + 1 + L(D+1) + N_t + \text{angular table size} + Lr,$$

with $(L-1)k_*$ more for the implicit sampler. These include the normalizing square roots; scalar arithmetic and comparisons are already charged above. Sum the actual primitive-call costs, or multiply each count by a valid per-call bound, and add each routine's simultaneous scratch to peak memory. Arbitrary analytic activations do not imply bounded-cost evaluators.

The source spaces and actual selected widths are identical under the coupling between the two executions. Their final learned-state bound remains $(L-1)q^2 + q(d+1) + m$, and the established sufficient all-retained inventory remains $1020(L+1)R^2 + 10m(d+1)$. The same runtime training, cache-refresh and single-query costs apply to either initializer.

Width specialization and practical interpretation

At $T = 32(m/\gamma) \log(en)$, $\eta = 1/n$, and separately fixed admissible parameters, the complete recipe yields

$$\begin{aligned}
J, D = & O(\log(en)^{3/2}), \quad K = O(\log(en)), \quad p+1, N_t = O(\log(en)^{5/2}), \\
N_x, H_{\text{sph}} = & O(\log(en)^{3(d-1)/2}), \quad N, R, r, q, k_* = O(\log(en)^{3d/2+1}).
\end{aligned} \tag{135}$$

Use the two-point convention at $d = 1$; for $d \geq 2$, $\ell_* = O(\log(en)^{3/2})$. These are sufficient orders, not optimality claims. Substitution proves the near-quadratic and near-linear bounds, including all temporary state. For the latter, the conservative Lnr^3 selector bounds the remaining width-proportional work. The full finite envelopes, not these specializations, apply to arbitrarily enlarged horizons, orders or accuracies.

The same setup exponents hold at the exact absolute target Y/n . Use (23); the bound (24) gives $u = O(\log(en))$, $T = O(\log(en))$, and $\log(1/\eta) = O(\log(en))$ at fixed positive problem parameters. Substitution into the full supplied-order certificates gives the same source cutoffs, quadratures, nodal tolerance and continuation orders, as calculated in Appendix B. This is distinct from the baseline $n^{-1+o(1)}$ error specialization.

An ordinary full-batch dense Euler rollout at physical step $0 < h \leq T$ has work $\Theta(mPT/h)$ under the same primitive convention. For the displayed horizon, the explicit initializer's work divided by this benchmark is $O(h \log(en)^{3d/2})$ at fixed parameters. This tends to zero for every fixed inverse-polynomial h ; it neither proves such an Euler step necessary nor certifies its accuracy. The setup may be slower than a dense solver using the same high-order integrator but doing no compression assembly. The implicit representation can also accelerate other algorithms.

Neither asymptotic result establishes the faster initializer at widths 1000–10000. The practical expectation, not a measured finding, is that well-batched explicit dense evolution can win when its arrays fit comfortably in memory; implicit execution is more attractive when its accumulated query spans and stored update factors stay far below n . A small final q does not itself ensure small setup history. Projection and selection are shared costs. No timing benchmark or finite-precision conditioner is proved here.

D.16.5 Deterministic orders for both efficient initializers

The following finite recipe supplies all orders used by the explicit and implicit local-continuation algorithms. It reuses the exact source and fitting coefficients already defined in this document. Every new radius, tolerance, degree and panel count is displayed here; their error and operation arguments are in the complete initializer proofs. No norm of an unknown trained path is an input. Confidence affects the inherited admissible-width qualification, whose source portion is still not numerically quantified.

The proof components use shorter locally scoped names. In this prescription $R_{\text{loc}}, J_{\text{loc}}, V_{\text{loc}}$ mean their local-continuation R, J, V ; $U_j^{\text{act}}, K_j^{\text{act}}, Q_j^{\text{act}}, C_F^{\text{act}}$ mean the activation component's U_j, K_j, Q_j, C_F ; and $B_{\text{act}}, \Delta_{\text{act}}$ mean its ellipse scale B and gap Δ . Quadrature names $a_t, h_{\text{ang}}, \sigma_{\text{ang}}, \ell_*, \mathcal{Y}$ mean its $a, h, \sigma, \ell_*, \mathcal{Y}_{\ell_*}$. These are exact renamings, with no rescaling of coefficients or changes to the inequalities.

Inputs and conventions

Supply integers $n, m, d \geq 1$, hidden depth $L \geq 2$, normalized data $\|x_a\|_2 = \sqrt{d}$, the layer activations, positive gap γ , label RMS $Y = \|y\|_2/\sqrt{m}$, confidence $0 < \delta < 1$, source horizon T , and source-coordinate tolerance η . Put

$$\lambda = \gamma/m, \quad S = 16Y/\lambda, \quad T_0 = 32\lambda^{-1} \log(en), \quad \eta_0 = \min(1, Y, S).$$

The nontrivial branch has $Y > 0$, $T \geq T_0$ and $0 < \eta \leq \eta_0$. For $Y = 0$, retain the exact zero-predictor branch and do not evaluate formulas dividing by Y . All logarithms are natural except $\log_2$. Impossible binomial coefficients are zero.

The activations are real on the real axis and holomorphic on $|\text{Im } z| < a$, with bounded first derivative on that full open strip, as in the original activation hypothesis. Use the exact bounds or certified upper bounds

$$b = \max_j |\phi_j(0)|, \quad s = \max\{1, \sup_{j, |\text{Im } z| \leq a/2} |\phi_j'(z)|\}, \quad t_2 = \max\{1, \sup_{j, |\text{Im } z| \leq a/2} |\phi_j''(z)|\},$$

$$\beta = \max\{10, 1 + b, 16/a, s, t_2\}.$$

The second-derivative bound is needed by the proof even though setup does not ask for second-derivative values. A bound on ϕ' only on the same closed half-strip is insufficient to infer that bound on its boundary. If a bound s_{full} on ϕ' on the full strip is supplied, one may use $t_2 = \max(1, 2s_{\text{full}}/a)$, by Cauchy. Alternatively a supplied valid β provides $t_2 \leq \beta$.

Using looser bounds may reduce a sufficient label allowance. To preserve the full original allowance for a given activation, evaluate the original recurrences with its stated original bounds rather than silently replacing them by worst-case β values.

The notation below distinguishes source coefficients by superscript src , the real stability coefficients by r , and the Legendre/runtime coefficients by their own labels. Unsuperscripted $H_j, P_j, B_j, G, J_{\text{loc}}$ introduced later belong only to the local complex continuation. These are exact translations of the component components' local notation.

Existing source constants and exact notation correspondence

Compute the source coefficients from the complete recurrences (S.5)–(S.10), (S.22), (S.24)–(S.25), and (S.30) in the source foundations, in their displayed order. Those recurrences already define every activation-only coefficient, the source allowance S_*^{src} , K_{src} , and the actual-label query radii r_t, r_q ; no source recurrence is duplicated here.

The notation correspondence is exact. Source-section $H_j, P_j, k_j, f_j, q_j, \tau_j$ become $H_j^{\text{src}}, P_j^{\text{src}}, k_j^{\text{src}}, f_j^{\text{src}}, q_j^{\text{src}}, \tau_j^{\text{src}}$. Write $H_{\text{max}}^{\text{src}} = \max_j H_j^{\text{src}}$, $k_*^{\text{src}} = \max_j k_j^{\text{src}}$, $f_*^{\text{src}} = \max_j f_j^{\text{src}}$, and $\tau_*^{\text{src}} = \max_j \tau_j^{\text{src}}$. The V of (S.25) is denoted V_{qry} here; $U, T_Q, T_J, G_d, K_{\text{src}}, c_t, c_q, \mathcal{K}$ retain their source meanings. The $S = 16Ym/\gamma$ appearing in those recurrences is the actual activity allowance, not its upper bound one. The a in c_t, c_q is the activation strip width. The source-budget tolerance in (S.10) is distinct from the requested approximation error η .

Full original label interval

Use the common exact label allowance, or the full Harmonic allowance when only the Harmonic theorem is needed. With $H_D = H_d$ and $F_D = F_d$ from (79), it is

$$0 < Y \leq \lambda \min\{(8H_D\sqrt{F_D})^{-1}, S_*^{\text{Leg}}/8, (16H_c\sqrt{F_c})^{-1}, S_*^{\text{src}}/16\}.$$

The Legendre entry is omitted for Harmonic alone. The actual activation Gaussian moments defining H_D are those of the dense fitting theorem; they are supplied certificates when an exact allowance is evaluated. The setup prescription adds no label restriction and does not replace this interval by the optional sufficient cap $Y \leq \lambda\beta^{-30L}$.

Width, confidence and inherited gates

Require $n \geq \max(m, d)$, the explicit dense initialization threshold $N_{\text{fit}}(\delta/32)$ from dense fitting, and the same source event at failure allowance at most $\delta/32$. The source threshold remains existential. The union bound needs no independence; the lower-bound and central-limit events are not used here. Retain every original radius, counting and analytic-extension gate in the Harmonic storage-and-gate statement, including the dimension-one gate and $n^{-1} \leq \eta_0$. These are explicit scalar comparisons already written there. They do not test sampled matrix norms.

For use below, define the late-state displacement bound from the all-horizon extension,

$$Z_n = (2Y/\sqrt{\lambda} + 8Y\sqrt{\mathcal{K}}r_t)(en)^{-16}.$$

The extension requires $Z_n < b_{\text{ext}}/2$, with the extension radius defined in the construction gates. It is distinct from the numerical restart radius r_{tube} defined below. The second carrier gate is also retained. The source amplitude and dimension coefficients used for the exact mode count are

$$M_{\text{amp}} = 10 \max(H_L^{\text{src}}, \tau_*^{\text{src}}), \quad b_d = 2d - 2, \quad D_d = 2^{d+1}d^{d-2} \quad (d \geq 2).$$

No deterministic gate supplies the missing numerical confidence-to-width threshold of the source event.

Retained modes, rank and selected budget

Put $M_n = M_{\text{amp}}\sqrt{n}$ and $\alpha_T = r_t/(4T)$. For $d \geq 2$, define

$$P_T = \frac{18D_db_d!2^{b_d+1}}{\alpha_T r_q^{b_d+1}}, \quad H(T, \eta) = 2 \log \frac{16M_n P_T}{\eta},$$

$$p = \left\lfloor \frac{H(T, \eta)}{\alpha_T} \right\rfloor, \quad \ell_* = \left\lfloor \frac{H(T, \eta)}{r_q} \right\rfloor, \quad h_\ell = \binom{\ell + d - 1}{d - 1} - \binom{\ell + d - 3}{d - 1},$$

$$p_\ell = \left\lfloor \frac{H(T, \eta) - r_q \ell}{\alpha_T} \right\rfloor, \quad N = \sum_{\ell=0}^{\ell_*} (p_\ell + 1) h_\ell, \quad H_{\text{sph}} = \sum_{\ell=0}^{\ell_*} h_\ell, \quad \mathcal{Y} = \max_{\ell \leq \ell_*} \sqrt{h_\ell}.$$

The retained set contains precisely $0 \leq k \leq p_\ell$ and all h_ℓ harmonics at degree ℓ . Set $R = 2m + d + 1 + 4N$. For $d = 1$, instead use

$$H_1(T, \eta) = \log \frac{64M_n}{\alpha_T \eta}, \quad N_1 = 1 + \lfloor H_1(T, \eta) / \alpha_T \rfloor, \quad p = N_1 - 1,$$

$$N = 2N_1, \quad H_{\text{sph}} = 2, \quad N_x = 2, \quad R = 2m + d + 1 + 8N_1.$$

Here N counts both point families; the quadrature accuracy formulas below use the per-point N_1 . There is no spherical degree at $d = 1$.

A deterministic sufficient supplied budget is $Q = \min(n, 9R)$. If $9R \geq n$, use the original exact full-width fallback, rather than assert compression. If the headline convention $Q \geq 18(2m + d + 9)$ is desired, use $Q = \min(n, \max\{9R, 18(2m + d + 9)\})$. The actual rank $r = \max_j \dim E_j \leq \min(n, R)$ may permit a smaller budget $9r$; that is computed after coefficient assembly and is not an order-only prediction. No preservation of accidental rank deficiencies is assumed. In the selected branch, the actual supports satisfy $q_j \leq 9r_j$, and the actual maximum $q = \max_j q_j$ is the quantity charged in the setup costs. Unused supplied budget Q is not padded.

Positive quadrature orders

For $d \geq 2$, define $I_0 = \pi$, $I_1 = 2$, $I_b = (b - 1)I_{b-2}/b$ for $b \geq 2$, and

$$A_d = \prod_{b=1}^{d-2} \frac{\pi}{I_b}, \quad D_d^{\text{ang}} = (d - 2)(d - 1)/2, \quad h_{\text{ang}} = \frac{r_q}{2(d - 1)}, \quad \sigma_{\text{ang}} = \text{arsinh}(2h_{\text{ang}}/\pi).$$

Empty products equal one. Put $a_t = \alpha_T/2$ and

$$\epsilon_c = \frac{\eta}{16N\mathcal{Y}}, \quad \delta_{\text{node}} = \frac{\eta}{64NA_d\mathcal{Y}^2},$$

$$B_{\text{int}} = 2M_n\mathcal{Y}A_d \exp[a_t p + \ell_*(d - 1)h_{\text{ang}}] (\cosh h_{\text{ang}})^{D_d^{\text{ang}}}.$$

Then take

$$N_t = \max\{1, \lceil a_t^{-1} \log(1 + 4dB_{\text{int}}/\epsilon_c) \rceil\},$$

$$N_\varphi = \max\{1, \lceil h_{\text{ang}}^{-1} \log(1 + 4dB_{\text{int}}/\epsilon_c) \rceil\}.$$

For $d \geq 3$, take

$$N_\theta = \max\left\{1, \left\lceil \sigma_{\text{ang}}^{-1} \log \frac{8dB_{\text{int}}}{\epsilon_c(1 - e^{-\sigma_{\text{ang}}})} \right\rceil \right\}, \quad N_x = N_\varphi N_\theta^{d-2}.$$

For $d = 2$, use $N_x = N_\varphi$ and no polar rule. For $d = 1$, use

$$\epsilon_c = \frac{\eta}{16N_1}, \quad \delta_{\text{node}} = \frac{\eta}{64N_1}, \quad B_{\text{int}} = 2M_n e^{a_t p}, \quad N_t = \max\{1, \lceil a_t^{-1} \log(1 + 4B_{\text{int}}/\epsilon_c) \rceil\}.$$

These are the original positive periodic/Fejér orders. No equalities $N_t = p + 1$ or $N_x = H_{\text{sph}}$ are used.

Complex local continuation constants

Define the real-stability coefficients first:

$$H_1^r = \max(1, b + 10s), \quad H_j^r = \max(1, b + 10sH_{j-1}^r), \quad H^r = H_L^r,$$

$$R^r = 1 + 2Y/\sqrt{\lambda}, \quad B^r = R^r s(10s)^{L-1}, \quad F_z^r = H^r(10s)^{L-1}, \quad P_{\text{layer}}^r = 1 + (L-1)H^r,$$

$$D^r(M) = (10s)^{L-1}[s(1 + B^r) + Lt_2 F_z^r M],$$

$$J^r(M) = \sqrt{L+1}[P_{\text{layer}}^r D^r(M) + (1 + (L-1)B^r)sF_z^r],$$

$$C^r(M) = sF_z^r \sqrt{L} + LB^r sF_z^r + \frac{1}{2}L^2 t_2 (F_z^r)^2 M.$$

Evaluate J^r, C^r at $M = 2K_{\text{src}}S\sqrt{\log(en)}$, then set

$$A_n = 4J^r Y/\lambda, \quad K_{\text{qry}}^r = R^r(10s)^{L-1},$$

$$C_{\text{src}} = 8\sqrt{L+1} \max \{ sF_z^r, (10s)^{L-1}[s(1 + B^r) + Lt_2 F_z^r K_{\text{qry}}^r] \}.$$

The different local complex coefficients are

$$R_{\text{loc}} = 1 + SH_L^{\text{src}}, \quad H_1 = \max(1, b + 11s), \quad H_j = \max(1, b + 11sH_{j-1}),$$

$$B_j = R_{\text{loc}} s(11s)^{L-j}, \quad P_1 = 1, \quad P_j = H_{j-1} + 11sP_{j-1}, \quad P_* = \max_j P_j,$$

$$\kappa_L(M) = 1,$$

$$\kappa_j(M) = 11s\kappa_{j+1}(M) + 11t_2 MP_{j+1} + B_{j+1},$$

$$D_j(M) = s\kappa_j(M) + t_2 MP_j,$$

$$\kappa_*(M) = \max_j \kappa_j(M),$$

$$G = H_L + B_1 + \sum_{j=2}^L B_j H_{j-1}, \quad J_{\text{loc}}(M) = sP_L + D_1(M) + \sum_{j=2}^L [H_{j-1} D_j(M) + B_j sP_{j-1}].$$

Put

$$M_0 = K_{\text{src}}S\sqrt{\log(en)}, \quad M_c = M_0 + \sqrt{n}\kappa_*(M_0)Z_n,$$

$$r_{\text{tube}} = \min\{1/4, a/(16\sqrt{n}P_*), 1/(\sqrt{n}\kappa_*(M_c))\},$$

$$L_n = 2G^2 + 2(2Y + G)J_{\text{loc}}(M_c + 1), \quad V_{\text{loc}} = 2G(2Y + G), \quad \widehat{R}_n = \min\{r_t/2, (8L_n)^{-1}\}.$$

The symbols $r_{\text{tube}}, V_{\text{loc}}$ here are the backend's local tube radius and velocity coefficient; they are not b_{ext} or V_{qry} .

One activation polynomial and complete local degree

For the strong source-jet interface use $\delta_* = \delta_{\text{node}}/(32\sqrt{n})$, the source-jet bridge's tightened backend nodal target. Define

$$e_d = e^{-A_n} \min \left\{ \frac{1}{4}, \frac{1}{\sqrt{8}(C^r + J^r)\sqrt{T}}, \frac{\delta_*}{4C_{\text{src}}n}, \frac{r_{\text{tube}}}{16} \right\},$$

$$\zeta_F = \min\{e_d/(2T), r_{\text{tube}}L_n/4, V_{\text{loc}}/2\}.$$

For the activation substitution put

$$U_0^{\text{act}} = 0, \quad U_j^{\text{act}} = 1 + 11sU_{j-1}^{\text{act}}, \quad U_*^{\text{act}} = \max_j U_j^{\text{act}}, \quad K_j^{\text{act}} = R_{\text{loc}}(11s)^{L-j},$$

$$Q_{L+1}^{\text{act}}(M) = 0, \quad Q_j^{\text{act}}(M) = 11(s+1)Q_{j+1}^{\text{act}}(M) + K_j^{\text{act}} + 11t_2MU_{j-1}^{\text{act}}.$$

Evaluate the following training constants at $M = M_c + 1$:

$$Q_\xi = U_L^{\text{act}} + Q_1^{\text{act}}(M) + \sum_{j=2}^L [Q_j^{\text{act}}(M)(H_{j-1} + 1) + B_j U_{j-1}^{\text{act}}], \quad Q_f = R_{\text{loc}} U_L^{\text{act}},$$

$$C_F^{\text{act}} = 2Q_f(G+1) + 2(2Y+G)Q_\xi.$$

For passive sources use

$$M_q = \sqrt{n} \max_j K_j^{\text{act}}, \quad C_p = 8\sqrt{n} \max\{U_*^{\text{act}}, \max_j Q_j^{\text{act}}(M_q)\},$$

$$\epsilon = \min \left\{ 1, (U_*^{\text{act}})^{-1}, \frac{a}{5632\sqrt{n}U_*^{\text{act}}}, Q_\xi^{-1}, \frac{\zeta_F}{C_F^{\text{act}}}, \frac{\delta_*}{2C_p} \right\}.$$

The activation ellipse scale and degree are

$$Z_* = (K_{\text{src}} + G_d H_{\text{max}}^{\text{src}} + 1 + US^2)\sqrt{\log(en)} + a/8 + \sqrt{n}P_*Z_n, \quad B_{\text{act}} = 8(Z_* + a + 1),$$

$$\tau_o = \text{arsinh} \frac{31a}{64B_{\text{act}}}, \quad \tau_i = \text{arsinh} \frac{29a}{64B_{\text{act}}}, \quad \Delta_{\text{act}} = \tau_o - \tau_i, \quad M_\phi = b + s(B_{\text{act}} + a),$$

$$C_a = \max\{1, 512/a, 2(512/a)^2\}, \quad \epsilon_p = \epsilon/C_a,$$

$$D = \max \left\{ 1, \left\lceil \Delta_{\text{act}}^{-1} \log \max\{e, 24M_\phi/(\epsilon_p \Delta_{\text{act}})\} \right\rceil \right\}, \quad N_\phi = 4(D+1).$$

Each layer's polynomial uses real activation values at $B_{\text{act}} \cos(2\pi r/N_\phi)$. If those values are inexact, the sufficient individual accuracy is

$$\epsilon_{\text{eval}} = \frac{\epsilon_p}{4(D+1)e^{D\tau_i}}.$$

Exact initialized source additions remain original-activation evaluations under the inherited exact-real convention. They are not replaced by these polynomial values.

Finally set $C_g = \max_j \{H_j + 1, B_j + 1\}$ and choose

$$K = \max \left\{ 8, \left\lceil 2 \log_2 \max\{1, 16TV_{\text{loc}}/e_d\} \right\rceil, \left\lceil \log_2 \max\{1, 8V_{\text{loc}}\hat{R}_n/r_{\text{tube}}\} \right\rceil, \right. \\ \left. \left\lceil \log_2 \max \left\{ 1, \frac{128V_{\text{loc}}\hat{R}_n C_{\text{src}} n^{3/2}}{\delta_{\text{node}}} \right\} \right\rceil, \left\lceil \log_2 \max \left\{ 1, \frac{64\sqrt{n}C_g}{\delta_{\text{node}}} \right\} \right\rceil \right\}.$$

Take exactly

$$J = \left\lceil \frac{2T}{\widehat{R}_n} \right\rceil = \lceil \max\{4T/r_t, 16TL_n\} \rceil$$

equal physical-time panels, and use degree K for both parameter and passive source jets. The first three cutoffs are backend (Setup-A.22); the final two are the source-jet bridge's source-jet cutoffs. This is an explicit sufficient choice, not a claim of minimal degree.

What is and is not expressed solely through beta

There is no suppressed structural parameter in the preceding recipe. Substituting inputs successively produces all $J, K, D, p, \ell_*, N_t, N_x, N, R, Q$. The dataset enters through m, d , its actual label size Y , its population gap γ , and, for exact qualification, its actual activation moments and source event.

The source power ledger provides useful bounds, already before its optional small-label specialization:

$$H_j^{\text{src}} \leq 3\beta^{2j}, \quad k_j^{\text{src}} \leq 3\beta^{4L-2j}, \quad \tau_j^{\text{src}} \leq 3\beta^{4L-2j+1}, \quad P_j^{\text{src}} \leq \beta^{3j-2}, \quad f_j^{\text{src}} \leq \beta^{3j-1},$$

$$K_{\text{src}} \leq \beta^{20L+2}, \quad T_Q \leq \beta^{14L-3}, \quad T_J \leq \beta^{8L-1}, \quad S_*^{\text{src}} \geq \beta^{-26L}.$$

The full label interval implies $S \leq 1$. Using this fact and the displayed source formulas gives conservative full-interval envelopes

$$U \leq \beta^{40L} \sqrt{d+3}, \quad V_{\text{qry}} \leq \beta^{36L} \sqrt{d+3},$$

$$c_t^{-1} \leq 4YS\beta^{40L+1} \sqrt{d+3}, \quad c_q^{-1} \leq \max\{8, \tfrac{1}{2}\beta^{36L+1} \sqrt{d+3}\}.$$

For example, the largest term inside the U_j bracket is bounded using $T_Q \leq \beta^{14L-3}$; multiplication by $sK_{\text{src}} \leq \beta^{20L+3}$ is well below the generous β^{40L} envelope. The $G_d q_{j-1}^{\text{src}}$ term is smaller. Likewise $sK_{\text{src}} T_J$ is below β^{28L+2} , leaving ample room for the second envelope. The radius bounds then use $a^{-1} \leq \beta/16$. These bounds describe dependence; replacing exact mode definitions by looser bounds may enlarge a sufficient budget and is not an assertion of unchanged minimal counts.

On the additional smaller cap $Y/\lambda \leq \beta^{-30L}$, the source ledger gives the sharper $U \leq \beta^{26L} \sqrt{d+3}$ and $V_{\text{qry}} \leq \beta^{2L} \sqrt{d+3}$. They are not used for the full interval recipe.

Beta alone cannot recover the actual Gaussian moments H_D , the data gap γ , the full original label interval, the scalar activation value algorithm, or the effective stochastic width threshold. Coarse bounds can replace some of these numerical coefficients conservatively, but cannot be represented as exact recovery of them.

The formulas display powers exponential in depth, factorial/binomial and tensor-product dependence on dimension, and possible exponential factors in stability constants. They do not assert polynomial dependence jointly on $m, d, L, \beta, \gamma^{-1}, Y^{-1}, \delta^{-1}$. In particular, $N_x = N_\varphi N_\theta^{d-2}$ is not a dimension-free polynomial bound. The qualitative source-width threshold prevents a complete numerical confidence-to-work theorem, even though the conditional deterministic orders at a supplied admissible width are explicit.

Confidence δ does not otherwise enter the order formulas at fixed admissible n ; the same source event covers every permitted T, η . No new tolerance-dependent probability event, label restriction, or unknown trained-state norm has been inserted.

D.16.6 Original initialization-jet warmup with all internal orders exposed

This subsection costs the original construction using only time-zero jets. It is an alternative to the two later local-continuation initializers, not an obligation imposed on those algorithms.

The independent supplied setup resolutions are:

Symbol	Supplied meaning
$K \geq 1$	Highest initial time-derivative order computed

Symbol	Supplied meaning
$p \geq 0$	Highest retained temporal Chebyshev degree
$\ell_* \geq 0$	Highest retained spherical-harmonic degree
$N_x \geq 1$	Total number of spatial quadrature nodes
$N_t \geq 1$	Number of temporal quadrature nodes

The source horizon T , continuation map, quadrature nodes/weights and retained joint mode set must also be specified. The default costed rule is the elementary angular Riemann rule used in the source proof; the additional construction and node/weight storage of a different rule must be charged separately. These are not additional runtime state after initialization. On the analytic branch of the forward theorem, the supplied q prescribes a horizon and source tolerance; a valid implementation must choose the setup resolutions to meet that tolerance. This cost analysis does not identify K, p, ℓ_*, N_x, N_t with q , or choose any of them. In particular $N_x = H$, $N_t = p + 1$, and $p \leq K$ are not asserted: the last relation need not hold after continuation in the nonlinear time coordinate.

For $d \geq 2$, put

$$h_j = \binom{j+d-1}{d-1} - \binom{j+d-3}{d-1}, \quad H = \sum_{j=0}^{\ell_*} h_j = \binom{\ell_*+d-1}{d-1} + \binom{\ell_*+d-2}{d-1}, \quad (136)$$

with impossible binomials zero. If the temporal cutoff at spherical degree j is $p_j \leq p$, count all h_j basis functions there. A sufficient source-generator count per layer is

$$R = 2m + d + 1 + 4 \sum_{j=0}^{\ell_*} (p_j + 1) h_j \leq 2m + d + 1 + 4(p + 1)H. \quad (137)$$

Use the actual smaller number of present source families at boundary layers when available. For $d = 1$, there are two input points, no nontrivial angular degree, and the corresponding bound is $R = 2m + d + 1 + 8(p + 1)$; one can use $H = 2$ in the rectangular cost bounds. The actual largest source rank is $r \leq \min(n, R)$. The deterministic selector guarantees the requested budget when $9r \leq q < n$; $9R \leq q$ is a stronger sufficient test, not a necessary rank condition. No successful compressed construction is asserted for an arbitrary incompatible combination of supplied orders.

The following counts assume all initial dense matrices are resident, training jets are shared, and spatial nodes are streamed. Here and below using K instead of $K + 1$ in big-O is valid because $K \geq 1$.

Detailed comparison

Jet representation: Materialized dense parameter coefficients

Arithmetic, excluding	$O(P(m + N_x)K^2)$
scalar activation	
composition	
Jet-stage peak memory	$O(PK + LmnK)$

Jet representation: Factored gradient increments, no contraction cache

Arithmetic, excluding	$O(P(m + N_x)K + Lnm(m + N_x)K^3)$
scalar activation	
composition	
Jet-stage peak memory	$O(P + LmnK)$

Jet representation: Factored increments with contraction caches

Arithmetic, excluding	$O(P(m + N_x)K + Lm(m + N_x)K^2(n + K))$
scalar activation	
composition	

Detailed comparison

Jet-stage peak memory $O(P + LmnK + Lm^2K^2)$

These are alternative executions of the same coefficient recursion. Choose an implementation for the available memory; no row is claimed uniformly fastest. No parameter-Hessian tensor is materialized. The N_x query jets reuse the actual empirical-training jets; designed query points do not replace training samples in the gradients. This particular execution uses derivatives at time zero and is data- and label-dependent. The local-continuation alternatives instead use computed anchors across the source horizon; their gradients use the same training data.

For arbitrary activations, define the following **implementation costs**, not approximation parameters. Let $a_{\text{on}}(K), b_{\text{on}}(K)$ bound arithmetic and peak workspace for one scalar activation and its derivative in the sequential training-jet recursion, including persistent series state and derivative-generation scratch; let $a_{\text{off}}(K), b_{\text{off}}(K)$ bound full-series composition and scratch for a passive query. Each includes the chosen activation's scalar-derivative generation algorithm. Then add

$$\mathcal{A}_{\text{time}} = Ln[ma_{\text{on}}(K) + N_x a_{\text{off}}(K)], \quad \mathcal{A}_{\text{memory}} = Lmnb_{\text{on}}(K) + b_{\text{off}}(K). \quad (138)$$

Take maxima over the finitely many layer activations if their implementations differ. A fixed-size differential recurrence, such as for tanh, permits $a_{\text{on}}, a_{\text{off}} = O(K^2)$ and $b_{\text{on}}, b_{\text{off}} = O(K)$. For a general analytic activation with its scalar Taylor coefficients supplied, a direct online power triangle costs $O(K^3)$ time and $O(K^2)$ persistent words; offline truncated Horner composition costs $O(K^3)$ time and $O(K)$ scratch. Scalar derivative generation is additional to these latter composition bounds. The theorem's analytic envelope does not bound that algorithmic cost.

Scalar geometry can be implemented without expanding all harmonics in Cartesian monomials. The recursive separated orthonormal basis in the cost proof gives the following sufficient overhead envelopes. They include the source proof's elementary Riemann nodes/weights, generated from supplied grid indices in $O(d)$ work per spatial node and $O(1)$ per temporal node. They do not bound construction of an arbitrary quadrature rule:

$$\mathcal{B}_{\text{time}} = \begin{cases} N_t + N_x, & d = 1, \\ N_x(\ell_* + 1) + N_t, & d = 2, \\ d(\ell_* + 1)^2 + dN_x[(\ell_* + 1)^2 + H] + N_t, & d \geq 3, \end{cases} \quad \mathcal{B}_{\text{memory}} = \begin{cases} 1, & d = 1, \\ \ell_* + 1, & d = 2, \\ d[(\ell_* + 1)^2 + H], & d \geq 3. \end{cases} \quad (139)$$

The $d = 1$ case uses the two sphere points, not an angular grid. The geometry envelopes are retained explicitly rather than hidden inside an activation or data constant.

For cached factored jets and temporal-first streamed projection, define the complete arithmetic and peak-memory envelopes

$$\begin{aligned} \mathcal{T}_H &= P + P(m + N_x)K + Lm(m + N_x)K^2(n + K) \\ &\quad + (p + 1)K(N_t + K) + LnN_x(p + 1)(K + H) \\ &\quad + LnRr + Lnr^3 + Ln^2r + Lq^2r + \mathcal{A}_{\text{time}} + \mathcal{B}_{\text{time}}, \end{aligned} \quad (140)$$

$$\begin{aligned} \mathcal{M}_H &= P + LmnK + Lm^2K^2 + LnR + Lq^2 + (p + 1)K + m(d + 1) \\ &\quad + \mathcal{A}_{\text{memory}} + \mathcal{B}_{\text{memory}}. \end{aligned} \quad (141)$$

All quantities on the right have been defined. The actual work and peak words are $O(\mathcal{T}_H)$ and $O(\mathcal{M}_H)$, plus the separately specified Gaussian sampling cost/scratch. These envelopes include creation of the dense reference, source construction, selection, model assembly and fixed metric inverse caches. They are not bounds solely in q .

The universal temporal map costs $O((p + 1)K(N_t + K))$ once, not once per neuron. Projection then costs $O(LnN_x(p + 1)(K + H))$. Orthogonalization must process all R generators, costing $O(LnRr)$; the proof's deterministic selector costs $O(Lnr^3)$. Dense basis-to-basis mixer compression costs $O(Ln^2r)$ in general even

though some paired actions can be reused. Metric and retained-matrix assembly costs $O(Lq^2r)$. The LnR memory term retains all source-coefficient blocks and therefore introduces no unreported query recomputation.

Alternative executions. Replace the cached-jet contributions by either other row of the jet table when advantageous. Spatial-first projection costs $O(LnK[N_xH + (p+1)H])$, with $O(LnKH)$ intermediate words instead of temporal-first projection. Coefficient blocks of size b can replace LnR storage by $Ln(r+b)$; without stored nodal jets this requires up to $\lceil R/b \rceil$ passes over the passive-query jet and temporal projection calculations, together with streamed node/basis evaluation unless separately cached, not repeated training-jet construction. On a uniform circle grid supporting the retained modes, FFT angular projection takes $O(Ln(p+1)N_x \log(eN_x))$ work after temporal projection; a direct batched implementation additionally stores $O(Ln(p+1)N_x)$ nodal values. These are explicit time-memory tradeoffs, not unproved simultaneous minima.

The initialization-only baseline branch can skip source expansions that it does not use. The full-width exact branch can simply retain the dense network and use the dense costs. The formulas above concern the genuine analytic compressed branch and are upper bounds, not necessary work for every special dataset.

D.16.7 What an inverse budget does and does not determine

Insert the inverse interface’s prescribed n or q into the forward cost table; no other substitution is licensed automatically. For Harmonic, the setup orders must additionally satisfy the time/spatial-tail, initial-jet and quadrature accuracy conditions in the source construction, and the resulting source rank must fit the budget. The origin-jet argument proves finiteness, not an efficient joint order choice. The local-continuation alternatives below supply a complete deterministic recipe and certified work/peak-memory bounds. Their finite formulas apply to the actual inverse horizon/tolerance. The baseline specialization proves the simpler width and epsilon exponents; (23)–(24) also establish them at the exact Y/n target. No near-linear or near-quadratic cost is inferred merely by renaming the origin-jet order as q .

Nor do arithmetic bounds certify finite-precision execution. Continuation coefficients, source rank tests and small positive selection weights can require additional precision. No sufficient bit precision, subsequent runtime integration step count, or end-to-end wall-clock efficiency follows from the retained-storage theorem. The new analysis changes no activation class, label cap, probability event or trajectory-error certificate.

E Complete proofs and constructions

The proofs are grouped by dependency, with every specialized construction used above included here. The dense initialization/fitting argument uses only finite-dimensional Gaussian estimates and deterministic energy. The source proof adds finite deletion, conditional Gaussian comparison, moment-budget removal and complex continuation. The later dense upper and lower proofs, and both compressed-model proofs, use that shared source.

The dependency order is not circular: dense fitting does not use source analyticity; the source starts from dense fitting; the deterministic late-time extension uses the source only through its original finite horizon. The Legendre comparison uses that extension’s real carrier bound, and Harmonic’s variable-horizon construction uses its full analytic domain. The complete extension is in the Harmonic proof and its explicit gates are in the common construction inventory.

E.1 Source foundations: explicit coefficients and the probabilistic bridge

The coefficients in this section are local proof coefficients, not extra model parameters or adjustable approximation orders. The scalar $\lambda = \gamma/m$ is never capped at one. The activation bounds are $b = \max_j |\phi_j(0)|$, $s = \max\{1, \sup_{j, |\operatorname{Im} z| \leq a/2} |\phi'_j(z)|\}$, and $t_2 = \max\{1, \sup_{j, |\operatorname{Im} z| \leq a/2} |\phi''_j(z)|\}$. They obey $1 + b, s, t_2, 16/a \leq \beta$. In numbered form the definitions are

$$b = \max_j |\phi_j(0)|, \quad s = \max\{1, \sup_{j, |\operatorname{Im} z| \leq a/2} |\phi'_j(z)|\}, \quad t_2 = \max\{1, \sup_{j, |\operatorname{Im} z| \leq a/2} |\phi''_j(z)|\}. \quad (\text{S.2})$$

There is an unavoidable distinction between an explicit error coefficient and an effective probability threshold. The original source argument proves probability tending to one by sending width to infinity at each fixed empirical moment degree and only then taking the infimum over those degrees. It does **not**

specify a numerical sufficient width as a function of confidence for that entire complex-sphere source event. The later finite training-source proof gives a separate explicit width for the Seeded decoder; it must not be substituted for the stronger query-tube event. The deterministic source coefficients, label allowance, radii and additional width inequalities below are explicit. The finite-deletion argument uses eventual domination of fixed logarithmic factors; it is the source of the remaining qualitative width quantifier.

E.1.1 Finite deletion and its local comparison

The local finite-deletion comparison below is proved on independently stopped complex query domains, for each fixed deletion count, with an eventual qualitative width onset. Its source-event closure is stated in Lemma E.1 and proved in §E.1.2. The separate training-only specialization in §E.5.1 has the explicit power-1100 width gate; it does not quantify the complex query-tube event. Throughout, retain the complete common label allowance, including (S.10). No restriction depending on deletion count or confidence is added.

1. Coordinates and the exact retained equation

Write $\ell = \log(en)$, $\lambda = \gamma/m$, and $S = 16Y/\lambda$. For $Y = 0$, the readout and all parameter velocities vanish and the predictor is zero; the argument with barred variables below is only for $Y > 0$. Use Euclidean/Frobenius norm on the direct sum of the mobility coordinates

$$\Theta = (A, H^{(2)}, \dots, H^{(L)}, w), \quad H^{(j)} = \sqrt{n}W^{(j)}.$$

Here $H^{(j)}$ is a rescaled matrix; the scalar RMS bound remains H_j . Write $v_a = x_a/\sqrt{d}$ for a normalized training input and $F_a = nf(t, x_a) = w^\top h_a^{(L)}$ for its unnormalized prediction. Delete activations in one layer, keeping normalization n in every rectangular matrix. For a deleted interior neuron i , its initialized incoming row transpose y_i and outgoing column x_i have law $N(0, I/n)$ and are independent of the entire retained initialization and of the other omitted root pairs. At the first layer only x_i enters the retained forcing; the initialized incoming row has law $N(0, I_d)$ and is used separately to evaluate its own preactivation. At the top there is no x_i .

Add forward ports $e_a^{(j)}$ to preactivations and set

$$\bar{u} = (\Theta - \Theta_0)/S, \quad \bar{F}_a = F_a(\Theta_0 + S\bar{u}, e_a)/S, \quad \bar{r}_a = \bar{F}_a/n - y_a/S, \quad \bar{k} = k/S, \quad \bar{\delta} = \delta/S. \quad (\text{IC.1})$$

Zero initial readout gives $\bar{F}_a = \bar{u}_w^\top h_a^{(L)}$. The scaled backward recursion and gradient blocks are

$$\begin{aligned} \bar{k}_a^{(L)} &= \bar{u}_w, \quad \bar{\delta}_a^{(j)} = \phi'_j(z_a^{(j)}) \odot \bar{k}_a^{(j)}, \quad \bar{k}_a^{(j)} = W^{(j+1)\top} \bar{\delta}_a^{(j+1)}, \\ \nabla_{\bar{u}_A} \bar{F}_a &= S \bar{\delta}_a^{(1)} v_a^\top, \quad \nabla_{\bar{u}_{H^{(j)}}} \bar{F}_a = \frac{S}{\sqrt{n}} \bar{\delta}_a^{(j)} h_a^{(j-1)\top}, \quad \nabla_{\bar{u}_w} \bar{F}_a = h_a^{(L)}. \end{aligned} \quad (\text{IC.2})$$

For deletion at layer j , the actual ports are

$$e_a = \sum_{i \in I} W_{:,i}^{(j+1)} h_{a,i}^{(j)}, \quad \bar{q}_a = \sum_{i \in I} W_{i,:}^{(j)\top} \bar{\delta}_{a,i}^{(j)}.$$

Differentiating through the deleted activation accounts for exactly the second summand in the following equation:

$$\dot{\bar{u}} = -\frac{2}{m} \sum_a \bar{r}_a \left[\nabla_{\bar{u}} \bar{F}_a + D_{\bar{u}} h_a^{(j-1)\top} \bar{q}_a \right]. \quad (\text{IC.3})$$

In particular $D_{\bar{u}} h = S D_{\Theta} h$. The reverse source does not enter the forward residual. At the top the additional scalar $\bar{d}_a = n^{-1} \sum_{i \in I} \bar{u}_{w,i} h_{a,i}^{(L)}$ is added to $\bar{r}_a$ and multiplies both terms in the bracket.

Each reference is the zero-source cavity with its own initialization test, real stopping time and complex stopping domain. Failed own initializations give identically zero coefficient paths. This convention is fixed

before any omitted root is integrated. On a successful stopped reference, the physical RMS bounds are (S.5)–(S.6), its budgets are at most $2\mathcal{B}$, its operator caps are ten, and

$$2 \int \bar{\rho} |dt| \leq 1, \quad \bar{\rho} = \left(m^{-1} \sum_a |\bar{r}_a|^2\right)^{1/2}, \quad \bar{\rho} \leq \lambda/8, \quad M_n = \eta^{-1} \log(2n\mathcal{B}). \quad (\text{IC.4})$$

For the original qualitative source, the short-contour conditions in (S.31) give (IC.4). For the quantitative training source, they will be proved directly in the quantitative training-source ledger at a smaller radius.

The independent complex stops and their extensions are defined as follows. For a fixed target rectangle $K = [-r_t, T + r_t] + i[-r_t, r_t]$, use the nested closed convex rectangles $K_s = sK$, $0 \leq s \leq 1$. Each cavity starts from its own initial germ and has its own first-exit level s_c : all its running budgets, maxima, pole and response stops are suprema on K_s . In the qualitative passive-query construction, take those suprema on K_s times the intrinsic query tube of thickness sr_q , retaining the full real sphere at every level. Thus the first level has only real initial queries. This is the same target domain and the same stops at $s = 1$, with a precise nested-prefix convention. The real fitting trajectory is already defined independently for all positive time. Before a stop, finite-width parameter bounds and the strict separation from the activation singularities permit local holomorphic ODE continuation. Compactness and uniqueness glue these extensions. At a first stopped level the coefficients have continuous values on its closed rectangle; the activation strip still has strict slack, so the same local continuation justifies the one-sided derivative bounds there. A failed own initialization is treated separately by the identically zero convention.

Let P_c be Euclidean projection onto the fixed rectangle K_{s_c} , explicitly clamping its real and imaginary coordinates to the two closed intervals. It is 1-Lipschitz, fixes this rectangle, and is measurable in that cavity's retained initialization only. Extend each of its coefficient paths b_c to the whole target rectangle by $\tilde{b}_c(z) = b_c(P_c z)$. A derivative bound on the convex rectangle gives the corresponding Lipschitz bound by integration on a segment, and composition with P_c preserves that bound. For passive-query grids also clamp the intrinsic imaginary-circle coordinate to $[-s_c r_q, s_c r_q]$; the real frame ranges over the full frame manifold. These extensions need not be holomorphic outside their own stopped domains. Holomorphic identities are used only on the common successful prefix, where the projections are identity.

In particular define the real-anchor map $a(z) = \min\{T, \max\{0, \text{Re } z\}\}$. The correct complex-minus-real coefficient is

$$b_c(P_c z) - b_c(P_c a(z)) = b_c(P_c z) - b_c(a(P_c z)). \quad (\text{IC.stop})$$

The two projections commute because both real intervals contain zero and the anchor interval is $[0, T]$. Both arguments lie in K_{s_c} , their distance is at most $|z - a(z)| \leq 2r_t$, and both argument maps are 1-Lipschitz. Thus the small-radius and global modulus bounds for this difference follow from the derivative bounds on the closed stopped rectangle. Its real anchors trace a monotone interval from zero to $\min\{s_c(T + r_t), T\}$, then freeze. The original real activity modulus and moment proof therefore apply to that real coefficient path. For two different cavities, extend each with its own projection before subtracting; the sum of their Lipschitz constants bounds the difference. On a successful full prefix, strict cavity-stop transfer proves s_c is at least its level, so these extended paths equal the actual paths there. This construction supplies the globally defined coefficient processes used in all later conditional Gaussian integrals.

2. Derivative maps, control moduli, and terminal interpolation

We give recurrences rather than a dimension-dependent estimate for the parameter Hessian. In augmented mobility/port coordinates put $Z_a^{(j)} U = D z_a^{(j)} [U]$. Then

$$Z_a^{(1)} U = U_A v_a + U_{e^{(1)}}, \quad Z_a^{(j)} U = \frac{U_{H^{(j)}}}{\sqrt{n}} h_a^{(j-1)} + W^{(j)} \text{diag}(\phi'_{j-1}) Z_a^{(j-1)} U + U_{e^{(j)}}. \quad (\text{IC.5})$$

The coefficients P_j, f_j in (S.6) bound these maps and their activated versions. The exact Hessian is

$$\begin{aligned}
D^2 F_a[U, V] &= U_w^\top \text{diag}(\phi'_L) Z_a^{(L)} V + V_w^\top \text{diag}(\phi'_L) Z_a^{(L)} U \\
&\quad + \sum_j (Z_a^{(j)} U)^\top \text{diag}(k_a^{(j)} \phi_j'') Z_a^{(j)} V \\
&\quad + \sum_{j \geq 2} \frac{\delta_a^{(j)\top}}{\sqrt{n}} \{U_{H^{(j)}} \text{diag}(\phi'_{j-1}) Z_a^{(j-1)} V + V_{H^{(j)}} \text{diag}(\phi'_{j-1}) Z_a^{(j-1)} U\}.
\end{aligned} \tag{IC.6}$$

Each cross term factors through a hidden space of dimension at most n . Each curvature term has exactly one carrier diagonal. Schatten ideal inequalities applied to these displayed factors give (S.13); the port identity $\partial_{e_a^{(j)}} F_a = \delta_a^{(j)}$ proves the same bounds for both endpoint blocks $D_\Theta \delta$.

The zero-source variational generator is

$$\mathcal{L} = -\frac{2}{mn} \sum_a g_a g_a^\top - \frac{2}{m} \sum_a r_a D_\Theta^2 F_a, \quad g_a = \nabla_\Theta F_a. \tag{IC.7}$$

Its first summand contracts along forward real time. Along disjoint short nonreal/backward pieces its accumulated norm cost is at most two. By (S.13) and (IC.4), its propagator $J(t, s)$ therefore satisfies

$$\|J(t, s)\| \leq 2e^{SA_* + (S^2 D_*/\eta) \log(2n\mathcal{B})} \leq J_0 n^\kappa, \quad J_0 = 2e^{1/4} (2\mathcal{B})^{1/4000}, \quad \kappa = 1/4000. \tag{IC.8}$$

The coefficient of $\log n$ is bounded using the unchanged label condition, before any width is enlarged.

For deterministic scalar controls $a_{a,i}, b_{a,i}$, put $e_a = \sum_i x_i a_{a,i}$, $\bar{q}_a = \sum_i y_i b_{a,i}$. The three derivative maps from these ports to (IC.3), at zero ports, are exactly

$$-\frac{2}{mn} \nabla_{\bar{u}} \bar{F}_a \bar{\delta}_a^{(j+1)\top}, \quad -\frac{2}{m} \bar{r}_a (D_{\bar{u}} \bar{\delta}_a^{(j+1)})^\top, \quad -\frac{2}{m} \bar{r}_a (D_{\bar{u}} h_a^{(j-1)})^\top. \tag{IC.9}$$

The first is required because the residual depends on the forward port. Let $\Gamma = \sqrt{\mathcal{K}}$, $\tau_* = \max \tau_j$, $f_* = \max f_j$, and take a contour length at most $T_* \ell$, where $T_* = 32/\lambda + 2c$ if its half-width is $c/\sqrt{\ell}$. Define

$$\begin{aligned}
M_0 &= \eta^{-1} [1 + \log(2\mathcal{B})], \quad A_0 = b + sM_0, \quad B_0 = sM_0, \quad E_0 = A_* + SD_* M_0, \\
v_0 &= J_0 \{A_0 (2T_* \Gamma \tau_* + E_0) + S f_* B_0\}, \\
z_j^0 &= P_j (S v_0 + A_0), \quad h_j^0 = s z_j^0, \quad k_L^0 = v_0, \\
d_j^0 &= s k_j^0 + t_2 M_0 z_j^0, \quad k_j^0 = 10 d_{j+1}^0 + S \tau_{j+1} v_0 + B_0 \quad (j < L).
\end{aligned} \tag{IC.10}$$

As maps from the concatenation g of the omitted roots, the mobility, preactivation and feature variations have norms at most $\sqrt{p} n^\kappa \ell^2 (v_0, z_j^0, h_j^0)$; the scaled carrier/response variations have norms at most $\sqrt{p} n^\kappa \ell^3 (k_j^0, d_j^0)$. Indeed integrate (IC.9), using $\|g_a\| \leq \sqrt{n} \Gamma$, $\|\bar{\delta}_a\| \leq \sqrt{n} \tau_*$, $\|D_{\bar{u}} \bar{\delta}_a\| \leq A_* + SD_* M_n$, $\|D_{\bar{u}} h_a\| \leq S f_*$, and (IC.4). Cauchy-Schwarz over roots gives $\sqrt{p}$. The subsequent forward/backward differentiation is, explicitly,

$$\begin{aligned}
z_{[1]}^{(j)} &= \frac{S V_{H^{(j)}}}{\sqrt{n}} h^{(j-1)} + W^{(j)} h_{[1]}^{(j-1)} + e_{[1]}^{(j)}, \\
h_{[1]}^{(j)} &= \phi_j' \odot z_{[1]}^{(j)}, \\
\bar{k}_{[1]}^{(j)} &= W^{(j+1)\top} \bar{\delta}_{[1]}^{(j+1)} + \frac{S V_{H^{(j+1)}}^\top}{\sqrt{n}} \bar{\delta}^{(j+1)} + \bar{q}_{[1]}^{(j)}, \\
\bar{\delta}_{[1]}^{(j)} &= \phi_j' \odot \bar{k}_{[1]}^{(j)} + \phi_j'' \odot \bar{k}^{(j)} \odot z_{[1]}^{(j)}.
\end{aligned} \tag{IC.11}$$

Only one forward and one reverse port is present in an actual deletion; allowing a port at every layer only enlarges these bounds. A logarithm is introduced by the reference carrier once, in the inhomogeneous last term, and is propagated by $10s$, not by a carrier maximum.

For a control difference of supremum norm ϵ , replace A_0, B_0 by one in the forcing coefficients of (IC.10), retaining M_0, E_0, J_0 . Explicitly set

$$v_c = J_0 (2T_* \Gamma \tau_* + E_0 + S f_*), \quad z_j^c = P_j (S v_c + 1), \quad h_j^c = s z_j^c,$$

$$k_L^c = v_c, \quad d_j^c = sk_j^c + t_2 M_0 z_j^c, \quad k_j^c = 10d_{j+1}^c + S\tau_{j+1}v_c + 1, \quad C_c = \max(v_c, z_j^c, h_j^c, k_j^c, d_j^c). \quad (\text{IC.12})$$

Linearity of the variational equation in its controls proves $\|\Delta L\| \leq C_c \sqrt{pn} \ell^2 \epsilon$ for every map just listed. Indeed the difference forcing has amplitude ϵ , rather than $A_0 \ell$ or $B_0 \ell$. Its mobility integral has only one factor ℓ , from the horizon or the reference Hessian bound. Forward differences retain that one factor. The backward gate multiplies by $M_0 \ell$ once, giving ℓ^2 . This is why the control modulus has one fewer logarithm than the amplitude bound.

All quadratic matrices and their moduli are now explicit. If E_i selects one root block and L maps roots to a lower feature or upper response variation, its pairing is

$$(E_i g)^\top L g = g^\top R_i g, \quad R_i = \frac{1}{2}(E_i^\top L + L^\top E_i). \quad (\text{IC.13})$$

Thus $\|R_i\| \leq \|L\|$ and $\|\Delta R_i\| \leq \|\Delta L\|$. Only the selected diagonal root block contributes to $\text{tr } R_i/n$. In particular an incoming/outgoing cross block is centered. On $\|g\| \leq 2\sqrt{2p}$, centered-form interpolation costs

$$|\Delta(g^\top R g - \text{tr } R/n)| \leq 10p\|\Delta R\|. \quad (\text{IC.14})$$

The trace term has been included; it costs at most $2p\|\Delta R\|$.

Here are terminal-time moduli, including the normalization needed for off-grid interpolation. Define

$$T_L^0 = 2sH_L + 2t_2 M_0 V_L, \quad T_j^0 = 10sT_{j+1}^0 + 2s\tau_{j+1}^2 H_j + 2t_2 M_0 V_j,$$

$$D_c = \max\{\lambda s \max_j V_j/4, \lambda \max_j T_j^0/8\},$$

$$a_j^z = \lambda S^2 V_j/4, \quad a_j^h = sa_j^z, \quad a_j^W = \lambda S^2 \tau_j H_{j-1}/4, \quad a_j^\delta = \lambda T_j^0/8,$$

$$a_L^k = \lambda H_L/4, \quad a_j^k = a_{j+1}^W \tau_{j+1} + 10a_{j+1}^\delta, \quad A_t = 2K + (\lambda/4)SE_0,$$

$$v_t = A_t v_0 + A_0(2\Gamma\tau_* + \lambda E_0/4) + (\lambda/4)Sf_* B_0. \quad (\text{IC.15})$$

Raw scalar control speeds are bounded by $D_c \sqrt{n} \ell$. The reference RMS speeds are bounded by a_j^z, a_j^h, a_j^W and $\ell a_j^\delta, \ell a_j^k$. Differentiating (IC.11) gives the following coefficient recurrences:

$$\begin{aligned} z_1^t &= Sv_t + D_c, \quad h_1^t = sz_1^t + t_2 a_1^z z_1^0, \\ z_j^t &= S(H_{j-1}v_t + a_{j-1}^h v_0) + a_j^W h_{j-1}^0 + 10h_{j-1}^t + D_c, \quad h_j^t = sz_j^t + t_2 a_j^z z_j^0, \\ k_L^t &= v_t, \quad k_j^t = a_{j+1}^W d_{j+1}^0 + 10d_{j+1}^t + S\tau_{j+1}v_t + Sa_{j+1}^\delta v_0 + D_c, \\ d_j^t &= sk_j^t + t_2 a_j^z k_j^0 + (t_3 M_0 a_j^z + t_2 a_j^k)z_j^0 + t_2 M_0 z_j^t, \quad t_3 = 16\beta/a \leq \beta^2. \end{aligned} \quad (\text{IC.16})$$

Forward derivatives have bound $\sqrt{pn} n^\kappa \ell^3$ times their coefficients; backward ones have $\sqrt{pn} n^\kappa \ell^4$ times theirs. The factor $\sqrt{n}$ comes from a reference coordinate speed and appears once. The homogeneous propagated derivative still costs $10s$. For the control-independent lower probes, differentiate (IC.5):

$$z_1^J = 0, \quad h_1^J = t_2 a_1^z P_1, \quad z_j^J = a_{j-1}^h + a_j^W f_{j-1} + 10h_{j-1}^J, \quad h_j^J = sz_j^J + t_2 a_j^z P_j. \quad (\text{IC.17})$$

Restriction to a lower port and transposition give the raw adjoint probes used below. Their operator time derivatives are bounded by $\sqrt{n} \max(z_j^J, h_j^J)$.

Let C_t be the maximum of all coefficients in (IC.16)–(IC.17) and v_t . A terminal mesh of size

$$h_t \leq \min \left\{ 1, \frac{n^{-1/10-1/2-\kappa}}{80p^{3/2}(1+C_t)\ell^4} \right\} \quad (\text{IC.18})$$

makes both the linear and centered-quadratic off-grid errors less than $n^{-1/10}/4$, using (IC.14). A rectangle of real length $32\ell/\lambda + 2r_t$ and height $2r_t$ has at most $[2 + (32\ell/\lambda + 2r_t)/h_t][2 + 2r_t/h_t]$ grid points. On complex rectangles these derivative identities apply to analytic actual controls. At grid points the Gaussian union uses deterministic one-dimensional control histories along the prescribed contour. Actual analytic controls are substituted only after that uniform event. No path independence is asserted for arbitrary two-dimensional controls.

3. Exact nonlinear remainder and reverse probes

Here is a deterministic local lemma that specifies every term used in the first-exit argument. Write the augmented displacement as $\eta_a = (V, e_a) + (U, 0)$, with

$$\|V\|, \|e_a\|, \|Dg_0(V, e_a)\|_2 \leq N, \quad \|Dg_0(V, e_a)\|_\infty \leq d_0, \quad \|U\| \leq u, \quad (\text{IC.19})$$

where the bound on the combined augmented norm may be obtained by enlarging N by a fixed factor. The quantities g here are all forward vectors and scaled backward vectors in (IC.11). Assume $N \geq 1$, $u, d_0 \leq 1$, $N + u \leq \sqrt{n}$, and put

$$R = d_0N + d_0u + u^2, \quad P = (N + u)^2/\sqrt{n}, \quad u + R + P \leq 1. \quad (\text{IC.20})$$

With reference scaled carrier maximum M , define $g_{[1]} = Dg_0\eta$ and $E_g = g_1 - g_0 - g_{[1]}$. Feature RMS, first forward derivative, and first backward derivative bounds on the joining segment/reference are, respectively, β^{3L} , $J_f = \beta^{10L}$, and $J_b = \beta^{20L}(1 + M)$. To verify the latter two, use (IC.5), then the backward derivative in (IC.11): its forcing is a single reference carrier times the forward derivative, and its homogeneous propagation is $10s$. Along a unit segment, physical operators are at most eleven and ports at most $\sqrt{n}$; the affine feature recurrence is bounded by $(13\beta)^L$.

The localized product bound is

$$\|z_{[1]} \odot z_{[1]}\|_2 \leq d_0N + 2d_0J_fu + J_f^2u^2 \leq 3J_f^2R. \quad (\text{IC.21})$$

The following identities are exact, so their use does not assume a bound on an uncontrolled changed carrier:

$$E_{z^{(1)}} = 0, \quad E_{z^{(j)}} = W_1^{(j)}E_{h^{(j-1)}} + \frac{S\Delta\bar{u}_{H^{(j)}}}{\sqrt{n}}h_{[1]}^{(j-1)}, \quad (\text{IC.22})$$

$$E_{\bar{k}^{(L)}} = 0, \quad E_{\bar{k}^{(j)}} = W_1^{(j+1)\top}E_{\bar{\delta}^{(j+1)}} + \frac{S\Delta\bar{u}_{H^{(j+1)}}^\top}{\sqrt{n}}\bar{\delta}_{[1]}^{(j+1)}, \quad (\text{IC.23})$$

$$E_{\bar{\delta}} = g_1 \odot E_{\bar{k}} + (g_1 - g_0) \odot \bar{k}_{[1]} + \bar{k}_0 \odot [g_1 - g_0 - \phi''(z_0)z_{[1]}], \quad g_i = \phi'(z_i). \quad (\text{IC.24})$$

For activation subtraction insert $z_0 + z_{[1]}$; this gives $\|E_h\| \leq \beta\|E_z\| + (\beta/2)\|z_{[1]}^{\odot 2}\|$. The cross term in (IC.22) is at most J_fP . Summing its geometric propagation gives

$$\max_j(\|E_z\|, \|E_h\|) \leq \beta^{30L}(R + P). \quad (\text{IC.25})$$

In (IC.24), the second term is bounded by

$$\beta\{3J_fJ_bR + \beta^{30L}(R + P)(d_0 + J_bu)\};$$

the last is at most $M[\beta\|E_z\| + (\beta^2/2)\|z_{[1]}^{\odot 2}\|]$. The matrix cross term in (IC.23) is at most J_bP . Homogeneous propagation uses only the perturbed mixer and slope, so

$$\max_j(\|E_{\bar{k}}\|, \|E_{\bar{\delta}}\|) \leq \beta^{60L}(1 + M)(R + P). \quad (\text{IC.26})$$

For a hidden gradient block its exact product remainder is

$$\frac{S}{\sqrt{n}}[E_{\bar{\delta}}h_1^\top + \bar{\delta}_{[1]}(h_1 - h_0)^\top + \bar{\delta}_0 E_h^\top]. \quad (\text{IC.27})$$

Use reference feature/response RMS in its first and last terms and $J_f J_b P$ in its middle term. The first-layer and readout blocks give the gradient remainder bound $\beta^{70L}(1+M)(R+P)$.

Applying (IC.26) to each initial part of the joining segment proves a carrier cap $\beta^{62L}(1+M)$ on that segment. Hence (IC.6) gives an augmented Hessian bound $\beta^{87L}(1+M)$, and $\|D\bar{F}\| \leq \beta^{13L}\sqrt{n}$. Thus

$$|\Delta\bar{r}| \leq \beta^{13L}(N+u)/\sqrt{n}, \quad |\Delta\bar{r} - D\bar{r}_0\eta| \leq \frac{1}{2}\beta^{87L}(1+M)(N+u)^2/n. \quad (\text{IC.28})$$

The constant labels y_a/S cancel from both differences. For $g = \nabla\bar{F}$, the product remainder is exactly

$$\bar{r}_0 E_g + (\Delta\bar{r} - D\bar{r}_0\eta)g_0 + \Delta\bar{r}(g_1 - g_0). \quad (\text{IC.29})$$

Its last two terms cost $2\beta^{100L}(1+M)P$.

For the reverse force, fix an omitted incoming root y_i and write $\psi_i = y_i^\top h_a^{(j-1)}$. Include in the Gaussian event all its lower adjoint carriers. Their Euclidean norms are bounded by the forward operator recurrences and their coordinate maxima by d_0 . For an aggregate $q = \sum_i y_i b_{a,i}$ with $\|y_i\| \leq 2$, $|b_{a,i}| \leq \beta M$, put $Q = 2p\beta M$ and $P_q = p\beta M d_0$. Subtracting the probe adjoint recursion along a parameter segment gives

$$\max_j \|k_{q,t}^{(j)} - k_{q,0}^{(j)}\|_2 \leq \beta^{20L}(P_q + Q/\sqrt{n})(N+u). \quad (\text{IC.30})$$

The changed gate multiplies the reference probe maximum P_q ; the changed mixer costs $Q(N+u)/\sqrt{n}$; the propagated difference costs $11s$. These are all three terms in the recursion. Apply (IC.6) to the scalar lower-network output ψ_q and integrate its Hessian to obtain

$$\|(Dh_1 - Dh_0)^\top q\| \leq \beta^{50L}(P_q + Q/\sqrt{n})[N+u + (N+u)^2]. \quad (\text{IC.31})$$

The exact extra nonlinear terms in (IC.3) are $\bar{r}_0(Dh_1 - Dh_0)^\top \bar{q}$ and $\Delta\bar{r} Dh_1^\top \bar{q}$. Equations (IC.28), (IC.31) bound both. Combining them with (IC.29) proves

$$\begin{aligned} \|\bar{\mathcal{V}}_1 - \bar{\mathcal{V}}_0 - D_{\bar{u}}\bar{\mathcal{V}}_0\Delta\bar{u} - D_e\bar{\mathcal{V}}_0e - D_{\bar{q}}\bar{\mathcal{V}}_0\bar{q}\| &\leq \beta^{120L}(1+p)(1+M) \\ &\cdot \{\bar{\rho}_0[R + d_0(N^2 + Nu + u^2)] + (1 + \bar{\rho}_0)P\}. \end{aligned} \quad (\text{IC.32})$$

For complex application, every scalar Taylor point must remain in the half-strip. Starting from imaginary parts at most $7a/16$, the explicit sufficient gate is

$$d_0 + \beta^{10L}u + \beta^{30L}(R+P) < a/32. \quad (\text{IC.33})$$

A first-exit argument using (IC.22) places the joining segments and the auxiliary points $z_0 + z_{[1]}$ within $15a/32$. Cauchy's formula for ϕ'' , on radius $a/64$, bounds the third derivative by $64\beta/a \leq 4\beta^2$. Replacing the coefficient β^{120L} in (IC.32) by β^{240L} pays for all complex versions above. Transposes remain algebraic; no complex energy inequality has been used.

There is no hidden division by S in this proof. More formally, the effective network with actual hidden matrices and readout $\bar{u}_w$ is the original architecture pulled back by the affine map whose derivative multiplies hidden parameter blocks by S , the readout by one, and ports by one. This derivative and its transpose are contractions. Its second and higher derivatives vanish. Thus every displayed scaled derivative is the corresponding effective derivative with a factor S per hidden parameter argument, and never a negative power of S .

Finally integrate the actual omitted row/column equations, not Gaussian surrogates. Their learned parts obey

$$\|\Delta W_{:,i}^{(j+1)}\| \leq S^2 \tau_* A_n / \sqrt{n}, \quad \|\Delta W_{i,:}^{(j)}\| \leq S^2 H_{\max} B_n / \sqrt{n}, \quad A_n = b + sM_n, \quad B_n = sM_n. \quad (\text{IC.34})$$

Multiplying by the actual controls gives forward and scaled reverse source errors at most $pS^2(\tau_*A_n^2 + H_{\max}B_n^2)/\sqrt{n}$. Differentiating (IC.3) in its ports using (IC.28) shows that their additional force is at most

$$S^2\beta^{260L}(1+p)^2(1+M_n)^3(1+\bar{\rho})/\sqrt{n}. \quad (\text{IC.35})$$

At the top, $|\bar{d}_a| \leq pM_nA_n/n$, and its multiplication of both retained terms costs at most $\beta^{32L}(1+p)^2(1+M_n)^3/\sqrt{n}$. This includes the reverse-offset product. Current row norms are at most three once the row increment in (IC.34) is below one.

4. Passive-query derivatives and the uniform Gaussian event

For a passive query q , set $C_q^{(j)} = D_\Theta h_q^{(j)}$. For evaluated query q and training driver a , define $\bar{R}_{qa}^{(j)} = S^{-1}D_\Theta z_q^{(j)} g_a$, $\bar{Q}_{qa}^{(j)} = \phi'_j(z_q^{(j)}) \odot \bar{R}_{qa}^{(j)}$. The exact response recurrences are

$$\bar{R}_{qa}^{(1)} = \bar{\delta}_a^{(1)} v_a^\top q, \quad \bar{R}_{qa}^{(j)} = \bar{\delta}_a^{(j)} c_{aq}^{(j-1)} + W^{(j)} \bar{Q}_{qa}^{(j-1)}, \quad c_{aq}^{(j-1)} = h_a^{(j-1)\top} h_q^{(j-1)} / n. \quad (\text{IC.36})$$

For a complex great circle $q(\zeta) = u \cos \zeta + v \sin \zeta$, write $J^{(j)} = \partial_\zeta z_q^{(j)}$ and $I^{(j)} = \phi'_j(z_q^{(j)}) \odot J^{(j)}$. Then

$$J^{(1)} = Aq', \quad J^{(j)} = W^{(j)} I^{(j-1)}. \quad (\text{IC.37})$$

These are passive derivatives; none changes the training equation.

The complete additional first-variation recurrences are

$$\begin{aligned} c_{aq,[1]} &= (h_{a,[1]}^\top h_q + h_a^\top h_{q,[1]})/n, \\ \bar{R}_{qa,[1]}^{(j)} &= \bar{\delta}_{a,[1]}^{(j)} c_{aq} + \bar{\delta}_a^{(j)} c_{aq,[1]} + \frac{SV_{H^{(j)}}}{\sqrt{n}} \bar{Q}_{qa}^{(j-1)} + W^{(j)} \bar{Q}_{qa,[1]}^{(j-1)} + e_{qa,[1]}^{R,j}, \\ \bar{Q}_{qa,[1]}^{(j)} &= \phi'_j \odot \bar{R}_{qa,[1]}^{(j)} + \phi''_j \odot \bar{R}_{qa}^{(j)} \odot z_{q,[1]}^{(j)}, \\ J_{[1]}^{(1)} &= SV_A q', \quad J_{[1]}^{(j)} = \frac{SV_{H^{(j)}}}{\sqrt{n}} I_{[1]}^{(j-1)} + W^{(j)} I_{[1]}^{(j-1)} + e_{[1]}^{J,j}, \\ I_{[1]}^{(j)} &= \phi'_j \odot J_{[1]}^{(j)} + \phi''_j \odot J^{(j)} \odot z_{q,[1]}^{(j)}. \end{aligned} \quad (\text{IC.38})$$

At the layer immediately above the deleted layer the extra ports are $e_{qa}^{R,j+1} = \sum_i x_i \bar{Q}_{qa,i}^{(j)}$ and $e^{J,j+1} = \sum_i x_i I_i^{(j)}$. At all other layers these ports are zero. There is also a direct lower reverse observable, which must be included in addition to (IC.38): below the deleted layer,

$$Q_{qa,\text{full}}^{(j-1)} = C_q^{(j-1)} [g_a + C_a^{(j-1)\top} q_a], \quad S^{-1}(Q_{qa,\text{full}} - C_q g_a) = C_q C_a^\top \bar{q}_a. \quad (\text{IC.39})$$

Thus its reference linear map includes $C_q^0 C_a^{0\top} y_i$ for every driver and omitted incoming root. Include the reference forward images of $\xi = C_a^{0\top} y_i$ at every lower layer in the same Gaussian coordinate event. This explicitly lists the additional probes. Their coefficient operators are measurable in retained initialization and independent of the omitted root; the probe vectors themselves depend linearly on that root. Their operator bounds therefore give the claimed centered Gaussian vector estimates.

Here are remainder bounds for these additions. The feature-pairing product identity gives

$$|c_{aq,[1]}| \leq \beta^{10L}(N+u)/\sqrt{n}, \quad |E_c| \leq \beta^{40L}\{(R+P)/\sqrt{n} + (N+u)^2/n\}. \quad (\text{IC.40})$$

The exact product/mixer remainders in (IC.36) are

$$\begin{aligned} E_{\bar{R}} &= E_{\bar{\delta}} c_1 + \bar{\delta}_{[1]}(c_1 - c_0) + \bar{\delta}_0 E_c + W_1 E_{\bar{Q}_{\text{prev}}} + \Delta W \bar{Q}_{[1]}^{\text{prev}} + E_{\text{ports}}, \\ E_{\bar{Q}} &= \phi'(z_1) E_{\bar{R}} + [\phi'(z_1) - \phi'(z_0)] \bar{R}_{[1]} + \bar{R}_0 [\phi'(z_1) - \phi'(z_0) - \phi''(z_0) z_{[1]}]. \end{aligned} \quad (\text{IC.41})$$

Equations (IC.23)–(IC.24) apply verbatim to J, I , with $E_{J(1)} = 0$, their displayed response ports, and J_0 in place of a reference carrier. In (IC.41), $\bar{\delta}_0 E_c$ uses response RMS $\sqrt{n}\tau_j$, canceling the first $1/\sqrt{n}$ in (IC.40).

Every product of two changed factors retains a width or localized-coordinate factor. Reference $\bar{R}$ and J coordinate caps appear only in additive gate remainders.

For (IC.39), subtract the directional forward recursions in the fixed direction ξ . The new forcing terms are $\xi_H \Delta h / \sqrt{n}$, $\Delta H D h_0[\xi] / \sqrt{n}$, and a changed activation gate multiplying $D z_0[\xi]$. On the enlarged Gaussian event that last vector has maximum d_0 . Propagation through the perturbed bounded mixers/slopes gives

$$\|(C_q^1 - C_q^0)\xi\| \leq \beta^{50L}(d_0 + n^{-1/2})(N + u). \quad (\text{IC.42})$$

The other product difference is $C_q^1(C_a^1 - C_a^0)^\top y_i$; use (IC.31) and $\|C_q^1\| \leq \beta^{10L}$. Multiplication by the actual controls and summation over roots gives

$$\|C_q^1 C_a^{1\top} \bar{q}_a - C_q^0 C_a^{0\top} \bar{q}_a\| \leq \beta^{70L}(1 + p)(1 + M_n)(d_0 + n^{-1/2})[N + u + (N + u)^2]. \quad (\text{IC.43})$$

The missing feature-pairing term is at most $p A_n^2/n$; the learned response and angular ports are bounded using (IC.34) times their actual stopped controls. These are additional $n^{-1/2}$ vector errors. This accounts for every term hidden by rectangular deletion in (IC.36)–(IC.39).

The doubled caps in the next estimate refer to full-network stops. For independently stopped cavities use their fourfold caps instead; these are fixed multipliers in the same polynomial bounds.

For fixed p, m, d, L , use the temporary query preactivation cap in S.6 and doubled response/angular caps from (S.25). Equations (IC.10)–(IC.17), (IC.38), and the product rule give operator bounds n^κ times fixed polynomials in ℓ for all the listed linear maps. For completeness, each differentiated gate is $\phi'' z_{[1]}$, each differentiated carrier gate is $\phi''' z_t z_{[1]} \bar{k} + \phi'' z_{[1]} \bar{k}_t + \phi'' \bar{k} z_{[1],t}$, and the response/angular gate has the same three terms with $\bar{R}$ or J . At most one raw reference coordinate derivative is used in each term. Its bound is $\sqrt{n}$ times a fixed polynomial in ℓ . All homogeneous propagated derivatives have multiplier $10s$. This proves terminal and frame derivative bounds $\sqrt{nn}^\kappa \text{poly}(\ell)$, with no power of n depending on depth. Frame derivatives of the input and of q' are bounded on a fixed tubular neighborhood of the orthonormal-frame manifold. Differentiating (IC.38) in those variables uses the same gate terms just listed. Polar normalization provides bounded local frame-coordinate derivatives.

The controls in (IC.38) have polynomial logarithmic amplitudes. Their time speeds cost one $\sqrt{n}$: for example $\partial_t \bar{Q} = \phi' \partial_t \bar{R} + \phi'' \bar{R} \partial_t z$, where the stopped coordinate cap on $\bar{R}$ and the RMS bound on $\partial_t z$ give a polynomial logarithmic RMS speed. The same calculation with J gives the angular-control speed. Thus the complete control net at accuracy $\tau = n^{-1/8}$ has logarithmic cardinality bounded by $n^{5/8} \text{poly}(\ell)$. In detail, q_c real component controls with amplitude M_c , Lipschitz constant $D'_c \sqrt{n}$, and contour length T_c admit a net with

$$\log N_{\text{ctrl}} \leq q_c(2 + 8T_c D'_c n^{5/8}) \log(1 + 8M_c n^{1/8}). \quad (\text{IC.44})$$

Sample each real/imaginary component at spacing $\tau/(8D'_c \sqrt{n})$, round on mesh $\tau/4$, and interpolate. This proves both its accuracy and cardinality.

For fixed controls, the conditional root covariance is I/n . After eventual absorption of the fixed logarithmic coefficients, every listed linear operator and quadratic matrix has norm at most $n^{1/200}$. A coordinate has variance at most $n^{-99/100}$. For a centered form, diagonalizing its real symmetric part and using

$$\begin{aligned} \mathbb{E} e^{t(G^\top R G - \text{tr } R)} &= \prod_{\alpha} e^{-t\lambda_{\alpha}} (1 - 2t\lambda_{\alpha})^{-1/2}, \\ -\log(1 - v) - v &\leq v^2/[2(1 - |v|)] \quad (|v| < 1) \end{aligned} \quad (\text{IC.45})$$

gives a tail bounded by the minimum of threshold squared divided by Hilbert–Schmidt variance and threshold divided by operator scale. Here $\|R\|_{\text{HS}} \leq \sqrt{2pn}\|R\|$. At threshold $n^{-1/10}/8$, real/imaginary splitting yields $4e^{-n^{0.78}}$ eventually. Independent-root bilinear forms are the symmetric off-diagonal block case of (IC.45). Root norms $\|g\| \leq 2\sqrt{2p}$ have failure at most e^{-pn} . They also give whole-vector image bounds below $n^{1/100}$ after coefficients are absorbed. Use a smaller fixed share of this bound for each of the finitely many augmented components, so their combined norm in (IC.19) is at most that radius.

Control interpolation uses (IC.12)–(IC.14); its error is $n^{-1/8+1/200} \text{poly}(\ell)$, smaller than $n^{-1/10}/4$. A time/query/frame mesh of size $n^{-2}/\text{poly}(\ell)$ gives the other $n^{-1/10}/4$. There are $2d + 3$ real frame/time/tube

coordinates, so for fixed parameters its point count is at most n^{4d+10} eventually. The entropy (IC.44), this grid, the coordinate/probe tests, and at most pLn^p deletion sets cost less than $n^{0.65}$ in logarithmic cardinality. The resulting uniform failure is at most $e^{-n^{0.7}}$ eventually. Only now substitute the adaptive training and passive-query controls.

Let V solve the exact linear variational equation and let U be the retained scaled displacement minus V . It satisfies

$$U(t) = \int_0^t J(t, s) [\mathcal{R}(\Delta \bar{u}(s), e(s), \bar{q}(s)) + \mathcal{R}_{\text{learned}}(s) + \mathcal{R}_{\text{top}}(s)] ds, \quad U(0) = 0, \quad (\text{IC.46})$$

where the three terms have exactly the bounds (IC.32), (IC.35), and the top-offset bound following it. Take $N = n^{1/100}$, $d_0 = n^{-1/10}$, $u_0 = n^{-1/25}$. On a first-exit prefix $\|U\| \leq u_0$, integration with (IC.4) and (IC.8) bounds (IC.46) by

$$n^\kappa \text{poly}(\ell) [n^{-8/100} + u_0^2 + n^{-48/100} + n^{-1/2}] = o(u_0). \quad (\text{IC.47})$$

This strictly improves the remainder stop. Equations (IC.25)–(IC.26) and (IC.40)–(IC.43) then give retained coordinate discrepancies at most $n^{-1/30}$ eventually, for preactivations, scaled carriers, responses and angular derivatives, and whole-vector discrepancies at most $2n^{1/100}$. The strict strip condition (IC.33) closes simultaneously. The reference is always its independently stopped cavity; this argument transfers its prefix rather than conditioning a Gaussian law on full-network survival.

5. The actual contractions behind (S.15) and the mixed endpoints

We now write the singleton expansion in unscaled mobility coordinates; all $1/n$ factors remain visible. At an interior omitted neuron, put

$$C_a(t) = D_\Theta h_a^{(j-1)}(t), \quad B_a(t) = D_\Theta \delta_a^{(j+1)}(t), \quad E_a(t) = D_{e_a^{(j+1)}} \delta_a^{(j+1)}(t).$$

All three are zero-source cavity derivatives. With deterministic controls $\alpha_a = h_{a,i}^{(j)}$, $b_a = \delta_{a,i}^{(j)}$, the linear mobility response is

$$V(t) = -\frac{2}{m} \sum_b \int_0^t J(t, s) \left[\frac{g_b(s) d_b(s)^\top}{n} x_i \alpha_b(s) + r_b(s) B_b(s)^\top x_i \alpha_b(s) + r_b(s) C_b(s)^\top y_i b_b(s) \right] ds, \quad d_b = \delta_b^{(j+1)}. \quad (\text{IC.48})$$

The two scalar linearized pairings are

$$z_{a,i} = y_i^\top h_a^0 + y_i^\top C_a V + (\Delta W_{i,:}) h_a + \varepsilon_z,$$

$$k_{a,i} = x_i^\top \delta_a^0 + x_i^\top B_a V + x_i^\top E_a x_i \alpha_a + (\Delta W_{:,i})^\top \delta_a + \varepsilon_k. \quad (\text{IC.49})$$

The remainders include the nonlinear vector errors and their root pairings; they are uniform small errors from (IC.46)–(IC.47). For the scaled statement apply the same identities before dividing the carrier terms by S , as in (IC.1)–(IC.35).

Substituting (IC.48) into (IC.49) identifies all nonzero Gaussian means:

$$\begin{aligned} \text{incoming reverse mean:} & \quad -\frac{2}{m} \sum_b \int_0^t r_b(s) b_b(s) \frac{\text{tr}[C_a(t) J(t, s) C_b(s)^\top]}{n} ds, \\ \text{outgoing Hessian mean:} & \quad -\frac{2}{m} \sum_b \int_0^t r_b(s) \alpha_b(s) \frac{\text{tr}[B_a(t) J(t, s) B_b(s)^\top]}{n} ds, \\ \text{direct outgoing mean:} & \quad \alpha_a(t) \text{tr} E_a(t)/n, \\ \text{adaptive-residual mean:} & \quad -\frac{2}{mn^2} \sum_b \int_0^t \alpha_b(s) d_b(s)^\top B_a(t) J(t, s) g_b(s) ds. \end{aligned} \quad (\text{IC.50})$$

Every remaining linearized pairing has distinct incoming/outgoing roots and hence zero conditional mean, or is a centered same-root form. Their fluctuations are already controlled by (IC.13)–(IC.45). In particular, the direct mean multiplies the evaluated sample's current α_a , whereas the integrated means retain a sum over driving samples. The last line has the additional $1/n$: using $\|d_b\|, \|g_b\| = O(\sqrt{n})$ bounds it by $n^{-1+\kappa} \text{poly}(\ell)$ on a logarithmic horizon. It belongs to the small remainder, not to a nonzero limiting trace coefficient.

Here is the complete trace bound for the second line. Split the propagator into its negative-Gram base and residual-Hessian insertions. An ordered term with h insertions has residual-activity integral at most $S^h/h!$. The product of base propagators costs at most two: their disjoint short contour pieces have total norm-growth integral at most $\log 2$, while forward real pieces contract. Normalized Schatten Hölder with exponent $h+2$ on the two endpoint Hessian blocks and all h inserted Hessians bounds its normalized trace by

$$\frac{2S^h}{h!} [A_* + (SD_*/\eta)(h+2)(2\mathcal{B})^{1/(h+2)}]^{h+2}. \quad (\text{IC.51})$$

The term $h=0$ is at most $2H_*^2$, using both endpoint Hilbert–Schmidt bounds. For $h \geq 1$, use $(x+y)^{h+2} \leq 2^{h+1}(x^{h+2} + y^{h+2})$. The bounded part sums to $4A_*^2(e^{2SA_*} - 1)$. For the carrier part, $h! \geq (h/e)^h$ and $(1+2/h)^h \leq e^2$ bound its sum by

$$\frac{8e^2 D_*^2 S^2 \mathcal{B}}{\eta^2} \sum_{h \geq 1} (2eD_* S^2/\eta)^h (h+2)^2 \leq \frac{576e^3 D_*^3 \mathcal{B} S^4}{\eta^3}. \quad (\text{IC.52})$$

The last inequality uses $q = 2eD_* S^2/\eta \leq 1/2$ and $\sum_{h \geq 1} q^h (h+2)^2 \leq 36q$, obtained by differentiating the geometric series twice. Equation (S.10) gives this condition.

For the first line of (IC.50), the zero term is at most $2f_*^2$. For $h \geq 1$, give one forward endpoint exponent two, the other infinity, and each Hessian exponent $2h$. The bounded part of the inserted series is controlled by $e^{2SA_*} - 1$, and the carrier part by $\sqrt{2\mathcal{B}} \sum_{h \geq 1} (4eD_* S^2/\eta)^h$. Both are at most one by (S.10). Their factors, together with the zero term, give

$$\frac{1}{n} |\text{tr}(C_a J C_b^\top)| \leq 8f_*^2. \quad (\text{IC.53})$$

These uses of Schatten Hölder have total reciprocal exponent one. The normalizations multiply to exactly n^{-1} , independent of the parameter dimension.

For the direct term, let $T_{r,j+1}$ be the forward derivative from the port in layer $j+1$ to preactivation in layer r . The port Hessian is exactly

$$E_a = \sum_{r=j+1}^L T_{r,j+1}^\top \text{diag}(k_a^{(r)} \phi_r'') T_{r,j+1}.$$

Here $\|T_{r,j+1}\| \leq (10s)^{r-j-1}$, and the nuclear norm per width of the diagonal is at most $t_2 S k_r$, by the carrier RMS. Thus $|\text{tr } E_a|/n \leq SE$ with exactly the coefficient E in (S.8).

The learned outgoing column paired with the current upper response has bound $s^2 k_*^2 S^3$ times the driving forward amplitude after the residual average. The learned incoming row paired with a current feature has bound $s S^2 U_i H_{j-1}^2$. Both follow by substituting their rank-one update integrals and applying RMS Cauchy–Schwarz to each retained vector. At the first layer its direct update has bound $s S^2 U_i$. Finally, for every driving time,

$$\frac{1}{m} \sum_b |r_b| |h_{b,i}| \leq \rho(b + sZ_i), \quad \frac{1}{m} \sum_b |r_b| |\delta_{b,i}| \leq \rho s S U_i. \quad (\text{IC.54})$$

These follow from the running sample RMS definitions (S.12), so they also bound all past integrands. Combine (IC.50)–(IC.54) and divide the backward inequality by S . The constants (S.8) give exactly

$$K_{a,i}/S \leq G_{\delta,a,i}/S + [D_0 + D_1 \mathcal{B} S^4/\eta^3](1 + Z_{a,i} + Z_i) + \varepsilon_n, \quad Z_{a,i} \leq G_{h,a,i} + C_F S^2 U_i + \varepsilon_n. \quad (\text{IC.55})$$

This is (S.15) with its contractions specified. For fixed parameters $\varepsilon_n \rightarrow 0$; under the finite gate below it is at most $n^{-1/40}$, in the scaled norms. At the top use the exact readout integral $\sup |w_i|/S \leq b + sZ_i$, taking

$G_{\delta,a,i} = 0$. At the first layer take $G_{h,a,i} = \sup |A_{0,i}v_a|$. These endpoint formulas prove the same absorption, without inventing absent roots.

The mixed endpoint constants in (S.24) can now be checked directly. For $Q_{qa} = C_q g_a$,

$$DQ_{qa} = C_q Dg_a + (DC_q)[\cdot]g_a. \quad (\text{IC.56})$$

The first term has normalized Hilbert–Schmidt bound $f_j H_*$. For the second, differentiating its forward recursion gives respectively the curvature diagonal $\text{diag}(\phi'' R)Dz$, the changed mixer $U_H Q/\sqrt{n}$, the term where the fixed gradient block acts on a changed feature, and the propagated lower derivative. Their normalized Hilbert–Schmidt coefficients are $t_2 r_j P_j$, $s q_{j-1}$, $s \tau_j H_{j-1} f_{j-1}$, and $10 s e_{j-1}$, respectively. This is precisely the recurrence defining e_j . For example $\|\text{diag}(R)Dz\|_{\text{HS}}/\sqrt{n} \leq P_j \|R\|_2/\sqrt{n}$; no response maximum is needed for this Hilbert–Schmidt estimate. Replacing R by J gives $t_2 j_j P_j + s(b_{j-1} + 10a_{j-1})$; at the first layer the map $U_A q'$ adds at most two. These are the coefficients a_j in (S.24). Applying the asymmetric estimate (IC.53) with these endpoints gives exactly T_Q, T_J .

For clarity, row insertion of (IC.36) has four noncentered terms: the direct feature pairing, the direct reverse observable (IC.39), the integrated mixed endpoint (IC.56), and the learned incoming row. After division by S , their coefficients are

$$sK_{\text{src}} H_{j-1}^2, \quad sK_{\text{src}} f_{j-1}^2, \quad sK_{\text{src}} S T_Q, \quad sK_{\text{src}} S^2 H_{j-1} q_{j-1}. \quad (\text{IC.57})$$

For (IC.37) there are the mixed endpoint and learned row, with coefficients $sK_{\text{src}} S^2 T_J$ and $sK_{\text{src}} S^2 H_{j-1} b_{j-1}$. The incoming-root Gaussian reference contributes $G_d q_{j-1}$ or $G_d b_{j-1}$. Every forward-source pairing uses one incoming and one outgoing root and is centered; the training reverse source supplies the same incoming root twice. This proves the enumeration and the coefficients (S.25), including the lower direct reverse term. Its factor two leaves strict margin for the uniform errors in the passive-query insertion argument. The Gaussian mesh count there and $G_d = 16\sqrt{d+3}$ give (S.26). No exponential query budget has been added.

E.1.2 Explicit source recurrences, budget removal and analytic domain

In this source section only, H_j, P_j, k_j, f_j, q_j are the source recurrence coefficients. They are denoted H_j^{src} , $P_j^{\text{src}}, k_j^{\text{src}}, f_j^{\text{src}}, q_j^{\text{src}}$ elsewhere in this document. In particular q_j here is not the supplied compression order q .

The dense rank-one equations, with $v_a = x_a/\sqrt{d}$, are

$$\dot{A} = -\frac{2}{m} \sum_a r_a \delta_a^{(1)} v_a^\top, \quad \dot{W}^{(j)} = -\frac{2}{mn} \sum_a r_a \delta_a^{(j)} h_a^{(j-1)\top}, \quad \dot{w} = -\frac{2}{m} \sum_a r_a h_a^{(L)}. \quad (\text{S.1})$$

The independently proved dense-fitting theorem supplies the physical operator caps below nine and, conservatively,

$$\rho(t) = \|r(t)\|_2/\sqrt{m} \leq Y e^{-\lambda t/4}. \quad (\text{S.3})$$

Put

$$S = 16Y/\lambda, \quad T = 32\lambda^{-1} \log(en), \quad d\nu = 2\rho |dz|. \quad (\text{S.4})$$

Here $d\nu$ is a contour integration measure, not a new training clock. Its real mass is at most $S/2$. The source-domain proof uses this T ; the later analytic-tail lemma extends this domain without changing its radii or introducing an order-dependent stochastic event.

Linear growth is the exact bound $|\phi_j(z)| \leq b + s|z|$ on the safe half-strip. If $s_{\text{full}} = \max\{1, \sup_{j, |\text{Im } z| < a} |\phi'_j(z)|\}$, Cauchy's formula gives the bound $(h-1)! s_{\text{full}} (4/a)^{h-1}$ for derivatives of order $h \geq 2$ on that half-strip. Those higher-derivative coefficients occur only in the qualitative local-width threshold; the explicit recurrences use b, s, t_2, a .

S.2. Explicit RMS, derivative, and trace coefficients

Use the complex operator caps ten and query input norm at most two. Define the forward feature RMS bounds

$$H_1 = \max(1, b + 20s), \quad H_j = \max(1, b + 10sH_{j-1}) \quad (j \geq 2). \quad (\text{S.5})$$

Indeed $\|Av\|_2/\sqrt{n} \leq 20$, and linear growth followed by the operator bound proves $\|h^{(j)}\|_2/\sqrt{n} \leq H_j$. Integrating the readout equation over a contour of ν -mass at most S gives $\|w\|_2/\sqrt{n} \leq SH_L$. Define

$$k_L = H_L, \quad k_j = 10sk_{j+1}, \quad \tau_j = sk_j, \quad P_1 = 3, \quad P_j = H_{j-1} + 10sP_{j-1} + 1, \quad f_j = sP_j. \quad (\text{S.6})$$

Thus carrier and response RMS bounds are $Sk_j, S\tau_j$. The derivative numbers P_j also cover additive preactivation ports at every layer: augment the mobility coordinates $\Theta = (A, \sqrt{n}W^{(2)}, \dots, \sqrt{n}W^{(L)}, w)$ by vectors $e^{(j)}$ entering additively in $z^{(j)}$, and evaluate derivatives at $e = 0$. The direct first map has norm at most $2 + 1$; every later direct map contributes $H_{j-1} + 1$. The recurrence proves the claimed augmented derivative bound. These ports are proof variables and are never retained at runtime.

For $F_a = nf_n(v_a)$, define

$$A_* = 2sP_L + 2s^2 \sum_{j=2}^L k_j P_{j-1}, \quad D_* = t_2 \sum_{j=1}^L P_j^2, \quad H_* = A_* + t_2 \sum_{j=1}^L P_j^2 k_j. \quad (\text{S.7})$$

There are two readout cross terms of rank at most n , two mixed terms per hidden matrix, and curvature terms $Dz_a^{(j)\top} \text{diag}(\phi_j'' k_a^{(j)}) Dz_a^{(j)}$. The rank-normalized Schatten bounds consequently use (S.7), as shown in §S.3 below. Crucially, $D_\Theta \delta_a^{(j)}$ is a subblock of this augmented Hessian, since $\partial_{e^{(j)}} F_a = \delta_a^{(j)}$. The same constants therefore cover both endpoints of the carrier trace; no additional unproved endpoint estimate is needed.

Let $f_* = \max_j f_j$, $k_* = \max_j k_j$, and $H_{\max} = \max_j H_j$. Define

$$\begin{aligned} E &= t_2 \sum_{j=1}^L (10s)^{2(j-1)} k_j, \quad T_0 = 2H_*^2 + 4A_*^2(e^2 - 1), \\ D_0 &= (1 + b + s)(1 + T_0 + E + s^2 k_*^2), \\ D_1 &= 576e^3(1 + b + s)D_*^3, \\ C_F &= s(8f_*^2 + H_{\max}^2 + 1), \\ C_{\text{abs}} &= 8(D_0 + 1). \end{aligned} \quad (\text{S.8})$$

The subscript in the last constant means the scalar absorption coefficient. It is unrelated to the initial first-weight matrix. All quantities in (S.5)–(S.8) depend only on depth and activation bounds.

For the real activity moduli, set

$$\begin{aligned} V_1 &= \tau_1, \quad V_j = \tau_j H_{j-1}^2 + 10sV_{j-1}, \\ G_L &= sH_L + 14t_2V_L + s, \quad G_j = 10sG_{j+1} + s^3 k_{j+1}^2 H_j + 14t_2V_j + s, \\ W_G &= 128\{1 + H_{\max} + s \max_j V_j + \max_j G_j\}. \end{aligned} \quad (\text{S.9})$$

Their interpretation and derivation are in §S.5 below. Finally choose in the stated order

$$\eta = \min\{1, (1024C_{\text{abs}}W_G)^{-1}\}, \quad \mathcal{B} = 1024e^2L, \quad \Lambda = \log(e + \mathcal{B}) + \log(1/\eta),$$

$$S_*^{\text{src}} = \min \left\{ 1, (4A_*)^{-1}, \sqrt{\eta/(4000D_*)}, \sqrt{\eta/[16eD_*\sqrt{2\mathcal{B}}]}, \sqrt{\eta/\Lambda}, (\eta^3/(D_1\mathcal{B}))^{1/4}, (8D_0C_F)^{-1/2} \right\}. \quad (\text{S.10})$$

Every denominator is strictly positive by (S.2), (S.5)–(S.9). The sufficient source condition is $S \leq S_*^{\text{src}}$, or $Y/\lambda \leq S_*^{\text{src}}/16$, together with the separate real-fitting condition. This is a finite evaluable recurrence, independent of m, d, γ, Y, n , confidence, and empirical moment degree.

S.3. Separate joint budgets retain samplewise Schatten control

For each training sample separately let

$$Z_{a,i}^{(j)} = \sup_z |z_{a,i}^{(j)}(z)|, \quad K_{a,i}^{(j)} = \sup_z |k_{a,i}^{(j)}(z)|, \quad \mathcal{H}_a = \frac{1}{n} \sum_{j=1}^L \sum_i e^{\eta(Z_{a,i}^{(j)} + K_{a,i}^{(j)})/S}. \quad (\text{S.11})$$

The suprema are over each current stopped time rectangle. Stop every $\mathcal{H}_a$ at $\mathcal{B}$, and each independently defined cavity budget at $2\mathcal{B}$. This includes the top carrier $k^{(L)} = w$. There is no maximum over samples inside an exponential. The sample RMS running amplitudes at a neuron are

$$Z_i^{(j)} = \left(m^{-1} \sum_a (Z_{a,i}^{(j)})^2 \right)^{1/2}, \quad U_i^{(j)} = \left(m^{-1} \sum_a (K_{a,i}^{(j)}/S)^2 \right)^{1/2}. \quad (\text{S.12})$$

These RMS quantities are not themselves the stopping budget. A single RMS budget would not automatically control each sample's Schatten norms; retaining (S.11) is essential.

For every sample, layer, and real $p \geq 2$, the elementary inequality $x^p \leq (p/e)^p e^x$ yields

$$\left(\frac{1}{n} \sum_i |k_{a,i}^{(j)}|^p \right)^{1/p} \leq (S/\eta)p(2\mathcal{B})^{1/p}, \quad \|k_a^{(j)}\|_\infty \leq (S/\eta) \log(2n\mathcal{B}).$$

The augmented Hessian decomposition and (S.7) then prove

$$\|D^2 F_a\|_{p,n} \leq A_* + (SD_*/\eta)p(2\mathcal{B})^{1/p}, \quad \|D^2 F_a\|_{\text{HS}}/\sqrt{n} \leq H_*,$$

$$\|D^2 F_a\|_{\text{op}} \leq A_* + (SD_*/\eta) \log(2n\mathcal{B}). \quad (\text{S.13})$$

Here $\|M\|_{p,n} = n^{-1/p} \|M\|_{S_p}$, also for rectangular maps, where $\|M\|_{S_p} = (\sum_i \sigma_i(M)^p)^{1/p}$ and $\sigma_i(M)$ are the singular values. The mixed blocks use the RMS of reference features; their action is $U_H h/\sqrt{n}$, which is bounded by $H_j \|U_H\|_F$. Only the carrier diagonals need exponential moments. Normalized residual mixtures retain (S.13), since $\sum_a |r_a|/(m\rho) \leq 1$.

The negative-Gram variational base contracts along the real solution. All short complex/backwards pieces together have norm cost at most two, after the explicit width condition in §S.8 below. The full propagator is therefore bounded by

$$2 \exp\{SA_* + (D_* S^2/\eta) \log(2n\mathcal{B})\}.$$

Condition (S.10) gives $D_* S^2/\eta \leq 1/4000$, leaving the $n^{1/1000}$ variational cap at sufficiently large width: the coefficient J_0 in (IC.8) is fixed, so $J_0 n^{1/4000} \leq n^{1/1000}$ eventually. This coefficient of $\log n$ has not been moved into a width threshold.

The local graph proved in finite deletion uses only polynomial logarithmic coordinate caps. Every samplewise cap in (S.11) gives such a cap without a factor m . Deleted forward amplitudes now have size $b + sZ_{a,i}$; learned columns and rows satisfy the inherited $n^{-1/2}$ polylog n source bounds. The exact top residual offset multiplies both the retained gradient and reverse force. Their norms are respectively $n^{-1/2}$ polylog n and n^{-1} polylog n . Thus the complete unbounded local proof, including its uniform control net and nonlinear remainders, applies on this stronger collection of stops. Its strict powers of width do not change. This invokes the local theorem above in its actual form; it does not assume independence after conditioning on full survival.

S.4. Actual-amplitude traces close in sample RMS

At an interior omitted neuron, the incoming row and outgoing column are independent $N(0, I/n)$ roots conditional on retained initialization. Let $G_{h,a,i}$ and $G_{\delta,a,i}$ be the suprema of their forward-feature and backward-response cavity pairings. Divide the latter by S below. First establish the trace bounds for deterministic control paths; the control-uniform Gaussian event above is uniform over those paths.

In the backward trace, the two endpoints and h Hessian factors use Schatten exponent $h + 2$. The zero-insertion term is at most $2H_*^2$. For $h \geq 1$, the integrated bound is

$$\frac{2S^h}{h!} [A_* + (SD_*/\eta)(h+2)(2\mathcal{B})^{1/(h+2)}]^{h+2}.$$

Split the power into its bounded and carrier parts. Their sums are bounded by $4A_*^2(e^{2SA_*} - 1)$ and

$$\frac{8e^2 D_*^2 S^2 \mathcal{B}}{\eta^2} \sum_{h \geq 1} (2eD_* S^2/\eta)^h (h+2)^2 \leq \frac{576e^3 D_*^3 \mathcal{B} S^4}{\eta^3}.$$

The last inequality uses $\sum_{h \geq 1} q^h (h+2)^2 \leq 36q$ for $q \leq 1/2$, and follows by summing the geometric series and its first two derivatives. Conditions (S.10) imply this restriction and $SA_* \leq 1$. The direct external trace is at most ES : its curvature contraction at layer j has nuclear norm per width at most $t_2(10s)^{2(j-1)} S k_j$. The learned outgoing column adds at most $s^2 k_*^2 S^3$ times its actual forward amplitude. These are exactly the terms in (S.8).

The substantive new point is to retain the sum over driving samples. For each evaluated sample a , each endpoint trace coefficient is bounded by the same number, uniformly in driving sample b . Hence

$$\frac{1}{m} \sum_b |r_b| |h_{b,i}| \leq \rho(b + sZ_i), \quad \frac{1}{m} \sum_b |r_b| |\delta_{b,i}| \leq \rho s S U_i. \quad (\text{S.14})$$

These inequalities follow from Cauchy–Schwarz and the normalized sample RMS in (S.12). They do not replace the time supremum by an instantaneous quantity: the running amplitudes dominate every past integrand.

For the forward reverse-force trace the two forward endpoints have operator and normalized Hilbert–Schmidt bound f_* . In every term with $h \geq 1$ insertions, give one endpoint exponent two, the other infinity, and each Hessian exponent $2h$. Its split series is bounded by $e^{2SA_*} - 1$ and $\sqrt{2\mathcal{B}} \sum_{h \geq 1} (4eD_* S^2/\eta)^h$. Under (S.10) each is at most one. Including the zero term gives the normalized trace bound $8f_*^2$. Its learned incoming row has norm at most $sS^2 U_i H_{j-1}/\sqrt{n}$; pairing against a retained feature costs at most $sS^2 U_i H_{j-1}^2$. The first layer’s direct update costs at most $sS^2 U_i$. Thus C_F bounds all forward feedback coefficients.

The explicit contractions (IC.48)–(IC.54), (S.8), and (S.14) now give, sample by sample,

$$K_{a,i}/S \leq G_{\delta,a,i}/S + [D_0 + D_1 \mathcal{B} S^4/\eta^3](1 + Z_{a,i} + Z_i) + o(1), \quad Z_{a,i} \leq G_{h,a,i} + C_F S^2 U_i + o(1). \quad (\text{S.15})$$

Independent incoming/outgoing cross forms are centered and their uniform fluctuations are already in the insertion event. The direct external trace at the evaluated sample multiplies its own current control $h_{a,i}$, not an average over driving samples. Its coefficient ES costs $ES(b + sZ_{a,i})$, explaining the separate $Z_{a,i}$ term. Only the integrated trace and learned-column terms use (S.14)’s driving-sample RMS. This distinction is essential.

At the first layer, $G_{h,a,i}$ is the initialized Gaussian row paired with v_a . At the top there is no outgoing root and $\sup |w_i|/S \leq b + sZ_i$ by (S.14); take $G_{\delta,a,i} = 0$. Thus both endpoints admit the same scalar argument when their direct terms are assigned as specified below.

Taking sample RMS in (S.15), set $\bar{G}_h = (m^{-1} \sum_a G_{h,a,i}^2)^{1/2}$ and $\bar{G}_\delta = (m^{-1} \sum_a (G_{\delta,a,i}/S)^2)^{1/2}$. Condition (S.10) gives backward coefficient at most $2D_0$, and forward coefficient $D = C_F S^2 \leq 1/(8D_0)$. Taking sample RMS keeps both forward amplitudes and gives

$$U_i \leq \bar{G}_\delta + 2D_0(1 + 2Z_i) + o(1), \quad Z_i \leq \bar{G}_h + D U_i + o(1).$$

The feedback product is $4D_0 D \leq 1/2$. Consequently

$$U_i \leq 2\overline{G}_\delta + 4D_0 + 8D_0\overline{G}_h + o(1), \quad Z_i \leq \frac{1}{2} + 2\overline{G}_h + \overline{G}_\delta/(4D_0) + o(1).$$

For an individual sample, $Z_{a,i} \leq G_{h,a,i} + \frac{1}{2} + \overline{G}_h + \overline{G}_\delta/(4D_0) + o(1)$. Substitution into its backward inequality gives coefficients at most $4D_0$ for the constant, $2D_0$ for $G_{h,a,i}$, $6D_0$ for $\overline{G}_h$, and one for $\overline{G}_\delta$. Adding the forward estimate proves the convenient envelope

$$Z_{a,i} + K_{a,i}/S \leq C_{\text{abs}}(1 + G_{h,a,i} + G_{\delta,a,i}/S + \overline{G}_h + \overline{G}_\delta) + o(1). \quad (\text{S.16})$$

The constants in (S.8) include the direct evaluated-sample term. The coefficient is independent of sample count and deletion count. The smallness choice precedes the empirical moment degree.

S.5. Sample-uniform Gaussian moments, with no assumed moment law

Parameterize a real cavity by its own activity $u = \nu([0, t])/S \in [0, 1]$, freezing after its endpoint. With $v = |u - u'|$, direct integration of (S.1) and (S.5)–(S.6) gives

$$\begin{aligned} \|\Delta w\|_2/\sqrt{n} &\leq H_L S v, \quad \|\Delta A\|_F/\sqrt{n} \leq \tau_1 S^2 v, \quad \|\Delta W^{(j)}\|_F \leq \tau_j H_{j-1} S^2 v, \\ \|\Delta z_a^{(j)}\|_2/\sqrt{n} &\leq V_j S^2 v, \quad \|\Delta h_a^{(j)}\|_2/\sqrt{n} \leq s V_j S^2 v. \end{aligned} \quad (\text{S.17})$$

For a changed backward gate, split the reference carrier at SR . The high part is bounded by

$$\|k\mathbf{1}_{|k|>SR}\|_2/\sqrt{n} \leq (4S/\eta)\sqrt{2\mathcal{B}}e^{-\eta R/4},$$

using $x^2 e^{-x/2} \leq 16$ and (S.11). Choose $R = (4/\eta) \log[8\sqrt{2\mathcal{B}}/(\eta v)]$ for $v > 0$. The high changed-gate contribution is at most sSv . The low part is at most $t_2 V_j S^3 R v$. Since $R \leq \eta^{-1}[10\Lambda + 4\log(1/v)]$, the condition $S^2\Lambda/\eta \leq 1$, together with $v \log(1/v) \leq 2\sqrt{v}/e$, bounds that part by $14t_2 V_j S\sqrt{v}$. The top carrier is treated by this same split; it does not require a coordinate bound on the readout.

At the top, the other contribution is $sH_L S v$. At a lower layer, the changed mixer costs $s^3 k_{j+1}^2 H_j S^3 v$, and propagation costs ten times the gate bound s . This proves the recurrence (S.9) and

$$\|\delta_a^{(j)}(u) - \delta_a^{(j)}(u')\|_2/(S\sqrt{n}) \leq G_j \sqrt{|u - u'|}. \quad (\text{S.18})$$

All constants are independent of $\eta, \mathcal{B}, m$ after the explicit smallness restriction; the displayed recurrences already include the top-carrier repair.

Condition on any autonomous cavity. For either of its scalar Gaussian references, its initial variance and its square-root activity modulus are bounded by (S.5), (S.9), (S.17)–(S.18). Dyadic grids of spacing 4^{-q} give at most $2 \cdot 4^q$ increments of standard deviation bounded by $2G2^{-q}$. Gaussian tails with thresholds proportional to $2^{-q}\sqrt{z^2 + 4(q+1)}$, summed over all levels and with the initial Gaussian value added, give

$$\Pr\{X > W_G(1+z) \mid \text{cavity}\} \leq 4e^{-z^2}. \quad (\text{S.19})$$

Here X can be $G_{h,a,i}$ or $G_{\delta,a,i}/S$. The coefficient 128 in (S.9) exceeds the sum of these dyadic thresholds and the initial-value threshold. Integrating (S.19), or dominating the positive excess by a variable with that tail, gives

$$\mathbb{E}[e^{X^2/(16W_G^2)} \mid \text{cavity}] < 2. \quad (\text{S.20})$$

For example, use $(1+z)^2 \leq 2(1+z^2)$ and $\mathbb{E}e^{Z^2/8} \leq 1 + 4/7$ for a tail at most $4e^{-z^2}$; the resulting $e^{1/8}(1 + 4/7)$ is below two.

For any collection of possibly dependent samplewise references, $\overline{X}^2 = m^{-1} \sum_a X_a^2$. Convexity of the exponential therefore gives the pointwise inequality

$$e^{\overline{X}^2/(16W_G^2)} \leq m^{-1} \sum_a e^{X_a^2/(16W_G^2)}.$$

Equation (S.20) thus holds for both RMS references in (S.16), with the same constant and no independence between samples. Completing the square gives $\mathbb{E}e^{qX} \leq 2e^{4q^2W_G^2}$ for each of the four references. Hölder for their sum and (S.16) imply a moment bound

$$\mathbb{E}e^{q(Z_{a,i}+K_{a,i}/S)} \leq 2 \exp\{qC_{\text{abs}} + 64q^2C_{\text{abs}}^2W_G^2\} + o(1). \quad (\text{S.21})$$

For $q \leq 8\eta$, the right side is below four eventually by (S.10). All higher fixed q also have finite moments. This is a consequence of the stopped equations and Gaussian roots, not a premise concerning trained coordinates.

Initial budgets are valid independently: at zero readout only forward preactivations appear. On bounded preceding covariance sets, their conditional second exponential moments are finite and their conditional means continuous. Conditional Chebyshev gives convergence in probability of each empirical layer budget. Its limiting mean is at most $2e^{\eta^2 H_{\text{max}}^2/2} < 3$, so every initial sample budget is below $\mathcal{B}/2$ with probability tending to one. A fixed sample union is allowed here. The preceding coordinate-small deletion comparison transfers this margin to every fixed-size cavity. It does not union-bound an $O(1/n)$ estimate over deletion subsets.

S.6. Maximum, mixed endpoints, and explicit query coefficients

Impose training-coordinate maximum stops in addition to (S.11). Define

$$C_G = 32 \max(1, H_{\text{max}}, \max_j \tau_j), \quad K_{\text{src}} = 16C_{\text{abs}}(1 + C_G). \quad (\text{S.22})$$

For cavity Gaussian references on a time rectangle of bounded width, a mesh of size n^{-2} has a fixed multiple of n^5 points. Neurons, layers and the fixed sample count increase this to at most n^7 eventually. Complex Gaussian splitting gives tail $4e^{-u^2/(4\sigma^2)}$. The RMS coefficients are at most H_{max} and $S \max \tau_j$, so (S.22) supplies strict Gaussian margins of order $\sqrt{\log(en)}$. Stopped derivatives are polynomial logarithmic in normalized norm; the corresponding raw coordinate derivatives are at most $\sqrt{n}$ polylog n , making the off-grid error vanish. The scalar inequalities (S.15)–(S.16) therefore improve the full stops to

$$\max_{a,j,i} |z_{a,i}^{(j)}| \leq K_{\text{src}} \sqrt{\log(en)}, \quad \max_{a,j,i} |k_{a,i}^{(j)}| \leq K_{\text{src}} S \sqrt{\log(en)}. \quad (\text{S.23})$$

This uses a polynomial Gaussian grid, not Gaussian supremum moments or budget removal. Hence it precedes the complex moment argument.

For a complex great circle $q(z) = u \cos z + v \sin z$, with real orthonormal u, v , define

$$R_a^{(j)} = D_{\Theta} z^{(j)}(q) \nabla_{\Theta} F_a, \quad Q_a^{(j)} = \phi'_j(z^{(j)}(q)) \odot R_a^{(j)}, \quad J^{(j)} = \partial_z z^{(j)}(q(z)).$$

These are residual-free response vectors, not physical velocities. For $|\text{Im } z| \leq 1/8$, the input and every fixed input derivative have norm below two. Put

$$g = \tau_1 + \sum_{j=2}^L \tau_j H_{j-1}, \quad r_j = P_j g, \quad q_j = f_j g, \quad j_j = 20(10s)^{j-1}, \quad b_j = s j_j,$$

$$e_1 = t_2 r_1 P_1, \quad e_j = t_2 r_j P_j + s(q_{j-1} + \tau_j H_{j-1} f_{j-1} + 10e_{j-1}),$$

$$a_1 = t_2 j_1 P_1 + 2s, \quad a_j = t_2 j_j P_j + s(b_{j-1} + 10a_{j-1}),$$

$$T_Q = 8f_* \max_j (f_j H_* + e_j), \quad T_J = 8f_* \max_j a_j. \quad (\text{S.24})$$

Here the layer coefficients r_j have a layer index, while the physical residual r_a has a sample index. RMS bounds on R, Q, J are Sr_j, Sq_j, j_j . To derive the mixed endpoint, differentiate $Q = Dh \nabla F$: the first term uses $Dh D^2 F$ and (S.13); the second term uses the curvature diagonal $\text{diag}(\phi'' R) Dz$, the map $U_H Q / \sqrt{n}$, and a rank-one weight-gradient term. Their normalized Hilbert–Schmidt coefficients are exactly the three forcing terms and propagation in e_j . The angular endpoint substitutes J for R ; its first-layer additional map

is $U_A q'$, whose normalized Hilbert–Schmidt norm is at most two. This gives a_j . Neither endpoint uses a lower-layer coordinate maximum. Applying the asymmetric trace from §S.4 with endpoint Hilbert–Schmidt norm $\max(f_j H_* + e_j)$ or $\max a_j$ proves (S.24).

Set $G_d = 16\sqrt{d+3}$, and define

$$U_1 = 4sK_{\text{src}},$$

$$U_j = 2\{sK_{\text{src}}[H_{j-1}^2 + f_{j-1}^2 + ST_Q + S^2 H_{j-1} q_{j-1}] + G_d q_{j-1} + 1\}, \quad j \geq 2,$$

$$V_1^{\text{qry}} = 2(2G_d + 2sK_{\text{src}}S^2 + 1),$$

$$V_j^{\text{qry}} = 2\{G_d b_{j-1} + sK_{\text{src}}S^2(T_J + H_{j-1}b_{j-1}) + 1\}, \quad j \geq 2, \quad U = \max_j U_j, \quad V = \max_j V_j^{\text{qry}}. \quad (\text{S.25})$$

The four deterministic forward-response terms are, in order, the direct feature pairing, direct reverse observable trace, exterior residual integral, and learned-row correction. Their normalized coefficients are the four terms inside the bracket. The centered Gaussian pairing gives $G_d q_{j-1}$. For the angular derivative both corrections have one activity factor and one training carrier, hence S^2 . These are the exact row-insertion terms from the endpoint insertion proof above with each bounded activation value replaced by its actual layer RMS (S.5).

Frames (u, v) form a bounded subset of $\mathbb{R}^{2d}$. A frame mesh and the three real time/tube coordinates have at most n^{4d+10} points eventually. The multiplier G_d dominates the complex Gaussian tail union. Nearby frames are joined by polar normalization of their linear interpolation; its derivative is bounded near the orthonormal-frame manifold. Thus the same off-grid estimates apply without chart multiplication. The local graph above, with a temporary passive-query preactivation cap $2[G_d H_{\max} + 1 + US^2 + a][\log(en)]^2$, has only polynomial logarithmic deleted controls. Query forward-control terms are centered independent-root forms; the only nonzero same-root query trace uses the training reverse control. The forward source enters only centered independent-root pairings; the reverse source uses the training root twice and gives the trace. This verifies the distinction directly. Consequently the improved coordinate estimates are

$$\max |R_a^{(j)}| \leq US\sqrt{\log(en)}, \quad \max |J^{(j)}| \leq V\sqrt{\log(en)}. \quad (\text{S.26})$$

The initial whole-sphere preactivation maximum is at most $(G_d H_{\max} + 1)\sqrt{\log(en)}$ eventually with probability tending to one, by fresh-row Gaussian tails on a polynomial sphere mesh: the conditional row variance is at most $H_{\max}^2$, the factor $G_d = 16\sqrt{d+3}$ covers the mesh union, and its off-grid error is eventually at most one. Real-time integration of (S.26) adds at most $US^2\sqrt{\log(en)}$, so the temporary passive-query size cap is strictly improved. No additional query-coordinate exponential budget is needed.

S.7. Sharp complex derivative and joint-budget removal

Equation (S.1) and (S.24)–(S.26) give

$$\|\dot{z}_a^{(j)}\|_2/\sqrt{n} \leq 2\rho S r_j, \quad \|\dot{z}_a^{(j)}\|_\infty \leq 2\rho S U_j \sqrt{\log(en)}.$$

Let $N_* = \max_j(r_j, U_j)$, and take any real $p \geq 4$. Interpolate the velocity to exponent $2p/(p-2)$, then apply Hölder with the carrier p -norm from (S.11). This yields

$$\|k_a^{(j)} \odot \dot{z}_a^{(j)}\|_2/\sqrt{n} \leq (2\rho S^2 N_*/\eta)p(2\mathcal{B}\log(en))^{1/p}.$$

For $\log(en) \geq \max(e^2, 2\mathcal{B})$, take $p = \max(4, \log(2\mathcal{B}\log(en)))$. The last exponential factor is at most e , and $p \leq 2\log(e + \log(en))$. This is a deterministic consequence of the stopped exponential budget at a growing real exponent; it does not require growing deletion sets.

Define

$$J_L^{\text{time}} = 2sH_L + 4et_2N_*, \quad J_j^{\text{time}} = 10sJ_{j+1}^{\text{time}} + 2s^3k_{j+1}^2H_j + 4et_2N_*, \quad J_*^{\text{time}} = \max_j J_j^{\text{time}}.$$

Differentiating the backward recursion gives respectively the readout, changed mixer, and changed gate contributions in this recurrence. Therefore

$$\|\dot{\delta}_a^{(j)}\|_2/\sqrt{n} \leq J_*^{\text{time}}\rho[1 + (S^2/\eta)\log(e + \log(en))]. \quad (\text{S.27})$$

The forward derivative is at most $2\rho S \max q_j$. Thus, on any time rectangle whose half-width is $c_t/\sqrt{\log(en)}$ with fixed $c_t > 0$, both normalized complex-minus-real Gaussian coefficient radii tend to zero. For the backward coefficient divided by S , the explicit bound is

$$D_n = (\lambda c_t J_*^{\text{time}}/4)[\log(en)]^{-1/2}[1 + (S^2/\eta)\log(e + \log(en))]. \quad (\text{S.28})$$

The two time-parameter Lipschitz coefficients have the same bracket times fixed constants. Their parameter interval has length $O(\log(en))$. The Gaussian rectangle estimate (IC.77), including its sample-RMS form, is proved independently below from finite Gaussian nets and (IC.59); its proof does not use the source event being established here. These nets therefore have mean $O([\log(en)]^{-1/2}[\log(e + \log(en))]^{3/2})$ and tail scale $O([\log(en)]^{-1/2}\log(e + \log(en)))$. Both vanish. Their exponential moments at every fixed order tend to one, as do fixed squared-exponential moments; the latter also transfer to RMS sample corrections by the Jensen argument of §S.5.

Each cavity is frozen or clamped on its own domain and assigned zero reference paths on failed own initialization. All Gaussian laws are then conditional on retained initialization alone. Common-cavity path differences are projected onto deterministic Euclidean balls of radius $n^{1/90}$, which dominates the fixed-deletion Euclidean error eventually. Projection is nonexpansive and preserves independence from the omitted root. Its Gaussian radius is $n^{-22/45}$, so the real activity moduli and the complex nets give vanishing moments at every fixed order. No full-network survival event is used as a Gaussian conditioning event.

For a fixed sample and a fixed layer, expand the p th empirical moment at fixed integer p . Distinct neurons have independent root pairs conditional on their common same-layer cavity. Equations (S.16), (S.21), and the vanishing corrections give a main moment base below sixteen. A fixed Hölder split handles projected errors at orders proportional to p ; these moments tend to one at each fixed p . Collision tuples have $O_p(n^{p-1})$ choices and finite higher fixed moments, hence vanishing normalized contribution. The local exceptional probabilities are superpolynomial and the stopped budgets are bounded.

The coordinate comparison transfers each individual joint budget:

$$\mathcal{H}_a^{\text{cavity}} \leq e^{\eta o(1)(1+1/S)}\mathcal{H}_a + O_p(1/n) < 2\mathcal{B}.$$

Remove full-event indicators only after bounding by the nonnegative, cavity-measurable Gaussian references. If any budget is hit, some sample and layer has empirical budget at least $\mathcal{B}/L$. Thus

$$\limsup_{n \rightarrow \infty} \Pr\{\text{a joint budget is hit}\} \leq mL(16L/\mathcal{B})^p. \quad (\text{S.29})$$

Take the width limit for each fixed p , then its infimum over positive integers. Since $16L/\mathcal{B} < 1$, the right side tends to zero. This removes all joint budgets. Sample count affects a fixed union and eventual width, not the source allowance in (S.10).

S.8. Label-sensitive time radius and the whole-sphere domain

Take

$$c_t = \frac{a}{64YSU}, \quad c_q = \min\{1/8, a/(8V)\}, \quad r_t = c_t/\sqrt{\log(en)}, \quad r_q = c_q/\sqrt{\log(en)}. \quad (\text{S.30})$$

For $d \geq 2$, the intrinsic query tube consists of $u \cosh h + iv \sinh h$, with real orthonormal u, v and $|h| \leq r_q$. It is exactly the complex quadric neighborhood $q^\top q = 1$, $\|\text{Im } q\|_2 \leq \sinh r_q$. For $d = 1$, retain the two queries ± 1 separately.

Set

$$\mathcal{K} = H_L^2 + S^2[\tau_1^2 + \sum_{j=2}^L \tau_j^2 H_{j-1}^2], \quad D_W = \max\{\tau_1, \max_{j \geq 2} \tau_j H_{j-1}\}.$$

The actual algebraic residual Gram divided by m , and the negative-Gram variational generator divided by its factor two, have operator norm at most $\mathcal{K}$. This follows by summing the squared normalized gradient-block norms, not by complex positivity. Impose the explicit eventual width conditions

$$n^{-1} \leq Y, \quad \sqrt{\log(en)} \geq c_t \max\{8, \lambda, 4\mathcal{K}/\log 2, 32YSD_W\}. \quad (\text{S.31})$$

Follow the real solution to the nearest real anchor in $[0, T]$, then at most two short pieces of total length $2r_t$. Their residual growth and base-propagator cost are at most $e^{4\mathcal{K}r_t} \leq 2$. The extra ν -activity is at most $8Yr_t \leq S/2$. The first-weight normalized and hidden-mixer increments are at most $8YSD_W r_t \leq 1/4$, preserving strict margins inside cap ten. No long horizontal complex contour is used.

The exact physical derivative is $\partial_t z^{(j)} = -(2/m) \sum_a r_a R_a^{(j)}$. On the short pieces, $\rho \leq 2Y$, so (S.26) bounds its coordinate magnitude by $4YSU\sqrt{\log(en)}$. The short time pieces and one intrinsic query segment therefore have total preactivation displacement at most

$$8c_t YSU + c_q V \leq a/8 + a/8 = a/4. \quad (\text{S.32})$$

Full pole stops at $3a/8$ and cavity stops at $7a/16$ have strict separation inside the derivative strip $a/2$. Equation (S.32) improves the full stop, and the coordinate comparison transfers its prefix to all fixed-size cavity stops. The passive-query size stop improves by §S.6. The logical order is local transfer, maximum improvement, mixed endpoint and response improvement, query size/pole improvement, complex Gaussian moments, and only then (S.29). Holomorphy holds on a neighborhood of the closed rectangle times the closed query tube.

The radius coefficient c_t need not be at most one: its actual radius shrinks at each fixed $Y > 0$. Condition (S.31) explicitly includes

$$\log(en) \geq 64c_t^2 = a^2\lambda^2/(16384Y^4U^2).$$

Thus the label-sensitive statement is not uniform as $Y \downarrow 0$. No large fixed radius coefficient is used at an unchanged width.

Lemma E.1 (Qualitative complex query-tube source event). *Consider the canonical Gaussian model of §2, with fixed admissible data, activation functions and depth, population Gram gap $\gamma > 0$, and $\lambda = \gamma/m$. Suppose*

$$0 < Y/\lambda \leq \min\{(8H_D\sqrt{F_D})^{-1}, S_*^{\text{src}}/16\}.$$

Use the source coefficients defined above, put $S = 16Y/\lambda$, $\ell = \log(en)$ and $T = 32\ell/\lambda$, and take r_t, r_q from (S.30). Define

$$K_n = [-r_t, T + r_t] + i[-r_t, r_t].$$

For $d \geq 2$, let

$$\mathcal{Q}_{r_q} = \{q \in \mathbb{C}^d : q^\top q = 1, \|\text{Im } q\|_2 \leq \sinh r_q\};$$

for $d = 1$, use the two queries $\{-1, 1\}$.

For every fixed $0 < \delta < 1$, at every sufficiently large width, with probability at least $1 - \delta$, the dense trajectory and its training fields extend holomorphically to a neighborhood of K_n , and its passive forward fields extend jointly to a neighborhood of $K_n \times \mathcal{Q}_{r_q}$, relative to the complex quadric when $d \geq 2$. On these domains the normalized first-weight and hidden-matrix operator norms are at most ten. The forward, training carrier and training response RMS bounds are respectively

$$\frac{\|h^{(j)}\|_2}{\sqrt{n}} \leq H_j, \quad \frac{\|k_a^{(j)}\|_2}{\sqrt{n}} \leq Sk_j, \quad \frac{\|\delta_a^{(j)}\|_2}{\sqrt{n}} \leq S\tau_j,$$

with coefficients from (S.5)–(S.6). The training-coordinate bounds (S.23) and the response and angular bounds (S.26) hold. Every training budget (S.11) is at most $\mathcal{B}$. The real trajectory has the global fitting and endpoint properties of §D.5.

The sufficient width includes (S.31) and $\ell \geq \max(e^2, 2\mathcal{B})$; its stochastic part is qualitative.

Proof. Dense fitting is independent of the source construction. The independently stopped finite-deletion comparison and Gaussian insertion are proved in §E.1.1. The recurrences and scalar absorption above give (S.23) and (S.26). The complex correction bounds (S.27)–(S.28) and the fixed-moment argument (S.29) remove the budgets. Their failure probabilities, and those of the initialization and local events, tend to zero, giving the stated fixed-confidence quantifier. Finally, (S.31)–(S.32) give strict operator, activity and pole margins, hence continuation across the closed domain. $\square$

E.1.3 Exact activation-power ledger on the explicit sufficient label cap

This subsection proves the numerical envelope; it does not replace the larger recurrence allowance. Here only put $z = Y/\lambda$ and suppose $z \leq \beta^{-30L}$. Then $S = 16z \leq \beta^{-26L}$. The references labelled S are the explicit recurrences above; the references labelled SP are the power inequalities in this subsection.

SP.1. Elementary absorptions and source base recurrences

All bounds below use $b \leq \beta - 1$, $s, t_2 \leq \beta$, $10s \leq \beta^2$, and

$$L \leq \beta^{L-1}, \quad L+1 \leq \beta^{L-1}, \quad 4j \leq \beta^j \quad (j \geq 1), \quad \sum_{j=1}^L \beta^{rj} \leq \frac{\beta^{rL}}{1 - \beta^{-r}} \quad (r > 0). \quad (\text{SP.3})$$

The first two inequalities follow by induction from $L = 2$; the third follows by induction from $j = 1$. Also $\log \beta \leq \beta$, and $30 \leq \beta^2$.

For the source forward recurrence, unrolling the affine map gives

$$H_j \leq 22\beta(10\beta)^{j-1} \leq 3\beta^{2j}.$$

For example its first value is at most 21β , and the subsequent additive contributions sum to at most $\beta(10\beta)^{j-1}/(10\beta - 1)$. Unrolling the augmented-port derivative recurrence similarly gives

$$P_j \leq 4j(10\beta)^{j-1} \leq \beta^{3j-2}.$$

Consequently, for $1 \leq j \leq L$,

$$H_j \leq 3\beta^{2j}, \quad k_j \leq 3\beta^{4L-2j}, \quad \tau_j \leq 3\beta^{4L-2j+1}, \quad P_j \leq \beta^{3j-2}, \quad f_j \leq \beta^{3j-1}. \quad (\text{SP.4})$$

These bounds use the actual source RMS recurrence and no global bound on the activation values.

SP.2. Trace constants and the numerical source allowance

Direct substitution of (SP.4) into source (S.7)–(S.8), using the geometric sum in (SP.3), gives the following ledger:

Detailed comparison

Source constant: A_*

Intermediate bound $7\beta^{5L-3}$

Power bound β^{5L-2}

Source constant: D_*

Intermediate bound $2\beta^{6L-3}$

Detailed comparison

Power bound β^{6L-2}

Source constant: H_*

Intermediate bound $4\beta^{8L-3}$

Power bound β^{8L-2}

Source constant: E

Intermediate bound $4\beta^{6L-3}$

Power bound β^{6L-2}

Source constant: T_0

Intermediate bound $3\beta^{16L-4}$

Power bound β^{16L-3}

Source constant: D_0

Intermediate bound $8\beta^{16L-3}$

Power bound β^{16L-2}

Source constant: D_1

Intermediate bound $9216e^3\beta^{18L-8}$

Power bound β^{18L-2}

Source constant: C_F

Intermediate bound $9\beta^{6L-1}$

Power bound β^{6L}

Source constant: C_{abs}

Intermediate bound $65\beta^{16L-3}$

Power bound β^{16L-1}

Here are the exponent checks for the terms that could otherwise be hidden by this table. The mixed summand in A_* is at most $6\beta^{4L+j-3}$; the curvature summand in H_* is at most $3\beta^{4L+4j-3}$. The summand in E is at most $3\beta^{4L+2j-3}$. In D_0 , its outer factor is at most 2β , and its inner sum is at most

$$1 + 3\beta^{16L-4} + 4\beta^{6L-3} + 9\beta^{8L-2} \leq 4\beta^{16L-4}.$$

For D_1 , the coefficient $9216e^3 < 200000 \leq \beta^6$. In C_F , the inner expression is at most $8\beta^{6L-2} + 9\beta^{4L} + 1 \leq 9\beta^{6L-2}$. All displayed comparisons hold already at $L = 2, \beta = 10$, and the ratios of the smaller powers decrease when either variable increases.

To check the real activity moduli of source (S.9), unroll both recurrences. The first one gives

$$V_j \leq 27j\beta^{4L+2j-3}, \quad \max_j V_j \leq \beta^{7L-2}. \quad (\text{SP.5})$$

Indeed every propagated forcing term has exponent $4L + 2j - 3$. For the reverse recurrence, its terminal value satisfies $G_L \leq 380L\beta^{6L-2}$. The propagated terminal contribution, mixed-weight contributions, and curvature contributions at level j are bounded respectively by

$$380L\beta^{8L-2j-2}, \quad 27L\beta^{8L-2j-1}, \quad 379L\beta^{8L-2j-6}.$$

The remaining constant contributions total at most $2\beta^{2L-2j-1}$. Hence

$$G_j \leq 66L\beta^{8L-2j-1}, \quad \max_j G_j \leq \beta^{9L-2}, \quad W_G \leq \beta^{9L+1}. \quad (\text{SP.6})$$

For the last bound, the braces in the definition of W_G are at most $2\beta^{9L-2}$, and $256 \leq \beta^3$.

The source choices of $\eta, \mathcal{B}, \Lambda$ therefore obey

$$\eta^{-1} \leq \beta^{25L+4}, \quad \mathcal{B} = 1024e^2L \leq \beta^{L+3}, \quad \Lambda \leq (26L+8) \log \beta \leq \beta^{2L}. \quad (\text{SP.7})$$

The first inequality uses $1024 \leq \beta^4$; the second uses $1024e^2 < 7600 \leq \beta^4$. For the last one, $(26L+8) \log \beta \leq 30L \log \beta \leq \beta^{L+2} \leq \beta^{2L}$.

Each nontrivial entry in the minimum defining S_*^{src} is at least β^{-p} , with the following exponents:

Entry in source (S.10)	p
$(4A_*)^{-1}$	$5L - 1$
$\sqrt{\eta/(4000D_*)}$	$(31L + 6)/2$
$\sqrt{\eta/(16eD_*\sqrt{2\mathcal{B}})}$	$(63L + 12)/4$
$\sqrt{\eta/\Lambda}$	$(27L + 4)/2$
$(\eta^3/(D_1\mathcal{B}))^{1/4}$	$(94L + 13)/4$
$(8D_0C_F)^{-1/2}$	$(22L - 1)/2$

These use $4000 \leq \beta^4$, $16e \leq \beta^2$, and $\sqrt{2\mathcal{B}} \leq \beta^{(L+4)/2}$. Every listed exponent is at most $26L$ for $L \geq 2$. We have proved

$$S_*^{\text{src}} \geq \beta^{-26L}. \quad (\text{SP.8})$$

In particular the sufficient source label allowance is at least $\lambda\beta^{-26L}/16$, and the common cap $Y/\lambda \leq \beta^{-30L}$ satisfies it.

SP.3. Carrier and query coefficients

source (S.22) and (SP.4) give

$$C_G \leq 96\beta^{4L-1}, \quad K_{\text{src}} \leq 1552\beta^{20L-2} \leq \beta^{20L+2} \leq \beta^{21L}. \quad (\text{SP.9})$$

For source (S.24), direct substitution and geometric summation give

$$g \leq 9L\beta^{4L-1} \leq \beta^{5L-1}, \quad r_j \leq \beta^{5L+3j-3}, \quad q_j \leq \beta^{5L+3j-2},$$

$$j_j \leq 2\beta^{2j-1}, \quad b_j \leq 2\beta^{2j}, \quad e_j \leq 2\beta^{5L+6j-4}, \quad a_j \leq 3\beta^{5j-2},$$

$$T_Q \leq \beta^{14L-3}, \quad T_J \leq \beta^{8L-1}. \quad (\text{SP.10})$$

For clarity, the three forcing terms in the recurrence for e_j are at most $\beta^{5L+6j-4}$, $\beta^{5L+3j-4}$, and $9\beta^{4L+3j-4}$; its propagation factor is at most β^2 . The leading exponents grow by six per layer, so their propagated sum is bounded by the factor two in (SP.10). The angular recurrence has leading forcing $2\beta^{5j-2}$, lower forcing $2\beta^{2j-1}$, and propagation factor β^2 ; factor three suffices, including $a_1 \leq 62\beta$. Finally $f_*H_* \leq \beta^{11L-3}$, and $8f_* \max(f_jH_* + e_j) \leq \beta^{14L-3}$; $24 \leq \beta^2$ proves the stated T_J bound.

Under $S \leq \beta^{-26L}$, both ST_Q and every $S^2H_{j-1}q_{j-1}$ are at most one. The braces multiplying sK_{src} in source (S.25) are at most $11\beta^{6L-8}$. Using the sharper bound $K_{\text{src}} \leq \beta^{20L+2}$ from (SP.9), we obtain for $j \geq 2$

$$U_j \leq 22\beta^{26L-5} + 32\sqrt{d+3}\beta^{8L-5} + 2.$$

Also $U_1 \leq 4\beta^{20L+3}$. These imply

$$U \leq \beta^{26L}\sqrt{d+3}. \quad (\text{SP.11})$$

For the query coefficient, the term $sK_{\text{src}}S^2(T_J + H_{j-1}b_{j-1})$ is at most $2\beta^{-24L+2} \leq 1$. Thus

$$V_j^{\text{qry}} \leq 64\sqrt{d+3}\beta^{2L-2} + 4 \leq \beta^{2L}\sqrt{d+3} \quad (j \geq 2).$$

The first-layer bound is at most $67\sqrt{d+3}$, and obeys the same envelope. Therefore

$$V \leq \beta^{2L}\sqrt{d+3}, \quad c_q^{-1} = \max\{8, 8V/a\} \leq \beta^{2L+1}\sqrt{d+3}. \quad (\text{SP.12})$$

E.2 Dense proofs

The initialization event first gives a global fitting tube and a uniform endpoint tail. On the trained source event, endpoint subtraction and signed parameter energy then give Gaussian concentration in the complete trajectory norm. Independently, the initialized Gram CLT and a distribution-free innovation inequality identify a fluctuating training derivative. The finite-query complex extension and polynomial derivative inequality convert that derivative into an actual positive-time witness.

E.2.1 Proof of the initialization event

A maximal r -separated set of a Euclidean unit sphere is an r -net of size at most $(1 + 2/r)^k$ in ambient dimension k : the disjoint radius- $r/2$ balls about its points lie in the ball of radius $1 + r/2$, and comparison of volumes gives that count. For two $1/4$ -nets, approximating a maximizing pair of unit vectors in a bilinear form gives $\|M\|_{\text{op}} \leq 2 \max_{u,v \text{ in nets}} |u^T M v|$. For a hidden initialized mixer each pairing has law $N(0, 1/n)$; for $A_0/\sqrt{n}$ it has the same law. The elementary Gaussian tail bound $\Pr\{|G| > z\} \leq 2e^{-z^2/2}$, followed by the net union, therefore gives

$$\Pr\{\|W_0^{(j)}\|_{\text{op}} > 8\} \leq 2e^{-(8-2\log 9)n}, \quad \Pr\{\|A_0\|_{\text{op}}/\sqrt{n} > 8\} \leq 2e^{-(8-\log 9)n+d\log 9}.$$

The scalar tail itself follows by applying Markov's inequality to e^{zG} and e^{-zG} , using $\mathbb{E}e^{zG} = e^{z^2/2}$. The displayed width terms make their combined loss smaller than $\alpha/2$.

For a positive semidefinite covariance C , let $\Psi_j(C)_{ab} = \mathbb{E}\phi_j(Z_a)\phi_j(Z_b)$, where $Z \sim N(0, C)$. Along a symmetric covariance direction E , differentiating the nonsingular Gaussian density and integrating twice by parts yields

$$(D\Psi_j(C)[E])_{ab} = \frac{1}{2}E_{aa}\mathbb{E}[\phi_j''(Z_a)\phi_j(Z_b)] + E_{ab}\mathbb{E}[\phi_j'(Z_a)\phi_j'(Z_b)] + \frac{1}{2}E_{bb}\mathbb{E}[\phi_j(Z_a)\phi_j''(Z_b)].$$

This is the Gaussian identity $DE F(Z)[E] = \frac{1}{2}\sum_{u,v} E_{uv}\mathbb{E}\partial_{uv}F(Z)$ for $F(z) = \phi_j(z_a)\phi_j(z_b)$. Linear growth and bounded first two derivatives make the boundary terms vanish and dominate the differentiated integrands by integrable Gaussian polynomials. For singular endpoints, add ηI to the covariance segment, integrate the identity on that segment, and let $\eta \downarrow 0$. The continuous positive square-root coupling and Gaussian moment domination give the same integrated identity on the closed positive semidefinite cone. For a diagonal entry it reads $E_{aa}\mathbb{E}[(\phi_j')^2 + \phi_j\phi_j'']$.

If both covariance diagonals are at most $2H_D^2$, then $\mathbb{E}|\phi_j(Z_a)| \leq b + \sqrt{2}sH_D$. Integrating along the segment thus gives

$$\|\Psi_j(C) - \Psi_j(C')\|_{\text{max}} \leq A_D\|C - C'\|_{\text{max}}.$$

Also $(u+v)^4 \leq 8(u^4 + v^4)$ and $\mathbb{E}Z_a^4 \leq 12H_D^4$ imply $\mathbb{E}|\phi_j(Z_a)|^4 \leq M_D$. Cauchy-Schwarz bounds the variance of each product $\phi_j(Z_a)\phi_j(Z_b)$ by M_D .

Conditionally on the preceding initialized layers, the next layer has n independent Gaussian rows with covariance equal to the preceding empirical feature Gram. For each of its m^2 training entries and the squared norm at each of the P_D query-net points, conditional Chebyshev at tolerance e_D/T_D has failure at most $M_D T_D^2/(ne_D^2)$. At the first layer the input covariance is deterministic and the same calculation is unconditional. If all preceding tests passed, the covariance error after layer j is at most

$$\frac{e_D}{T_D} \sum_{i=0}^{j-1} A_D^i \leq e_D.$$

Its input diagonal is at most $H_D^2 + e_D < 2H_D^2$, so the conditional variance condition closes inductively. A stopped union bound over the L layers has failure at most

$$\frac{L(m^2 + P_D)M_DT_D^2}{ne_D^2} \leq \alpha/2.$$

The top training covariance error has operator norm at most $me_D \leq \gamma/2$, proving the initial Gram gap after division by m . On the matrix event, the initialized layer- j feature map, measured in RMS, is $(8s)^j$ -Lipschitz in v . A net-query feature norm is at most $\sqrt{H_D^2 + e_D} \leq \sqrt{5/4}H_D$; the off-net addition is at most $(8s)^L h_D \leq H_D/4$. Since $\sqrt{5/4} + 1/4 < 3/2$, this proves the sphere bound and finishes the initialization probability calculation.

E.2.2 Proof of global fitting and the uniform tail

Stop at the first failure of mixer cap nine, sphere feature cap $2H_D$, or normalized readout-Gram gap $\lambda/4$. Backward propagation in this tube gives $\|\delta_a^{(j)}\|_2/\sqrt{n} \leq D_j^D \|w\|_2/\sqrt{n}$. Cauchy-Schwarz in the sample index bounds each hidden block's normalized Frobenius velocity by $2\rho U_j^D \|w\|_2/\sqrt{n}$.

In normalized parameter coordinates the flow is minus the gradient of ρ^2 . Therefore

$$-\frac{d}{dt}\rho^2 = \|\dot{\theta}\|_{\text{par}}^2.$$

The readout block alone contributes at least $4(\lambda/4)\rho^2 = \lambda\rho^2$, proving the residual decay on the stopped interval. While $\rho > 0$,

$$\left(\int \|\dot{\theta}\|_{\text{par}} dt\right)^2 \leq \left(\int \frac{\|\dot{\theta}\|_{\text{par}}^2}{\rho} dt\right) \left(\int \rho dt\right) \leq (2Y)(2Y/\lambda).$$

The first factor is $-2 \int d\rho \leq 2Y$. If the residual reaches zero, every velocity becomes zero, so the estimates continue without dividing by zero. Since $w_0 = 0$, the path-length bound controls its RMS by $2Y/\sqrt{\lambda}$. Integrating the hidden velocities then yields the displayed hidden displacement bounds.

For an arbitrary unit query, forward subtraction gives the first-layer feature error at most $s\|A - A_0\|_F/\sqrt{n}$, and at a later layer at most

$$s \left[2H_D \|W^{(j)} - W_0^{(j)}\|_F + 9\|h^{(j-1)} - h_0^{(j-1)}\|_2/\sqrt{n} \right].$$

This is precisely the recurrence defining F_j^D . The label allowance gives

$$\frac{8F_D Y^2}{\lambda^{3/2}} \leq \frac{\sqrt{\lambda}}{8H_D^2} \leq \frac{1}{8H_D}.$$

The matrix cap thus improves to at most $8 + 1/8$; the feature cap improves to at most $3H_D/2 + H_D/8$. The training feature matrix divided by $\sqrt{mn}$ changes in operator norm by at most $\sqrt{\lambda}/(8H_D^2) \leq \sqrt{\lambda}/8$. Its smallest singular value stays above $\sqrt{\lambda}(1/\sqrt{2} - 1/8) > \sqrt{\lambda}/2$, improving the stopped Gram bound strictly. Hence no finite first exit exists.

At fixed n , bounded parameter norm on every finite interval and local Lipschitz continuity of the vector field give global continuation. The finite total path length makes the parameters Cauchy as $t \rightarrow \infty$; the residual decay proves fitting. Finally,

$$\|\dot{w}\|_2/\sqrt{n} \leq 4H_D\rho, \quad \|\dot{h}^{(L)}(v)\|_2/\sqrt{n} \leq 2\rho F_D \|w\|_2/\sqrt{n},$$

so

$$|\dot{f}_n(t, v)| \leq 8(H_D^2 + F_D Y^2/\lambda) \rho(t).$$

Integration gives the first endpoint tail. The second follows from $F_D Y^2/\lambda \leq \lambda/(64H_D^2) \leq 1/64$ and $H_D \geq 1$. All estimates are uniform on the sphere.

The simpler sufficient real-fitting cap is $Y \leq \lambda\beta^{-5L}$. Indeed $H_D \leq (2\beta)^L$, the geometric sum gives $F_D \leq (21/20)H_D^2 s^2 (9s)^{2L-2}$, and hence

$$8H_D\sqrt{F_D} \leq H_D^2(9s)^L \leq 36^L\beta^{3L} \leq \beta^{5L}.$$

This sufficient cap does not replace the larger recurrence allowance.

E.2.3 Endpoint subtraction and the signed parameter energy

Use Euclidean/Frobenius normalized coordinates $\theta = (A/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$. For two actual good states, write $e_\theta = \theta - \theta'$, $d_\theta = \|e_\theta\|_2$, and

$$D_h = \|A - A'\|_F/\sqrt{n} + \sum_{j=2}^L \|W^{(j)} - W'^{(j)}\|_F, \quad D_w = \|w - w'\|_2/\sqrt{n}.$$

Cauchy-Schwarz gives $D_h \leq \sqrt{L}d_\theta$ and $D_h + D_w \leq \sqrt{L+1}d_\theta$. Forward subtraction first gives $\|\Delta z^{(1)}\|_2/\sqrt{n} \leq D_h$. At each later layer use $\Delta z^{(j)} = W^{(j)}\Delta h^{(j-1)} + \Delta W^{(j)}h'^{(j-1)}$. The operator caps, feature RMS bounds, and s -Lipschitz activations then give, uniformly on the query sphere,

$$\|\Delta z^{(j)}\|_2/\sqrt{n} \leq F_z D_h, \quad \|\Delta h^{(j)}\|_2/\sqrt{n} \leq sF_z D_h.$$

The coefficient $F_z = 2H_D T^{L-1}$ dominates every propagated block coefficient because $T \geq 9$, $H_D \geq 1$, and the block increments are summed in D_h .

For a training sample, the gate difference in backward subtraction is bounded by $t_2 M_n F_z D_h$, using the carrier of one actual endpoint. The remaining terms are a changed readout, or a changed mixer multiplying a reference backward vector, followed by propagation with factor T . To make the recursion explicit, put $b_j^{\text{err}} = \|\delta_a^{(j)} - \delta_a'^{(j)}\|_2/\sqrt{n}$. It obeys

$$b_L^{\text{err}} \leq sD_w + t_2 M_n F_z D_h, \quad b_j^{\text{err}} \leq T b_{j+1}^{\text{err}} + sB_\delta \|\Delta W^{(j+1)}\|_F + t_2 M_n F_z D_h.$$

Unrolling it gives

$$b_j^{\text{err}} \leq sT^{L-j}D_w + sB_\delta \sum_{i=j+1}^L T^{i-j-1} \|\Delta W^{(i)}\|_F + t_2 M_n F_z \sum_{r=0}^{L-j} T^r D_h.$$

The three coefficients are at most sT^{L-1} , $sB_\delta T^{L-1}$, and $Lt_2 M_n F_z T^{L-1}$, respectively. Thus

$$\|\delta_a^{(j)} - \delta_a'^{(j)}\|_2/\sqrt{n} \leq D_\delta(M_n)(D_h + D_w).$$

The gradient of one prediction has the exact block form

$$\xi_a := \nabla_\theta f_n(v_a) = \left(\frac{\delta_a^{(1)} v_a^T}{\sqrt{n}}, \left(\frac{\delta_a^{(j)} (h_a^{(j-1)})^T}{n} \right)_{j=2}^L, \frac{h_a^{(L)}}{\sqrt{n}} \right).$$

Its block norms sum to at most G_* . Subtracting each block gives one first-layer backward difference, $L-1$ hidden products each bounded by $2H_D D_\delta(D_h + D_w) + B_\delta sF_z D_h$, and a readout feature difference $sF_z D_h$. Thus

$$\|\xi_a\|_2 \leq G_*, \quad \|\xi_a - \xi_a'\|_2 \leq J(M_n)(D_h + D_w) \leq \sqrt{L+1}J(M_n)d_\theta.$$

The Taylor remainder can also be evaluated entirely at actual endpoints. Anchor derivatives and carriers at the unprimed endpoint, and define

$$a^{(j)} = \Delta h^{(j)} - \text{diag}(\phi_j'(z^{(j)}))\Delta z^{(j)}, \quad e^{(j)} = \Delta h^{(j)} - D_\theta h^{(j)}[e_\theta].$$

Scalar Taylor's formula gives $|a_i^{(j)}| \leq (t_2/2)|\Delta z_i^{(j)}|^2$. Direct differentiation and subtraction yield

$$e^{(1)} = a^{(1)}, \quad e^{(j)} = \text{diag}(\phi'_j(z^{(j)})) [W^{(j)} e^{(j-1)} - \Delta W^{(j)} \Delta h^{(j-1)}] + a^{(j)}.$$

Expanding this recurrence backward from the readout gives exactly

$$\begin{aligned} f_n(\theta, v_a) - f_n(\theta', v_a) - \langle \xi_a(\theta), e_\theta \rangle &= \frac{1}{n} \sum_{j=1}^L (k_a^{(j)})^T a_a^{(j)} \\ &\quad - \frac{1}{n} \sum_{j=2}^L (\delta_a^{(j)})^T \Delta W^{(j)} \Delta h_a^{(j-1)} - \frac{1}{n} \Delta w^T \Delta h_a^{(L)}. \end{aligned}$$

The last term is bounded by $sF_z D_w D_h \leq sF_z \sqrt{L} d_\theta^2$. The middle sum is at most $B_\delta s F_z D_h^2 \leq L B_\delta s F_z d_\theta^2$. The first sum is at most $(L t_2 M_n / 2) F_z^2 D_h^2 \leq (L^2 t_2 F_z^2 M_n / 2) d_\theta^2$. This proves the remainder bound $T_{\text{rem}}(M_n) d_\theta^2$. Let $v_a^{\text{err}} = r_a - r'_a$ and $\|v^{\text{err}}\|_m^2 = m^{-1} \sum_a |v_a^{\text{err}}|^2$. Since $\theta = -(2/m) \sum_a r_a \xi_a$,

$$\frac{1}{2} \frac{d}{dt} d_\theta^2 = -\frac{2}{m} \sum_a v_a^{\text{err}} \langle e_\theta, \xi_a \rangle - \frac{2}{m} \sum_a r'_a \langle e_\theta, \xi_a - \xi'_a \rangle.$$

Insert the remainder identity into the first pairing and use $\|v^{\text{err}}\|_m \leq \rho + \rho'$. The result is

$$\frac{1}{2} \frac{d}{dt} d_\theta^2 \leq -2 \|v^{\text{err}}\|_m^2 + 2 \{T_{\text{rem}}(M_n)(\rho + \rho') + \sqrt{L+1} J(M_n) \rho'\} d_\theta^2.$$

Discard the nonpositive prediction-difference square for the upper bound. The remaining Taylor-remainder and Jacobian-variation terms are weighted by residuals. Both residual integrals are at most Y/κ , hence

$$\sup_{t \geq 0} d_\theta(t) \leq d_\theta(0) e^{E_{\text{en}}(M_n)}.$$

At a vanishing discrepancy this follows by the regularized norm or by uniqueness; no division by a zero discrepancy is needed.

E.2.4 Residual damping and the readout factor

The tangent Gram is $K_{ab} = \langle \xi_a, \xi_b \rangle$. Its normalized gap is at least $\lambda/4$, because its readout block is the top feature Gram and its other blocks are positive semidefinite. The preceding endpoint estimates give

$$\|(K - K')/m\|_{\text{op}} \leq 2G_* J(M_n)(D_h + D_w).$$

Indeed each entry is bounded by that quantity, and the operator norm of an $m \times m$ matrix divided by m is at most its maximum entry magnitude. Subtract the exact equations $\dot{r} = -(2/m)Kr$. The homogeneous propagator contracts at rate κ , and the initial residual difference is zero. Variation of constants and Tonelli therefore give

$$\int_0^\infty \|r - r'\|_2 / \sqrt{m} dt \leq \frac{4G_* J(M_n)}{\kappa} \int_0^\infty \rho(D_h + D_w) dt,$$

with ρ chosen from the path multiplying the kernel difference. The exact readout subtraction, with equal zero initial readouts, gives

$$D_w(t) \leq 4H_D \int_0^t \|r - r'\|_2 / \sqrt{m} ds + 2sF_z \int_0^t \rho D_h ds.$$

Combining these with the already closed energy estimate proves

$$\sup_t D_w(t) \leq \sqrt{L+1} \left(\frac{16H_D G_* J(M_n)}{\kappa} + 2sF_z \right) \frac{Y}{\kappa} d_\theta(0) e^{E_{\text{en}}(M_n)}.$$

The sphere prediction difference is at most $2H_D D_w + R s F_z D_h$. Substitution yields

$$\|f - f'\|_* \leq \sqrt{L+1} Y C_Y(M_n) d_\theta(0) e^{E_{\text{en}}(M_n)}.$$

This derivation uses the reciprocal gap only after energy stability is closed; it is not fed back into the Gronwall exponent.

To recover the original unsigned certificate, put $D = D_h + D_w$. Blockwise parameter subtraction gives

$$D(t) \leq D(0) + 2G_* \int_0^t \|r - r'\|_2 / \sqrt{m} ds + 2J(M_n) \int_0^t \rho D ds.$$

Insert the damped residual integral above and apply Gronwall to obtain $\sup_t D(t) \leq D(0) \exp\{2J(M_n)(1 + 4G_*^2/\kappa)Y/\kappa\}$. Prediction subtraction costs at most $(2H_D + R_s F_z)D$, proving the stated older full-range coefficient.

E.2.5 Gaussian extension, exact confidence, and the all-time sphere norm

Let $\mathbf{G} = (A_0, \sqrt{n}W_0^{(2)}, \dots, \sqrt{n}W_0^{(L)})$ be the vector of independent standard Gaussian initialization coordinates. Both readouts initially vanish, so $d_\theta(0) = \|\mathbf{G} - \mathbf{G}'\|_2 / \sqrt{n}$. The preceding estimate proves a pairwise Lipschitz bound on the actual good set, with constant $\mathcal{L}_n$, for every scalar prediction at a fixed time and query. Its McShane extension is

$$F_{t,v}(g) = \inf_{h \text{ in the good set}} \{f_n(h; t, v) + \mathcal{L}_n \|g - h\|_2\}.$$

The triangle inequality proves that this is globally $\mathcal{L}_n$ -Lipschitz, and the pairwise bound proves equality with the actual prediction on the good set. Truncation to $[-2H_D R, 2H_D R]$ preserves both properties and bounds the extension.

Here is the Gaussian concentration fact and its needed proof. If F is globally a_F -Lipschitz on standard Gaussian Euclidean space with $a_F > 0$, then for $z > 0$,

$$\mathbb{P}\{|F - \mathbb{E}F| > z\} \leq 2e^{-z^2/(2a_F^2)}.$$

For a smooth function h bounded above and away from zero, define the Ornstein–Uhlenbeck averaging operator $P_t h(x) = \mathbb{E}h(e^{-t}x + \sqrt{1 - e^{-2t}}G)$. Its Gaussian entropy is $\text{Ent}(h) = \mathbb{E}[h \log h] - (\mathbb{E}h) \log(\mathbb{E}h)$. Differentiation, Gaussian integration by parts, and invariance of the Gaussian law give

$$\text{Ent}(h) = \int_0^\infty \mathbb{E} \frac{\|\nabla P_t h\|_2^2}{P_t h} dt.$$

To verify the identity, differentiate $\mathbb{E}[(P_t h) \log(P_t h)]$, use generator $\Delta - x \cdot \nabla$, integrate by parts, and integrate from $t = 0$ to ∞ , where $P_t h \rightarrow \mathbb{E}h$. The commutation $\nabla P_t h = e^{-t} P_t \nabla h$, followed by weighted Cauchy–Schwarz, bounds the integrand by $e^{-2t} \mathbb{E} P_t (\|\nabla h\|_2^2 / h)$. Integration and Gaussian invariance prove $\text{Ent}(h) \leq \frac{1}{2} \mathbb{E}(\|\nabla h\|_2^2 / h)$. Apply this to $h = e^{zF}$ and integrate the resulting inequality for $z^{-1} \log \mathbb{E} e^{zF}$. It gives $\mathbb{E} e^{z(F - \mathbb{E}F)} \leq e^{z^2 a_F^2 / 2}$. Markov’s inequality optimized at $z = u/a_F^2$, and the same argument for $-F$, give the displayed two-sided tail. Smooth approximation and truncation extend it to every Lipschitz F ; Gaussian exponential integrability of its linear growth justifies the limits. If $a_F = 0$, the function is constant and the tail assertion is trivial.

For two independent roots, the difference of the identical scalar extensions has mean zero and Lipschitz constant $\sqrt{2}\mathcal{L}_n$ on the product Gaussian space. Thus its two-sided tail at level z is at most $2e^{-z^2/(4\mathcal{L}_n^2)}$. This argument does not condition the Gaussian law on the good event or differentiate its indicator.

Use the proof coordinate $u = 1 - e^{-\kappa t}$, including $u = 1$ for the fitted limit. The fitting speed bound gives time modulus $K_t |u - u'|$. The actual matrix caps give a feature-map Lipschitz coefficient at most T^L in feature RMS on the entire input space, so prediction increments on the sphere are at most $K_x \|v - v'\|_2$. A time grid of $n + 1$ points and sphere $1/n$ -net of size at most $(1 + 2n)^d$ have at most N_n joint points. At threshold $2\mathcal{L}_n \sqrt{\log(4N_n/\delta)}$, the Gaussian union costs at most $\delta/2$. The two actual paths’ off-grid errors total at most $2(K_t + K_x)/n$.

For each path, take fitting failure at most $\delta/8$ using its explicit width, and source failure at most $\delta/8$ using the source’s eventual width. The two path failures total at most $\delta/2$. Adding this to the net failure

proves exactly the displayed probability $1 - \delta$. The proof coordinate is only used for a covering argument; neither training clock is changed. Uniform fitting tails pass the estimate to $t = \infty$.

E.2.6 The beta-only envelope and its actual-label factors

For this paragraph only, put $z = Y/\lambda$ and $h = 1 + 1/\lambda$. The source power ledger in the shared foundation gives $S_*^{\text{src}} \geq \beta^{-26L}$ and $K_{\text{src}} \leq \beta^{21L}$. The sufficient cap $z \leq \beta^{-30L}$ implies $S = 16z \leq \beta^{-29L}$, so both source and real-fitting allowances hold. Linear growth gives

$$H_D \leq (2\beta)^L \leq \beta^{3L/2}, \quad \lambda \leq \beta^{3L}, \quad F_D \leq \beta^{7L}, \quad F_z \leq \beta^{4L}, \quad M_n \leq \beta^{22L} z u_n.$$

The geometric sum for F_D and $T \leq \beta^2$ prove its stated power. The elementary inequalities $L \leq \beta^{L-1}$, $2, 8, 32 \leq \beta^L$, and $128 \leq \beta^{2L}$, valid for $L \geq 2, \beta \geq 10$, then give

$$R \leq \beta^{-28L}, \quad B_\delta \leq 1, \quad P, G_* \leq \beta^{3L}, \quad D_\delta(M_n) \leq \beta^{8L}(1 + M_n), \quad J(M_n) \leq \beta^{12L}(1 + M_n).$$

For example, the two terms of D_δ are bounded by β^{3L} and $\beta^{7L}M_n$; in J , its second term is at most β^{5L} . The three Taylor-remainder terms are dominated by $\beta^{11L}(1 + M_n)$. Substitution therefore gives

$$E_{\text{en}}(M_n) \leq \beta^{14L} z (1 + M_n) \leq \beta^{14L} z + \beta^{36L} z^2 u_n.$$

For the prefactor, expand it without suppressing gap powers:

$$C_Y(M_n) = \frac{128H_D^2 G_* J(M_n)}{\lambda^2} + \frac{8H_D s F_z}{\lambda} + \frac{2s F_z}{\sqrt{\lambda}} \leq \beta^{21L} h^2 (1 + M_n).$$

The three terms are bounded respectively by $\beta^{20L} h^2 (1 + M_n)$, $\beta^{8L} h$, and $\beta^{6L} \sqrt{h}$, which proves the last inequality.

The actual coefficients satisfy $\beta^{14L} z \leq 1$ and $\beta^{36L} z^2 \leq 1$. Convexity gives $e^x \leq 1 + 2x$ on $[0, 1]$ and $e^{\vartheta u} \leq 1 + \vartheta(e^u - 1)$ for $0 \leq \vartheta \leq 1$. Consequently

$$e^{E_{\text{en}}(M_n)} \leq (1 + 2\beta^{14L} z)[1 + \beta^{36L} z^2 (e^{u_n} - 1)].$$

Since $u_n \geq 1$ and $z \leq 1$, expansion proves

$$(1 + M_n)(1 + 2\beta^{14L} z) \leq 1 + \beta^{38L} z u_n.$$

Indeed the three added coefficients are at most $\beta^{22L} + 2\beta^{14L} + 2\beta^{36L} \leq \beta^{38L}$, after using $z^2 \leq z$. Thus

$$2\mathcal{L}_n \leq \frac{\beta^{23L} Y h^2}{\sqrt{n}} (1 + \beta^{38L} z u_n) [1 + \beta^{36L} z^2 (e^{u_n} - 1)].$$

The physical coefficients satisfy $2(K_t + K_x) \leq \beta^{5L} h Y$, by $F_D Y^2 / \lambda \leq 1$ and the displayed bounds on H_D, T, R . Since every bracket and the logarithmic square root is at least one, and $n^{-1} \leq n^{-1/2}$, this term is absorbed by enlarging the leading exponent from $23L$ to $24L$. Finally replace the three powers $\beta^{24L}, \beta^{38L}, \beta^{36L}$ by $B_D = \beta^{100L}$, and $4N_n$ by $8N_n$. This proves the stated beta-only certificate without hiding numerical constants.

E.2.7 Proof of the numerical accuracy inversion

To verify this, use $N_n \leq 23^d n^{d+1}$, hence $\log(8N_n/\delta) \leq D_{d,\delta} \log(en)$, and

$$1 + B_D(Ym/\gamma)u_n \leq (1 + B_D Y m/\gamma)u_n, \quad 1 + B_D(Ym/\gamma)^2(e^{u_n} - 1) \leq (1 + B_D(Ym/\gamma)^2)e^{u_n}.$$

In fact $v_\varepsilon \geq 1$, $\log v_\varepsilon \leq \sqrt{v_\varepsilon}$, and the ceiling is at most twice its argument. Therefore

$$\log(e\bar{n}_\varepsilon) \leq 2v_\varepsilon + 10\sqrt{v_\varepsilon} + 18 \leq \min\{(\sqrt{2v_\varepsilon} + 5)^2, 30v_\varepsilon\}.$$

The lower bound for the same ceiling gives the ratio of the envelope to ε at most $30e^{-2\sqrt{v\varepsilon}-3} \leq 30e^{-5} < 1$. Finally $\ell e^{\sqrt{\ell}} e^{-(\ell-1)/2}$ is decreasing for $\ell \geq 4$, because its logarithmic derivative is $1/\ell + 1/(2\sqrt{\ell}) - 1/2 \leq 0$; the chosen width is already in that range. This proves the assertion for all larger widths.

E.2.8 Proof of the initialized Gram central limit theorem

To prove it, condition on the preceding initialized layers. The next Gram is the average of n independent matrices $\phi_j(Z_i)\phi_j(Z_i)^T$, with Gaussian covariance $K_n^{(j-1)}$, and conditional mean $\Psi_j(K_n^{(j-1)})$. On bounded covariance sets, linear growth gives uniform moments of every fixed order. Conditional Chebyshev first proves $K_n^{(j)} \rightarrow Q^{(j)}$ in probability by induction.

For a fixed linear functional u on symmetric matrices, center one conditional summand and call its contraction X . Taylor's integral remainder for the scalar exponential gives

$$\left| \mathbb{E}[e^{iX/\sqrt{n}} | \mathcal{F}_{j-1}] - 1 + \frac{\mathbb{E}[X^2 | \mathcal{F}_{j-1}]}{2n} \right| \leq \frac{\mathbb{E}[|X|^3 | \mathcal{F}_{j-1}]}{6n^{3/2}}.$$

The right side is uniformly of order $n^{-3/2}$ on each bounded covariance set. Raising the characteristic function to its n th power therefore gives the Gaussian innovation characteristic function with covariance B_j , in probability. Both characteristic functions have magnitude at most one, so convergence also holds in L^1 . Multiplication by any bounded characteristic function of the preceding fluctuations proves joint convergence with an independent new innovation.

It remains to linearize the conditional mean. The integrated Gaussian identity proved in dense fitting, regularized by adding ηI to the entire covariance segment, gives the displayed map T_j . Its coefficient expectations are continuous at singular covariances: use their continuous positive square roots to couple the Gaussians, and dominate by a fixed Gaussian polynomial using linear growth and bounded derivatives. Integration on a feasible segment then gives

$$\Psi_j(Q + E) - \Psi_j(Q) - T_j(Q)E = o(\|E\|)$$

as $E \rightarrow 0$ with $Q + E \succeq 0$. Applied to the preceding tight $n^{-1/2}$ fluctuation, its remainder is $o_{\mathbb{P}}(n^{-1/2})$. Combining with the independent conditional innovation proves the recursion and completes the induction. This proof does not need an invertible intermediate covariance or a quantitative central-limit error bound.

E.2.9 Proof of the innovation inequality and onset variance

Here is its complete proof for a random vector with finite fourth moment and $Q = \mathbb{E}HH^T \succ 0$. Set $c = Qy \neq 0$, $A = \|c\|_2$, $T_Q = \text{tr } Q$, $M_4 = \mathbb{E}\|H\|_2^4$, and $D = T_Q - c^T Q c / A^2 > 0$. The squared area satisfies pointwise

$$J(H) := A^2 \|H\|_2^2 - (c^T H)^2 = \|HS_y - c\|_2^2 \|H\|_2^2 - [(HS_y - c)^T H]^2 \leq \|HS_y - c\|_2^2 \|H\|_2^2.$$

Also $0 \leq J(H) \leq A^2 \|H\|_2^2$ and $\mathbb{E}J(H) = A^2 D$. For any $R_0 > 0$, the contribution from $\|H\|_2 > R_0$ is at most $A^2 M_4 / R_0^2$, whereas the other part is at most $R_0^2 V$, where $V = \mathbb{E}\|HS_y - c\|_2^2$. Therefore

$$R_0^2 V \geq A^2 (D - M_4 / R_0^2).$$

Choosing $R_0^2 = 2M_4 / D$ gives

$$V \geq \frac{A^2 D^2}{4M_4} = \frac{[y^T Q^2 (T_Q I - Q)y]^2}{4M_4 \|Qy\|_2^2}.$$

All denominators are positive. In the current model, $M_4 \leq m^2 \mu_4$ by Cauchy-Schwarz and $T_Q = m\mu_{2,L}$. Every eigenvalue x of Q lies in $[\gamma, T_Q - (m-1)\gamma] \subset [\gamma, T_Q - \gamma]$. Consequently, if $N = y^T Q^2 (T_Q I - Q)y$, then

$$N \geq (m-1)\gamma \|Qy\|_2^2.$$

For a second bound, the identity $x(T_Q - x) - \gamma(T_Q - \gamma) = (x - \gamma)(T_Q - \gamma - x) \geq 0$, followed by multiplication by $x \geq \gamma$, gives $N \geq \gamma^2 (T_Q - \gamma) \|y\|_2^2$. Multiplying these two positive lower bounds and substituting into the

moment inequality yields its first displayed version. Finally $\gamma \leq \mu_{2,L}$ and $m\mu_{2,L} - \gamma \geq (m-1)\mu_{2,L}$ give the remaining versions. The weaker bound

$$V \geq \frac{(m-1)^2\gamma^4}{4m^2\mu_4}\|y\|_2^2$$

also follows directly from $D \geq (m-1)\gamma$ and $\|Qy\|_2 \geq \gamma\|y\|_2$.

For the optional bounded-value subclass $|H_a| \leq K$, the same area inequality can be integrated without truncation, since $\|H\|_2^2 \leq mK^2$. It gives

$$V \geq \frac{\gamma^2(m\mu_{2,L} - \gamma)}{mK^2}\|y\|_2^2, \quad \max_a \text{Var}(H_a S_y) \geq \frac{m-1}{m} \frac{\gamma^2\mu_{2,L}}{K^2} Y^2.$$

Boundedness is used only in this optional improvement.

The qualitative obstruction has a short exact form. If $V = 0$, then $HS_y = Qy$ almost surely. Pairing with y makes $S_y^2 = y^T Q y > 0$ deterministic; hence $H = (Qy)/S_y$ lies in one fixed line almost surely. Its second-moment matrix would have rank at most one, contradicting $m \geq 2$ and $Q \succ 0$.

At least one deterministic training index a_* , chosen from the fixed data and labels, now satisfies

$$\text{Var}(H_{a_*} S_y) \geq \frac{\gamma^3\mu_{2,L}}{16\mu_4} Y^2.$$

At zero readout every hidden initial velocity vanishes, so exactly

$$\dot{f}_n(0, v_a) = \frac{2}{m} \sum_b K_{n,ab}^{(L)} y_b.$$

The initialized Gram central limit theorem and independent second run therefore imply

$$\sqrt{n}[\dot{f}_n(0, v_{a_*}) - \tilde{\dot{f}}_n(0, v_{a_*})] \implies N(0, \sigma_{a_*}^2),$$

$$\sigma_{a_*}^2 = \frac{8}{m^2} \sum_{b,c} y_b (C_L)_{a_*b, a_*c} y_c \geq \frac{8}{m^2} \text{Var}(H_{a_*} S_y) \geq \frac{\gamma^3\mu_{2,L} Y^2}{2m^2\mu_4}.$$

The factor eight is $(2/m)^2$ times the two independent fluctuation variances. The inequality follows from $C_L = T_L C_{L-1} T_L^T + B_L \succeq B_L$; earlier layers cannot cancel the final innovation. Duplicate query coordinates and singular augmented Gaussian covariances are covered by the CLT proof above.

E.2.10 Proof of the finite-query source localization and label simplification

For a fixed finite collection of real unit queries, including the training inputs and the selected witness, replace only the passive frame-mesh union in the source construction. The training sample budgets, independently stopped cavities, actual-amplitude traces, and activity cap are unchanged. In the residual-free response recurrence, the sphere coefficient $16\sqrt{d+3}$ came from the frame mesh; for this finite collection it is replaced by 64, producing precisely U_j^{fin} above. There are no angular derivatives or complex query segments.

For clarity, the response at a query is

$$R_a^{(j)}(v) = D_{\Theta} z^{(j)}(v) \nabla_{\Theta} (n f_n(v_a)), \quad \Theta = (A, \sqrt{n} W^{(2)}, \dots, \sqrt{n} W^{(L)}, w).$$

The source's endpoint and insertion identities require only query norm at most two, so they still apply. A mesh of spacing n^{-2} on the time rectangle has at most n^5 points eventually; its length is proportional to $\log(en)$ and its width is bounded. Including root/neuron indices and the fixed layer, sample, and finite-query indices leaves at most n^{10} points eventually. The normalized centered complex Gaussian tail is at most $4e^{-u^2/4}$, so the union at $u = 64\sqrt{\log(en)}$ is bounded by

$$4n^{10} e^{-1024 \log(en)} \longrightarrow 0.$$

The unchanged stopped derivative bounds are $\sqrt{n}$ times a fixed power of $\log(en)$; hence the n^{-2} off-grid error vanishes. Fixed queries have no spatial off-grid term. Their initialized maxima have the same fresh-row Gaussian bounds. The source's passive-query size stop improves by the real increment $U_{\text{fin}} S^2 \sqrt{\log(en)}$. There is no union over deletion subsets and no new exponential query budget. Applying the shared insertion theorem proved above gives, with probability tending to one,

$$\max_{v,a,j} \|R_a^{(j)}(v)\|_\infty \leq S U_{\text{fin}}(S) \sqrt{\log(en)}.$$

Following the real path to its nearest anchor and then at most two short pieces of combined length $2r_n$, these gates give residual growth at most two, additional activity at most $S/2$, and hidden parameter increments at most $1/4$. The exact velocity identity $\partial_t z^{(j)} = -(2/m) \sum_a r_a R_a^{(j)}$, with $\rho \leq 2Y$ on these pieces, bounds total preactivation displacement by $8cY S U_{\text{fin}} \leq a/8$. There is no angular displacement. Thus the pole stops retain strict margins inside the half-strip.

The complex-minus-real Gaussian coefficient radii from the shared source proof tend to zero for each fixed c . Its empirical moment argument, with $\mathcal{B} = 1024e^2 L$, has the unchanged estimate

$$\limsup_{n \rightarrow \infty} \Pr\{\text{a training budget is hit}\} \leq mL(16L/\mathcal{B})^p$$

for each fixed positive integer p . First taking the width limit, then the infimum over p , makes this zero. One may take $c = \chi(S)/\lambda$ on the full recurrence range. On the simpler cap $Y/\lambda \leq \beta^{-30L}$, the source power ledger specializes to

$$U_1^{\text{fin}} \leq 4\beta^{20L+3}, \quad U_j^{\text{fin}} \leq 22\beta^{26L-5} + 128\beta^{8L-5} + 2 \leq \beta^{26L}.$$

The first quantity is also at most β^{26L} , since $4\beta^{-6L+3} \leq 4 \cdot 10^{-9}$. For the second, division by β^{26L} gives at most $22 \cdot 10^{-5} + 128 \cdot 10^{-41} + 2 \cdot 10^{-52} < 1$. Using $a \geq 16/\beta$ and the actual activity cap yields

$$\frac{a}{4S^2 U_{\text{fin}}(S)} = \frac{a\lambda^2}{1024Y^2 U_{\text{fin}}(S)} \geq \frac{\beta^{34L-1}}{64} > 1.$$

Thus $\chi(S) = 1$ on that cap and $r_n = m/[\gamma\sqrt{\log(en)}]$. This simplification is not imposed on the full recurrence range.

E.2.11 The deterministic derivative-to-trajectory inequality

Suppose a function g is holomorphic on a neighborhood of the filled parameter-two Bernstein ellipse for $[0, r]$, and $|g| \leq M$ there. For every integer $N \geq 1$,

$$\sup_{0 \leq t \leq r} |g(t)| \geq \frac{r}{2N^2} |g'(0)| - 24M2^{-N}.$$

To prove it, send $[0, r]$ to $[-1, 1]$ by $t = r(x+1)/2$. Under the Joukowski map $x = (z+z^{-1})/2$, holomorphy on the ellipse gives a Laurent expansion on an annulus containing $1/2 \leq |z| \leq 2$. Cauchy's coefficient bound gives Chebyshev coefficients $|a_k| \leq 2M2^{-k}$. The degree- N truncation has uniform tail at most $2M2^{-N}$, since $|T_k(x)| \leq 1$ on $[-1, 1]$. At the physical endpoint $t = 0$, its derivative tail is at most

$$\frac{4M}{r} \sum_{k > N} k^2 2^{-k} = \frac{4M}{r} 2^{-N} (N^2 + 4N + 6).$$

The identity follows by writing $k = N + j$ and using $\sum_{j \geq 1} 2^{-j} = 1$, $\sum_{j \geq 1} j 2^{-j} = 2$, and $\sum_{j \geq 1} j^2 2^{-j} = 6$.

For completeness, the endpoint polynomial inequality $|P'(-1)| \leq N^2 \sup_{[-1,1]} |P|$ also holds for complex coefficients. Interpolate at the Chebyshev extrema $x_j = \cos(j\pi/N)$, $0 \leq j \leq N$. The derivative weights $\ell'_j(-1)$ of the Lagrange basis have alternating signs, with $\ell'_j(-1)(-1)^j$ all of the same sign; this follows by counting the signs in their products, including $\ell'_N(-1) = \sum_{j < N} (-1 - x_j)^{-1} < 0$. Since $T_N(x_j) = (-1)^j$, their absolute sum is $|\sum_j \ell'_j(-1)(-1)^j| = |T'_N(-1)| = N^2$. The last equality follows by differentiating $T_N(\cos \theta) = \cos(N\theta)$ and taking the endpoint limit. The triangle inequality now proves the claimed polynomial bound.

Apply it to the truncated series, multiply by $2/r$ for the physical derivative, and add the two tails. After rearrangement, the error coefficient is $4 + 8/N + 12/N^2 \leq 24$, giving the stated inequality.

E.2.12 Applying the inequality to the actual nonlinear predictions

Take $g_n(t) = f_n(t, v_{a_*}) - \tilde{f}_n(t, v_{a_*})$. The parameter-two ellipse for $[0, r_n]$ has real projection $[-r_n/8, 9r_n/8]$ and imaginary projection $[-3r_n/8, 3r_n/8]$. The finite-query rectangle contains it: the numerical source gates give $r_n \leq 1/\lambda$, whereas $T_n = 32 \log(en)/\lambda$. Choose

$$N_n^{\text{Ch}} = \left\lceil \frac{2 \log(en)}{\log 2} \right\rceil \leq 4 \log(en), \quad 2^{-N_n^{\text{Ch}}} \leq n^{-2}.$$

On the two localized source events the preceding deterministic inequality therefore gives

$$\|f_n - \tilde{f}_n\|_* \geq \frac{c}{32 \log(en)^{5/2}} |g'_n(0)| - \frac{768 \beta^{6L} Y}{\lambda n^2}.$$

This estimate concerns the actual nonlinear trajectories with the original labels. It has no fixed nonzero Taylor remainder that could dominate the $n^{-1/2}$ fluctuation.

Write $\sigma = \sigma_{a_*} > 0$. For a fixed $u > 0$, the explicit deterministic gate

$$n^{3/2} \geq \frac{49152 \beta^{6L} Y}{\lambda c u \sigma} \log(en)^{5/2}$$

makes the remainder at most half the leading term whenever $|g'_n(0)| \geq u\sigma/\sqrt{n}$. The initialized CLT has no atom at either threshold $\pm u\sigma$, and both source failures tend to zero. Thus, without requiring independence of these events,

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left\{ \|f_n - \tilde{f}_n\|_* \geq \frac{u c \sigma}{64 \sqrt{n} \log(en)^{5/2}} \right\} \geq 2[1 - \Phi(u)].$$

Take $c = \chi(S)/\lambda$, insert $\sigma \geq \gamma^{3/2} Y \sqrt{\mu_{2,L}/\mu_4}/(\sqrt{2}m)$, and use $1/(64\sqrt{2}) \geq 1/128$. This gives the displayed coefficient, with $\chi(S) \geq \chi_{\text{act}}$. Finally choose $u = \Phi^{-1}(1/2 + \delta/4)$. The limit lower probability is $1 - \delta/2$, leaving a strict margin for the source failures and CLT convergence. Hence the probability is at least $1 - \delta$ at every sufficiently large individual width. The width is not effective because neither the insertion probability nor the CLT remainder is quantified. Since $g_n(0) = 0$, a positive lower threshold is attained only at a strictly positive time in the compact witness interval.

E.2.13 Exact exceptions and the dense storage consequence

For $Y = 0$, both predictors are identically zero. For $m = 1$, the last innovation is

$$y_1^2 [\mu_4 - \mu_{2,L}^2] = y_1^2 \text{Var}(\phi_L(\sqrt{\mu_{2,L-1}} G)^2).$$

If $\mu_{2,L-1} > 0$, continuity and full Gaussian support make this variance zero exactly when ϕ_L^2 is constant on the real line. A continuous real function with positive constant square must itself be constant, because the real line is connected. If $\mu_{2,L-1} = 0$, the feature is also deterministic. These cases are permitted by a positive one-sample uncentered gap. In particular, for $\phi_L \equiv c \neq 0$, every hidden gradient vanishes and

$$f_n(t, v) = y_1(1 - e^{-2c^2 t})$$

for every width, initialization, and query. Thus the general lower theorem cannot include $m = 1$.

The zero-preactivation-variance exception is deterministic as well. Starting from $\mu_{2,0} = 1$, take the last transition from a positive moment to zero before layer L . Gaussian full support forces that activation to be identically zero. All subsequent moments up to $\mu_{2,L-1}$ are zero, so the intervening activations vanish at zero. The final feature is the constant $\phi_L(0)$, and the same deterministic formula applies. Outside these effective-constant cases, the scalar innovation is positive, but no coefficient uniform in the gap and β alone

follows: with identity earlier activations and $\phi_L(t) = \sqrt{1 - \eta^2} + \eta t$, $0 < \eta < 1$, one has $\mu_{2,L-1} = \mu_{2,L} = \gamma = 1$, a common $\beta = 10$, and $\mu_4 - \mu_{2,L}^2 = 4\eta^2 - 2\eta^4 \rightarrow 0$.

Fix $0 < \delta < 1/2$ and a nondegenerate $m \geq 2$ task. If an independent dense pair at a sufficiently large width satisfies $\Pr\{\|f_n - \tilde{f}_n\|_* \leq \varepsilon\} \geq 1 - \delta$, then its accuracy event intersects the lower event, since their failure probabilities sum to less than one. Therefore necessarily

$$n[\log(en)]^5 \geq \frac{c_{\phi,L,\delta}^2 Y^2 \gamma}{\varepsilon^2},$$

and its per-run dense parameter count obeys the explicit implicit bound

$$S_D(n) \geq (L-1)n^2 \geq \frac{(L-1)c_{\phi,L,\delta}^4 Y^4 \gamma^2}{\varepsilon^4 [\log(en)]^{10}}.$$

At a fixed task this implies the necessary order $\varepsilon^{-4}/\log(1/\varepsilon)^{10}$ up to a positive fixed-task coefficient: if $n \leq \varepsilon^{-4}$ and $\varepsilon \leq e^{-1}$, then $\log(en) \leq 5\log(1/\varepsilon)$; if $n > \varepsilon^{-4}$, its quadratic storage is already larger than that order for all sufficiently small ε . This is a restriction on the canonical independent dense family, not a storage or bit-complexity lower bound against arbitrary compressed representations.

E.3 Legendre proofs

The proof first establishes the polynomial projection identities and all-order physical fitting. It then controls the reconstruction defect by signed parameter energy, proves both forward certificates, and justifies the scalar accuracy inverses.

E.3.1 Projection identities, with the numerical endpoint bounds

Let P_j be the Legendre polynomial defined by Rodrigues' formula $P_j(x) = (2^j j!)^{-1} (d/dx)^j (x^2 - 1)^j$. On $[0, A]$, write $p_j^A(\xi) = P_j(2\xi/A - 1)$, and let Π_q^A be orthogonal projection in Lebesgue $L^2([0, A])$ onto polynomials of degree below q . Integration by parts in Rodrigues' formula gives

$$\int_0^A p_i^A p_j^A d\xi = \frac{A}{2j+1} \mathbf{1}_{i=j}, \quad -\frac{d}{d\xi} [\xi(A-\xi)(p_j^A)'] = j(j+1)p_j^A.$$

The normalized derivative polynomials are orthogonal in the weighted derivative inner product. Bessel's inequality there, followed by the ordinary orthogonal expansion of u , proves

$$\|(I - \Pi_q^A)u\|_{L^2}^2 \leq \frac{1}{q(q+1)} \int_0^A \xi(A-\xi) \|u'(\xi)\|^2 d\xi. \quad (142)$$

One first applies this calculation to finite polynomial expansions; polynomial density and the weighted derivative Bessel inequality give the displayed bound for absolutely continuous histories with the finite energy appearing on its right. It also holds for Hilbert-valued histories by applying the scalar argument to finitely many orthogonal coordinates and passing to their increasing squared norms.

The derivative identity $P_j' = \sum_{i < j, j-i \text{ odd}} (2i+1)P_i$, obtained by integration by parts against the orthogonal polynomials, telescopes to the endpoint projection kernel

$$K_q^A(A, \xi) = \frac{1}{A} \sum_{j=0}^{q-1} (2j+1) p_j^A(\xi) = \frac{(p_q^A)'(\xi) + (p_{q-1}^A)'(\xi)}{2}.$$

Its integral is one. Integrating by parts, using $p_q^A(0) + p_{q-1}^A(0) = 0$, proves

$$u(A) - (\Pi_q^A u)(A) = \frac{1}{2} \int_0^A [p_q^A(\xi) + p_{q-1}^A(\xi)] u'(\xi) d\xi.$$

The squared norm of this scalar multiplier is $A[(2q+1)^{-1} + (2q-1)^{-1}]/4 \leq A/(3q)$. Thus

$$\|u(A) - (\Pi_q^A u)(A)\| \leq \sqrt{A/(3q)} \|u'\|_{L^2}. \quad (143)$$

Here is a direct proof of the additional bound

$$\|(\Pi_q^A u)(A)\| \leq 64\sqrt{q} \|u\|_{L^\infty}. \quad (144)$$

For $k \geq 1$, put $v(\theta) = \sqrt{\sin \theta} P_k(\cos \theta)$ and $a_k(\theta) = (k + 1/2)^2 + (4 \sin^2 \theta)^{-1}$. The Legendre differential equation gives $v'' + a_k v = 0$. Therefore

$$\frac{d}{d\theta} \left(v^2 + \frac{(v')^2}{a_k} \right) = -\frac{a'_k}{a_k^2} (v')^2 \geq 0 \quad (0 < \theta < \pi/2).$$

At the center, $P_{2j}(0) = (-1)^j (2j)/4^j$ and $P'_{2j+1}(0) = (2j + 1)P_{2j}(0)$. The elementary induction $((2j)/4^j)^2 \leq 1/(j + 1)$ bounds the central energy by $2/k$, and hence by $4/k$. Taking square roots and differentiating $P_k = v/\sqrt{\sin \theta}$ gives

$$|\partial_\theta P_k(\cos \theta)| \leq 3\sqrt{k}(\sin \theta)^{-1/2} + 2k^{-1/2}(\sin \theta)^{-3/2}.$$

On $[1/k, \pi/2]$, the two integrals are at most $3\pi\sqrt{k}$ and $4(\pi/2)^{3/2} < 8$, respectively, using $\sin \theta \geq 2\theta/\pi$. On $[0, 1/k]$, $|P'_k(x)| \leq k(k + 1)/2$ gives an integral at most $(k + 1)/(4k) \leq 1/2$. To verify that derivative bound, use the derivative expansion above and $|P_i(x)| \leq 1$ on $[-1, 1]$. The latter follows, for example, from

$$P_j(\cos \theta) = \frac{1}{\pi} \int_0^\pi (\cos \theta + i \sin \theta \cos \psi)^j d\psi;$$

the identity follows by comparing its geometric-series generating function with $(1 - 2tx + t^2)^{-1/2}$, and the integrand has modulus at most one. Reflection around $\pi/2$ now proves $\text{Var}_{[-1,1]} P_k \leq 36\sqrt{k}$. The endpoint kernel formula has L^1 norm at most $36\sqrt{q} \leq 64\sqrt{q}$. The case $q = 1$ is the constant averaging kernel. This proves (144), including vector-valued histories.

For a fixed history g , let $E_g(A) = \int_0^A \|g - \Pi_q^A g\|^2 d\xi$. Orthogonality to $\partial_A \Pi_q^A g$, which is still a polynomial of degree below q , gives the exact growing-interval identity

$$E'_g(A) = \|g(A) - \Pi_q^A g(A)\|^2. \quad (145)$$

In clock coordinates define $h_a(\tau(t)) = \hat{h}_a(t)$ and $b_a(\tau(t)) = (\hat{r}_a/\hat{\rho})\hat{\delta}_a(t)$, with their specified prefixes. Since $\widehat{W}^{(L+1)}(0) = 0$, the backward history is continuous at the prefix junction. The moments in (56) equal $\int_0^{\tau(t)} p_j^{\tau(t)} h_a$ and $\int_0^{\tau(t)} p_j^{\tau(t)} b_a$. Differentiating those integrals proves the stated ODEs, using the derivative expansion already proved. By orthogonality, reconstruction subtracts $2/(mn) \sum_a \int (\Pi_q b_a)(\Pi_q h_a)^T$. Differentiating this bilinear projection integral gives its full endpoint product minus the product of its two endpoint errors. Hence the exact physical defect is

$$\dot{\widehat{W}}^{(\ell)} = -\frac{2}{mn} \sum_a \hat{r}_a \hat{\delta}_a^{(\ell)} (\hat{h}_a^{(\ell-1)})^T + \mathcal{E}_\ell, \quad \mathcal{E}_\ell = \frac{2\hat{\rho}}{mn} \sum_a e_{b,a}^{(\ell)} (e_{h,a}^{(\ell-1)})^T. \quad (146)$$

Here $e_g = g(\tau) - \Pi_q^\tau g(\tau)$. These identities first hold before a residual zero, with ordinary clock differentiation; the runtime is continuous at zero, and its stationary continuation gives the corresponding stopped limits.

E.3.2 All-order fitting and the physical bounds

For this proof only set $\lambda = \gamma/m$, $S = 8Y/\lambda$, $R = S\sqrt{\lambda}$, and $\kappa = \lambda/4$. Assume (61) and $S \leq S_*^{\text{Leg}}$. Stop at total clock activity S , readout RMS R , any operator norm nine, or sphere feature RMS $2H$. Before that stop, backward propagation gives $\|\hat{\delta}_a^{(\ell)}\|_2/\sqrt{n} \leq D_\ell R$. Projection contraction in the reconstruction, sample Cauchy–Schwarz, and the zero backward prefix imply

$$\|\widehat{W}^{(\ell)} - W_0^{(\ell)}\|_F \leq 4HD_\ell R\sqrt{(1+S)S}, \quad \frac{\|\widehat{W}^{(1)} - W_0^{(1)}\|_F}{\sqrt{n}} \leq 2D_1 RS.$$

For the first bound the forward history norm is at most $2H\sqrt{1+S}$, and the aggregate backward norm at most $D_\ell R\sqrt{S}$, both with neuron RMS and sample RMS normalizations. Since $\sqrt{\lambda} \leq H$, $S \leq 1$, and $64H^2D_1S \leq 1$, both increments are strictly less than one. Thus no operator boundary is reached.

For each layer let Z_ℓ be the supremum over sphere queries of $n^{-1} \int_1^{\tau(t)} \|\partial_\xi h^{(\ell)}(\xi, x)\|_2^2 d\xi$, also taking the supremum over the current stopped time interval. The endpoint evaluation norm on polynomial $L^2([0, A])$ is $q/\sqrt{A}$, because $K_q^A(A, A) = q^2/A$. Consequently the aggregate backward endpoint error is at most $(q+1)D_\ell R$. Integrate the square of (146), use (145), and apply (142) with $\xi(A-\xi) \leq A^2/4$. The prefix error is zero, and $(q+1)/q \leq 2$, giving

$$\int_0^t \hat{\rho} \|\mathcal{E}_\ell / \hat{\rho}\|_F^2 du \leq 2(1+S)^2 D_\ell^2 R^2 Z_{\ell-1}. \quad (147)$$

The first-layer chain rule gives $Z_1 \leq 4Ss^2 D_1^2 R^2$. At each later layer the three contributions to its preactivation derivative are the dense gradient term, the defect term and the previous feature derivative. Their squared integrals are bounded respectively by $64SH^4 D_\ell^2 R^2$, $32H^2 D_\ell^2 R^2 Z_{\ell-1}$, and $81Z_{\ell-1}$. Using $\|u+v+w\|^2 \leq 3(\|u\|^2 + \|v\|^2 + \|w\|^2)$ therefore gives

$$Z_\ell \leq 3s^2 \{64SH^4 D_\ell^2 R^2 + [81 + 32H^2 D_\ell^2 R^2] Z_{\ell-1}\} \leq C_\ell S R^2. \quad (148)$$

The last inequality uses $32H^2 D_\ell^2 R^2 \leq 32H^4 D_1^2 S^2 < 1$. The recurrence C_ℓ increases with ℓ . Thus every sphere feature moves by at most $\sqrt{SZ_\ell} \leq \sqrt{C_\ell} S^2 \sqrt{\lambda} \leq \sqrt{\lambda}/8$, by the third entry in S_*^{Leg} . This is also at most $H/8$, excluding the feature boundary. The normalized top training feature matrix moves in operator norm by at most $\sqrt{\lambda}/8$. Its minimum singular value stays above $(1/\sqrt{2} - 1/8)\sqrt{\lambda} > \sqrt{\lambda}/2$, proving the mean readout-Gram gap $\lambda/4$.

For a pointwise defect estimate use the sharper endpoint kernel: the backward endpoint error is at most $65\sqrt{q}D_\ell R$, while the forward endpoint error is at most $\sqrt{(1+S)/(3q)}\sqrt{Z_{\ell-1}}$, with the same RMS normalizations. Since $S \leq 1$, (146) gives

$$e_{\mathcal{E}}(t) := \sum_{\ell=2}^L \|\mathcal{E}_\ell(t)\|_F \leq E\sqrt{SR^2}\hat{\rho}(t). \quad (149)$$

Indeed $2 \cdot 65\sqrt{2/3} < 130$. The prediction Jacobian applied to hidden-matrix defects has sample RMS at most $2HD_1 Re_{\mathcal{E}}$. The fourth entry in S_*^{Leg} gives

$$\|\mathcal{J}\mathcal{E}\|_2/\sqrt{m} \leq 2HD_1 ES^{7/2} \lambda^{3/2} \hat{\rho} \leq \lambda \hat{\rho}/4.$$

The residual equation is $\dot{\hat{r}} = -2\Gamma\hat{r} + \mathcal{J}\mathcal{E}$, where Γ is the mean full tangent Gram and $\Gamma \succeq \lambda I/4$. It follows that $\dot{\hat{\rho}} \leq -\kappa\hat{\rho}$. Thus $\hat{\rho}(t) \leq Y e^{-\kappa t}$ and $\int_t^\infty \hat{\rho} \leq \hat{\rho}(t)/\kappa$; before any stopped endpoint the same statement holds with the integral only up to that endpoint. In particular the activity never exceeds $Y/\kappa = S/2$.

Let $\mathcal{F}$ be the dense gradient field evaluated at the reconstructed parameters, in the physical Hilbert norm

$$\|U\|_{\text{par}}^2 = \|U^{(1)}\|_F^2/n + \sum_{\ell=2}^L \|U^{(\ell)}\|_F^2 + \|U^{(L+1)}\|_2^2/n.$$

Its squared norm is $4\hat{r}^T \Gamma \hat{r}/m \geq \lambda \hat{\rho}^2$. Therefore

$$-\frac{d}{dt} \hat{\rho}^2 = \|\mathcal{F}\|_{\text{par}}^2 - 2\hat{r}^T \mathcal{J}\mathcal{E}/m \geq \frac{1}{2} \|\mathcal{F}\|_{\text{par}}^2.$$

Multiplication by $e^{\kappa t}$ and integration by parts show $\int e^{\kappa t} \|\mathcal{F}\|_{\text{par}}^2 dt \leq 4Y^2$. Weighted Cauchy-Schwarz then yields $\int \|\mathcal{F}\|_{\text{par}} dt \leq 2Y/\sqrt{\kappa} = 4Y/\sqrt{\lambda} = R/2$. The readout has no defect, so this excludes its boundary. All stops have strict margins. At fixed finite n, q , the moment integral formulas and $1 \leq \tau \leq 1+S$ bound every stored coordinate on finite intervals, proving global continuation. Local uniqueness excludes a finite first residual zero when $Y > 0$, because zero residual is an equilibrium of the entire stored-state ODE. The finite path lengths of $\mathcal{F}$ and $\mathcal{E}$ give convergent physical parameters, and the moment ODEs also have integrable velocities. Residual decay proves interpolation. This establishes (62) for every order on one event.

Finally (149) and the gradient velocities give the coefficients T_ℓ, V_ℓ in (67). In particular $\|\hat{h}^{(\ell)}(x)\|_2/\sqrt{n} \leq V_h \hat{\rho}$. The tangent Gram's trace is at most $4H^2 + D_1^2 R^2 + 4H^2 R^2 \sum_{\ell=2}^L D_\ell^2$. Differentiating $\hat{c} = \hat{r}/\hat{\rho}$, using the residual equation and the triangle inequality, proves $\|\hat{c}\|_2/\sqrt{m} \leq C_c$, with exactly the displayed coefficient.

E.3.3 Signed stability and the projection comparison

Use Euclidean physical coordinates $\theta = (W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, W^{(L+1)}/\sqrt{n})$. Let $e_\theta = \hat{\theta} - \theta_D$, $e = \|e_\theta\|$, and let $D(t)$ be the sum of the $L+1$ block norms of $e_\theta(t)$. Then $D(t) \leq \sqrt{L+1}e(t)$.

On the parameter segment $\theta_v = \theta_D + v e_\theta$, $0 \leq v \leq 1$, convexity preserves the matrix and readout bounds. Linear growth of the activation gives feature RMS at most Q . Subtracting forward passes recursively gives

$$\max_\ell \frac{\|z_v^{(\ell)} - z_D^{(\ell)}\|_2}{\sqrt{n}} \leq F_{\text{seg}} v D(t), \quad \max_\ell \frac{\|h_v^{(\ell)} - h_D^{(\ell)}\|_2}{\sqrt{n}} \leq s F_{\text{seg}} v D(t).$$

The first bound starts with the first parameter block; each later layer has recurrence $\Delta z_\ell \leq 9s \Delta z_{\ell-1} + Q \Delta W_\ell$. For backward subtraction use

$$\delta_v - \delta_D = \phi'(z_v) \odot (k_v - k_D) + [\phi'(z_v) - \phi'(z_D)] \odot k_D.$$

Only the dense carrier maximum M_n is used. Propagation by $9s$, the changed matrix acting on a dense response of RMS at most B_δ , and at most L changed gates give

$$\max_\ell \frac{\|\delta_v^{(\ell)} - \delta_D^{(\ell)}\|_2}{\sqrt{n}} \leq (d_0 + d_1 M_n) v D(t).$$

There is no carrier assumption on the segment or the closure. The gradient of one training output has blocks

$$g_{a,1} = \delta_a^{(1)} v_a^T / \sqrt{n}, \quad g_{a,\ell} = \delta_a^{(\ell)} (h_a^{(\ell-1)})^T / n, \quad g_{a,L+1} = h_a^{(L)} / \sqrt{n}.$$

Subtracting each hidden product in the order $(\delta_v - \delta_D) h_v^T + \delta_D (h_v - h_D)^T$ proves

$$\|g_a(\theta_v) - g_a(\theta_D)\| \leq (K_0 + K_1 M_n) v e.$$

Thus, with $K = K_0 + K_1 M_n$, segment integration yields

$$\hat{r} - r_D = J_D e_\theta + \mathcal{R}, \quad \|\mathcal{R}\|_2 / \sqrt{m} \leq K e^2 / 2, \quad \|(J_{\hat{\theta}} - J_D) e_\theta\|_2 / \sqrt{m} \leq K e^2.$$

The physical equations are $\dot{\theta}_D = -2J_D^* r_D$ and $\dot{\hat{\theta}} = -2J_{\hat{\theta}}^* \hat{r} + \mathcal{E}$, with adjoints using the sample inner product. Putting $v_r = \hat{r} - r_D$ gives the exact signed energy identity

$$\begin{aligned} \frac{1}{2} \frac{d}{dt} e^2 &= -2 \|v_r\|_2^2 / m + 2 v_r^T \mathcal{R} / m - 2 \hat{r}^T (J_{\hat{\theta}} - J_D) e_\theta / m + \langle e_\theta, \mathcal{E} \rangle \\ &\leq K(\rho_D + 3\hat{\rho}) e^2 + e \|\mathcal{E}\|. \end{aligned}$$

Regularize e by $\sqrt{e^2 + \eta^2}$ and then let $\eta \downarrow 0$ to justify the scalar inequality at zero. The integrated residual bounds are $2Ym/\gamma$ and $4Ym/\gamma$. Gronwall and the common initial state imply, on every finite horizon $[0, T]$,

$$\sup_{t \leq T} D(t) \leq \mathcal{A}_n \mathcal{D}_T, \quad \mathcal{D}_T = \int_0^T \sum_{\ell=2}^L \|\mathcal{E}_\ell(t)\|_F dt. \quad (150)$$

The coefficient is exactly (68).

The forward history has clock derivative RMS at most V_h . Since $\tau \leq 1 + a_0$ and $\int_1^A \xi(A - \xi) d\xi \leq A(A-1)^2/2$, the weighted projection bound gives forward tail norm at most F_h/q . For the backward history record the dense response in the closure clock, $\tilde{b}_a(\tau(t)) = \hat{c}_a(t) \delta_{D,a}(t)$, with zero prefix. Its physical derivative

has aggregate RMS at most $C_c B_\delta + V_\delta \rho_D$. The dense speed constants V_z, V_δ in (67) follow by differentiating forward and backward recursions: the backward terms are the readout speed, changed hidden matrices and changed gates, respectively. The changed-gate term uses only M_n .

Freeze $\tilde{b}$ after physical time $u = 2 \log q / \kappa$, or after T if earlier. For $A = \tau(T)$, the residual tail estimate gives $A - \tau(t) \leq \hat{\rho}(t) / \kappa$. In the weighted clock derivative energy this cancels the inverse clock speed, so

$$\int_0^A \xi(A - \xi) \|\partial_\xi \tilde{b}^{\text{frozen}}\|^2 d\xi \leq \frac{1 + a_0}{\kappa} \left[2u C_c^2 B_\delta^2 + 2V_\delta^2 \int_0^u \rho_D(t)^2 dt \right].$$

The aggregate norm here is $(mn)^{-1/2} (\sum_a \|\cdot\|_2^2)^{1/2}$. Since $\int \rho_D^2 \leq Y^2 / (2\kappa)$, its projected tail is at most the first two terms of $(b_0 + b_1 \sqrt{\log q}) / q$. The freezing remainder has norm at most $2B_\delta \sqrt{a_0} e^{-\kappa u / 2} = 2B_\delta \sqrt{a_0} / q$. This includes $q = 1$, when $u = 0$ and the frozen response is zero.

Forward and backward subtraction directly between the two trajectories uses $2H$ instead of Q , giving response difference RMS at most $B_d D(t)$. Multiplication by $\hat{c}$, whose sample RMS is one, and integration over a clock interval of length at most a_0 give an additional history error $B_d \sqrt{a_0} \sup_{t \leq T} D(t)$.

To control the integral of the absolute defect, not merely its signed integral, apply sample Cauchy–Schwarz to (146) and then Cauchy–Schwarz in $dA = \hat{\rho} dt$. Both prefix energies vanish. Equation (145) yields for each interface $\int_0^T \|\mathcal{E}_\ell\|_F dt \leq 2\sqrt{E_b(\tau(T))E_h(\tau(T))}$. Consequently

$$\mathcal{D}_T \leq 2(L-1)F_h \left[\frac{b_0 + b_1 \sqrt{\log q}}{q^2} + \frac{B_d \sqrt{a_0} \sup_{t \leq T} D(t)}{q} \right]. \quad (151)$$

If $q \geq q_{\text{abs}}$, substituting (150) absorbs the last term with factor at most one half. Whole-sphere forward and readout subtraction give prediction discrepancy at most $C_f D$. Thus

$$\|f_{\text{Leg},n,q} - f_n\|_* \leq \frac{4(L-1)C_f \mathcal{A}_n F_h}{q^2} (b_0 + b_1 \sqrt{\log q}) \leq C_n \frac{\sqrt{\log(eq)}}{q^2} \quad (q \geq q_{\text{abs}}). \quad (152)$$

The finite-horizon constants are independent of T ; taking its increasing supremum and then the physical limits proves the stated norm. For every order, (62) and Cauchy–Schwarz give $\|f_{\text{Leg},n,q} - f_n\|_* \leq U$. When $q < q_{\text{abs}}$, the second term in (70) exceeds U ; when $q \geq q_{\text{abs}}$, its first term suffices by (152). This proves the full-range all-order certificate without an order gate.

E.3.4 The unchanged simple-cap coefficient also covers small orders

Only in this algebraic proof, write $X = \beta^L \geq 100$, $z = Ym/\gamma$, $\chi = 1 + m/\gamma$, and $u = \sqrt{\log(en)} \geq 1$. Here z is a scalar proof abbreviation, never a preactivation. The simple cap says $z \leq X^{-30}$.

First the cap implies the full fitting and source restrictions. The Gaussian recursion gives $H \leq (2\beta)^L \leq \beta^{3L/2}$. The geometric sum for F gives $8H\sqrt{F} \leq \beta^{5L}$. Moreover $D_\ell \leq \beta^{2L}$, $\sqrt{C_\ell} \leq \beta^{8L}$, and $E \leq \beta^{12L}$. For the middle bound, unroll $C_\ell = 192s^2 H^4 D_\ell^2 + 246s^2 C_{\ell-1}$, use $192s^2, 246s^2 \leq \beta^5$, $H^4 D_\ell^2 \leq \beta^{10L}$, and sum at most L terms; the result is below β^{16L} . For E , use $130L \leq (\beta^L)^2$. Substitution in every entry of S_*^{Leg} gives $S_*^{\text{Leg}} \geq \beta^{-8L}$. Thus $8z \leq S_*^{\text{Leg}}$. The shared source power proof gives $S_*^{\text{src}} \geq \beta^{-26L}$ and $K_{\text{src}} \leq \beta^{21L}$; hence $16z \leq S_*^{\text{src}}$. No extra label restriction has entered.

For the remaining algebra use the weaker $H \leq X^2$, $\gamma/m \leq X^4$, $D_\ell \leq X^2$, and $\sqrt{C_\ell} \leq X^8$. The cap gives $R \leq X^{-27}$, $S \leq 1$, $a_0 \leq 1$, and $B_\delta \leq X^3 R \leq 1$. The affine segment recurrence gives $Q \leq X^2$, whence

$$F_{\text{seg}} \leq X^4, \quad d_0 \leq 2X^3, \quad d_1 \leq LX^7 \leq X^8, \quad p \leq X^4, \quad K_0 \leq X^9, \quad K_1 \leq X^{13}.$$

Using $M_n \leq 32X^{21}zu$, $14 \leq X$ and $448 \leq X^2$, the exact stability exponent obeys

$$14(K_0 + K_1 M_n)z \leq X^{10}z + X^{36}z^2u. \quad (153)$$

Both $X^{10}z \leq X^{-20}$ and $X^{36}z^2 \leq X^{-24}$ are at most one. The elementary convexity inequalities $e^v \leq 1 + 2v$ for $0 \leq v \leq 1$, and $e^{\vartheta u} \leq 1 + \vartheta(e^u - 1)$ for $0 \leq \vartheta \leq 1$, therefore show

$$\mathcal{A}_n \leq X\Xi_n, \quad \Xi_n = (1 + X^{100}z)[1 + X^{100}z^2(e^u - 1)]. \quad (154)$$

For explicit verification of the remaining coefficient assembly, the speed recurrence has $T_1 \leq X^3 R$, $T_\ell \leq (4X^4 + 130X^{10})R \leq X^{12}R$, forcing at most $2X^{15}R$, and total propagation at most $(9s)^L \leq X^2$. Summing at most L terms proves $V_h \leq X^{20}R$. The displayed tangent trace gives $C_c \leq X^{11}$. Direct substitution then gives

$$\begin{aligned} F_z &\leq X^5, \quad P_0 \leq X^4, \quad F_h \leq X^{24}z^2, \quad F_h/Y \leq X^{22}z\sqrt{\chi}, \\ B_d &\leq X^{33}(1+zu), \quad V_z \leq 2X^{12}R, \quad V_\delta \leq X^{40}(1+zu), \\ b_1 &\leq X^{16}z\sqrt{\chi}, \quad b_0 \leq X^{42}z(1+zu), \quad C_f \leq X^7. \end{aligned} \tag{155}$$

For the label-sensitive terms, the explicit substitutions are $F_h \leq 32X^{20}z^2\sqrt{\gamma/m}$ and $F_h/Y \leq 32X^{20}z\sqrt{m/\gamma}$. The two terms of B_d are at most $2X^3$ and $32LX^{29}zu$. The non-carrier and carrier parts of V_δ are at most $4LX^5$ and $64LX^{36}Rzu$. Finally $b_1 \leq 96X^{14}z\sqrt{m/\gamma}$; the two terms of b_0 are at most $6X^{40}z(1+zu)$ and $32X^5z^{3/2}$. Since $z \leq 1$, $L \leq X$, and $X \geq 100$, these imply each entry of the ledger without discarding the actual label factors.

In particular $b_0 + b_1\sqrt{\log q} \leq X^{43}z\sqrt{\chi}(1+zu)\sqrt{\log(eq)}$. Combining the ledger with $4(L-1) \leq X^2$ and (154) shows that the numerator of (152), divided by Y , is at most

$$X^{75}z^2\chi(1+zu)\Xi_n\sqrt{\log(eq)}.$$

Since $(1+zu)(1+X^{100}z) \leq 3(1+X^{100}zu)$, this is at most $P_n\sqrt{\log(eq)}$. Thus the unchanged coefficient works whenever $q \geq q_{\text{abs}}$.

It remains to cover $1 \leq q < q_{\text{abs}}$ with this same coefficient. Retain the factor $\sqrt{a_0} = 2\sqrt{z}$ in the exact threshold:

$$q_{\text{abs}} \leq 2X^{60}z^{5/2}(1+zu)\exp\{X^{10}z + X^{36}z^2u\}.$$

Put $\alpha = X^{10}z$ and $\vartheta = X^{36}z^2$. Then $e^{2\alpha} \leq 2$, $2\vartheta \leq 1/2$, and, for $u \geq 1$, the exponential series gives $u^2e^{u/2} \leq 16(e^u - 1)$. Consequently

$$(1+zu)^2e^{2\alpha+2\vartheta u} \leq 4 + (8\vartheta + 64z^2)(e^u - 1) \leq 4[1 + X^{100}z^2(e^u - 1)].$$

Indeed expand $(1+zu)^2 \leq 2 + 2z^2u^2$, apply convexity to $e^{2\vartheta u}$, and use $18X^{36} \leq X^{100}$. Since $12H\sqrt{m/\gamma} \leq X^3\sqrt{\chi}$, this proves

$$12H\sqrt{m/\gamma}q_{\text{abs}}^2 \leq 16X^{123}z^5\sqrt{\chi}[1 + X^{100}z^2(e^u - 1)] \leq P_n.$$

The final comparison is exactly $16X^{23}z^3 \leq 16X^{-67} \leq 3 \leq 3\sqrt{\chi}(1 + X^{100}zu)$. Therefore, if $q < q_{\text{abs}}$,

$$\|f_{\text{Leg},n,q} - f_n\|_* \leq U \leq YP_n/q_{\text{abs}}^2 \leq YP_n\sqrt{\log(eq)}/q^2.$$

If $q_{\text{abs}} < 1$, the already proved large-order case covers every admissible integer. This completes (65).

For (66), the cap and $\gamma/m \leq X^4$ give $Y \leq X^{-26} \leq 1$, so $z \leq m/\gamma \leq \chi$. Thus the two bracket factors in P_n are at most $2X^{100}\chi u$ and $2X^{100}\chi^2e^u$. Together with $z^2 \leq \chi^2$, this gives $P_n \leq 12X^{300}\chi^6ue^u$, as asserted.

E.3.5 Proof of the scalar inverses and their storage rates

For $g(q) = \sqrt{\log(eq)}/q^2$, $g'/g = [2q\log(eq)]^{-1} - 2/q < 0$ on $[1, \infty)$, and $g(q) \rightarrow 0$. Hence the minimum exists and every larger integer works.

For $r > 1$, let $\ell = \log(e+r) > 1$. The chosen order lies between $2\sqrt{r}\ell^{1/4}$ and $3\sqrt{r}\ell^{1/4}$, and $\log(eq_{\text{suff}}) \leq 4\ell$. The latter follows from $\log(3e) \leq 2\log(e+1) \leq 2\ell$, $\log r \leq \ell$, and $\log \ell \leq \ell$. Therefore $r\sqrt{\log(eq_{\text{suff}})}/q_{\text{suff}}^2 \leq 1/2$. In particular this formula even leaves half of the error budget unused.

In the full-range certificate the other summand is Uq_{abs}^4/q^4 , which is decreasing and tends to zero. Thus its sum with $C_ng(q)$ has a finite least integer inverse. Allocating $\varepsilon/2$ to each summand gives (76) directly, including ratios at most one. Storage is affine and strictly increasing in q , so the floor in (77) is the largest

feasible order. At fixed positive labels, all finite recurrence coefficients except M_n and $\mathcal{A}_n$ are bounded by polynomials in $\sqrt{\log(en)}$, and $\log \mathcal{A}_n$ is affine in that quantity. Therefore $C_n, q_{\text{abs}} = n^{o(1)}$. Substitution in the two sufficient inverse branches gives the stated polynomial-accuracy orders and the exact storage formula gives their moving-state powers.

For (78), put $a = \log(e + Q_n) > 1$. Then $4Q_n a^{1/4} \leq q_n \leq 5Q_n a^{1/4}$ and $\log(eq_n) \leq 4a$, since $Q_n \geq 3$, $\log(5e) \leq 2a$, $\log Q_n \leq a$, and $\log a \leq a$. The forward certificate is therefore at most $YP_n/(8Q_n^2) \leq Y/(8\sqrt{n})$, as stated.

E.4 Harmonic proofs

E.4.1 Proof of the source expansion and finite coefficient count

The argument first obtains decay in spherical degree from holomorphy on the intrinsic tube. Temporal Fourier decay then produces a weighted simplex of coefficients. This proves a finite-dimensional approximation without evaluating any trained trajectory during runtime.

For $d \geq 2$, the intrinsic complex sphere tube of radius r is

$$\mathcal{Q}_r = \{u \cosh h + iv \sinh h : u, v \in \mathbb{R}^d, \|u\| = \|v\| = 1, u^T v = 0, |h| \leq r\}.$$

It equals $\{z : z^T z = 1, \|\text{Im } z\| \leq \sinh r\}$: write $z = b + ic$, use $b^T c = 0$ and $\|b\|^2 - \|c\|^2 = 1$, and put $h = \text{arsinh } \|c\|$. The case $c = 0$ permits any unit tangent vector. This gives all intrinsic imaginary directions.

Let F be scalar, holomorphic near $\mathcal{Q}_r$, and bounded there by M . Use normalized real-sphere measure. Harmonic polynomials of degree j , restricted to the sphere, form a space $\mathcal{H}_j$ of dimension

$$h_j = \binom{j+d-1}{d-1} - \binom{j+d-3}{d-1} \leq 2 \binom{j+d-2}{d-2}.$$

Indeed every homogeneous polynomial decomposes uniquely into terms $\|x\|^{2k} H_{j-2k}(x)$ with harmonic H_{j-2k} : induction uses $\Delta(\|x\|^{2k} H_l) = 2k(2l + 2k + d - 2)\|x\|^{2k-2} H_l$. The coefficients are nonzero for $k \geq 1$; dimension counting gives the formula. The polar Laplacian gives eigenvalue $-j(j+d-2)$, so integration by parts makes different degrees orthogonal.

Let P_j be the orthogonal projection onto $\mathcal{H}_j$. For a real orthonormal basis $Y_{j,b}$, rotational invariance and the trace of its reproducing kernel give $\sum_b Y_{j,b}(x)^2 = h_j$. Consequently

$$\|P_j g\|_\infty \leq h_j \|g\|_\infty, \quad |Y_{j,b}(x)| \leq \sqrt{h_j}. \quad (156)$$

To extract exponential decay, average over tangent directions:

$$(A_\zeta F)(x) = \int_{v \perp x, \|v\|=1} F(x \cos \zeta + v \sin \zeta) d\sigma_x(v).$$

This is holomorphic and bounded by M for $|\text{Im } \zeta| \leq r$. For $d \geq 3$, put $\nu = (d-2)/2$ and define $C_j^\nu(t)$ by the generating function $(1 - 2tw + w^2)^{-\nu}$. Differentiation of that generating function gives

$$(1 - t^2)(C_j^\nu)'' - (d-1)t(C_j^\nu)' + j(j+d-2)C_j^\nu = 0, \quad C_j^\nu(1) = (2\nu)_j/j!,$$

where $(b)_j = b(b+1) \cdots (b+j-1)$ and $(b)_0 = 1$. Averaging a degree- j harmonic over rotations fixing x gives the unique zonal harmonic with its value at x : write an invariant homogeneous polynomial in powers of $x^T y$ and $\|y - (x^T y)x\|^2$; its harmonic equation determines every coefficient from the leading one. The resulting sphere equation is the displayed differential equation. Hence on $\mathcal{H}_j$, A_ζ acts by $C_j^\nu(\cos \zeta)/C_j^\nu(1)$.

For real ζ , the joint law of $(x, x \cos \zeta + v \sin \zeta)$ is symmetric, since it is the rotationally invariant law on pairs with the specified inner product. Thus A_ζ is self-adjoint. Apply this identity to each harmonic coefficient, and continue the scalar analytic identity to $\zeta = ir$, to obtain

$$P_j A_{ir} F = \frac{C_j^\nu(\cosh r)}{C_j^\nu(1)} P_j F.$$

No harmonic expansion at complex points was used here. Factoring the generating function into $(1 - e^r w)^{-\nu}(1 - e^{-r} w)^{-\nu}$ and retaining one positive term gives $C_j^\nu(\cosh r) \geq e^{rj}(\nu)_j/j!$. Moreover

$$\frac{(2\nu)_j}{(\nu)_j} = 2 \prod_{k=1}^{j-1} \left(1 + \frac{\nu}{\nu + k}\right) \leq 2 \exp\left(\nu \int_0^{j-1} \frac{dx}{\nu + x}\right) \leq 2^d(j+1)^d.$$

The first equality applies for $j \geq 1$; the last bound also holds at zero. Together with the projector bound and $h_j \leq 2d^{d-2}(j+1)^{d-2}$, this proves

$$\|P_j F\|_\infty \leq MD_d(j+1)^{b_d} e^{-rj}. \quad (157)$$

For $d = 2$, averaging the two tangent directions has multiplier $\cosh(jr) \geq e^{jr}/2$; the same inequality holds with $D_2 = 8, b_2 = 2$. The bound makes the real-sphere harmonic series absolutely uniformly convergent. Its sum equals F : restrictions of polynomials contain constants and separate points and are dense by the real Stone–Weierstrass theorem; harmonic decomposition then makes a continuous function with all harmonic coefficients zero vanish.

For a coordinate of any source family, substitute $t = T(1 + \cos u)/2$. The resulting function is even and periodic in u . Since $T \geq T_0$, the counting gate (109) gives $0 < \alpha_T \leq \alpha_{T_0} \leq 1$. If $|\operatorname{Im} u| \leq \alpha_T$, its imaginary time displacement is at most $T\alpha_T = r_t/4$ and its real overshoot is at most $T\alpha_T^2/2 \leq r_t$. Fourier contour translation, first in time and then the proved harmonic estimate, gives

$$\|P_j G_k\|_\infty \leq M_n D_d(j+1)^{b_d} e^{-\alpha_T |k| - r_q j}.$$

For $0 < r \leq 1$, comparison with the integral over successive unit intervals proves

$$\sum_{j \geq 0} (j+1)^{b_d} e^{-rj/2} \leq e^r b_d! (2/r)^{b_d+1} \leq 3b_d! (2/r)^{b_d+1},$$

and $\sum_{k \in \mathbb{Z}} e^{-\alpha_T |k|/2} \leq 6/\alpha_T$. Split the omitted exponential into two halves. Outside $\alpha_T |k| + r_q j \leq H(T, \eta)$, the total tail is at most $M_n P_T e^{-H(T, \eta)/2} = \eta/16$.

Evenness leaves one real cosine coefficient for each $k \geq 0$, and each spherical degree has exactly h_j real coefficients. This is the count in the coefficient formula. Using $h_j \leq 2 \binom{j+d-2}{d-2}$, the count is at most twice the number of nonnegative integer d -tuples satisfying $\alpha_T k + r_q \sum_{i=1}^{d-1} b_i \leq H(T, \eta)$. The disjoint unit cubes based at these tuples lie in the weighted simplex enlarged by $\alpha_T + (d-1)r_q$. Its volume yields

$$N(T, \eta) \leq \frac{2[H(T, \eta) + \alpha_T + (d-1)r_q]^d}{d! \alpha_T r_q^{d-1}}. \quad (158)$$

For $d = 1$, positive cosine coefficients are bounded by $2M_n e^{-\alpha_T k}$. Since $1 - e^{-\alpha_T} \geq \alpha_T/2$, omission above $H_1(T, \eta)/\alpha_T$ costs at most $4M_n \alpha_T^{-1} e^{-H_1(T, \eta)} = \eta/16$. The two points of the sphere require two temporal families, so the dimension bound is $B + 8N_1$; it is $B + 4N$ when $d \geq 2$.

E.4.2 Finite setup from the initialized network

The exact integral coefficients used above serve to prove the error bound. Their finite approximations can be computed from initial derivatives. The following explicit map supplies this fact for every finite T , even when the needed setup work is very large. For this paragraph only put

$$\begin{aligned} \chi &= \frac{\pi T}{8r_t}, \quad b_0 = \tanh \chi, \quad b_1 = \tanh(\chi + \pi/4), \quad \vartheta = b_0/b_1, \quad \psi(\xi) = \frac{\xi - \vartheta}{1 - \vartheta \xi}, \\ \mathfrak{t}(\xi) &= \frac{T}{2} + \frac{2r_t}{\pi} \log \frac{1 + b_1 \psi(\xi)}{1 - b_1 \psi(\xi)}. \end{aligned} \quad (159)$$

The logarithm is the analytic branch zero when its fraction is one. The disk automorphism has $|\psi| < 1$. For $|w| < b_1$, the real part of $\log((1+w)/(1-w))$ has magnitude less than $2 \operatorname{arctanh} b_1 = 2\chi + \pi/2$, and its

imaginary part has magnitude less than $\pi/2$. Thus the map lies inside the source rectangle, has $\mathfrak{t}(0) = 0$, and maps the real interval $[0, \xi_*]$ onto $[0, T]$, where $\xi_* = 2\vartheta/(1 + \vartheta^2) < 1$.

For a fixed real query and any source coordinate g , its composed Taylor coefficients are

$$a_j = \sum_{k=0}^j \frac{\partial_t^k g(0)}{k!} [\xi^j] \mathfrak{t}(\xi)^k. \quad (160)$$

They depend on only finitely many initial derivatives. Those derivatives are obtained by finite differentiation of the dense ODE and passive backward recursion, using the initial matrices and labels. Cauchy's bound gives $|a_j| \leq M_n$. A degree- K Taylor sum therefore has error at most $M_n \xi_*^{K+1}/(1 - \xi_*)$ at every required real time. This tends to zero and has an explicit finite cutoff for any positive nodal tolerance.

Choose each real harmonic basis by harmonic decomposition and Gram-Schmidt of monomials. Its polynomial coefficients and sphere integrals are fixed, finite data. Each retained coefficient is an integral against a cosine and a harmonic polynomial. Finite Riemann quadrature in the time angle and the ordinary real sphere angles approximates it to arbitrary accuracy. This can be certified by Cauchy derivative bounds in the time strip and along great circles, together with the known polynomial derivatives and sphere Jacobian on compact angle boxes. Finitely many coefficients permit one common quadrature mesh and one common initial-jet cutoff.

If $J = \lfloor H(T, \eta)/r_q \rfloor$, the real basis functions have magnitude at most $\max_{j \leq J} \sqrt{h_j}$. Choosing each coefficient error at most $\eta/[16N(T, \eta) \max_{j \leq J} \sqrt{h_j}]$ makes their total reconstruction error at most $\eta/16$, including the factor two converting positive Fourier modes to cosines in the coefficient tolerance. Add the tail $\eta/16$; there is ample slack to obtain the asserted coordinate error η . The dimension-one case omits the harmonic factor.

For a paired initialized image, differentiation gives exactly $\partial_t^k(W_0 g) = W_0 \partial_t^k g$, and the same holds for W_0^T . Apply identical scalar linear operations in the Taylor formula, quadrature and coefficient truncation to both members. Their coefficient vectors then satisfy the image identity exactly. Each member separately satisfies its coordinate error bound, since the common nodal accuracy can be chosen for all members. No operator-norm conversion of one member's coordinate error is used. This proves the initialization-only paired-source construction, with finite real setup.

E.4.3 Coordinate selection and the exact source metric

The finite selection fact needed here is: if vectors $v_i \in \mathbb{R}^r$ satisfy $\sum_i v_i v_i^T = I_r$, then there are nonnegative weights, at most $9r$ of them positive, for which $I_r \preceq \sum_i s_i v_i v_i^T \preceq 4I_r$. It is the sparsity-nine specialization of Batson, Spielman and Srivastava, Theorem 3.1. For completeness the required specialization has the following direct barrier proof.

Starting at $A = 0$, lower barrier $l = -3r$, and upper barrier $u = 6r$, maintain

$$lI \prec A \prec uI, \quad \Phi_l(A) = \text{tr}(A - lI)^{-1} \leq 1/3, \quad \Phi^u(A) = \text{tr}(uI - A)^{-1} \leq 1/6.$$

The barriers are moved by one and two, respectively, at each step. For $l' = l + 1$, $u' = u + 2$, define

$$U(v) = \frac{v^T(u'I - A)^{-2}v}{\Phi^u(A) - \Phi^{u'}(A)} + v^T(u'I - A)^{-1}v,$$

$$L(v) = \frac{v^T(A - l'I)^{-2}v}{\Phi_{l'}(A) - \Phi_l(A)} - v^T(A - l'I)^{-1}v.$$

Both denominators are positive. Also $A - l'I$ is positive: $\Phi_l \leq 1/3$ forces every eigenvalue's distance above l to be at least three. The elementary rank-one inverse identity

$$(B + \theta v v^T)^{-1} = B^{-1} - \frac{B^{-1} v v^T B^{-1}}{\theta^{-1} + v^T B^{-1} v}$$

is verified by multiplication. Applied with either sign, it shows that $U(v) \leq \theta^{-1} \leq L(v)$, $\theta > 0$, preserves both potential bounds at the shifted barriers. The upper denominator is strictly positive because the first

term in $U(v)$ is positive for nonzero v ; hence the upper spectral barrier is preserved as well. The lower barrier is preserved by positivity of the rank-one addition.

Such a vector and step size always exist. Summing over i uses $\sum_i v_i v_i^T = I$. In the upper sum, each eigenvalue a_j satisfies

$$\frac{1}{u - a_j} - \frac{1}{u' - a_j} = \frac{2}{(u - a_j)(u' - a_j)} \geq \frac{2}{(u' - a_j)^2}.$$

Thus $\sum_i U(v_i) \leq 1/2 + \Phi^{u'}(A) \leq 2/3$. For the lower sum put $a'_j = 1/(a_j - l')$ and $b'_j = 1/(a_j - l) = a'_j/(1 + a'_j)$, and let $D = \Phi_{l'} - \Phi_l = \sum_j a'_j b'_j$. Weighted Cauchy–Schwarz and $\sum b'_j \leq 1/3 \leq 1$ give

$$D^2 \leq \left(\sum_j (a'_j)^2 b'_j \right) \sum_j b'_j \leq \sum_j (a'_j)^2 b'_j = \sum_j (a'_j)^2 - D.$$

It follows that $\sum_i L(v_i) = \sum_j (a'_j)^2 / D - \Phi_{l'} \geq 1 - \Phi_l \geq 2/3$. Hence some nonzero v_i has $L(v_i) \geq U(v_i) > 0$. Choose a positive θ with reciprocal in that interval and update A by $\theta v_i v_i^T$. After $9r$ steps the barriers are $6r$ and $24r$. Dividing all accumulated weights by $6r$ proves the desired selection fact. Repeated choices only reduce the final support size.

Apply this fact at a layer with $v_i = (U_j)_{i,:}^T / \sqrt{n}$. Its hypothesis is exactly $U_j^T U_j / n = I$. Let $P = (U_j)_{I_j}$ and $D = \text{diag}(s_i/n)$ on the positive support. Then $G = P^T D P$ has spectrum in $[1, 4]$. Set $Z = D^{1/2} P$, and define

$$M = D^{1/2} [Z G^{-2} Z^T + I - Z G^{-1} Z^T] D^{1/2}. \quad (161)$$

The last two terms give the orthogonal projector on $\text{ran } Z^\perp$; on $\text{ran } Z$ the first term has the eigenvalues of G^{-1} . Therefore $D/4 \preceq M \preceq D$. Multiplication gives $P^T M P = G G^{-2} G + G - G G^{-1} G = I$. This is exact source isometry. Since the constant is a source member, its empirical squared norm is one, giving $\mathbf{1}^T M \mathbf{1} = 1$ and $\mathbf{1}^T D \mathbf{1} \leq 4$.

If D is any coordinate multiplication operator, the diagonal metric and the two metric comparisons imply $\|D\|_{M \rightarrow M} \leq 2\|D\|_{\infty \rightarrow \infty}$. Thus activations are $2s$ -Lipschitz in this metric, and $\|\phi_j(z)\|_M \leq 2b + 2s\|z\|_M$. These estimates have no minimum-weight factor. The metric adjoint of D is $M^{-1} D M$, which need not equal D .

The initialized mixer in the initialization formula is the compression of $W_0^{(j)}$ between two isometric source spaces, so its norm is at most $\|W_0^{(j)}\|$. If $g \in E_{j-1}$ and $W_0^{(j)} g \in E_j$, direct substitution gives its exact forward action on $g_{I_{j-1}}$. Its metric adjoint gives the exact reverse action whenever $d \in E_j$, $W_0^{(j)T} d \in E_{j-1}$. The initial additions therefore preserve the training forward pass by induction over layers, and exact top isometry preserves its Gram. First-weight columns give the first operator bound. This establishes every metric and initialization property used by the Harmonic runtime.

E.4.4 Independent fitting and endpoint control

Work until the first exit from operator caps nine, sphere feature cap $2H_c$, or normalized Gram margin $Q_C \succeq \lambda I/4$. The local vector field is smooth on this open Gram-positive set. Give raw parameter tuples the squared norm

$$\|\theta_C\|_{\text{par}}^2 = \text{tr}(A_C^T M_1 A_C) + \sum_{j=2}^L \|M_j^{1/2} B_C^{(j)} M_{j-1}^{-1/2}\|_F^2 + w_C^T M_L w_C.$$

Every term of K_C is a Gram matrix, using tensor products for the hidden terms. Expanding the raw velocities, including all cross-sample products, gives

$$-\frac{d}{dt} \rho_C^2 = 4c_C^T K_C c_C / m^2 = \|\dot{\theta}_C\|_{\text{par}}^2, \quad \rho_C = \|c_C\|_2 / \sqrt{m}, \quad K_C / m \succeq Q_C. \quad (162)$$

This identity is algebraic and does not require gradient backpropagation. The stopped gap implies $-\dot{\rho}_C \geq \lambda \rho_C / 2$. Since $\|\dot{\theta}_C\|^2 = 2\rho_C(-\dot{\rho}_C)$, $\|\dot{\theta}_C\| \leq 2(-\dot{\rho}_C)/\sqrt{\lambda}$. Hence

$$\rho_C(t) \leq Y e^{-\lambda t/2}, \quad \int_0^t \rho_C \leq 2z, \quad \int_0^t \|\dot{\theta}_C\| \leq 2\alpha, \quad \|w_C(t)\| \leq 2\alpha. \quad (163)$$

If a zero deficit is reached, all velocities vanish, so the same bounds continue with the constant solution.

Let $P_C = V_C Q_C^{-1} V_C^*$, the orthogonal projector in the top metric. The corrected readout is the orthogonal sum $(I - P_C)w_C + V_C Q_C^{-1}(y - c_C)/\sqrt{m}$. The two squared norms are at most $4\alpha^2$ and $16\alpha^2$, respectively, proving $\|\hat{w}_C\| \leq \sqrt{20}\alpha < 5\alpha$. The gate and mixer bounds then give $\|\delta_{C,a}^{(j)}\| \leq d_j^c R_c$. Each hidden block velocity is at most $2\rho_C R_c U_j^c$. Its integral is at most $20U_j^c Y^2/\lambda^{3/2}$. Subtracting the initialized forward pass propagates with factor $18s$, and the direct matrix term costs $2s(2H_c)$; these are exactly the recurrence defining F_j^c . Therefore every feature displacement is at most $20F_j^c Y^2/\lambda^{3/2}$.

The full label interval makes these bounds at most $5\sqrt{\lambda}/(64H_c^2)$, hence less than $1/(8H_c)$. Initialized features were bounded by H_c , and operators by eight, so both stopped caps improve strictly. The normalized training-feature matrix changes in operator norm by less than $\sqrt{\lambda}/8$. Its initial smallest singular value is at least $\sqrt{\lambda}/2$, so it remains above $(1/\sqrt{2} - 1/8)\sqrt{\lambda} > \sqrt{\lambda}/2$. The stopped Gram margin also improves. No finite first exit exists. Bounded raw parameters and a strict Gram margin permit continuation at every finite time. Finite total path length gives parameter convergence; the readout formula gives effective-readout convergence; and $c_C \rightarrow 0$ gives exact fitting.

For the endpoint coefficient set $T_C = V_C Q_C^{-1}$ and $b_C = (y - c_C)/\sqrt{m}$. Differentiating the inverse and collecting projectors yields

$$\dot{T}_C = (I - P_C)\dot{V}_C Q_C^{-1} - T_C \dot{V}_C^* T_C, \quad \dot{P}_C = (I - P_C)\dot{V}_C T_C^* + T_C \dot{V}_C^* (I - P_C). \quad (164)$$

The bounds are $\|T_C\| \leq 2/\sqrt{\lambda}$, $\|\dot{T}_C\| \leq 8\|\dot{V}_C\|/\lambda$, $\|\dot{P}_C\| \leq 4\|\dot{V}_C\|/\sqrt{\lambda}$, and $\|\dot{V}_C\| \leq 2R_c F_c \rho_C$. The Gram formula gives $\|K_C/m\| \leq G_c$, hence $\|\dot{b}_C\| \leq 2G_c \rho_C$, while $\|\dot{b}_C\| \leq 2Y$, $\|\dot{w}_C\| \leq 4H_c \rho_C$. Differentiate $\hat{w}_C = (I - P_C)w_C + T_C b_C$ to obtain $\|\dot{\hat{w}}_C\| \leq B_w \rho_C$. For every unit query, $\|\dot{h}_C^{(L)}\| \leq 2R_c F_c \rho_C$, so $|\dot{f}_C(v)| \leq B_f \rho_C$. Integration proves

$$\sup_{\|v\|=1} |f_C(\infty, v) - f_C(t, v)| \leq 2B_f^c z e^{-\lambda t/2}. \quad (165)$$

Together with the dense source bound $|\dot{f}_n(v)| \leq 2\mathcal{K}\rho_n$ and its fitting tail, comparing the two flows to their values at T costs at most $\mathcal{D}e^{-\lambda T/4}$. Neither flow is frozen. Finally $\|\hat{w}_C\| \leq 5\alpha$, $\|h_C\| \leq 2H_c$, $\|w_n\|_2/\sqrt{n} \leq 2\alpha$, and $\|h_n\|_2/\sqrt{n} \leq 2H_d$ prove the baseline certificate.

E.4.5 Source metric estimates and reference energy

Fix a source horizon and coordinate tolerance $\eta \leq \eta_0$. All norms of selected vectors in this proof use their layer metric; dense vector norms always mean the explicitly normalized Euclidean quantity $\|u\|_2/\sqrt{n}$. If a vector u has an approximant p in the source space with $\|u - p\|_\infty \leq e$, then

$$\frac{\|u - p\|_2}{\sqrt{n}} \leq e, \quad \|(u - p)_I\|_M \leq 2e, \quad \|u_I\|_M \leq \frac{\|u\|_2}{\sqrt{n}} + 3e. \quad (166)$$

The first two statements use the unit empirical mass and the diagonal mass at most four; the last uses exact isometry of p . For two vectors with approximation errors e_u, e_v and dense RMS bounds U, V , insertion of their approximants in both pairings gives

$$|\langle u_I, v_I \rangle_M - u^T v/n| \leq 3(e_u V + e_v U) + 9e_u e_v. \quad (167)$$

For example the dense pairing error is at most $e_u V + e_v U + e_u e_v$, and the selected error at most $2e_u V + 2e_v U + 8e_u e_v$. Thus feature pairings cost at most $P_h \eta$, and response pairings at most $SP_\delta \eta$, using $\eta \leq 1$ and $\eta \leq S$, respectively.

For proof only, define $A_R = (A_n)_{I_1}$, $w_R = (w_n)_{I_L}$, and

$$B_R^{(j)}(t) = B_C^{(j)}(0) + \int_0^t \frac{2}{m} \sum_a c_{n,a}(s) \delta_{n,a,I_j}^{(j)}(s) h_{n,a,I_{j-1}}^{(j-1)}(s)^T M_{j-1} ds. \quad (168)$$

The true restricted features and responses need not be the forward and backward pass through these matrices. The initial paired actions have defect at most $(8 \cdot 2 + 2)\eta = 18\eta$. The exact dense rank-one integral and the pairing estimate show that the learned forward defect is at most $S^2(\tau + 3)P_h\eta$, and the learned reverse defect at most $S^2H_rP_\delta\eta$. Their total bounds are therefore $A_f\eta, A_b\eta$. Integrating the top-feature pairing defect in the raw readout equation gives the observation bound

$$\sup_{\|v\|=1} |\langle w_R, h_{n,I_L}^{(L)}(v) \rangle_{M_L} - f_n(v)| \leq 16zP_h\eta. \quad (169)$$

This estimate also applies to the normalized vector of training observations, with no sample-count factor.

Let V_n have columns $h_{n,a}^{(L)}/\sqrt{m}$, acting into the dense RMS space, and let V_R have the restricted columns in the selected space. The normalized column-error Hilbert–Schmidt norms are at most η and 2η . Thus, for every sample vector,

$$\|V_R\xi\| \leq \|V_n\xi\| + 3\eta\|\xi\|_2.$$

Set $\nu(t) = \|V_Rc_n/\sqrt{m}\|$. The dense path length and $\dot{w}_n = 2V_n c_n/\sqrt{m}$ give $\int_0^T \nu \leq \alpha + 6\eta z$. Integrating the existing top-feature approximants against $2c_{n,a}/m$ gives a vector in the fixed source space within coordinate error $4\eta z$ of w_n . Finite Riemann sums and closedness of the finite-dimensional source space justify this integral. The norm transfer therefore gives $\|w_R\| \leq 2\alpha + 12\eta z$. Because $\eta \leq Y$ and $\eta/\sqrt{\lambda} \leq \alpha \leq 1/16$, these imply

$$\int_0^T \nu \leq 2\alpha, \quad \|w_R\| \leq 3\alpha. \quad (170)$$

The same transfer applied to true dense responses gives

$$\|\delta_{n,a,I_j}^{(j)}\| \leq 2s(9s)^{L-j}\alpha + 3\eta \leq d_j^E\alpha. \quad (171)$$

Here $\eta \leq Y = \sqrt{\lambda}\alpha \leq H_c\alpha$. This step is why no assumption $\lambda \leq 1$ is needed. No source error was differentiated in any of these arguments.

E.4.6 Full-range readout and deficit cancellation

Let $\mathcal{J}_C$ map sample space into the direct sum of hidden parameter Hilbert spaces, with columns divided by $\sqrt{m}$ equal to

$$\left(\delta_{C,a}^{(1)} v_a^T, (\delta_{C,a}^{(j)} h_{C,a}^{(j-1)T} M_{j-1})_{j=2}^L \right).$$

Define $\mathcal{J}_R$ using the true restricted dense features and responses. A rank-one map $uv^T M$ has Hilbert–Schmidt norm $\|u\|\|v\|$, so the definitions give exactly

$$\begin{aligned} \dot{\theta}_{h,C} &= 2\mathcal{J}_C c_C / \sqrt{m}, & \dot{\theta}_{h,R} &= 2\mathcal{J}_R c_n / \sqrt{m}, & K_C/m &= V_C^* V_C + \mathcal{J}_C^* \mathcal{J}_C, \\ \|\mathcal{J}_C\| &\leq J_C \alpha = 5\alpha \sqrt{F_c}, & \|\mathcal{J}_R\| &\leq J_R \alpha. \end{aligned} \quad (172)$$

The norm bounds use the normalized sum of squared column norms, which dominates the operator norm and introduces no factor $\sqrt{m}$.

Write

$$a_h = \|\theta_{h,C} - \theta_{h,R}\|, \quad e = (c_C - c_n)/\sqrt{m}, \quad u = \|e\|_2, \quad z_w = w_C - w_R, \quad T_C = V_C Q_C^{-1}, \quad P_C = T_C V_C^*,$$

$$p = T_C e, \quad \zeta = z_w + p, \quad b_e = \|p\| + \|\zeta\|, \quad E = a_h + b_e.$$

These are proof variables and are initially zero. The right inverse satisfies $T_C^* T_C = Q_C^{-1}$, $V_C^* T_C = I$, $T_C V_C^* = P_C$, and $\|T_C\| \leq 2/\sqrt{\lambda}$. Forward subtraction with the paired action defect gives

$$\sup_{j,v} \|h_C^{(j)}(v) - h_{n,I_j}^{(j)}(v)\| \leq F(a_h + \eta), \quad \|V_C - V_R\| \leq F(a_h + \eta). \quad (173)$$

Layer one costs a_h in preactivation; later layers cost 9 times the lower feature difference, $H_r a_h$ for the changed mixer, and $A_f \eta$ for the action defect. Multiplication by the gate bound $2s$ proves the defining recurrences for F_j^z, F_j^h , and hence this inequality.

Let $\Delta V = V_C - V_R$ and $d_R = V_R^* w_R - (y - c_n)/\sqrt{m}$. Substitution in the corrected readout gives the exact identity

$$\widehat{w}_C - w_R = (I - P_C)z_w - p - T_C \Delta V^* w_R - T_C d_R.$$

Since $(I - P_C)p = 0$, the preceding energy and observation bounds yield

$$\|\widehat{w}_C - w_R\| \leq b_e + 6zF(a_h + \eta) + 32zP_h\eta/\sqrt{\lambda}. \quad (174)$$

In particular, for every query,

$$|f_C - f_n| \leq H_C \|\widehat{w}_C - w_R\| + 3\alpha F(a_h + \eta) + 16zP_h\eta. \quad (175)$$

The source Gram defect $D = V_R^* V_R + \mathcal{J}_R^* \mathcal{J}_R - K_n/m$ obeys $\|D\| \leq D_s \eta$. To see every term, feature pairings cost $P_h \eta$, response pairings cost $SP_\delta \eta$, and the hidden products are split as $a_R b_R - ab = (a_R - a)b_R + a(b_R - b)$. Their bounds are $SP_\delta H_r^2 \eta$ and $S^2 \tau^2 P_h \eta$. The first-layer input pairing is at most one. Summing gives exactly D_s . A matrix whose entries have magnitude at most d has operator norm at most md ; the normalization by m removes this factor. Writing $\Delta \mathcal{J} = \mathcal{J}_C - \mathcal{J}_R$, the exact factorization is

$$\Delta \mathcal{K} := K_C/m - K_n/m = V_C^* \Delta V + \Delta V^* V_R + \mathcal{J}_C^* \Delta \mathcal{J} + \Delta \mathcal{J}^* \mathcal{J}_R + D. \quad (176)$$

Its second feature term will be controlled by the actual selected readout velocity ν , retaining its energy scale.

Subtracting the exact deficit and raw-readout equations gives

$$\dot{e} = -2(Q_C + \mathcal{J}_C^* \mathcal{J}_C)e - 2\Delta \mathcal{K} c_n/\sqrt{m}, \quad \dot{z}_w = 2V_C e + 2\Delta V c_n/\sqrt{m}.$$

Using $T_C Q_C = V_C$, the terms $2V_C e$ cancel after lifting the deficit error:

$$\dot{\zeta} = 2\Delta V c_n/\sqrt{m} + \dot{T}_C e - 2T_C \mathcal{J}_C^* \mathcal{J}_C e - 2T_C \Delta \mathcal{K} c_n/\sqrt{m}. \quad (177)$$

Only the actual Harmonic feature map is differentiated. The derivative identity already proved and $Q_C^{-1}e = T_C^* p$ imply

$$\|\dot{T}_C e\| \leq 40zF_c \rho_C \|p\|.$$

In the equation for $p = T_C e$, pairing with p gives the negative term $-2\langle p, V_C e \rangle = -2u^2$. The hidden Gram need not be dissipative in this lifted metric. Its absolute contribution is bounded by

$$2\|p\|\|T_C\|\|\mathcal{J}_C\|^2 u \leq 8\|\mathcal{J}_C\|^2 u^2/\lambda \leq 200z^2 F_c u^2 \leq \frac{25}{32H_c^2} u^2. \quad (178)$$

At least $39u^2/32$ remains dissipative. Thus this absorption uses precisely the full runtime label allowance. Put $\mathcal{F} = 2\|T_C \Delta \mathcal{K} c_n/\sqrt{m}\|$ and

$$I(t) = \int_0^t [40zF_c \rho_C b_e + \mathcal{F}] ds.$$

Replace $\|p\|$ by $(\|p\|^2 + \vartheta^2)^{1/2}$ in the norm calculation, integrate, and let $\vartheta \downarrow 0$. The nonnegative damping integrands increase to $u^2/\|p\|$, defined as zero when $p = 0$, since then $e = V_C^* p = 0$. Monotone convergence proves

$$\|p(t)\| + \int_0^t \frac{u^2}{\|p\|} ds \leq I(t), \quad \int_0^t u ds \leq \frac{2}{\sqrt{\lambda}} I(t). \quad (179)$$

The second inequality uses $\|p\| \leq 2u/\sqrt{\lambda}$. In the equation for ζ , the integrated hidden-Gram term therefore costs at most $200z^2F_cI \leq 25I/32$. Consequently

$$b_e(t) \leq 2F \int_0^t \rho_n(a_h + \eta) ds + 3I(t). \quad (180)$$

Let $M = 1 + 32zK_{\text{src}}\sqrt{\log(en)}$; the all-time carrier extension proved below bounds every actual dense training carrier by M . Subtract backward responses by the exact identity

$$\delta_C^{(j)} - \delta_R^{(j)} = \phi'_j(z_C^{(j)}) \odot (k_C^{(j)} - k_R^{(j)}) + [\phi'_j(z_C^{(j)}) - \phi'_j(z_R^{(j)})] \odot k_R^{(j)}.$$

The second term costs at most $2t_2MF_j^z(a_h + \eta)$. The first term propagates the upper response difference with coefficient $18s$, adds the changed mixer times a true response of norm at most $d_{j+1}^E\alpha$, and adds $2sA_b\eta$. At the top use the effective-readout estimate. These terms are exactly the recurrence for B_j . Subtracting each rank-one hidden direction next gives

$$\|\Delta\mathcal{J}\| \leq B_h[b_e + M(a_h + \eta) + \eta/\sqrt{\lambda}]. \quad (181)$$

The factor M is added in the gate forcing, so it appears once, rather than being multiplied at every layer.

Apply the factored Gram difference in its displayed order. The projector identity $T_C V_C^* = P_C$ handles the first feature term, and the second term uses ν . The result is

$$\begin{aligned} \mathcal{F} &\leq 2F\rho_n(a_h + \eta) + 4F\nu(a_h + \eta)/\sqrt{\lambda} \\ &\quad + 4z(J_C + J_R)B_h\rho_n[b_e + M(a_h + \eta) + \eta/\sqrt{\lambda}] + 4D_s\eta\rho_n/\sqrt{\lambda}. \end{aligned} \quad (182)$$

The hidden-parameter equation and the lifted-error integral give

$$a_h(t) \leq \frac{5}{4}I(t) + 2B_h \int_0^t \rho_n[b_e + M(a_h + \eta) + \eta/\sqrt{\lambda}] ds,$$

because $4zJ_C = 20z\sqrt{F_c} \leq 5/(4H_c) \leq 5/4$. Adding the two error bounds and enlarging $17I/4$ to $5I$ gives

$$E(t) \leq \int_0^t [200zF_c\rho_C + K_1M\rho_n + 20F\nu/\sqrt{\lambda}][E + \eta(1 + \lambda^{-1/2})] ds. \quad (183)$$

Here $b_e + M(a_h + \eta) + \eta/\sqrt{\lambda} \leq M[E + \eta(1 + \lambda^{-1/2})]$. Both source-error terms are required; replacing their sum by a maximum without a factor would be incorrect. For a nonnegative coefficient $b(t)$, the integral inequality $E \leq \int b(E + c)$, with $E(0) = 0$, is bounded by the solution $c(e^{\int b} - 1)$: differentiate its integral majorant and apply the integrating factor $e^{-\int b}$. Using $\int \rho_C, \int \rho_n \leq 2z$ and $\int \nu \leq 2\alpha$ therefore proves

$$E(t) \leq \eta(1 + \lambda^{-1/2})(e^{\mathcal{B}_n} - 1). \quad (184)$$

Insert this bound into the query-output inequality. Its coefficients are precisely $\mathcal{A}_n$, proving the finite-horizon error certificate. The separately proved endpoint tails give the all-time certificate, including the limiting predictors.

E.4.7 Why the comparison exponent has numerical coefficients

This verification retains the existing source and runtime label caps. Only in this calculation put $r = L - 1$, $r_0 = 10s$, $u_0 = H_L^{\text{src}}$, $p_0 = P_L^{\text{src}}$, $\kappa = (9/5)^r$, and $A_0 = (18s)^r = \kappa r_0^r$. The positive source recurrences give

$$u_0 \geq 20sr_0^r, \quad p_0 \geq 3r_0^r, \quad \tau = su_0r_0^r, \quad H_c \leq 2\kappa u_0.$$

The last comparison is an induction from $2b + 16s \leq 2(b + 20s)$ and $18s = (9/5)10s$. The last entry of the source allowance implies

$$z \leq \frac{1}{16\sqrt{8D_0C_F}} \leq \frac{1}{16\sqrt{8}\tau u_0\sqrt{s}}. \quad (185)$$

Here the source definitions give $D_0 \geq \tau^2$ and $C_F \geq su_0^2$. Since $P_h \leq 7u_0$, $P_\delta \leq 7\tau$, $H_r \leq 2u_0$, the corrections to 18 in A_f, A_b are at most $14\tau u_0/(8s\tau^2 u_0^2) < 1$. Thus $A_f, A_b \leq 19$.

Unroll the forward subtraction recurrence and use $2s/(18s-1) \leq 1/8$ to get

$$F \leq A_0 [2s + (H_r + A_f)/8] \leq u_0 A_0, \quad \max_j F_j^z \leq F/(2s).$$

The last step uses $H_r + A_f \leq u_0 + 22$ and $u_0 \geq 20s \geq 20$. Also $d_j^E \leq 7\kappa u_0$, so the same smallness bound gives

$$(\max_j d_j^E) z H_c \leq 1, \quad 6zF \leq 1, \quad 32zP_h \leq 1.$$

For the first inequality it suffices to use $14\kappa^2/(16\sqrt{8}sr_0^r) \leq 1$, since $\kappa^2/r_0^r \leq (3.24/10)^r$. The other two have the additional factors $u_0, \tau \geq 20$ in their denominators. The backward recurrence now has terminal value at most $6s + t_2 F/s$ and forcing at most $40s + t_2 F/s$. Its geometric sum therefore gives

$$\max_j B_j \leq A_0(9s + 2t_2 F/s) \leq 3t_2 u_0 A_0^2/s.$$

Here $9sA_0 \leq t_2 u_0 A_0^2/s$, because $u_0 \geq 20s$, $A_0 \geq 18s$, and $t_2 \geq 1$. Substituting in B_h , with $H_C \leq 4\kappa u_0$, gives

$$B_h \leq 13\sqrt{L}\kappa^3 t_2 u_0^2 r_0^{2r}/s \leq t_2 u_0^2 p_0^3.$$

For the last inequality, $\sqrt{L}\kappa^3 \leq r_0^r$: the ratio at $L = 2$ is $\sqrt{2}(5.832/10) < 1$, and each next step multiplies it by at most $\sqrt{3/2}(5.832/10) < 1$. Now use $p_0 \geq 3r_0^r$ and $13/(27s) \leq 1$.

The runtime allowance gives $zJ_C \leq 5/(16H_c) \leq 5/16$. Moreover $J_R \leq 14\sqrt{L}\kappa u_0^2$, whence $zJ_R \leq 14/(16\sqrt{8}) < 1/3$, using $\sqrt{L}(1.8/r_0)^r \leq 1$. Therefore $z(J_C + J_R) < 1$. Substitute the sharper smallness bound, including $\sqrt{s}$, in the three terms defining D_s . They are bounded by $7u_0$, $10Lu_0$, and $(3+L)/u_0$, respectively; the last term alone is at most $7L/(8su_0)$. Thus $D_s \leq 15Lu_0$, and

$$K_1 \leq 400t_2 u_0^2 p_0^3. \tag{186}$$

Indeed the coefficients of the terms bounded by $t_2 u_0^2 p_0^3$ total at most $12 + 22 + 300 = 334 < 400$.

The stronger source entry also gives

$$D_0 \geq 2t_2^2 u_0^2 p_0^4, \quad D_* \geq t_2 p_0^2, \quad W_G \geq u_0^3.$$

The first two use the top terms of the source trace recurrences. For the third, $W_G \geq 128sV_L \geq 128s^2 u_0 (H_{L-1}^{\text{src}})^2$ and $u_0 \leq (10s+1)H_{L-1}^{\text{src}} \leq 11sH_{L-1}^{\text{src}}$, so the lower bound is at least $(128/121)u_0^3$. Write η_{src} for the exponential-budget coefficient in the source allowance. Its definition implies $\eta_{\text{src}}^{-1} \geq 8192D_0W_G$, and $D_1\mathcal{B} \geq D_*^3$. The fourth-root source entry yields

$$z \leq \frac{1}{16}(8192D_0W_GD_*)^{-3/4}.$$

Combining these inequalities proves

$$zK_1 \leq \frac{25}{2^{21/2}} t_2^{-5/4} u_0^{-7/4} p_0^{-3/2} \leq 1.$$

Also $C_G = 32\tau$, $C_{\text{abs}} = 8(D_0 + 1) \leq 16D_0$, and $\tau \geq 1$, so $K_{\text{src}} \leq 8448D_0\tau$. Consequently

$$\begin{aligned} z^2 K_1 K_{\text{src}} &\leq \frac{13200}{8192^{3/2}} \frac{t_2 u_0^2 p_0^3 \tau}{\sqrt{D_0} W_G^{3/2} D_*^{3/2}} \\ &\leq \frac{13200}{\sqrt{2} 8192^{3/2}} \frac{\tau}{t_2^{3/2} p_0^2 u_0^{7/2}} \leq 1. \end{aligned}$$

The last step substitutes $\tau = su_0 r_0^r$, $p_0 \geq 3r_0^r$, $u_0 \geq 20s$; the remaining numerical coefficient is less than one. Finally the first smallness bound and $F \leq u_0 A_0$ give $zF \leq \kappa/(16\sqrt{8}su_0) \leq 1$. We have proved

$$zK_1 \leq 1, \quad zF \leq 1, \quad z^2K_1K_{\text{src}} \leq 1. \quad (187)$$

Together with $400z^2F_c \leq 25/(16H_c^2)$, these imply

$$\mathcal{B}_n \leq 44 + 64\sqrt{\log(en)}. \quad (188)$$

No structural factor has been moved into a width threshold.

For explicit power envelopes put $X = \beta^L \geq 100$ only in this calculation. Unrolling the positive source recurrences gives

$$H_j^{\text{src}} \leq 3\beta^{2j}, \quad P_j^{\text{src}} \leq \beta^{3j-2}, \quad k_j^{\text{src}} \leq 3\beta^{4L-2j}, \quad \tau_j^{\text{src}} \leq 3\beta^{4L-2j+1}.$$

For example the feature affine recurrence is bounded by $22\beta(10\beta)^{j-1}$; the derivative recurrence is bounded by $4j(10\beta)^{j-1}$. Use $10s \leq \beta^2$ and $4j \leq \beta^j$. Substitution in the trace recurrences gives $A_* \leq \beta^{5L-2}$, $D_* \leq \beta^{6L-2}$, $H_* \leq \beta^{8L-2}$, $E \leq \beta^{6L-2}$, and $D_0 \leq \beta^{16L-2}$. These follow by summing geometric powers with ratio at most β^{-1} ; the outer factor in D_0 is at most 2β , and its inner sum is at most $4\beta^{16L-4}$. Thus $C_{\text{abs}} \leq \beta^{16L-1}$ and $C_G \leq 96\beta^{4L-1}$, proving $K_{\text{src}} \leq 1552\beta^{20L-2} \leq X^{21}$. The runtime recurrence gives $H_c \leq X^3$ and $F_c \leq X^{15}$. The preceding comparison bounds give

$$u_0, p_0, H_c \leq X^3, \quad F \leq X^6, \quad K_1 \leq X^{17}, \quad K_{\text{src}} \leq X^{21}.$$

For K_1 , use $400t_2u_0^2p_0^3 \leq 400X^{31/2} \leq X^{17}$, since $t_2 \leq \beta \leq X^{1/2}$ and $X \geq 100$. The explicit formula for $\mathcal{B}_n$ now gives

$$\mathcal{B}_n/z \leq 71X^{38}(1 + \sqrt{\log(en)}).$$

Indeed its four coefficients are bounded by $25X^{15}, 2X^{17}, 40X^6, 4X^{38}\sqrt{\log(en)}$. The coefficient of $(e^{\mathcal{B}_n} - 1)(1 + \lambda^{-1/2})$ in $\mathcal{A}_n$ is at most $17X^3$; its remaining terms total at most $575zX^9(1 + \lambda^{-1/2})$. Using $e^b - 1 \leq be^b$ proves

$$\mathcal{A}_n \leq 2000e^{44}\beta^{42L}\frac{Ym}{\gamma}\left(1 + \sqrt{\frac{m}{\gamma}}\right)(1 + \sqrt{\log(en)})e^{64\sqrt{\log(en)}}. \quad (189)$$

For the tails, the runtime cap gives $G_c \leq 5H_c^2$, $B_w^c \leq 25H_c^2 \max(1, \lambda^{-1/2})$, and $B_f^c \leq 51H_c^3 \max(1, \lambda^{-1/2})$. For example $48R_cF_cz \leq 15/(16H_c)$, and $2R_c^2F_c \leq 50/256$, which verify the latter two bounds term by term. The source smallness bound gives $\mathcal{K} \leq 2u_0^2$: its hidden sum is at most $LS^2\tau^2u_0^2 \leq L/(8s) \leq u_0^2$. Therefore

$$\mathcal{D} \leq 236\beta^{9L}\frac{Ym}{\gamma}\left(1 + \sqrt{\frac{m}{\gamma}}\right). \quad (190)$$

The optional original-tolerance refinement follows from the same coefficients. On $z \leq X^{-30}$, let $\mathcal{B}_n^{(0)}$ have coefficient 32 in its last term. The preceding powers give

$$\mathcal{B}_n^{(0)} \leq 1 + \sqrt{\log(en)}, \quad \mathcal{B}_n^{(0)}/z \leq X^{18}(1 + \sqrt{\log(en)}).$$

For the first inequality the four terms are at most $400X^{-45}, 2X^{-13}, 40X^{-24}, 32X^{-22}\sqrt{\log(en)}$. For the second they are at most $400X^{-15}, 2X^{17}, 40X^6, 32X^8\sqrt{\log(en)}$. These sums have the displayed bounds already at $X = 100$. The exact output coefficient is therefore at most

$$\mathcal{A}_n^{(0)} \leq [17eX^{21} + 575X^9]z(1 + \lambda^{-1/2})e^{2\sqrt{\log(en)}} \leq X^{22}z(1 + \lambda^{-1/2})e^{2\sqrt{\log(en)}}.$$

The tail coefficient is at most $236X^9z(1 + \lambda^{-1/2})$, and $e^{-8\log(en)} \leq 1/n$. These inequalities imply the displayed coefficient $10X^{40}$, with considerable slack. Finally, for $r = \sqrt{\log(en)}$, $2r \leq r^2/2 + 2$, so $n^{-1}e^{2r} \leq e^{5/2}n^{-1/2}$. Use $\lambda^{-1}(1 + \lambda^{-1/2}) \leq 2(1 + \lambda^{-1})^2$ and $20e^{5/2} < 250$ to obtain the stated numerical root-width bound. This specialization does not impose its smaller cap on the full-range Harmonic theorem.

E.4.8 All finite source horizons on the same event

The original source event supplies joint time/query holomorphy through T_0 , with the radii r_t, r_q , and preactivation imaginary parts at most $a/4$. We extend this domain deterministically using the dense late-time path length. Use the dense parameter norm

$$\|\theta\|_{\text{par}}^2 = \|A\|_F^2/n + \sum_{j=2}^L \|W^{(j)}\|_F^2 + \|w\|_2^2/n,$$

with its usual complex Euclidean extension. The dense energy identity and gap imply, for every real t ,

$$\int_t^\infty \|\dot{\theta}(s)\|_{\text{par}} ds \leq 2\rho_n(t)/\sqrt{\lambda} \leq 2\alpha e^{-\lambda t/2}. \quad (191)$$

This is the same pointwise bound $\|\dot{\theta}\| \leq 2(-\dot{\rho}_n)/\sqrt{\lambda}$ used in fitting.

Consider the complex parameter ball of radius b_{ext} around $\theta(T_0)$. The dense real operator bounds improve below $8 + 1/8$, so the ball stays inside cap ten. The real readout is at most $2\alpha \leq SH_L^{\text{src}}/8$, and the ball keeps it below SH_L^{src} . On a straight parameter segment stopped at a first query-strip exit, the source forward derivative recurrence bounds preactivation RMS differentials by $P_j^{\text{src}}\|d\theta\|_{\text{par}}$: the first direct map costs at most two, later direct maps at most H_{j-1}^{src} , and propagation at most 10s. Coordinate differentials are bounded by $\sqrt{n}P_j^{\text{src}}\|d\theta\|_{\text{par}}$. The radius definition limits the change to $a/8$, so all query preactivations stay below imaginary part $3a/8$. This strictly improves the stopped boundary. Linear growth now gives the original complex RMS source bounds throughout the ball.

On that ball the complex algebraic residual Gram has norm at most $\mathcal{K}$, and $\|\dot{\theta}\|_{\text{par}} \leq 2\sqrt{\mathcal{K}}\rho_n$. These estimates use absolute Cauchy–Schwarz, not positivity of a complex Gram. Fix any real anchor $t \geq T_0$, and continue its finite analytic ODE along a complex segment of length at most $2r_t$, stopped in this same ball. The deficit equation gives

$$\rho_n(\zeta) \leq \rho_n(t)e^{2\mathcal{K}|\zeta-t|} \leq 2\rho_n(t), \quad \|\theta(\zeta) - \theta(t)\|_{\text{par}} \leq 8r_t\sqrt{\mathcal{K}}\rho_n(t).$$

The original radius gate supplies $4\mathcal{K}r_t \leq \log 2$. The late-time parameter tail and the first analytic-extension gate therefore keep this solution strictly within half of the fixed ball around $\theta(T_0)$.

For clarity, local holomorphic existence here follows by Picard iteration on a smaller closed parameter ball: the holomorphic vector field has bounded derivative there, so its integral map is a contraction on a sufficiently small complex time disk. The strict interior bound allows repeated continuation to the whole disk of radius $2r_t$. A finite-radius obstruction would have a bounded state a positive distance from the boundary, where the same local argument extends it. Uniqueness glues overlapping germs. The disks at all real anchors cover every late source rectangle with strict slack, and the parameter ball leaves strict margin below the activation half-strip. The earlier source domain already has a holomorphic neighborhood. Compactness of each finite closed time/query domain gives such a neighborhood for every finite T , without reducing either radius. All anchors use one fixed ball, so no accumulated continuation error or new stochastic event is involved. The four source families retain coordinate bound M_n .

The comparison requires a coordinate maximum only for the actual real training carriers. Between two real states in the operator tube, forward subtraction bounds preactivation RMS changes by $P_j^{\text{src}}\|\Delta\theta\|_{\text{par}}$. Converting this to a coordinate maximum costs $\sqrt{n}$. In the backward recursion, the changed mixer costs $S\tau_{j+1}^{\text{src}}\|\Delta\theta\|$, the propagated carrier difference costs 10s, and the changed gate costs $10t_2SP_{j+1}^{\text{src}}k_{j+1}^{\text{src}}\sqrt{n}\|\Delta\theta\|$. The top carrier is just the readout. Induction gives exactly the recurrence for C_j^k , and a final RMS-to-coordinate conversion proves

$$\max_{a,j} \|k_{n,a}^{(j)}(t) - k_{n,a}^{(j)}(T_0)\|_\infty \leq C_{\text{carrier}}n\|\theta(t) - \theta(T_0)\|_{\text{par}}.$$

The second extension gate and the late parameter tail make this at most $K_{\text{src}}S\sqrt{\log(en)}$. Adding the source maximum at T_0 gives, simultaneously for all real times,

$$\max_{a,j,i} |k_{n,a,i}^{(j)}(t)| \leq 2K_{\text{src}}S\sqrt{\log(en)}. \quad (192)$$

This proves the value of M used in the comparison. The first extension gate is eventual because its left side is a fixed multiple of n^{-16} , whereas b_{ext} is bounded below by a fixed positive multiple of $n^{-1/2}$. The second is eventual by the same comparison of powers. Neither changes the label interval.

E.4.9 Budget count, inversion and boundary cases

The simplex count and $\lambda^{-1}c_t^{-1} = 1024(U/a)z^2$ give, for $d \geq 2$,

$$\dim E_j \leq B + \frac{2^{15}}{d!} \frac{U}{a} z^2 (\lambda T) c_q^{-(d-1)} \log(en)^{d/2} [H(T, \eta) + \alpha_T + (d-1)r_q]^d. \quad (193)$$

The factor four counts the four source families. In dimension one,

$$\dim E_j \leq B + 8 + 2^{15}(U/a)z^2(\lambda T)\sqrt{\log(en)}H_1(T, \eta). \quad (194)$$

The extra eight retain all temporal zero modes.

Set $T(u) = T_0 + 4u/\lambda$, $\eta(u) = \eta_0 e^{-u}$, for $u \geq 0$. Since P_T is proportional to T ,

$$H(T(u), \eta(u)) = H(T_0, \eta_0) + 2u + 2\log(1 + u/[8\log(en)]) \leq H(T_0, \eta_0) + 9u/4.$$

Also $\alpha_{T(u)} \leq \alpha_{T_0}$ and $\lambda T(u) = 32\log(en) + 4u \leq 4(h_0 + u)$. The variable rank bound therefore gives

$$\dim E_j \leq B_* + \mathcal{C}_d(h_0 + u)^{d+1}. \quad (195)$$

For $d = 1$, $H_1(T(u), \eta(u)) \leq H_1(T_0, \eta_0) + 9u/8$, so the same bound holds with slack in the factor $9/4$. The error certificate gives $\mathcal{A}_n \eta(u) + \mathcal{D}e^{-\lambda T(u)/4} = \mathcal{E}_1 e^{-u}$.

For a supplied q , the nonnegative value of u in the construction formula makes this rank bound at most $q/9$. Since actual dimension is an integer, it is at most $\lfloor q/9 \rfloor$ without first rounding the real upper bound upward. The selection theorem then uses at most q coordinates. When that formula would give negative u , retain only the exact initialization additions; they have rank at most B , fit independently, and obey the baseline. This proves the piecewise certificate. For $q \geq 18B_*$, the weaker choice $u = (q/(18\mathcal{C}_d))^{1/(d+1)} - h_0$, when nonnegative, gives $B_* + q/18 \leq q/9$. It proves the simpler certificate. The inverse prescription is the same calculation with u chosen to make $\mathcal{E}_1 e^{-u} \leq \varepsilon$, followed by the final integer ceiling. The exact coefficient counts give the stated sharper discrete alternative.

For this activation-only calculation set $X = \beta^L$. The source recurrences imply

$$H_{\max}^{\text{src}}, \max_j(sP_j^{\text{src}}) \leq X^3, \quad K_{\text{src}} \leq X^{21}, \quad T_Q \leq X^{14}, \quad T_J \leq X^8, \quad \max_j q_j^{\text{src}} \leq X^8, \quad \max_j b_j^{\text{ang}} \leq X^3.$$

Here q_j^{src} and b_j^{ang} are the response and angular coefficients in the source recurrence, not selected widths. To check these last powers, its formulas give $g \leq \beta^{5L-1}$, $q_j^{\text{src}} \leq \beta^{5L+3j-2}$, $b_j^{\text{ang}} \leq 2\beta^{2j}$, $e_j \leq 2\beta^{5L+6j-4}$, and $a_j \leq 3\beta^{5j-2}$. The leading forcing grows by six powers per layer in e_j , and by five in a_j , while propagation is at most β^2 ; geometric summation gives the stated factors two and three. Substitution into the source definitions of T_Q, T_J gives $T_Q \leq \beta^{14L-3}$, $T_J \leq \beta^{8L-1}$, which imply the displayed bounds. Using only $S \leq 1$, the bracket in U_j is at most $2X^{14}$. Its full higher-layer expression is then bounded by $4X^{36} + 32\sqrt{d+3}X^8 + 2$, and the corresponding query expression by $32\sqrt{d+3}X^3 + 4X^{30} + 2$. The first-layer expressions are smaller. Since $X \geq 100$,

$$U \leq X^{37}\sqrt{d+3}, \quad V \leq X^{31}\sqrt{d+3}, \quad U/a \leq X^{38}\sqrt{d+3}, \quad c_q^{-1} \leq X^{32}\sqrt{d+3}.$$

Consequently

$$\mathcal{C}_d \leq \frac{2^{17}(9/4)^d}{d!} \beta^{(32d+6)L} (d+3)^{d/2} (Ym/\gamma)^2 \log(en)^{d/2}. \quad (196)$$

Integrating $\log x$ over $[1, d]$ gives $d! \geq (d/e)^d$. Together with $d+3 \leq 4d$, $d^{1/(d+1)} \leq 2$, and $2\sqrt{2}e < 8$, this implies

$$\frac{(d!)^{1/(d+1)}}{(d+3)^{d/(2d+2)}} \geq \frac{\sqrt{d}}{8}.$$

Use also $(18 \cdot 2^{17})^{1/(d+1)} \leq 1536$, $(9/4)^{d/(d+1)} \leq 9/4$, and $(32d+6)/(d+1) \leq 32$. The resulting numerical rate is

$$\left(\frac{q}{18\mathcal{C}_d}\right)^{1/(d+1)} \geq \frac{\sqrt{d}}{32768\beta^{32L}} \left(\frac{q}{(Ym/\gamma)^2 \log(en)^{d/2}}\right)^{1/(d+1)}. \quad (197)$$

The original counting gates and $\eta_0 \geq 1/n$ give $H(T_0, \eta_0) \leq 7 \log(en)$, hence $h_0 \leq 9 \log(en)$, for $d \geq 2$. In dimension one, the stated temporal gate gives $H_1(T_0, \eta_0) \leq (7/2) \log(en)$, since $(3/2) \log \log(en) \leq \log(en)$; hence $h_0 = 8 \log(en)$.

For $D = (q/(18\mathcal{C}_d))^{1/(d+1)}$,

$$\min\{1, e^{9 \log(en) - D}\} \leq e^{\log(en) - D/9}.$$

If $D \leq 9 \log(en)$, the right side is at least one; otherwise compare its exponent with $9 \log(en) - D$. Thus the simplified certificate is at most $en\mathcal{E}e^{-D/9}$. The numerical prefactor and tail bounds, $\eta_0 \leq 1$, $H_d \leq X^2$, $\sqrt{\lambda} \leq H_c \leq X^3$, and $1 + \sqrt{\log(en)} \leq e^{\sqrt{\log(en)}}$ give

$$\mathcal{E} \leq 4096e^{44}\beta^{42L}\frac{Ym}{\gamma} \left(1 + \sqrt{\frac{m}{\gamma}}\right) e^{65\sqrt{\log(en)}}.$$

Combine this with the negative exponential rate divided by nine to obtain the numerical budget certificate, including its constant 294912.

If $q \geq n$, take the full source space and $M_j = I_n/n$, and retain the original dense matrices. Along the dense trajectory, $w_C = w_n$, $c_C = y - f_n$, and the readout correction is zero. Metric adjoints are ordinary transposes; the hidden rank-one update has exactly the factor $1/n$ supplied by M_{j-1} . Every Harmonic equation therefore equals its dense counterpart. The positive Gram margin and local uniqueness identify the trajectories for all time and at the limit. This branch needs only fitting and exact full retention.

The learned count follows by counting the entries of the moving arrays. The fixed metric and data counts follow by their matrix dimensions. For the upper inventory, put $P = dq_1 + \sum_{j=2}^L q_j q_{j-1}$ only in this count. Two copies of these blocks, at most three full metric-related matrices per layer, four training arrays per layer, four top-layer vectors, four sample Gram/solve matrices and two sample vectors use at most

$$2P + 3 \sum_j q_j^2 + 4m \sum_j q_j + 4q_L + 4m^2 + 2m + m(d+1)$$

real coordinates. This is a conservative upper count for the named copies and caches, not a compulsory cache layout. Since each selected dimension is at most $9R$, while $m, d \leq R$ on the exact initialization event when $n \geq d$, this is bounded by $(441L - 102)R^2 + m(d+1)$, and hence by $1020(L+1)R^2 + 10m(d+1)$. Diagonal comparison matrices may be stored within the allowance of three full metric-related matrices. In particular the factor $1020/81 < 13$ proves the supplied-budget inventory. The exact positive top Gram forces $q_L \geq m$; exact first-weight columns force $q_1 \geq \text{rank } A_0 = \min(n, d)$ almost surely. No conclusion for budgets below these ranks is inferred.

E.4.10 Scope and claim boundary

The resulting deterministic Harmonic implication is complete relative to the shared source and dense initialization/fitting theorems. Its success width still includes the source theorem's unquantified stochastic threshold. The approximation measures predictor fidelity on known inputs, not test risk for unknown labels. No claim of optimal rate, bounded-precision compression, or ordinary-gradient Harmonic training follows. Efficient setup requires the separate local-continuation/coupling arguments now integrated below; its costs do not follow from retained storage alone.

E.5 Seeded decoder proofs

The proof has four finite steps. The first constructs a physical source at an explicit width. The second replaces its independent row tape by a short seed without changing the law of its finite scalar transcript by more than the allocated variation error. The third regenerates those rows to answer an unseen query and amplifies complete source experiments. The fourth counts words and word operations. Every probability below includes the source, metric replay, finite arithmetic and query construction; there is no uncounted ideal-real computation.

E.5.1 The finite source event and independent reference

The local identities (IC.1)–(IC.57) are proved in the source foundation above. The following quantitative specialization uses only training queries, includes its own stronger initialization concentration argument, and proves the reference certificate at the displayed finite width.

6. A sharper initialization gate, needed for the finite width claim

The older displayed N_{fit} has a sphere-net cardinality outside a logarithm. It cannot be used to justify a width polynomial in dimension. The following replacement has the same event and label cap. Here only write $H = H_D$ for the population RMS coefficient of the existing dense-fitting theorem, and define

$$R_0 = 2H, \quad K_0 = b + sR_0, \quad D_0^{\text{init}} = 2sb + 4s^2R_0, \quad T_1 = \sum_{j=0}^{L-1} s^j, \quad T_2 = \sum_{j=0}^{L-1} s^{2j},$$

$$C_0 = T_2(2K_0 + 1 + D_0^{\text{init}}T_1), \quad \epsilon_0 = \min\{(8T_1)^{-1}, \lambda/(2C_0)\}, \quad h_0 = \min\{1/2, H/[4(8s)^L]\}.$$

For failure $\alpha \in (0, 1)$, set

$$\Xi_0 = \log[16L(m+1)^2/\alpha] + d\log(1 + 2/h_0),$$

$$N_{\text{fit}}^{\text{exp}}(\alpha) = \left\lceil \max \left\{ 1, \frac{\log(8L/\alpha)}{8 - 2\log 9}, \frac{d\log 9 + \log(8/\alpha)}{8 - \log 9}, \frac{32s^2R_0^2\Xi_0}{\epsilon_0^2} \right\} \right\rceil. \quad (\text{IC.58})$$

For every $n \geq N_{\text{fit}}^{\text{exp}}(\alpha)$, with probability at least $1 - \alpha$, initial operators are at most eight, all initial sphere feature RMS norms are at most $11H/8$, and the normalized training Gram gap is at least $\lambda/2$.

We supply the concentration proof. If F is a_0 -Lipschitz in a standard Gaussian vector, then

$$\log \mathbb{E} e^{t(F - \mathbb{E}F)} \leq a_0^2 t^2 / 2, \quad \text{Var } F \leq a_0^2. \quad (\text{IC.59})$$

One proof uses the Gaussian semigroup $P_t g(x) = \mathbb{E} g(e^{-t}x + \sqrt{1 - e^{-2t}}G)$. Differentiating its invariant entropy and integrating by parts gives

$$\text{Ent}_\gamma(g) = \int_0^\infty \mathbb{E} \frac{|\nabla P_t g|^2}{P_t g} dt \leq \int_0^\infty e^{-2t} \mathbb{E} P_t (|\nabla g|^2 / g) dt = \frac{1}{2} \mathbb{E} |\nabla g|^2 / g.$$

The inequality is weighted Cauchy–Schwarz and $\nabla P_t g = e^{-t} P_t \nabla g$. Apply it to $g = e^{uF}$ and integrate the differential inequality for $u^{-1} \log \mathbb{E} e^{uF}$. Truncation and smoothing extend the bounded smooth calculation to Lipschitz F ; its Gaussian exponential integrability follows from linear growth. This proves (IC.59), its variance statement by differentiation at zero, and tails $2e^{-t^2/(2a_0^2)}$. For nonnegative F , its mean differs from $\sqrt{\mathbb{E} F^2}$ by at most a_0 . Therefore at tolerance $\epsilon \geq 2a_0$ its deviation from that RMS has probability at most $2e^{-\epsilon^2/(8a_0^2)}$.

Conditioned on preceding layers, an individual empirical feature RMS is $sR_0/\sqrt{n}$ -Lipschitz. The empirical RMS of $\phi(Z_a) \pm \phi(Z_b)$ is $2sR_0/\sqrt{n}$ -Lipschitz, including singular pair covariances. Polarization of these two norms shows that simultaneous RMS errors at most $\epsilon_0 \leq 1$ give covariance-entry error at most $(2K_0 + 1)\epsilon_0$ relative to the conditional covariance transform.

Two distinct propagation estimates avoid a depth-squared power. The population scalar RMS transform is s -Lipschitz in its input standard deviation, by coupling with the same scalar Gaussian. If two Gaussian pair covariances have diagonal standard deviations at most R_0 , differing by at most Δ , their activation covariances differ by at most

$$s^2|C_{ab} - C'_{ab}| + (2sb + 4s^2R_0)\Delta. \quad (\text{IC.60})$$

To prove this, shrink both marginal standard deviations to their common minimum, preserving correlations. The total scaling cost is at most $2sK_0\Delta$, by subtracting product factors and applying Cauchy–Schwarz. At fixed marginal standard deviations τ_a, τ_b , Gaussian density differentiation and two integrations by parts give correlation derivative $\tau_a\tau_b\mathbb{E}[\phi'(Z_a)\phi'(Z_b)]$, of modulus at most $s^2\tau_a\tau_b$. Direct subtraction of the two correlations gives $\tau_a\tau_b|\varrho - \varrho'| \leq |C_{ab} - C'_{ab}| + 2R_0\Delta$. These estimates prove (IC.60). Degenerate variances and correlations ± 1 follow by Gaussian dominated convergence, using linear activation growth; no variance is divided out in the final bound.

Take a sphere h_0 -net of cardinality at most $(1 + 2/h_0)^d$, obtained from disjoint balls centered at a maximal separated set. At each layer test its individual norms, the training individual norms, and the two polarization norms for every training pair. If Δ_j is the largest individual RMS error and E_j the largest training covariance-entry error, the stopped induction is

$$\Delta_j \leq \epsilon_0 + s\Delta_{j-1}, \quad E_j \leq (2K_0 + 1)\epsilon_0 + s^2E_{j-1} + D_0^{\text{init}}\Delta_{j-1}.$$

It gives $\Delta_j \leq 1/8$, $E_j \leq \lambda/2$, and conditional input standard deviations below $H + 1/8 < R_0$. There are at most $L[3m^2 + (1 + 2/h_0)^d]$ tests. Their union failure is at most

$$2L[3m^2 + (1 + 2/h_0)^d]e^{-n\epsilon_0^2/(32s^2R_0^2)} \leq \alpha/2.$$

The two one-quarter operator nets have scalar threshold exponent $-8n$ and cardinalities 9^{2n} or 9^{n+d} . The second and third gates in (IC.58) pay their total failure $\alpha/2$. On that operator event, normalized features are $(8s)^j$ -Lipschitz in the query. Their off-net increment is at most $H/4$, so their sphere RMS is at most $H + 1/8 + H/4 \leq 11H/8$. Finally $\|\mathbf{H}^\top \mathbf{H}/n - Q^{(L)}\| \leq mE_L \leq \gamma/2$. This proves the asserted event.

The deterministic fitting proof in §D.5 applies to this stronger event with its unchanged condition $Y \leq \lambda/(8H_D\sqrt{F_D})$. Its stop improvements use only the initial operator/RMS/Gram bounds just proved. In particular $\rho(t) \leq Ye^{-\lambda t/2}$, and the weaker decay (S.3) is available. Direct finite-sum estimates give

$$H \leq \beta^{2L}, \quad R_0, K_0 \leq \beta^{3L}, \quad D_0^{\text{init}} \leq \beta^{4L}, \quad T_1 \leq \beta^{2L}, \quad T_2 \leq \beta^{3L}, \quad C_0 \leq \beta^{10L},$$

$$\epsilon_0^{-1} \leq \beta^{11L}(1 + \lambda^{-1}), \quad 32s^2R_0^2 \leq \beta^{8L}, \quad \log(1 + 2/h_0) \leq 3L \log \beta.$$

Thus a convenient sufficient envelope for (IC.58) is

$$n \geq \beta^{32L}(1 + \lambda^{-1})^2[dL \log \beta + \log(16L(m + 1)^2/\alpha)] + 1. \quad (\text{IC.61})$$

This gate replaces, rather than assumes domination of, the older exponential-in-dimension initialization gate in the Seeded proof.

7. Quantitative training-source coefficient and width ledger

Fix failure $0 < \rho < 1/4$ for this one source event, and set

$$p = \max \left\{ 1, \left\lceil \frac{\log(4emL/\rho)}{\log(64e^2)} \right\rceil \right\}, \quad Q = m + d + p + 1, \quad \mathcal{A} = \beta^{2000L}(1 + \lambda^{-1})^4 Q^4. \quad (\text{IC.62})$$

The time domain is the neighborhood of the closed rectangle

$$[-r_t, 32\ell/\lambda + r_t] + i[-r_t, r_t], \quad r_t = \frac{1}{p\beta^{100L}(1 + \lambda)\sqrt{\ell}}. \quad (\text{IC.63})$$

Only the m training queries and m^2 training responses are used in the finite union. Define U_j^{fin} by (S.25) with G_d replaced by 64. There is no dimension mesh in this source event. The exact response, reverse-observable and probe calculations are (IC.36)–(IC.43).

Here is a numerical coefficient ledger from those displayed recurrences. All powers of $\ell, n, \sqrt{p}$ stated separately below are removed before bounding a coefficient.

Coefficient	Sufficient bound
H_j, P_j, k_j, τ_j	$\beta^{3L}, \beta^{3L}, \beta^{5L}, \beta^{6L}$
$A_*, D_*, H_*, D_0, C_{\text{abs}}, W_G$	$\beta^{11L}, \beta^{8L}, \beta^{13L}, \beta^{30L}, \beta^{32L}, \beta^{23L}$
η^{-1}, M_0, A_0, B_0	β^{80L}
$\max U_j^{\text{fin}}$	β^{72L}
(IC.10)–(IC.17), including control/time moduli	$\beta^{300L}(1 + \lambda^{-1})$
Additional response/probe maps (IC.38)–(IC.43)	$B = \beta^{1000L}(1 + \lambda^{-1})^2$
Integrated local, learned-port and top-offset coefficients	$B(1 + p)^2$

To check the depth dependence in this ledger, every homogeneous layer propagation is bounded by $10s \leq \beta^2$, and every finite layer sum costs at most $L \leq \beta^L$. In (S.7), substitute $P_j \leq \beta^{3L}$, $k_j \leq \beta^{5L}$: the mixed, curvature and squared trace coefficients then have the exponents in the first two rows. The sums (S.9) give $V_* \leq \beta^{15L}$, $G_* \leq \beta^{20L}$, hence the displayed W_G . Inserting these into $\eta^{-1} = 1024C_{\text{abs}}W_G$ and multiplying by $1 + \log(2\mathcal{B})$ gives the third row with numerical slack. For (S.24), its terms give $g \leq \beta^{10L}$, $r_j \leq \beta^{13L}$, $q_j \leq \beta^{14L}$, $e_j \leq \beta^{21L}$, and $T_Q \leq \beta^{27L}$. Substitution into (IC.57) gives the 72L response envelope. The largest operation in (IC.16) is one additional reference-carrier factor times a forward time derivative; the entire recurrence has degree at most four in M_0 and degree one in T_* . This gives its 300L envelope. To check this last power without repeatedly using the deliberately loose 80L row, retain the sharper intermediate bounds $\eta^{-1} \leq \beta^{57L}$, $M_0, A_0, B_0 \leq \beta^{60L}$, and $E_0 \leq \beta^{69L}$. They give $v_0, v_c \leq \beta^{133L}(1 + \lambda^{-1})$, forward coefficients at most $\beta^{137L}(1 + \lambda^{-1})$, backward coefficients at most $\beta^{202L}(1 + \lambda^{-1})$, $A_t \leq \beta^{76L}$, and forward time coefficients at most $\beta^{217L}(1 + \lambda^{-1})$. The final backward time forcing adds at most 61L, and its geometric propagation at most 3L, remaining below 300L. These estimates also use $\lambda \leq \beta^L$, so no positive-gap power is left uncharged.

More explicitly, the enlargement from base to response maps has coefficient at most

$$\beta^{100L}(1 + M_0 + 4U_*^{\text{fin}})^2(1 + C_*)^2, \quad C_* = \max\{1, \text{coefficients in (IC.10)–(IC.17)}\}. \quad (\text{IC.64})$$

In (IC.38), its four preactivation forcing terms are respectively a backward variation times a bounded scalar, a reference response times the pairing variation, a changed mixer times response RMS, and a lower propagated variation. The new gate term adds one reference response cap. The additional reverse map (IC.39) is a product of two bounded forward derivatives. Time differentiation adds one reference speed, and control subtraction is linear in the control differences. Thus (IC.64) covers the map, control modulus, and time modulus with respective powers $\sqrt{p}n^\kappa \ell^5$, $\sqrt{p}n^\kappa \ell^4 \epsilon$, and $\sqrt{p}n^\kappa \ell^8$. It is at most the stated B , since its numerical/depth factors cost less than $100 + 160 + 600 + 20 = 880$ powers of β^L . The response remainder from (IC.40)–(IC.43) is bounded by

$$\beta^{240L}(1 + p)^2(1 + M_n + 4U_*^{\text{fin}}\sqrt{\ell})^3[R + P + d_0(N + u + (N + u)^2)]. \quad (\text{IC.65})$$

The homogeneous terms in (IC.41) cost $10s$; the reference cap occurs only in forcing. This proves the fixed cutoff degree three. The coefficient in (IC.35), contour length, and (IC.8) also fit the last ledger row with room below $1000L$. No depth-dependent power of ℓ or cutoff is absorbed into that row.

There is a sharper bound useful to the numerical decoder: $K_{\text{src}} \leq \beta^{21L}$, and hence certainly $K_{\text{src}} \leq \beta^{40L}$. It follows directly from the S-independent source recurrence ledger: $C_{\text{abs}} \leq \beta^{16L-1}$, $H_j \leq 3\beta^{2j}$, $\tau_j \leq 3\beta^{4L-2j+1}$, and (S.22). These bounds concern source constants and do not require a stronger label cap.

The useful width conditions are

$$\mathcal{A}\ell^{16} \leq n^{1/1000}, \quad n \geq 64mL/\rho. \quad (\text{IC.66})$$

They are implied by the factorized gate

$$n \geq \lceil \max \{ [2\mathcal{A}(2000 \log(2\mathcal{A}))^{16}]^{1000}, 64mL/\rho \} \rceil, \quad (\text{IC.67})$$

which is in turn implied by $n \geq \lceil (\mathcal{A}/\rho)^{1100} \rceil$. To prove this without an implicit logarithmic onset, put $u = \log(2\mathcal{A}) > 9000$. The increasing function $u/320 - \log(2000u)$ is positive at 9000, so $x_0 = 1000[u + 16 \log(2000u)] \leq 1050u$, and $1+x_0 \leq 2000u$. Hence $\mathcal{A}(1+x_0)^{16} \leq \frac{1}{2}e^{x_0/1000}$. The function $x/1000 - 16 \log(1+x)$ increases for $x > 15999$, proving (IC.66) at every larger width. Also $1050 \log(2\mathcal{A}) \leq 1100 \log \mathcal{A}$ and $\mathcal{A} \geq 64mL$, proving the power-1100 envelope.

For explicit checking of the remaining inequalities, (IC.66) implies

$$\log n > 9 \cdot 10^6, \quad p, m, d \leq n^{1/4000}, \quad L \leq n^{1/2000000}, \quad \log(1/\rho) \leq \log n, \quad \log(64nmL/\rho) \leq 2 \log n. \quad (\text{IC.68})$$

Every numerical comparison of logarithms with positive powers below is valid already at this displayed lower endpoint and improves with width.

The following table gives the actual finite Gaussian-union and remainder bounds. It instantiates Sections 2–4 at the coefficients just proved.

Step	Bound or sufficient inequality
Linear/quadratic map norm	$B\sqrt{pn}^{1/4000}\ell^5 \leq n^{3/4000} < n^{1/200}$
Quadratic fixed-grid exponent at $n^{-1/10}/8$	$\min\{n^{.79}/(8192p), n^{.895}/128\} > n^{.78}$
Centered control interpolation	$80Bp^{3/2}\ell^4 \leq n^{99/4000}$
Terminal mesh	$h_t = n^{-2}/(1 + \mathcal{A}\ell^8)$, at most n^6 terminals
Terminal error	$10\sqrt{2}Bp^{3/2}n^{1/2+1/4000}\ell^8 h_t < n^{-1/10}/8$
Control entropy	$\log N_{\text{ctrl}} \leq \mathcal{A}\ell^8 n^{5/8} \leq n^{.626}$
Remaining coordinate/probe/form tests	at most $\mathcal{A}n^3$ per deletion set and terminal
Local total failure	$4pL\mathcal{A}n^{p+9}e^{n^{.626}-n^{.78}} + pLn^pe^{-pn} < e^{-n^{.7}} < \rho/16$
Integrated nonlinear force	$\mathcal{A}\ell^8 n^{1/4000}[n^{-.08} + u_0^2 + n^{-.48} + n^{-1/2}] < u_0/2$
Coordinate transfer	$16\mathcal{A}\ell^8[d_0 + u_0 + R + P + d_0(N + u_0 + (N + u_0)^2)] < n^{-1/30}$
Scalar trace error	$32\mathcal{A}\ell^8 n^{-1/30} < n^{-1/40}$

For example, $B = \sqrt{\mathcal{A}}/Q^2$, so the control interpolation left side is at most $80n^{1/2000}$. The nonlinear-force left side is at most $4n^{-.07875}$, whereas $u_0/2 = n^{-.04}/2$. The coordinate-transfer left side is at most $256n^{-.039}$, whose exponent is strictly smaller than $-1/30$. These verify the margins; the exceptional probability row follows by taking logarithms and (IC.68). The terminal mesh count uses side lengths at most $34(1 + \lambda^{-1})\ell$, and control count at most $4p(m+1)^2$ real components. Their speeds are at most $\sqrt{n}\ell^3$ times the ledger coefficient, as shown by (IC.38). Thus the powers used in this table do not presume the desired Gaussian event.

Full budgets stop at $\mathcal{B}$, cavity budgets at $2\mathcal{B}$; temporary full coordinate and response caps are twice their claimed values and cavity caps four times. Full and cavity pole stops are $3a/8$ and $7a/16$. The coordinate table gives, on their common prefix,

$$\max |\Delta z| + \max |\Delta \bar{k}| \leq n^{-1/30}, \quad \max |\Delta \bar{R}| \leq n^{-1/30}, \quad \mathcal{H}_a^{-I} \leq e^{2\eta n^{-1/30}} \mathcal{H}_a + p/n < 2\mathcal{B}. \quad (\text{IC.69})$$

The local size, learned-row norm, and strip gates (IC.20), (IC.33) hold because $R \leq 3n^{-.08}$, $P \leq 4n^{-.48}$, and $a^{-1} \leq \beta/16$. Thus no cavity stop occurs first.

There is no exponential short-contour gate at (IC.63). The physical coefficient bounds are $\mathcal{K} \leq \beta^{20L}$, $D_W \leq \beta^{9L}$, $U_*^{\text{fin}} \leq \beta^{72L}$, and $\lambda \leq H_D^2 \leq \beta^{6L}$. With $c = [p\beta^{100L}(1+\lambda)]^{-1}$, each of

$$8c, \quad \lambda c, \quad 4\mathcal{K}c/\log 2, \quad 32YSD_Wc$$

is below one for every $n \geq 1$. Follow a real solution to its nearest real anchor and then at most two short pieces. Residual growth is at most $e^{4\mathcal{K}r_t} \leq 2$, extra activity at most $8Yr_t \leq S/2$, hidden normalized increments at most $8YSD_Wr_t \leq 1/4$, and, once the response cap is improved, the preactivation displacement is at most

$$8YSU_*^{\text{fin}}c \leq a/8. \quad (\text{IC.70})$$

This improves the pole stop. It uses $64YS = 4\lambda S^2$ and $S \leq 1$, so no inverse label amplitude occurs. The source rectangle therefore supplies exactly the decoder radius $\lambda r_t = [p\beta^{100L}(1+\lambda^{-1})\sqrt{\ell}]^{-1}$. Using this source in a Taylor construction requires the corresponding factor p in the number of time patches; it does not license the former larger complex radius.

8. Initializing every cavity and improving the finite stops

Apply (IC.58) at failure $\rho/32$. Gate (IC.66) implies its sufficient envelope (IC.61), since the latter is at most $\mathcal{A}\ell$. Conditional initial Gaussian tails, before conditioning on the current operator event, also give

$$Z_0 = 2H_D\sqrt{2\log(64nmL/\rho)} \leq 4H_D\sqrt{\ell} \quad (\text{IC.71})$$

as a simultaneous initial training-preactivation cap, with failure at most $\rho/32$. Initial exponential budgets are controlled directly: $\eta(2H_D) \leq 1$, so for a centered Gaussian with standard deviation at most $2H_D$, $\mathbb{E}e^{\eta|Z|} < 4$ and $\mathbb{E}e^{2\eta|Z|} < 15$. Conditional Chebyshev at layer average eight gives failure $15/(16n)$. The stopped layer/sample union therefore bounds all initial budgets by $8L$ with failure at most $15mL/(16n) < \rho/32$. Initial carriers are zero. Gaussian norm concentration gives all initialized hidden rows/columns norm at most two, with failure at most $4nLe^{-n/2} < \rho/32$; the first layer's incoming rows are not needed for a reverse source and are excluded from this norm bound.

Delete at most p neurons at one layer. Zero-embedding its rectangular features, the omitted initialized feature vector has norm at most $\sqrt{p}(b + sZ_0)$. Every subsequent initialized feature difference has norm at most

$$D_{\text{init}} = (8s)^L \sqrt{p}(b + sZ_0) \leq \beta^{10L} \sqrt{p\ell}. \quad (\text{IC.72})$$

Restrictions do not increase initial operator norms. Gate (IC.66) implies $D_{\text{init}}/\sqrt{n} \leq \min\{H_D/8, \sqrt{\lambda}/32\}$. Thus each cavity starts with training feature RMS below $3H_D/2$ and least singular value of its normalized feature matrix at least $(1/\sqrt{2} - 1/32)\sqrt{\lambda}$. Under the unchanged label condition its later feature-matrix motion is at most $\sqrt{\lambda}/8$, by the existing deterministic fitting calculation. Its Gram gap stays strictly above $\lambda/4$. This proves the fitting tube for all cavities from one full initialization event, without union-bounding a cavity Gram probability. In particular (IC.58) is not applied literally to a rectangular cavity. Its empirical norm, if estimated directly, would have mean-square factor $(n-p)/n$, hence an RMS bias at most pH_D/n . The zero-embedded difference bound (IC.72) already includes the missing coordinates themselves and dominates this bias. The strict RMS and Gram margins above therefore pay for original normalization n without any false claim that a cavity's empirical covariance has the full-network expectation. The n^p multiplicity is used only in the coordinate comparison (IC.73), with its exponential Gaussian tail.

For initial coordinate localization at a layer above the deletion, condition on the lower initialized layers and project their complete feature difference onto the deterministic ball of radius (IC.72). The current row

is still independent Gaussian, so each pairing has variance at most D_{init}^2/n . At threshold $n^{-1/10}$ its tail is $2e^{-n^{4/5}/(2D_{\text{init}}^2)}$. The union over deletion sets and tests costs at most

$$2pmL^2n^{p+1}e^{-n^{4/5}/(2D_{\text{init}}^2)} < \rho/32. \quad (\text{IC.73})$$

The negative exponent is at least $n^{799}/2$, while the logarithm of the prefactor is at most $10n^{1/4000}\ell$, by (IC.68). At the deleted layer retained coordinates agree. Hence each cavity starts with budget at most $e^{\eta n^{-1/10}}8L + p/n < \mathcal{B}/2$. These initial failure allocations sum to less than $\rho/4$. Each cavity nevertheless uses its own initialization test (including the just specified initial Gram margin), with the zero-reference convention preceding (IC.stop). This is needed when the full initial-event indicator is subsequently dropped.

On the local uniform event in the quantitative training-source ledger, (IC.55) and the unchanged scalar absorption (S.16) give

$$Z_{a,i} + K_{a,i}/S \leq C_{\text{abs}}[1 + G_{h,a,i} + G_{\delta,a,i}/S + \bar{G}_{h,i} + \bar{G}_{\delta,i}] + n^{-1/40}. \quad (\text{IC.74})$$

Its coefficient remains the singleton coefficient, independent of the moment order p . Each Gaussian reference has coefficient RMS at most $V_* = \max(1, H_{\text{max}}, \tau_*)$. Its raw time derivative costs $\sqrt{n}$ times the proved coefficient and a power of ℓ , so a mesh of size n^{-2} has vanishing interpolation error and at most n^5 points under (IC.66). Use scalar grid threshold $31V_*\sqrt{\ell}$, reserving $V_*\sqrt{\ell}$ for interpolation. Real/imaginary splitting gives tail $4e^{-31^2\ell/8}$. There are at most $\mathcal{A}n^6$ tests. Their union is below n^{-100} . Consequently (IC.74) improves the temporary full maxima to exactly

$$\max_{a,j,i} |z_{a,i}^{(j)}| \leq K_{\text{src}}\sqrt{\ell}, \quad \max_{a,j,i} |k_{a,i}^{(j)}| \leq SK_{\text{src}}\sqrt{\ell}. \quad (\text{IC.75})$$

Apply the explicit response contractions (IC.57) in increasing layer order and use the Gaussian multiplier 64. The m^2 driver/evaluated training pairs and the same mesh have total additional failure below n^{-100} , and the response stops improve to

$$\max_{a,b,j,i} |R_{ba,i}^{(j)}| \leq SU_j^{\text{fin}}\sqrt{\ell}. \quad (\text{IC.76})$$

The direct reverse mean here is explicitly $\bar{\delta}_{a,i} \text{tr}(C_b C_a^\top)/n$; its integrated term is (IC.56), and its cross forms are centered. Thus this step includes the reverse observable instead of inferring a response cap from a feature cap. Equations (IC.70), (IC.76) now improve the full pole stop. Budget removal has not yet been invoked.

9. Finite-order common-cavity moments and collisions

We first state an elementary Gaussian rectangle estimate, including its constants. If a deterministic complex coefficient map b_z on a rectangle with sides at most H_0 satisfies $\sup \|b_z\| \leq D$ and $\|b_z - b_{z'}\| \leq K\|z - z'\|$, then for a standard real Gaussian vector G ,

$$X = \sup_z |G^\top b_z|, \quad M = 256D\sqrt{\log(e + H_0K/D)}, \quad \log \mathbb{E}e^{uX} \leq uM + 2u^2D^2. \quad (\text{IC.77})$$

If $D = 0$, interpret the process as zero. To prove the mean bound, use dyadic meshes of size proportional to $D2^{-j}/K$, with at most $4(1 + H_0K/D)^{2^j}$ points. Connect each point to the preceding mesh; increment standard deviations are at most $3D2^{-j}$. The bound $\mathbb{E} \max_{r \leq N} |Z_r| \leq \sigma\sqrt{2\log(2N)}$ for Gaussian variables of standard deviation at most σ follows by their exponential moments and a union bound. Summing uses $\sum 2^{-j} = 1$, $\sum 2^{-j}\sqrt{j} \leq 2$. The constant 128 covers the real initial grid and increments; using real and imaginary parts gives 256. The supremum is at most $2D$ -Lipschitz in G , so (IC.59) proves its exponential moment. For samplewise processes sharing the same root, their RMS $\bar{X}$ is also $2D$ -Lipschitz, and $\mathbb{E}\bar{X} \leq M + 2D$ by the variance estimate. Thus (IC.77) holds for the sample RMS with M replaced by $M + 2D$, without independence between samples. For a root with covariance I/n , use coefficient norms divided by $\sqrt{n}$.

For a set $I \ni i$, both the singleton cavity and the common I -cavity omit the particular root against which their difference is paired. Their complete independently stopped coefficient paths, including their

own failed-initialization conventions, are independent of that root. Project the whole difference path onto a deterministic Euclidean ball of radius $4n^{1/100}$, before restricting to the successful full prefix. Projection is nonexpansive, preserves root independence, and agrees with the difference on that prefix by the $2n^{1/100}$ comparison proved above. Its normalized Gaussian radius is

$$v_n = 4n^{-49/100}.$$

The raw normalized derivative recurrences give rectangle modulus at most $\beta^{300L}(1+\lambda)\ell^3$; its side length is at most $(64/\lambda + 4)\ell$. Applying (IC.77), and its RMS version, to the sum of the two cavity moduli yields

$$\log \mathbb{E} e^{uE_i} \leq u\mu_n + 2u^2v_n^2, \quad \mu_n \leq 256v_n \sqrt{\log \left(e + \frac{\beta^{400L}(1+\lambda^{-1})\ell^4}{v_n} \right)} + 2v_n. \quad (\text{IC.78})$$

The two cavity paths need not be independent of each other. Equations (IC.66)–(IC.68) imply

$$p\mu_n + p^2v_n^2 \leq 1032n^{-.48975}\sqrt{\ell} + 16n^{-.9795} < 10^{-3}. \quad (\text{IC.79})$$

We also need finite moments for complex-minus-real corrections, rather than a limit at fixed moment degree. The sharp derivative calculation yielding (S.27), now using (IC.76), gives

$$\|\partial_t \delta_a^{(j)}\|_{2,n} \leq J\rho \left[1 + \frac{S^2}{\eta} (4 + \log(2\mathcal{B}) + \log \ell) \right], \quad J = \beta^{80L}. \quad (\text{IC.80})$$

To see that the constant is finite with this power, interpolate the RMS and maximum bounds on $\dot{z}$, then use the carrier Schatten bound at real exponent $r = \max\{4, \log(2\mathcal{B}\ell)\}$. The product $k \odot \dot{z}$ is at most $(2\rho S^2 N_*/\eta)r(2\mathcal{B}\ell)^{1/r}$ in RMS, where $N_* \leq \beta^{72L}$. The last exponential is at most e . Differentiate the backward pass: the propagated upper derivative costs $10s$, changed mixer costs $2s^3 k_{j+1}^2 H_j$, and changed gate costs $4et_2 N_*$. Each forcing coefficient is at most β^{75L} , and its geometric sum at most β^{78L} . Enlarging to J also pays for the forward derivative and fixed contour factors. This proves (IC.80) for every width, retaining the fixed budget logarithm instead of assuming it is below $\log \ell$.

Call the bracket in (IC.80) A_ℓ . The unchanged condition $S^2\Lambda/\eta \leq 1$ gives $A_\ell \leq 8[1 + \log(e + \ell)]$. Subtract the coefficient at the nearest real anchor. With $c = [p\beta^{100L}(1+\lambda)]^{-1}$, the radius coefficient in (IC.63), its normalized radius, Lipschitz coefficient, and rectangle side are bounded by

$$D = \frac{\lambda c J A_\ell}{\sqrt{\ell}}, \quad K = \lambda J A_\ell, \quad H_0 = 64\ell/\lambda + 4c/\sqrt{\ell}. \quad (\text{IC.81})$$

These bounds also hold after a reference's own clamping; the nearest real-anchor map is Lipschitz. Set $r = \lambda^{-1}$ only in the following arithmetic. Then

$$D \leq \frac{\beta^{-20L}}{p(1+r)} \frac{A_\ell}{\sqrt{\ell}}, \quad H_0 K / D = 64p\beta^{100L}(1+r)\ell^{3/2} + 4, \quad D \leq 32\beta^{-20L}/p.$$

The logarithm in (IC.77) is at most $6 + 100L \log \beta + \log p + \log(1+r) + 2\log(e + \ell)$. Use $(2+u)e^{-u/2} \leq 2$ and $(2+u)^{3/2}e^{-u/2} \leq 4$ for $u = \log \ell \geq 0$, $\sqrt{\log p/p} \leq 1$, and $\sqrt{\log(1+r)/(1+r)} \leq 1$. Substitution gives $M + 2D \leq 2^{16}\beta^{-18L}$. Therefore for a complex-minus-real correction X , including its sample RMS,

$$(\mathbb{E} e^{apX})^{1/p} \leq \exp\{a^{16}\beta^{-18L} + 2^{11}a^2\beta^{-40L}/p\} \leq e^{\beta^{-10L}}, \quad 0 \leq a \leq 8. \quad (\text{IC.82})$$

This holds at every width for the stopped references. It is why the factor p in (IC.63) suffices without an exponential width in the moment order or the inverse gap.

On the common cavity, the real Gaussian references have the squared exponential bound (S.20), including the sample RMS form. Completing the square and Hölder for the four references in (IC.74) give a main one-neuron exponential moment below four at exponent at most 8η , exactly as in (S.21). Distinct omitted root pairs are conditionally independent on that common cavity. Bound each full singleton by its common-cavity real reference plus the projected correction (IC.78) and complex correction (IC.82), then drop the event indicator from these nonnegative bounds. Cauchy–Schwarz separates the product of main references from all corrections. Hölder over the at most p roots and four reference families pays for the corrections: their

required coefficient is at most eight because $\eta C_{\text{abs}} \leq (1024W_G)^{-1}$. Equation (IC.79) makes the projected factor smaller than $e^{1/10}$, (IC.82) makes the complex factor smaller than $e^{1/10}$, and the scalar error in (IC.74) contributes less than $e^{1/10}$. These margins are per root; together with the main bound below four they are below sixteen. Thus, with E the common initialization, local and maximum event,

$$\mathbb{E} \left[1_E \prod_{i \in I} e^{\eta(Z_{a,i}^{(j)} + K_{a,i}^{(j)}/S)} \right] \leq 16^{|I|}, \quad 1 \leq |I| \leq p. \quad (\text{IC.83})$$

At no point was E , or full survival, a Gaussian conditioning event. Samples and correction processes have not been assumed independent.

Repeated indices require separate treatment. The unchanged constants obey

$$2\eta K_{\text{src}} \leq \frac{1 + C_G}{32W_G} \leq 33/4096 < 1/100.$$

Indeed $W_G \geq 128 \max(1, H_{\text{max}}, \tau_*)$, since $V_1 = \tau_1 = \max_j \tau_j$. On (IC.75), put $B_i = e^{\eta(Z_{a,i}^{(j)} + K_{a,i}^{(j)}/S)}$; then

$$0 \leq B_i \leq e^{\sqrt{\ell}/100} \leq e^{1/100} n^{1/100} = W_n. \quad (\text{IC.84})$$

Expand $(n^{-1} \sum_i B_i)^p$. For a partition of its ordered positions into k equal-index blocks, retain one factor per distinct index and bound the repeated factors by W_n^{p-k} . The retained expectation is at most 16^k by (IC.83), and there are at most n^k distinct index assignments. Partitions with $j = p - k$ identifications are at most $\binom{p}{j}$: map each block to the star joining its least position to all others; the resulting j edges determine the partition. Consequently

$$\mathbb{E}[1_E (n^{-1} \sum_i B_i)^p] \leq 16^p \exp\{\binom{p}{2} W_n/(16n)\} \leq e16^p, \quad (\text{IC.85})$$

where $n \geq 4p^3$, implied by (IC.68), suffices for the last inequality. This does not request an exponential moment at coefficient $p\eta$ of a repeated main root.

If any full budget reaches $\mathcal{B}$, some sample/layer average reaches $\mathcal{B}/L$. Markov and the mL union give

$$\Pr(\text{a budget is hit}, E) \leq emL(16L/\mathcal{B})^p = emL(64e^2)^{-p} \leq \rho/4. \quad (\text{IC.86})$$

Initialization costs less than $\rho/4$, the local event less than $\rho/16$, and the two maximum events together cost $2n^{-100} < \rho/16$. All stops are therefore removed with total failure less than ρ . Their strict margins give holomorphy on a neighborhood of the closed rectangle (IC.63). The original RMS and operator bounds, (IC.75)–(IC.76), the real fitting trajectory and endpoint all hold on this event.

For the Seeded decoder, use $\rho = 2^{-20}\delta$ and $p = p_{\text{ref}}$ for the independent dense reference. A constructed member uses $\rho = 2^{-20}$, giving the stated implemented order p independent of δ . The reference gate dominates the member gate because both p and ρ^{-1} are larger for the reference. Whole-experiment amplification, proved in the decoder section, pays for member failure. It is not a union-bound requirement that every member source succeed. Substitution in (IC.67) and its envelope gives exactly

$$n \geq \left\lceil \left[\frac{2^{20} \beta^{2000L}}{\delta} (1 + m/\gamma)^4 (m + d + p_{\text{ref}} + 1)^4 \right]^{1100} \right\rceil. \quad (\text{IC.87})$$

Additional numerical-precision or decoder-work gates remain their separately stated conditions. In particular this source proof does not remove a separately chosen $n \geq Y^{-1}$ precision convention, and it imposes no positive lower label bound of its own.

The independent dense-reference certificate also follows from this training-only source, with no qualitative complex-sphere event. Here is the exact bridge. Put $T_0 = 32\ell/\lambda$ and define

$$C_L^k = 1, \quad C_j^k = S\tau_{j+1} + 10sC_{j+1}^k + 10t_2SP_{j+1}k_{j+1}, \quad C_{\text{carrier}} = \max_j C_j^k. \quad (\text{IC.88})$$

For two real states in the physical operator/RMS tube, forward subtraction bounds a preactivation RMS difference by $P_j \|\Delta\theta\|_{\text{par}}$. In the backward subtraction, the changed mixer costs $S\tau_{j+1} \|\Delta\theta\|_{\text{par}}$, propagation

costs $10s$, and the changed gate costs $10t_2SP_{j+1}k_{j+1}\sqrt{n}\|\Delta\theta\|_{\text{par}}$. The first $\sqrt{n}$ is the preactivation coordinate conversion; after the final carrier coordinate conversion this gives

$$\max_{a,j} \|k_a^{(j)}(t) - k_a^{(j)}(T_0)\|_\infty \leq C_{\text{carrier}}n\|\theta(t) - \theta(T_0)\|_{\text{par}}.$$

Apply the real fitting energy/path-length inequality starting at T_0 : the rightmost parameter difference is at most $2\rho(T_0)/\sqrt{\lambda} \leq 2Y(en)^{-16}/\sqrt{\lambda}$. Consequently the exact sufficient carrier-tail gate is

$$\frac{2C_{\text{carrier}}Y}{\sqrt{\lambda}} n(en)^{-16} \leq K_{\text{src}}S\sqrt{\ell}. \quad (\text{IC.89})$$

It is implied by

$$n \geq \max \left\{ 1, \left(\frac{C_{\text{carrier}}\sqrt{\lambda}}{8K_{\text{src}}} \right)^{1/15} \right\} \quad \text{and hence by} \quad n \geq \max\{1, (\beta^{23L}/8)^{1/15}\}. \quad (\text{IC.90})$$

Indeed $S = 16Y/\lambda$, $K_{\text{src}} \geq 1$, $C_{\text{carrier}} \leq \beta^{20L}$ by (IC.88)'s single geometric sum, and $\sqrt{\lambda} \leq \beta^{3L}$. Gate (IC.87) dominates (IC.90). Adding (IC.75) at T_0 gives the exact all-time training-carrier bound used in dense comparison, $M_n = 2K_{\text{src}}S\sqrt{\ell}$, also at the fitted endpoint.

Define the real dense good set by the fitting operator/RMS/Gram tube, its endpoint tail, and this training-carrier bound. Its definition is independent of p , the complex radius, and confidence. Both the fixed-confidence members and the high-confidence independent reference belong to that same set on their respective source events. The dense endpoint-subtraction proof uses coordinate carrier maxima only for training samples: its arbitrary-query step is forward RMS subtraction in the real operator tube. Its Gaussian extension then uses a real sphere net, a real compactified-time grid, and the real fitting tail. Thus the signed dense Lipschitz constant, its common extension center, and the whole-sphere/all-time certificate B_n are unchanged when this finite-training event replaces the original qualitative source event. No analyticity or carrier maximum at passive complex queries is a hypothesis of that comparison. This explicitly closes the reference-event dependency of the Seeded decoder.

E.5.2 Finite physical program, selected metric and unseen queries

1. Physical variables and the precise source-event interface

Write $v_a = x_a/\sqrt{d}$, so $\|v_a\|_2 = 1$, and retain the dense forward and backward recursions

$$\begin{aligned} z_a^{(1)} &= Av_a, & z_a^{(j)} &= W^{(j)}h_a^{(j-1)}, & h_a^{(j)} &= \phi_j(z_a^{(j)}), & f_a &= w^\top h_a^{(L)}/n, \\ k_a^{(L)} &= w, & \delta_a^{(j)} &= \phi'_j(z_a^{(j)}) \odot k_a^{(j)}, & k_a^{(j)} &= W^{(j+1)\top} \delta_a^{(j+1)}. \end{aligned} \quad (\text{FC.1})$$

The residual is $r_a = f_a - y_a$, and the loss is $m^{-1} \sum_a r_a^2$. In physical time the exact equations are

$$\dot{A} = -\frac{2}{m} \sum_a r_a \delta_a^{(1)} v_a^\top, \quad \dot{W}^{(j)} = -\frac{2}{mn} \sum_a r_a \delta_a^{(j)} h_a^{(j-1)\top}, \quad \dot{w} = -\frac{2}{m} \sum_a r_a h_a^{(L)}. \quad (\text{FC.2})$$

Thus these are the original mobilities $(n, 1, \dots, 1, n)$. Initially $A_{ik} \sim N(0, 1)$, $W_{ik}^{(j)} \sim N(0, 1/n)$ independently and $w = 0$. Every label restriction below is a consequence of the original common label allowance; none replaces it by a smaller cap.

Let

$$\lambda = \gamma/m, \quad r = \lambda^{-1}, \quad Y = \|y\|_2/\sqrt{m} > 0, \quad S = 16Yr \leq 1, \quad \ell = \log(en), \quad B = \beta^{100L}. \quad (\text{FC.3})$$

The scalar r is distinct from the indexed residual r_a . The zero-label branch is the exact constant zero predictor. The nonzero finite-word branch retains $nY \geq 1$. The source/fitting bounds give $Y \leq \beta^{3L}$ and $r^{-1} \leq \beta^{6L}$. All estimates use the existing

$$Z = \ell + \log \left(e + \frac{(m+d+2)\beta^{100L}(1+r)}{\delta} \right).$$

The implemented moment order is p ; it has fixed source confidence and is prescribed in the integrated theorem, with $\log(p+2) \leq CZ$.

Here is the exact event used from the internal finite-source lemma. The training trajectory, its forward/backward fields and the training response fields are holomorphic on a neighborhood of

$$[-r_t, 32\ell/\lambda + r_t] + i[-r_t, r_t], \quad r_t = \frac{1}{pB(1+\lambda)\sqrt{\ell}}. \quad (\text{FC.4})$$

The initialized operator bounds, the fitted real trajectory, and all-time real feature bounds hold there with their stated real/complex distinction. On the complex domain, hidden operators are at most ten, training feature RMS is at most β^{3L} , training backward RMS is at most $S\beta^{6L}$, carrier maximum is at most $S\beta^{40L}\sqrt{\ell}$, and the preactivation imaginary part is at most $a/4$. The parameter-gradient response satisfies

$$\max_{a,b,j} \|R_{ba}^{(j)}\|_\infty \leq S\beta^{72L}\sqrt{\ell}. \quad (\text{FC.5})$$

Here $\Theta = (A, \sqrt{n}W^{(2)}, \dots, \sqrt{n}W^{(L)}, w)$ and $R_{ba}^{(j)} = D_\Theta z_b^{(j)} \nabla_\Theta (nf_a)$, with the Euclidean/Frobenius gradient in these blocks. This definition includes all mobility and width factors and contains no residual. For normalized time $\tau = \lambda t$, a disk of radius $\lambda r_t/2$ about a real anchor τ_j has training residual RMS at most $2Ye^{-\tau_j/4}$. These are direct conclusions of that source lemma at the full label allowance. In particular, no lower bound of the form $\sqrt{\ell} \geq c(m, \gamma, \beta, L)$ is imported here.

For a displacement $u = (A - A_0, W^{(2)} - W_0^{(2)}, \dots, w)$, use

$$\begin{aligned} \|u\|_\Sigma &= \|A - A_0\|_F/\sqrt{n} + \sum_{j=2}^L \|W^{(j)} - W_0^{(j)}\|_F + \|w\|_2/\sqrt{n}, \\ \|u\|_{\mathcal{H}}^2 &= \|A - A_0\|_F^2/n + \sum_{j=2}^L \|W^{(j)} - W_0^{(j)}\|_F^2 + \|w\|_2^2/n. \end{aligned} \quad (\text{FC.6})$$

Then $\|u\|_{\mathcal{H}} \leq \|u\|_\Sigma \leq c_L \|u\|_{\mathcal{H}}$, where $c_L = \sqrt{L+1}$. Set $\bar{u}(\tau) = u(r\tau)/Y$. This normalization changes proof coordinates only. The normalized vector field is denoted by $\bar{F}$.

2. Analytic tube and real stability, with explicit panel orders

Choose the normalized source horizon and the equal-panel mesh by

$$\begin{aligned} T_0 &= 2\{10 \log n + \log(1 + 66Br)\}, \\ T_0 &\leq T \leq T_0 + 1/4, \\ h_0 &\leq \frac{1}{128pB(1+r)\sqrt{\ell}}, \\ H &= \lceil T/h_0 \rceil, \\ h &= T/H. \end{aligned} \quad (\text{FC.7})$$

Choose a certified dyadic upper approximation T in the indicated interval and the largest dyadic h_0 below its bound. The numerical horizon gate gives $T_0 < 24\ell$, hence $T < 32\ell$. Now T/h_0 is rational, its ceiling is an exact integer operation, and the equal-panel length $h = T/H$ is rational. Scalar integration weights using this rational are rounded only at their declared local arithmetic step. Thus $h_0/3 \leq h \leq h_0$, and

$$H \leq CpB(1+r)Z\sqrt{\ell}. \quad (\text{FC.8})$$

Every radius- $4h$ panel disk lies in (FC.4). Put

$$q(\tau) = 2^{-\lfloor \tau/4 \rfloor}, \quad q_j = q(jh), \quad q_T = q(T), \quad d_n = \frac{\beta^{-200L} q_T}{(1+r)^2 \sqrt{n}},$$

$$M = B(1+r), \quad \Lambda_j = B(1+r)(1+q_j \sqrt{\ell}). \quad (\text{FC.9})$$

We first verify the tube needed for numerical restarts. A physical parameter perturbation of sum norm at most Yd_n changes a training preactivation by RMS at most $\beta^{4L}Yd_n$, by forward subtraction of (FC.1). Indeed, each changed-matrix term is bounded by the old feature RMS times its matrix operator norm, and each unchanged hidden propagation multiplies the previous RMS error by at most $11\beta \leq \beta^3$. Its coordinate error is therefore at most $\sqrt{n}\beta^{4L}Yd_n < a/16$. Stopping the segment at a first half-strip exit proves that no exit occurs. The operator cap increases by at most one.

Backward subtraction at a reference source state uses

$$\Delta\delta^{(j)} = \phi'_j(\tilde{z}^{(j)}) \odot \Delta k^{(j)} + [\phi'_j(\tilde{z}^{(j)}) - \phi'_j(z^{(j)})] \odot k^{(j)}, \quad \Delta k^{(j)} = \widetilde{W}^{(j+1)\top} \Delta\delta^{(j+1)} + \Delta W^{(j+1)\top} \delta^{(j+1)}. \quad (\text{FC.10})$$

Only the changed-gate term uses a reference carrier maximum. Summing the downward linear recursion gives RMS at most $\beta^{50L}(1+S\sqrt{\ell})Yd_n$. There is one carrier-maximum factor, not a product of such factors across layers. Conversion to a coordinate bound, using $Y/S = 1/(16r)$, $r^{-1} \leq \beta^{6L}$, and (FC.9), keeps the carrier maximum within twice its source bound. Prediction subtraction on the training complex tube has coefficient at most β^{8L} in the physical sum norm. At real parameter states the same bound holds for every real unit input, by the real all-sphere feature bound and forward subtraction; no passive complex-time analyticity is used. The residual on this tube is consequently at most $3Yq_j$ in sample RMS.

Subtracting the three factors in each gradient product of (FC.2), and using the forward and backward subtractions just proved, bounds the complex normalized Jacobian by Λ_j . For clarity, before normalization the residual-difference terms contribute at most β^{70L} , and the backward/feature-difference terms contribute at most $\beta^{70L}Yq_j(1+S\sqrt{\ell})$. Normalization multiplies this by r ; $Yr \leq 1/16$ and the unused powers from 70 to 100 give (FC.9). The normalized true derivative is at most M . Thus the unclipped field is holomorphic and Λ_j -Lipschitz on the complex normalized sum-norm tube of radius d_n .

The sharper real estimate is essential. In the Euclidean coordinates $P = (A/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$, (FC.2) is $\dot{P} = -\nabla_P \mathcal{L}$. The blocks of $g_a = \nabla_P f_a$ are $\delta_a^{(1)} v_a^\top / \sqrt{n}$, $\delta_a^{(j)} h_a^{(j-1)\top} / n$, and $h_a^{(L)} / \sqrt{n}$. Differentiating the forward recursion in a physical direction gives RMS derivatives of z, h at most $\beta^{9L} \|e\|_\Sigma$. Differentiating (FC.10), with the unchanged point as reference, gives backward derivative RMS at most $\beta^{55L}(1+S\sqrt{\ell}) \|e\|_\Sigma$. Each hidden block of $Dg_a[e]$ is the sum of the two outer products $D\delta_a[e] h_a^\top / n + \delta_a D h_a[e]^\top / n$; their Frobenius norms are products of the respective RMS norms. Summing the at most $L+1$ blocks, and using $\|e\|_\Sigma \leq c_L \|e\|_\mathcal{H}$, proves

$$\|D_P^2 f_a\|_{\mathcal{H} \rightarrow \mathcal{H}} \leq \beta^{70L}(1+S\sqrt{\ell}). \quad (\text{FC.11})$$

The real normalized Jacobian is exactly

$$D\bar{F} = -\frac{2r}{m} \sum_a \{g_a g_a^\top + r_a D_P^2 f_a\}.$$

Its first quadratic form is nonpositive. Using the tube residual bound and (FC.11) on its second term gives

$$\langle e, D\bar{F}e \rangle_\mathcal{H} \leq \mu_j \|e\|_\mathcal{H}^2, \quad \mu_j = \beta^{80L} q_j (1+S\sqrt{\ell}), \quad E_+ = 9\beta^{80L} (1+S\sqrt{\ell}). \quad (\text{FC.12})$$

Indeed $m^{-1} \sum_a |r_a| \leq 3Yq_j$ and $rY \leq 1/16$. Since $\int_0^\infty q = 8$, its decreasing left sums obey $\sum_j h q_j \leq 8+h \leq 9$, so $\sum_j h \mu_j \leq E_+$. This estimate holds throughout each real tube and hence on segments between the compared states. It is not applied on complex segments.

Choose

$$\varepsilon = \min\{d_n/(2^{12}c_L), n^{-10}/(2^{12}Bc_L)\},$$

$$K = 8 + \left\lceil \frac{4E_+ + \log[2^{20}(c_L + 1)(M + 1)(T + 1)/\varepsilon] + \log(1 + 16c_L H)}{\log 2} \right\rceil,$$

$$\delta_0 = \frac{\beta^{-150L} S q_T}{(1 + r)^2 \sqrt{n}}, \quad C_N = B^2(1 + r)^2 \sqrt{\ell}, \quad \delta_{\dagger} = \min \left\{ \delta_0, \frac{\varepsilon e^{-4E_+ - 10}}{2^{20} C_N (T + 1)} \right\}. \quad (\text{FC.13})$$

These are constructive choices, with fixed fractions of $\delta_{\dagger}$ allocated below. In particular

$$K + \log(1/\delta_{\dagger}) + \log(1/\varepsilon) + \log(H + 1) + \log(h^{-1}) \leq CBZ. \quad (\text{FC.14})$$

To check the small scales, $S \geq 16\beta^{-6L}/n$, $q_T^{-1} \leq 2e^{T \log 2/4}$, and $T \leq CZ$; every remaining factor in (FC.13) has logarithm at most CZ , apart from $E_+ \leq CB\sqrt{\ell}$. No algebraic factor r enters the degree or numerical word length.

3. Causal coefficients, real activation values, and physical forcing

Use $\xi = (\tau - jh)/h$; brackets $[k]$ denote its power-series coefficients. The current physical parameters have coefficients through degree K , and the normalized recurrence is

$$\bar{u}[0] = \bar{u}_j, \quad \bar{u}[k + 1] = \frac{h}{k + 1} [\xi^k] \bar{F} \left(\sum_{b=0}^k \bar{u}[b] \xi^b \right), \quad 0 \leq k < K. \quad (\text{FC.15})$$

At a fixed order evaluate forward layers upward, backward layers downward, residuals, and then gradient coefficients. Write $D^{(j)} = W^{(j)} - W_0^{(j)}$. The initialized actions are exactly

$$z_a^{(j)}[k] = W_0^{(j)} h_a^{(j-1)}[k] + \sum_{b+c=k} D^{(j)}[b] h_a^{(j-1)}[c],$$

$$k_a^{(j)}[k] = W_0^{(j+1)\top} \delta_a^{(j+1)}[k] + \sum_{b+c=k} D^{(j+1)}[b]^\top \delta_a^{(j+1)}[c]. \quad (\text{FC.16})$$

The first layer uses its d root columns and fixed input coordinates; the top backward carrier is $w[k]$. For $g_a^{(j)}[b] = \sum_{c+e=b} r_a[c] \delta_a^{(j)}[e]$, the physical hidden coefficients satisfy

$$D^{(j)}[k + 1] = -\frac{2hr}{mn(k + 1)} \sum_a \sum_{b+c=k} g_a^{(j)}[b] h_a^{(j-1)}[c]^\top. \quad (\text{FC.17})$$

The analogous first and readout formulas are obtained directly from (FC.2). A learned matrix is represented by these actual stored factors and scalar weights; no orthogonalization or inverse Gram enters them. Committing a panel appends their integrated weights. All old fields keep their original scalar arguments.

Here is a finite value-based realization of the activation coefficients. Let $\alpha = \min(1, a/16)$, so $\alpha^{-1} \leq \beta$. At a real order-zero preactivation x , interpolate J_a values of $\phi(x + \alpha t_i)$, where $t_i = -1 + 2i/(J_a - 1)$. Each supplied value has absolute error at most ν . Use its polynomial p_x in the centered variable $(z - x)/\alpha$, and use the actual derivative of this same polynomial for the backward gate.

For completeness, subtract $\phi(x)$, which interpolation reproduces. On $|s| = 4$, the resulting function has modulus at most $4\beta\alpha$. The residue formula with $\omega(s) = \prod_i (s - t_i)$ gives interpolation error at most $C\beta\alpha 2^{-J_a}$ on $|s| \leq 1/2$: numerator factors are at most $3/2$ and denominator factors at least three. The sum of the absolute cardinal polynomials on that disk is at most

$$\frac{[3(J_a - 1)/2]^{J_a - 1}}{(J_a - 1)!} \leq 5^{J_a}.$$

Cauchy's formula on circles of radius $1/4$, followed by rescaling, therefore proves simultaneous value/first/second derivative errors at most $C\beta^2(2^{-J_a} + 5^{J_a}\nu)$ on $|z - x| \leq \alpha/4$. Choose

$$J_a \geq \max\{K + 2, \lceil \log_2(C\beta^2/\delta_{\dagger}) \rceil\}, \quad \nu \leq \delta_{\dagger}/(C\beta^2 5^{J_a}), \quad (\text{FC.18})$$

with the least admissible orders and fixed additional error fractions. Then $J_a \leq CK$ and $\log \nu^{-1} \leq CBZ$. Only real values are requested; their evaluator costs remain charged.

The scalar disk is available at each computed center. From (FC.2), (FC.5) and residual RMS at most $2Y$, $\|\partial_\tau z_a^{(j)}\|_\infty \leq S^2 \beta^{72L} \sqrt{\ell}/4$. A radius- $4h$ panel changes the reference preactivation by at most $hS^2 \beta^{72L} \sqrt{\ell} < \alpha/4096$. The tube changes it by at most $\sqrt{n} \beta^{4L} Y d_n < \alpha/4096$. The small node defects below leave the same strict margin. A first-exit argument therefore puts all polynomial evaluations in their interpolation disks.

In the following bounds R is the deterministic enclosing field/call count explicitly constructed in (FC.25); that recipe depends only on the orders already chosen. Every initialized action is replaced by $W_0 v + \sigma \zeta$ or $W_0^\top u + \sigma \zeta$, with fresh hidden independent $\zeta \sim N(0, I_n)$. On the event that each such vector has RMS at most two, its errors through one panel through coefficient $K - 1$ form a polynomial of RMS at most $\sigma 4^K$ on $|\xi| \leq 4$. Only those orders are needed by (FC.15). Choose

$$\sigma \leq \frac{1}{64} \delta_{\dagger} 4^{-K}. \quad (\text{FC.19})$$

Every positive tolerance specified by an upper bound here is chosen as the largest dyadic below the stated minimum, after the fixed budget fractions. The precision upper bounds refer to those choices, not arbitrary smaller numbers. The common noise level is refined in the precision schedule before setup. The probability of a raw-noise RMS violation is at most $Re^{-n/2}$. Indeed exponential Markov with parameter $3/8$ gives $\Pr(\|\zeta\|_2^2 > 4n) \leq \exp[-(3 - \log 4)n/2] \leq e^{-n/2}$.

Physical scalar pair calls use the actual created operands and return $\hat{p}(u, v) = n^{-1} u^\top v + e$, with $|e| \leq \epsilon_{\text{pair}}$. If an actual stored matrix is $D = n^{-1} \sum_\mu c_\mu a_\mu b_\mu^\top$, its approximate forward action differs from its exact action by precisely

$$\sum_\mu c_\mu a_\mu e(b_\mu, v). \quad (\text{FC.20})$$

This compares the same realized matrix on both sides. Let $N_{\text{sum}} = C(R + K + 1)^3$ bound all rank/convolution summand counts, and let $A_* \geq 2$ bound their scalar weights, factor RMS, $n + 1, m, L, K, Y^{-1}, h^{-1}$, and inverse strict guard margins. Use finite bounds from the explicit formulas with a factor-four slack; interpolation scratch has size e^{CJ_a} times a polynomial in these scales. Thus $\log A_* \leq CBZ$. Equation (FC.20) has RMS at most $N_{\text{sum}} A_*^2 \epsilon_{\text{pair}}$. Readout coefficients are sums of pairs and have scalar error at most $N_{\text{sum}} \epsilon_{\text{pair}}$. The exact rank-matrix norm is

$$\|D\|_F^2 = \sum_{\mu, \nu} c_\mu c_\nu \langle a_\mu, a_\nu \rangle_n \langle b_\mu, b_\nu \rangle_n;$$

its pair-error change is at most $3N_{\text{sum}}^2 A_*^4 \epsilon_{\text{pair}}$. Converting integrated-coefficient errors to velocity errors multiplies by $(k + 1)/(hY)$, already in the polynomial allowance. Consequently all these physical errors, including normalized residual errors, extend to forcing polynomials bounded by $C4^K N_{\text{sum}}^2 A_*^8 \epsilon_{\text{pair}}$. Choose

$$\epsilon_{\text{pair}} \leq \delta_{\dagger} / (C4^K N_{\text{sum}}^2 A_*^{10}). \quad (\text{FC.21})$$

Principal local arithmetic outputs obey the same allowance. Their finite precision uses only the logarithms in (FC.14), (FC.18), (FC.21): centered polynomial interpolation has coefficient sums at most e^{CJ_a} , and truncated convolution is submultiplicative in coefficient ℓ^1 . It never raises the large real center to power J_a .

Freeze the realized coefficient errors into these forcing polynomials. This is a proof construction, not advance observation of future noises. The resulting holomorphic field $\bar{F}_j^{\text{num}}(\xi, U)$, with fixed polynomial activation coefficients and fixed forcings, obeys

$$\|\bar{F}_j^{\text{num}} - \bar{F}\|_\Sigma \leq C_N \delta_{\dagger} =: \eta_{\dagger}, \quad \|D_U \bar{F}_j^{\text{num}}\| \leq 2\Lambda_j. \quad (\text{FC.22})$$

To verify the label scaling, forward error is at most $\beta^{4L} \delta_{\dagger}$; backward error is at most $\beta^{60L} (1 + S\sqrt{\ell}) \delta_{\dagger}$; readout RMS is at most $C\beta^{3L} S$; hence residual error is at most $\beta^{10L} S \delta_{\dagger}$. Subtraction of the residual/backward/feature factors gives physical field error at most $C\beta^{80L} [Y(1 + S\sqrt{\ell}) + S + S^2] \delta_{\dagger}$. Multiplication by r/Y , using $S/Y = 16r$, gives (FC.22). An independently added residual error of RMS $Y \delta_{\dagger}$ has the same bound.

Differentiating the same recursions uses only the first two polynomial derivatives from (FC.18) and the same tube bounds, proving the second inequality. No derivative of an activation center is taken.

If the starting Hilbert error is e_j with $c_L e_j \leq d_n/8$, the difference equation on $|z| < 4h$ is a contraction in the sum-norm ball of radius $d_n/2$: its contraction factor is $8h\Lambda_j \leq 1/4$, and its forcing is bounded by $c_L e_j + 4h\eta_{\dagger}$. Thus both local flows exist on that disk, and their difference is at most $2c_L e_j + 8h\eta_{\dagger}$. On its real diameter, (FC.12) instead gives Hilbert error $(e_j + s\eta_{\dagger})e^{s\mu_j}$. Cauchy's coefficient estimate at radius $2h$ then gives the endpoint recurrence

$$e_{j+1} \leq (1 + 2h\mu_j + 2c_L 2^{-K})e_j + 4h\eta_{\dagger} + 2Mh2^{-K}. \quad (\text{FC.23})$$

The extra unit in the forcing coefficient pays separately rounded endpoints; they are additive errors, not forcings asserted to leave the polynomial unchanged. The same calculation controls interior times. Products of the multipliers are at most $\exp(2E_+ + 2c_L H 2^{-K})$. Substitution of (FC.13) bounds the total error by $\varepsilon/8$, provided

$$e_0 \leq \varepsilon e^{-4E_+ - 10}/2^{20}. \quad (\text{FC.24})$$

The proof closes the tube assumptions inductively. Strict-slack guards are justified causally: at a first potentially active guard, freeze the preceding defects and complete the remaining panel with zero defects. The contraction just proved supplies its local solution. Formal coefficient induction identifies the already computed prefix with that solution, whose Cauchy bounds lie strictly inside the guard. The proposed first activation is impossible. The argument applies to coefficient caps and approximate norm tests using (FC.20)–(FC.21), and inserts no residual projection into the holomorphic recurrence.

Uniform real prediction subtraction and the original fitting tail at T now give a parameter-defined predictor of error at most CYn^{-10} , for the entire sphere and all physical times, when parameters are frozen after T . This includes the dense fitted endpoint.

4. Actual named fields and the finite Gaussian law

Fix the source instructions before drawing randomness. In each panel and for each sample/layer/order name the preactivation, activation, gate, backward carrier, backward response, residual-weighted backward response, and each initialized answer in (FC.16). Also name the interpolation sample values and its needed centered coefficients, the readout coefficients, the constant field, and the d first-layer root columns. Each initialized call names one innovation field. Coefficient products internal to one coordinatewise interpolation or convolution are temporary scalar work; they never become operands of empirical reductions. Residuals and rank weights are shared scalars. This is a literal finite field recipe, of size

$$R_0 = 1 + d + C_0 m L H (K + J_a + 1), \quad R = R_0 + C_0 (L + 1), \quad (\text{FC.25})$$

where a fixed integer C_0 , for example 128, covers the listed field types and reserved query fields. Naming a reused field twice is unnecessary. The bounds (FC.8), (FC.14), (FC.18), and $L \leq \beta^L$ give

$$R \leq Cp\beta^{201L}(m + d + 2)(1 + r)Z^{5/2}. \quad (\text{FC.26})$$

There are $O(mLHK^2)$ rank weights, at most CR^2 ; there are not that many distinct row factors. Acquisition of every missing unordered pair is performed when both fields exist, at the end of a complete matrix call when necessary. Mandatory innovation contractions remain inside their own call. Thus at most $P \leq C_1 R^2$ scalar acquisitions occur. Here P includes each scalar component of every mandatory innovation contraction, ordinary complete-table pair, and appended query reduction. Reserve distinct scalar marks for all of them, including an innovation pair that may also have an ordinary pair entry. This counts every finite scalar sampler used in the Gaussian cutoff below. All scalar means use the constant field. Norms of rank matrices use products of acquired pairs as displayed after (FC.20), not additional fourth moments.

One fully explicit oversized local cap for the physical forcing construction is

$$A_* = 2^{128(J_a + K + 1)} [(n + 1)(m + L + d + R + K + 2)(B + 1)(1 + r)(1 + Y^{-1})(1 + h^{-1})(1 + d_n^{-1})(1 + S^{-1})]^{128}. \quad (\text{FC.27})$$

It exceeds the physical Cauchy coefficient bounds, rank weights, all interpolation intermediate coefficient sums, and the inverse fixed guard margins. For example, endpoint rank weights are bounded by a polynomial in rT, K, m ; interpolation weights by e^{CJ_a} ; and principal coordinate bounds by $\sqrt{n}$ times their RMS bounds. Each displayed operation has bounded product degree or its explicit centered interpolation coefficient bound. Enlarging the fixed factor 128, if required by a chosen arithmetic implementation, changes no parameter exponent. Its logarithm is at most CBZ . Take $b = A_*$ as a common loose RMS cap for all physical query/raw-answer fields; interpolation-only fields do not enter physical posterior Gram solves. Their finite coordinate caps are retained separately for metric selection.

We now derive the initialized-matrix calls, including their two orientations. For a single hidden matrix W with iid $N(0, 1/n)$ entries, suppose the previous raw observations are

$$Y_f = WV + \sigma\Xi, \quad X_b = W^\top U + \sigma Z_b. \quad (\text{FC.28})$$

Here $V \in \mathbb{R}^{n \times a}$, $U \in \mathbb{R}^{n \times b_h}$ are predictable old forward and reverse queries. The noise vectors are fresh independent standard Gaussians and are hidden. The letters Y_f, X_b denote answer matrices and are unrelated to the label scale Y . At a complete observable history, put

$$Q = V^\top V/n, \quad K_h = U^\top U/n, \quad H_c = U^\top Y_f/n, \quad J_c = X_b^\top V/n, \quad \Delta = \sigma^2.$$

The likelihood times prior density is proportional to

$$\exp\{-\frac{n}{2}\|W\|_F^2 - \frac{1}{2\Delta}(\|Y_f - WV\|_F^2 + \|X_b - W^\top U\|_F^2)\}.$$

Predictability permits this expression even for adaptive calls: each query is fixed after the preceding answer history is fixed. Completing its square shows that the posterior mean solves

$$\Delta \bar{W} + (UU^\top/n)\bar{W} + \bar{W}(VV^\top/n) = (Y_f V^\top + U X_b^\top)/n. \quad (\text{FC.29})$$

The covariance operator is $(\Delta/n)(\Delta I + \mathcal{L}_{UU^\top/n} + \mathcal{R}_{VV^\top/n})^{-1}$, where left and right multiplication have their literal meanings. Its inverse is well defined since the quadratic form of its denominator is at least $\Delta\|M\|_F^2$. Likelihoods factor by matrix label, preserving conditional independence of different hidden matrices.

Define $C = (\Delta I + Q)^{-1}$, $D = (\Delta I + K_h)^{-1}$, and solve

$$(\Delta I + K_h)E + EQ = -H_c C - D J_c. \quad (\text{FC.30})$$

Diagonalizing the two real symmetric coefficients shows that all Sylvester divisors are $\Delta + \lambda_i(K_h) + \lambda_j(Q) \geq \Delta$. Direct substitution in (FC.29) verifies $\bar{W} = Y_f C V^\top/n + U D X_b^\top/n + U E V^\top/n$. For the next forward query q , let $v = V^\top q/n$, $x = X_b^\top q/n$, $d_q = q^\top q/n$. Its conditional mean is

$$m_q = Y_f C v + U(Dx + Ev). \quad (\text{FC.31})$$

For the next reverse query u , it is $V(CY_f^\top u/n + E^\top U^\top u/n) + XD(U^\top u/n)$, with $X = X_b$. Set

$$f(a) = \frac{\Delta}{\Delta + a} \{d_q - v^\top [(\Delta + a)I + Q]^{-1} v\}, \quad a \geq 0. \quad (\text{FC.32})$$

Diagonalize $UU^\top/n$ and $VV^\top/n$ in the posterior covariance operator. The variance of Wq along an eigenvector of the former with eigenvalue a is $(\Delta/n)q^\top [(\Delta + a)I + VV^\top/n]^{-1}q$. The identity

$$(cI + VV^\top/n)^{-1} = c^{-1}[I - V(cI + Q)^{-1}V^\top/n]$$

is checked by multiplication and turns this variance into (FC.32). Therefore the next noisy answer has covariance $\Gamma = \Delta I + f(UU^\top/n)$. In particular $0 \leq f(a) \leq d_q$ and $\Gamma \succeq \sigma^2 I$.

Let $f_0 = f(0)$, $c = \sqrt{\Delta + f_0}$, and define the rational divided difference

$$h_f(a) = \frac{-f_0 + \Delta v^\top C[(\Delta + a)I + Q]^{-1}v}{\Delta + a}, \quad T_q(a) = \frac{h_f(a)}{\sqrt{\Delta + f(a)} + c}. \quad (\text{FC.33})$$

Resolvent subtraction gives $ah_f(a) = f(a) - f_0$, including zero by continuity; no division by a possibly zero Gram eigenvalue occurs. Along each singular direction of $U/\sqrt{n}$, $c + aT_q(a) = \sqrt{\Delta + f(a)}$. Hence

$$\Gamma^{1/2} = cI + UT_q(K_h)U^\top/n. \quad (\text{FC.34})$$

This is the positive square root. Every inverse above has floor σ^2 , and the denominator in (FC.33) has floor 2σ . The reverse formula exchanges the forward and reverse histories. Empty histories simply omit their blocks.

All these coefficients are computable with matrices of order at most CR . In eigenbases of Q, K_h , the Sylvester solve divides entry (i, j) by $\Delta + k_i + q_j$, and the two contractions needed in (FC.33) are

$$\sum_j \frac{\tilde{v}_j^2}{\Delta + k_i + q_j}, \quad \sum_j \frac{\tilde{v}_j^2}{(\Delta + q_j)(\Delta + k_i + q_j)}. \quad (\text{FC.35})$$

There is no materialized Kronecker matrix. Away from genuine moment arrays, symmetrize the whole physical-field Gram, project it onto the PSD cone and cap its Frobenius norm at $8Rb^2$. This protection fixes a genuine Gram under the cap. Euclidean projection on a closed convex set is nonexpansive: the two minimizing variational inequalities imply $\|Px - Py\|^2 \leq \langle Px - Py, x - y \rangle$. The protected augmented block $\begin{pmatrix} Q & v \\ v^\top & d_q \end{pmatrix}$ is a Gram, so its $f(a)$ is nonnegative by the same variance formula. Positive-part protections on any finitely computed f preserve the explicit floors.

These coefficient maps have fixed polynomial local sensitivity. For a checkable envelope let $z = C(2 + R + b + \sigma^{-1})$. Genuine/protected Gram norms are at most z^4 ; inverse norms are at most z^2 . For perturbation size e in the moment entries, Frobenius conversions cost at most z^2 . The inverse identity gives error at most z^6e . Subtracting (FC.30) gives E -error at most $z^{18}e$, including its bounded right side. Resolvent contractions (FC.35), or their equivalent gapped tensor formulas, have errors at most $z^{36}e$. The square-root bound $\|A^{1/2} - B^{1/2}\|_F \leq (2\sigma)^{-1}\|A - B\|_F$ follows by solving $A^{1/2}X + XB^{1/2} = A - B$. Multiplying the remaining factors and applying the final gapped inverse bounds the correction and mean coefficient errors by $z^{60}e$. Thus z^{100} is a common coefficient and Lipschitz cap, with ample dimension and constant slack. This calculation is for one call, not a product across earlier calls.

5. One finite pre-setup precision schedule and chronological coupling

The finite row packet contains its d first roots and one innovation coordinate for every prescribed training call and every reserved query call. All coordinates are independent. Scalar-noise marks are a separate independent finite bit string. A finite Gaussian sampler can be coupled to a standard Gaussian g with coordinate error at most ϵ_G when $|g| \leq T_G$: use the midpoint of the b_G -bit interval containing $\Phi(g)$, then a certified inverse CDF clipped to $[-T_G, T_G]$. The interval index is uniform. The clipped inverse has Lipschitz constant at most $\sqrt{2\pi}e^{T_G^2/2}$; consequently

$$b_G \log 2 \geq T_G^2/2 + \log(C/\epsilon_G) \quad (\text{FC.36})$$

suffices, with inverse-evaluation error at most $\epsilon_G/2$. The finite arithmetic backend supplies the certified evaluation. For $N_G = n(d + R) + P + 1$, choose $T_G = \lceil 2 + \sqrt{2 \log(1024N_G/\rho_0)} \rceil$, with $\rho_0 = 2^{-20}$. The union of coupled sampler failures is below $\rho_0/512$. The implemented algorithm draws bits directly.

All named row fields are dyadic. An ordinary pair acquisition uses exact dyadic products and exact integer summation, retains division by n as a rational until rounding, and records

$$C_a = Q_{h_s} \left(n^{-1} \sum_i u_a(i) v_a(i) + \eta \hat{e}_a \right). \quad (\text{FC.37})$$

Here Q_{h_s} is fixed nearest-grid rounding with a fixed tie convention, and $h_s \leq \eta$ is the scalar grid, distinct from the panel length. For a forward matrix call compute the protected mean coefficients, symmetric correction $\tilde{T}$, and scalar $\tilde{c}$ before reading its fresh innovation. Put $A_c = U^\top/n$, $B_c = U\tilde{T}$. Then execute

$$\hat{t} = Q_{h_t}(A_c \hat{g} + \eta \hat{e}), \quad \hat{y} = Q_{h_y}(\tilde{m} + B_c \hat{t} + \tilde{c} \hat{g}). \quad (\text{FC.38})$$

Small coefficient arrays are rounded once to dyadics, symmetrically where required; final row combinations have exact dyadic arithmetic before the indicated rounding. The guard $\tilde{c} \geq \sigma/2$ is inactive on the successful range. No other matrix call or new coefficient solve occurs between these two operations. Reverse calls exchange orientations. The subsequent complete-table pairs are acquired after the answer is formed.

At a fixed past consider, only for the proof, the affine Gaussian shadow

$$t = A_c g + \eta e, \quad y = \tilde{m} + B_c t + \tilde{c} g. \quad (\text{FC.39})$$

Its answer covariance is $\tilde{\Gamma} = \tilde{S}^2 + \eta^2 B_c B_c^\top$, where $\tilde{S} = \tilde{c}I + B_c A_c$ is symmetric. The map is invertible since $\eta, \tilde{c} > 0$:

$$g = (y - \tilde{m} - B_c t)/\tilde{c}, \quad e = (t - A_c g)/\eta. \quad (\text{FC.40})$$

Therefore the entire finite call (FC.38), including every finite innovation mark used later, is a deterministic function $J_H(y, t)$: reconstruct (FC.40), apply the fixed sampler functions and run the finite arithmetic. This possibly discontinuous map is used only in the proof. The algorithm executes (FC.38) directly and pays no numerical cost for (FC.40).

Let $Q_H(dy, dt)$ be the shadow kernel. Couple it to the physical call $y = W_0 q + \sigma \zeta$, with hidden fresh noise, by attaching $t \mid (H, y, W_0) \sim Q_H(dt \mid y)$, then applying the same J_H . This attachment is conditionally independent of every initialized matrix given (H, y) ; its likelihood cancels in Bayes' formula. Consequently the Gaussian posterior derived in the finite Gaussian law remains valid at the next complete call, even if the finite marks are reused. Moreover

$$\|Q_H(dy, dt) - P_H(dy)Q_H(dt \mid y)\|_{\text{TV}} = \|Q_H(dy) - P_H(dy)\|_{\text{TV}}. \quad (\text{FC.41})$$

Integration against the common conditional kernel gives one inequality; projection onto y gives the reverse. Deterministic finite postprocessing cannot increase this distance. The proof filtration includes raw answers, but excludes hidden physical noise, future innovation coordinates, and the private selected metric. No Gaussian-posterior claim is made at the intermediate prefix that reveals only $\hat{t}$.

Here are quantitative finite choices. Let $\mathcal{M} = (n+1)z^{120}$. This bounds every row-affine norm in (FC.38). Direct subtraction from (FC.39) gives

$$\|\hat{t} - t\|_\infty \leq (\mathcal{M} + 1)\epsilon_G + h_t/2 =: d_t, \quad \|\hat{y} - y\|_\infty \leq h_y/2 + \mathcal{M}d_t + \mathcal{M}\epsilon_G. \quad (\text{FC.42})$$

Choose $h_t \leq \min(\eta, h_y/(64\mathcal{M}))$ and $\epsilon_G \leq \min(h_t/(64\mathcal{M}), h_y/(64\mathcal{M}^2))$. Then the answer error is below h_y . At a shared raw history the old finite and raw answers differ by at most h_y . Their moments differ by at most $2bh_y + h_y^2$; acquired-pair noise adds at most $\eta(T_G + 1) + h_s$. Thus the coefficient input error is at most

$$e_{\text{mom}} = C\{(b+1)h_y + h_y^2 + \eta(T_G + 1) + h_s\}. \quad (\text{FC.43})$$

Both queries are the same actual finite vectors at this history; there is no comparison with recomputed queries at a different history. If finite coefficient error is ϵ_c , put $e_c = e_{\text{mom}} + z^{10}\epsilon_c$. The local estimates in the finite Gaussian law give $\|\tilde{m} - m\|_2 \leq \sqrt{n}z^{110}e_c$ and $\|\tilde{S} - S\|_{\text{op}} \leq z^{110}e_c$, including the old answer-to-finite-answer change in the row combinations. Also $\|B_c B_c^\top\|_F \leq nRb^2z^{200}$.

Since $S \succeq \sigma I$, a sufficient smallness condition is

$$\kappa = z^{230}(\sqrt{n}e_c + n\eta^2) \leq 1/32. \quad (\text{FC.44})$$

It implies $\tilde{S} \succeq \sigma I/2$. Expanding $\tilde{S}^2 - S^2 = (\tilde{S} - S)\tilde{S} + S(\tilde{S} - S)$ bounds the Frobenius covariance error. To verify the TV conclusion, write $E = \Gamma^{-1/2}(\tilde{\Gamma} - \Gamma)\Gamma^{-1/2}$. The Gaussian density integral gives relative entropy

$$\frac{1}{2}\{\|\Gamma^{-1/2}(\tilde{m} - m)\|^2 + \text{tr } E - \log \det(I + E)\}.$$

For $\|E\| \leq 1/2$, diagonalization and $x - \log(1+x) \leq x^2$ bound this by half the sum of squared σ^{-1} -scaled mean and σ^{-2} -scaled covariance errors. Split the density integral over an event attaining total variation and its complement. Convexity of $u \log u$ bounds relative entropy below the resulting Bernoulli relative

entropy. Its second derivative in its first argument is $1/[a(1-a)] \geq 4$, and it and its first derivative vanish at equality; integrating twice gives $\text{KL} \geq 2\text{TV}^2$. Thus $\text{TV} \leq \sqrt{\text{KL}/2}$. The exponents in (FC.44) dominate these expressions, so the one-call TV is at most κ , with all positivity hypotheses included.

Make the choices once, before generating any source or metric. A passive answer tolerance sufficient for the unseen-query construction is

$$a_{\text{qry}} = n^{-10}/[C\beta^{6L}(1+r)].$$

First choose a dyadic σ below (FC.19) and $a_{\text{qry}}/16$, common to training and query calls. Then compute $z, \mathcal{M}, T_G$, and

$$\xi_0 = \frac{\rho_0}{2^{20}(R+1)(n+1)z^{240}},$$

$$\eta \leq \frac{\min(\epsilon_{\text{pair}}, \xi_0)}{64(T_G+1)}, \quad h_s \leq \eta/4, \quad h_y \leq \min\{a_{\text{qry}}/64, \epsilon_{\text{pair}}/64, \xi_0/[64(b+1)]\}, \quad \epsilon_c \leq \xi_0/(64z^{10}). \quad (\text{FC.45})$$

Take each dyadic within a factor two below its bound. Further lower h_y to the initialized-answer fraction of (FC.21), so its polynomial forcing is also allocated there. Choose h_t, ϵ_G from (FC.42), and add

$$\eta\epsilon_G \leq \frac{\rho_{\text{acq}} \min(h_s, h_t)}{64(P+1)}, \quad \epsilon_G \leq \frac{Y\epsilon e^{-4E_+ - 10}}{2^{20}(\sqrt{d}+1)}. \quad (\text{FC.46})$$

The first pays for literal metric replay below; the second implies (FC.24) since the finite first-layer displacement has normalized Hilbert norm at most $\sqrt{d}\epsilon_G/Y$. All other local rank/query arithmetic is made finer than its allocated fraction of a_{qry} , divided by the explicit local polynomial caps.

Every tolerance logarithm above is at most CBZ . Indeed this is a fixed number of products/minima of (FC.14), A_* , counts, and local noise floors; it never multiplies precision by the number of earlier calls. Exact empirical summation adds $\log n$ bits. The prescribed finite backend can therefore use

$$w \leq C\beta^{110L}Z \quad (\text{FC.47})$$

bits per word, including integer ranges, coefficient scratch and counters. This claim retains the charged activation/input interfaces of the theorem.

At complete-call boundaries maximally couple the shadow and physical raw answers. On equality attach the same conditional augmentation and use the same finite postprocessing. The finite tapes then agree literally until failure. Summing (FC.44) over at most R calls costs at most $\rho_0/64$. Sampler failure is charged on the iid-packet marginal, and raw-noise/source failure on the physical marginal. Before a mismatch, both histories are identical, so these probabilities add. On the surviving event, finite answers have form $W_0q + \sigma\zeta + d$ with $\|d\|_{2,n} \leq h_y$, current-operand pairs meet (FC.21), and the strict-slack proof in the physical forcing construction applies. This proves the finite physical-source coupling without a global row-history Lipschitz assumption and without changing the scientific Gaussian initialization.

6. Selected packets and literal replay at local precision

Use only the completed finite training table in the permitted offline setup. Let $V_* \in \mathbb{R}^{n \times s}$, $s \leq CR$, have its named dyadic fields as columns, including the constant one. If its exact dyadic rank is q , then $1 \leq q \leq \min(n, s)$. Choose independent columns J and rows I for which $H_I = V_*[I, J]$ is invertible; define

$$C_I = V_*[:, J]H_I^{-1}, \quad M_I = C_I^\top C_I/n. \quad (\text{FC.48})$$

Every column lies in the span of those selected columns. Restriction to I identifies its coordinates, proving

$$V_* = C_I V_*[I, :], \quad C_I[I, :] = I_q, \quad V_*^\top V_*/n = V_*[I, :]^\top M_I V_*[I, :]. \quad (\text{FC.49})$$

In particular $M_I \succeq I_q/n$, and the constant column gives $\mathbf{1}^\top M_I \mathbf{1} = 1$.

The selected matrix can be bounded without any subspace-gap hypothesis. If $|(C_I)_{ij}| > 2$, replace selected row j by original row i . Multilinearity gives $|\det H_{I, \text{new}}| = |(C_I)_{ij}| |\det H_I| > 2 |\det H_I|$. The new

row cannot duplicate a different selected row, since its coefficient there would be zero. Repeat until every entry of C_I is at most two in modulus. For a table with fractional precision b_f and coordinate cap B_f , a nonzero starting determinant is at least 2^{-qb_f} , and Hadamard bounds every selected determinant by $q^{q/2}B_f^q$. There are therefore fewer than $q[b_f + \log_2 B_f + \frac{1}{2} \log_2 q] + 1$ swaps. This is a finite exact integer algorithm on the scaled dyadic table; its work and temporary expanded-integer storage are charged in the word-backend lemma. At termination

$$|(M_I)_{ij}| \leq 4, \quad \|M_I\|_{\text{op}} \leq 4q. \quad (\text{FC.50})$$

To make every unsuccessful branch finite as well, cap each named field coordinate at a fixed dyadic B_f larger than $8(n+1)z^{300}$, choosing a dyadic within a factor two of that bound. The physical local bounds and the sampler event keep these guards inactive. Off the successful range this specifies a total algorithm. Its cap logarithm is at most CBZ ; finite endpoint failure flags are deterministic scalar instructions. Rows with identical finite packet/prefix arguments always produce identical field bits, including on failed branches.

Every operational guard is computable from the finite row fields or scalar state: coordinate caps are rowwise; learned-matrix norm tests use (FC.20)'s pair formula; residual norm tests use the m scalar outputs; coefficient floors and caps use their small finite arrays. The initialized operator and scientific source events are used only in the proof. The algorithm does not test an unmaterialized initialized operator norm or recognize a source-good event for free. If a finite operational guard fails, its deterministic failure flag and dummy output follow the finite prescribed instruction schedule.

Round symmetric metric entries with error at most ϵ_M , producing M_0 , and retain

$$\widehat{M} = M_0 + q\epsilon_M I_q.$$

Then $\widehat{M} \succeq M_I$ and $\|\widehat{M} - M_I\|_{\text{op}} \leq 2q\epsilon_M$. For selected operand vectors bounded by B_f ,

$$|u^\top (\widehat{M} - M_I)v| \leq 2q^2 B_f^2 \epsilon_M. \quad (\text{FC.51})$$

No inverse metric is used in training or querying.

We prove the replay claim despite the private metric depending on the entire completed source and its noises. For a fixed real a , Gaussian e , grid $g > 0$, and $0 < v \leq g/2$,

$$\Pr\{\text{dist}(a + \eta e, g(\mathbb{Z} + 1/2)) \leq v\} \leq 2v(g^{-1} + \eta^{-1}). \quad (\text{FC.52})$$

If f is the density of $a + \eta e$, then $\int |f'| = 2/(\eta\sqrt{2\pi}) \leq \eta^{-1}$. Integration of $f(s + kg) \leq f(x) + \int_{s+kg}^{s+(k+1)g} |f'|$ over each interval, then summation, gives $\sum_k f(s + kg) \leq g^{-1} + \eta^{-1}$. Integrating over an interval of length $2v$ proves (FC.52), uniformly in a . The same proof works conditionally when a is known and the fresh e remains independent.

Fix $\rho_{\text{acq}} = 2^{-12}$, let $g_* = \min(h_s, h_t)$, and set

$$\zeta = \rho_{\text{acq}} g_* / [64(P+1)], \quad \epsilon_M \leq \zeta / (2q^2 B_f^2). \quad (\text{FC.53})$$

In setup copy the selected packets and scalar marks exactly. Compact training replaces the empirical pair in (FC.37) by $u(I; C_{<a})^\top \widehat{M} v(I; C_{<a})$, and uses the same finite mark, grid and finite coefficient instructions. This bilinear form is formed with exact dyadic products and summation, then the specified outer rounding; an additional local arithmetic error up to ζ would also suffice.

Condition on the entire packet array and all preceding scalar marks. The next exact empirical mean is then fixed and the fresh raw scalar Gaussian is independent. Since both scalar grids are at most η , (FC.52) with $v = 4\zeta$ bounds the chance of being within 4ζ of this update's boundary by $16\zeta/g_*$. A union over the P updates costs less than $\rho_{\text{acq}}/4$. Condition (FC.46) and the sampler event give simultaneous scalar-mark change at most ζ . If compact and source prefixes agree, their selected operand bits agree. Equations (FC.49), (FC.51), (FC.53) put the metric error below ζ . The source and compact pre-rounding values are then within 3ζ of the same raw Gaussian value, and lie in its same rounding cell. Induction gives literal equality of all scalar prefixes with failure at most ρ_{acq} .

This proof conditions on the packet array, not on the privately chosen metric. It applies to any random packet array independent of the fresh scalar marks; independence among different packets is unnecessary.

Thus it also applies after the inner short-seed replacement. It asserts equality on the realized tape, and does not assert uniform quadrature over changed prefixes. The selected packet coordinates, rounded metric, finite scalar-noise marks, current scalar history and coefficient arrays occupy CR^2 words. The full field table, exact rational metric intermediates, and all future scalar answers are discarded after offline setup. Future answers are recomputed causally during compact training, not retained as advice.

7. An unseen query streams the original empirical contractions

At physical time $t = r\tau \leq rT$, take the already acquired panel containing τ and evaluate its finite parameter polynomial. A panel is acquired as a whole by the causal coefficient recurrence at its left endpoint; requests within it do not acquire the next panel. At $\tau \geq T$ use its frozen final state. The current first-layer increment and readout are scalar combinations of old named fields. A hidden increment has exactly the finite rank-list form

$$D = T_r A_r S_r^\top / n, \quad (\text{FC.54})$$

where columns of T_r, S_r are old row fields and A_r is the current finite matrix of scalar rank weights. The subscripts here identify the rank representation and are not the inverse-gap scalar r .

For a new unit input v , regenerate one row packet at a time from the retained inner seed and evaluate the old finite field DAG at its immutable creation-time scalar arguments. No scalar training reduction is recomputed. The first-layer value is the dot product of its d finite initial roots with v , plus the exact finite learned increment. At a hidden layer the learned action is computed from the actual empirical contraction

$$c_h = S_r^\top h / n, \quad Dh = T_r A_r c_h. \quad (\text{FC.55})$$

Stream all n rows to form every entry of c_h . These are exact dyadic sums followed by allocated finite scalar rounding; no population expectation is substituted. Similarly stream the new field's pairings with each old forward and reverse query/answer field. Use these and the already acquired old moments in (FC.30)–(FC.35), then perform (FC.38) using the reserved finite innovation coordinates and scalar marks. In particular the entire $U\tilde{T}\tilde{t}$ covariance correction remains. The final readout is likewise its empirical pair with the top feature.

A constant number of passes per layer suffices: a pass obtains the new query moments and rank contractions; a pass obtains the innovation contractions; a pass evaluates the answer/activation and its next needed pairs. Multiple quantities in a pass have separate exact accumulators. Previous query fields are reevaluated rowwise at their already computed scalar arguments. The coefficient arrays are prepared once per fixed query context, never per regenerated row. A query uses CR^2 words of scalar history, coefficients, accumulators and row scratch, besides its seed scratch.

Each query starts its own local transcript at the current training prefix. The reserved coordinates are reused as a deterministic function of the fixed member seed; no query augments the retained training history, and no previous query answer is inserted as a new matrix constraint. This gives one well-defined finite prediction for every external code. The analysis below couples each fixed code separately; its final simultaneous event covers adaptive choices.

For a fixed input/time code, run the iid source only through that acquired prefix and append the passive calls just described. Do not reveal the discarded future training answers first. The chronological coupling applies to this chronological source-plus-query computation with the same pre-setup precision, because the old training moments and new query moments satisfy the same tolerances. Refining only the new query would not justify that step. The private selected metric is not added to the Gaussian posterior filtration. Metric replay is used afterward to identify the actual retained prefix.

In the physical coupled process, the finite parameter state is within the all-sphere bound from the physical forcing construction. On its operator/feature event, forward subtraction for an additional answer/arithmetic defect of RMS at most a_{qry} gives

$$e_j \leq \beta(11e_{j-1} + Ca_{\text{qry}}), \quad e_j = \|\hat{h}^{(j)} - h^{(j)}\|_2 / \sqrt{n}. \quad (\text{FC.56})$$

The first-layer input/initial-root arithmetic has its own allocated fraction of the same tolerance. Since $11\beta \leq \beta^{5/2}$, the geometric sum in (FC.56) is at most $C\beta^{3L}a_{\text{qry}}$. The readout RMS is at most $CY\beta^{3L}(1+r)$,

so the choice in the precision schedule makes the query-only prediction change at most CYn^{-10} , with an absolute total allocation after increasing the fixed constant in a_{qry} . Fresh physical noise RMS failures have total probability at most $CLe^{-n/2}$, already included when R counts the appended calls. For one member impose $Re^{-n/2} \leq 2^{-20}$; the clean branch's stated implementation gate is a stronger sufficient bound. The source failure at 2^{-20} , this noise failure, the coupling at $\rho_0/64$, the sampler at $\rho_0/512$, and metric replay at 2^{-12} total less than $1/1024$. Thus allocating at most $1/64$ each to the internal dense-center and inner-generator errors still leaves the combined failure strictly below $1/16$. These allocations give the fixed-code one-member physical conclusion required by the internal dense-center and short-seed lemmas. There is no passive bias of size R/n or $\sqrt{R/n}$: (FC.55) has computed its exact finite empirical target.

8. Physical sphere/time codes, tails, and adaptive queries

The finite decoder is permitted to be discontinuous. Its input mesh uses regularity of the physical reference alone. On the real operator/readout event, successive forward subtraction gives

$$|f_n(t, v) - f_n(t, v')|/Y \leq C\beta^{20L}(1+r)\|v - v'\|_2. \quad (\text{FC.57})$$

The first normalized operator has bounded norm; each hidden propagation costs at most 11β ; Cauchy-Schwarz pairs the last feature difference with readout RMS at most $CY\beta^{3L}(1+r)$. Differentiating the physical prediction and using (FC.2) gives

$$|\partial_\tau f_n(r\tau, v)|/Y \leq C\beta^{20L}(1+r) \quad (\text{FC.58})$$

through the finite horizon. For example the physical vector-field norm is at most $Y\beta^{12L}$, and prediction sensitivity at most β^{8L} ; multiplication by r gives the stated bound. These estimates use real states and real inputs only.

Choose an integer

$$k_{\text{ext}} = \lceil \log_2[Cn^{10}\beta^{20L}(1+r)(d+2)(T+2)] \rceil. \quad (\text{FC.59})$$

For a unit v , approximate its coordinates by a lattice vector z of step $2^{-k_{\text{ext}}-6-\lceil \log_2(d+1) \rceil}$, then use the mathematical sphere point $z/\|z\|_2$. The lattice approximation can ensure $\|z - v\|_2 \leq 2^{-k_{\text{ext}}-3}$; consequently $\|z\|_2 \geq 1/2$ and $\|z/\|z\|_2 - v\|_2 \leq 2^{-k_{\text{ext}}-2}$. Its code is the finite integer vector z ; normalization is evaluated to the finer internal precision, with that error allocated in the precision schedule. Approximate input access may choose any qualifying code; no exact rounding tie on an arbitrary real input must be decided.

Within its current acquired panel, approximate the fractional time from below, round down at mesh $2^{-k_{\text{ext}}}$, and clamp to $[0, 1]$. Using a certified lower approximation with error at most one mesh width changes it by at most two mesh widths. The code and original time remain in the same acquired panel. At a boundary either already acquired adjacent certified panel can be used; both compare to the same physical value. Include all panel labels and their two endpoints and one frozen-tail code. The mesh is never enumerated or stored. Since $\log H \leq CZ$,

$$\log N_{\text{ext}} \leq C\{d[k_{\text{ext}} + \log(d+2)] + k_{\text{ext}} + \log(H+1)\} \leq C(d+1)Z. \quad (\text{FC.60})$$

Equations (FC.57)–(FC.59) make the physical input/time rounding error a fixed fraction of Yn^{-10} . Beyond T , compare to the physical prediction at T , adding the fitting tail $CYH_D^2 re^{-T/2} \leq Yn^{-10}/4$. This proves the same scope at $t = \infty$.

The internal dense-center lemma and the unseen-query construction therefore give, for each fixed code, a one-member bad-output probability at most $1/16$, after the fixed source/coupling/replay/generator allocations. The internal whole-member amplification lemma makes all coded median outputs accurate simultaneously with the specified confidence. To transfer to an arbitrary query, compare its returned coded value with the physical value at that code, then use (FC.57)–(FC.59) and the tail. No continuity estimate for the finite decoder is used. Because this is one event for every code, queries may be chosen after inspecting the retained model and previous answers. Their dependence creates no new union bound or posterior premise.

The resulting comparison with the independent dense reference is $2B_n + A_{\text{num}}Yn^{-10}$, with A_{num} an absolute allocated constant. Its explicit mesh coefficient gives $B_n \geq 32Y/n$, so the stated numerical gate

absorbs the remainder into $3B_n$. The centers and source events used in this argument are proof objects; the retained model never receives them or any unknown test label.

9. Small positive labels and scope of the finite interfaces

For completeness the separate branch $0 < nY < 1$ has the following explicit recipe. Put $Z_Y = Z + \log(1/(nY))$. Keep the same physical horizon and source domain, replace the panel count by $H = \lceil T/h_0 \rceil + \lceil BZ_Y \rceil$, and then compute (FC.13) with this H , adding $\lceil \log_2(1/(nY)) \rceil$ to its Taylor degree. Choose $J_a, \delta_{\dagger}$ and every subsequent tolerance from the actual Y, S by the same formulas. There is no circular choice: Z_Y, H, K, J_a, R are fixed in that order before the pair and sampler precisions. The extra degree makes $J_a = O(K)$ valid even when δ_0 is tiny, and the extra panels ensure $K \leq CH$. All short-complex-panel estimates improve as the step shrinks.

Here $\log(1/Y) = \log n + \log(1/(nY)) \leq CZ_Y$. Therefore every precision estimate above holds with Z_Y , and

$$R_Y \leq Cp\beta^{201L}(m+d+2)(1+r)Z_Y^{5/2}, \quad w_Y \leq C\beta^{110L}Z_Y. \quad (\text{FC.61})$$

The source-event width has no new lower label condition. The finite implementation must additionally keep the explicitly enlarged Gaussian-RMS gate

$$R_Y e^{-n/2} \leq 2^{-20}, \quad \text{equivalently} \quad n \geq 2 \log(2^{20} R_Y), \quad (\text{FC.62})$$

where R_Y is the constructed integer field/call count. This is a finite numerical gate, not an unquantified stochastic threshold. Using the right side of (FC.61) as a deterministic certificate gives a directly checkable sufficient version. The finite row-root cutoff and code count are recomputed with these orders. Replacing only the word precision would have missed the activation interpolation orders and the raw-noise union. All claimed numerical storage/work envelopes use Z_Y in this branch; the displayed clean headline remains its $nY \geq 1$ branch.

Finite interfaces

Equations (FC.25)–(FC.26) define the actual field/call recipe and establish the claimed parameter dependence. Equations (FC.13), (FC.18), (FC.21), (FC.36), (FC.42), (FC.45)–(FC.46), and (FC.53) define one shared finite precision schedule before setup; their logarithms give (FC.47). The selected metric has at most R selected packets, all empirical communication is through CR^2 scalar words, and its exact replay is on the finite tape it actually acquired. New queries regenerate the original finite rows and use all two-orientation covariance terms. These are the interfaces required by the separate internal word-backend and short-seed lemmas; no additional scientific hypothesis or undeclared history sensitivity is needed here.

For the short-seed interface the entire scalar transcript has $O(R^2 w)$ bits. This count includes acquired pairs, shared scalar branch outputs consumed by later field definitions, and the final row coefficient vector at each field's creation. Row-local branches and internal spectral pivots or CDF bisection decisions are deterministic computations from the packet and candidate scalar inputs, and are recomputed rather than retained as transcript entries. A matrix call does not retain its whole correction matrix forever: after $\hat{t}$ is acquired it multiplies $\hat{T}\hat{t}$, combines that vector with the mean coefficients, retains at most CR coefficients for its new affine row field, and discards the R -square temporary matrices. There are CR field creations, so these vectors occupy CR^2 words. The temporary spectral/Sylvester matrices are recomputed or discarded during fixed-context coefficient preparation.

For a candidate-transcript verifier, all creation-time scalar arguments and these coefficient vectors are hardwired proof data. One pass through the packets accumulates at most CR^2 exact dyadic sums. Each accumulator needs only a constant multiple of w bits, since its products have bounded arity and $\log n$ is included in w . After the stream, the verifier checks every scalar rounding instruction and coefficient-vector consistency in chronological order from those sums, reusing CR^2 words of small-matrix scratch. It need not retain each earlier spectral matrix. Induction then identifies a passing candidate with the actual transcript. Thus both transcript length and between-packet verifier state are $O(R^2 w)$, not $O(R^3 w)$. At an intermediate training time only the already acquired coefficient vectors are retained; the future vectors used in offline selection have been discarded with the future-answer table.

Initialization remains offline: its selected metric may depend on the entire virtual finite training trajectory. Its full source table is temporary. The retained state contains seeds, selected finite packets, the rounded metric, scalar-noise marks, current acquired history and finite instructions. It contains no dense hidden matrix, full row table or future-answer table. Training is causal scalar acquisition. Querying does not advance training. The numerical finite source approximates dense gradient flow; it is not asserted itself to fit exactly. The comparison remains to an independent dense reference over the whole sphere, all physical times and the fitted endpoint. The analytic upper certificate remains an absolute guarantee. The same finite construction also supports the intrinsic-quantile comparison proved below, with the original numerical remainder.

E.5.3 A short seed preserves the finite source transcript

Consider the actual finite source program. Its random input consists of n independent finite row blocks and a separate string of scalar-noise marks. Communication between rows occurs only through finitely many exact dyadic reductions, followed by prescribed rounding and guarded scalar instructions. Include the reductions, rounded answers and branch decisions in one finite transcript.

Fix a candidate transcript. A one-pass verifier hardwires that candidate, reads each row once, evaluates every row contribution at the candidate’s creation-time scalar arguments, and checks at the end that every exact sum, rounding and branch reproduces the candidate. Induction over the scalar instructions proves that the verifier accepts exactly when the program’s actual transcript equals that candidate.

If the transcript has at most B bits, apply the finite-space generator proved in (WB.1)–(WB.4), with error $\varepsilon 2^{-B-1}$ for each candidate. Summing over at most 2^B candidates gives

$$\|\text{Law}(T_{\text{seed}}) - \text{Law}(T_{\text{iid}})\|_{\text{TV}} \leq \varepsilon. \quad (198)$$

The algorithm does not enumerate these candidates; they occur only in this probability proof. The recursive affine-Toeplitz construction and its full error estimate are proved in the finite backend. A depth-first traversal emits the same ordered blocks using fewer than twice as many hashes as rows, while retaining only its seed and recursion stack. No cryptographic assumption is used.

For the present program, the transcript and its exact accumulators use at most CR^2w bits, where the local field count and word length satisfy

$$R \leq Cp\beta^{201L}(m+d+2)(1+m/\gamma)Z^{5/2}, \quad w = \lceil C_{\text{word}}\beta^{110L}Z \rceil. \quad (199)$$

Consequently an inner block of CR^2w bits and its streamed short seed suffice. The selected-packet metric identity is deterministic for every finite packet table. Its replay proof needs fresh scalar noise to be independent of that table, not independence among generated packets, so the seed substitution preserves literal acquisition.

E.5.4 Exact unseen queries and simultaneous amplification

At any training time a learned matrix increment has the retained finite rank-list form

$$D = TAS^\top/n.$$

For an unseen input the decoder regenerates rows and streams the actual empirical contraction $S^\top h/n$; it does not replace it by a population expectation. The initialized Gaussian part uses both orientations of every old constraint:

$$\hat{t} = Q_{h_t}(U^\top \hat{g}/n + \eta \hat{e}), \quad \hat{y} = Q_{h_y}(\tilde{m} + U\tilde{T}\hat{t} + c\hat{g}). \quad (200)$$

In particular the $U\tilde{T}\hat{t}$ correction is not dropped. The coefficients come from the acquired finite Gram table with its stated noise floors. A constant number of row streams per layer computes every new contraction and innovation. Previous query fields are reevaluated at their already fixed scalar arguments, so scalar training is not replayed.

Choose one common precision schedule for the source and the appended query. The finite coupling and the physical forward recurrence then put one complete member, at every fixed input/time code, within

$$B_n + A_{\text{num}} Y n^{-10}$$

of the deterministic dense proof center with bad probability at most $1/16$. This probability already includes failure of the member's entire physical source; it is not conditioned on an unrecognized good-source event.

Let N_{ext} be the finite input/time code count. The physical grid gives

$$\log N_{\text{ext}} \leq C(d+1)Z.$$

Use the smallest odd integer J at least $\lceil \log_2(16N_{\text{ext}}/\delta) \rceil$; then $J \leq C(d+1)Z$. For independent members the probability of a bad median is at most $2^J(1/16)^{J/2} = 2^{-J}$. An outer copy of the same finite block generator fools the bad-median test to error $\delta/(16N_{\text{ext}})$. A union over the codes, followed by the physical space/time modulus and the frozen fitted tail, gives

$$\|f_{\text{Seed},n} - f_n^{\text{ind}}\|_* \leq 2B_n + A_{\text{num}} Y n^{-10}.$$

The explicit mesh term satisfies $B_n \geq 32Y/n$. Hence the numerical gate $n \geq (A_{\text{num}}/32)^{1/9}$ absorbs the last term and proves (117). This also proves the statement for a query chosen adaptively after seeing the retained model, because the final event is simultaneous over the complete external code set before the query is chosen.

E.5.5 Word storage and operation counts

The exact modular count uses

$$J \leq C(d+1)Z, \quad a_{\text{blk}} \asymp R^2 w, \quad b_{\text{mem}} \leq C\{R^2 w \log(en) + \log(N_{\text{ext}}/\delta)\},$$

where a_{blk} is the inner generator block length and b_{mem} is the random input length of one complete member. These are local cost variables, not model orders. The retained and live training/query memory is

$$C \left\{ JR^2 + \left\lceil \frac{b_{\text{mem}}}{w} \right\rceil \log(J+2) \right\} \quad (201)$$

words. Initialization processes members sequentially and adds only $C(nR + R^3)$ temporary words.

Packed Toeplitz hashing computes a t -bit affine hash in $Ct\lceil t/w \rceil$ word operations. Depth-first generator traversal, cached exact metric products and indexed exact spectral pivots therefore give

$$\begin{aligned} W_{\text{init}} &\leq CJ\{nR^5 w + (nR + R^2)w^2 + R^4 w \log(R+2) + R^3 w^2 + R^4 + nR^2 + na_{\text{blk}}\lceil a_{\text{blk}}/w \rceil\} + W_{\text{outer}}, \\ W_{\text{train}} &\leq CJ\{R^4 w \log(R+2) + R^3 w^2 + R^4\}, \\ W_{\text{query}} &\leq CJ(L+1)\{n\lceil a_{\text{blk}}\rceil\lceil a_{\text{blk}}/w \rceil + R w^2 + R^2\} + R^3 w \log(R+2) + R^2 w^2 + R^3 \\ &\quad + W_{\text{outer}} + CJ \log(J+2), \end{aligned} \quad (202)$$

with $W_{\text{outer}} = C J b_{\text{mem}} \lceil b_{\text{mem}}/w \rceil$. Every generator stack, exact accumulator, median value and temporary member block is present in these formulas.

Substitute (199) and $\log N_{\text{ext}} \leq C(d+1)Z$. The two leading memory terms become JR^2 and $R^2 \log(en) \log(J+2)$, which give exactly (118). The leading initialization term is $JnR^5 w$; the leading complete-training term is $JR^4 w \log(R+2)$; and the two leading query terms are the streamed inner and outer hashes. Their substitutions are precisely the three rows of (120). This proves six powers of Z , with one outer logarithm, in numerical words. Multiplying by the word length adds one power of Z in bits. It does not turn the bit bound into a sixth-power claim.

Finally, for fixed problem parameters $B_n = n^{-1/2+o(1)}$. Thus the smallest admissible reference width satisfying $3B_n \leq \varepsilon$ is $n = \varepsilon^{-2+o(1)}$. Substitution in the word count proves the inverse logarithmic-storage statement. There is no order q to optimize, and no conclusion that the decoder error is negligible relative to the realized dense-versus-dense discrepancy.

E.5.6 Finite generators and numerical backends

All parameters in this section are local algorithmic resources. In particular, the field count below is not a new compression order.

A. An explicit finite-space generator, including its error proof

An a -bit affine Toeplitz hash is $h(x) = Tx + b$ over the field with two elements. Its seed consists of the $2a - 1$ diagonals of T and the a bits of b , independently uniform. For distinct x, y , the pair $h(x), h(y)$ is uniform on two independent a -bit strings. Indeed, for a nonzero vector $x - y$, let j be its first nonzero coordinate. In the successive coordinates of $T(x - y)$, the diagonal indexed by $i - j$ occurs with coefficient one; all other participating diagonals have smaller index. Prescribing the remaining diagonals and solving in increasing i proves surjectivity. The independent offset then proves the assertion about the pair.

For independent hashes, define recursively

$$G_0(x) = x, \quad G_k(x) = G_{k-1}(x) \parallel G_{k-1}(h_k(x)),$$

where the two children share the same lower-level hashes. This is the affine-hash recursive construction associated with Nisan's finite-space generator. The following direct expectation argument proves the precise version needed here, without invoking that paper's probability theorem.

Consider a deterministic machine with at most $s = 2^u$ states between blocks. Its block counter is included in these states, so the transition rule may depend on the position. The machine reads N blocks; pad its execution to 2^k , where $k = \lceil \log_2 N \rceil$, by absorbing states. It may perform arbitrary computation within a block. If

$$a \geq \max\{\text{raw block length}, 4u + 2k + 2\lceil \log_2(1/\varepsilon) \rceil + 4\}, \quad (\text{WB.1})$$

then the acceptance probabilities under G_k and independent uniform blocks differ by at most ε . The seed has exactly $a + k(3a - 1)$ bits.

Here is a proof. Write μ for uniform measure on a -bit strings. Pairwise independence gives, for arbitrary subsets A, B ,

$$\mathbb{E}_h [\mu\{x \in A : h(x) \in B\} - \mu(A)\mu(B)]^2 = 2^{-a} \mu(A)\mu(B)(1 - \mu(B)). \quad (\text{WB.2})$$

Fix all lower-level hashes. Let A_{ij} be the seeds that send machine state i to state j through the lower-level generator. For each i , these sets partition the seed space. Let M_{k-1} be its transition matrix averaged over its root seed. Its square has entries $\sum_l \mu(A_{il})\mu(A_{lj})$. The transition matrix M_k instead has entries $\sum_l \mu\{x \in A_{il} : h_k(x) \in A_{lj}\}$. By (WB.2), Cauchy-Schwarz, and the partition identities,

$$\begin{aligned} \mathbb{E}_{h_k} \|M_k - M_{k-1}^2\|_\infty &\leq 2^{-a/2} \sum_{i,l,j} \sqrt{\mu(A_{il})\mu(A_{lj})} \\ &\leq s^2 2^{-a/2}. \end{aligned} \quad (\text{WB.3})$$

Here $\|M\|_\infty = \max_i \sum_j |M_{ij}|$. If N_{k-1} is the transition matrix under independent blocks, then $N_k = N_{k-1}^2$. Stochastic matrices have this norm one, so

$$\|M^2 - N^2\|_\infty \leq 2\|M - N\|_\infty.$$

Consequently

$$\mathbb{E} \|M_k - N_k\|_\infty \leq (2^k - 1) 2^{2u-a/2} \leq \varepsilon/4. \quad (\text{WB.4})$$

An acceptance event is a subset of final states; its probability difference is bounded by this norm. Averaging also over the root seed proves (WB.1). In particular no claim that generated blocks are independent, no cryptographic hypothesis, and no unproved generator lemma is needed.

For a finite scalar transcript of at most B bits, fix both a candidate transcript and all independent scalar-noise marks. The one-pass verifier recomputes its row reductions at the candidate's creation-time arguments and checks its guarded scalar instructions. Induction over those instructions shows that it accepts

exactly the candidate transcript. Apply (WB.1) to each candidate with error $\varepsilon 2^{-B-1}$, and sum the absolute probability differences over at most 2^B candidates. The total variation distance is at most ε . The same bound holds after averaging the independently retained noise marks. The proof uses the candidates only as tests; the algorithm never enumerates them.

For the finite program below, the verifier's state and transcript have at most CR^2w bits, including exact dyadic accumulators, row counters, rounding decisions and guarded branches. Thus (WB.1) permits an inner block of CR^2w bits. A full member, including its independent noise marks, needs at most

$$C\{R^2w \log(en) + \log(N_{\text{ext}}/\delta)\} \quad (\text{WB.5})$$

random bits. The outer application reads one complete member per block. For a fixed external input/time code, its test executes the member, compares its finite answer to the fixed success interval, discards its workspace, and increments a bad-member counter. Thus the counter test needs $CR^2w + \log(J+2)$ between-block bits, not a table of all member answers. Taking per-test error $\delta/(16N_{\text{ext}})$, (WB.1) gives the stated outer seed. The fixed interval may have real endpoints: on the finite answer alphabet the comparison is just a finite Boolean transition table in this probability test, not a real-number oracle in the implemented algorithm.

For independent members with bad probability at most $1/16$, an odd $J \geq \lceil \log_2(16N_{\text{ext}}/\delta) \rceil$ has bad median probability at most $2^J(1/16)^{J/2} = 2^{-J}$. The outer generator, then a union over codes, proves the simultaneous claim. This argument amplifies complete source experiments, not conditionally successful members sharing an unamplified source failure.

B. Exact word model and generator streaming

A word contains w bits. The counted primitives are word reads and writes, comparisons, addition and subtraction, Boolean operations, bounded shifts, a full product returned in two words, and integer quotient/remainder on a constant number of words. Constant-factor wider scalars occupy a constant number of words. Integers with $O(Rw)$ bits occupy $O(R)$ words. There is no matrix-function, Gaussian, activation, unbounded-integer or carryless-multiplication oracle.

For a t -bit Toeplitz hash, each input bit selects a contiguous t -bit window of the packed diagonal string to XOR into the output. Each window word uses at most two stored words and bounded shifts. Including the affine offset, the work is at most $Ct\lceil t/w \rceil$, with $C\lceil t/w \rceil$ scratch words. Traverse the generator depth first, storing pending right-child arguments. The full padded tree has fewer than $2N$ internal nodes. It therefore produces its ordered stream with

$$CNt\lceil t/w \rceil \text{ word operations,} \quad C\lceil t/w \rceil \log(N+2) \text{ words.} \quad (\text{WB.6})$$

This space includes the seed, traversal stack, flags and counters. Every query restarts the same ordered stream a constant number of times per layer; no sparse random-access regeneration is assumed. An outer stream is suspended, with its stack retained, while its current member is used.

C. Exact metric selection, including large integers

The finite field table is an n -by- s dyadic matrix V , $s \leq CR$, with entries of $O(w)$ bits. Include its constant field. Let its exact rank be $r \leq \min(n, s)$. Select independent columns and an invertible r -row square block H of these columns. Put

$$C_V = V_{:,J}H^{-1}, \quad M = C_V^\top C_V/n. \quad (\text{WB.7})$$

Then $V = C_V V_{I,:}$, and every required empirical field product is exactly a product of selected values with M . The selected rows of C_V form the identity, whence $M \succeq I/n$. Its constant-field identity is $\mathbf{1}^\top M \mathbf{1} = 1$.

If some coefficient has absolute value greater than two, replace the corresponding row of H by that row. Multilinearity shows that the absolute determinant more than doubles. With a common $O(w)$ -bit dyadic denominator, a nonzero determinant is at least 2^{-Crw} ; Hadamard's bound is $r^{r/2}2^{Crw}$. Because $w \geq \log_2(r+2)$, there are at most Crw swaps. At termination every coefficient is at most two in absolute value. This bound concerns selected row values; it does not assume a lower singular-value gap in the original data.

For completeness, exact rank and inverse arithmetic can be performed by fraction-free elimination. For a pivot a_{kk} , update

$$a_{ij} \leftarrow \frac{a_{kk}a_{ij} - a_{ik}a_{kj}}{a_{k-1,k-1}}, \quad i, j > k, \quad (\text{WB.8})$$

with denominator one at the first step and row/column pivoting when necessary. The two-by-two minor identity proves by induction that the division is exact and the intermediate entries are signed minors of the original integer matrix. Applying the same elimination to augmented right sides yields the adjugate and determinant. All these integers have $O(rw)$ bits, even when the determinant is small.

Schoolbook arithmetic on k words takes $O(k^2)$ word operations. For division, use radix $2^{\lfloor w/4 \rfloor}$ and normalize the leading divisor digit to at least half the radix. The two-leading-digit quotient estimate is too large by at most two: the omitted divisor tail changes its trial product by less than twice the normalized divisor. Correcting and subtracting therefore takes $O(k)$ work per quotient digit and $O(k^2)$ overall. These steps use only the declared constant-word products and divisions.

A rank test or adjugate consequently costs $O(r^5)$ word work. A coefficient scan costs $O(nr^4)$; $r \leq n$ absorbs one adjugate per scan. Including the initial exact rank scan, at most Crw swaps, and the final metric scan gives

$$W_{\text{metric}} \leq CnR^5w, \quad M_{\text{metric}} \leq C(nR + R^3) \text{ words.} \quad (\text{WB.9})$$

The algorithm stores the original $O(w)$ -bit dyadic table V , not the expanded rational table C_V . Each row of C_V is computed, used for the swap test or exact metric accumulator, and discarded. Only the inverse/adjugate and the r -square large-integer accumulators use $O(R^3)$ words. This is essential to the stated peak memory. For a final metric rounded entrywise to error at most ϵ_M , add $r\epsilon_M I$. The spectral norm of the rounding perturbation is at most $r\epsilon_M$, so this is positive semidefinite, and any two selected fields bounded by B have pairing error at most $2r^2B^2\epsilon_M$. The adjacent replay proof chooses this tolerance before setup and charges its noise-boundary probability.

D. Finite spectral routines without eigenvalue separation

Let a symmetric r -square dyadic matrix have norm at most $M \geq 1$ and p -bit entries. Suppose the requested absolute error is 2^{-b} . For inverse or a gapped square root let the stated positive gap be $a \in (0, 1]$. Set locally

$$v = 2 + p + b + \lceil \log_2(r + 2) \rceil + \lceil \log_2(M + 2) \rceil + \lceil \log_2(1/a) \rceil. \quad (\text{WB.10})$$

For positive part, omit the last term. A sufficiently large universal multiple of v working bits suffices. This is already included in the integrated word length; it is not R times that word length.

Here are a terminating algorithm and its error analysis. Use a largest off-diagonal Jacobi pivot with a fixed lexicographic tie rule. If e is the Frobenius off-diagonal norm, a pivot has square at least $e^2/[r(r-1)]$. An exact rotation removes twice that square from e^2 . For a target diagonalization residual $t \leq 1$, stop at $e \leq t/16$, or perform at most

$$K = \left\lceil 8r(r-1) \log \frac{64r(M+1)}{t} \right\rceil \quad (\text{WB.11})$$

rotations. With local rounding errors at most $t/[2^{16}Kr^3(M+1)^2]$, the rounded recurrence satisfies

$$e_{j+1} \leq \sqrt{1 - 2/[r(r-1)]} e_j + t/[2^{10}Kr(M+1)].$$

Its geometric sum, and the product telescoping estimates for the rotations and updated basis, bound both the final off-diagonal norm and the accumulated similarity error by $t/8$, and bound the orthogonality defect by $t/[8(M+1)]$, after reducing local errors by another fixed numerical factor if needed. Thus the iteration cap always reaches the prescribed residual scale. Every local tolerance has logarithm $O(v)$, since $K = O(r^2v)$.

The two-dimensional rotation is computed from its two diagonal entries and pivot. The stable quadratic formula chooses a tangent of magnitude at most one, followed by a positive scalar square root for its cosine. Before stopping the pivot has magnitude at least $t/[16r]$; hence these divisions require only $O(v)$ guard bits,

not the inverse of an eigenvalue separation. Scalar square roots use binary search on an integer square, with $O(w)$ constant-word operations.

For the computed basis V_0 , its exact polar correction $O = V_0(V_0^\top V_0)^{-1/2}$ exists when the orthogonality defect is less than $1/2$, and satisfies $\|O - V_0\|_F \leq 2\|V_0^\top V_0 - I\|_F$. This follows by scalar diagonalization of $V_0^\top V_0$ on $[1/2, 3/2]$. It is a proof device, not a further implemented matrix operation. The preceding bounds therefore give a nearby exactly orthogonally diagonalized matrix within t of the input.

Choose $t \leq 2^{-b-8}a^2$ for inversion, and $t \leq 2^{-b-8}\min(a, \sqrt{a})$ for a gapped square root. The identity $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$ bounds the first error. For square roots, their difference solves a Sylvester equation; the inverse is the integral of left and right multiplication by $e^{-sA^{1/2}}$ and $e^{-sB^{1/2}}$, bounded by $1/(2\sqrt{a})$ in Frobenius norm when both gaps are at least a . Using the actual half-gap in this estimate covers the nearby matrix. For positive part take $t \leq 2^{-b-8}$. Positive part is the Euclidean projection onto the positive-semidefinite cone; the two projection variational inequalities, added together, prove it is Frobenius nonexpansive. This proof uses no spectral gap at zero.

Evaluate the scalar functions on the approximate diagonal, clamping small negative entries where a positive part is requested. Round all nonnegative diagonal function values to nonnegative dyadics and form $V_0 \text{diag}(f_i)V_0^\top$ by exact dyadic products and sums. This preserves positive semidefiniteness. The final scalars have only a constant multiple of w bits, including the sum over r . No unsafe final entrywise rounding is used.

Store pivots in an indexed max-heap. One rotation changes only $O(r)$ matrix entries, so heap maintenance costs $O(r \log(r+2))$. The exact dyadic squared off-diagonal norm is updated by subtracting old squares and adding new squares, also in $O(r)$ operations. Updating the basis costs $O(r)$; scalar roots cost $O(w)$. With $O(r^2w)$ rotations, the whole macro costs

$$C\{r^3w \log(r+2) + r^2w^2\} \text{ word operations,} \quad Cr^2 \text{ words.} \quad (\text{WB.12})$$

For $r = 1$, use just the scalar routine. Exact heap ordering and exact residual accumulation give the same pivots and stop as a full-scan finite algorithm. No hidden r^2 scan occurs per rotation.

The decoder's coefficient solve also avoids a Kronecker-sized system. For its two Gram matrices $Q, K \succeq 0$, diagonalize them separately. In these bases a Sylvester equation

$$(\sigma^2 I + K)E + EQ = F$$

is solved entrywise by division by $\sigma^2 + k_j + q_i \geq \sigma^2$. Scalar divided differences of square roots are evaluated as $(\sqrt{a} - \sqrt{b})/(a - b) = 1/(\sqrt{a} + \sqrt{b})$, including $a = b$. All denominator floors are the already fixed noise floors, so the normwise perturbation bounds in the finite construction apply to nearby matrices without attempting to match individual eigenvectors. This uses a constant number of (WB.12) macros and ordinary $O(r^3)$ scalar arithmetic per call.

E. Finite Gaussian and activation values

Each Gaussian coordinate uses a dyadic uniform midpoint and inverse normal CDF on $[-T, T]$, with $T^2 \leq Cw$. Choose its dyadic uniform precision so that

$$\sqrt{2\pi} e^{T^2/2} 2^{-b_U} \leq \epsilon_G/4. \quad (\text{WB.13})$$

Clamp the midpoint to the CDF image of $[-T, T]$ before numerical inversion; mass moved by this clamp is charged to the displayed tail event. Couple the midpoint to a uniform variable in its cell. Except on the two Gaussian tails, of probability at most $2e^{-T^2/2}$, the inverse-CDF Lipschitz bound gives coordinate error at most $\epsilon_G/4$, before numerical evaluation. The cutoff and failure allocation used in the finite construction make $b_U = O(w)$.

There is a finite implementation of that evaluation in Cw^2 word operations and $O(1)$ scalar scratch. Indeed, integrate the Taylor series of $e^{-x^2/2}$ term by term on the cutoff interval. Since $T^2 = O(w)$, Cw terms have a remainder below $2^{-C'w}$, with suitable fixed constants by the factorial bound $j! \geq (j/e)^j$. Intermediate terms have magnitude at most e^{Cw} , so Cw guard bits suffice. Its recurrence and summation use $O(w)$ word operations. Bisection of the coordinate takes $O(w)$ CDF evaluations. Use enclosing rational intervals throughout: if a comparison overlaps, the positive density lower bound converts the overlap into a coordinate interval. Thus no exact comparison of transcendental numbers is assumed. The normalizing constant can be

computed to this precision from $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, whose identity follows by the tangent addition formula and the angle ranges. The alternating series take $O(w)$ terms; integer square root supplies the remaining normalization. Its cost is smaller than the CDF bisection.

For $\tanh$ there is also an $O(w)$ -word-work value evaluator, using only $O(1)$ scalar scratch. By oddness take $x \geq 0$. For absolute error 2^{-p} , return one if $x \geq p + 4$; its error is at most $2e^{-2x}$. Otherwise scale $z = 2x/2^k \leq 1$, with $2^k \leq 4(p + 4)$. The exponential series for e^{-z} , truncated after $16(p + 1)$ terms, has error far below 2^{-p} . Use $p + C \log_2(p + 2)$ working bits, clip to $[0, 1]$, and square k times. On $[0, 1]$, each squaring amplifies error by at most two, so the total amplification is at most 2^k . The guard bits cover it and the final rational map $(1 - e^{-2x})/(1 + e^{-2x})$, whose denominator is at least one. All operations are finite.

For a different analytic activation its supplied finite evaluator's actual work and workspace are added at every recorded value call. Analyticity bounds approximation errors but does not give a finite description or complexity bound for evaluating an arbitrary function. This is an explicit computational interface, not a probabilistic source assumption, and is necessary for the original activation scope.

F. Complete modular ledger and substitution

There are at most CR spectral calls of size at most CR . Every selected field is created at immutable scalar arguments. Cache both its selected value vector and its exact product with the fixed metric; form all pairings by exact dyadic sums before their prescribed rounding. There are $O(R^2)$ cached words and $O(R^3)$ ordinary scalar work. The finite construction counts the same fields and calls for an appended query; old fields are evaluated at their creation-time arguments, not retrained.

Write locally $a_{\text{blk}} \asymp R^2 w$ and let b_{mem} be (WB.5). For J complete ensemble members, retained and live training/query storage is

$$C [JR^2 + \lceil b_{\text{mem}}/w \rceil \log(J + 2)] \text{ words.} \quad (\text{WB.14})$$

Setup treats members sequentially, so its additional peak is $C(nR + R^3)$, not J times this temporary inventory. Including all finite Gaussian coordinates, metric selection, coefficient preparation, source-row streams and both generator levels gives

$$\begin{aligned} W_{\text{outer}} &= C J b_{\text{mem}} \lceil b_{\text{mem}}/w \rceil, \\ W_{\text{init}} &\leq C J \{ n R^5 w + (nR + R^2) w^2 + R^4 w \log(R + 2) + R^3 w^2 + R^4 + n R^2 + n a_{\text{blk}} \lceil a_{\text{blk}}/w \rceil \} + W_{\text{outer}}, \\ W_{\text{train}} &\leq C J \{ R^4 w \log(R + 2) + R^3 w^2 + R^4 \}, \\ W_{\text{query}} &\leq C J (L + 1) \{ n \lceil a_{\text{blk}} \rceil \lceil a_{\text{blk}}/w \rceil + R w^2 + R^2 \} + R^3 w \log(R + 2) + R^2 w^2 + R^3 \\ &\quad + W_{\text{outer}} + C J \log(J + 2). \end{aligned} \quad (\text{WB.15})$$

The last term includes an ordinary sorting computation of the median. All exact accumulators, temporary member blocks, suspended generator stacks and selected-field caches are counted. Input/certificate access and the supplied activation evaluator are added as stated in the detailed-statement appendix.

Finally insert the already proved bounds

$$R \leq C p \beta^{201L} (m + d + 2) (1 + m/\gamma) Z^{5/2}, \quad w \leq C \beta^{110L} Z, \quad J \leq C (d + 1) Z. \quad (\text{WB.16})$$

The leading memory terms are JR^2 and $R^2 \log(en) \log(J + 2)$. They are bounded by

$$C p^2 \beta^{402L} (m + d + 2)^2 (1 + m/\gamma)^2 Z^6 [d + 1 + \log(e + (d + 1)Z)]. \quad (\text{WB.17})$$

The leading setup term $JnR^5 w$ gives the exponents $1115L$ and $29/2$. The leading training term $JR^4 w \log(R + 2)$ gives $914L$ and 12 . Inner and outer query hashing give, respectively, $(L + 1)nZ^{12}$ and Z^{14} with the same $p^4 \beta^{914L} (m + d + 2)^4 (1 + m/\gamma)^4 (d + 1)$ coefficient. These are the supplementary Seeded decoder's resource bounds. Word length is separate: multiplying (WB.17) by w adds one logarithmic power to the retained bit count.

When $0 < nY < 1$, every appearance of Z in this construction and ledger is replaced by $Z + \log_+(1/(nY))$, together with the stated finite Gaussian-RMS gate. This includes field counts and external codes, not just word length. Zero labels use the exact zero branch.

E.6 Actual dense variability: the decoder comparison

The conclusion is

$$\begin{aligned} \Pr\{\|f_{\text{Seed},n} - f_n^{\text{ind}}\|_* \leq 2b_n(\delta/32) + A_{\text{num}}Yn^{-10}\} &\geq 1 - \delta, \\ b_n(\alpha) = \inf\{b \geq 0 : \Pr(\|f_n - \tilde{f}_n\|_* \leq b) &\geq 1 - \alpha\}, \quad 0 < \alpha < 1. \end{aligned} \quad (\text{VD.1})$$

Here the two dense trajectories in the definition are independent, ordinary width- n dense runs. Thus $b_n(\delta/32)$ is an actual variability quantile, with confidence fixed as width changes. It is not an analytic upper certificate. Neither this quantile nor the deterministic center used in the proof is an input to the implemented decoder.

The additive numerical term is necessary in the full admissible class. The analytic envelope satisfies $B_n \geq 32Y/n$, but this does not imply the same lower bound for the actual quantile $b_n(\delta/32)$. A verified example below has positive labels and zero actual dense variability.

E.6.1 Setup and precise statement

Use the original dense architecture and normalization. For $v = x/\sqrt{d} \in \mathbb{S}^{d-1}$, its forward pass is

$$z^{(1)} = Av, \quad z^{(j)} = W^{(j)}h^{(j-1)}, \quad h^{(j)} = \phi_j(z^{(j)}), \quad f_n = w^\top h^{(L)}/n.$$

The initial entries of A are independent $N(0, 1)$, those of each hidden mixer are independent $N(0, 1/n)$, all blocks are independent, and $w(0) = 0$. Gradient flow minimizes $m^{-1} \sum_a (f_n(x_a) - y_a)^2$ with mobilities $(n, 1, \dots, 1, n)$. Set $Y = \|y\|_2/\sqrt{m}$, $\lambda = \gamma/m$, where $\gamma > 0$ is the minimum eigenvalue of the last-layer population uncentered feature second-moment matrix.

Retain all original decoder assumptions: analytic activations with the stated strip and derivative bounds, $L \geq 2$, normalized inputs, $m \geq d$, spanning training inputs, and the entire original common label interval. In particular no new label cap, covariance nonsingularity, or actual-variability lower bound is assumed. Fix $0 < \delta < 1/4$. Keep the existing decoder orders, complete explicit enclosing width gate (or its stated factorized alternative with all physical and implementation gates), finite-word interfaces, and numerical error allocation. For the clean branch use $nY \geq 1$. The small-label extension is recorded below.

The norm remains

$$\|f - g\|_* = \sup_{t \in [0, \infty]} \sup_{\|x\|_2 = \sqrt{d}} |f(t, x) - g(t, x)|,$$

including the fitted endpoint. To define a quantile before restricting to a success event, assign discrepancy $+\infty$ if either trajectory lacks a complete trajectory with the uniform fitted limit required by this norm. For every pair where these objects exist, use its actual discrepancy, including pairs outside the proof's source-good event. This convention does not condition or truncate the dense law and changes no success claim. On successful trajectories, continuity and the uniform tail make the supremum measurable by a countable dense set of finite times and inputs.

Under these assumptions (VD.1) holds for the existing decoder. In fact the proof below bounds its failure by $3\delta/16$, leaving unused budget for a joint statement. The decoder is independent of the comparison dense run. Its event covers all times, all sphere inputs, the endpoint, and inputs selected after inspecting the model or earlier answers.

E.6.2 A regular deterministic center exists at the actual quantile

Write $\alpha = \delta/32$, and let F, F' denote independent dense trajectories. The finite training-source theorem with reference failure $\rho_{\text{ref}} = 2^{-20}\delta$ supplies a regularity event E_{ref} of probability at least $1 - \rho_{\text{ref}}$. Its physical consequences include global fitting, the real operator and all-sphere feature bounds, and the uniform endpoint tail. The usual readout and feature estimates give

$$\sup_{t,x} |F(t, x)| \leq (2Y/\sqrt{\lambda})(2H_D) = 4YH_D/\sqrt{\lambda} \quad \text{on } E_{\text{ref}},$$

where H_D is the existing dense population feature RMS envelope. Consequently

$$\Pr\{\|F - F'\|_* \leq 8YH_D/\sqrt{\lambda}\} \geq 1 - 2\rho_{\text{ref}} > 1 - \alpha.$$

The quantile $b = b_n(\delta/32)$ is therefore finite without using any dense concentration rate. By continuity of probability along decreasing events, its definition implies $\Pr\{\|F - F'\|_* > b\} \leq \alpha$, including when $b = 0$.

For each deterministic dense trajectory f , define the ball failure

$$a(f) = \Pr\{\|F - f\|_* > b\}.$$

Fubini's theorem gives $\mathbb{E}[a(F')] = \Pr\{\|F - F'\|_* > b\} \leq \alpha$. Markov's inequality yields $\Pr\{a(F') > 2\alpha\} \leq 1/2$. Since $\Pr(E_{\text{ref}}) > 1/2$, there exists a realization $F' = f_c$ in E_{ref} with

$$\Pr\{\|F - f_c\|_* > b\} \leq 2\alpha = \delta/16 < 1/64. \quad (\text{VD.2})$$

Fix this realization for the proof. The center f_c inherits the physical input and normalized-time moduli (FC.57)–(FC.58) and the fitting tail. These are deterministic bounds with precisely the coefficients already used in the decoder grid. Choosing a regular center is essential: an arbitrary center obtained from averaging would not automatically have these moduli. No algorithm locates, stores, or evaluates f_c .

E.6.3 The finite law certificate transfers each fixed-code test

Fix an external input/time code c . It specifies an input on the sphere and a time in an acquired panel, or the frozen-tail time. The existing finite source law has the following sufficient interface:

1. Run the source only through the code's acquired training prefix and append the reserved passive-query calls. At this fixed code the iid finite program can be coupled to one ordinary Gaussian dense initialization. On its physical source, sampler, coupling, and raw-noise events, its scalar prediction differs from that dense prediction by at most e_{loc} , an allocated multiple of Yn^{-10} .
2. The source-plus-query scalar transcript has $O(R^2w)$ bits, and its one-pass candidate verifier has $O(R^2w)$ between-row bits. With inner error $\varepsilon_{\text{in}} = 1/64$, its law under the short seed differs from the iid transcript law by at most that total variation. In particular this holds for any fixed success interval of its final scalar answer.
3. Literal selected-metric replay fails with probability at most 2^{-12} , also after replacing iid packets by the short seed. It only requires the separate fresh scalar marks to remain independent of the packet array. It does not require independence among packets.

These interfaces are proved in FC Sections 3–7 and its final transcript inventory, and WB Section A. Their derivations matter here. The complete Gaussian matrix call has the joint shadow augmentation (FC.39)–(FC.41); conditioning only at complete-call boundaries preserves the two-orientation posterior. Old and new moments use the same pre-setup precision. The unseen-query contractions (FC.54)–(FC.55) are exact empirical reductions, and the complete covariance correction is retained. Thus no population replacement bias enters e_{loc} .

The private metric is excluded from the Gaussian posterior filtration. One first transfers the virtual source transcript and then uses literal replay to identify the compact prefix. Revealing the completed source's future answers before the passive call, or asking for total variation of the metric together with an entire dense path, would not be justified by the certificate and is unnecessary here.

By (VD.2), the coupled physical dense answer is within b of $f_c(c)$, except with probability at most $\delta/16$. This follows from a ball event for that coupled dense marginal; it does not require the same coupling for two different codes. Apply transcript total variation to the fixed interval

$$I_c = [f_c(c) - b - e_{\text{loc}}, f_c(c) + b + e_{\text{loc}}].$$

The probability that one seeded, selected, complete member returns a value outside I_c is at most

$$\frac{\delta}{16} + \frac{1}{64} + \underbrace{(2^{-20} + 2^{-20} + 2^{-26} + 2^{-29} + 2^{-12})}_{< 1/1024} < \frac{1}{16}. \quad (\text{VD.3})$$

The five bracketed allocations are source failure, raw-noise RMS failure, chronological Gaussian coupling failure, finite sampler failure, and metric replay failure. All are unconditional complete-experiment failures. No common unamplified source event is imposed across members.

The real endpoints of I_c cause no computational addition. The final scalar answer has a finite alphabet; membership in I_c is a fixed Boolean transition table in the proof test. WB Section A permits arbitrary deterministic within-block computation. The implemented decoder only computes the median and never tests membership in I_c .

E.6.4 Complete-member amplification and all-query transfer

Let N_{ext} be the finite external code count, with $\log N_{\text{ext}} \leq C(d+1)Z$ as in (FC.60). Retain the existing smallest odd ensemble size

$$J \geq \lceil \log_2(16N_{\text{ext}}/\delta) \rceil.$$

For independent complete members, if their median lies outside I_c , at least $(J+1)/2$ members lie outside I_c . Taking a union over such subsets and using (VD.3) gives the conservative bound

$$\Pr\{\text{median}_j G_j(c) \notin I_c\} \leq 2^J (1/16)^{J/2} = 2^{-J} \leq \delta/(16N_{\text{ext}}).$$

The outer generator fools this fixed-code bad-member-counter test with error at most $\delta/(16N_{\text{ext}})$. It needs only the already counted member workspace and counter between member blocks; no list of real center values is supplied to the actual model. A union over codes proves, with failure at most $\delta/8$,

$$|f_{\text{Seed},n}(c) - f_c(c)| \leq b + e_{\text{loc}} \quad \text{for every external code } c. \quad (\text{VD.4})$$

The decoder answers an arbitrary query through one of the existing qualifying codes. The physical regularity of f_c , the space/time mesh, and the frozen tail give

$$|f_c(t, x) - f_c(c)| \leq e_{\text{mesh}} + e_{\text{tail}}.$$

This holds also for $t = \infty$, and for either permitted acquired panel at a boundary. By the pre-existing numerical allocations, $e_{\text{loc}} + e_{\text{mesh}} + e_{\text{tail}} \leq A_{\text{num}} Y n^{-10}$. Therefore (VD.4) implies

$$\|f_{\text{Seed},n} - f_c\|_* \leq b + A_{\text{num}} Y n^{-10}.$$

An independent dense reference satisfies $\|f_n^{\text{ind}} - f_c\|_* \leq b$ except with probability $\delta/16$, by (VD.2). The triangle inequality and union bound give (VD.1), with total failure at most $\delta/8 + \delta/16 = 3\delta/16 < \delta$. No full-process coupling of a finite member to a dense path, and no total-variation statement for an infinite family of queries, has been used. Adaptively chosen queries are covered because (VD.4) is simultaneous before any query is selected.

E.6.5 Resources, confidence, and boundary cases

All implemented orders remain those of the existing construction:

$$R \leq Cp\beta^{201L}(m+d+2)(1+m/\gamma)Z^{5/2}, \quad w \leq C\beta^{110L}Z, \quad J \leq C(d+1)Z.$$

Nothing in these choices depends on $b_n(\delta/32)$ or on the chosen center. Thus the retained/live word count remains

$$Cp^2\beta^{402L}(m+d+2)^2(1+m/\gamma)^2Z^6[d+1+\log(e+(d+1)Z)],$$

the initialization work retains its $1115L$ activation power and $Z^{29/2}$ factor, the training work retains $914L$ and Z^{12} times its displayed logarithm, and query work retains $914L$ with $(L+1)nZ^{12} + Z^{14}$. Every interface cost and setup peak inventory remains charged. The explicit decoder width gates suffice; their numerical absorption gate is retained as a sufficient condition, although the additive intrinsic comparison does not need it for absorption.

For $0 < nY < 1$, use the existing enlarged orders, code counts, and Gaussian-RMS gate (FC.61)–(FC.62), replacing Z everywhere by $Z + \log(1/(nY))$. The same argument and error formula apply. For $Y = 0$, all dense and compressed predictions are identically zero, so the quantile and the error are exactly zero.

If a benchmark instead uses one universal failure level $0 < \alpha_0 \leq 1/128$, independent of both n and the requested δ , the same center argument gives a dense-ball failure at most $2\alpha_0$. Assuming that quantile is finite, the unchanged amplification gives success at least $1 - 2\alpha_0 - \delta/8$, not arbitrary $1 - \delta$. A fixed universal quantile cannot support arbitrarily high confidence for an independent target without additional tail information. In (VD.1), the confidence level $\delta/32$ is fixed in width and is explicit.

The nonzero-label zero-variability case is admissible. Take $m = d = 1$, $x_1 = 1$, $L \geq 2$, and $\phi_j(z) \equiv 1$. Then $Q^{(L)} = [1]$, $\gamma = 1$, the input spans, and a sufficiently small nonzero label satisfies the full original allowance. All hidden gradients vanish, while

$$\dot{w}_i = -2(f_n - y), \quad \dot{f}_n = -2(f_n - y), \quad f_n(t) = y(1 - e^{-2t}).$$

Every dense realization is the same nonconstant path, so $b_n(\delta/32) = 0$. The prescribed finite time-code decoder has finitely many scalar output values through its finite horizon and cannot equal this nonconstant continuous function at every time. A pure bound $Cb_n(\delta/32)$ would require exact equality and is therefore false for this decoder over the complete original admissible class.

Whenever a separately justified lower bound gives $b_n(\delta/32) \geq A_{\text{num}}Yn^{-10}$, (VD.1) does imply the pure factor-three bound. The existing actual-trajectory lower theorem for fixed admissible problems with $m \geq 2$ and $Y > 0$ eventually supplies such domination: its $cY\sqrt{\gamma}/(\sqrt{n}\log^{5/2}(en))$ lower scale dominates Yn^{-10} . This consequence inherits that theorem's unquantified eventual width. It must not be substituted into the explicit finite decoder gate or asserted for the $m = 1$ case. Refining numerical precision to an arbitrary unknown quantile would instead introduce a new dependence on its scale and would require fresh resource accounting.

E.7 Sphere-model size derivations and supplementary decoder comparison

This subsection derives the Legendre and Harmonic sizes and the separate seeded-decoder comparison from their complete certificates. It is used by, but is not by itself the proof of, the current unified theorem; its new finite-panel component is assembled in Appendix F.7. Only in this subsection, c, C may depend on fixed activations and depth. The confidence allocations are fixed numerical values and these constants hide no dependence on m, n, d, Y, γ . Assume $m/\gamma \geq 1$, $m \geq 2$, and fixed $Y > 0$ in the common label interval. The main text uses the simpler shared subinterval $Y \leq c\gamma/m$. No label inequality is used to reduce a displayed sample/gap power.

Let $b_n(10^{-4}) = \inf\{b \geq 0 : \Pr(\|f_n - \tilde{f}_n\|_* \leq b) \geq 0.9999\}$, the actual independent-dense discrepancy quantile. The source-good amplitude bound in the preceding proof makes it finite at qualifying widths. The dense lower theorem, used with failure $1/200$, gives

$$a_n = \frac{cY\sqrt{\gamma}}{\sqrt{n}[\log(en)]^{5/2}}, \quad \Pr(\|f_n - \tilde{f}_n\|_* \geq a_n) \geq 199/200. \quad (203)$$

If $b < a_n$, then $\Pr(\|f_n - \tilde{f}_n\|_* \leq b) \leq 1/200$, so such a b cannot attain confidence 0.9999. Consequently $b_n(10^{-4}) \geq a_n$. Applying the dense upper theorem at failure 10^{-4} also gives

$$b_n(10^{-4}) \leq CY(m/\gamma)^5 \sqrt{d/n} \log(en) e^{\sqrt{\log(en)}}. \quad (204)$$

The factor $1 + m/\gamma$ has been replaced by at most $2m/\gamma$. Both conclusions are eventual. These same lower and upper scales simultaneously bound a realized dense pair with probability at least $1 - 1/200 - 10^{-4} = 0.9949$. The upper inequality is not used to certify the decoder's intrinsic accuracy.

For Legendre take the internal integer order

$$q = \left\lceil C(m/\gamma)^3 n^{1/4} [\log(en)]^2 e^{\sqrt{\log(en)}/2} \right\rceil. \quad (205)$$

At fixed structural parameters, $\log(eq) \leq C \log(en)$ eventually. The all-order forward certificate therefore yields, at every fixed confidence $1 - \delta$ for sufficiently large width,

$$\begin{aligned} \|f_{\text{Leg},n,q} - f_n\|_* &\leq CY(m/\gamma)^6 \frac{\sqrt{\log(en)\log(eq)}}{q^2} e^{\sqrt{\log(en)}} \\ &\leq \frac{CY}{\sqrt{n}[\log(en)]^3}. \end{aligned} \quad (206)$$

The ratio of this last bound to a_n is at most $C/(\sqrt{\gamma}\sqrt{\log(en)})$, and hence tends to zero. The exact learned count is $n(d+1) + 1 + 2(L-1)mnq$. Substitution gives the stated $Cm(m/\gamma)^3 n^{5/4} [\log(en)]^2 e^{\sqrt{\log(en)}/2}$ bound, since the nd and rounding terms are absorbed at a sufficiently large width for each fixed problem. This absorption needs no assumption $m \geq d$. The additional fixed inventory is exactly $(L-1)n^2$. The logarithmic margin supplies the main theorem's negligible-relative-error conclusion via the arbitrary-confidence argument in Appendix F.7; no optimality of that margin is asserted.

For Harmonic use the exact inverse in the supplied-budget theorem at absolute target $\varepsilon = Y/n$. The exact minimum/ceiling branches remain those of that theorem; no new branch is introduced here. At sufficiently large width, their sufficient order obeys

$$q \leq C(m+d) + \left(\frac{C}{\sqrt{d}}\right)^{d+1} (Ym/\gamma)^2 [\log(en)]^{3d/2+1}. \quad (207)$$

This follows by inserting $\varepsilon = Y/n$ in the full inverse logarithm, keeping Y fixed, and absorbing fixed structural addends into $\log(en)$ only after enlarging the eventual width. The label-explicit inverse (46) and its factorial-elimination proof give the displayed d coefficient while retaining the factor Y^2 . The exact retained inventory is $13(L+1)q^2 + 10m(d+1)$. Squaring the two budget terms, using $(u+v)^2 \leq 2u^2 + 2v^2$, and absorbing the data inventory gives

$$C(m+d)^2 + \left(\frac{C}{d}\right)^{d+1} (Ym/\gamma)^4 [\log(en)]^{3d+2}. \quad (208)$$

Its certified error is at most Y/n . Its ratio to a_n is at most $C[\log(en)]^{5/2}/\sqrt{\gamma n}$, which tends to zero. The exact full-retention branch remains available at smaller widths; it does not alter the eventual statement.

For Seeded use the existing finite construction at failure $\delta = 1/100$, with its internal orders and width gate unchanged. The fixed-quantile version of the preceding center proof applies with $\alpha_0 = 10^{-4}$: the regular center has dense-ball failure at most $2\alpha_0 = 1/5000$, and the median/code failure is at most $\delta/8 = 1/800$. The single-member failure remains below $1/16$. Hence

$$\Pr \{ \|f_{\text{Seed},n} - f_n^{\text{ind}}\|_* \leq 2b_n(10^{-4}) + A_{\text{num}}Yn^{-10} \} \geq 1 - \frac{1}{5000} - \frac{1}{800} > 0.99. \quad (209)$$

The exact A_{num} is the absolute coefficient of the original numerical allocation. Its ratio to the lower scale obeys

$$\frac{A_{\text{num}}Yn^{-10}}{a_n} = \frac{A_{\text{num}}}{c\sqrt{\gamma}} n^{-19/2} [\log(en)]^{5/2} \longrightarrow 0. \quad (210)$$

It is thus at most $b_n(10^{-4})$ eventually, proving the supplementary Seeded decoder's $3b_n(10^{-4})$ comparison, not the main theorem's negligible-error conclusion. No computed quantile, center values, or new precision rule is required. The extra eventual threshold is only for this pure multiplicative conclusion, not for (209).

It remains to simplify its inventory without hiding structural powers. The explicit enclosing gate and fixed failure probability give $Z \leq C \log(en)$, $p \leq C \log(em)$, and $nY \geq 1$. The method's own $m \geq d$, together with $m/\gamma \geq 1$, gives $m+d+2 \leq 4m$ and $1+m/\gamma \leq 2m/\gamma$. Finally

$$d+1 + \log(e + (d+1)Z) \leq C[d + \log \log(e^e + n)]. \quad (211)$$

Indeed the logarithm splits into a numerical constant, $\log(d+1)$ and $\log(e+Z)$, with $\log(d+1) \leq d$. Inserting these inequalities in (118) yields the stated $Cm^2(m/\gamma)^2 [\log(em)]^2 [\log(en)]^6 [d + \log \log(e^e + n)]$ complete word bound. Word length is at most $C \log(en)$. All ensemble members, seed levels, data, metrics, caches and live query workspace remain counted.

These arguments establish the sphere models' size bounds and the supplementary decoder's fixed-confidence comparison. Legendre and Harmonic attain arbitrary fixed confidence at eventual width; their negligible-error assembly with finite-panel Logarithmic is in Appendix F.7. A specified finite collection uses split failure budgets without changing the displayed orders. No uniform event over infinitely many widths is asserted. For $Y = 0$, the zero predictor is exact; for $m = 1$, use the additive intrinsic theorem rather than the nondegenerate lower argument.

E.8 Derivation of the headline inverses and storage corollaries

All finite-width certificates are the exact formulas in the detailed-statement appendix. This subsection explains the simplifications, without replacing a deterministic coefficient by an unspecified constant.

For dense, use the numerical scalar prescription in the exact inverse and its proof below the concentration argument. The selected width must also exceed the source and, when used, lower-bound onset. For fixed problem and confidence, its logarithm is $2 \log(1/\varepsilon) + o(\log(1/\varepsilon))$. The exact learned count $(L - 1)n^2 + n(d + 1)$ therefore has exponent $4 + o(1)$ in $1/\varepsilon$.

For Legendre, the function $\sqrt{\log(eq)}/q^2$ is decreasing on $[1, \infty)$. The exact simple-cap and full-range inverses in the inverse statement allocate error explicitly and then round upwards. Substitution in $n(d + 1) + 1 + 2(L - 1)m n q$ is the complete finite-width storage certificate. The structural sufficient-order formula there retains $[\log(en) \log(en/\varepsilon)]^{1/4}$; no identification of these two logarithms is needed. At $n = \varepsilon^{-2+o(1)}$, that order is $\varepsilon^{-1/2+o(1)}$, so the moving count is $\varepsilon^{-5/2+o(1)}$. The fixed mixers still have $(L - 1)n^2 = \varepsilon^{-4+o(1)}$ coordinates.

For Harmonic, its exact inverse and exact inventory provide the unsuppressed storage formula

$$13(L + 1) \left[\max \{ 18(2m + d + 9), \lceil 9 [2m + d + 9 + \mathcal{C}_d(h_0 + u_\varepsilon)^{d+1}] \rceil \} \right]^2 + 10m(d + 1) \quad (212)$$

whenever the analytic branch is used. The definitions of $\mathcal{C}_d, h_0, u_\varepsilon$ are exactly those in the supplied-budget statement. For the loose-target branch use its initialization-only budget instead; for the full-width branch use the exact retained dense arrays rather than forcing a selected-rank estimate. Both branches are already explicitly constructed in the Harmonic theorem.

For polynomial accuracy at fixed problem, $h_0 = O(\log(en))$ and $u_\varepsilon = O(\log(en/\varepsilon))$. The numerical rank envelope proved above gives $\mathcal{C}_d$ proportional to $\log(en)^{d/2}$; hence the sufficient Harmonic budget is $O(\log(en)^{3d/2+1})$. The inventory is quadratic in that budget, giving $O(\log(en)^{3d+2})$. The separate m, d, L, γ factors in the headline follow by squaring the exact rank envelope and using

$$d! \geq (d/e)^d, \quad \frac{(d + 3)^d}{(d!)^2} \leq \frac{(4e^2)^d}{d^d} \leq \frac{(8e^2)^d}{d^{d+1}} \quad (d \geq 1). \quad (213)$$

The last step uses $d \leq 2^d$; it does not set the structural parameters equal. Numerical powers are absorbed only in the headline activation-depth exponent. Taking $n = \varepsilon^{-2+o(1)}$ then gives the stated Harmonic $O(\log(1/\varepsilon)^{3d+2})$ storage.

For the dense necessary-storage statement, restrict to widths beyond the lower theorem's onset, choose lower confidence strictly below one half, and use the positive coefficient in the actual-trajectory theorem. If $n \log(en)^5$ is below its displayed necessary bound, the success event at accuracy ε and the strict lower event are disjoint while their probabilities sum to more than one. This gives the scalar necessity. To obtain the storage consequence, split into $n \geq \varepsilon^{-2}$, where the claim is immediate, and $n < \varepsilon^{-2}$, where $\log(en) \leq 1 + 2 \log(1/\varepsilon)$. Thus n^2 is bounded below by a positive fixed-problem multiple of $\varepsilon^{-4}/\log(1/\varepsilon)^{10}$ eventually. This argument applies only inside the independently initialized dense family above that onset; it does not rule out exceptional smaller widths.

Finally fix an arbitrary failure tolerance. Apply the dense lower theorem at half that tolerance and the compression upper theorem at the other half. Harmonic's original $n^{-1+o(1)}$ specialization divided by $n^{-1/2} \log(en)^{-5/2}$ tends to zero. For Legendre use the strict root-width order in (78), multiplied by $\log(en)^{3/2}$ and rounded up. Its squared denominator gains $\log(en)^3$; the ratio of the new to old logarithmic numerator tends to one, since $\log q = (1/4 + o(1)) \log n$. Its ratio to the dense lower bound is therefore bounded by a fixed-problem, fixed-confidence multiple of $\log(en)^{-1/2}$, which tends to zero. The union bound proves a limsup probability at most the arbitrary failure tolerance for every positive ratio threshold. Letting that tolerance tend to zero proves convergence in probability. This step uses neither event independence nor a confidence-uniform lower coefficient, and makes no fitted-endpoint lower claim.

E.9 Proof of the computational-cost bounds

The cost contract and local symbols are those of the detailed statements. This proof counts explicit executions, not all possible algorithms. Matrices use classical multiplication: an a -by- b matrix acting on c vectors costs $O(abc)$ arithmetic. Rectangular factors and nodes can be streamed as specified. All comparisons here concern arithmetic identities of the defined models, not a new numerical approximation theorem.

E.9.1 Dense and Legendre execution

Drawing/filling the P dense coordinates has the stated arithmetic and sampling counts. A forward/backward pass over m inputs and the outer-product updates cost $O(mP)$, plus the listed activation calls. Save $O(Lmn)$ features and signals, along with parameters, directions and data. A query uses two width- n buffers, with $O(P)$ arithmetic; normalization of the first preactivation can follow multiplication by the supplied input.

For Legendre, initialize only the $L-1$ feature layers used as zeroth moments. This takes $mnd + (L-2)mn^2$ matrix arithmetic, $(L-1)mn$ nonlinear evaluations, and the associated scalar operations. Write the other $2(L-1)mnq$ moment entries, including zeros; their allocation/filling cost is not omitted.

Each reconstructed hidden mixer is its fixed dense matrix plus mq outer-product corrections. For a batch of m columns, first take their inner products with one stored feature vector, then accumulate the scaled response vector. Across all factors this costs $O(nm^2q)$ per interface. Streaming those operations does not require an m^2q -word intermediate. The transpose action has the same count. All sums $\sum_{i < j} (2i+1)\bar{h}_{a,i}$ and their response analogues are formed with one running prefix per sample and interface; there are $O(Lnmq)$, not $O(Lnmq^2)$, operations. The first-layer/readout updates and residual calculation fit $O(mP)$. Moment storage dominates Lmn because $q \geq 1, L \geq 2$.

At one query, each correction contributes one scalar inner product times one stored vector. Accumulate directly into the next layer's feature buffer, costing $O(Lnmq)$ work and $O(n)$ extra words. Explicitly summing all outer products to cache reconstructed matrices costs $O((L-1)n^2mq)$ and creates the stated dense cache. It must be refreshed after its defining state changes.

E.9.2 Exact factored initial-jet execution

For this paragraph, $[g]_s = \partial_t^s g(0)/s!$ denotes a normalized Taylor coefficient. Put $u_a^{(j)} = r_a \delta_a^{(j)}$, where the residual and backward response are the actual dense-training ones. Coefficient comparison in the dense ODE gives, for $s \geq 1$,

$$[A]_s = -\frac{2}{ms} \sum_a [u_a^{(1)}]_{s-1} v_a^T, \quad [W^{(j)}]_s = -\frac{2}{mns} \sum_a \sum_{b+c=s-1} [u_a^{(j)}]_b [h_a^{(j-1)}]_c^T. \quad (214)$$

Thus initial matrices and $O(LmnK)$ training coefficients determine all parameter jets without retaining K dense mixers. For a batch $X(t)$ of b query vectors, write U_i, H_j for the n -by- m weighted-response and feature coefficients. The increment's coefficient of degree s is

$$-\frac{2}{mn} \sum_{i+j+k=s-1} \frac{U_i (H_j^T X_k)}{i+j+1}. \quad (215)$$

Directly contracting every triple costs $O(nmbK^3)$. Alternatively, cache $H_j^T X_k$: all these inner products cost $O(nmbK^2)$; their weighted sums for each outer index cost $O(mbK^3)$; the final vector combinations cost $O(nmbK^2)$. Initialized matrix actions cost $O(n^2bK)$. The transpose calculation interchanges the two training factors and has identical counts. Every increment at order s uses lower orders $i, j, k < s$, so this cached calculation is causal during the training recursion. New order- s contractions are saved for later orders.

Use $b = m$ once for the shared training calculation and $b = 1$ for each streamed spatial query. Training contractions occupy $O(Lm^2K^2)$ words, and a query can reuse $O(mK^2)$ scratch layerwise. First-layer coefficients can be formed one at a time and discarded, or applied through the training input inner products; their cost fits the stated envelopes when $n \geq m, d$. Pointwise residual/response convolutions and readout operations fit the same bounds. Saving the initialized-matrix contributions already computed in forward/backward propagation also supplies the paired source images. The materialized alternative stores all parameter coefficients and

performs ordinary two-index convolutions, yielding the $P(m + N_x)K^2$ count. None of these recursions advances physical training time.

For a general activation, composition with a scalar Taylor series can be executed online by a triangular table of power coefficients: there are $O(K^2)$ entries, each updated by at most $O(K)$ terms. This gives the stated K^3 work and K^2 persistent words. With the full query series available, truncated Horner composition uses $O(K)$ scratch and $O(K^3)$ work. Fixed-size differential identities, when available for the chosen activation, instead use $O(K^2)$ convolution work and $O(K)$ words. These statements account for composition only; the backend cost definition separately includes scalar derivative generation.

E.9.3 Compiling the temporal map and accumulating spatial coefficients

For the actual continuation map in the initial-jet proof, write locally

$$\mathbf{t}(\xi) = c[\log(1 + u\xi) - \log(1 - v\xi)], \quad c = 2r_t/\pi, \quad u = \frac{b_1 - \vartheta}{1 - b_0}, \quad v = \frac{b_1 + \vartheta}{1 + b_0}.$$

The constant logarithm cancels because $\mathbf{t}(0) = 0$. The source quantities b_0, b_1, ϑ were defined in the initial-jet proof. This form gives

$$[1 + (u - v)\xi - uv\xi^2](\mathbf{t}^k)' = kc(u + v)\mathbf{t}^{k-1}.$$

Writing $A_{jk} = [\xi^j]\mathbf{t}(\xi)^k$, coefficient comparison yields

$$(j + 1)A_{j+1,k} = kc(u + v)A_{j,k-1} - (u - v)jA_{jk} + uv(j - 1)A_{j-1,k}. \quad (216)$$

Use $A_{00} = 1$, zero other entries in column zero and row zero, and zero negative-index entries. Generate each column from the preceding one, using $O(K)$ scratch and $O(K^2)$ total work.

For temporal quadrature nodes, let ξ_ℓ be their coordinates under the inverse map and let $\omega_{a\ell}$ include quadrature weight, temporal cosine and its normalization. Form

$$E_{aj} = \sum_{\ell=1}^{N_t} \omega_{a\ell} \xi_\ell^j, \quad B_{ak} = \sum_{j=0}^K E_{aj} A_{jk}, \quad 0 \leq a \leq p. \quad (217)$$

Form E in $O(N_t(p + 1)K)$ work. As each column of A is generated, multiply it by E and discard it. This adds $O((p + 1)K^2)$ work and uses $O((p + 1)K)$ words for E, B and the streamed columns. A source at a spatial node then has temporal coefficient $\sum_k B_{ak}[g]_k$. This is exactly the chosen finite initial-jet reconstruction followed by the chosen quadrature, not a replacement approximation.

Apply that transform to the source coordinates at one spatial node, then accumulate the angular coefficients. This proves temporal-first work $O(LnN_x(p + 1)(K + H))$ and $O(LnR)$ coefficient storage. Reversing the two finite sums proves the spatial-first alternative and its $LnKH$ intermediate. The same scalar maps are used for paired initialized-matrix images, so their exact linear identities are preserved. Blocking complete coefficient columns and recomputing nodal jets proves the stated alternative peak memory and pass count.

E.9.4 Geometric basis overhead

The circle basis is $1, \sqrt{2}\cos(j\theta), \sqrt{2}\sin(j\theta)$; angle-addition recurrences evaluate all degrees through ℓ_* in $O(\ell_* + 1)$ arithmetic. In higher dimension split $v = (\cos\theta, \sin\theta u)$, $u \in S^{d-2}$. From a degree- ℓ harmonic on S^{d-2} , use

$$(\sin\theta)^\ell C_{j-\ell}^{\ell+(d-2)/2}(\cos\theta) Y_{\ell,b}(u), \quad 0 \leq \ell \leq j, \quad (218)$$

with its scalar normalization. Substitution into the separated sphere Laplacian shows eigenvalue $-j(j + d - 2)$. Orthogonality in u separates unequal lower modes; for equal lower modes the angular integral is the Gegenbauer inner product with weight $(1 - z^2)^{\ell+(d-3)/2}$. Dimension counting by the harmonic decomposition already proved above makes this a complete basis. The three-term Gegenbauer recurrence, obtained by differentiating its generating function, is

$$(s+1)C_{s+1}^\lambda(z) = 2(s+\lambda)zC_s^\lambda(z) - (s+2\lambda-1)C_{s-1}^\lambda(z).$$

For each intermediate dimension there are $O((\ell_* + 1)^2)$ degree/lower-degree pairs. Their polynomial values and normalization ratios can be generated recursively, without a gamma-function oracle. For the probability-sphere angular measure, denote the squared norm of the angular factor locally by $N_{\ell,s}$, where $\lambda = \ell + (d-2)/2$. The beta integral and the displayed Gegenbauer recurrence give

$$N_{0,0} = 1, \quad \frac{N_{\ell+1,0}}{N_{\ell,0}} = \frac{2\ell+d-1}{2\ell+d}, \quad \frac{N_{\ell,s+1}}{N_{\ell,s}} = \frac{(s+2\lambda)(s+\lambda)}{(s+1)(s+\lambda+1)}.$$

Inverse square roots supply the normalization constants. Combining with the lower-dimensional basis costs at most $O(H)$ additional products per dimension and node. This proves (139), including $O(d(\ell_* + 1)^2)$ reusable scalar data and $O(dN_x)$ node generation. The case $d = 1$ is the two-point sphere. No numerical conditioning of these recurrences is inferred from their arithmetic count.

E.9.5 Source selection and assembly

Process the R generators by rank-aware orthogonalization against at most r current basis columns, costing $O(nRr)$ per layer. Exact arithmetic distinguishes dependence; a numerical rank tolerance would require its own error allowance. Each of the selector's $9r$ barrier iterations can form its shifted r -by- r inverses in $O(r^3)$, then test n row vectors by $O(r^2)$ quadratic forms each. Thus the work is $O(nr^3 + r^4) = O(nr^3)$, since $r \leq n$.

Using the proof's selected basis matrix P_j and diagonal weights D_j , locally suppress the layer index and put $G = P^T D P$. The prescribed metric and its inverse satisfy

$$M = D + D P (G^{-2} - G^{-1}) P^T D, \quad M^{-1} = D^{-1} + P (I - G^{-1}) P^T, \quad P^T M = G^{-1} P^T D. \quad (219)$$

For verification, put $Z = D^{1/2} P$, decompose into $\text{ran } Z$ and its orthogonal complement, and use $Z^T Z = G$; multiplication gives each identity. Form the r -dimensional Gram/inverse and materialize both metric caches in $O(qr^2 + r^3 + q^2 r) = O(q^2 r)$ work.

The core initialized map $U_j^T W_0^{(j)} U_{j-1}/n$ costs $O(n^2 r + nr^2)$ in general. Paired source actions may save some matrix calls, but do not cover the initialized mixer on the entire source space. Its remaining assembly through selected bases and the last metric identity costs $O(q^2 r + qr^2)$. These counts give the assembly terms of $\mathcal{T}_H$. Source bases fit inside the LnR envelope and the final arrays inside Lq^2 . Adding the live training jets/caches, scalar transform, data and geometry bounds proves $\mathcal{M}_H$. Freeing phase-specific work arrays can reduce this safe peak upper bound.

E.9.6 Harmonic training without cubic hidden-matrix operations

Let H_j, Δ_j have training-feature and backward-response columns, and let X have normalized training inputs as columns. The prescribed directions can be evaluated as

$$\dot{A}_C = \frac{2}{m} (\Delta_1 \text{diag } c_C) X^T, \quad \dot{B}_C^{(j)} = \frac{2}{m} (\Delta_j \text{diag } c_C) (M_{j-1} H_{j-1})^T, \quad \dot{w}_C = \frac{2}{m} H_L c_C.$$

Apply metrics to the batch arrays first. Multiplying an already formed q -by- q update by a dense metric would introduce an unnecessary q^3 cost. Backward propagation similarly applies M_{j+1} , then $B_C^{(j+1)T}$, then the cached M_j^{-1} to the batch, rather than constructing dense metric adjoints.

The feature Gram is $Q_C = H_L^T M_L H_L / m$; its construction costs $O(mq_L^2 + q_L m^2)$, and an ordinary fresh factorization costs $O(m^3)$. Its positivity is inherited from the fitting theorem. This moving solve is distinct from the fixed metric inverse caches.

The update Gram need not be formed. Direct expansion of the directions above gives, sample by sample,

$$\begin{aligned} \frac{2}{m} (K_C c_C)_a &= h_{C,a}^{(L)T} M_L \dot{w}_C + \delta_{C,a}^{(1)T} M_1 \dot{A}_C v_a \\ &\quad + \sum_{j=2}^L \delta_{C,a}^{(j)T} M_j \dot{B}_C^{(j)} h_{C,a}^{(j-1)}. \end{aligned} \quad (220)$$

For example, expansion of the j -th hidden term produces $(2/m) \sum_b c_{C,b} \langle \delta_{C,a}^{(j)}, \delta_{C,b}^{(j)} \rangle_{M_j} \langle h_{C,a}^{(j-1)}, h_{C,b}^{(j-1)} \rangle_{M_{j-1}}$, exactly that term of $2K_C c_C/m$. Set $\dot{c}_C$ to the negative of these contractions. This is an algebraic identity, not an assertion that the optimizer is a gradient or that the formula is a Jacobian-vector product of the corrected predictor.

Forward passes, these ordered backward/update operations, the feature-Gram solve and state directions give (124) and its peak-memory count. A final post-update readout refresh repeats no more than the forward pass and top Gram solve, so it has the same order and the stated activation calls. With $M_L \hat{w}_C$ resident, query evaluation is the ordinary forward pass with rolling buffers, which proves (126).

E.9.7 Cost consequences of the existing inverse choices

Substitution of an inverse width/order is arithmetic only; it does not provide a numerical-integration or roundoff bound. For fixed data/activations/confidence and $n = \varepsilon^{-2+o(1)}$, dense P is $\varepsilon^{-4+o(1)}$. The Legendre inverse has $q = \varepsilon^{-1/2+o(1)}$, so its $Lnmq$ storage/action terms are $\varepsilon^{-5/2+o(1)}$ and the fixed dense mixers dominate the total warmup, training and query arithmetic envelopes. Its additional query workspace remains $\varepsilon^{-2+o(1)}$. The Harmonic inverse has $q = O(\log(1/\varepsilon)^{3d/2+1})$ for fixed structural parameters. Its runtime arithmetic and training peak memory are therefore $O(\log(1/\varepsilon)^{3d+2})$, and its query workspace is $O(\log(1/\varepsilon)^{3d/2+1})$, under the separate scalar-evaluation convention. Substituting an inverse q alone does not eliminate the setup resolutions, primitive costs or precision requirements. For the efficient warmup rows, use the separately certified $T = 32(m/\gamma) \log(en)$, $\eta = 1/n$ construction. At $n = \varepsilon^{-2+o(1)}$, its error is $\varepsilon^{2+o(1)} = o(\varepsilon)$, and its sufficient selected budget has the same logarithmic storage order. Its explicit setup work and peak memory are $\varepsilon^{-4+o(1)}$; its implicit setup work and peak memory are $\varepsilon^{-2+o(1)}$. These are sufficient stronger-accuracy choices, not minimal-budget or polylogarithmic-warmup claims.

E.10 Proofs of the efficient Harmonic initializers

The explicit initializer constructs source coefficients by certified local continuation through the full source horizon. The implicit initializer executes that same finite calculation through exact adaptive Gaussian actions. The argument proceeds from signed defect stability and positive quadrature to local numerical continuation, activation replacement, source assembly, and finally the joint-law coupling and its operation count. The legacy initialization-jet construction remains a separate valid alternative.

All notation referring to the dense model retains its original meaning: $A = W^{(1)}$, $w = W^{(L+1)}$, $v = x/\sqrt{d}$, and $\phi_j = \phi^{(j)}$. The parameter vector $\theta = (A/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$ uses ordinary Euclidean/Frobenius norms and the original mean-square loss and mobilities. This is the normalized coordinate form of the same dense flow. The positive-label analytic branch has $L \geq 2$, $Y > 0$, $T \geq 32(m/\gamma) \log(en)$, and $0 < \eta \leq \min(1, Y, 16Ym/\gamma)$, with the full original label and width/source qualifications. Zero labels and full-width retention keep their separate definitions.

Each proof component scopes its auxiliary coefficient names locally. In the quadrature component, $a = \alpha_T/2$ is the temporal angular strip radius and ℓ_* the largest spherical degree; the continuation components use a for the original activation strip width, and the assembly/execution components use J for the number of time panels and ℓ_* for spherical degree. Their exact correspondence to the single order interface is given in the deterministic prescription. In the sampler component, Y, R, U, V, p, q are stored matrices and query-span dimensions, not network labels, source budgets or selected widths.

The source recurrences, source event, source expansion, exact selector and runtime comparison are proved elsewhere in this same document. Every new deterministic and probabilistic execution argument needed for the two initializers is included below. No source-event test or new failure allowance is introduced. Exact joint-law equality refers to the explicit local finite initializer with the same scalar routines and deterministic conventions; it does not assert equality with the alternative origin-jet coefficient program.

E.10.1 Signed defect stability

Deterministic setup

Let $v = x/\sqrt{d}$, so $\|v\|_2 = 1$, and retain the dense network

$$z^{(1)} = Av, \quad z^{(j)} = W^{(j)}h^{(j-1)}, \quad h^{(j)} = \phi_j(z^{(j)}), \quad f_n = w^T h^{(L)}/n.$$

Its parameter coordinates with Euclidean mobility metric are

$$\theta = (A/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n}).$$

Write $r_a = f_n(v_a) - y_a$, $\rho = \|r\|_2/\sqrt{m}$, $Y = \|y\|_2/\sqrt{m}$, and $\lambda = \gamma/m$. With $\xi_a(\theta) = \nabla_\theta f_n(v_a)$, the exact vector field is

$$F(\theta) = -\frac{2}{m} \sum_{a=1}^m r_a \xi_a(\theta), \quad \dot{\theta} = F(\theta). \quad (\text{Setup-S.1})$$

The initial readout is zero. All blocks train with the loss and mobilities of the preceding Harmonic construction; no frozen-feature model replaces (Setup-S.1).

Condition on the existing real-fitting and source event, including the stated analytic-extension gates when all-time carrier control is used. The complete original common label allowance is unchanged. The imported deterministic conclusions needed here are

$$\|A(t)\|_{\text{op}}/\sqrt{n} < 9, \quad \|W^{(j)}(t)\|_{\text{op}} < 9, \quad \|w(t)\|_2/\sqrt{n} \leq 2Y/\sqrt{\lambda},$$

$$\rho(t) \leq Y e^{-\lambda t/2}, \quad \int_0^\infty \rho(t) dt \leq 2Y/\lambda,$$

$$\max_{a,j,i,t \geq 0} |k_{a,i}^{(j)}(t)| \leq M_n, \quad M_n = 2K_{\text{src}} S \sqrt{\log(en)}, \quad S = 16Y/\lambda. \quad (\text{Setup-S.2})$$

Here $k^{(L)} = w$ and $k^{(j)} = (W^{(j+1)})^T \delta^{(j+1)}$, with $\delta^{(j)} = \phi_j'(z^{(j)}) \odot k^{(j)}$. The carrier maximum in (Setup-S.2) concerns only the true training trajectory. No carrier maximum for perturbed trajectories or parameter segments is assumed.

Take $b = \max_j |\phi_j(0)|$, and let $s \geq 1$, $t_2 \geq 1$ bound the first and second activation derivatives on the safe strip, as in the source recurrences. Define explicit deterministic coefficients

$$H_1 = \max(1, b + 10s), \quad H_j = \max(1, b + 10sH_{j-1}), \quad H = H_L,$$

$$R = 1 + 2Y/\sqrt{\lambda}, \quad B = Rs(10s)^{L-1}, \quad F_z = H(10s)^{L-1}, \quad P_{\text{layer}} = 1 + (L-1)H, \quad G = H + P_{\text{layer}}B,$$

$$D(M) = (10s)^{L-1} \{s(1+B) + Lt_2 F_z M\},$$

$$J(M) = \sqrt{L+1} \{P_{\text{layer}} D(M) + [1 + (L-1)B] s F_z\},$$

$$C(M) = s F_z \sqrt{L} + L B s F_z + \frac{1}{2} L^2 t_2 F_z^2 M. \quad (\text{Setup-S.3})$$

These constants expose dependence on depth, activation bounds and $Y/\sqrt{\lambda}$; $\lambda = \gamma/m$ exposes the sample/gap dependence. The parameter norm and these bounds are independent of input dimension once the input norm is one. The original source event still has its stated problem-dependent eventual-width qualification.

Every real state u within Euclidean distance one of $\theta(t)$ has mixer and normalized first-matrix operator caps ten and readout RMS at most R . Linear activation growth therefore gives feature RMS at most H on the entire input sphere. These statements do not require bounded activation values.

Endpoint estimates with only one controlled carrier endpoint

Let $e = u - \theta$, $E = \|e\|_2 \leq 1$, and let D_h be the sum of the first normalized Frobenius discrepancy and the hidden-matrix Frobenius discrepancies; let D_w be the readout RMS discrepancy. Then $D_h \leq \sqrt{L}E$ and $D_h + D_w \leq \sqrt{L+1}E$. Forward subtraction through matrices with operator norm at most ten gives

$$\sup_{v,j} \frac{\|z^{(j)}(u, v) - z^{(j)}(\theta, v)\|_2}{\sqrt{n}} \leq F_z D_h, \quad \sup_{v,j} \frac{\|h^{(j)}(u, v) - h^{(j)}(\theta, v)\|_2}{\sqrt{n}} \leq s F_z D_h. \quad (\text{Setup-S.4})$$

Indeed the first direct coefficient is one; every later direct coefficient is at most H , and each propagation contributes $10s$. The sum of the parameter block discrepancies is already in D_h .

For a training sample, split the backward difference using the true carrier:

$$\delta^{(j)}(u) - \delta^{(j)}(\theta) = \phi'_j(z^{(j)}(u)) \odot [k^{(j)}(u) - k^{(j)}(\theta)] + [\phi'_j(z^{(j)}(u)) - \phi'_j(z^{(j)}(\theta))] \odot k^{(j)}(\theta).$$

The second term in RMS is bounded by $t_2 M_n F_z D_h$. The first propagates with factor $10s$, and changed mixers cost at most $sB\|\Delta W\|_F$. The top readout difference costs sD_w . Summing the resulting finite geometric recursion gives

$$\frac{\|\delta_a^{(j)}(u) - \delta_a^{(j)}(\theta)\|_2}{\sqrt{n}} \leq D(M_n)(D_h + D_w).$$

The exact gradient blocks are

$$\xi_a = \left(\frac{\delta_a^{(1)} v_a^T}{\sqrt{n}}, \left(\frac{\delta_a^{(j)} h_a^{(j-1)T}}{n} \right)_{j=2}^L, \frac{h_a^{(L)}}{\sqrt{n}} \right).$$

Bounding and subtracting these blocks gives

$$\|\xi_a(u)\|_2, \|\xi_a(\theta)\|_2 \leq G, \quad \|\xi_a(u) - \xi_a(\theta)\|_2 \leq J(M_n)E. \quad (\text{Setup-S.5})$$

There is also a Taylor remainder anchored at the true state:

$$|f_n(u, v_a) - f_n(\theta, v_a) - \langle \xi_a(\theta), e \rangle| \leq C(M_n)E^2. \quad (\text{Setup-S.6})$$

For completeness set $\Delta h = h(u) - h(\theta)$, $\Delta z = z(u) - z(\theta)$, and $a^{(j)} = \Delta h^{(j)} - \phi'_j(z^{(j)}(\theta)) \odot \Delta z^{(j)}$. Scalar Taylor's formula gives $|a_i^{(j)}| \leq t_2 |\Delta z_i^{(j)}|^2 / 2$. Expanding the network discrepancy by forward differentiation and then backward substitution expresses its scalar remainder as

$$\frac{1}{n} \sum_j k_a^{(j)}(\theta)^T a_a^{(j)} + \frac{1}{n} \sum_{j=2}^L \delta_a^{(j)}(\theta)^T \Delta W^{(j)} \Delta h_a^{(j-1)} + \frac{1}{n} \Delta w^T \Delta h_a^{(L)}.$$

The three absolute bounds are respectively $L^2 t_2 M_n F_z^2 E^2 / 2$, $LB s F_z E^2$, and $s F_z \sqrt{L} E^2$, proving (Setup-S.6). No Hessian bound on an uncontrolled interpolating state has been inserted.

Defect stability lemma

Let $u : [0, T] \rightarrow \mathbb{R}^P$ be an absolutely continuous approximate parameter path, where $P = (L-1)n^2 + n(d+1)$. Its additive ODE defect is

$$d(t) = \dot{u}(t) - F(u(t)), \quad E_0 = \|u(0) - \theta(0)\|_2 + \int_0^T \|d(t)\|_2 dt.$$

Write $J = J(M_n)$, $C = C(M_n)$, and $A_n = 4JY/\lambda$. Suppose

$$E_0 \leq e^{-A_n} / 4, \quad 8(C + J)^2 T e^{2A_n} E_0^2 \leq 1. \quad (\text{Setup-S.7})$$

Then

$$\sup_{0 \leq t \leq T} \|u(t) - \theta(t)\|_2 \leq 2e^{A_n} E_0. \quad (\text{Setup-S.8})$$

Proof. Stop at discrepancy one. Put $e_{f,a} = f_n(u, v_a) - f_n(\theta, v_a)$, so the perturbed residual is $r_a + e_{f,a}$. The exact vector-field subtraction in (Setup-S.1) gives

$$\frac{1}{2} \frac{d}{dt} E^2 = -\frac{2}{m} \sum_a e_{f,a} \langle e, \xi_a(\theta) \rangle - \frac{2}{m} \sum_a (r_a + e_{f,a}) \langle e, \xi_a(u) - \xi_a(\theta) \rangle + \langle e, d \rangle.$$

Using (Setup-S.5)–(Setup-S.6) and the ordinary sample RMS gives

$$\frac{1}{2} \frac{d}{dt} E^2 \leq -2 \frac{\|e_f\|_2^2}{m} + 2(C + J) \frac{\|e_f\|_2}{\sqrt{m}} E^2 + 2J\rho E^2 + E\|d\|_2.$$

Maximizing the first two terms over their nonnegative scalar argument bounds them by $(C + J)^2 E^4 / 2$. Consequently, in the upper-derivative sense,

$$E' \leq 2J\rho E + \frac{1}{2}(C + J)^2 E^3 + \|d\|_2. \quad (\text{Setup-S.9})$$

At zero discrepancy use $(E^2 + \varepsilon^2)^{1/2}$ and pass to zero ε ; all coefficients and the defect are integrable. Let $a(t) = 2J \int_0^t \rho(s) ds \leq A_n$ and $z(t) = e^{-a(t)} E(t)$. As long as $z \leq 2E_0$, integration of (Setup-S.9) gives

$$z(t) \leq E_0 + 4(C + J)^2 T e^{2A_n} E_0^3 \leq \frac{3}{2} E_0.$$

Thus a first crossing of $2E_0$ is impossible. The first condition in (Setup-S.7) also keeps $E \leq 1/2$, so the geometric stop cannot occur. The case $E_0 = 0$ follows by uniqueness or the same regularization. This proves (Setup-S.8).

At fixed admissible m, d, L, γ, Y and activations, (Setup-S.3) is affine in M_n , hence

$$A_n = a_0 + a_1 \sqrt{\log(en)}$$

for explicitly defined nonnegative coefficients a_0, a_1 . The amplification is therefore subpolynomial in width. Its exponent has no factor T : the training residual is integrable and the negative prediction-error square was retained before estimating it.

This is a finite-horizon defect theorem. It does not assert that an arbitrary perturbed flow fits labels, nor that a persistent nonzero defect is harmless over an infinite horizon. The original Harmonic endpoint theorem is unchanged.

Polynomial precision for all source coordinates

The source families contain backward fields at every passive query. Their carrier maxima are not supplied by (Setup-S.2). A deterministic RMS-to-coordinate conversion is sufficient for this precision estimate.

Set

$$K_q = R(10s)^{L-1}, \quad C_{\text{src}} = 8\sqrt{L+1} \max \{ sF_z, (10s)^{L-1} [s(1+B) + Lt_2 F_z K_q] \}.$$

Every query carrier has RMS at most K_q , hence maximum at most $\sqrt{n}K_q$. Repeat the backward subtraction above with that maximum. A final RMS-to-coordinate conversion and $\sqrt{n} \leq n$ show that each feature or response family differs between $u(t)$ and $\theta(t)$ in coordinate norm by at most $C_{\text{src}} n E(t)$. The factor eight also includes its initialized forward/reverse image because $\|W_0\|_{\text{op}} \leq 8$. Consequently source accuracy η is guaranteed if, in addition to (Setup-S.7),

$$E_0 \leq \frac{\eta e^{-A_n}}{2C_{\text{src}} n}. \quad (\text{Setup-S.10})$$

For $\eta = n^{-a}$, fixed $a > 0$, and $T = O((m/\gamma) \log n)$, the logarithm of the required inverse defect tolerance is $O(\log n + \sqrt{\log n})$, with the explicit coefficients above. Thus polynomial precision suffices at the original

$\eta = 1/n$ specialization. This is a statement about the required numerical tolerance, not the bit complexity of an activation evaluation or the condition number of coordinate selection.

The source coefficient construction must still apply identical scalar linear operations to both members of every initialized-mixer pair. Approximate ODE solutions do not by themselves preserve that algebraic identity. Forming the image from the computed base coefficient preserves it exactly in the same exact-real operation model as the preceding Harmonic construction; finite precision needs its own roundoff allowance.

E.10.2 Positive coefficient quadrature

Precise input and conclusion

Fix a permitted width and source event in the preceding Harmonic construction, a finite source horizon $T \geq T_0 = 32(m/\gamma) \log(en)$, and source-coordinate tolerance $0 < \eta \leq \eta_0 = \min(1, Y, 16Ym/\gamma)$. These are the existing Harmonic source qualifications; in particular the retained set below is nonempty. At hidden layer j , the existing source families are

$$h_n^{(j)}(t, v), \quad W_0^{(j)} h_n^{(j-1)}(t, v), \quad \delta_n^{(j)}(t, v), \quad W_0^{(j+1)T} \delta_n^{(j+1)}(t, v),$$

with the same omissions at the first and last layers as in the preceding Harmonic construction. Here $v = x/\sqrt{d} \in S^{d-1}$, W_0 is an initialized dense mixer, and δ_n is the existing passive-query backward field. Consider one scalar coordinate $g(t, v)$ of any of these families. Put

$$G(u, v) = g(T(1 + \cos u)/2, v), \quad \alpha = \alpha_T = \frac{r_t}{4T}, \quad a = \alpha/2. \quad (\text{Setup-Q.1})$$

The source theorem supplies holomorphy on a neighborhood of $\{|\operatorname{Im} u| \leq \alpha\} \times \mathcal{Q}_{r_q}$ and the uniform coordinate bound $|G| \leq M_n$, where

$$\mathcal{Q}_r = \{z \in \mathbb{C}^d : z^T z = 1, \|\operatorname{Im} z\|_2 \leq \sinh r\}, \quad M_n = 10 \max(H_L^{\text{src}}, \tau) \sqrt{n}. \quad (\text{Setup-Q.2})$$

As in the original source proof, $\alpha \leq 1$. The choice of the smaller strip a merely leaves an analytic margin. All bounds below are deterministic on that same source event.

For $d \geq 2$, use the real orthonormal spherical harmonics $Y_{\ell, b}$ under probability-sphere measure σ . Their degrees are denoted ℓ , to distinguish them from hidden layers. Write

$$h_\ell = \binom{\ell + d - 1}{d - 1} - \binom{\ell + d - 3}{d - 1}, \quad \mathcal{Y}_{\ell_*} = \max_{0 \leq \ell \leq \ell_*} \sqrt{h_\ell}, \quad H = \sum_{\ell=0}^{\ell_*} h_\ell. \quad (\text{Setup-Q.3})$$

Impossible binomials are zero. Retain **exactly** the original joint set

$$\Lambda = \{(k, \ell, b) : k \geq 0, \alpha k + r_q \ell \leq H(T, \eta), 1 \leq b \leq h_\ell\}, \quad N = |\Lambda| = N(T, \eta), \quad (\text{Setup-Q.4})$$

and put $p = \max_\Lambda k$, $\ell_* = \max_\Lambda \ell$. The exact retained cosine coefficients are

$$c_{k\ell b} = \gamma_k \int_0^{2\pi} \int_{S^{d-1}} G(u, v) \cos(ku) Y_{\ell, b}(v) d\sigma(v) \frac{du}{2\pi}, \quad \gamma_0 = 1, \quad \gamma_k = 2 \ (k \geq 1). \quad (\text{Setup-Q.5})$$

Thus the factor two for positive modes is part of the definition and of every error estimate below. The original expansion proof gives coordinate error at most $\eta/16$ for the omitted modes.

Define the target error for each retained coefficient by

$$\epsilon_c = \frac{\eta}{16N\mathcal{Y}_{\ell_*}}. \quad (\text{Setup-Q.6})$$

The explicit rule constructed below gives exact-value quadrature error at most $\epsilon_c/2$. If supplied nodal values have the common coordinate accuracy in (Setup-Q.24), their contribution is at most $\epsilon_c/2$. Consequently every retained coefficient has error at most ϵ_c , and its finite reconstruction has error at most

$$\frac{\eta}{16} + N\mathcal{Y}_{\ell_*} \epsilon_c = \frac{\eta}{8} < \eta. \quad (\text{Setup-Q.7})$$

The first term is the original truncation tail. Every member of every source pair is treated with its own coordinate guarantee.

Sphere coordinates and their complex neighborhood

For $d \geq 3$, use the usual successive polar angles $\theta_1, \dots, \theta_{d-2} \in [0, \pi]$ and final azimuth $\varphi \in [0, 2\pi]$:

$$v = (\cos \theta_1, \sin \theta_1 \cos \theta_2, \dots, \prod_{i=1}^{d-2} \sin \theta_i \cos \varphi, \prod_{i=1}^{d-2} \sin \theta_i \sin \varphi). \quad (\text{Setup-Q.8})$$

The intermediate coordinates follow the same displayed pattern. For $d = 2$, use only $v = (\cos \varphi, \sin \varphi)$. Introduce

$$I_b = \int_0^\pi \sin^b \theta \, d\theta, \quad I_0 = \pi, \quad I_1 = 2, \quad I_b = \frac{b-1}{b} I_{b-2} \quad (b \geq 2). \quad (\text{Setup-Q.9})$$

The recurrence follows by integration by parts. Differentiating (Setup-Q.8) gives orthogonal coordinate tangent vectors: the vector for each successive angle has length equal to the product of the preceding polar sines. Multiplying these lengths gives the Jacobian $\prod_{i=1}^{d-2} \sin^{d-1-i} \theta_i$. Its successive integrals are the factors I_{d-1-i} , so normalization proves that probability-sphere integration becomes product integration under the uniform measures $d\theta_i/\pi$ and $d\varphi/(2\pi)$, with the multiplier

$$W(\theta) = \prod_{i=1}^{d-2} \frac{\pi}{I_{d-1-i}} \sin^{d-1-i} \theta_i. \quad (\text{Setup-Q.10})$$

Empty products are one. In particular this formula includes $d = 2$. Write

$$A_d = \prod_{b=1}^{d-2} \frac{\pi}{I_b}, \quad D_d^{\text{ang}} = \frac{(d-2)(d-1)}{2}, \quad h = \frac{r_q}{2(d-1)}, \quad \sigma = \text{arsinh}(2h/\pi). \quad (\text{Setup-Q.11})$$

These constants expose the dependence on dimension. The angular multiplier satisfies $0 \leq W \leq A_d$ on real angles and

$$|W(\theta)| \leq A_d (\cosh h)^{D_d^{\text{ang}}} \quad \text{if all } |\text{Im } \theta_i| \leq h. \quad (\text{Setup-Q.12})$$

Indeed $|\sin(x + iy)|^2 = \sin^2 x + \sinh^2 y \leq \cosh^2 y$.

To check the source domain, represent (Setup-Q.8) as successive coordinate-plane rotations applied to a real unit vector. Real rotations preserve $z^T z = 1$ and $\|\text{Im } z\|_2$. For $z = b + ic$ on that quadric there is $s \geq 0$ with $\|b\|_2 = \cosh s$, $\|c\|_2 = \sinh s$. Applying a coordinate-plane rotation of imaginary angle iy gives

$$\|\text{Im}(R(iy)z)\|_2 \leq \sinh |y| \|b\|_2 + \cosh |y| \|c\|_2 = \sinh(s + |y|). \quad (\text{Setup-Q.13})$$

For the unchanged coordinates the real part of this rotation is the identity, whose norm is at most $\cosh |y|$; its imaginary part has norm at most $\sinh |y|$, which proves the inequality. Decompose each complex angle into its commuting real and imaginary rotations and apply (Setup-Q.13) successively. If all $d-1$ angles have imaginary parts of magnitude at most h , the resulting point lies in $\mathcal{Q}_{(d-1)h} = \mathcal{Q}_{r_q/2}$. Real parts may lie outside the fundamental angle boxes without changing this argument.

The angular composition of a degree- ℓ harmonic polynomial is a trigonometric polynomial of degree at most ℓ separately in every angle. If a trigonometric polynomial P has degrees between $-\ell$ and ℓ and satisfies $|P| \leq C$ on the real axis, then

$$|P(x + iy)| \leq C e^{\ell|y|}. \quad (\text{Setup-Q.14})$$

For $y \geq 0$, multiply its Laurent polynomial by z^ℓ , apply the maximum modulus principle inside $|z| \leq 1$, and set $z = e^{i(x+iy)}$. For $y \leq 0$, use the reversed Laurent polynomial. Iterating this argument over the angles, starting from the real-sphere bound $|Y_{\ell,b}| \leq \sqrt{h_\ell}$, gives

$$|Y_{\ell,b}(v)| \leq \mathcal{Y}_{\ell,*} e^{\ell_*(d-1)h} \quad (\ell \leq \ell_*). \quad (\text{Setup-Q.15})$$

Finally $|\cos(ku)| \leq e^{ka}$ in the time strip. Every coefficient integrand, including γ_k , is therefore holomorphic on the product domain just described and bounded there by

$$B_{\text{int}} = 2M_n \mathcal{Y}_{\ell_*} A_d \exp\{ap + \ell_*(d-1)h\} (\cosh h)^{D_d^{\text{ang}}}. \quad (\text{Setup-Q.16})$$

This is an integrand bound, not an additional source or rank parameter.

Explicit positive quadrature and its error

Periodic variables

For a 2π -periodic function F , use $Q_N F = N^{-1} \sum_{r=0}^{N-1} F(2\pi r/N)$. If F is holomorphic near $|\text{Im } z| \leq b$ and bounded there by B , contour translation gives Fourier coefficient bound $|\widehat{F}_k| \leq B e^{-b|k|}$. Absolute convergence permits termwise quadrature. The discrete average of e^{ikz} is one when N divides k , zero otherwise; therefore

$$\left| \int_0^{2\pi} F(u) \frac{du}{2\pi} - Q_N F \right| \leq 2B \sum_{r=1}^{\infty} e^{-brN} = \frac{2B}{e^{bN} - 1}. \quad (\text{Setup-Q.17})$$

Use this rule for time with $b = a$, and for azimuth with $b = h$. Both rules have positive weights summing to one.

Polar variables

The following construction supplies the nodes and weights; no Gaussian quadrature construction is assumed. For an integer $M \geq 1$, let

$$\omega_r = \frac{(2r+1)\pi}{2M}, \quad x_r = \cos \omega_r, \quad r = 0, \dots, M-1, \\ w_r = \frac{1}{M} \left[1 - 2 \sum_{k=1}^{\lfloor (M-1)/2 \rfloor} \frac{\cos(2k\omega_r)}{4k^2 - 1} \right]. \quad (\text{Setup-Q.18})$$

These are Fejér's first interpolatory weights for the normalized integral $\frac{1}{2} \int_{-1}^1$. The discrete cosine identities at these nodes give the interpolant

$$I_{M-1} F(x) = \frac{1}{M} \sum_r F(x_r) \left[1 + 2 \sum_{k=1}^{M-1} \cos(k\omega_r) T_k(x) \right]. \quad (\text{Setup-Q.19})$$

To verify the identity, sum the elementary geometric series for $e^{ij\omega_r}$, or use $2 \cos j\omega \cos k\omega = \cos((j-k)\omega) + \cos((j+k)\omega)$. The constant mode has squared discrete norm M ; all nonconstant modes through $M-1$ have squared norm $M/2$, and unequal modes are orthogonal. Evaluating (Setup-Q.19) at the nodes then gives their prescribed values. Substitution $x = \cos \omega$ in the elementary integral gives

$$\frac{1}{2} \int_{-1}^1 T_k(x) dx = \begin{cases} 1, & k = 0, \\ 0, & k \text{ odd}, \\ -1/(k^2 - 1), & k \geq 2 \text{ even}. \end{cases}$$

Integrating (Setup-Q.19) proves (Setup-Q.18) and exactness for every polynomial of degree at most $M-1$. Moreover

$$2 \sum_{k=1}^K \frac{1}{4k^2 - 1} = 1 - \frac{1}{2K+1} < 1$$

for finite K , by telescoping. Thus every $w_r > 0$, including the empty-sum case, and exactness for constants gives $\sum_r w_r = 1$.

Suppose F is holomorphic near the filled Bernstein ellipse $x = (z + z^{-1})/2$, $|z| = e^\sigma$, and bounded there by B . Its Laurent expansion in z is symmetric under $z \mapsto z^{-1}$. Cauchy's coefficient formula therefore gives Chebyshev coefficients of magnitude at most $2B e^{-k\sigma}$ for $k \geq 1$. The truncated degree- $M-1$ series has

uniform real-interval error at most $2Be^{-M\sigma}/(1 - e^{-\sigma})$. Both the exact normalized integral and the positive quadrature have operator norm one on real continuous functions. Subtracting that polynomial from each proves

$$\left| \frac{1}{2} \int_{-1}^1 F(x) dx - \sum_r w_r F(x_r) \right| \leq \frac{4Be^{-M\sigma}}{1 - e^{-\sigma}}. \quad (\text{Setup-Q.20})$$

Apply this rule to $\theta = \pi(1 + x)/2$. The ellipse has $|\text{Im } \theta| \leq (\pi/2) \sinh \sigma = h$, by (Setup-Q.11). It is therefore inside the previously verified angular domain. The normalization becomes $d\theta/\pi$, as required in (Setup-Q.10).

Tensor rule and explicit sufficient sizes

Each univariate exact integral and quadrature is a contraction in the supremum norm. Write their product difference as a telescoping sum, changing one factor at a time. Apply (Setup-Q.17) or (Setup-Q.20) to that factor while the remaining variables are real, and then apply the other contraction operators. There are d variables in total: time, azimuth and $d - 2$ polar variables. This gives

$$|c_{k\ell b} - Qc_{k\ell b}| \leq B_{\text{int}} \left[\frac{2}{e^{aN_t} - 1} + \frac{2}{e^{hN_\varphi} - 1} + \frac{4(d-2)e^{-\sigma N_\theta}}{1 - e^{-\sigma}} \right]. \quad (\text{Setup-Q.21})$$

Here $Qc_{k\ell b}$ denotes application of the tensor rule to the integrand of (Setup-Q.5) after the angular change of variables. The signs of the harmonic and cosine do not affect positivity of the underlying integration rule. Choose

$$\begin{aligned} N_t &= \max \left\{ 1, \left\lceil \frac{1}{a} \log \left(1 + \frac{4dB_{\text{int}}}{\epsilon_c} \right) \right\rceil \right\}, \\ N_\varphi &= \max \left\{ 1, \left\lceil \frac{1}{h} \log \left(1 + \frac{4dB_{\text{int}}}{\epsilon_c} \right) \right\rceil \right\}, \\ N_\theta &= \max \left\{ 1, \left\lceil \frac{1}{\sigma} \log \left(\frac{8dB_{\text{int}}}{\epsilon_c(1 - e^{-\sigma})} \right) \right\rceil \right\} \quad (d \geq 3). \end{aligned} \quad (\text{Setup-Q.22})$$

Each of the d contributions in (Setup-Q.21) is then at most $\epsilon_c/(2d)$. The total spatial node count is

$$N_x = N_\varphi N_\theta^{d-2} \quad (d \geq 3), \quad N_x = N_\varphi \quad (d = 2). \quad (\text{Setup-Q.23})$$

The dependence on d , α , r_q , p , ℓ_* , M_n , N , and η is explicit in (Setup-Q.3), (Setup-Q.6), (Setup-Q.9), (Setup-Q.11), (Setup-Q.16), and (Setup-Q.22). In particular no equality $N_x = H$ or $N_t = p + 1$ is assumed.

Inexact nodal values and exact source pairing

At every time/spatial node, suppose an initialization-based procedure supplies all needed vector sources with coordinate error at most

$$\delta_{\text{node}} = \frac{\epsilon_c}{4A_d \mathcal{Y}_{\ell_*}}. \quad (\text{Setup-Q.24})$$

All quadrature weights before the multiplier W are positive and have total mass one. On real nodes $W \leq A_d$, $|Y_{\ell,b}| \leq \mathcal{Y}_{\ell_*}$, $|\cos ku| \leq 1$, and $\gamma_k \leq 2$. Therefore the coefficient error caused by these nodal errors is at most

$$2A_d \mathcal{Y}_{\ell_*} \delta_{\text{node}} = \epsilon_c/2. \quad (\text{Setup-Q.25})$$

This supplies the second half of the error allocation in (Setup-Q.7). It does not assume holomorphy of the numerical nodal approximation; the quadrature analysis applies to the exact analytic source, followed by the separate finite-sum perturbation estimate (Setup-Q.25).

For exact preservation of initialized-matrix pairing, more than separate numerical accuracy is required. In the following pairing identities only, let g denote the full $\mathbb{R}^n$ -valued source, so the preceding coordinate

estimates apply to each of its entries. Let A be the fixed initialized matrix in one pair: either $W_0^{(j)}$ or $W_0^{(j+1)T}$. The two input families are g and Ag . Require their nodal approximants to have the exact algebraic form

$$\widetilde{Ag}(u_r, v_s) = A\widetilde{g}(u_r, v_s), \quad (\text{Setup-Q.26})$$

while **each side separately** has coordinate error at most δ_{node} relative to its own exact source. Applying identical real quadrature, cosine normalization and retained-mode restriction to both sides gives, in exact arithmetic,

$$\widetilde{c}_{k\ell b}^{Ag} = \sum_{r,s} \beta_{k\ell b,rs} A\widetilde{g}(u_r, v_s) = A \sum_{r,s} \beta_{k\ell b,rs} \widetilde{g}(u_r, v_s) = A\widetilde{c}_{k\ell b}^g. \quad (\text{Setup-Q.27})$$

The scalar coefficients β are the same in both sums. The reconstructed approximants consequently satisfy $p_{Ag} = Ap_g$ at every real time/query, exactly. The source-space membership and action identities used in the preceding Harmonic construction are unchanged.

Exact source evaluation satisfies (Setup-Q.26). The original common finite initial-jet reconstruction also satisfies it: differentiation, scalar continuation, truncation and this quadrature all commute with a fixed matrix. Its sufficiently large common jet cutoff can certify the required coordinate errors separately for every member, because each member has the same source bound. This establishes finite realizability without establishing an efficient cutoff. Independently computed approximate nodal values do not automatically satisfy (Setup-Q.26), even when they are each accurate. Deriving the image's error from the feature's coordinate error through a dimension-dependent matrix norm is not used.

No quadrature node is a generator of E_j . The generators remain the coefficient vectors indexed by Λ , plus the original exact initialized vectors. Thus

$$\dim E_j \leq R = 2m + d + 1 + 4N, \quad (\text{Setup-Q.28})$$

with the existing boundary-layer improvements. This is exactly the old dimension certificate. The original budget prescription proves its own bound $9R \leq q$; replacing the quadrature does not enlarge it. For a supplied alternative mode set, the existing actual-rank test $9r \leq q$, $r = \max_j \dim E_j$, is still the applicable selector guarantee. Numerical perturbations can change actual rank, so the safe argument uses (Setup-Q.28), not an assertion that an old accidental rank deficiency persists. The selected runtime, its learned and fixed storage, and the dense-comparison certificate therefore retain their existing bounds whenever the original count certificate applies. The exact initialized training vectors, first-weight columns and constant are still inserted by their original exact construction. Baseline-only, zero-label and full-width exact-retention branches need no quadrature and are unaffected.

Dimensions one and two

For $d = 2$, equations (Setup-Q.3)–(Setup-Q.28) apply with no polar angles, $A_2 = 1$, $D_2^{\text{ang}} = 0$, and $h = r_q/2$. The basis is $1, \sqrt{2}\cos(\ell\varphi), \sqrt{2}\sin(\ell\varphi)$. Hence $\mathcal{Y}_{\ell_*} = \sqrt{2}$ if $\ell_* \geq 1$, and $\mathcal{Y}_0 = 1$. The spatial rule is just the positive circle trapezoid, with $N_x = N_\varphi$.

For $d = 1$, the source domain has precisely the two inputs $v = -1, 1$. Perform their cosine expansions separately, with $N_1 = N_1(T, \eta)$, $p = N_1 - 1$, and

$$\epsilon_c = \frac{\eta}{16N_1}, \quad B_{\text{int}} = 2M_n e^{ap}, \quad N_t = \max \left\{ 1, \left\lceil \frac{1}{a} \log \left(1 + \frac{4B_{\text{int}}}{\epsilon_c} \right) \right\rceil \right\}. \quad (\text{Setup-Q.29})$$

Use the time trapezoid only. Its coefficient error is at most $\epsilon_c/2$. The common nodal tolerance is $\delta_{\text{node}} = \epsilon_c/4$, which gives the other half because $\gamma_k \leq 2$. The original separate time tails are at most $\eta/16$, so the same reconstruction bound (Setup-Q.7) holds at both inputs. There are $N_x = 2$ spatial evaluations, not an angular grid. The rank certificate remains $R = 2m + d + 1 + 8N_1$, and the same exact linear pairing proof applies.

Node generation, arithmetic and peak setup memory

All counts below use the exact-real arithmetic contract of the preceding Harmonic construction. Elementary trigonometric and inverse/hyperbolic scalar evaluations are reported as scalar calls; their bit complexity is not claimed to be constant. The needed one-dimensional rule orders are part of the input to the following execution, determined by the explicit certificates above.

For $d \geq 3$, build the N_θ nodes and weights in (Setup-Q.18). At each node use angle-addition recurrences to evaluate the $O(N_\theta)$ cosine terms in its weight. This costs $O(N_\theta^2)$ arithmetic and $O(N_\theta)$ words. The same normalized unweighted rule is reused in every polar coordinate; only the Jacobian power changes. Computing the constants (Setup-Q.9) and the normalizations (Setup-Q.10) costs $O(d)$ arithmetic. The time and azimuth rules cost $O(N_t + N_\varphi)$ arithmetic and words if cached. Enumerating the spatial tensor product and mapping each point to the sphere costs $O(dN_x)$. Powers in (Setup-Q.10) can be computed using $O(d)$ multiplications per node after powers of each cached one-dimensional sine through $d - 2$ are prepared, or directly in $O(d^2)$ per node. Use the former execution and charge $O(dN_\theta)$ preparation and storage.

Consequently a sufficient rule-only bound is

$$\mathcal{G}_{\text{rule,time}} = O(N_\theta^2 + dN_\theta + N_t + N_\varphi + dN_x + d),$$

$$\mathcal{G}_{\text{rule,memory}} = O(dN_\theta + N_t + N_\varphi + d), \quad d \geq 3. \quad (\text{Setup-Q.30})$$

There are $O(N_\theta + N_t + N_\varphi)$ elementary scalar trigonometric calls with direct one-dimensional tables. The orders in (Setup-Q.22) require a fixed number of further elementary calls. For $d = 2$ delete all polar terms and use $O(N_t + N_x)$ time and cached words; for $d = 1$ use $O(N_t)$. Streaming the periodic tables can reduce their memory, but the displayed cached implementation is enough.

At each spatial node, the separated harmonic basis from the complete geometric-basis argument in the preceding Harmonic construction can still be used. In addition to (Setup-Q.30), charge

$$\mathcal{G}_{\text{basis,time}} = \begin{cases} 0, & d = 1, \\ O(N_x(\ell_* + 1)), & d = 2, \\ O(d(\ell_* + 1)^2 + dN_x[(\ell_* + 1)^2 + H]), & d \geq 3, \end{cases}$$

$$\mathcal{G}_{\text{basis,memory}} = \begin{cases} O(1), & d = 1, \\ O(\ell_* + 1), & d = 2, \\ O(d[(\ell_* + 1)^2 + H]), & d \geq 3. \end{cases} \quad (\text{Setup-Q.31})$$

That recurrence includes scalar normalization generation; this route does not assume a precomputed dense table of harmonics at all spatial nodes. Let $\mathcal{G}_{\text{time}}, \mathcal{G}_{\text{memory}}$ be the sums of the rule and basis bounds.

Polylogarithmic quadrature regime and claim boundary

The explicit estimates hold for every finite permitted T, η . The following asymptotic statement has additional stated conditions: fix d, L, m , activation/source coefficients and a positive allowed label size, and let

$$T = O(\log(en)), \quad \log(1/\eta) = O(\log(en)), \quad r_t^{-1}, r_q^{-1} = O(\sqrt{\log(en)}). \quad (\text{Setup-Q.34})$$

These conditions cover the original $(T_0, 1/n)$ specialization and the existing polynomial-accuracy inverse prescriptions whenever their analytic compressed branch is used. They do not cover every arbitrarily large supplied budget under an unchanged claim of polylogarithmic horizon.

Then $\alpha^{-1} = O(\log(en)^{3/2})$, $H(T, \eta) = O(\log(en))$, and the unchanged cutoffs satisfy

$$p = O(\log(en)^{5/2}), \quad \ell_* = O(\log(en)^{3/2}), \quad N = O(\log(en)^{3d/2+1}). \quad (\text{Setup-Q.35})$$

The logarithm of M_n is $O(\log(en))$, that of $\mathcal{Y}_{\ell_*}$ is $O_d(\log \log(en))$, and $ap + \ell_*(d - 1)h = O(\log(en))$. It follows directly from (Setup-Q.6), (Setup-Q.11), and (Setup-Q.16) that $\log(B_{\text{int}}/\epsilon_c) = O_d(\log(en))$. Since σ is comparable to h for the current bounded radii, (Setup-Q.22) gives

$$N_t = O_d(\log(en)^{5/2}), \quad N_\varphi, N_\theta = O_d(\log(en)^{3/2}), \quad N_x = O_d(\log(en)^{3(d-1)/2}) \quad (d \geq 2). \quad (\text{Setup-Q.36})$$

For $d = 1$, (Setup-Q.29) gives $N_t = O(\log(en)^{5/2})$ and $N_x = 2$. Thus even the **total** number of real time/query pairs is polylogarithmic for fixed dimension:

$$N_t N_x = O_d(\log(en)^{3d/2+1}). \quad (\text{Setup-Q.37})$$

Node generation and harmonic evaluation have the explicit costs (Setup-Q.30)–(Setup-Q.31). The continuation, source projection, selection and assembly costs are proved below in the complete source-jet bridge. All quadrature and dense setup arrays are discarded under the original retained-storage convention.

For the arbitrary supplied-budget construction in the preceding Harmonic construction, substitute its actual $T = T_0 + 4mu/\gamma$, $\eta = \eta_0 e^{-u}$ into (Setup-Q.22). If u grows faster than $\log n$, the claim (Setup-Q.36) need not hold. The finite quantitative rule and preservation of that budget's rank/error certificate still hold. There is no silent restriction of the forward theorem to polynomial target accuracies.

E.10.3 Local Taylor continuation from computed anchors

Model, imported event, and claimed output

Write $v = x/\sqrt{d}$, so the query sphere is $\|v\|_2 = 1$. The network, its backward fields, and the normalized parameter coordinates are

$$z^{(1)} = Av, \quad z^{(j)} = W^{(j)} h^{(j-1)}, \quad h^{(j)} = \phi_j(z^{(j)}), \quad f_n = w^T h^{(L)}/n,$$

$$k^{(L)} = w, \quad \delta^{(j)} = \phi'_j(z^{(j)}) \odot k^{(j)}, \quad k^{(j)} = (W^{(j+1)})^T \delta^{(j+1)},$$

$$\theta = (A/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n}) \in \mathbb{R}^P, \quad P = nd + (L-1)n^2 + n. \quad (\text{Setup-C.1})$$

Let $r_a = f_n(v_a) - y_a$, $\rho = \|r\|_2/\sqrt{m}$, $Y = \|y\|_2/\sqrt{m}$, $\lambda = \gamma/m$, and $S = 16Y/\lambda$. With $\xi_a = \nabla_\theta f_n(v_a)$, the mean-square-loss flow with the prescribed mobilities is

$$\dot{\theta} = F(\theta), \quad F(\theta) = -\frac{2}{m} \sum_a r_a \xi_a(\theta). \quad (\text{Setup-C.2})$$

The complex extension of this equation uses the algebraic transpose and no complex conjugations. Norm bounds below are ordinary complex Euclidean or Frobenius bounds; they do not use positivity of a complex Gram matrix.

Assume $Y > 0$, the full original common label interval, and the original source and analytic-extension width gates. All activations are the original strip-holomorphic activations, with possibly unbounded values. Let $b = \max_j |\phi_j(0)|$, and let $s \geq 1, t_2 \geq 1$ bound their first and second derivatives on $|\operatorname{Im} z| \leq a/2$. The imported event gives, on the real trajectory,

$$\|A\|_{\text{op}}/\sqrt{n} < 9, \quad \|W^{(j)}\|_{\text{op}} < 9, \quad \|w\|_2/\sqrt{n} \leq 2Y/\sqrt{\lambda},$$

$$\rho(t) \leq Y e^{-\lambda t/2}, \quad \int_0^\infty \rho(t) dt \leq 2Y/\lambda, \quad \max_{a,j,i,t \geq 0} |k_{a,i}^{(j)}(t)| \leq 2K_{\text{src}} S \sqrt{\log(en)}. \quad (\text{Setup-C.3})$$

For every finite source horizon, the exact trajectory is holomorphic on the source time neighborhood with radius $r_t = c_t/\sqrt{\log(en)}$, including its endpoint neighborhoods. Its joint passive-query extension has preactivation imaginary parts at most $3a/8$, operator caps ten, feature RMS bounds H_j^{src} , and readout RMS at most SH_L^{src} . Along every complex disk of radius $r_t/2$ centered on a nonnegative real time, its residual RMS is at most $2Y$. These are the source-domain and late-extension conclusions of the preceding Harmonic construction.

The task of this component is to supply at any finite list of real time/query nodes all four families

$$h^{(j)}, \quad W_0^{(j)} h^{(j-1)}, \quad \delta^{(j)}, \quad W_0^{(j+1)T} \delta^{(j+1)}, \quad (\text{Setup-C.4})$$

with the existing boundary-layer omissions, with coordinate error at most a given $\delta_{\text{node}} > 0$, and with exact initialized-matrix pairing. For the certified quadrature route one uses

$$\delta_{\text{node}} = \frac{\eta}{64N A_d \mathcal{Y}_{\ell_*}^2} \quad (d \geq 2), \quad \delta_{\text{node}} = \frac{\eta}{64N_1} \quad (d = 1). \quad (\text{Setup-C.5})$$

Here $N, N_1, A_d, \mathcal{Y}_{\ell_*}$ are exactly the quantities defined in positive coefficient quadrature. In its fixed-parameter polynomial accuracy regime, $\log(1/\delta_{\text{node}}) = O(\log(en))$.

Complex endpoint estimates with explicit coefficients

The constants in this section apply to complex parameters with operator caps eleven, readout RMS at most $R = 1 + SH_L^{\text{src}}$, and preactivations in the safe half-strip. The extra margin is used only for the numerical restart proof. Define

$$\begin{aligned} H_1 &= \max(1, b + 11s), & H_j &= \max(1, b + 11sH_{j-1}), \\ B_j &= Rs(11s)^{L-j}, & P_1 &= 1, \quad P_j = H_{j-1} + 11sP_{j-1}, \quad P_* = \max_j P_j. \end{aligned} \quad (\text{Setup-C.6})$$

For $M \geq 0$, define

$$\begin{aligned} \kappa_L(M) &= 1, \\ \kappa_j(M) &= 11s\kappa_{j+1}(M) + 11t_2MP_{j+1} + B_{j+1}, \\ D_j(M) &= s\kappa_j(M) + t_2MP_j, \quad \kappa_*(M) = \max_j \kappa_j(M), \\ G &= H_L + B_1 + \sum_{j=2}^L B_j H_{j-1}, \\ J(M) &= sP_L + D_1(M) + \sum_{j=2}^L \{H_{j-1}D_j(M) + B_j sP_{j-1}\}. \end{aligned} \quad (\text{Setup-C.7})$$

Every coefficient is explicit, and κ_*, D_j, J are bounded by affine functions of M , with fixed structural coefficients.

Let u, v be two such states, $E = \|u - v\|_2$, and suppose only the carriers at v have coordinate bound M . Subtracting the forward recursion gives

$$\frac{\|z^{(j)}(u) - z^{(j)}(v)\|_2}{\sqrt{n}} \leq P_j E, \quad \frac{\|h^{(j)}(u) - h^{(j)}(v)\|_2}{\sqrt{n}} \leq sP_j E. \quad (\text{Setup-C.8})$$

For the first layer this uses $\|\Delta A\|_F/\sqrt{n} \leq E$. For a later layer use

$$z^{(j)}(u) - z^{(j)}(v) = W^{(j)}(u)\Delta h^{(j-1)} + \Delta W^{(j)}h^{(j-1)}(v),$$

whose two normalized bounds are $11sP_{j-1}E$ and $H_{j-1}E$. The straight scalar segment between each pair of preactivations remains in the half-strip, justifying the complex derivative bound used here.

For the backward subtraction use the carrier at v :

$$\Delta\delta^{(j)} = \phi'_j(z^{(j)}(u)) \odot \Delta k^{(j)} + [\phi'_j(z^{(j)}(u)) - \phi'_j(z^{(j)}(v))] \odot k^{(j)}(v).$$

The second term has RMS at most t_2MP_jE . For $j < L$,

$$\Delta k^{(j)} = (W^{(j+1)}(u))^T \Delta\delta^{(j+1)} + (\Delta W^{(j+1)})^T \delta^{(j+1)}(v).$$

The two costs are $11D_{j+1}(M)E$ and $B_{j+1}E$, while the top readout cost is E . Induction proves

$$\frac{\|\Delta k^{(j)}\|_2}{\sqrt{n}} \leq \kappa_j(M)E, \quad \frac{\|\Delta \delta^{(j)}\|_2}{\sqrt{n}} \leq D_j(M)E. \quad (\text{Setup-C.9})$$

The gradient blocks are

$$\xi_a = \left(\delta_a^{(1)} v_a^T / \sqrt{n}, (\delta_a^{(j)} h_a^{(j-1)T} / n)_{j=2}^L, h_a^{(L)} / \sqrt{n} \right).$$

Their norms are bounded by $B_1, B_j H_{j-1}, H_L$. In a hidden block the difference splits into $\Delta \delta h(u)^T + \delta(v) \Delta h^T$; (Setup-C.8)–(Setup-C.9) then give

$$\|\xi_a(u)\|_2 \leq G, \quad \|\xi_a(u) - \xi_a(v)\|_2 \leq J(M)E. \quad (\text{Setup-C.10})$$

If the parameter segment from v to u is in the same admissible domain, integration of the first gradient bound also gives $|f_a(u) - f_a(v)| \leq GE$. Consequently

$$\|F(u) - F(v)\|_2 \leq [2G^2 + 2\rho(v)J(M)]\|u - v\|_2. \quad (\text{Setup-C.11})$$

Indeed subtract (Setup-C.2) as $-2m^{-1} \sum_a [(r_a(u) - r_a(v))\xi_a(u) + r_a(v)(\xi_a(u) - \xi_a(v))]$, and apply sample Cauchy–Schwarz to both sums. Equations (Setup-C.8)–(Setup-C.11) require a carrier bound at one endpoint, not a hypothesized bound on an unknown numerical trajectory.

Uniform complex carriers for the true reference

The early source event supplies the complex carrier maximum

$$M_0 = K_{\text{src}} S \sqrt{\log(en)}$$

through $T_0 = 32\lambda^{-1} \log(en)$, including the short complex time pieces. The late-extension proof puts every later complex disk in the fixed safe parameter ball about $\theta(T_0)$ and gives

$$\|\theta(\zeta) - \theta(T_0)\|_2 \leq Z_n, \quad Z_n = \left(\frac{2Y}{\sqrt{\lambda}} + 8Y\sqrt{\mathcal{K}} r_t \right) (en)^{-16}. \quad (\text{Setup-C.12})$$

Here $\mathcal{K}$ is the explicit source Gram bound in the preceding Harmonic construction. The existing extension gate makes this distance less than half the radius of that ball. Its straight parameter segments are in the safe half-strip. Use (Setup-C.9), anchored at $\theta(T_0)$, and convert RMS to a coordinate bound. Thus at every real anchor's disk of radius $r_t/2$, for all training carriers,

$$\max_{a,j,i} |k_{a,i}^{(j)}(\zeta)| \leq M_c, \quad M_c = M_0 + \sqrt{n} \kappa_*(M_0) Z_n. \quad (\text{Setup-C.13})$$

At fixed positive admissible labels and other structural parameters, $M_c = O(\sqrt{\log(en)})$. This derivation fills the possible late-complex-carrier gap without another stochastic event or another label restriction. It does not assert that arbitrary points in the full late parameter ball have that maximum.

A complex restart disk from a nearby real anchor

Define the parameter tube radius, Lipschitz coefficient, disk radius, and velocity bound by

$$r_{\text{tube}} = \min \left\{ \frac{1}{4}, \frac{a}{16\sqrt{n}P_*}, \frac{1}{\sqrt{n}\kappa_*(M_c)} \right\},$$

$$L_n = 2G^2 + 2(2Y + G)J(M_c + 1), \quad R_n = \min\{r_t/2, (4L_n)^{-1}\}, \quad V = 2G(2Y + G). \quad (\text{Setup-C.14})$$

The numerical restart radius r_{tube} is distinct from the late-extension radius b_{ext} in the Harmonic construction. At fixed parameters,

$$r_{\text{tube}}^{-1} = O(\sqrt{n \log(en)}), \quad L_n = O(\sqrt{\log(en)}), \quad R_n^{-1} = O(\sqrt{\log(en)}). \quad (\text{Setup-C.15})$$

Fix a real time $t \geq 0$. On $|\zeta| \leq R_n$, consider the moving parameter balls centered at $\theta(t + \zeta)$ with radius r_{tube} . Every such ball lies in the activation domain. To prove this, follow a straight parameter segment from its center, stopped at first half-strip exit. The forward bounds (Setup-C.8) apply before that exit, and each coordinate changes by at most $\sqrt{n}P_*r_{\text{tube}} \leq a/16$. Its center has imaginary parts at most $3a/8$, so the result stays below $7a/16 < a/2$ and the alleged first exit cannot occur. Operator caps eleven and readout cap R also hold since the change is at most $1/4$.

Equation (Setup-C.9) and (Setup-C.13) show that every state in each moving ball has carrier maximum at most $M_c + \sqrt{n}\kappa_*(M_c)r_{\text{tube}} \leq M_c + 1$. The segment between any two states in the ball remains in it. Their residual RMS is at most $2Y + Gr_{\text{tube}} \leq 2Y + G$. Hence (Setup-C.11) proves the uniform, genuine pairwise Lipschitz bound L_n throughout each ball. This is where control at the exact complex endpoint is converted into a usable neighborhood for numerical restarts.

Let the real restart value u_0 satisfy

$$\|u_0 - \theta(t)\|_2 \leq r_{\text{tube}}/4. \quad (\text{Setup-C.16})$$

On the Banach space of vector functions continuous on the closed disk and holomorphic inside it, with supremum norm at most r_{tube} , define

$$(\mathcal{T}e)(\zeta) = u_0 - \theta(t) + \int_0^\zeta [F(\theta(t+z) + e(z)) - F(\theta(t+z))] dz. \quad (\text{Setup-C.17})$$

The integrand is holomorphic, so its integral is independent of path in the disk; bounding it along the straight radius gives

$$\|\mathcal{T}e\|_\infty \leq r_{\text{tube}}/4 + L_n R_n r_{\text{tube}} \leq r_{\text{tube}}/2, \quad \|\mathcal{T}e - \mathcal{T}\tilde{e}\|_\infty \leq \frac{1}{4}\|e - \tilde{e}\|_\infty.$$

Iteration is Cauchy, converges uniformly on the disk, and its limit remains holomorphic by uniform convergence on every smaller circle and the Cauchy integral formula. Passing to the integral equation proves existence of its fixed point; the same contraction inequality proves uniqueness. Consequently $v_t(\zeta) = \theta(t + \zeta) + e(\zeta)$ solves (Setup-C.2) and starts at u_0 . Moreover,

$$\|v_t - \theta(t + \cdot)\|_\infty \leq \frac{4}{3}\|u_0 - \theta(t)\|_2 \leq r_{\text{tube}}/3, \quad \|v'_t\|_\infty \leq V. \quad (\text{Setup-C.18})$$

This proof establishes the complex restart disk directly; it does not assume that analyticity about the exact anchor automatically transfers to a perturbed anchor. Conjugating the equation and using uniqueness shows v_t is real on the real diameter. The exact reference is only a proof device: the algorithm below computes coefficients from u_0 and the initialized data.

Taylor tail, continuous defect, and global error budget

Let p_t be the degree- K Taylor polynomial of v_t about zero. Since $\|v_t(\zeta) - u_0\|_2 \leq VR_n$, its coefficients obey $\|[\zeta^k]v_t\|_2 \leq VR_n^{1-k}$ for $k \geq 1$, by the vector-valued Cauchy integral formula. On $0 \leq h \leq R_n/2$, summing the resulting geometric series and its derivative gives

$$\|p_t(h) - v_t(h)\|_2 \leq VR_n 2^{-K}, \quad \|p'_t(h) - v'_t(h)\|_2 \leq V(2K+4)2^{-K}. \quad (\text{Setup-C.19})$$

If $VR_n 2^{-K} \leq r_{\text{tube}}/3$, the polynomial and the exact restarted solution stay in the same moving ball, so (Setup-C.11) applies to them. Since $L_n R_n \leq 1/4$, their continuous ODE defect satisfies

$$\|p'_t(h) - F(p_t(h))\|_2 \leq V(2K+5)2^{-K}. \quad (\text{Setup-C.20})$$

For clarity, the real defect-stability constants imported from signed defect stability are reproduced here with a superscript r so they are not confused with (Setup-C.6)–(Setup-C.7). Put

$$H_1^r = \max(1, b + 10s), \quad H_j^r = \max(1, b + 10sH_{j-1}^r), \quad H^r = H_L^r,$$

$$R^r = 1 + 2Y/\sqrt{\lambda}, \quad B^r = R^r s(10s)^{L-1}, \quad F_z^r = H^r(10s)^{L-1}, \quad P_{\text{layer}}^r = 1 + (L-1)H^r,$$

$$D^r(M) = (10s)^{L-1} \{s(1 + B^r) + Lt_2 F_z^r M\},$$

$$J^r(M) = \sqrt{L+1} \{P_{\text{layer}}^r D^r(M) + [1 + (L-1)B^r] s F_z^r\},$$

$$C^r(M) = s F_z^r \sqrt{L} + L B^r s F_z^r + \frac{1}{2} L^2 t_2 (F_z^r)^2 M. \quad (\text{Setup-C.21})$$

Use $M = 2K_{\text{src}} S \sqrt{\log(en)}$, and abbreviate the last two values by J^r, C^r . Define

$$A_n = 4J^r Y/\lambda, \quad K_q = R^r(10s)^{L-1},$$

$$C_{\text{src}} = 8\sqrt{L+1} \max\{s F_z^r, (10s)^{L-1} [s(1 + B^r) + Lt_2 F_z^r K_q]\}. \quad (\text{Setup-C.22})$$

The real lemma states: for an absolutely continuous numerical path u , let $d = u' - F(u)$ and $E_0 = \|u(0) - \theta(0)\|_2 + \int_0^T \|d\|_2 dt$. If

$$E_0 \leq e^{-A_n}/4, \quad 8(C^r + J^r)^2 T e^{2A_n} E_0^2 \leq 1, \quad (\text{Setup-C.23})$$

then $\sup_{[0,T]} \|u - \theta\|_2 \leq 2e^{A_n} E_0$. Its proof keeps the negative sample prediction-error square before applying the scalar energy inequality; it is not the exponential of a worst-case Jacobian times T . The signed-stability proof above proves each coordinate of (Setup-C.4), including initialized images, has error at most $C_{\text{src}} n \|u - \theta\|_2$ on the whole real query sphere. This uses only deterministic query RMS bounds, and hence does not need passive-query carrier maxima.

For $T > 0$, choose the following total allowed defect:

$$\varepsilon_d = e^{-A_n} \min \left\{ \frac{1}{4}, \frac{1}{\sqrt{8}(C^r + J^r)\sqrt{T}}, \frac{\delta_{\text{node}}}{2C_{\text{src}} n}, \frac{r_{\text{tube}}}{8} \right\}. \quad (\text{Setup-C.24})$$

Choose the explicit integer

$$K = \max \{8, \lceil 2 \log_2 \max\{1, 4TV/\varepsilon_d\} \rceil, \lceil \log_2 \max\{1, 3VR_n/r_{\text{tube}}\} \rceil \}. \quad (\text{Setup-C.25})$$

For $K \geq 8$, $2K + 5 \leq 4 \cdot 2^{K/2}$: it holds at eight, and the ratio of the left side at $K + 1$ to its value at K is less than $\sqrt{2}$. Thus (Setup-C.25) ensures both

$$TV(2K + 5)2^{-K} \leq \varepsilon_d, \quad VR_n 2^{-K} \leq r_{\text{tube}}/3. \quad (\text{Setup-C.26})$$

Set $N_{\text{step}} = \lceil 2T/R_n \rceil$, and divide $[0, T]$ into that many equal panels. Initialize $u(0) = \theta(0)$. On each panel compute the degree- K Taylor polynomial from its already computed initial value, and take its endpoint as the next initial value. This creates a continuous, piecewise polynomial, absolutely continuous real path.

This procedure is well defined through the whole horizon. Inductively, all already constructed panels have defect bounded by (Setup-C.20), so their total defect is at most ε_d . The real stability lemma, applied to that prefix with the conservative full-horizon constants in (Setup-C.23)–(Setup-C.24), gives

$$\|u(t) - \theta(t)\|_2 \leq 2e^{A_n} \varepsilon_d \leq r_{\text{tube}}/4 \quad (\text{Setup-C.27})$$

at its final endpoint. Thus the next panel satisfies the restart hypothesis (Setup-C.16), and (Setup-C.19)–(Setup-C.20) give the same defect bound for that panel. The initial case is exact. Finite induction constructs every panel; no assumption about the yet-uncomputed numerical path has been used. Applying the same estimate on the completed interval proves

$$\sup_{0 \leq t \leq T} \|u(t) - \theta(t)\|_2 \leq \frac{\delta_{\text{node}}}{C_{\text{src}} n}. \quad (\text{Setup-C.28})$$

For polynomial nodal accuracy and $T = O(\log(en))$, (Setup-C.15), (Setup-C.21)–(Setup-C.24) give

$$\log(1/\varepsilon_d) = O(\log(en)), \quad K = O(\log(en)), \quad N_{\text{step}} = O((\log(en))^{3/2}). \quad (\text{Setup-C.29})$$

The full dependence on depth, sample count, gap, labels, strip width and activation bounds remains in the displayed constants. No additional smallness condition on the labels was imposed. The $Y = 0$ case has stationary zero readout and zero predictor and uses the original exact zero-label branch.

Computable Taylor coefficients and paired source values

On a panel write $u(h) = \sum_{k=0}^K u_k h^k$. Compute successively

$$u_{k+1} = \frac{[h^k]F(\sum_{i=0}^k u_i h^i)}{k+1}, \quad 0 \leq k < K. \quad (\text{Setup-C.30})$$

At stage k , every input coefficient on the right side has already been computed. The forward pass, backward pass and rank-one updates in (Setup-C.1)–(Setup-C.2) give its coefficient by finite additions, scalar multiplications and truncated series compositions. Induction identifies these coefficients with the Taylor coefficients of the unique local solution proved above. The algorithm needs neither exact anchors nor samples of the unknown exact path.

At a requested real time node, evaluate the current panel polynomial by Horner's rule and perform the ordinary dense forward/backward pass at each requested passive query. Form every paired image by multiplying that computed base vector by its initialized matrix. For example, define

$$\tilde{h}^{(j)} = h^{(j)}(u(t), v), \quad \widetilde{W_0^{(j)} h^{(j-1)}} = W_0^{(j)} \tilde{h}^{(j-1)}, \quad (\text{Setup-C.31})$$

and do the same for the reverse image of $\tilde{\delta}$. Equations (Setup-C.22), (Setup-C.28) give the required separate coordinate error δ_{node} for every family. Pairing is an exact algebraic identity. Applying identical scalar quadrature, cosine normalization and mode restriction to both members preserves it. Passive-query Taylor jets are not needed.

The numerical path need not itself be globally analytic across restart points. Quadrature accuracy is proved for the exact source function, and its finite sum is then perturbed by the nodal error (Setup-C.5). Therefore the piecewise representation does not invalidate the analytic coefficient-quadrature proof.

E.10.4 Real-value activation backend

Domain to be approximated

Use $P_*, G, L_n, r_{\text{tube}}, V, M_c, Z_n$ from equations (Setup-C.6)–(Setup-C.14) of the local-continuation proof, with the $r_{\text{tube}} \leq a/(16\sqrt{n}P_*)$. Thus the exact network throughout every moving parameter ball has preactivation imaginary parts at most $7a/16$.

We need a bound on preactivation real parts for every real query, including when the parameters are on a complex time disk. Let $H_{\text{max}}^{\text{src}} = \max_j H_j^{\text{src}}$, $G_d = 16\sqrt{d+3}$, and let U be the explicit source coefficient in (S.25) of the preceding Harmonic construction. One sufficient bound at the exact disk centers is

$$Z_* = (K_{\text{src}} + G_d H_{\text{max}}^{\text{src}} + 1 + US^2) \sqrt{\log(en)} + a/8 + \sqrt{n}P_*Z_n. \quad (\text{Setup-A.1})$$

Indeed the initial whole-sphere coordinate bound is $(G_d H_{\text{max}}^{\text{src}} + 1) \sqrt{\log(en)}$. The real velocity bound $|\partial_t z_i^{(j)}| \leq 2\rho S U \sqrt{\log(en)}$, integrated using the source activity bound $\int 2\rho dt \leq S/2$, adds at most $US^2 \sqrt{\log(en)}$. The short complex time pieces add at most $8YSUr_t \sqrt{\log(en)} = a/8$. After T_0 , compare with $\theta(T_0)$ in the late safe ball; the forward endpoint estimate adds at most $\sqrt{n}P_*Z_n$. The extra $K_{\text{src}} \sqrt{\log(en)}$ in (Setup-A.1) also covers the early training-coordinate maximum. Only real query vectors are needed here; no polynomial is evaluated on a complex query sphere.

Every original preactivation at a state in one of the moving balls therefore satisfies

$$|\operatorname{Re} z| \leq Z_* + a/16, \quad |\operatorname{Im} z| \leq 7a/16. \quad (\text{Setup-A.2})$$

At fixed structural parameters, $Z_* = O(\sqrt{\log(en)})$. Define

$$B = 8(Z_* + a + 1), \quad \tau_o = \operatorname{arsinh} \frac{31a}{64B}, \quad \tau_i = \operatorname{arsinh} \frac{29a}{64B}, \quad \Delta = \tau_o - \tau_i. \quad (\text{Setup-A.3})$$

For $\tau > 0$, write $\mathcal{E}_\tau$ for the filled ellipse with boundary $z = B(w + w^{-1})/2$, $|w| = e^\tau$. Its semiaxes are $B \cosh \tau$ and $B \sinh \tau$. The outer ellipse is strictly inside the activation half-strip because its imaginary semiaxis is $31a/64 < a/2$. Moreover $B \geq 8a$, so

$$0 < \Delta < 1, \quad \Delta \geq \frac{a}{64B}, \quad \tau_o < 1. \quad (\text{Setup-A.4})$$

For the lower bound, integrate the derivative of $\operatorname{arsinh} x$ between $29a/(64B)$ and $31a/(64B)$; it is at least $1/2$ on that interval.

The inner ellipse contains every disk of radius $a/512$ around every point at distance at most $a/512$ from a point of (Setup-A.2). Its real coordinate divided by its real semiaxis is at most $1/8$, while its imaginary coordinate divided by the imaginary semiaxis is at most $(7/16 + 2/512)/(29/64) = 226/232$. Their squared sum is less than one. This verifies the domain for activation substitution and derivative estimation below, with strict margin.

Explicit real-node Chebyshev construction

Let $b = \max_j |\phi_j(0)|$ and let s be the original first-derivative bound on the half-strip. On $\mathcal{E}_{\tau_o}$,

$$|\phi_j(z)| \leq M_\phi, \quad M_\phi = b + s(B + a). \quad (\text{Setup-A.5})$$

Integrate the bounded derivative on the segment from zero to z , which lies in the half-strip, and use $|z| \leq B \cosh \tau_o \leq B + a$.

Fix a desired value/first/second derivative accuracy $\epsilon > 0$, and put

$$C_a = \max\{1, 512/a, 2(512/a)^2\}, \quad \epsilon_p = \epsilon/C_a,$$

$$D = \max\{1, \lceil \Delta^{-1} \log \max\{e, 24M_\phi/(\epsilon_p \Delta)\} \rceil\}, \quad N_\phi = 4(D + 1). \quad (\text{Setup-A.6})$$

At the real nodes $x_r = B \cos(2\pi r/N_\phi)$, evaluate the original activation. Define

$$\begin{aligned} \tilde{c}_0 &= \frac{1}{N_\phi} \sum_{r=0}^{N_\phi-1} \phi_j(x_r), \quad \tilde{c}_k = \frac{2}{N_\phi} \sum_{r=0}^{N_\phi-1} \phi_j(x_r) \cos(2\pi k r/N_\phi), \quad 1 \leq k \leq D, \\ \psi_j(z) &= \sum_{k=0}^D \tilde{c}_k T_k(z/B), \end{aligned} \quad (\text{Setup-A.7})$$

where $T_k(\cos u) = \cos(ku)$. Thus ψ_j has real coefficients and is computed from N_ϕ real calls to ϕ_j .

Here is the error estimate, including the finite coefficient rule. For $g(u) = \phi_j(B \cos u)$, shifting a Fourier contour within $|\operatorname{Im} u| \leq \tau_o$ gives $|\hat{g}_k| \leq M_\phi e^{-|k|\tau_o}$. The function is even. Its Chebyshev coefficients are $c_0 = \hat{g}_0$ and $c_k = 2\hat{g}_k$ for $k \geq 1$. Also $|T_k(z/B)| \leq e^{k\tau_i}$ on $\mathcal{E}_{\tau_i}$, by its Laurent formula on the ellipse boundary and the maximum modulus principle. Consequently the degree- D exact coefficient tail is at most

$$\frac{2M_\phi e^{-(D+1)\Delta}}{1 - e^{-\Delta}} \leq \frac{4M_\phi}{\Delta} e^{-(D+1)\Delta}. \quad (\text{Setup-A.8})$$

Absolute Fourier convergence permits interchange with the finite sum. The discrete Fourier coefficient is $\sum_{q \in \mathbb{Z}} \hat{g}_{k+qN_\phi}$, since the discrete sum of $e^{i\ell u}$ vanishes unless N_ϕ divides ℓ . For $0 \leq k \leq D < N_\phi/2$, this gives

$$|\tilde{c}_k - c_k| \leq \frac{4M_\phi e^{-(N_\phi-D)\tau_o}}{1 - e^{-N_\phi\tau_o}}.$$

The factor four safely includes the constant coefficient. Since $(D+1)\Delta \geq 1$, the denominator is at least $1/2$. Summing the coefficient error on the inner ellipse and using $N_\phi = 4(D+1)$ yields

$$\left| \sum_{k=0}^D (\tilde{c}_k - c_k) T_k(z/B) \right| \leq 8M_\phi(D+1)e^{-2(D+1)\tau_o} \leq \frac{8M_\phi}{\Delta} e^{-(D+1)\Delta}. \quad (\text{Setup-A.9})$$

For the last inequality use $xe^{-x} \leq 1$ with $x = (D+1)\tau_o$, and $\tau_o \geq \Delta$. Equations (Setup-A.6), (Setup-A.8)–(Setup-A.9) show that $\sup_{\mathcal{E}_{\tau_i}} |\psi_j - \phi_j| \leq \epsilon_p/2$.

The reserved half also permits inexact real value calls: if each value in (Setup-A.7) has absolute error at most

$$\epsilon_{\text{eval}} = \frac{\epsilon_p}{4(D+1)e^{D\tau_i}}, \quad (\text{Setup-A.10})$$

the resulting extra polynomial error on the inner ellipse is at most $\epsilon_p/2$. Bound every coefficient perturbation by $2\epsilon_{\text{eval}}$ and sum $D+1$ terms.

Cauchy's integral formula on the disks checked after (Setup-A.4) gives, at every point at distance at most $a/512$ from the rectangle (Setup-A.2),

$$|\psi_j - \phi_j| \leq \epsilon, \quad |\psi'_j - \phi'_j| \leq \epsilon, \quad |\psi''_j - \phi''_j| \leq \epsilon. \quad (\text{Setup-A.11})$$

The derivative factors are $512/a$ and $2(512/a)^2$, precisely those included in C_a . Exact calls make (Setup-A.10) unnecessary; finite-accuracy calls only require polynomially small error in the regime analyzed below.

Error in the training vector field

Replace every activation by ψ_j only in a disposable setup network, and call its gradient-flow vector field $\tilde{F}$. We compare this polynomial field to F at the same parameter value u in a moving ball of the local-continuation proof. Use the core coefficients H_j, B_j , and define

$$U_0 = 0, \quad U_j = 1 + 11sU_{j-1}, \quad U_* = \max_j U_j, \quad K_j = R(11s)^{L-j}. \quad (\text{Setup-A.12})$$

Superscripts 0 and p below distinguish the original and polynomial activation evaluations at this one parameter state. If

$$\epsilon \leq \min \left\{ 1, U_*^{-1}, \frac{a}{5632\sqrt{n}U_*} \right\}, \quad (\text{Setup-A.13})$$

then induction over the layers gives

$$\frac{\|h^{p,(j)} - h^{0,(j)}\|_2}{\sqrt{n}} \leq U_j \epsilon, \quad \frac{\|z^{p,(j)} - z^{0,(j)}\|_2}{\sqrt{n}} \leq 11U_{j-1} \epsilon. \quad (\text{Setup-A.14})$$

The first preactivations agree. At the next layer the unchanged matrix costs eleven, and the activation difference splits into its error at the polynomial argument plus the original activation's s -Lipschitz change. Every preactivation coordinate discrepancy is at most $11\sqrt{n}U_*\epsilon \leq a/512$, so (Setup-A.11) applies at every induction step. Also $\|h^{p,(j)}\|_2/\sqrt{n} \leq H_j + 1$.

For $M \geq 0$, define backward error coefficients by descending recursion:

$$Q_{L+1}(M) = 0, \quad Q_j(M) = 11(s+1)Q_{j+1}(M) + K_j + 11t_2MU_{j-1}. \quad (\text{Setup-A.15})$$

If the original carriers at u have maximum M , then

$$\frac{\|\delta^{p,(j)} - \delta^{0,(j)}\|_2}{\sqrt{n}} \leq Q_j(M)\epsilon. \quad (\text{Setup-A.16})$$

The changed carrier costs $11Q_{j+1}\epsilon$ at $j < L$, while at $j = L$ it is zero. The polynomial gate has modulus at most $s + 1$. The gate discrepancy multiplied by the original carrier contributes $\epsilon K_j + t_2 M(11U_{j-1}\epsilon)$, by (Setup-A.11) and (Setup-A.14). These are exactly the terms in (Setup-A.15).

For the training vector field take $M = M_c + 1$. Set

$$Q_\xi = U_L + Q_1(M) + \sum_{j=2}^L \{Q_j(M)(H_{j-1} + 1) + B_j U_{j-1}\}, \quad Q_f = R U_L, \\ C_F = 2Q_f(G + 1) + 2(2Y + G)Q_\xi. \quad (\text{Setup-A.17})$$

Subtracting the gradient blocks gives $\|\tilde{\xi}_a - \xi_a\|_2 \leq Q_\xi \epsilon$. Subtracting the predictions gives $|\tilde{f}_a - f_a| \leq Q_f \epsilon$. If also $Q_\xi \epsilon \leq 1$, the polynomial gradient norm is at most $G + 1$. Splitting the factors in the residual-gradient product in (Setup-C.2) of the local-continuation proof, and using the original residual bound $2Y + G$, proves

$$\sup_{\text{moving ball}} \|\tilde{F} - F\|_2 \leq C_F \epsilon. \quad (\text{Setup-A.18})$$

All training constants in (Setup-A.15)–(Setup-A.18) grow at most like $O(\sqrt{\log(en)})$ at fixed structural parameters.

Modified restart and a combined defect budget

Keep the real signed-stability constants A_n, C^r, J^r and C_{src} from the local-continuation proof. For a positive source horizon T and target nodal error δ_{node} , define

$$e_d = e^{-A_n} \min \left\{ \frac{1}{4}, \frac{1}{\sqrt{8}(C^r + J^r)\sqrt{T}}, \frac{\delta_{\text{node}}}{4C_{\text{src}}n}, \frac{r_{\text{tube}}}{16} \right\}, \\ \zeta_F = \min\{e_d/(2T), r_{\text{tube}}L_n/4, V/2\}, \quad \hat{R}_n = \min\{r_t/2, (8L_n)^{-1}\}. \quad (\text{Setup-A.19})$$

Choose ϵ satisfying (Setup-A.13), $Q_\xi \epsilon \leq 1$, and $C_F \epsilon \leq \zeta_F$. A further source-output requirement is imposed in the next section. Construct the activation polynomials once using (Setup-A.6)–(Setup-A.7).

There is a uniform derivative bound for the field perturbation in each ball of radius $r_{\text{tube}}/2$ about the exact complex reference:

$$\|D(\tilde{F} - F)\|_{\text{op}} \leq 2\zeta_F/r_{\text{tube}}. \quad (\text{Setup-A.20})$$

For a unit complex parameter direction, the disk of radius $r_{\text{tube}}/2$ about any such state remains in the original ball of radius r_{tube} . Apply the vector-valued Cauchy integral formula to $\tilde{F} - F$, using (Setup-A.18). This proves (Setup-A.20) without a dimension-dependent conversion of coordinate derivatives. The polynomial field is therefore $2L_n$ -Lipschitz on that smaller ball.

The exact restarted solution of $\dot{v} = \tilde{F}(v)$ exists on the complex disk of radius $\hat{R}_n$ whenever its real initial anchor has distance at most $r_{\text{tube}}/8$ from the true original trajectory. Repeat the core Picard map for the error, now with integrand $\tilde{F}(\theta + e) - F(\theta)$ and error ball of radius $r_{\text{tube}}/2$. Its contraction factor is at most $2L_n\hat{R}_n \leq 1/4$. Its forcing at $e = 0$ is at most ζ_F . The map sends that ball to radius at most

$$r_{\text{tube}}/8 + \hat{R}_n\zeta_F + (1/4)(r_{\text{tube}}/2) \leq 9r_{\text{tube}}/32 < r_{\text{tube}}/2.$$

The fixed point has error at most $(r_{\text{tube}}/8 + r_{\text{tube}}/32)/(1 - 1/4) = 5r_{\text{tube}}/24 < r_{\text{tube}}/4$. Its velocity is at most $V + \zeta_F \leq 2V$. Thus its degree- K Taylor polynomial, on a half-radius real panel, has tail at most $2V\hat{R}_n 2^{-K}$. If this is at most $r_{\text{tube}}/4$, the polynomial remains inside the smaller ball. Its defect against the polynomial field is at most $2V(2K + 5)2^{-K}$, by the same differentiated geometric series as the core proof. Its defect against the original field is therefore at most

$$2V(2K + 5)2^{-K} + \zeta_F. \quad (\text{Setup-A.21})$$

Choose

$$K = \max \left\{ 8, \lceil 2 \log_2 \max\{1, 16TV/e_d\} \rceil, \left\lceil \log_2 \max\{1, 8V\widehat{R}_n/r_{\text{tube}}\} \right\rceil \right\}. \quad (\text{Setup-A.22})$$

The first error term in (Setup-A.21), integrated over the horizon, is at most $e_d/2$, because $2K + 5 \leq 4 \cdot 2^{K/2}$ for $K \geq 8$. The second term contributes at most $e_d/2$ by (Setup-A.19). Use $N_{\text{step}} = \lceil 2T/\widehat{R}_n \rceil$ panels and the explicit coefficient recurrence (Setup-C.30) of the local-continuation proof, with $\widehat{F}$ in place of F .

The prefix induction from the core proof now gives

$$\sup_{[0,T]} \|u(t) - \theta(t)\|_2 \leq 2e^{A_n} e_d \leq \min\{r_{\text{tube}}/8, \delta_{\text{node}}/(2C_{\text{src}}n)\}. \quad (\text{Setup-A.23})$$

It justifies every successive restart and reserves half the nodal error for evaluating sources with the activation polynomials.

Source values without derivative calls to the original activations

The original passive-query carrier RMS is at most K_j throughout the operator/readout tube. Hence its coordinate maximum is at most

$$M_q = \sqrt{n} \max_j K_j.$$

Use (Setup-A.15) with $M = M_q$, and define

$$C_p = 8\sqrt{n} \max\{U_*, \max_j Q_j(M_q)\}. \quad (\text{Setup-A.24})$$

Choose, finally, the single sufficient scalar tolerance

$$\epsilon = \min \left\{ 1, U_*^{-1}, \frac{a}{5632\sqrt{n}U_*}, Q_\xi^{-1}, \frac{\zeta_F}{C_F}, \frac{\delta_{\text{node}}}{2C_p} \right\}. \quad (\text{Setup-A.25})$$

All denominators are positive; Q_ξ, C_F in this expression are the training constants from (Setup-A.17).

At each time/query node evaluate the network at $u(t)$, using ψ_j, ψ'_j . Equations (Setup-A.14), (Setup-A.16), and RMS-to-coordinate conversion show that the coordinate error from replacing the activations is at most $\sqrt{n} \max(U_*, \max_j Q_j(M_q))\epsilon$. Initialized image actions cost at most eight in Euclidean norm, so every family, including each image, has error at most $C_p\epsilon \leq \delta_{\text{node}}/2$. The original-activation source at $u(t)$ differs from its true source at $\theta(t)$ by at most the other half, by (Setup-A.23) and the core source comparison. Every family has the required total coordinate tolerance.

Form initialized images directly from their computed base vectors. Pairing therefore remains exact, and identical scalar coefficient quadrature preserves it. The approximated training field and passive evaluations do not introduce independent approximations to members of a pair.

The exact initialized additions to the source spaces are a separate operation: evaluate their features with the original ϕ_j , and form their original matrix images exactly in the same exact-real model as the preceding Harmonic construction. These take $O(Lmn)$ real activation-value calls and $O(mP)$ arithmetic. Initial backward fields vanish because the readout is zero. No derivative call to the original activation is required by this setup backend. The final compressed network still evaluates the original activation and derivative during its specified runtime; the polynomials are discarded after setup.

Degree, cost, and the ordinary Euler comparison

Fix the full admissible structural parameters, take $T = O(\log(en))$, and let $\log(1/\delta_{\text{node}}) = O(\log(en))$. Then (Setup-A.19), (Setup-A.24)–(Setup-A.25) imply $\log(1/\epsilon) = O(\log(en))$: $C_p = O(n)$, the training error coefficients are $O(\sqrt{\log(en)})$, and $e^{A_n} = \exp(O(\sqrt{\log(en)}))$. Equations (Setup-A.3)–(Setup-A.6) give

$$D = O(\log(en)^{3/2}), \quad N_\phi = O(\log(en)^{3/2}), \quad K = O(\log(en)), \quad N_{\text{step}} = O(\log(en)^{3/2}). \quad (\text{Setup-A.26})$$

Also $D\tau_i = O(\log(en))$, so the sufficient value-call accuracy (Setup-A.10) is polynomially small in n . This is a precision requirement, not a proof about a particular floating-point library.

Computing the coefficients in (Setup-A.7) by cosine recurrences costs $O(LD^2)$ arithmetic and $O(LD)$ retained coefficient words. It uses $O(LD)$ scalar activation calls and trigonometric calls at real arguments. No complex activation values or high derivatives are requested.

Convert each Chebyshev polynomial to ordinary polynomial coefficients, or evaluate its truncated series by the three-term Chebyshev recurrence. Both give $O(DK^2)$ arithmetic for one online degree- K scalar composition, and $O(DK)$ sufficient workspace. Each of the D recurrence products is a convolution; accumulating its successive coefficients through K costs $\sum_{k \leq K} O(k) = O(K^2)$. The derivative polynomial can be prepared in $O(D^2)$ arithmetic and evaluated with the same bound.

Let N_t, N_x be the certified time/spatial quadrature node counts. Including polynomial passive-query evaluations, a sufficient source-value arithmetic bound is

$$O(LD^2 + N_{\text{step}}[mPK^2 + LmnDK^2] + PKN_t + PN_tN_x + LnDN_tN_x + mP). \quad (\text{Setup-A.27})$$

In addition, there are $O(LD + Lmn)$ original scalar activation-value calls and $O(LD)$ elementary trigonometric calls. A sufficient streamed memory bound, before adding the unchanged source coefficient/selection arrays, is

$$O(PK + LmnDK + LD + Ln). \quad (\text{Setup-A.28})$$

At fixed parameters all arithmetic and call counts in (Setup-A.27)–(Setup-A.28), and the certified quadrature/selection/assembly costs, are $n^{2+o(1)}$. If scalar activation values are unit-cost primitives, this is the total arithmetic bound. If a concrete value routine is supplied, add its explicitly requested costs at the arguments and tolerances above. Nothing assumes an arbitrary analytic activation has an efficient representation merely because it satisfies a strip bound.

For the specified Euler comparison, ordinary explicit Euler with numerical step h over physical horizon T takes $\Theta(mPT/h)$ arithmetic before query/source processing, up to its activation costs. If its accuracy requirement calls for $h = n^{-1/2}$ at fixed structural parameters, that work is $n^{5/2+o(1)}$; the present high-order setup has $n^{2+o(1)}$ work. This is a conditional comparison using the stated Euler step requirement. The theorem does not claim that every dense solver must use such a step, or prove a lower bound against high-order dense solvers. It shows that covering the full physical source horizon need not carry the width-dependent first-order time-step cost.

E.10.5 Streamed explicit assembly

Conditional input and unchanged output

Let L, m, d, n denote hidden depth, training sample count, input dimension and hidden width. Inputs are $v = x/\sqrt{d} \in S^{d-1}$. Retain the exact dense architecture, loss, mobilities, zero initial readout and full original label allowance from the preceding Harmonic construction. Thus, with $A = W^{(1)}$ and $w = W^{(L+1)}$,

$$z^{(1)} = Av, \quad z^{(j)} = W^{(j)}h^{(j-1)}, \quad h^{(j)} = \phi_j(z^{(j)}), \quad f_n = w^T h^{(L)}/n,$$

$$\delta^{(L)} = w \odot \phi'_L(z^{(L)}), \quad \delta^{(j)} = \phi'_j(z^{(j)}) \odot W^{(j+1)T} \delta^{(j+1)}.$$

A passive query uses the current trained parameters and does not enter the training residuals or parameter updates. Write

$$P = (L-1)n^2 + n(d+1)$$

for the dense parameter count. We use the existing cost convention $n \geq \max(m, d)$, numerical big-O constants, and exact-real arithmetic. Activation and Gaussian-generation costs are accounted for separately.

Take a permitted source horizon $T \geq T_0 > 0$ and tolerance $0 < \eta \leq \eta_0$. To avoid a collision with the number of panels, let ℓ_* denote the largest retained spherical degree, and put

$$h_\ell = \binom{\ell+d-1}{d-1} - \binom{\ell+d-3}{d-1}, \quad H_{\text{sph}} = \sum_{\ell=0}^{\ell_*} h_\ell, \quad \mathcal{Y} = \max_{\ell \leq \ell_*} \sqrt{h_\ell}$$

when $d \geq 2$. The original weighted simplex is retained exactly:

$$\Lambda = \{(k, \ell, b) : k \geq 0, \alpha_T k + r_q \ell \leq H(T, \eta), 1 \leq b \leq h_\ell\}, \quad N = |\Lambda|, \quad p = \max_{\Lambda} k. \quad (\text{Setup-E.1})$$

The source-generator bound, actual rank and selected-width condition remain

$$R = 2m + d + 1 + 4N, \quad r = \max_j \dim E_j \leq \min(n, R), \quad 9r \leq q < n. \quad (\text{Setup-E.2})$$

The original sufficient budget uses $9R \leq q$. For $d = 1$, use the two separate queries $v = \pm 1$, $N_x = 2$, $H_{\text{sph}} = 2$, $p + 1 = N_1$, and $R = 2m + d + 1 + 8N_1$; below N in rectangular work bounds may be replaced by $2N_1$. No angular quadrature is needed then.

The positive time/sphere rule of the quadrature proof has N_t time nodes $u_a = 2\pi a/N_t$, physical times $t_a = T(1 + \cos u_a)/2$, and N_x spatial nodes v_s . Its spatial weights, including the real angular Jacobian, are denoted ω_s . They are positive and satisfy

$$\sum_s \omega_s \leq A_d, \quad \epsilon_c = \frac{\eta}{16N\mathcal{Y}}, \quad \delta_{\text{node}} = \frac{\epsilon_c}{4A_d\mathcal{Y}}. \quad (\text{Setup-E.3})$$

Here A_d is the explicit product of polar normalization factors in the quadrature proof; it equals one in dimension two. In dimension one the existing separate-point definitions are $\epsilon_c = \eta/(16N_1)$ and $\delta_{\text{node}} = \epsilon_c/4$. Exact-value quadrature already has coefficient error at most $\epsilon_c/2$, and the exact source tail is at most $\eta/16$.

The conditional integrator supplies consecutive real panels

$$0 = \tau_0 < \tau_1 < \dots < \tau_J = T, \quad \Delta_b = \tau_{b+1} - \tau_b,$$

and an approximate dense anchor at each panel's left endpoint. The following are the precise required properties.

1. At panel b , normalized training Taylor coefficients through degree $K \geq 1$, generated from that anchor using the actual dense ODE, are available or can be generated by the recursion counted below. The same coefficients advance the anchor to the next panel. Only one anchor and its current panel data need remain resident.
2. For every needed passive query, the corresponding forward and backward source polynomials can be generated from those shared training jets:

$$\tilde{g}_b(t, v_s) = \sum_{c=0}^K G_{b,c}(v_s) ((t - \tau_b)/\Delta_b)^c, \quad g \in \{h^{(j)}, \delta^{(j)}\}. \quad (\text{Setup-E.4})$$

At each assigned quadrature node they satisfy

$$\|\tilde{g}_b(t_a, v_s) - g(t_a, v_s)\|_2 \leq \delta_{\text{node}}/8. \quad (\text{Setup-E.5})$$

This error includes anchor error, local-series truncation, and any approximation by the activation backend. A sufficient coordinatewise interface is $\delta_{\text{node}}/(8\sqrt{n})$ for every coordinate of each base source.

3. Panel construction, local error certification and any corrective work not included in the explicit recursions have total work C_{cert} and additional peak workspace M_{cert} . They must be proved small; neither implicit solves nor rejected panels are free.

Only the finite node set requires (Setup-E.5). A uniform local source estimate is a stronger sufficient input. The local-continuation and activation proofs above provide the numerical path; the source-jet bridge below proves this stronger Euclidean source interface for its polynomial-activation jets. Thus this conditional assembly lemma applies after that explicit interface replacement, with no extra numerical-path hypothesis.

Exact pairing with the initialized mixers

At a learned anchor the degree-zero mixer is $W_b^{(j)}$, whereas source pairing requires $W_0^{(j)}$. In general

$$W_b^{(j)} G_{b,c} \neq W_0^{(j)} G_{b,c}.$$

Consequently the saved dense forward action in the local recursion cannot be relabelled as an initialized-mixer image. This is the key difference from a single Taylor recursion at time zero.

Compute global coefficients only for the base fields $h^{(j)}$ and $\delta^{(j)}$. After all panels have been summed, set

$$\tilde{c}_{k\ell b}^{W_0^{(j)} h^{(j-1)}} = W_0^{(j)} \tilde{c}_{k\ell b}^{h^{(j-1)}}, \quad \tilde{c}_{k\ell b}^{W_0^{(j+1)T} \delta^{(j+1)}} = W_0^{(j+1)T} \tilde{c}_{k\ell b}^{\delta^{(j+1)}}. \quad (\text{Setup-E.6})$$

Every coefficient is a finite scalar linear combination of nodal polynomial values. Thus (Setup-E.6) is exactly what would be obtained by applying W_0 or its transpose to every base nodal polynomial before quadrature. This proves exact coefficient and reconstructed-source pairing without storing image jets or performing initialized-mixer actions at every node.

Accuracy for the image remains a separate requirement. On the inherited event $\|W_0^{(j)}\|_{\text{op}} \leq 8$, (Setup-E.5) gives

$$\|W_0(\tilde{g} - g)\|_{\infty} \leq \|W_0\|_{\text{op}} \|\tilde{g} - g\|_2 \leq \delta_{\text{node}}. \quad (\text{Setup-E.7})$$

The base source also has coordinate error at most δ_{node} . Apply the quadrature proof's positive-weight perturbation estimate separately to each family, using its own exact analytic source. Its coefficient error is at most $\epsilon_c/2$ from nodal approximation and $\epsilon_c/2$ from quadrature. The source reconstruction error is therefore at most

$$\eta/16 + N\mathcal{Y}\epsilon_c = \eta/8 < \eta. \quad (\text{Setup-E.8})$$

This deliberately pays a factor $\sqrt{n}$ if the available source estimate is only coordinatewise. It avoids an invalid dimension-free inference from coordinate accuracy to image accuracy. At polynomial target accuracy this additional factor changes the constant multiplying $\log n$ in the precision/degree requirement, not its order. Alternatively one can certify both polynomial families directly, but their exact relationship must still be imposed.

There are at most $2(L-1)N$ image columns, so (Setup-E.6) costs

$$O(Ln^2N) \quad (\text{Setup-E.9})$$

arithmetic. Applying initialized mixers instead to every local source jet would cost $O(JLn^2N_xK)$. Either bound is legitimate; (Setup-E.9) avoids that work during continuation and is convenient for streaming. No claim that (Setup-E.9) is always smaller is required.

Panel-major streaming and the local-to-global transform

Partition the time-node indices into disjoint sets I_b , assigning each t_a to exactly one panel. Repeated physical times from the cosine map may remain separate quadrature nodes. Their scalar weights are simply both counted. Sort the nodes in $O(N_t \log(2 + N_t))$ work if needed; the monotonic halves of the cosine grid also allow a linear merge. The continuation proceeds in increasing physical time, regardless of the original quadrature ordering.

For each panel form the scalar moment table

$$B_{kc}^{(b)} = \frac{\gamma_k}{N_t} \sum_{a \in I_b} \cos(ku_a) ((t_a - \tau_b)/\Delta_b)^c, \quad \gamma_0 = 1, \quad \gamma_k = 2 \ (k \geq 1). \quad (\text{Setup-E.10})$$

For one base source the global coefficient is exactly

$$\tilde{c}_{k\ell b'}^g = \sum_{b=0}^{J-1} \sum_{s=1}^{N_x} \omega_s Y_{\ell, b'}(v_s) \sum_{c=0}^K B_{kc}^{(b)} G_{b,c}^g(v_s). \quad (\text{Setup-E.11})$$

Only $\lambda = (k, \ell, b') \in \Lambda$ is accumulated. In particular, the J local polynomial descriptions are not added as J new source-space families. They are integrated into the original global coefficients, so they do not multiply the rank bound.

The execution order is: generate training jets for one panel; process all spatial nodes against those jets; add that panel's contribution to (Setup-E.11); advance the dense anchor once; discard the panel jets. This requires one dense continuation pass. Spatial-node-major execution without storing all panels would repeat the dense training computation N_x times; it is not the execution costed here.

Powers in (Setup-E.10) and cosine values can be generated by scalar recurrences. A safe total moment-table cost, including clearing every panel table, is

$$T_{\text{moments}} = O((N_t + J)(p + 1)K), \quad M_{\text{moments}} = O((p + 1)K + N_t + J). \quad (\text{Setup-E.12})$$

No conformal-map power composition or $JK^2(p + 1)$ term is needed: the local polynomials are evaluated directly in the local affine coordinate. Panels containing no quadrature nodes require no passive queries or transform, although their training continuation still has to be performed.

There are two useful orders for the sums in (Setup-E.11).

Temporal first. For a streamed spatial query, transform its $K + 1$ jets to all $p + 1$ temporal modes, then update just the retained coefficients. Over all panels the work is

$$T_{\text{proj,t}} = O(LnJN_x[(p + 1)K + N]). \quad (\text{Setup-E.13})$$

Beyond the jets and final coefficient arrays, the vector buffer is $O(Ln(p + 1))$, which fits within $O(LnR)$.

Spatial first. For the current panel accumulate spherical coefficients of each local jet,

$$S_{c\ell b'}^g = \sum_s \omega_s Y_{\ell, b'}(v_s) G_{b, c}^g(v_s).$$

Then transform only the final retained pairs using (Setup-E.10). The work and extra vector buffer are

$$T_{\text{proj,x}} = O(LnJK[N_x H_{\text{sph}} + N]), \quad M_{\text{proj,x}} = O(LnKH_{\text{sph}}). \quad (\text{Setup-E.14})$$

These are alternative implementations, with the displayed memory attached to the chosen one. They are not simultaneous minima. Degree-dependent temporal cutoffs are used when applying the final transform. Both methods produce the same finite coefficients in exact arithmetic. In dimension one, transform the two point families separately; (Setup-E.13)–(Setup-E.14) remain valid as upper bounds under the conventions following (Setup-E.2).

The positive-rule construction and separated harmonic recurrences are exactly those in the quadrature proof. Let $G_{\text{rule}}, G_{\text{basis}}$ denote its time bounds (Setup-Q.30)–(Setup-Q.31), with spherical cutoff ℓ_* . Scalar basis values may be recomputed on each panel for work $G_{\text{rule}} + JG_{\text{basis}}$ and the original geometry workspace, or cached once for work $G_{\text{rule}} + G_{\text{basis}}$ and additional storage $O(N_x(H_{\text{sph}} + d))$. Either choice avoids dense-path storage and is polynomial in the displayed orders. Time-node sorting, panel moment tables, and their workspace are charged separately in (Setup-E.12).

Dense local jets and anchor advancement

The factorization used at time zero remains algebraically valid at an arbitrary anchor; zero readout is not required. Temporarily use unscaled normalized Taylor coefficients $[\cdot]_c$ with respect to $t - \tau_b$, and set $u_a^{(j)} = r_a \delta_a^{(j)}$. For $s \geq 1$, the dense ODE gives

$$\begin{aligned} [A]_s &= -\frac{2}{ms} \sum_a [u_a^{(1)}]_{s-1} v_a^T, \\ [W^{(j)}]_s &= -\frac{2}{mns} \sum_a \sum_{i+k=s-1} [u_a^{(j)}]_i [h_a^{(j-1)}]_k^T. \end{aligned} \quad (\text{Setup-E.15})$$

The constant mixer in this identity is the current $W_b^{(j)}$. Training jets shared by every passive query therefore determine all higher parameter jets without storing K dense matrices. The scalar change to coefficients in (Setup-E.4) costs only a multiplication by Δ_b^c for each coefficient.

For a query series X , let U_i, H_k be the n -by- m training coefficient matrices. The increment action of degree s is

$$-\frac{2}{mn} \sum_{i+k+c=s-1} \frac{U_i(H_k^T X_c)}{i+k+1}. \quad (\text{Setup-E.16})$$

Caching all $H_k^T X_c$ first gives the same causal operation bound as the initial-jet construction. Only lower orders occur in an order- s increment. Training uses a batch of m vectors once per panel, and passive queries use one vector at a time. Transpose actions interchange the two training factors. The resulting non-activation work is

$$T_{\text{train}} = O(J[PmK + Lm^2K^2(n+K)]), \quad (221)$$

$$T_{\text{passive}} = O(JN_x[PK + LmK^2(n+K)]). \quad (\text{Setup-E.17})$$

The live jets/contraction caches need

$$O(P + LmnK + Lm^2K^2) \quad (\text{Setup-E.18})$$

words, including a dense anchor and the resident initial arrays up to a numerical constant. The P term cannot be dropped merely because the retained runtime model is small. At most one spatial query's jets are live.

Advancing the dense anchor also has to be charged. Its hidden-matrix endpoint increment can be grouped as

$$\Delta W^{(j)} = -\frac{2}{mn} \sum_{i=0}^{K-1} U_i V_i^T, \quad V_i = \sum_{k=0}^{K-1-i} \frac{\Delta_b^{i+k+1}}{i+k+1} H_k. \quad (\text{Setup-E.19})$$

Forming the V_i costs $O(nmK^2)$ per layer. Applying the K rank-at-most- m outer products to the dense anchor costs $O(n^2mK)$. First-matrix and readout updates fit the bound

$$T_{\text{advance}} = O(J[PmK + LnmK^2]). \quad (\text{Setup-E.20})$$

This is absorbed by (221), but (Setup-E.19) shows why it does not require an omitted $O(Jn^2mK^2)$ materialization. One can form one V_i , update the anchor, and discard it after the panel's source processing is complete.

A simpler materialized implementation is also valid: generate all dense parameter jets and use ordinary series convolutions for training and passive queries. Its non-activation bound is

$$O(JP(m+N_x)K^2), \quad O(PK + LmnK) \quad (\text{Setup-E.21})$$

for time and live jet memory, with anchor evaluation included. Equations (221) and (Setup-E.17) give the more economical training and passive-query construction used below; (Setup-E.21) is not needed to establish its validity.

Let $a_{\text{on}}(K), b_{\text{on}}(K)$ be time and workspace for one scalar activation/derivative in sequential training-jet generation, and $a_{\text{off}}(K), b_{\text{off}}(K)$ the corresponding passive-query series costs, including scalar derivative generation. The separate activation terms are

$$T_{\text{act}} = JLn[ma_{\text{on}}(K) + N_x a_{\text{off}}(K)],$$

$$M_{\text{act}} = Lmnb_{\text{on}}(K) + b_{\text{off}}(K). \quad (\text{Setup-E.22})$$

Maxima over layer-specific backends suffice. Fixed-size differential recurrences, such as tanh, permit quadratic series arithmetic and linear workspace. General analytic regularity does not imply a fast derivative oracle. A cubic composition algorithm with supplied scalar derivatives still leaves derivative generation separately charged through (Setup-E.22).

Selection, final initialization, total work and memory

Save the exact initialized training features and their initialized forward images during the first training pass at $\tau_0 = 0$. Include the constant vector and initial first-weight columns. These additions are exact; later approximate anchors cannot replace them. Their direct work is at most $O(mP)$, absorbed by (221), plus the already charged scalar activations. These initial values use the inherited exact-real activation primitives; finite-precision exact initialization is a separate issue.

After computing (Setup-E.11) and (Setup-E.6), discard dense continuation anchors and panel workspace. Orthogonalize the complete source-generator list, select coordinates, and assemble the original source metrics and initial compressed matrices. The existing costs are

$$T_{\text{select+assemble}} = O(LnRr + Lnr^3 + Ln^2r + Lq^2r). \quad (\text{Setup-E.23})$$

The Ln^2r term compresses W_0 on the complete source bases. The paired image coefficients do not generally give its action on every basis vector, so this term remains even after (Setup-E.9). The final model starts at the original $A_0, W_0, w_0 = 0$, with the original exact training Gram and optimizer. The last continuation anchor is not its initialization.

Let T_{proj} and M_{proj} denote either (Setup-E.13) with buffer $O(Ln(p+1))$, or (Setup-E.14). Let G_{time} , G_{memory} denote one of the explicit geometry choices after (Setup-E.14). A sufficient complete envelope is

$$\begin{aligned} T_{\text{setup}} = & O(P + J\{P(m + N_x)K + Lm(m + N_x)K^2(n + K)\} \\ & + (N_t + J)(p + 1)K + T_{\text{proj}} + Ln^2N \\ & + LnRr + Lnr^3 + Ln^2r + Lq^2r \\ & + T_{\text{act}} + G_{\text{time}} + N_t \log(2 + N_t) + C_{\text{cert}}), \end{aligned} \quad (\text{Setup-E.24})$$

$$\begin{aligned} M_{\text{setup}} = & O(P + LmnK + Lm^2K^2 + LnR + Lq^2 \\ & + (p + 1)K + N_t + J + M_{\text{proj}} + m(d + 1) \\ & + M_{\text{act}} + G_{\text{memory}} + M_{\text{cert}}). \end{aligned} \quad (\text{Setup-E.25})$$

These bounds do not store JP anchors, JPK dense jets, or $N_t N_x$ dense source arrays. Their source coefficients require $O(LnR)$ words, which are discarded after assembly. The formula is a safe peak upper bound; freeing stage-specific buffers can lower it. Fresh initialization additionally requires exactly $nd + (L - 1)n^2$ Gaussian draws and the sampler's cost/scratch.

By (Setup-E.6)–(Setup-E.8), the spaces meet the original approximation and pairing assumptions with the original rank bound (Setup-E.2). Therefore the selected neural dynamics, learned-state count

$$(L - 1)q^2 + q(d + 1) + m,$$

and retained inventory bound

$$1020(L + 1)R^2 + 10m(d + 1) \quad (\text{Setup-E.26})$$

remain valid with the same qualifications as in the preceding Harmonic construction. The original all-time prediction-error certificate is unchanged because every source error is below its original allowance. Neither quadrature nodes nor local panel degrees become runtime state. Baseline-only, zero-label and full-width branches retain their separate existing constructions.

E.10.6 Source-jet compatibility and complete explicit costs

Precision and source-jet compatibility

Let $\delta_{\text{node}} > 0$ be the global quadrature's coordinate-error allowance. It is defined in assembly equation (Setup-E.3), with its separate dimension-one convention. Run the polynomial-activation backend with the stricter requested nodal tolerance

$$\tau = \frac{\delta_{\text{node}}}{32\sqrt{n}}. \quad (\text{Setup-B.1})$$

All its scalar approximation and global defect budgets use this τ in place of its local symbol δ_{node} . Let $\hat{R}_n, V, C_{\text{src}}$ be the explicit radius, velocity and original-source comparison coefficient of that backend and the local-continuation proof. Set

$$C_g = \max_j \{H_j + 1, B_j + 1\}, \quad (\text{Setup-B.2})$$

where H_j, B_j are the core's complex forward-feature and backward response RMS bounds. Increase the backend's training Taylor degree, if necessary, to the maximum of its equation (Setup-A.22) and

$$\left\lceil \log_2 \max \left\{ 1, \frac{128V\hat{R}_n C_{\text{src}} n^{3/2}}{\delta_{\text{node}}} \right\} \right\rceil, \quad \left\lceil \log_2 \max \left\{ 1, \frac{64\sqrt{n}C_g}{\delta_{\text{node}}} \right\} \right\rceil. \quad (\text{Setup-B.3})$$

Use that same degree K for training and passive sources. Increasing it preserves the previous tail/defect inequalities. All additional inverse tolerances have logarithm $O(\log(en))$ in the polynomial-accuracy regime, so $K = O(\log(en))$ is unchanged.

To verify the assembly's Euclidean error interface, consider one panel and one passive real unit query. Write v for the exact local solution of the polynomial-activation setup field, p for its computed degree- K parameter Taylor polynomial, and g_K for the degree- K Taylor polynomial of a base source $h^{(j)}$ or $\delta^{(j)}$ along v , using the setup activations. Denote original and setup activation evaluation by g^0 and g^p , respectively. The desired source is $g^0(\theta(t))$, on the original dense trajectory.

On the local complex disk, both original feature and response RMS are bounded by H_j, B_j . The activation replacement estimate applies there at each fixed real query: every original carrier has RMS bounded by the core coefficient, hence coordinate maximum at most $\sqrt{n}$ times that coefficient, exactly as used in backend equations (Setup-A.15), (Setup-A.24). Equation (Setup-A.25) with (Setup-B.1) makes the additional source RMS smaller than one. Thus $\|g^p(v)\|_2 \leq \sqrt{n}C_g$ throughout the complex disk. Vector-valued Cauchy and the half-radius panel give

$$\|g_K - g^p(v)\|_2 \leq \sqrt{n}C_g 2^{-K} \leq \delta_{\text{node}}/64. \quad (\text{Setup-B.4})$$

The remaining differences are split before estimating:

$$g_K - g^0(\theta) = [g_K - g^p(v)] + [g^p(v) - g^0(v)] + [g^0(v) - g^0(p)] + [g^0(p) - g^0(\theta)]. \quad (\text{Setup-B.5})$$

The second term has each coordinate at most $\tau/2$, by backend (Setup-A.24)–(Setup-A.25); its Euclidean norm is therefore at most $\sqrt{n}\tau/2 = \delta_{\text{node}}/64$.

For the third term the original-source comparison gives a coordinate bound $C_{\text{src}}n\|v - p\|_2$. It applies to these two real states: both remain within the core's real operator/readout neighborhood, and its passive-query proof uses only those caps and the RMS-to-coordinate estimate, not a probabilistic maximum for the perturbed state. The Taylor tail gives $\|v - p\|_2 \leq 2V\hat{R}_n 2^{-K}$. Conversion to Euclidean norm and (Setup-B.3) thus give at most $\delta_{\text{node}}/64$.

For the fourth term the backend's global estimate (Setup-A.23), with tolerance τ , gives original-source coordinate error at most $\tau/2$. Its Euclidean norm is again at most $\delta_{\text{node}}/64$. Combining the four terms proves

$$\|g_K - g^0(\theta)\|_2 \leq \delta_{\text{node}}/16 < \delta_{\text{node}}/8. \quad (\text{Setup-B.6})$$

This is precisely the stronger nodal interface needed for assembly (Setup-E.5). It permits forming initialized-matrix image coefficients only after global accumulation: their coordinate error is bounded by eight times the base Euclidean error. The two computed members satisfy the exact pairing by linearity, while their error is measured against their own original analytic source functions. The unchanged quadrature and source-tail allocation then gives source-coordinate approximation below the original tolerance η .

Why the factored execution still applies

The assembly's original statement names Taylor jets of the actual dense field F . Here they are jets of the certified disposable field $\tilde{F}$, with activations ψ_j . This is an explicit interface replacement, justified by (Setup-B.6) and the following exact identity.

The polynomial-activation system has the same rank-one gradient equations, loss normalization and mobilities; only its scalar activation functions are different. Coefficient comparison therefore gives the identical equations (Setup-E.15)–(Setup-E.20) of the assembly proof, using its actual residuals, responses and features. The cached action formula and grouped dense-anchor advancement remain exact for those jets. The proof of their arithmetic count uses only these finite sums, not a special activation or zero readout at a restart. There is no extra dense matrix factor hidden in changing F to $\tilde{F}$.

For one scalar degree- D polynomial ψ , the Chebyshev recurrence is linear in a product by the scalar input series. At time order k , computing every recurrence index in increasing polynomial degree uses already known lower time coefficients and the just computed order- k coefficients at the preceding recurrence indices. Each product costs $O(k+1)$; there are $O(D)$ such indices. Summing over $0 \leq k \leq K$ gives $O(DK^2)$ work and $O(DK)$ sufficient persistent words. Preparing ψ' costs $O(D^2)$ once and its series composition has the same bound. Thus both online training and offline passive-query backend costs satisfy

$$a_{\text{on}}(K), a_{\text{off}}(K) = O(DK^2), \quad b_{\text{on}}(K), b_{\text{off}}(K) = O(DK). \quad (\text{Setup-B.7})$$

The exact original initialized features and their images must be computed separately using ϕ_j , as prescribed by the backend. They must not be replaced by the constant-order ψ_j features. This costs $O(mP)$ arithmetic and $O(Lmn)$ original value calls and restores the exact initialized Gram and action conditions. The final compressed network uses the original activations and optimizer at original time zero; neither the last dense anchor nor the setup polynomials survive as runtime state.

Complete costs and headline specialization

Use the supplied internal counts and geometry from the assembly proof. Let J be the number of local panels in this section. Its complete time envelope (Setup-E.24) and memory envelope (Setup-E.25) apply after the substitutions (Setup-B.7), with the following explicit additions and qualifications:

- Add $O(LD^2 + mP)$ arithmetic for the polynomial backend and exact original initialized features, and $O(LD)$ polynomial coefficient words.
- Add $O(LD + Lmn)$ original real activation-value calls. There are no high derivative, original first-derivative, complex activation or trained-reference oracles. The activation-value precision allowed by backend (Setup-A.10) is polynomially small in width in this regime.
- Panel counts, degrees and tolerances are explicit scalar formulas in the backend and (Setup-B.1)–(Setup-B.3). Their evaluation fits its displayed scalar preprocessing. There are no adaptive rejections, implicit nonlinear solves or uncharged certification runs; take the assembly's $C_{\text{cert}}, M_{\text{cert}}$ to be this elementary scalar overhead.
- Fresh initialization adds the original Gaussian draws and sampler costs. All arithmetic counts otherwise have numerical, implementation-only big-O constants. Finite precision, numerical rank reliability and bit complexity are not implied by these exact-real counts.

In particular, with spatial-first projection, a sufficient complete non-activation bound is

$$\begin{aligned} &O(P + LD^2 + mP + JP(m + N_x)K + JLn(m + N_x)K^2(n + K) \\ &\quad + JLn(m + N_x)DK^2 + (N_t + J)(p + 1)K \\ &\quad + LnJK(N_x H_{\text{sph}} + N) + Ln^2N \\ &\quad + LnRr + Ln r^3 + Ln^2r + Lq^2r + G_{\text{time}} + N_t \log(2 + N_t)), \end{aligned} \quad (\text{Setup-B.8})$$

where $P = (L - 1)n^2 + n(d + 1)$. A sufficient peak count in real words is

$$\begin{aligned} &O(P + LmnDK + Lm^2K^2 + LnR + Lq^2 + LnKH_{\text{sph}} + LD \\ &\quad + (p + 1)K + N_t + J + m(d + 1) + G_{\text{memory}}). \end{aligned} \quad (\text{Setup-B.9})$$

The quadrature geometry terms in these formulas are the explicitly specified recomputed-per-panel or cached choice in the assembly proof; the chosen time and memory versions must be used together. No source or dense trajectory array is retained after selection.

For each separately fixed admissible dataset, depth, activation, positive label size and confidence, at the original source tolerance $1/n$ and horizon $32(m/\gamma) \log(en)$, the certified choices satisfy

$$D = O(\log(en)^{3/2}), \quad K = O(\log(en)), \quad J = O(\log(en)^{3/2}),$$

$$N_t, p+1 = O(\log(en)^{5/2}), \quad N_x, H_{\text{sph}} = O(\log(en)^{3(d-1)/2}), \quad N, R, r = O(\log(en)^{3d/2+1}). \quad (\text{Setup-B.10})$$

Use the assembly's two-point conventions at $d = 1$. With actual selected width $q = O(R)$, every term of (Setup-B.8) is bounded by

$$O\left(n^2 \log(en)^{3d/2+1} + n \log(en)^{9d/2+3} + \text{polylog}(n)\right). \quad (\text{Setup-B.11})$$

For example the new activation arithmetic is at most $n \log(en)^{3d/2+7/2}$, which is bounded by the second term for every fixed integer $d \geq 1$. The n^2 terms include source-image formation and initialized mixer assembly, not merely the dense integrator. The n -proportional terms are eventually smaller than the displayed quadratic term, giving the convenient headline

$$\text{setup work} = O\left(n^2 \log(en)^{3d/2+1}\right), \quad \text{peak setup memory} = O(n^2). \quad (\text{Setup-B.12})$$

Both statements are at fixed admissible structural parameters. The explicit separate m, d, L , order and source-radius dependence is in (Setup-B.8)–(Setup-B.10) and the backend's degree/step formulas; it has not been identified or discarded. Bounded-cost scalar evaluations and Gaussian sampling, or a supplied initialization for the latter, are required to include those costs inside the work shorthand in (Setup-B.12); otherwise add their exposed costs. At $d = 1$, (Setup-B.12) has logarithmic exponent $5/2$.

The original whole-sphere/all-time error and retained inventory are unchanged: source approximation is below $1/n$, so the original comparison gives $n^{-1+o(1)}$ error and $O(\log(en)^{3d+2})$ retained words. No statement of optimality among all setup algorithms or finite-precision implementations follows from these sufficient bounds.

For ordinary dense Euler at physical step h , full source horizon $T = 32(m/\gamma) \log(en)$, and the same scalar-cost convention, the work is $\Theta(mPT/h)$. At fixed parameters the setup/Euler ratio is therefore at most

$$O\left(h \log(en)^{3d/2}\right). \quad (\text{Setup-B.13})$$

In particular $h = n^{-1/2}$ gives a ratio tending to zero, and the same holds for every fixed inverse-polynomial step. This comparison does not assert that such an Euler step is necessary or sufficient for a particular numerical accuracy. The construction may cost more than a dense solver using the same high-order continuation, which omits source sampling and compression assembly. It provides the stated Euler comparison, not a lower-bound separation from all dense training methods.

E.10.7 Exact adaptive Gaussian sampling

Statement and computational model

Fix an integer width $n \geq 1$, a finite set of layer indices $\mathcal{I}$, and deterministic nonnegative query budgets k_ℓ , $\ell \in \mathcal{I}$. In the dense reference model,

$$W^{(\ell)} \in \mathbb{R}^{n \times n}, \quad W_{ij}^{(\ell)} \text{ are jointly independent } N(0, 1/n) \quad (\ell \in \mathcal{I}, 1 \leq i, j \leq n).$$

Let ξ denote all independent randomness used by the querying algorithm; it is independent of these matrices. At each step the algorithm either stops or chooses, measurably from ξ and the global past transcript, a layer, a direction of multiplication, and a finite vector in $\mathbb{R}^n$. The reply is $W^{(\ell)}v$ for a forward query or $(W^{(\ell)})^T u$ for a transpose query. The transcript includes query choices and replies. There are at most k_ℓ queries to layer ℓ , including zero, repeated, and linearly dependent queries. The stopping time is at most $\sum_\ell k_\ell$. No observation of a reference matrix outside this interface is allowed.

The sampler below receives the queries, draws fresh standard Gaussian scalar variables as needed, and returns replies. Its Gaussian draws are independent of ξ ; its private random tape is not an extra observation made available to the querying algorithm.

Theorem. For every such querying algorithm, the sampler has exactly the same joint law of ξ , queries, replies, stopping time, and every measurable quantity computed from them as the dense reference model. On an extension of the sampler's probability space, matrices $\widehat{W}^{(\ell)}$ can be defined so that:

1. Their entries are jointly independent $N(0, 1/n)$, independently of ξ .
2. Every reply actually returned by the sampler equals the corresponding product with $\widehat{W}^{(\ell)}$, simultaneously and almost surely.
3. The joint law including these matrices is the dense reference joint law.

This completion is only a proof coupling. The sampler does not initialize a dense reference matrix or evaluate the dense Gaussian remainder in that coupling. Its stored matrix information consists only of query bases and their answers, as described below.

For a layer receiving $k \geq 0$ queries, the following bounds hold. Here a scalar addition, subtraction, multiplication, division, or square root costs one arithmetic operation; generating one exact standard Gaussian scalar costs one Gaussian draw. One exact real scalar or integer/address occupies one word. These are idealized real-arithmetic and Gaussian-generation primitives, not claims about finite-bit implementations.

$$\begin{aligned} \text{arithmetic operations} &\leq 4nk^2 + 6nk + 2, \\ \text{Gaussian draws} &\leq n \min\{k, 2n - 1\} \leq nk, \\ \text{sampler words} &\leq 2n \min\{k, 2n\} + 8n + 5 \min\{k, 2n\} + 64. \end{aligned}$$

There are at most k zero tests and k further rank comparisons. These bounds exclude the querying algorithm's own work and storage, including any transcript it elects to retain. They include one current query/reply workspace. For $k \geq 1$, they are $O(nk^2)$ arithmetic and $O(nk)$ words. The two initialization arithmetic operations can be shared between all layers of the same width. In particular, globally the arithmetic bound is

$$2 + \sum_{\ell \in \mathcal{I}} (4nk_\ell^2 + 6nk_\ell),$$

and the Gaussian-draw bounds add across layers. A sum of the displayed per-layer memory bounds is a valid global bound; the vector workspace can also be shared. When all query budgets are zero the sampler need not allocate vector workspace.

The budgets may depend arbitrarily on n . There is no asymptotic assertion and no assumption that the input vectors are well conditioned.

The proof maintains the conditional Gaussian law after every query. An orthogonal split isolates the newly observed Gaussian vector from the unobserved matrix remainder. This proves the answer kernels, preserves the conditional product law across layers, and supplies the stopped completion coupling.

Stored state and the online algorithm

Suppress the layer superscript while describing one layer. The sampler stores four matrices, initially with zero columns:

$$V, Y \in \mathbb{R}^{n \times p}, \quad U, R \in \mathbb{R}^{n \times q}.$$

The columns of V form an orthonormal basis of the span of the forward query vectors seen so far; the columns of U form an orthonormal basis of the span of the transpose query vectors seen so far. The stored answers have the meaning $Y = WV$, $R = W^T U$. This meaning will be supplied by the coupling proof; an actual W is not an input to the algorithm. The state always satisfies

$$V^T V = I_p, \quad U^T U = I_q, \quad U^T Y = R^T V. \quad (\text{Setup-G.1})$$

Empty matrix products are zero. Precompute $\sigma = 1/\sqrt{n}$. Every normalization and every zero test below is exact. In particular, a small positive norm is not rounded to zero.

Forward query $v \in \mathbb{R}^n$

Using the old state, compute

$$a = V^T v, \quad v_\perp = v - Va, \quad \alpha = \sqrt{v_\perp^T v_\perp}.$$

If $\alpha = 0$, return Ya , leave the state unchanged, and draw no randomness. Otherwise compute

$$e = v_\perp / \alpha, \quad b = R^T e.$$

If $q < n$, draw $g \in \mathbb{R}^n$ with independent $N(0, 1)$ coordinates and set

$$h = Ub + \sigma\{g - U(U^T g)\}. \quad (\text{Setup-G.2})$$

If $q = n$, set $h = Ub$ without drawing g . Return

$$Ya + \alpha h, \quad (\text{Setup-G.3})$$

then append the column pair (e, h) to (V, Y) . The returned expression uses the old V, Y and old coefficient a . The order of appending and returning can be reversed in an implementation that preserves those values.

Transpose query $u \in \mathbb{R}^n$

Using the old state, compute

$$b = U^T u, \quad u_\perp = u - Ub, \quad \beta = \sqrt{u_\perp^T u_\perp}.$$

If $\beta = 0$, return Rb , leave the state unchanged, and draw no randomness. Otherwise compute

$$f = u_\perp / \beta, \quad a = Y^T f.$$

If $p < n$, draw fresh $g \in \mathbb{R}^n$ with independent $N(0, 1)$ coordinates and set

$$h = Va + \sigma\{g - V(V^T g)\}. \quad (\text{Setup-G.4})$$

If $p = n$, set $h = Va$ without drawing g . Return

$$Rb + \beta h, \quad (\text{Setup-G.5})$$

then append (f, h) to (U, R) .

For multiple layers, retain a separate quadruple (V, Y, U, R) per layer and apply the corresponding rule to the chosen layer. All fresh g 's, across all times and layers, are independent. The algorithm forms only matrix-vector products with the stored columns. In particular, neither an $n \times n$ projector nor the conditional-mean matrix used in the proof is formed.

Elementary Gaussian splitting used in the proof

The needed Gaussian fact can be verified directly. If $Z \in \mathbb{R}^d$ has independent $N(0, \tau^2)$ coordinates and A, B are deterministic linear maps with $AB^T = 0$, then AZ and BZ are independent centered Gaussian vectors. Indeed their joint characteristic function at (s, t) is

$$\mathbb{E} e^{i(s^T AZ + t^T BZ)} = \exp\left(-\frac{\tau^2}{2} \|A^T s + B^T t\|_2^2\right).$$

The cross term is $2s^T AB^T t = 0$, so this factors as the product of the two marginal characteristic functions. This proves independence even when either covariance is singular. Applying this to the vector of matrix entries justifies every orthogonal Gaussian split below. No nonsingular covariance or matrix inverse is needed.

For fixed state satisfying (Setup-G.1), define, for proof purposes only,

$$P = VV^T, \quad Q = UU^T, \quad M = YV^T + UR^T(I - P). \quad (\text{Setup-G.6})$$

The compatibility identity in (Setup-G.1) implies

$$MV = Y, \quad M^T U = VY^T U + (I - P)R = PR + (I - P)R = R. \quad (\text{Setup-G.7})$$

The matrices preserving homogeneous versions of these observations are exactly

$$\{A : AV = 0, A^T U = 0\} = \{(I - Q)B(I - P) : B \in \mathbb{R}^{n \times n}\}. \quad (\text{Setup-G.8})$$

For the forward inclusion, $AV = 0$ implies $AP = 0$, and $A^T U = 0$ implies $QA = 0$. For the reverse inclusion use $(I - P)V = 0$ and $(I - Q)U = 0$. Each summand of M in (Setup-G.6) is Frobenius-orthogonal to the space in (Setup-G.8): the first has right support in $\text{span}(V)$, and the second has left support in $\text{span}(U)$.

Thus the candidate conditional matrix law at this state is

$$W = M + (I - Q)G(I - P), \quad G_{ij} \text{ independent } N(0, 1/n). \quad (\text{Setup-G.9})$$

The induction below proves (Setup-G.9) as a conditional-law identity after adaptive queries. It does not infer adaptive conditioning merely by treating random query directions as fixed in an unconditional formula.

One query updates the conditional law exactly

Assume (Setup-G.9) holds conditional on the full past. Once that past and the independent querying randomness are conditioned on, the next query direction and selected layer are fixed.

For a forward query, the algorithm's orthogonal decomposition is $v = Va + \alpha e$ when $\alpha > 0$. If $\alpha = 0$, (Setup-G.7) gives the deterministic answer Ya ; receiving that answer conveys no additional information. If $\alpha > 0$, then $\|e\|_2 = 1$, $Pe = 0$, and

$$We = UR^T e + (I - Q)Ge. \quad (\text{Setup-G.10})$$

The coordinates of Ge are independent $N(0, 1/n)$: distinct coordinates use independent rows of G , and the variance in each row is $\|e\|_2^2/n = 1/n$. Consequently (Setup-G.2) has exactly the conditional law of We , and (Setup-G.3) has that of Wv . If $q = n$, $Q = I$, so the random term is identically zero and skipping it preserves the law.

It remains to verify that the remaining randomness has the claimed form after the new answer. Let

$$P' = P + ee^T, \quad E = (I - Q)Ge, \quad D = (I - Q)G(I - P').$$

The remainder in (Setup-G.9) splits as

$$(I - Q)G(I - P) = Ee^T + D. \quad (\text{Setup-G.11})$$

The random vector E and matrix D are independent. One direct covariance calculation is

$$\text{Cov}(E_i, D_{ab}) = \frac{1}{n}(I - Q)_{ia} [e^T(I - P')]_b = 0,$$

because $P'e = e$. They are jointly Gaussian linear functions of the entries of G , so the characteristic-function argument in the exact adaptive gaussian sampling subsection 3 proves the asserted independence, including all singular cases. The covariance and mean of D are those of $(I - Q)G'(I - P')$ for fresh iid $G'_{ij} \sim N(0, 1/n)$.

Observing the reply is equivalent to observing $h = We$, since $h = (Wv - Ya)/\alpha$ and $\alpha > 0$. By (Setup-G.10) this specifies $E = h - UR^T e$ while leaving D independent and unchanged in law. The new stored columns satisfy $U^T h = R^T e$, since the noise is annihilated by U^T . Hence (Setup-G.1) remains valid. The new mean computed by (Setup-G.6) is

$$\begin{aligned} M' &= YV^T + he^T + UR^T(I - P - ee^T) \\ &= M + (h - UR^T e)e^T = M + Ee^T. \end{aligned}$$

Together with (Setup-G.11), this is exactly the updated kernel $M' + (I - Q)G'(I - P')$.

For a transpose query with $\beta = 0$, (Setup-G.7) gives the deterministic answer Rb . With $\beta > 0$, $Qf = 0$ and $\|f\|_2 = 1$, so

$$W^T f = VY^T f + (I - P)G^T f. \quad (\text{Setup-G.12})$$

Thus (Setup-G.4) is its conditional law. Put

$$Q' = Q + ff^T, \quad E = (I - P)G^T f, \quad D = (I - Q')G(I - P).$$

Then $(I - Q)G(I - P) = fE^T + D$, and

$$\text{Cov}(E_i, D_{ab}) = \frac{1}{n}(I - P)_{ib}[f^T(I - Q')]_a = 0.$$

The same characteristic-function argument shows that E and D are independent. The observed column $h = W^T f$ specifies $E = h - VY^T f$, while D keeps the law of $(I - Q')G'(I - P)$. Compatibility is preserved because $V^T h = Y^T f$. Expanding the new formula (Setup-G.6), with $U' = [U \ f]$, $R' = [R \ h]$, gives

$$M' = M + fh^T(I - P).$$

Compatibility yields $Ph = VY^T f$, so $h^T(I - P) = (h - VY^T f)^T = E^T$. Therefore $M' = M + fE^T$, precisely the updated conditional mean. When $p = n$, $P = I$ and the random term in (Setup-G.12) vanishes. This completes the one-query proof in both directions.

Arbitrary interleaving, stopping, and completion

Let the full past sigma-field include ξ and all queries and replies up to the current global step. The induction invariant in the dense model is: conditional on this sigma-field, the matrices in different layers are independent, with layer ℓ having kernel

$$M_\ell + (I - Q_\ell)G_\ell(I - P_\ell), \quad (\text{Setup-G.13})$$

where the G_ℓ 's in this conditional representation are mutually independent iid Gaussian matrices. At time zero, all stored lists are empty, $M_\ell = P_\ell = Q_\ell = 0$, and (Setup-G.13) is the assumed initialization law.

At a subsequent step, the query choice is measurable from the conditioned past. It therefore reveals no extra information once that past is fixed. Only the selected matrix determines the next answer. The one-query calculation in the exact adaptive gaussian sampling subsection 4 gives its answer law and posterior kernel. The conditional product structure in (Setup-G.13) implies that all other matrix kernels are unchanged: the joint conditional law before the answer is the product of the selected matrix/answer law and the other matrix laws. Conditioning that product on the answer changes only the selected factor. This proves the induction invariant.

These calculations can equivalently be read as conditional-expectation identities for bounded measurable test functions; they do not require giving positive probability to any particular real-valued transcript.

The sampler uses exactly the answer transition kernel just derived, at every history. Its initial independent randomness ξ has the same law as in the dense model. Induction over the finite maximal number of queries therefore gives equality of the complete transcript laws. Querying another layer using an earlier reply causes no difficulty: conditional on the global past the new query is already determined. It is the remaining matrices that are conditionally independent; the outputs themselves need not be independent.

Stopping adds no further conditioning beyond the transcript: whether the algorithm stops at the current step is measurable from that transcript and ξ . Formally, for a bounded stopping time T , partition by the finitely many events $\{T = t\}$. On each such event the fixed-time conditional identity already proved applies, with the state at time t . Summing those identities proves (Setup-G.13) at time T . Alternatively, make stopping absorbing and pad the remaining global steps with no observations.

To construct the stated coupling, run the sampler through its stopping time, then on an extended probability space take independent matrices G_ℓ , independent also of the entire sampler run, with iid $N(0, 1/n)$ entries. Define mathematically

$$\widehat{W}^{(\ell)} = M_\ell + (I - Q_\ell)G_\ell(I - P_\ell) \quad (\text{Setup-G.14})$$

using the final stored states. The conditional law in (Setup-G.14) is exactly the dense model's conditional law given the same stopped transcript and ξ . Integrating this kernel against the identical transcript laws shows equality of the full joint laws, including ξ and all completed matrices. In particular, their unconditional entries are jointly independent $N(0, 1/n)$, and the completed matrices are independent of ξ .

Equation (Setup-G.7) and the projected remainder imply $\widehat{W}^{(\ell)} V_\ell = Y_\ell$ and $(\widehat{W}^{(\ell)})^T U_\ell = R_\ell$. Every earlier forward query is in the span of the final V_ℓ , with its returned reply equal to the same linear combination of stored columns of Y_ℓ ; the new independent column is appended when that query is processed. Later appends preserve all previous columns and relations. The transpose argument uses U_ℓ, R_ℓ . Thus all earlier replies are simultaneously products with the single completed matrix for their layer. This is pathwise equality on the constructed coupling, not only a separate marginal-law assertion for each answer.

Construction (Setup-G.14) is not part of the online algorithm or its operation count. Even if the algorithm later resumes querying, it can continue from the stored quadruples with fresh Gaussian vectors: the same conditional kernel gives the correct continuation law. One must not first realize an independent completion and then claim that unrelated fresh sampler draws reproduce that particular completion; pathwise agreement requires the joint coupling just established.

Cost calculation

At a local query to one layer, let p, q be the old basis sizes. A length- n dot product costs at most $2n$ arithmetic operations. Therefore a product by an $n \times p$ matrix or its transpose costs at most $2np$. The following count assumes a new forward direction and allows all displayed operations even when an empty or full span would permit skipping some.

Computation	Arithmetic bound
$a = V^T v, v_\perp = v - Va$	$4np + n$
$\alpha = \sqrt{v_\perp^T v_\perp}$	$2n + 1$
$e = v_\perp / \alpha$	n
$b = R^T e$	$2nq$
$g = U(U^T g)$	$4nq + n$
$h = Ub + \sigma\{g - U(U^T g)\}$	$2nq + 2n$
after the projected vector is ready	
reply $Ya + \alpha h$	$2np + 2n$

The sum is $6np + 8nq + 9n + 1 \leq 8n(p + q) + 10n$, since $n \geq 1$. A dependent forward query costs at most $6np + 3n + 1$, which is smaller. The transpose bound exchanges p and q and has the same upper bound $8n(p + q) + 10n$. There is one zero test per query and at most one further rank comparison. Vector copying and bookkeeping take $O(n + p + q)$ word accesses per query, so also preserve the stated asymptotic work bound if word accesses are charged separately.

Before the j -th local query, $p + q \leq j - 1$. Summing gives

$$\sum_{j=1}^k \{8n(j-1) + 10n\} = 4nk(k-1) + 10nk = 4nk^2 + 6nk.$$

Computing $\sigma = 1/\sqrt{n}$ uses one square root and one division. This proves the claimed arithmetic bound. No inverse, linear solve, Gram-matrix eigenvalue bound, or condition-number assumption occurs in this calculation.

Every nonzero new query direction increases exactly one of p, q by one. Both remain at most n . Gaussian vectors are needed only until one of the two ranks first reaches n ; thereafter queries enlarging the other side have deterministic new columns. Immediately before that first full rank, both ranks are at most $n - 1$, so at most $2n - 1$ enlargements can draw a Gaussian vector. If neither rank reaches n , there are at most $2n - 2$

enlargements. There are also at most k enlargements in total. Each drawing enlargement uses n scalar draws, proving $n \min\{k, 2n - 1\}$.

The four stored matrices contain $2n(p + q)$ scalar words. Store their columns as linked lists of separate vectors with $2(p + q)$ next-column addresses; this allows append without holding a second full copy of the state. Eight length- n scratch vectors and $3(p + q)$ coefficient words are a conservative workspace bound: one needs only the current vector/residual, a Gaussian vector, a new answer, an output, temporary matrix-vector storage, and the three coefficient lists $a, b, U^T g$ (or their transpose counterparts). Sixty-four additional scalar or integer/address words suffice for the scale, norms, loop indices, dimensions, and list roots in an ordinary implementation. Newly retained column vectors are charged to the stored matrices. Thus

$$2n(p + q) + 8n + 5(p + q) + 64 \leq 2n \min\{k, 2n\} + 8n + 5 \min\{k, 2n\} + 64.$$

This is a bound for the sampler state, not for arbitrary client-side storage or for the imaginary dense matrices in the proof. When k is comparable to n , the bound is itself quadratic in n ; the theorem makes no stronger compression claim in that regime. For example, querying all n standard basis vectors forwards returns the entire matrix, and the retained Y then equals that matrix. Thus “implicit” specifies how reference randomness is generated and queried; it cannot mean that an unrestricted oracle transcript never reveals the whole matrix. A requirement of subquadratic storage must also constrain the query budget, for example $k = o(n)$.

Exact scope and limitations

The construction samples a new reference matrix implicitly with the correct joint iid law. It does not receive a previously realized concrete matrix or the seed of a specified dense-entry generator, and it does not reproduce products for such a supplied realization. Equality in law and the existence of the coupling (Setup-G.14) do not supply that additional service. To serve a fixed matrix or a prescribed seed-to-matrix map would require a separate input model and algorithm; no complexity bound for it follows here.

The theorem does permit exact finite- n distributional reasoning for any algorithm using the specified oracle, including nonlinear or discontinuous adaptive choices and data-dependent stopping within the budgets. It preserves the forward/transpose response terms, the dependence between reused products, and independence of the *unconditional* layer matrices. It makes no claim of coordinate independence for reused outputs.

Exact zero tests and normalizations are part of the mathematical algorithm. Near-dependent directions can cause finite-precision difficulties. Replacing zero tests by tolerances or ordinary orthogonalization by an approximate implementation requires a separate error analysis. Likewise, exact real words and exact Gaussian draws are primitives of the stated cost model, not hidden claims that infinitely precise numbers fit in fixed-bit machine words.

Finally, this sampler result by itself proves no stability, training-time, width-limit, mean-field, state-evolution, or approximation theorem. Such uses must supply their own query budgets and additional mathematical arguments.

E.10.8 Exact implicit execution and full costs

Inputs, notation, and precise sampler dependency

Let $L \geq 2$ and $m, d, n \geq 1$ be hidden depth, sample count, input dimension and hidden width, with $n \geq \max(m, d)$. Training inputs are

$$v_a = x_a / \sqrt{d} \in S^{d-1}, \quad 1 \leq a \leq m.$$

The first matrix $W^{(1)} \in \mathbb{R}^{n \times d}$ has independent standard normal entries. For $2 \leq j \leq L$, the initialized hidden matrix

$$W_0^{(j)} \in \mathbb{R}^{n \times n}, \quad (W_0^{(j)})_{uv} \sim N(0, 1/n) \quad (\text{Setup-I.1})$$

has independent entries, independently across layers and of $W_0^{(1)}$. The stored readout $w = W^{(L+1)} \in \mathbb{R}^n$ is initially zero. This is the zero-readout convention of the dense model. The ordinary forward and backward coordinates are

$$z^{(1)} = W^{(1)}v, \quad z^{(j)} = W^{(j)}h^{(j-1)}, \quad h^{(j)} = \phi_j(z^{(j)}), \quad f_n = w^T h^{(L)}/n,$$

$$\delta^{(L)} = w \odot \phi'_L(z^{(L)}), \quad \delta^{(j)} = \phi'_j(z^{(j)}) \odot W^{(j+1)T} \delta^{(j+1)}. \quad (\text{Setup-I.2})$$

Residuals are $r_a = f_n(v_a) - y_a$. The loss is

$$\mathcal{L}_n = m^{-1} \sum_a r_a^2.$$

The prescribed mobilities give the rank-one equations

$$\dot{W}^{(1)} = -\frac{2}{m} \sum_a r_a \delta_a^{(1)} v_a^T, \quad \dot{W}^{(j)} = -\frac{2}{mn} \sum_a r_a \delta_a^{(j)} h_a^{(j-1)T}, \quad \dot{w} = -\frac{2}{m} \sum_a r_a h_a^{(L)}. \quad (\text{Setup-I.3})$$

The setup replaces each ϕ_j in (Setup-I.2)–(Setup-I.3) by its computed degree- $D \geq 1$ polynomial ψ_j , but does not replace the exact initial source additions or the final runtime activations.

Supply a finite panel partition

$$0 = \tau_0 < \tau_1 < \dots < \tau_J = T, \quad \Delta_b = \tau_{b+1} - \tau_b,$$

with $J \geq 1$, a common Taylor degree $K \geq 1$, temporal degree $p \geq 0$, largest spherical degree ℓ_* , a finite retained mode set Λ , and its cardinality $N = |\Lambda|$. For $d \geq 2$, set

$$h_\ell = \binom{\ell + d - 1}{d - 1} - \binom{\ell + d - 3}{d - 1}, \quad H_{\text{sph}} = \sum_{\ell=0}^{\ell_*} h_\ell, \quad R = 2m + d + 1 + 4N. \quad (\text{Setup-I.4})$$

Impossible binomials are zero. The original weighted simplex is one allowed Λ ; its actual supplied cutoffs, rather than a rectangular enlargement, are used in every retained coefficient. There are $N_t \geq 1$ temporal nodes and $N_x \geq 1$ spatial nodes. For the positive angular rule of the quadrature proof, $N_x = N_\varphi N_\theta^{d-2}$ when $d \geq 3$, and $N_x = N_\varphi$ when $d = 2$.

For $d = 1$, use two point families, $N_x = H_{\text{sph}} = 2$, $N = 2(p + 1) = 2N_1$, and the same formula $R = 2m + d + 1 + 4N$. This is exactly $2m + d + 1 + 8N_1$; there is no angular quadrature.

Let $r_j = \dim E_j$ be the actual source-space ranks after coefficient construction, and let q_j be the actual selected support sizes. Put

$$r = \max_j r_j \leq \min(n, R), \quad q = \max_j q_j. \quad (\text{Setup-I.5})$$

The deterministic selector gives $r_j \leq q_j \leq 9r_j$, hence $r \leq q \leq 9r$. A supplied budget Q is sufficient when $9r \leq Q$; the original safe count uses $9R \leq Q < n$. Unused budget is not padded with extra coordinates. All occurrences of q in the costs below mean the actual retained maximum, not unused capacity. The constant source ensures $r_j \geq 1$.

Sampler contract. For each matrix (Setup-I.1), a stateful procedure accepts a finite adaptive sequence of requests $W_0^{(j)}x$ and $W_0^{(j)T}y$, with $x, y \in \mathbb{R}^n$. Queries may depend on all previous responses, the data, and the other layers' transcripts. Its joint responses admit a coupling with mutually independent matrices (Setup-I.1), under which every response is the corresponding exact action. The action procedure must handle both directions of one matrix consistently, including dependent and zero queries. For at most k vector requests at a layer, assume numerical upper bounds

$$O(nk^2) \text{ arithmetic}, \quad O(nk) \text{ words}, \quad \text{at most } nk \text{ independent standard normal draws.} \quad (\text{Setup-I.6})$$

It suffices to draw at most n normals for each newly exposed independent direction and none for dependent requests. Square roots used by normalization are accounted for separately below. The coupling must apply to globally interleaved adaptive requests, not just predetermined query vectors. The adaptive Gaussian theorem

proved above supplies this contract, including its deterministic bounded-stopping and global-interleaving hypotheses. Apply it to the hidden-matrix family indexed by $j = 2, \dots, L$, and include the independently generated first-layer matrix $W_0^{(1)}$ in the independent auxiliary randomness ξ . Thus queries depending on first-layer responses preserve the required conditional Gaussian law.

All arithmetic is exact real arithmetic, with exact zero/rank comparisons and the explicitly counted scalar primitives. Gaussian variates are ideal real normal samples. Activation evaluation costs and Gaussian-generation costs are not inferred from analyticity or hidden in arithmetic big-O constants.

Exact low-rank anchors and current-panel coefficients

At panel b , use normalized Taylor coefficients in the unscaled local variable $t - \tau_b$. Write

$$U_i^{(j)} = ([r_a \delta_a^{(j)}]_i)_{a=1}^m \in \mathbb{R}^{n \times m}, \quad H_k^{(j-1)} = ([h_a^{(j-1)}]_k)_{a=1}^m \in \mathbb{R}^{n \times m}.$$

Here the residuals and fields use the polynomial activations. Comparing coefficients in (Setup-I.3), for $1 \leq s \leq K$, gives

$$[W^{(j)}]_s = -\frac{2}{mn s} \sum_{i+k=s-1} U_i^{(j)} H_k^{(j-1)T}. \quad (\text{Setup-I.7})$$

The degree-zero term is the current anchor $W_b^{(j)}$. Its endpoint update is exactly

$$W_{b+1}^{(j)} - W_b^{(j)} = -\frac{2}{mn} \sum_{i=0}^{K-1} U_i^{(j)} V_i^{(j-1)T}, \quad V_i^{(j-1)} = \sum_{k=0}^{K-1-i} \frac{\Delta_b^{i+k+1}}{i+k+1} H_k^{(j-1)}. \quad (\text{Setup-I.8})$$

For each panel append the mK columns of the $U_i^{(j)}$, with the factor $-2/(mn)$, to a factor $A_b^{(j)}$, and the corresponding $V_i^{(j-1)}$ columns to $B_b^{(j)}$. These factor symbols are not the first weight matrix. They have shapes $n \times bmK$, and

$$W_b^{(j)} = W_0^{(j)} + A_b^{(j)} B_b^{(j)T}, \quad \text{rank}(W_b^{(j)} - W_0^{(j)}) \leq bmK. \quad (\text{Setup-I.9})$$

No independence, smallness, or low-rank property of the initialized matrix is used in (Setup-I.9). The rank bound holds for the finite numerical anchor increment because its endpoint is the specified degree- K polynomial. It is not a rank bound for an exact continuous trained increment.

For a current-panel query series $X(t) = \sum_{c=0}^K X_c(t - \tau_b)^c$, each coefficient of the constant-anchor action is computed as

$$W_b^{(j)} X_c = \text{Action}_j(X_c) + A_b^{(j)} (B_b^{(j)T} X_c). \quad (\text{Setup-I.10})$$

The transpose uses

$$W_b^{(j)T} X_c = \text{TransposeAction}_j(X_c) + B_b^{(j)} (A_b^{(j)T} X_c). \quad (\text{Setup-I.11})$$

The higher-parameter-jet contribution to the coefficient of order $s \geq 1$ is still the finite sum

$$-\frac{2}{mn} \sum_{i+k+c=s-1} \frac{U_i^{(j)} (H_k^{(j-1)T} X_c)}{i+k+1}. \quad (\text{Setup-I.12})$$

The cached-contraction implementation in the assembly proof computes (Setup-I.12) exactly. All its factors at order s have lower orders than s , so it remains causal during training-jet generation. Equations (Setup-I.10)–(Setup-I.12) give precisely the same local coefficient as multiplication by a materialized anchor and materialized parameter jets. Polynomial Chebyshev composition is also a finite exact recurrence. Induction over coefficient order, then over layers in the forward/backward passes, identifies every computed training and passive coefficient with its counterpart in the explicit algorithm.

The first matrix is stored explicitly in nd words. It does not require ndK persistent coefficient words. With $V = (v_a)_{a=1}^m \in \mathbb{R}^{d \times m}$, its higher coefficients act by

$$[W^{(1)}]_s x = -\frac{2}{ms} U_{s-1}^{(1)} (V^T x). \quad (\text{Setup-I.13})$$

Precompute $V^T V$ for training and compute $V^T x$ for each streamed passive query. To advance its anchor first sum $\sum_{i < K} \Delta_b^{i+1} U_i^{(1)} / (i+1)$, then multiply by $-2V^T/m$. Readout coefficients and endpoint updates are ordinary vectors obtained from $\dot{w} = -2 \sum_a r_a h_a^{(L)} / m$. Both calculations are exact rearrangements of the same finite sums.

Exhaustive matrix-access inventory and output coupling

Fix generator ordering, Gram–Schmidt sign conventions, deterministic selector tie-breaking, and all scalar quadrature/backend formulas. These choices select one concrete version of the explicit finite construction. Its only accesses to a hidden initialized matrix are the following.

1. **Exact original initialization.** Compute all training features at time zero with the original ϕ_j , saving the forward images $W_0^{(j)} h_0^{(j-1)}(v_a)$. This uses m forward actions at each hidden mixer. The backward fields are zero because $w_0 = 0$; no original derivative calls are required. Insert these exact feature/image vectors, the first-weight columns at layer one, and the constant vector into the source spaces. The degree-zero ψ_j features cannot replace them.
2. **Every panel’s forward/backward coefficients.** For each coefficient through K , the training batch uses at most m forward and m transpose actions per mixer through (Setup-I.10)–(Setup-I.11). Each passive query uses at most one of each. All current-anchor corrections and nonconstant parameter coefficients use retained factors and finite contractions, not new matrix entries. Panels with no assigned time nodes can omit passive queries, but the bound below permits all N_x queries on all J panels.
3. **Initialized images of completed global coefficients.** Accumulate only the base coefficient vectors for $h^{(j)}$ and $\delta^{(j)}$. For each retained mode form $W_0^{(j)} \tilde{c}^{h^{(j-1)}}$ and $W_0^{(j)T} \tilde{c}^{\delta^{(j)}}$. This uses at most N forward and N transpose actions per mixer. It is exact because all preceding projection operations are scalar finite sums shared by each source/image pair.
4. **Final basis-to-basis mixer.** Having constructed bases $U_j \in \mathbb{R}^{n \times r_j}$, with $U_j^T U_j / n = I$, request $W_0^{(j)} U_{j-1}$ column by column. This uses at most $r_{j-1} \leq r$ further forward actions. Paired source-image information is not assumed to cover these basis columns. Form $U_j^T W_0^{(j)} U_{j-1} / n$ from these returned vectors.

Consequently a sufficient number of vector requests per hidden mixer is

$$k_* = m + 2J(m + N_x)(K + 1) + 2N + r. \quad (\text{Setup-I.14})$$

The random realized rank in this pathwise cost bound is not a deterministic stopping budget. To invoke the sampler theorem, use the deterministic bound $m + 2J(m + N_x)(K + 1) + 2N + \min(n, R)$ at each layer, since $r \leq \min(n, R)$. Its online operation count may still be evaluated at the smaller actual request count and bounded by (Setup-I.14).

This counts requests even when their vectors vanish, repeat, or lie in a previously queried span. The adaptive sampler can save work in those cases; the sufficient envelope does not require such savings. Scalar data geometry, orthogonalization, selection, and metric formation do not access W_0 except through the four listed uses. In particular there is no request for its entries, Frobenius norm, singular values, or full operator norm.

To prove coupling of the full outputs, take a joint realization supplied by the sampler contract. Materialize its completed matrices only in the proof and run the explicit finite algorithm with those matrices. The original initialized features agree by induction over layers. Assume current anchors, stored jets, and accumulated global coefficients agree before one computation step. If the step requests a matrix action, (Setup-I.6), (Setup-I.10), and (Setup-I.11) give the exact explicit answer. If it performs a scalar operation, polynomial composition, contraction, or endpoint update, the operands and the operation are the same exact real quantities, with (Setup-I.8), (Setup-I.12), and (Setup-I.13) justifying any change in grouping. Thus the next state agrees.

Finite induction over all panels proves equality of the completed base coefficients, then the image coefficients in item 3.

The source generators and their order therefore agree. Exact orthogonalization returns the same ranks and bases. The selector sees the same row vectors and barrier comparisons, so the prescribed tie-breaking gives the same selected indices and weights. Item 4 then gives the same final compressed mixers. The metric formula and first-weight restriction also agree. In detail, with $P_j = (U_j)_{I_j}$, the initialized output is exactly

$$W_C^{(1)}(0) = (W_0^{(1)})_{I_1}, \quad w_C(0) = 0, \quad c_C(0) = y,$$

$$B_C^{(j)}(0) = P_j \frac{U_j^T W_0^{(j)} U_{j-1}}{n} P_{j-1}^T M_{j-1}. \quad (\text{Setup-I.15})$$

Any deterministic scalar failure branch in the explicit execution is reproduced as well. The algebraic statement holds for all finite supplied orders; source accuracy additionally requires the certified degree, panel, and quadrature certificates. This argument is equality under a coupling, not merely equality of marginal output laws. It preserves the original all-time comparison event and its probability without an additional failure budget.

The fresh-reference contract matters. This proof does not reproduce a previously materialized matrix, or a previously specified entrywise pseudorandom stream, without paying that reference's access costs. It supplies a fresh reference with the prescribed independent Gaussian law.

Scalar certificates and what is not tested

The finite construction takes admissible numeric structural certificates as inputs: a positive population gap γ , activation strip/derivative bounds, and the fitting/source constants required by the proved continuation. Non-elementary population moments used in the exact label allowance are also supplied if that allowance is to be evaluated exactly. An arbitrary analytic activation does not provide an algorithm for its Gaussian moment integrals. Obtaining such external certificates is not charged as free arithmetic here.

Given those quantities, the source constants (S.5)–(S.10), (S.22)–(S.25), the fitting/comparison recurrences, the restart constants, and the backend formulas are finite scalar recurrences. Their layer-indexed arrays have length $O(L)$; integer powers/factorials for the dimension use $O(d)$ multiplications. The label RMS costs $O(m)$ arithmetic and one square root. Thus their scalar preparation and deterministic gate comparisons cost $O(L + d + m + 1)$ arithmetic, elementary calls, and words as a safe common bound. Angular basis and mode enumeration costs are separately charged below.

The explicit width, radius, count, and analytic-extension gates are comparisons of these scalars and n . Their evaluation never tests a sampled hidden matrix's norm. Likewise the local Taylor construction selects its panels and degree from the explicit bounds; it does not form an n^2 -coordinate defect vector, solve an implicit equation, test membership in a complex parameter ball, or reject and recompute panels. Those norms and domains occur in the proof of its deterministic error bound. The underlying source probability event is an inherited theorem hypothesis, not a computable acceptance test, and its width threshold remains partly unquantified. There is no rejection sampling conditioned on that event.

If desired, the exact initial top feature Gram can be formed and factored in $O(nm^2 + m^3)$ arithmetic and $O(m^2)$ additional words. This is also a sufficient charge for preparing the final initial readout solve cache: exact source isometry makes the selected initial Gram the same Gram. It does not require querying any further initialized-matrix directions. The formulas below include this charge, even if a particular output omits that cache.

Complete supplied-order arithmetic and peak storage

All big-O constants in this section are numerical and implementation-dependent only. They hide no dependence on $L, m, d, n, J, K, D, p, \ell_*, N_t, N_x, N, R, r, q$, structural certificate values, confidence, activation cost, Gaussian cost, or precision. A bound is an operation count for this specified execution; it is not an optimality statement. Write k_* only for the explicit expression (Setup-I.14).

First account for the two terms introduced by implicit execution. The Gaussian action work is

$$O((L-1)nk_*^2), \quad O((L-1)nk_*) \text{ words.} \quad (\text{Setup-I.16})$$

At panel b , each forward/transposed application of the accumulated increment has cost $O(nbmK)$ per vector. There are at most $2(m + N_x)(K + 1)$ such vectors per mixer. Since $\sum_{b=0}^{J-1} b = J(J-1)/2$, their total work is

$$O((L-1)nm(m + N_x)J(J-1)K(K+1)). \quad (\text{Setup-I.17})$$

Retaining both rank factors uses $O((L-1)nJmK)$ words. Forming the grouped factors (Setup-I.8) costs $O(J(L-1)nmK^2)$. No dense rank-one outer-product update is executed.

The first matrix, its actions (Setup-I.13), and data pairings cost, sufficiently,

$$O(nd(1+m) + m^2d + J\{nd(m + N_x) + dmN_x + nm(m + N_x)(K+1)\}). \quad (\text{Setup-I.18})$$

Here nd creates the first matrix, ndm evaluates its original initialized training actions, and m^2d prepares the training input Gram. Each panel charges its direct first-matrix actions, passive input inner products, and all higher first-matrix coefficients. Its endpoint advancement fits the same bound. Spatial input inner products are recomputed per panel to avoid storing an m -by- N_x table.

The current-panel contractions (Setup-I.12), readout/residual series arithmetic, backward gate multiplication, and grouped factor formation are bounded by

$$O(JLm(m + N_x)(K+1)^2(n + K+1)). \quad (\text{Setup-I.19})$$

For (Setup-I.12), caching all $H_k^T X_c$ costs $O(nmb(K+1)^2)$ for a batch of b columns; their scalar weighted sums cost $O(mb(K+1)^3)$, and the final vector combinations cost $O(nmb(K+1)^2)$. Take $b = m$ for training and $b = 1$ for each passive query. The first-layer higher-coefficient term in (Setup-I.18) is absorbed by (Setup-I.19). Its separate display explains where it is paid.

The degree- D polynomial activation backend costs

$$O(L(D+1)^2 + JLn(m + N_x)(D+1)(K+1)^2) \quad (\text{Setup-I.20})$$

arithmetic, including coefficient construction and preparation of derivative polynomials. Online training composition needs $O(Lmn(D+1)(K+1))$ words. One offline scalar composition may reuse $O((D+1)(K+1))$ scratch. No original derivative calls occur.

Use spatial-first projection. On panel b , with time-node set I_b , form

$$\mathcal{B}_{kc}^{(b)} = \frac{\gamma_k}{N_t} \sum_{a \in I_b} \cos(ku_a)((t_a - \tau_b)/\Delta_b)^c, \quad \gamma_0 = 1, \quad \gamma_k = 2 \ (k > 0). \quad (\text{Setup-I.21})$$

These scalar tables cost $O((N_t + J)(p+1)(K+1))$ arithmetic and $O((p+1)(K+1) + N_t + J)$ words. For every spatial query accumulate the spherical coefficient of each local source coefficient, then transform only the retained mode pairs. Here the local source coefficients must first be converted from the unscaled convention in the exact implicit execution and full costs subsection 2 to the affine coordinate used by (Setup-I.21): for a base field g , define

$$G_{b,c}^g(v_s) = \Delta_b^c [g(\tau_b + \cdot, v_s)]_c, \quad \tilde{g}_b(t, v_s) = \sum_{c=0}^K G_{b,c}^g(v_s)((t - \tau_b)/\Delta_b)^c.$$

Consequently (Setup-I.21) acts on $G_{b,c}^g$, not on the raw Taylor coefficients. Generate the powers of Δ_b once per panel and rescale before spatial accumulation. This costs $O(JLnN_x(K+1))$ arithmetic, absorbed by the following projection bound because $H_{\text{sph}} \geq 1$, and requires no additional vector buffer. The full projection therefore costs

$$O(LnJ(K+1)(N_x H_{\text{sph}} + N)), \quad O(Ln(K+1)H_{\text{sph}}) \text{ extra words.} \quad (\text{Setup-I.22})$$

The completed global coefficient blocks use $O(LnR)$ words. They are not multiplied by the number of panels. Exact initialized-image actions have already been paid in (Setup-I.14)–(Setup-I.16), including their answer vectors. The optional sorting of time nodes costs $O(N_t \log(2 + N_t))$; a merge of the two cosine-grid halves can improve it.

For clarity, the following geometry choice recomputes spherical values on each panel. With the polar/azimuth orders defined after (Setup-I.4), take

$$G_{\text{time}} = \begin{cases} N_t + J, & d = 1, \\ N_t + N_x + JN_x(\ell_* + 1), & d = 2, \\ N_\theta^2 + dN_\theta + N_t + N_\varphi + d + d(\ell_* + 1)^2 + JdN_x[1 + (\ell_* + 1)^2 + H_{\text{sph}}], & d \geq 3, \end{cases} \quad (\text{Setup-I.23})$$

$$G_{\text{memory}} = \begin{cases} N_t + 1, & d = 1, \\ N_t + N_x + \ell_* + 1, & d = 2, \\ dN_\theta + N_t + N_\varphi + d[1 + (\ell_* + 1)^2 + H_{\text{sph}}], & d \geq 3. \end{cases} \quad (\text{Setup-I.24})$$

The rule and separated harmonic recurrences have numerical bounds $O(G_{\text{time}})$, $O(G_{\text{memory}})$. The term JdN_x includes repeated spatial coordinate/weight generation; it is not charged only on the first panel. The normalization constants are prepared once. The first case merely evaluates the two fixed points. If a different geometry implementation is chosen, its matched time and memory bounds must replace both (Setup-I.23) and (Setup-I.24).

Source orthogonalization, the conservative deterministic barrier selector, small basis contractions, and retained metric/matrix formation cost

$$O(LnRr + Lnr^3 + (L - 1)nr^2 + L(qr^2 + r^3 + q^2r) + qd). \quad (\text{Setup-I.25})$$

For orthogonalization, process each of at most R columns against at most r basis columns. At each of $9r_j$ selector steps, compute the shifted r_j -dimensional inverses in $O(r_j^3)$ and scan all n row vectors in $O(nr_j^2)$. Since $r_j \leq n$, their sum is $O(nr_j^3)$. This proves the selector term, with no assumed fast spectral-selection routine. After item 4 of the inventory, multiplying the returned basis images by U_j^T/n costs $O(nr^2)$ per mixer.

For the metric, with selected rows P , diagonal weights D , and $G = P^T D P$, use

$$M = D + DP(G^{-2} - G^{-1})P^T D, \quad M^{-1} = D^{-1} + P(I - G^{-1})P^T, \quad P^T M = G^{-1}P^T D. \quad (\text{Setup-I.26})$$

All inverses in (Setup-I.26) are on the r_j -dimensional positive Gram. The displayed matrix products give $O(qr^2 + r^3 + q^2r)$ work for each layer and its retained caches; final mixer multiplication has the same bound. Because $q \geq r$, this term is at most $O(Lq^2r)$, but (Setup-I.25) records its sources explicitly. First-weight restriction costs $O(qd)$. No initialized-matrix entry is needed in this stage.

Combining (Setup-I.16)–(Setup-I.25), one complete non-sampling, non-original-activation arithmetic envelope is

$$\begin{aligned} T_{\text{setup}} = & O(nd(1 + m) + m^2d + nm^2 + m^3 + L + d + m + 1 + L(D + 1)^2 \\ & + (L - 1)nk_*^2 + (L - 1)nm(m + N_x)J(J - 1)K(K + 1) \\ & + J[nd(m + N_x) + dmN_x] + JLn(m + N_x)(K + 1)^2(n + K + 1) \\ & + JLn(m + N_x)(D + 1)(K + 1)^2 \\ & + (N_t + J)(p + 1)(K + 1) + LnJ(K + 1)(N_x H_{\text{sph}} + N) \\ & + LnRr + Lnr^3 + (L - 1)nr^2 + L(qr^2 + r^3 + q^2r) + qd \\ & + G_{\text{time}} + N_t \log(2 + N_t)). \end{aligned} \quad (\text{Setup-I.27})$$

For peak resident real words, including phase-specific buffers safely by their sum, the matched sufficient envelope is

$$\begin{aligned}
M_{\text{setup}} = & O(nd + (L-1)n[k_* + JmK] + Lmn(D+1)(K+1) \\
& + Lm^2(K+1)^2 + LnR + Ln(K+1)H_{\text{sph}} \\
& + Lq^2 + q(d+Lm+1) + m^2 + m(d+1) \\
& + L(D+1) + (p+1)(K+1) + N_t + J \\
& + G_{\text{memory}} + L + d + m + 1).
\end{aligned} \tag{Setup-I.28}$$

The qLm term permits final training feature/backward caches. They can also be initialized directly by restriction of exact source features, with zero backward fields. Copying/zeroing their arrays costs $O(Lqm)$, which fits $LnRr$ because $q \leq 9r$, $m \leq R$. Other final state arrays fit (Setup-I.25); initial readout-Gram construction is already included in the first line of (Setup-I.27). The $Lmn(D+1)(K+1)$ term includes ordinary training jets, and $Lm^2(K+1)^2$ includes contraction caches. One passive query's vector jets fit the same envelope because $m, D \geq 1$. Small basis-selector arrays fit LnR and Lq^2 .

Unlike an explicit dense execution, (Setup-I.27)–(Setup-I.28) contain no mandatory hidden matrix initialization, anchor materialization, source-image multiplication, or final basis-image charge proportional to n^2 . All four are replaced by the explicitly charged transcript operations. This is not a uniform promise of subquadratic work at arbitrary orders: if k_* , r , or q are large, the displayed bounds can exceed n^2 . A branch returning full dense matrices necessarily has a quadratic output inventory. The present claim concerns the analytic selected construction under the fresh implicit reference contract.

Primitive calls and supplementary costs

The explicit first matrix requires exactly nd independent standard normal draws. By the sampler contract, the remaining draws are bounded by

$$N_{\text{Gaussian}} \leq nd + n \sum_{j=2}^L k_j \leq nd + (L-1)nk_*. \tag{Setup-I.29}$$

The count can be smaller than this because repeated/dependent directions need no fresh innovation. There are no Gaussian readout draws and no need to draw the as-yet unexposed part of any matrix. Sampling that part is only a possible mathematical completion used to define the coupled reference.

The activation backend uses $4(D+1)$ real scalar value calls per layer. Exact original initialized features use nm value calls per layer. Thus a sufficient original-activation call count is

$$N_{\phi} = 4L(D+1) + Lnm. \tag{Setup-I.30}$$

Add at most L calls if values $\phi_j(0)$ needed to compute a declared envelope have not been supplied. No high derivative, original first-derivative, complex-activation, or trained-reference calls occur. The backend's approximate value calls may use its specified accuracy; the original initialized features use exact real values in the current model. The equality proof requires the implicit and explicit executions to use the same value oracle or the same supplied approximate backend samples.

For a reproducible separation of elementary evaluations from additions, multiplications, divisions, and comparisons, define the angular table count

$$A_{\text{elem}} = \begin{cases} 0, & d = 1, \\ N_x, & d = 2, \\ N_{\theta} + N_{\varphi} + d(\ell_* + 1)^2, & d \geq 3. \end{cases}$$

A sufficient total count of square roots, trigonometric and fixed logarithmic/exponential/inverse-hyperbolic evaluations is

$$N_{\text{elem}} = O(L + d + m + 1 + L(D+1) + N_t + A_{\text{elem}} + (L-1)k_* + Lr). \tag{Setup-I.31}$$

The terms respectively cover scalar certificates, backend real cosine nodes, time nodes, angular tables and normalization square roots, query-basis normalizations in the sampler, and source orthogonalization.

A Cholesky factorization of the initial m -by- m Gram uses at most m additional square roots, already covered. Shifted selector inverses and the metric inverse can use elimination without spectral decompositions or further special functions. Integer rounding/floor operations used to choose the orders and enumerate the degree-dependent temporal cutoffs add $O(L + \ell_* + 1)$ scalar operations; their arithmetic/memory fits the geometry and mode inventory. The angular Gegenbauer normalizations are prepared once, and the cosine/harmonic recurrences do not call trigonometric functions anew at each coefficient or panel. Gaussian primitives themselves are counted only in (Setup-I.29); an implementation of the normal generator may add its own elementary calls.

If each class of scalar primitive has respective maximum work costs c_G, c_ϕ, c_e over the actual requested arguments and accuracies, add

$$c_G N_{\text{Gaussian}} + c_\phi N_\phi + c_e N_{\text{elem}} \quad (\text{Setup-I.32})$$

to arithmetic work, with the optional L activation calls just described. Add the maximum simultaneous scratch required by these routines to (Setup-I.28). One may instead sum the actual per-call costs. The usual bounded-cost primitive model makes (Setup-I.32) an operation-count shorthand, not a computability theorem for arbitrary analytic functions or a bit-complexity result.

Fixed-parameter specialization, proved from the full counts

Only in this section fix $L, m, d, \gamma, Y > 0$, admissible activation/source bounds and confidence. At the certified original specialization $T = 32(m/\gamma) \log(en)$, $\eta = 1/n$, the backend and bridge give

$$J = O(\log(en)^{3/2}), \quad K = O(\log(en)), \quad D = O(\log(en)^{3/2}),$$

$$N_t, p + 1 = O(\log(en)^{5/2}), \quad N_x, H_{\text{sph}} = O(\log(en)^{3(d-1)/2}),$$

$$N, R, r, q = O(\log(en)^{3d/2+1}). \quad (\text{Setup-I.33})$$

For $d = 1$, use the fixed two-point convention in place of angular growth; it gives the same last exponent $5/2$. The constants in (Setup-I.33) may now depend on the fixed admissible structural parameters. They were not hidden in (Setup-I.27).

Substitution into (Setup-I.14) first gives

$$k_* = O(\log(en)^{3d/2+1}), \quad JmK = O(\log(en)^{5/2}). \quad (\text{Setup-I.34})$$

The two new width-dependent terms are therefore, respectively,

$$nk_*^2 = O(n \log(en)^{3d+2}), \quad nm(m + N_x)J(J-1)K(K+1) = O(n \log(en)^{3d/2+7/2}). \quad (\text{Setup-I.35})$$

The activation term has the second exponent in (Setup-I.35). The local contraction term proportional to n has exponent at most $3d/2 + 2$. The spatial-first projection terms have exponents at most $3d - 1/2$ and $3d/2 + 7/2$. Orthogonalization and small basis contractions have exponents at most $3d + 2$. Finally,

$$nr^3 = O(n \log(en)^{9d/2+3}). \quad (\text{Setup-I.36})$$

For every fixed integer $d \geq 1$, $3d+2 \leq 9d/2+3$, $3d/2+7/2 \leq 9d/2+3$, and $3d-1/2 \leq 9d/2+3$. Thus every width-proportional term in (Setup-I.27) is bounded by (Setup-I.36). The remaining terms, including final $q^2 r$ assembly and geometry, are fixed powers of $\log(en)$, hence also bounded by (Setup-I.36) for sufficiently large n . This proves, with bounded-cost scalar primitives,

$$T_{\text{setup}} = O(n \log(en)^{9d/2+3}). \quad (\text{Setup-I.37})$$

For memory, the transcript and global coefficients are $O(n \log(en)^{3d/2+1})$. Rank-history and online activation storage are $O(n \log(en)^{5/2})$, which fit that bound because $d \geq 1$. The spatial-first vector buffer is

$O(n \log(en)^{3d/2-1/2})$. All remaining terms in (Setup-I.28) are either $O(n)$ at fixed parameters or fixed powers of $\log(en)$. Therefore

$$M_{\text{setup}} = O(n \log(en)^{3d/2+1}). \quad (\text{Setup-I.38})$$

The primitive inventories specialize to $N_{\text{Gaussian}} = O(n \log(en)^{3d/2+1})$, $N_\phi = O(n + \log(en)^{3/2})$, and a fixed power of $\log(en)$ for (Setup-I.31). Additional costs from nonunit scalar backends remain governed by (Setup-I.32), not automatically by (Setup-I.37).

The exact retained Harmonic state, metric, initial-matrix, and cache inventory is unchanged, including its bound $1020(L+1)R^2 + 10m(d+1)$, hence $O(\log(en)^{3d+2})$ at (Setup-I.33). The Gaussian transcripts, rank histories, first matrix, source coefficients, bases, and local backend arrays are all discarded after (Setup-I.15) and the final caches are formed. The reference is a coupled mathematical Gaussian network after disposal; the retained model does not include an oracle or hidden reference access.

The proved source approximation and original all-time comparison then give the same $n^{-1+o(1)}$ prediction error under the same event and label allowance. This component changes execution, not that error coefficient. For an arbitrary supplied budget whose actual horizon or required accuracy grows faster than in (Setup-I.33), only the finite bounds (Setup-I.14), (Setup-I.27)–(Setup-I.32) are asserted. They do not imply the logarithmic specializations (Setup-I.37)–(Setup-I.38).

F Residual-adapted finite-panel compression

This appendix proves the Logarithmic row of Theorem 3.1. It uses the same selected-coordinate optimizer as Harmonic compression, but constructs its source spaces from finitely many response curves. The new argument concerns the complex-time domain of those curves. The coordinate-selection, fitting, and cancellation arguments are reused only after their hypotheses have been verified below. All references in this appendix are to proofs included in this paper.

F.1 Contract, exact coefficients, and width qualification

Fix the training inputs and p additional inputs before initialization. Only the first m inputs have labels or enter training sums. Write $v_i = x_i/\sqrt{d}$, so $\|v_i\|_2 = 1$, and use the canonical Gaussian initialization, zero readout, mobilities, and physical time of the main text. The passive labels are neither required nor used. The activation assumptions are those of §E.1: holomorphy on $|\text{Im } z| < a$, reality on the real axis, and bounded first derivative on that strip. Activation values need not be bounded. Put

$$\begin{aligned} b &= \max_j |\phi_j(0)|, & s &= \max\{1, \sup_{j, |\text{Im } z| \leq a/2} |\phi'_j(z)|\}, \\ t_2 &= \max\{1, \sup_{j, |\text{Im } z| \leq a/2} |\phi''_j(z)|\}, & \beta &= \max(10, 1 + b, s, t_2, 16/a). \end{aligned}$$

The second derivative bound is finite by the strip hypothesis and Cauchy's formula. Throughout this appendix the local abbreviations are

$$\lambda = \gamma/m, \quad Y = \|y\|_2/\sqrt{m} > 0, \quad S = 16Y/\lambda, \quad \ell = \log(en), \quad \nu = \lambda/4, \quad T_0 = 32\ell/\lambda.$$

The case $Y = 0$ has the exact stationary zero predictor and needs no division by Y . The full label allowance is

$$\frac{Y}{\lambda} \leq \min \left\{ \frac{1}{8H_d\sqrt{F_d}}, \frac{1}{16H_c\sqrt{F_c}}, \frac{S_*^{\text{src}}}{16} \right\}. \quad (222)$$

Here H_d, F_d, H_c, F_c are exactly the finite recurrences in (79)–(80), and S_*^{src} is the explicit minimum (S.10). Thus (222) is precisely the existing Harmonic allowance, not a new smaller cap. In particular $S \leq 1$. The additional Legendre allowance is imposed only when all three methods are asserted together.

For an explicit finite prescription we recall the source coefficients. All symbols in the following recurrence are fixed scalars, not model coordinates or approximation orders:

$$\begin{aligned}
H_1 &= b + 20s, & H_j &= b + 10sH_{j-1}, & k_j &= H_L(10s)^{L-j}, & \tau_j &= sk_j, \\
P_1 &= 3, & P_j &= H_{j-1} + 10sP_{j-1} + 1, & f_j &= sP_j, \\
g &= \tau_1 + \sum_{j=2}^L \tau_j H_{j-1}, & r_j^{\text{src}} &= P_j g, & q_j^{\text{src}} &= f_j g, \\
A_* &= 2sP_L + 2s^2 \sum_{j=2}^L k_j P_{j-1}, & H_* &= A_* + t_2 \sum_{j=1}^L P_j^2 k_j, \\
E_* &= t_2 \sum_{j=1}^L (10s)^{2(j-1)} k_j.
\end{aligned}$$

Continue with

$$\begin{aligned}
D_0 &= (1 + b + s) \{1 + 2H_*^2 + 4A_*^2(e^2 - 1) + E_* + s^2 \max_j k_j^2\}, \\
K_{\text{src}} &= 128(D_0 + 1) \{1 + 32 \max(1, \max_j H_j, \max_j \tau_j)\}, \\
e_1 &= t_2 r_1^{\text{src}} P_1, \\
e_j &= t_2 r_j^{\text{src}} P_j + s(q_{j-1}^{\text{src}} + \tau_j H_{j-1} f_{j-1} + 10e_{j-1}), \\
T_Q &= 8 \max_j f_j \max_j (f_j H_* + e_j), \\
U &= U_{\text{fin}}(S) = \max \{4sK_{\text{src}}, \\
&\quad \max_{j \geq 2} 2 [sK_{\text{src}}(H_{j-1}^2 + f_{j-1}^2 + ST_Q + S^2 H_{j-1} q_{j-1}^{\text{src}}) + 64q_{j-1}^{\text{src}} + 1] \}.
\end{aligned}$$

These are the coefficients of §E.1.2 and §D.10; the superscripts on $r_j^{\text{src}}, q_j^{\text{src}}$ prevent confusion with residuals and selected widths. Define

$$\begin{aligned}
\mathcal{K} &= H_L^2 + S^2 \left[\tau_1^2 + \sum_{j=2}^L \tau_j^2 H_{j-1}^2 \right], & M_0 &= 10 \max_j (H_j, \tau_j), \\
r_n &= \frac{a}{64YSU\sqrt{\ell}}, & D_W &= \max\{\tau_1, \max_{j \geq 2} \tau_j H_{j-1}\}.
\end{aligned} \tag{223}$$

In the source budget (S.10), denote its exponential coefficient by η_{src} and its ceiling by $\mathcal{B} = 1024e^2L$, to distinguish them from an approximation tolerance. Both retain their exact definitions there.

The theorem is eventual for each fixed admissible problem and each fixed confidence. In addition to the unchanged initialization and source-event onsets, the numerical gates used below are

$$n^{-1} \leq \min(1, Y, S), \quad \ell \geq \max(e^2, 2\mathcal{B}), \quad \sqrt{\ell} \geq \frac{a}{64YSU} \max\{8, \lambda, 4\mathcal{K}/\log 2, 32YSD_W\}, \tag{224}$$

$$4\mathcal{K}r_n + \frac{2C_g}{\sqrt{\mathcal{K}}} B_n Y r_n < \log(3/2), \quad B_n = 1 + \frac{S^2}{\eta_{\text{src}}} \log(e + \ell), \tag{225}$$

where the finite coefficient C_g is given in (228) below. The fitted-tail and degree gates are (236) and (238). Every displayed gate holds eventually at fixed parameters. Already $\ell \geq 2\mathcal{B}$ requires $n \geq \exp(2048e^2L - 1)$, which exceeds 10^{13143} at the minimum depth $L = 2$. This deterministic requirement is far beyond practical widths; the remaining gates and the probability onset may require still larger widths. The probability onset is qualitative: no effective polynomial threshold, uniform growing-panel statement, or simultaneous event over infinitely many independent widths is asserted.

F.2 The enlarged analytic domain

The independently proved real fitting theorem gives $\rho(t) = \|r(t)\|_2/\sqrt{m} \leq Y e^{-\nu t}$, physical operator caps strictly below nine, and $2 \int_0^\infty \rho \leq S/2$. These facts also hold for each fixed-size rectangular cavity on its own initialization event. This last statement uses the zero-embedding calculation (IC.71)–(IC.73); it does not identify a rectangular cavity covariance with the full one.

Put

$$h(t) = \frac{1}{2\mathcal{K}} \log(1 + 2\mathcal{K}r_n e^{\nu t}), \quad h_0 = h(0), \quad d_h = \frac{\nu}{2\mathcal{K}}, \quad T_+ = T_0 + h(T_0) + r_n. \quad (226)$$

Since $\lambda \leq H_L^2 \leq \mathcal{K}$, $0 \leq h' \leq d_h \leq 1/8$, $h_0 \leq r_n$, and $h(t) \leq r_n + d_h t$. The inequality $\lambda \leq H_L^2$ follows by the population RMS recurrence and $\gamma \leq Q_{aa}^{(L)}$. For $0 \leq s_0 \leq 1$ use

$$D_{s_0} = \{x + iy : -s_0 h_0 \leq x \leq s_0 T_+, \quad |y| \leq s_0 h(\max(x, 0))\}.$$

These nested, vertically fibered domains are not assumed convex. Each point is reached by a real segment from zero and then a vertical segment. For negative real part the combined exceptional length is at most $2h_0$.

Each cavity is stopped independently at its own first-exit level s_c . Use the source stops of §E.1.1: full and cavity exponential budgets $\mathcal{B}, 2\mathcal{B}$, provisional maxima and response caps twice and four times their claimed values, pole margins $3a/8, 7a/16$, and the unchanged operator, activity, row-size and passive-preactivation stops. A failed own initialization gives identically zero reference coefficients. This convention precedes integration over omitted roots.

Here is the globally defined extension needed for conditional Gaussian arguments. For real-interval clipping write

$$X_{s_0}(x) = \text{clip}_{[-s_0 h_0, s_0 T_+]}(x), \quad P_{s_0}(x + iy) = X_{s_0}(x) + i \text{clip}_{[-s_0 h(X_{s_0}(x)_+), s_0 h(X_{s_0}(x)_+)]}(y).$$

This is a retraction onto D_{s_0} , measurable in s_0 . With $a_0(z) = \text{clip}_{[0, T_+]}(\text{Re } z)$, direct clipping gives

$$P_{s_0} a_0(z) = a_0(P_{s_0} z), \quad \text{Lip}(P_{s_0}) \leq 1 + d_h. \quad (227)$$

For the Lipschitz assertion the coordinate increments satisfy $|\Delta X| \leq |\Delta x|$ and $|\Delta \text{Im } P| \leq |\Delta y| + d_h |\Delta x|$; the resulting shear has norm at most $1 + d_h$. For a coefficient with derivative bound on D_{s_0} , integrate along the image under P_{s_0} of a straight segment. This proves the corresponding global Lipschitz bound without convexity. Extend each stopped coefficient as $b_c(P_{s_c} z)$ on the deterministic rectangle $E_n = [-h_0, T_+] + i[-h(T_+), h(T_+)]$. Its real anchors traverse a monotone real prefix and then freeze. Equation (227) supplies the same commuting-anchor identity as (IC.stop).

Local holomorphic ODE continuation and uniqueness apply on every strict prefix: finite parameter bounds and a strict activation-strip margin permit extension, and overlapping real-anchor/vertical patches agree. The domain retracts onto its real interval and is simply connected. At a first exit, the strict margin below the half-strip permits one-sided derivative estimates. A strict improvement of all stops therefore extends the solution across the compact boundary. The clipped extensions themselves need not be holomorphic outside their own stopped domains.

F.3 The derivative estimate and the noncircular propagator bound

Work first under each cavity's provisional response stop $\max |R_{ba,i}^{(j)}| \leq 4SU\sqrt{\ell}$, where $R_{ba}^{(j)} = D_\Theta z_b^{(j)} \nabla_\Theta F_a$, $F_a = n f_n(v_a)$, and $\Theta = (A, \sqrt{n}W^{(2)}, \dots, \sqrt{n}W^{(L)}, w)$. The exact derivative equations give

$$\frac{\|\dot{z}_a^{(j)}\|_2}{\sqrt{n}} \leq 2\rho S r_j^{\text{src}}, \quad \|\dot{z}_a^{(j)}\|_\infty \leq 8\rho S U \sqrt{\ell}.$$

Set $N_* = \max_j \{r_j^{\text{src}}, 4U\}$. Interpolation and the stopped exponential carrier budget imply, for $u \geq 4$,

$$\frac{\|\dot{z}_a^{(j)}\|_{2u/(u-2)}}{n^{(u-2)/(2u)}} \leq 2\rho S N_* \ell^{1/u}, \quad \frac{\|k_a^{(j)}\|_u}{n^{1/u}} \leq \frac{S}{\eta_{\text{src}}} u(2\mathcal{B})^{1/u}.$$

Hölder consequently bounds the normalized 2-norm of $k_a^{(j)} \odot \dot{z}_a^{(j)}$ by $2\rho S^2 N_* u (2\mathcal{B}\ell)^{1/u} / \eta_{\text{src}}$. Choose $u = \max(4, \log(2\mathcal{B}\ell))$. Under (224), the exponential factor is at most e and $u \leq 2\log(e + \ell)$. Differentiation of the backward recursion then gives

$$\begin{aligned} J_L &= 2sH_L + 4et_2N_*, & J_j &= 10sJ_{j+1} + 2s^3k_{j+1}^2H_j + 4et_2N_*, \\ J_* &= \max_j J_j, & \frac{\|\dot{h}_a^{(j)}\|_2}{\sqrt{n}} &\leq 2\rho Sq_j^{\text{src}}, & \frac{\|\dot{\delta}_a^{(j)}\|_2}{\sqrt{n}} &\leq J_*B_n\rho. \end{aligned}$$

The four terms are respectively the changed readout, propagated upper derivative, changed mixer, and changed activation gate. This derivation uses only the provisional stops, not their later Gaussian improvement.

For the normalized gradient matrix $G = (mn)^{-1/2}[\nabla_{\Theta}F_1, \dots, \nabla_{\Theta}F_m]$, its column blocks are $m^{-1/2}$ times $\delta_a^{(1)}v_a^\top/\sqrt{n}$, $\delta_a^{(j)}h_a^{(j-1)\top}/n$, and $h_a^{(L)}/\sqrt{n}$. Differentiating these products proves

$$\begin{aligned} \|G\|_{\text{op}} &\leq \sqrt{\mathcal{K}}, & \|\dot{G}\|_{\text{op}} &\leq C_g B_n \rho, \\ C_g &= J_* \left(1 + \sum_{j=2}^L H_{j-1} \right) + 2S^2 \sum_{j=2}^L \tau_j q_{j-1}^{\text{src}} + 2Sq_L^{\text{src}}. \end{aligned} \quad (228)$$

Bounding the Frobenius norm after division of each column by $\sqrt{m}$ introduces no sample factor.

The exact residual equation is $\dot{r} = -2G^\top Gr$, and the residual-free variational base is $\dot{P} = -2GG^\top P$. Transposes here are algebraic. On a vertical segment from a real anchor $t \geq 0$, norm differentiation yields

$$\rho(t \pm iv) \leq Y e^{-\nu t} e^{2\mathcal{K}v}, \quad \int_0^{h(t)} \rho(t \pm iv) dv \leq Y r_n. \quad (229)$$

The integral follows by substituting $e^{2\mathcal{K}h(t)} = 1 + 2\mathcal{K}r_n e^{\nu t}$. It also implies $\rho \leq Y + 2\mathcal{K}Y r_n \leq 2Y$.

At the real anchor $A_t = 2G(t)G(t)^\top$ is real symmetric; therefore $-iA_t$ generates unitary motion. Freeze this generator along the vertical segment. By (228),

$$\|2G(t \pm iv)G(t \pm iv)^\top - A_t\|_{\text{op}} \leq 4\sqrt{\mathcal{K}}C_g B_n \int_0^v \rho(t \pm iu) du.$$

Conjugating by the frozen unitary gives a propagator norm at most

$$\exp \left(4\sqrt{\mathcal{K}}C_g B_n \int_0^{h(t)} (h(t) - v) \rho(t \pm iv) dv \right) \leq \exp \left(\frac{2C_g}{\sqrt{\mathcal{K}}} B_n Y r_n \right). \quad (230)$$

Indeed the double integral is at most $Y r_n / (2\mathcal{K})$ by integrating the exponential in (229). Products on ordered disjoint subsegments have the same bound: use the same frozen generator and add their nonnegative growth integrals. Forward real segments contract. The negative-real corner instead has length at most $2r_n$ and costs at most $e^{4\mathcal{K}r_n}$. A common bound for the resulting products is therefore

$$e^{4\mathcal{K}r_n} \exp \left(\frac{2C_g}{\sqrt{\mathcal{K}}} B_n Y r_n \right) = \exp \left(4\mathcal{K}r_n + \frac{2C_g}{\sqrt{\mathcal{K}}} B_n Y r_n \right) < \frac{3}{2}, \quad (231)$$

by (225). This is strictly within the factor two required in (IC.8) and (IC.51). This is established *before* Gaussian insertion.

The extra activity is at most $8Y r_n$. The original gates give $8Y r_n \leq S/2$ and $8YSD_W r_n \leq 1/4$, preserving activity cap S , strict operator margins inside ten, and $\rho/S \leq \lambda/8$. These are all the physical hypotheses of (IC.4). Since $r_n = O(\ell^{-1/2})$ and $B_n r_n = O(\log(e + \ell)/\sqrt{\ell})$, both exponents in (231) tend to zero at fixed parameters. The summed propagator gate is therefore eventual without a new label restriction.

F.4 Transfer of the source proof to the larger domain

We now verify the domain-dependent hypotheses of the complete insertion proof §E.1.1. Its parameter derivative, nonlinear-remainder, and Gaussian-form identities are unchanged. First, canonical anchor contours

have length at most $(64/\lambda + 4r_n\sqrt{\ell})\ell$. Replacing the contour-length coefficient in (IC.10) by this fixed number leaves the derivative recurrences (IC.5)–(IC.17) and the passive response recurrences (IC.36)–(IC.43) of size $n^{1/4000}\text{poly}(\ell)$, with an additional $\sqrt{n}$ for raw terminal derivatives. Their inputs are precisely the unchanged operator/RMS/budget bounds, residual activity, provisional maxima, and base product bound just verified.

Second, the one-dimensional control nets of (IC.44) still have logarithmic cardinality $n^{5/8}\text{poly}(\ell)$: control amplitudes and speeds remain respectively polynomial-logarithmic and $\sqrt{n}\text{poly}(\ell)$ along those contours. The globally clipped coefficient paths have the same moduli up to $1 + d_h \leq 9/8$. A deterministic endpoint mesh in E_n , whose sides are $O(\ell)$, has polynomial size. The finite panel and any fixed deletion-count unions therefore cost at most $n^{0.65}$ in logarithmic cardinality eventually, whereas the conditional Gaussian linear/quadratic tails in (IC.45) have exponent $n^{0.78}$. The same uniform event consequently fails with probability at most $e^{-n^{0.7}}$ eventually. Deterministic one-dimensional control histories are netted before substituting actual analytic histories; no arbitrary two-dimensional control extension is being assumed.

Third, the remainder identities (IC.22)–(IC.43) use joining segments inside the same safe activation half-strip. Their Duhamel bound (IC.46) uses the same activity and propagator bound. With the original cutoffs $N = n^{1/100}$, $d_0 = n^{-1/10}$, $u_0 = n^{-1/25}$, its remainder remains

$$n^{1/4000}\text{poly}(\ell)[n^{-8/100} + u_0^2 + n^{-48/100} + n^{-1/2}] = o(u_0).$$

The common-prefix comparisons are therefore still coordinate discrepancy $n^{-1/30}$ and whole-vector discrepancy $2n^{1/100}$. The direct passive reverse observable in (IC.39) is included; it is not replaced by an independent reverse operator.

Fourth, the trace expansion (IC.48)–(IC.57) requires products over ordered disjoint base-propagator segments, exactly as verified in (231). Its ordered residual integrals remain bounded by $S^h/h!$. Thus its Schatten coefficients, direct exterior trace, learned row/column terms, and sample-RMS factors are unchanged. In particular the scalar absorption (S.16) and the coefficient U_{fin} above are unchanged.

The scalar Gaussian maximum grid now has at most $O(n^4\ell^2) \leq n^5$ points eventually. Fixed panel, layer, driver, and root indices fit the original enclosing count n^{10} . The tail union $4n^{10}e^{-1024\ell}$ and off-grid error $n^{-3/2}\text{poly}(\ell)$ vanish. Hence the full provisional caps improve to

$$\max |z_{a,i}^{(j)}| \leq K_{\text{src}}\sqrt{\ell}, \quad \max |k_{a,i}^{(j)}| \leq SK_{\text{src}}\sqrt{\ell}, \quad \max |R_{ba,i}^{(j)}| \leq SU\sqrt{\ell}. \quad (232)$$

The last estimate has each training driver and each declared evaluated input. Passive backward budgets are unnecessary. The passive preactivation stop improves by its initial finite-panel Gaussian maximum and its activity integral. Combining (229) and (232) bounds displacement from a real anchor by $a/32$; the negative corner gives $a/8$. The pole stop improves strictly.

A cavity cannot stop first on the successful full prefix: coordinate comparison transfers its maxima and pole margin, while its budget is at most $e^{2\eta_{\text{src}}n^{-1/30}}\mathcal{H}_a + O_u(1/n) < 2\mathcal{B}$ for fixed deletion count u . Operator and activity margins were already proved deterministically. This verifies the strict independent-cavity transfer of (IC.stop). No Gaussian law has been conditioned on full-network survival.

It remains to remove the budgets on complete independently stopped paths, not just that prefix. For a forward coefficient normalized by $\sqrt{n}$, its complex-minus-real radius is at most $2S\max_j q_j^{\text{src}}Yr_n$. For a backward coefficient normalized by $S\sqrt{n}$, it is at most $J_*B_nYr_n/S$. The negative corner costs at most four times these amounts. Commutation (227) thus gives the complete-path radius

$$D_n = 4Yr_n \max_j \{2S\max_j q_j^{\text{src}}, J_*B_n/S\} = O(\log(e + \ell)/\sqrt{\ell}).$$

Its global normalized modulus is $O(B_n)$. Applying the proved Gaussian rectangle estimate (IC.77) gives mean $O(\log(e + \ell)^{3/2}/\sqrt{\ell})$, tail radius $2D_n$, and every fixed exponential moment tending to one. Its sample-RMS version requires no independence among samples.

For singleton/common-cavity differences, extend each path with its own clipping map, subtract, and project onto the deterministic ball of radius $4n^{1/100}$. Both paths omit the root paired with that difference; projection preserves this independence and is the identity on the successful prefix. The normalized Gaussian radius is $4n^{-49/100}$, its modulus is polynomial-logarithmic, and (IC.77) gives fixed exponential moments tending to one, as in (IC.78).

The real reference paths retain their original activity modulus and squared-exponential estimate (S.20). Distinct omitted roots are conditionally independent given the common cavity. Separate their main-reference products from all correction products by Cauchy–Schwarz and then Hölder at each fixed moment order. The one-neuron main moment and scalar absorption retain the limiting base sixteen of (S.29). Joint independence of correction paths is neither used nor asserted. Indicators of the full event are dropped only from nonnegative, globally defined cavity bounds. For a tuple of moment order u with only $k < u$ distinct roots, the normalized multiplicity is $O_u(n^{k-u})$. Repeated weights cost at most $e^{2u\eta_{\text{src}}K_{\text{src}}\sqrt{\ell}} = n^{o(1)}$ by (232), so their contribution vanishes. The local exceptional events have superpolynomially small probability. Markov’s inequality and the fixed sample/layer union now give

$$\limsup_{n \rightarrow \infty} \Pr\{\text{a full budget is hit}\} \leq mL(16L/\mathcal{B})^u.$$

Take the width limit at each fixed positive integer u , then the infimum over u . Since $16L/\mathcal{B} < 1$, the probability tends to zero. All other stops have strict improvements. We have therefore proved holomorphy on a neighborhood of D_1 , with the original coordinate bound $M_0\sqrt{n}$, for all required finite-panel sources, with probability tending to one. This proves the new source event, rather than assuming it from the smaller rectangular event.

F.5 Adaptive polynomial sources and initialization-only coefficients

At each layer include the curves

$$h^{(j)}(t, v_i), W_0^{(j)} h^{(j-1)}(t, v_i) \quad (i \leq m+p), \quad \delta_a^{(j)}(t), W_0^{(j+1)\top} \delta_a^{(j+1)}(t) \quad (a \leq m),$$

omitting nonexistent boundary images. There are at most $2(2m+p)$ curves per layer. Linear activation growth $|\phi_j(z)| \leq b + s|z|$, the source RMS bounds, and initialized operator norm eight imply their coordinate bound $M_0\sqrt{n}$.

Set $\varrho(t) = h(t)/[2(1+d_h)]$ for $0 \leq t \leq T_0$. Its closed disk about t lies in D_1 : for nonnegative real coordinate x , $h(x) \geq h(t) - d_h\varrho(t) \geq \varrho(t)$. If the disk reaches a negative coordinate, then $t < \varrho(t)$ and $h(t) \leq h_0 + d_h t$ give $\varrho(t) < h_0$. The right endpoint is below $T_0 + h(T_0) < T_+$. The radius is positive, nondecreasing, and Lipschitz with constant $d_h/[2(1+d_h)]$.

Starting at $t_0 = 0$, set $t_{b+1} = \min(T_0, t_b + \varrho(t_b)/2)$ until $t_J = T_0$. For any nondecreasing radius with Lipschitz constant C_r , every full panel contributes at least $1/(2+C_r)$ to $\int dt/\varrho(t)$; hence $J \leq 1 + (2+C_r) \int_0^{T_0} dt/\varrho(t)$. For our radius this yields

$$J \leq 1 + 40960 \frac{U}{a} \left(\frac{Y}{\lambda} \right)^2 \sqrt{\ell} + 80 \frac{\mathcal{K}}{\lambda} \log \left(1 + \frac{4\ell}{\log 2} \right). \quad (233)$$

For completeness, it suffices to bound J by $1 + 5 \int_0^{T_0} dt/h(t)$. Split where $2\mathcal{K}r_n e^{\nu t} = 1$. Before that point use $\log(1+x) \geq x/2$; afterward its logarithm grows at least with slope $\nu/2$. Thus the integral is at most $2/(\nu r_n) + (4\mathcal{K}/\nu) \log(1 + \nu T_0/(2 \log 2))$, giving the displayed constants.

Given source tolerance $0 < \eta \leq \min(1, Y, S)$, choose

$$K = \max \left(0, \left\lceil \log_2 \frac{8M_0\sqrt{n}}{\eta} \right\rceil \right). \quad (234)$$

On panel b , the coefficients $\varrho(t_b)^k u^{(k)}(t_b)/k!$ have coordinate norm at most $M_0\sqrt{n}$, by Cauchy’s formula on its disk. Since $|t - t_b|/\varrho(t_b) \leq 1/2$, the degree- K tail is at most $M_0\sqrt{n} 2^{-K} \leq \eta/8$. All coefficients span one time-independent space of dimension at most $J(K+1)$ per curve. Adjacent approximants need not agree at a panel boundary: each approximates the same source there, and the comparison proof uses source values, never derivatives of the piecewise approximant.

These coefficients can be computed from finite initialization jets, including derivatives requested at later anchors. Indeed D_1 contains a uniform rectangle about $[0, T_0]$ of radius $\varrho(0) > 0$. Use this radius in the

explicit conformal map (159), denoted here $t = \mathbf{t}(\xi)$. It satisfies $\mathbf{t}(0) = 0$, has nonzero derivative, maps the unit disk into that rectangle, and maps $[0, \xi_*]$, $\xi_* < 1$, onto $[0, T_0]$. For $G(\xi) = u(\mathbf{t}(\xi))$,

$$[\xi^j]G = \sum_{k=0}^j \frac{u^{(k)}(0)}{k!} [\xi^j] \mathbf{t}(\xi)^k.$$

Every coefficient therefore uses finitely many zero-time derivatives. They are obtained by differentiating the dense ODE at initialization; in normalized form, if $\dot{\Theta} = F(\Theta)$, the recurrence is $\Theta_{j+1} = (j+1)^{-1}[t^j]F(\sum_{k=0}^j \Theta_k t^k)$. It uses initialized weights, training data/labels, passive inputs, and the specified activation-derivative evaluator, not trained weights.

To see that later derivatives are also finite computations, choose $\xi_* < \rho_* < 1$ and $0 < \delta_* < \rho_* - \xi_*$. The degree- N Taylor sum G_N obeys

$$\sup_{|\xi| \leq \rho_*} |G - G_N| \leq \frac{M_0 \sqrt{n} \rho_*^{N+1}}{1 - \rho_*}, \quad |(G - G_N)^{(k)}(\xi_b)| \leq \frac{k! M_0 \sqrt{n} \rho_*^{N+1}}{\delta_*^k (1 - \rho_*)}.$$

The second inequality is Cauchy's derivative formula. Applying $(t'^{-1} \partial_\xi)^k$ converts these to physical derivatives; its coefficients are finite at the finitely many anchors for $k \leq K$. A finite common N attains any prescribed coefficient accuracy. In particular, coordinate tolerance $\eta/(128\sqrt{n})$ for each scaled coefficient guarantees the Euclidean tolerance used below. This proves finite initialization-only provenance, not cheap arithmetic or well-conditioned numerical recovery.

Approximate each scaled base coefficient in Euclidean norm to $\eta/128$, and form its paired initialized image by the exact initialized matrix action. The base reconstruction error is at most $(\eta/128) \sum_{k \geq 0} 2^{-k} = \eta/64$; initialized norm eight makes its image error at most $\eta/8$. Adding Taylor tails keeps both source errors below η , while their image relation is exact. Thus no small coordinate error is incorrectly multiplied by a width-dependent infinity norm. An invertible conversion of each local polynomial to Chebyshev coordinates changes neither its vector span nor these paired actions; it is optional.

Add the constant vector, exact initialized training features and their images, and all first-weight columns. The resulting source dimension per layer is at most

$$R = 2m + d + 1 + 2(2m + p)J(K + 1). \quad (235)$$

The selection proof §E.4.3 supplies $q_j \leq 9R$, exact normalized source isometry, and metrics $D_j/4 \preceq M_j \preceq D_j$, $\mathbf{1}^\top D_j \mathbf{1} \leq 4$. The same selected initialized mixer and its metric adjoint preserve the forward and reverse image pairs. Exact training additions preserve the initial training Gram. These verify every initialization hypothesis of §D.14.2.

F.6 Autonomous runtime, all-time error, and retained inventory

Use the moving arrays and corrected readout (86)–(89), with training indices only in the deficit and Gram updates. The training prediction is exactly $y - c_C$; the passive inputs require neither deficits nor a Gram inverse. The fitting proof §E.4.4 applies because the metric bounds, initial operator bounds, exact Gram, and label allowance have just been verified. In particular its normalized training Gram remains above $\lambda/4$, residuals decay, and parameters converge.

For clarity, the comparison does not follow from fitting alone. Source coordinate error η has dense RMS at most η and selected metric norm at most 2η . Exact isometry therefore gives, for source RMS bounds A_1, A_2 , pairing error at most $3\eta(A_1 + A_2) + 9\eta^2$, and paired initialized-action error at most 18η . Learned actions integrate those defects against residuals. The source-energy proof §E.4.5 controls the selected readout by $O(Y/\sqrt{\lambda})$. With $V_C = [h_{C,a}^{(L)}/\sqrt{m}]_a$, $Q_C = V_C^* V_C$, $T_C = V_C Q_C^{-1}$, and $e = (c_C - c_n)/\sqrt{m}$, the error variable $w_C - R_L w_n + T_C e$ has the exact cancellation $+2V_C e - 2T_C Q_C e = 0$. This is (177), not the false assertion that $T_C V_C^*$ is the identity on all features. Remaining terms are handled by signed damping and residual-weighted integrals as in §E.4.6. Backward terms and Gram inverses involve training points only; passive forward subtraction uses the declared source curves and a maximum over them introduces no factor p .

Consequently the identical recurrence certificate (94) gives

$$\sup_{t \in [0, \infty]} \max_{i \leq m+p} |f_C(t, x_i) - f_n(t, x_i)| \leq \mathcal{A}_n \eta + \mathcal{D} e^{-8\ell}.$$

Here $\mathcal{A}_n$ is exactly (93), with coefficient 64 in its $\sqrt{\ell}$ exponent, and $\mathcal{D}$ is (82). The entire label interval permits the completely numerical substitutes

$$\begin{aligned}\bar{\mathcal{A}}_n &= 2000e^{44}\beta^{42L}\frac{Y}{\lambda}(1+\lambda^{-1/2})(1+\sqrt{\ell})e^{64\sqrt{\ell}}, \\ \bar{\mathcal{D}} &= 236\beta^{9L}\frac{Y}{\lambda}(1+\lambda^{-1/2}),\end{aligned}$$

by (189)–(190). For the stronger target Y/n , take

$$\eta = \min\{1, Y, S, Y/(2n\bar{\mathcal{A}}_n)\}, \quad \bar{\mathcal{D}}e^{-8\ell} \leq Y/(2n). \quad (236)$$

Source comparison is used only through T_0 . Afterward each flow is compared to its own value at T_0 , using the separate fitted tails (165). Both flows continue to evolve; neither is frozen or played back. Their limits give the same endpoint bound.

The complete retained real-coordinate inventory is at most

$$1020(L+1)R^2 + 36R + 10m(d+1) + pd + (m+p) + D_{\text{alg}}. \quad (237)$$

Indeed (107) counts moving matrices, initialized copies, metrics and inverse caches, training/Gram buffers, readout workspace, and training data. Four reusable neuron buffers add at most $36R$; passive input and panel output storage add $pd + (m+p)$. The separately charged D_{alg} contains the scalar evaluator, program, and evaluator workspace; it may not conceal width-dependent source arrays. Dense weights, coefficient vectors, jets, panel anchors, and selected index lists are discarded. There is no retained temporal forcing or panel schedule. The count concerns real coordinates, not bits, conditioning, setup work, or numerical integration step count.

An exact input-span reduction improves the dimension overhead. Let $U_0 \in \mathbb{R}^{d \times k}$ have orthonormal columns spanning all declared inputs, $k \leq \min(d, m+p)$. Replace inputs by $U_0^\top v_i$ and the first matrix by AU_0 . Every first-layer update lies in this span, so $A(t)(I - U_0U_0^\top) = A(0)(I - U_0U_0^\top)$. The reduced and original coupled trajectories agree on the panel by induction through the layers. Each Gaussian row of $A(0)U_0$ has covariance I_k , and all panel inner products and the population Gram are unchanged. Use $R = 2m + k + 1 + 2(2m+p)J(K+1)$ and replace d by k in (237); a safe additional charge for the map and both original/transformed panel data is $3(m+p)d$. This is exact on the declared panel, not a whole-sphere reduction.

Finally impose the directly checkable eventual degree gate

$$K+1 \leq 8 \log(en). \quad (238)$$

It follows eventually from (234) and (236). Squaring the three terms of (233) with $(u+v+w)^2 \leq 3(u^2+v^2+w^2)$, and using the exact input-span count, gives a sufficient envelope

$$\begin{aligned}C\beta^{CL}(m+p)^2 \left[\left(\frac{Ym}{\gamma} \right)^4 [\log(en)]^3 \right. \\ \left. + \left(\frac{m}{\gamma} \right)^2 [\log(en)]^2 [\log(e + \log(en))]^2 \right] + C(m+p)d + D_{\text{alg}}.\end{aligned} \quad (239)$$

Here only the final envelope uses an absolute numerical C . To verify its parameter dependence directly, the recurrences above give $H_j \leq \beta^{3j}$, $\tau_j \leq \beta^{5L-2j+1}$, $U/a \leq \beta^{60L}$, and $\mathcal{K} \leq \beta^{14L}$. For example the intermediate exponent bounds $A_* \leq \beta^{8L}$, $H_* \leq \beta^{11L}$, $D_0 \leq \beta^{24L}$, $K_{\text{src}} \leq \beta^{32L}$, $\max e_j \leq \beta^{17L}$, $T_Q \leq \beta^{22L}$, and $U \leq \beta^{57L}$ follow by geometric unrolling, using $10s \leq \beta^2$ and $L \leq \beta^L$. Also $\mathcal{K}/\lambda \geq 1$, so the lower-order initial ranks and ℓ^2 term are absorbed in the second displayed term. The bound $S \leq 1$ is used only for these source coefficients; the explicit factor $(Ym/\gamma)^4$ has not been removed by spending the label allowance. The two sample/gap factors in (239) must not be identified or compared by treating m and $1/\gamma$ as equal.

F.7 Assembly of the headline theorem

For $m \geq 2$, the dense-variability proof in §D.9 and §D.11 selects a deterministic *training index*, not an arbitrary sphere point. To recall the reason, for one initialized feature vector $H \in \mathbb{R}^m$, the positive Gram gap and fourth-moment argument give

$$\text{tr Cov}(H(y^\top H)) \geq \frac{\gamma^3 \mu_{2,L}}{16\mu_4} \|y\|_2^2.$$

At least one training coordinate carries this variance divided by m . The initialized Gram CLT and the independent second run then give an onset-velocity fluctuation there; the finite-query time-analytic argument in §D.11 converts it into an actual positive-time discrepancy. Thus for every fixed $0 < \delta_0 < 1$, eventually

$$\Pr \left\{ \|f_n - \tilde{f}_n\|_{\mathcal{X}} \geq c_{\phi,L,\delta_0} \frac{Y\sqrt{\gamma}}{\sqrt{n}[\log(en)]^{5/2}} \right\} \geq 1 - \delta_0$$

for the declared panel $\mathcal{X}$, with exactly the positive coefficient in §D.11. This is not an endpoint lower bound. Apply the absolute-error theorem above with failure probability δ_0 as well. On the intersection of the upper and lower events, whose probability is at least $1 - 2\delta_0$,

$$\frac{\|f_C - f_n\|_{\mathcal{X}}}{\|f_n - \tilde{f}_n\|_{\mathcal{X}}} \leq \frac{[\log(en)]^{5/2}}{c_{\phi,L,\delta_0} \sqrt{\gamma n}} \rightarrow 0.$$

For every fixed positive ratio tolerance, the limsup of its failure probability is therefore at most $2\delta_0$. Since δ_0 is arbitrary, this proves convergence in probability, not merely a constant-factor comparison. The successful source events also supply the fitted limits, and the lower event excludes a zero denominator, so the stated failure convention causes no extra term. Independence between the compressed construction and the lower event is unnecessary; a union bound suffices.

This proves the Logarithmic row and its all-time comparison in Theorem 3.1. The other two rows use their unchanged proofs: Legendre’s signed-stability and optimized-order arguments are in §E.3.3 and §E.3.5; Harmonic’s joint time/input source count and inverse are in §E.4.1 and §E.4.9, using the label-explicit inverse (46) rather than its optional small-label simplification. Their full-sphere errors also become negligible relative to the corresponding dense-pair discrepancy at the displayed orders. More explicitly, Legendre’s error bound (206) divided by the dense lower scale is at most a fixed-problem, fixed-confidence multiple of $[\log(en)]^{-1/2}$; Harmonic’s Y/n bound gives the same ratio estimate as above. Their upper events attain every fixed confidence at eventual width, so the same arbitrary-failure-budget argument proves (4) for both sphere models. These intersections cost only a change in the qualitative sufficient width, not in the storage powers. The query domains remain distinct: a declared-panel certificate is not an unseen-input theorem. No input-span condition, passive-label use, bounded-activation-value assumption, or additional label cap was added.

For $m = 1$ the additive Y/n theorem still holds, but positive uncentered Gram alone need not give nonzero dense variability (a constant final activation is a counterexample). The multiplicative comparison is therefore stated for $m \geq 2$. The strict initialization-only compiler above may be expensive; neither its retained-coordinate theorem nor its finite-jet provenance asserts near-linear or near-quadratic setup for this new panel construction.